

The mass gap over the compact simple groups

Colin Bundschu¹

¹University of California, Berkeley

September 18, 2026

1 The label calculus

The section constructs the label data over the scalar-pair stage $\mathbb{P}$ (Definition 1); the classical reading is Remark 1's.

Definition 1 (The ground data). The ground data are the positive naturals, generated from one by the successor and closed under the sum and the product, each injective in each argument ($a + c = b + c$ forces $a = b$, and $ac = bc$ forces $a = b$). A *read* of an object is an operation on its data given by formula in the displayed operations, returning ground data and decidable at every argument; an object *reads* the value the operation returns, and every ground object is natural data with decidable reads. Order is gap-existence: $q < p$ is the existence of the *gap*, the natural g at $q + g = p$, one value by the sum's injectivity, and of two naturals exactly one of three holds, a gap one way, the equality $q = p$, or a gap the other way, the trichotomy read, a sum exceeding its first summand by the shape of its data. A *count* reads how often a stated condition holds over a finite family: the naturals joined by the *vacant read*, the sum's unit, a summand at that unit fixing the other member and a factor at it reading it back, the sum injective in each argument with the product so at an occupied factor, the trichotomy running with the unit below every natural, a count *occupied* at a natural's read, and the *difference* of two counts reading the gap at the trichotomy's below outcome and the vacant read at the further two: the counts are a carrier with the displayed sum's unit. The *cofactor* c at $Yc = X$ is one value by the product's injectivity where it exists; the *division* of a natural K at an occupied natural d is the pair of its *quotient* q and *remainder* r at the naming identity $K = dq + r$ with r below d , one pair by the trichotomy and the sum's injectivity (two quotients at distinct values put a multiple of d at or beyond d inside the remainders' gap), the cofactor its read at the remainder the sum's unit; the *least natural* at a comparison *stepping* up the naturals, a decidable comparison held at a natural and so at its successor, from a stated start and at a stated natural at or beyond the start holding it, the comparison's witness, is the held natural whose predecessor refuses, or the start where the start holds, one value by the trichotomy and the stepping (two such naturals at distinct values put the lesser's successor's hold against the greater's predecessor's refusal); it reads by halving: on a segment refusing at its foot and holding at its top, the ends' sum's quotient at two sits strictly inside a segment of two or beyond, the comparison at that quotient moving the end it matches, the halving closing on a held natural whose predecessor refuses, one comparison per halving, and at a comparison reading the natural's multiple of a stated natural against a stated natural the division's quotient at its successor holds it, the halving's own witness; order and factor statements are witness existence in this vocabulary, and a witness is a field of its statement: the gap a field of a stated order, the cofactor of a stated factorization, and

a derived order's witness composed by formula through the sum and the product; a comparison is *priced* where its witness is so composed and displayed, the display its price. The scalar datum is a *pair*, written $[a:b]$ at $a, b \in \mathbb{N}$, the colon a separator of its data. The reads are displayed once and are the pair's whole interface:

$$[a:b] + [c:d] = [ad + cb : bd], \quad [a:b] [c:d] = [ac : bd],$$

the order $[a:b] < [c:d]$ is the natural read $ad < cb$, and two pairs are one value exactly at $ad = cb$, the orientation read of two rays. A natural enters as $[a:1]$; a finite family sums by the fold over its index order, two fold orders equal by the ground identities. A later site may name a pair directly from two natural expressions; a naming that enters a value read maps every representative of its arguments to one value, the homogeneity principle's own test. On pairs the cofactor is total, the construction at the components' cross products, and the gap is the order's own witness at the cross-multiplied read. A comparison that runs on both sides of an equality is the *balance pair* $\langle u:v \rangle$'s, two ground data read jointly, the colon a separator of its data, the reads displayed once: sums add componentwise, the product reads cross-added, $\langle u:v \rangle \langle u':v' \rangle = \langle uu' + vv' : uv' + vu' \rangle$, the order $\langle u:v \rangle < \langle u':v' \rangle$ is the cross-added read $u + v' < u' + v$, and two balance pairs are one value exactly at the equality $u + v' = u' + v$, a summand on both members fixing the value; the equal-membered class is the sum's unit, that fixing its law, and a fold over a finite family seeds at the sum's unit and adds the members, total at every family, in this carrier and in every carrier with its displayed sum's unit; a family multiplies by the fold as well, seeded at a stated datum, in every carrier whose displayed product exchanges and regroupes, two fold orders and seed placements one value at those reads, and at a key occupied once in the family the fold with that key's factor withdrawn, against the key's own factor, reads the whole fold, the factor rejoining across the exchanges; a ground datum enters at a one-member site, $\langle u:v \rangle + w = \langle u + w : v \rangle$, two data's sum and product entering at their sites' sum and product, one value each by the displays' entries, and the displayed product distributes across the displayed sum, $\langle u:v \rangle (\langle u':v' \rangle + \langle u'':v'' \rangle) = \langle u:v \rangle \langle u':v' \rangle + \langle u:v \rangle \langle u'':v'' \rangle$, each member the ground identities' read at the displays' entries. The pair's *side* is its two members' own trichotomy read, the equality among the outcomes, the exceeding member's gap its margin, and the gap of an ordered pair of balance pairs is the cross-added data's own, w at $(u + v') + w = u' + v$; matched lists of ground data pair componentwise, one balance pair per coordinate, the members equal exactly where every coordinate's are, and the *dot* of two matched lists, in this carrier and in every carrier with the displayed sum's unit and product, is the fold of their coordinate products, the coordinate reads' one contraction, exchanging its two lists at each product's exchange. The balance-pair displays read at every datum with the displayed sum, product and order reads, the scalar pairs among them, and a later site may name a pair at a balance-pair first datum: such a composite is the balance pair of its members' pairs at the one shared second datum, a natural expression,

$$[\langle u:v \rangle : c] = \langle [u:c] : [v:c] \rangle,$$

its reads the two displays' compositions, displayed once,

$$\begin{aligned} [\langle u:v \rangle : c] + [\langle u':v' \rangle : c'] &= [\langle uc' + u'c : vc' + v'c \rangle : cc'], \\ [\langle u:v \rangle : c] [\langle u':v' \rangle : c'] &= [\langle uu' + vv' : uv' + vu' \rangle : cc'], \end{aligned}$$

the order $[\langle u:v \rangle : c] < [\langle u':v' \rangle : c']$ the cross-added read $uc' + v'c < u'c + vc'$ with one value exactly at its equality, the side the first datum's own at every second datum, and a scalar pair entering at a one-member site through the members' display. $\mathbb{P}$ denotes the scalar pairs with their reads, the stage glyph. Every statement reads pairs through the displayed reads alone, under the

one *homogeneity principle*: every read is degree-known in each of its data, so a rescaling of any datum by ground data composes a comparison's witness by formula, one factor on each side, and the trichotomy reads the rescaled comparison back: a comparison holds exactly at its rescaling. Here it reads the representative freedom to one value, as it reads the form's scale and the family's: at $ad = cb$, rescaling both components of $[a : b]$ by d and of $[c : d]$ by b yields one pair, so two pairs at the identity read one value against every third. A formula in the displayed operations over stated data, ground data or pairs, reads its *collected display*, the coefficient family over its monomials at the sorted exponent lists, the ground identities' fold at the displays' entries, and two formulas at one collected display read one value at every datum list. The harmonic numbers are $H_k = \sum_{j=1}^k [1 : j]$; the *factorial* $k!$ is the product fold over the naturals from one to k , one at the vacant range; a pair of reads is componentwise data. Every later object reads the ground data through these operations.

Definition 2 (The ground polynomials). A polynomial over the ground pairs, or over their balance pairs, is the recursion $P = c + zQ$, a constant read with a shifted family, finitely deep; equivalently its occupancy family of coefficients at the variable's monomials, the words $1, z, z^2, \dots$ under the product with 1 their unit, the constant the unit monomial's read. The sum adds componentwise over the monomial keys, the product convolves, each coefficient one fold over the key pairs at its monomial, and the evaluation is the Horner read on the recursion. The *division* at a monic S , over the balance pairs, is the pair (Q, R) , the *quotient* and the *remainder*, at the naming identity

$$P = SQ + R ,$$

R 's monomial keys below S 's top, one value by the descent at the monic top, the identity descending the monomials from the top; at a linear factor $\langle z:r \rangle$, the variable against a value r , the remainder is the Horner read at r , and a *root* is a value at which the remainder's two members are equal, its multiplicity the count of iterated divisions there. The division reads exact at a stated factorization besides: a polynomial's *top* is its largest key at a coefficient off the sum's unit, one key by the trichotomy over the finitely many keys, and at factors of occupied tops the product's top sits at the summed key, the key's fold reading one summand at the two tops with every further summand reading a factor beyond its top, the sum's unit there, so the product's top coefficient is the tops' product, off the sum's unit by the entry carrier's product injectivity; a product reading the sum's unit at every key therefore puts one factor's every coefficient at that unit, and a shared factor of occupied top withdraws at two products of one value, the further factors' difference multiplying to the sum's unit and reading it key by key with the factors one value. The *exact division* at a divisor of occupied top and a dividend stated as the divisor's product descends at the tops: the quotient gains the entry carrier's cofactor at the two top coefficients on the cleared monomial at the keys' gap, the withdrawn dividend, the dividend joined to that multiple's balance partner, keeps its occupied keys below the prior top, and the descent closes at the dividend's top count with the quotient one value at the injectivity; the reads run over the ground pairs, over their balance pairs, and over the polynomials as the entry carrier in turn, each level's cofactor and injectivity the level below's own. The *derivative* P' is the successor-weighted shift, $P'_h = (h+1)P_{h+1}$ at every key, additive, with the Leibniz read $(PQ)' = P'Q + PQ'$ one convolution identity per key $((i+j)P_iQ_j)$ collected at either weighting over the key pairs joining to the successor). At a monic S the *remainder lists* are the coefficient lists at the monomial keys below S 's top, sums componentwise and the product the convolution's remainder at the division by S .

Definition 3 (The ground eliminations). A matrix over the ground data is its occupancy family of entries at row and column keys, total over stated finite key lists, the entries scalar pairs or

their balance pairs (Definition 1). A *minor* at matched row and column lists is a formula in the entries, the first-row fold: at one row the entry, and at rows (i, I) and columns J the fold over J 's splittings into a column j and a co-list K , the two joining to J , of the entry a_{ij} against the minor at $(I; K)$, the summand's side the count of J 's columns before j , even at the first member and odd at the second. Iterating the splittings, a minor at matched lists collects to the *assignment fold*: one summand per assignment of a distinct column to each row, at the product of the assigned entries, the summand's side the assignment's swap grading (Construction 1), the steps' splitting sides joining to the count of the assignment's inversion list at the withdrawn co-lists. The Gauss elimination is the pivot descent: the first step's entries are the entries themselves, the second's the two-list minors at their naming $b_{ij}^{(1)} + a_{i1}a_{1j} = a_{11}a_{ij}$, and Sylvester's identity names each later step from the two before it, at every $k \geq 1$,

$$P_k b_{ij}^{(k+1)} + b_{i,k+1}^{(k)} b_{k+1,j}^{(k)} = b_{k+1,k+1}^{(k)} b_{ij}^{(k)},$$

$b_{ij}^{(k)}$ the pivot-bordered minor at the lists $(1..k, i; 1..k, j)$ and P_k the leading minor at $(1..k)$, one identity of minor formulas per step, the new entry the identity's cofactor: the descent closes on the sum and the product, each division exact at the prior pivot, the identity's read at every prior pivot outright: an equal-membered pivot reads the division's both members at the sum's unit, the identity collecting its join there with the cofactor's product beside it. The identity derives from the fold's key reads. A minor is additive and scale-reading in one row or one column, the assignment fold reading that key's entry once per summand, so a minor at a column read as a stated combination expands over the members, one summand per member at its coefficient; a minor at two row keys, or two column keys, of one value reads equal members, the keys' transposition pairing the assignments at exchanged sides (Construction 1); the exchanged row and column lists read one determinant, each assignment paired with its flip, the inverse permutation (Construction 1), at one term, the entries' product reindexed along the assignment, and at one class, the grading additive over composition with the identity even so the flip shares its assignment's class; and the assignment fold regroups at any row's column choice, one family per column of the entry against the withdrawn lists' minor at the sides' join. The *adjugate* of a square list is the cofactor family, the entry at (i, j) the minor at the withdrawn row j and column i on the side of the count $i + j$: a row's fold against its own cofactors collects the regrouped assignment fold, the determinant, and against a further row's cofactors it reads equal members, the fold a determinant at that row repeated, so the list against its adjugate reads the determinant's diagonal, the *adjugate identity*, and the adjugate against the list reads that diagonal as well: the exchanged row and column lists read one determinant at every struck pair, so a column's fold against its own cofactors collects the exchanged assignment fold and a further column's reads equal members. Sylvester's identity then reads at the shifted bordered list: the variable t joins the diagonal over the balance-pair polynomials, a carrier with the displayed reads (Definition 2), and the shifted bordered determinant is monic of degree $k + 2$ in t , the identity assignment's product reading the top at the unit with every further assignment at two or more entries off the diagonal, the moved places a pair at the least, two degrees below. Join the two *solved columns*, the shifted list's fold against its adjugate's two border columns, each the shifted bordered determinant at its own border key alone by the adjugate identity; the determinant at the border columns read to the solved ones expands twice, the two-column multilinear fold keeping the border pair's selections alone with every further selection a column repeated at equal members, against the direct fold along the two determinant-scaled unit columns: the two reads join the adjugate corner's two-by-two determinant against the shifted bordered determinant to that determinant's square against the shifted leading minor, the border cofactors' sides joining even at each pair, and the monic determinant withdraws at the division's one value (Definition 2). The

evaluation at the equal-membered pair reads each shifted minor at the stated entries, the minor a formula in the displayed operations with the evaluation reading sums to sums and products to products (Definition 2): the identity holds at the entries themselves. A step pivots at the first uncrossed row holding an entry with unequal members, at that row's least such key, a row or column exchange swapping the members of the minors keyed at both. The descent reads the determinant with the rank: the terminal bordered minor at the last key pair is the minor at the full lists, the determinant, each step's entry the identity's own naming. At a leading minor off the sum's unit, the prior pivot's own read, a pivot column whose every bordered minor reads equal members reads the determinant at them as well: each bordered minor expands along its border row, one shared cofactor family over the leading columns with the off-unit leading minor the border key's own coefficient, a leading row's bordered minor a repeated-key read outright, so the column joins the leading columns' cleared combination at that off-unit coefficient, and the determinant, expanded at that column over the combination's members, reads a repeated column key per summand, equal members throughout with the off-unit coefficient withdrawing at the product's injectivity. The step keeps the span and the kernel: the update display reads the prior pivot's multiple of each new row as the pivot's multiple of the entry row against the entry's border multiple of the pivot row, and the entry rows read back at the divisions' witnesses, the two pivots off the unit, so the step's rows and the entry rows sit in one span at cleared combinations; a *kernel* member, a coordinate vector over the balance pairs pairing every row at the sum's unit, extends across the step at the pivot's clearing: the stepped member scales by the pivot and the pivot coordinate enters as the stepped row read's balance partner, the pivot row's own pairing collecting to the pivot's multiple of that read against its partner, the sum's unit, and the extension pairs the entry rows at the sum's unit exactly where the stepped member pairs the stepped rows, the update display's read at the cleared pivots, with the column exchanges permuting the coordinates as stated data. The descent names the *rank*, the pivot count, and the *kernel list* enters per pivot-free column by the back solve: at the terminal step every entry reads equal members and the unit family with one at the column's own key seeds the member, each earlier step extending at the pivot's clearing as above. The members pair every row at the sum's unit, the solves' reads; the list is independent, each member off the unit at its own pivot-free coordinate, the crossed pivots' product, and at the unit at the further pivot-free ones; and every kernel member joins the list's span, the pivot coordinates one value at the off-unit pivots by the product's injectivity with the cleared combination at the pivot-free coordinates reading the member whole: the list's length, the pivot-free count, is the *kernel dimension*. At a square list whose every leading minor sits off the sum's unit the descent pivots at the leading keys in order: each step's stored entries are the bordered minors at the crossed keys, the leading row past the crossed ones reading at an earlier key a repeated-key read of equal members and at its own key the next leading minor, so the step's pivot is that key's own; bordered by a stated family's balance partners, one column per member, the list's pivot-free columns are the border's, and the kernel member at border key k pairs the list against the member's leading coordinates at the crossed pivots' product on the family's member k , so the determinant's multiple of the leading coordinates is the pivots' product's multiple of the adjugate against that member, the adjugate identity at the product's injectivity: at the unit family the adjugate reads off one descent, each entry the cofactor at the pivots' product, and at one stated pairing vector, the family that vector alone, the determinant's multiple of the kernel member's leading coordinates is the pivots' product's multiple of the adjugate against the vector, the Gram-dual solve's cleared coordinates, each coordinate the cofactor at the product. At such a list symmetric at its entries, grown by one row and its column at the row's own entries, the grown list's descent is the list's own with the grown row stepped through the stored pivot rows in order: a stored entry at exchanged keys reads the exchanged row and column lists' one determinant at the symmetric entries, so the pivot row's entry at the grown

key is the grown row's own at the pivot key, the grown row's step is Sylvester's identity against the pivot row so extended, and the terminal entry is the grown determinant; the grown list's leading minors below the grown key are the list's own, the leading lists there the list's, and at the grown key the grown determinant, so at a grown determinant off the sum's unit every leading minor of the grown list sits off the sum's unit, the grown list a square list of this clause's own. A list whose Gram splits at a partition of its members, every cross pairing of equal members, reads its adjugate blockwise: each block's own adjugate at the further blocks' determinants, an assignment placing a member off its block's columns reading a cross entry of equal members, so the assignment fold collects over the blocks' own assignments alone with the determinant the blocks' determinants' product; and its Gram-dual solve at a stated pairing vector runs one block at a time, each block's adjugate column against its own determinant, the further blocks' determinants the shared scale. The *resultant* $\text{Res}(p, q)$ of two polynomials at tops of unequal members, over the ground data or any carrier with their displayed reads (Definition 2), is the site-datum determinant of their *shift matrix*, rows the monomial shifts $x^a q$ at a below p 's degree and $x^b p$ at b below q 's, columns the monomial keys below the degrees' sum, one minor read; the adjugate row solves

$$Ap + Bq = \text{Res}(p, q)$$

at polynomials of degrees below the other's, the solved witness's whole verification; a kernel vector reads the *cross pair* (A, B) at those degrees with Ap the balance partner of Bq , so the resultant reads equal members exactly at an occupied cross pair, the descent's kernel read, and at a Bézout read $up + vq = 1$ it reads unequal members: at $\check{c}$ the memberwise swap of uB , the identity maps a cross pair's A to $(Av + \check{c})q$ at q 's degree beyond A 's, every cross pair the sum's unit there.

Definition 4 (The deck families). The *deck relation* at a variable y is the monic quadratic $y^2 + 1 = wy$, its coefficient the *chord* w , and its *shift read* is $y^j + y^{j+2} = wy^{j+1}$, the relation's y^j -multiple, one identity per key; a *root pair* of the relation is two data at product one joining to the chord. The *deck families* are the polynomials in the chord at the relation's recursion,

$$p_j + p_{j+2} = w p_{j+1}, \quad Q_j + Q_{j+2} = w Q_{j+1},$$

over the balance-pair carrier (Definition 2), each later member the join's datum, at the seeds $p_1 = w$, $p_2 + 2 = w^2$ and $Q_1 = 1$, $Q_2 = w$: p_b is monic of degree b and Q_b of degree e at $e + 1 = b$, each of its degree's parity in the chord, the recursion moving one parity per step from the seeds: the recursion's running pair reads its two degrees with the unit tops jointly, the join's degree the chord multiple's own at one beyond its factor's and the join's top that multiple's, the withdrawn member two below. The families' column reads at the relation close by induction on the shift read, one cross-multiplied identity per step:

$$y^{2b} + 1 = y^b p_b(w), \quad \langle y^{2b} : 1 \rangle = \langle y^2 : 1 \rangle y^e Q_b(w),$$

the *plus read* and the *word read*, the word the geometric telescope's, $\sum_{c+d+1=b} y^{2d} = y^e Q_b(w)$ at $\langle y^{2b} : 1 \rangle = \langle y^2 : 1 \rangle \sum_{c+d+1=b} y^{2d}$; at a root pair the plus read collects to the matched powers' sum, $z^b + z'^b = p_b(w)$, the product's b -th power one.

Construction 1 (Places, shapes, and the column exhibit). The declared scalar of the development is one natural, the *residue* r , unconstrained, declared here once with its successor name, the *fundamental count* $d_f := r + 1$ (Definition 1): d_f is read wherever a site counts the standard list below (a loop's closure, a trace, a row bound), r wherever a site reads the ordering data (the beta-sets' gaps, the fundamentals, a reduced shape's depth), the residue name the pairing's own

read. $V := \mathbb{P}^{d_f}$ with its standard pairing and standard list $v_1, \dots, v_{d_f}$; the power $V^{\otimes k}$, with the pairing of monomial tensors and the *place action* of S_k , $\sigma(x_1 \otimes \dots \otimes x_k)$ the monomial reading x_j at place $\sigma(j)$, each σ a permutation matrix on the monomial tensors, orthogonal at the transpose identity $\sigma^\top \sigma = 1$, the transpose the *inverse permutation's* matrix, the assignment flipped, its value at j the witness k at $\sigma(k) = j$. A permutation's *inversion list* is the occupancy family of the place pairs $i < j$ at $\sigma(j) < \sigma(i)$; composing with an adjacent transposition moves the list by exactly one pair (the adjacent pair's read flips, and the pairs meeting the two places exchange theirs), so the descent at a reversed adjacent pair per step ends with every place pair sorted and writes σ as a word of adjacent transpositions, one letter per pair of the list. The *swap grading* is additive over composition, two words concatenating, *even* and *odd* its two classes: a word's letters flip one place pair's read each, the others exchanging theirs, so along a word from the sorted assignment the list moves by one pair per letter and the letter count is the witness sum $L = c + 2w$, c the inversion list's count and w the count of letters returning a pair: every word for σ shares the list's parity, one class per permutation, the identity's words at the paired letters alone, $L = 2w$, even outright; a transposition is odd, its inversion list its own place pair with two pairs per place between its two, one pair beyond an even family. The permutations are the words of unit content, one letter at every place once, one word per permutation at the assignment's own reads and at the factorial's count, the enumeration blocking at the head letter, one block per occupied letter at the withdrawn content's words, the unit content folding one letter out per step, the product fold's own read (Definition 1); and the *permuted display* of a list is its reads at a permutation's places, one entry per place. A permutation's *cycles* are its orbit words, from each least unwalked place the iterated images to the start's return: a walk's members are distinct until the return, two equal iterates reading equal predecessors down to the start at the assignment's own injectivity, so each walk closes within the place count, every place sits on one word, and the *cycle count* is the words' count; a permutation composed between a second permutation and its inverse permutation walks the composed orbits at the images, each iterate the second permutation's image of the walk's own at the flipped place, one orbit word per orbit either way, so the cycle count is one value across the composition. The *grading* is by content: the monomial tensor at indices $(i_1, \dots, i_k)$ has *content* the occupancy family μ over the letters, a member letter at the count of places reading it; $V^{\otimes k}$ is the orthogonal sum of its content summands, a homogeneous member read at its coordinate family over its own content's monomials, the pairing the coordinate fold within one summand, and two members at distinct contents pairing at equal members outright, the orthogonality's read. A *shape* is its *column multiset*: finitely many columns, each of a positive length at most d_f , an occupancy family over the lengths; the *row list* is the derived display, λ_i the count of columns of length at least i , the occupancies the row list's consecutive gaps, one shape per display, the rows sorted at the occupied depths with the *degree* k the box total. The shape's *beta-set* X_λ is the fold over the column multiset's occupancies n_ℓ , columns of length exactly ℓ : descending from $x_{d_f} := n_{d_f} + 1$ by $x_i = x_{i+1} + n_i + 1$, a set of d_f distinct positive naturals with sorted display $x_1 > \dots > x_{d_f}$, each consecutive gap x_i against x_{i+1} the occupancy successor $n_i + 1$, and the rows' read $x_i = \lambda_i + g + 1$ at the join $i + g = d_f$; the unit shape's is the *unit set* $U = \{1, \dots, d_f\}$, a full column moves every member by one successor, so a shape's class at full columns is its gap word $(n_1, \dots, n_r)$ alone, and a shape of c columns reads its *complement's* set, the gap columns at $\ell + \ell' = d_f$, at the interval complement, $x_i(\lambda^*) + x_{i'}(\lambda) = c + d_f + 1$ at every index pair joining to $d_f + 1$. The *column exhibit* is the wedge tensor

$$t_\lambda := w_{\lambda'_1} \otimes w_{\lambda'_2} \otimes \dots, \quad w_\ell := \left\langle \sum_{\sigma \text{ even}} v_{\sigma(1)} \otimes \dots \otimes v_{\sigma(\ell)} : \sum_{\sigma \text{ odd}} v_{\sigma(1)} \otimes \dots \otimes v_{\sigma(\ell)} \right\rangle,$$

one factor per column at the sorted lengths $\lambda'_1 \geq \lambda'_2 \geq \dots$, a column of one box reading its

vector, $w_1 = v_1$, and a deeper column the balance pair of the two graded sums over S_ℓ , read componentwise on the monomial coordinates (Definition 1), each monomial on its permutation's side; the monomials are distinct, so the two members' supports are disjoint and the members differ, of content the columns' own read, letter i at the count of columns of length at least i : the row list λ ; a wedge with a repeated vector has equal members, the repeat's transposition an odd move fixing every monomial and mapping either side's list onto the other's. The shapes compose by column union, the row lists adding, and the *unit shape* $\mathbf{1}$ is the union's unit; the powers compose by the tensor product with the scalars' line its unit, two spanning lists' tensors spanning the composite, a tensor of two combinations collecting to the pairs at the coefficients' products with a content summand of the composite reading its list at the pairs whose factor contents join to it, the grading's split; and $\mathbf{1}$'s exhibit is the scalar one on that line.

Lemma 1 (A transpose-closed action splits orthogonally). *At a set of operators on a finite-dimensional space with positive pairing, closed under transpose, the orthogonal complement of an invariant subspace is invariant. The place action is such a set ($\sigma^\top \sigma = 1$, Construction 1), and so is every set of operators listed together with its transposes.*

Proof. At $u' \perp U$, g in the set and $u \in U$, $\langle gu', u \rangle = \langle u', g^\top u \rangle$ with $g^\top u \in U$, so the value is one of u' 's pairings against U , of equal members by the complement's defining datum. $\square$

Construction 2 (The unit action). The matrix units act on V by $E_{ij}v_m = \delta_{jm}v_i$ and on a power by the Leibniz sum $\sum_{\text{places}} 1 \otimes \cdots \otimes E_{ij} \otimes \cdots \otimes 1$, written E_{ij} again; the pairing reads $E_{ij}^\top = E_{ji}$ place by place, and the commutators close on the table

$$[E_{ab}, E_{cd}] + \delta_{da} E_{cb} = \delta_{bc} E_{ad},$$

read on the standard list and extended to powers by the Leibniz rule, cross-place terms commuting. A *unit space* is a finite-dimensional space with positive pairing, operators E_{ij} obeying the table and the transpose identity, and the joint E_{ii} -grading by contents; the powers are unit spaces, and so are sums, tensor products (by the Leibniz rule), and invariant subspaces, a composite power's fold splitting at the arrangements' join, each factor's places collecting to its own fold read against the other factor's arrangement, the composition's letter actions the factors' own (Construction 1). The E_{ii} commute with integer grading, so the projection onto one content is a polynomial in them (one Lagrange interpolant per coordinate through the occupied integers), and an invariant subspace is graded: each of its elements' content components is its member. E_{ij} at $i < j$ is a *raising*, its content move the letter-pair move at (i, j) , the count at i raised by one and the count at j lowered by one, strictly up the dominance order; $i > j$ is a *lowering*, strictly down; E_{ii} reads the grading.

Construction 3 (The generator table). A *table space* is a finite-dimensional space with positive pairing and one interface of data, its fields. The *coordinates* are a finite key list, a *content* is a matched list over them (Definition 1), and the space is the orthogonal sum of its content summands, the occupied contents one *class*, every two joined by a balance-pair fold over the simple list. The *simple list* is finitely many contents $\alpha_1, \dots, \alpha_s$; the *root list* is finitely many contents, each a natural fold $\sum_i c_i \alpha_i$ occupied off the unit fold; the *dominance order* is the fold's witness read, $\mu \preceq \lambda$ at the join $\mu + \sum_i c_i \alpha_i = \lambda$, the coefficients one solve at the simple folds' injectivity below. Per root the table reads a *raising* E_α at the content move $+\alpha$, its transpose the *lowering* $F_\alpha := E_\alpha^\top$ at the move's balance partner; per simple index, the *triple* (E_i, F_i, H_i) at the displays $[E_i, F_i] = H_i$ and $[F_i, H_i] = 2F_i$, H_i reading a content summand at its *coroot pair* $\mu(\alpha_i^\vee)$, one balance pair per content, additive over the contents' sum, at $\alpha_i(\alpha_i^\vee) = 2$; the simple folds' coroot reads are injective, a balance-pair fold over the simple list reading equal members at every coroot pair the unit fold, so

two occupied contents at one value under every coroot pair are one content (their joining fold's reads equal members throughout), the joint H -grading is the content grading, and the projection onto one content is a polynomial in the H 's, one Lagrange interpolant per coroot read through the occupied values; the *reflection* $s_i\mu$ is the content at the join $\mu = s_i\mu + \mu(\alpha_i^\vee)\alpha_i$, and a content is *dominant* where every coroot pair reads its upper side or equal members; the *closure*, the commutator of two field operators a span fold over the list at the moved content, a raising's or the H -span's at a root or the unit move and a lowering's below it; the *form*, one symmetric pair read per content pair, additive in each argument over the contents' sums, its doubled read against a root clearing at the root's length, the *root coroot pair* $\mu(\alpha^\vee)$ the cofactor at $\langle\alpha, \alpha\rangle\mu(\alpha^\vee) = 2\langle\mu, \alpha\rangle$ with the simple triples' reads its instances, the *root reflection* $s_\alpha\mu$ the content at the join $\mu = s_\alpha\mu + \mu(\alpha^\vee)\alpha$; the *root triples*, per positive root the display $[E_\alpha, F_\alpha] = H_\alpha$ at H_α reading a content summand at its root coroot pair, the further display $[F_\alpha, H_\alpha] = 2F_\alpha$ the grading's own read, the commutator moving by α at $\alpha(\alpha^\vee) = 2$; and the *Casimir*,

$$C := Q + \sum_{\alpha} [\langle\alpha, \alpha\rangle : 2] (E_\alpha F_\alpha + F_\alpha E_\alpha) ,$$

the fold over the positive list at Q the diagonal read of the form's square $q(\mu)$ per content, commuting with every field operator: a content- λ vector whose every raising image is the sum's unit reads C at $q(\lambda) + \langle\lambda, 2\rho\rangle$, the fold's own evaluation, each $E_\alpha F_\alpha$ the commutator's read there and the weights collecting to the root fold's, $[\langle\alpha, \alpha\rangle : 2]\lambda(\alpha^\vee) = \langle\lambda, \alpha\rangle$. Sums of table spaces at one field set and one shared class are table spaces, and tensor products, the operators by the Leibniz rule, the coroot pairs additive and the classes adding; and so is an invariant subspace, graded by the projections. A member's table enters with every field a stated read, a series' tables polynomial pairs in the series' residue and a fixed member's finite data; the powers with the matrix units are the first table, the coordinates the letters, the roots the letter-pair moves with the consecutive moves simple, the triples the units' own displays, and the Casimir list the units themselves.

Construction 4 (The member weight tables). A member's weight data instantiate the content fields of Construction 3: the coordinates, the simple and root lists with their folds, the coroot pairs, the form, and the root fold $2\rho := \sum_{\alpha} \alpha$ over the positive list. Two reads are the series' shared derivation, run on their displayed positive lists. *The reflection permutes the positive list*: the list is reflection-closed and a family member at one occupied key is its own simple root, every further member occupied off any one key, two reads per family of the displayed folds; $s_i\alpha_i$ is α_i 's balance partner, and every further positive root keeps a positive fold coefficient off α_i (a one-key fold is its simple root's, the singleton check), kept by s_i , so s_i permutes the positive list off α_i with α_i 's own key fixed under the image assignment, a further member's image at α_i 's balance partner reading its root-fold dot on the lower side while every member's dot sits positive at its fold, $\langle\alpha, 2\rho\rangle = \sum_k c_k \langle\alpha_k, \alpha_k\rangle$, and its square positive at its family's displayed length, the form's fold read, and the fold reads $s_i(2\rho) + 2\alpha_i = 2\rho$: the root fold's coroot pairs read $(2\rho)(\alpha_i^\vee) = 2$ at every simple index. *The residue is one fold*: at the highest root's fold $\theta = \sum_i c_i \alpha_i$ and the form's read $\langle\theta, \theta\rangle = 2$,

$$\langle\theta, 2\rho\rangle = \sum_i c_i \langle\alpha_i, \alpha_i\rangle = 2r ,$$

the first equality the form's fold read at $\langle\alpha_i, 2\rho\rangle = \langle\alpha_i, \alpha_i\rangle$, the coroot read above cleared at the length, and the second the member's *derived residue* r , one fold per member. A third read joins them, the *reflection grading*, at any member whose positive list holds the permutation read, the first shared read's, s_i permuting the list off α_i . The member's *Weyl list* is the words' finite list of positive-list images: a word in the s_i acts on contents through the reflection joins, each letter an

involution ($s_i s_i \mu = \mu$, the moved coroot pair its own balance partner); a reflection keeps the form, the form's additivity read at the two joins,

$$\langle \mu, \nu \rangle = \langle s_i \mu, s_i \nu \rangle + \nu(\alpha_i^\vee) \langle s_i \mu, \alpha_i \rangle + \mu(\alpha_i^\vee) \langle \alpha_i, s_i \nu \rangle + \mu(\alpha_i^\vee) \nu(\alpha_i^\vee) \langle \alpha_i, \alpha_i \rangle ,$$

the middle terms collecting with the last to the sum's unit at the flipped coroot pairs, the halved products' read, so $\langle s_i \mu, s_i \nu \rangle = \langle \mu, \nu \rangle$; and a word fixing each member of the positive list keeps every content's coroot pairs, the kept form against the kept simples, so the member's coroot reads return the content, the displayed cofactors' solve with the fixed members' contents the coroot lists outright: two words at one image are one content map, and the list is finite at the images. A word's element reads its *inversion list*, the occupancy family of the positive list at the images of the positive list's balance partners; one further letter s_i moves the list by exactly one member, α_i 's read flipping and the further members' exchanging along s_i 's permutation of the positive list, so a word's letter count is the witness sum $L = c + 2w$ at c the inversion list's count, one parity per element, *even* and *odd* the two classes, the grading additive over composition, Construction 1's argument at the member's list with the place pairs' inversion lists the matrix units' instance; and the grading's fold reads $\rho, w\rho + \sum_{\alpha \in N} \alpha = \rho$ at N the inversion list, one letter per induction step, $s_i \rho + \alpha_i = \rho$ at the coroot read one, the halved $(2\rho)(\alpha_i^\vee) = 2$, the letter's permutation of the positive list mapping the further members across. The grading closes on four reads. *The transport*: at an image root $w\beta$ the reflection is the transported one, $s_{w\beta} \mu = w s_\beta w^{-1} \mu$, the coroot pair the kept form's own read, $\mu((w\beta)^\vee) = (w^{-1} \mu)(\beta^\vee)$; a reversed word's element shares its parity class at the one letter count. *The creation read*: along a word each letter moves the inversion list by one member, entering or leaving at the letter's own root and exchanging at the further members, so a list member names its *creating letter*, the member the later letters' image of that letter's simple root. *The deletion*: a word whose letter count exceeds its element's inversion list's count holds a leaving flip, the leaving member some earlier letter's created one, and the transport rewrites the word at two fewer letters, the leaving letter crossing to its creating one; so every word reduces to the list's count, and the element whose inversion list is the unit family is the identity, its reduced word letter-free. *The regular reads*: $w\rho = \rho$ forces w the identity, the ρ -fold's members at positive root-fold dots reading the list to the unit family; a dominant content κ at every simple coroot pair beyond the unit reads every positive pair beyond the unit, the fold's members positive, and its images are one per element with at most one dominant: at u off the identity the inversion list is occupied, a member γ the image of a partner, $u\beta = \tilde{\gamma}$ at a positive β , so $\langle u\kappa, \gamma \rangle$ is the balance partner of $\langle \kappa, \beta \rangle$, on its lower side, and $u\kappa$ sits off the dominant reads. A series member's list is the signed place permutations of its coordinates, every sign vector at B_ℓ and C_ℓ and the even flip counts at D_ℓ , the displayed list: the s_i generate it, the transpositions the place part with the last letter the flip at B_ℓ and C_ℓ and the flip pair at D_ℓ , a flip's transported images the further flips, and distinct signed permutations read distinct coordinate images, one list member each.

The series, at the rank ℓ , the coordinate count, the contents matched lists (Definition 1), the B and D lists at doubled coordinates, every read one value across the doubling by the homogeneity principle, their contents the coroot cofactors' lists at one parity each. Per series the coroot and form displays read the halved products, $2 \langle \mu, \alpha_i \rangle = \mu(\alpha_i^\vee) \langle \alpha_i, \alpha_i \rangle$, one identity per simple class off the displayed data. B_ℓ at $\ell \geq 2$: the simple list α_i , 2 on key i against its balance partner on key $i+1$ at $i < \ell$, and α_ℓ , 2 on the last key; the positive list in three families at their folds, the difference family $c_k = 1$ on $i \leq k < j$, the sum family $c_k = 1$ on $i \leq k < j$ with 2 on $j \leq k \leq \ell$, and the short family $c_k = 1$ on $i \leq k \leq \ell$; the form $[1 : 4]$ at the coordinate dot, the lengths 2 at $i < \ell$ and 1 at the last; the coroot pairs $2\mu(\alpha_i^\vee) = \langle \mu_i : \mu_{i+1} \rangle$ at $i < \ell$ and $\mu(\alpha_\ell^\vee) = \mu_\ell$; the root fold's key- i read $4g + 2$ at the join $i + g = \ell$; and θ the sum family's first member, 2 on keys one and two, its fold $c = (1, 2, \dots, 2)$, the residue fold $2r = 2 + 4g' + 2$ at $g' + 2 = \ell$: $r = 2g$ at $g + 1 = \ell$. C_ℓ at $\ell \geq 3$:

the simple list the letter-pair moves at $i < \ell$ and $\alpha_\ell, 2$ on the last key; the difference family $c_k = 1$ on $i \leq k < j$, the sum family $c_k = 1$ on $i \leq k < j$ with 2 on $j \leq k < \ell$ and 1 at the last, and the long family $2e_i$ at $c_k = 2$ on $i \leq k < \ell$ with 1 at the last; the form $[1 : 2]$ at the dot, the lengths 1 at $i < \ell$ and 2 at the last; the coroot pairs $\mu(\alpha_i^\vee) = \langle \mu_i : \mu_{i+1} \rangle$ and $\mu(\alpha_\ell^\vee) = \mu_\ell$; the root fold's key- i read $2g$ at $i + g = \ell + 1$; and θ the long family's first member, its fold $c = (2, \dots, 2, 1)$, the residue fold $2r = 2g' + 2$ at $g' + 1 = \ell$: $r = \ell$. D_ℓ at $\ell \geq 4$: the simple list the doubled letter-pair moves at $i < \ell$ and $\alpha_\ell, 2$ on keys $\ell - 1$ and ℓ ; the difference family $c_k = 1$ on $i \leq k < j$, the sum family $c_k = 1$ on $i \leq k < j$, 2 on $j \leq k$ at the joins $k + 2 \leq \ell$, and 1 on the last two keys, its $j = \ell$ member $c_k = 1$ on $i \leq k$ at $k + 2 \leq \ell$ with 1 on the last key; the form $[1 : 4]$ at the dot, every length 2; the coroot pairs the cofactors $2\mu(\alpha_i^\vee) = \langle \mu_i : \mu_{i+1} \rangle$ at $i < \ell$ and $2\mu(\alpha_\ell^\vee) = \mu_{\ell-1} + \mu_\ell$; the root fold's key- i read $4g$ at $i + g = \ell$ with the last key's read equal members; and θ the sum family's first member, the residue fold at its coefficient sum, $r = 1 + 2g' + 2$ at $g' + 3 = \ell$: $r = w$ at $w + 3 = 2\ell$.

The fixed members enter at their simple keys, the contents the coroot lists, $\mu(\alpha_i^\vee)$ the key- i read, the Cartan reads $\alpha_j(\alpha_i^\vee)$ and the length list the member's displayed data with the form's simple reads the halved products $\langle \alpha_j, \alpha_i \rangle = [1 : 2] \alpha_j(\alpha_i^\vee) \langle \alpha_i, \alpha_i \rangle$, the root fold at its coroot list, 2 on every key, the doubled fundamental fold $2 \sum_j \omega_j$ at the adjugate rows below, one content at those coroot pairs by the simple folds' injectivity, and the residue the one fold off the halved products: G_2 , two keys at the edge read $\alpha_2(\alpha_1^\vee)$ the balance partner of 3 and $\alpha_1(\alpha_2^\vee)$ of 1, lengths $[2 : 3]$ and 2, θ 's fold $(3, 2)$, the residue fold $2r = 3[2 : 3] + 2 \cdot 2$: $r = 3$. F_4 , four keys at the chain edges with the middle edge doubled toward the short pair, lengths $(2, 2, 1, 1)$, θ 's fold $(2, 3, 4, 2)$, the residue fold $2r = 4 + 6 + 4 + 2$: $r = 8$. E_6, E_7, E_8 , every length 2 at the simply-laced edge lists (the chain 1-3-4-5-6 with the branch 2-4; the chain grown by 7, and by 7-8), θ 's folds $(1, 2, 2, 3, 2, 1)$, $(2, 2, 3, 4, 3, 2, 1)$ and $(2, 3, 4, 6, 5, 4, 3, 2)$, the residue folds twice the coefficient sums: $r = 11, r = 17, r = 29$.

The adjugate rows. The fixed members' fundamental folds are the cleared displays

$$e\omega_i = \sum_k a_{ik} \alpha_k,$$

at the *fold key* e , one at G_2, F_4 and E_8 , three at E_6 and two at E_7 , the rows a_i the five members' tables in their stated order,

$$\begin{pmatrix} 2 & 1 \\ 3 & 2 \end{pmatrix}, \quad \begin{pmatrix} 2 & 3 & 4 & 2 \\ 3 & 6 & 8 & 4 \\ 2 & 4 & 6 & 3 \\ 1 & 2 & 3 & 2 \end{pmatrix}, \quad \begin{pmatrix} 4 & 3 & 5 & 6 & 4 & 2 \\ 3 & 6 & 6 & 9 & 6 & 3 \\ 5 & 6 & 10 & 12 & 8 & 4 \\ 6 & 9 & 12 & 18 & 12 & 6 \\ 4 & 6 & 8 & 12 & 10 & 5 \\ 2 & 3 & 4 & 6 & 5 & 4 \end{pmatrix},$$

$$\begin{pmatrix} 4 & 4 & 6 & 8 & 6 & 4 & 2 \\ 4 & 7 & 8 & 12 & 9 & 6 & 3 \\ 6 & 8 & 12 & 16 & 12 & 8 & 4 \\ 8 & 12 & 16 & 24 & 18 & 12 & 6 \\ 6 & 9 & 12 & 18 & 15 & 10 & 5 \\ 4 & 6 & 8 & 12 & 10 & 8 & 4 \\ 2 & 3 & 4 & 6 & 5 & 4 & 3 \end{pmatrix}, \quad \begin{pmatrix} 4 & 5 & 7 & 10 & 8 & 6 & 4 & 2 \\ 5 & 8 & 10 & 15 & 12 & 9 & 6 & 3 \\ 7 & 10 & 14 & 20 & 16 & 12 & 8 & 4 \\ 10 & 15 & 20 & 30 & 24 & 18 & 12 & 6 \\ 8 & 12 & 16 & 24 & 20 & 15 & 10 & 5 \\ 6 & 9 & 12 & 18 & 15 & 12 & 8 & 4 \\ 4 & 6 & 8 & 12 & 10 & 8 & 6 & 3 \\ 2 & 3 & 4 & 6 & 5 & 4 & 3 & 2 \end{pmatrix},$$

each row's witness the coroot identities $\sum_k a_{ik} \alpha_k(\alpha_j^\vee) = e\delta_{ij}$ over the member's Cartan reads, one fold per key pair, and one row the θ -fold's e -multiple, the highest root the fundamental at

that key. The rows' form reads close the fundamental data: the fold at the halved products $2 \langle \alpha_k, \omega_j \rangle = \delta_{jk} \langle \alpha_k, \alpha_k \rangle$, the coroot displays' read, gives

$$2e \langle \omega_i, \omega_j \rangle = a_{ij} \langle \alpha_j, \alpha_j \rangle, \quad e \langle \omega_i, 2\rho \rangle = \sum_k a_{ik} \langle \alpha_k, \alpha_k \rangle,$$

the second at the root fold's all-twos coroot list, every read positive by its shape at the natural entries, and $C_2(\omega_i) = \langle \omega_i, \omega_i + 2\rho \rangle$ is the member's fundamental Casimir list, one fold per key; two contents are one class (Construction 3) exactly at one remainder family of their cleared folds at the key, the remainder families additive over the contents' sums, every content the unit's class at key one.

The fixed members' positive lists. A fixed member's root list enters at doubled coordinates, its members matched lists (Definition 1) with the coroot pairs the halved products at the displayed form scale, and the simple list's halved products read the member's displayed Cartan reads, one identity per key pair. The three E -members share eight coordinates at the form $[1 : 4]$ against the dot: the simple list is α_1 , one on the first and eighth keys with its balance partner on the six between; α_2 , two on the first two keys; and the letter-pair moves α_{j+1} at $2 \leq j \leq 7$, two on key j with its balance partner on the key before. The positive list in three families: the *difference family*, two on a key with its balance partner on an earlier key; the *sum family*, two on two keys; and the *half-sum family*, one on the eighth key and one or its balance partner on each earlier key, the partners even in count, at the counts 28, 28 and 64, the join $28 + 28 + 64 = 120$. E_8 reads the whole list; E_7 reads the members at the seventh and eighth entries' sum of equal members, the counts $16 + 15 + 32 = 63$; E_6 reads those with the sixth and eighth entries' sum of equal members as well, the counts $10 + 10 + 16 = 36$; a member letter's own selecting sums read equal members, so the member's letters keep its selection, and a selected member's families read at its own keys, the selection's fixed partners. The families close under the letters, the permutation read: a letter-pair move permutes its two coordinates, each family to itself off its own letter; the α_2 letter reads the first two entries to their swapped balance partners, exchanging the difference and sum members meeting the first pair and keeping the half-sums at two moved entries, the partner count moved by an even read; and the α_1 letter moves a member by α_1 at its coroot pair, the pair one, its balance partner or the unit read at every member off α_1 , a difference or sum member at pair one to the half-sum at its moved entries and a half-sum back, the partner parity kept. Every member off the simple list holds a simple key at the coroot pair one: a difference member its top key's letter-pair move or, on the eighth key, α_1 at partners off the first key and the second key's letter-pair move at it; a sum member its lower key's letter-pair move off the first key, its top key's move on it, and α_1 at the first with the eighth; a half-sum member an ascent's letter-pair move, an earlier key's balance partner against its successor's one, α_2 at a leading plus pair, or α_1 , the trichotomy complete by the sign list: off an ascent the list descends, off a leading plus pair at most the first entry is one, and the even partner count leaves α_1 alone. The *fold descent* moves at that key: the image is a member by the closure reads, its dot against the *height vector*

$$(0, 4, 8, 12, 16, 20, 24, 92)$$

falling by the letter's length at the pair, so the walk ends on the member's simple list and the fold witness collects along it, one natural multiple per step: the root list is the families at their natural folds (Construction 3's field). The height dots are positive per family, the entries increasing up the keys for the differences, two entries' sum at one positive member for the sums, and the last entry clearing the earlier seven's sum for the half-sums, the join $84 + 8 = 92$; and the vector is the root fold's, its coroot pairs two on every key, one content with the doubled fundamental fold $2 \sum_j \omega_j$

at the simple folds' injectivity. F_4 enters at four coordinates, the form $[1 : 4]$ against the dot: the simple list is α_1 , two on the second key with its balance partner on the third; α_2 , two on the third with its balance partner on the fourth; α_3 , two on the fourth key; and α_4 , one on the first key with balance partners on the rest; the families are the *singles*, two on a key, the *difference and sum family*, two on a key with its balance partner or two on a later key, and the *half-sums*, one on the first key and one or its balance partner on each further key, the counts $4 + 12 + 8 = 24$, the singles and half-sums at length one and the pair family at two. The third letter flips the fourth entry, the letter-pair moves permute their coordinate pairs, and the α_4 letter moves a member by α_4 at its coroot pair: the families close, and every member off the simple list holds a key at a positive pair, twenty reads off the displayed families; the height vector is $(22, 10, 6, 2)$, the first entry clearing the rest's sum at the join $18 + 4 = 22$, its coroot pairs two on every key, and the fold descent reads the natural folds along it. G_2 enters at its simple keys outright, the six displayed folds

$$(1, 0), \quad (0, 1), \quad (1, 1), \quad (2, 1), \quad (3, 1), \quad (3, 2),$$

lengths $[2 : 3]$ at the first, third and fourth members and 2 at the rest, the form's fold reads; the closure is the displayed reflection reads, s_1 keeping the second key with the first key's image at the join $x + c_1 = 3c_2$, and s_2 keeping the first with the second's image at the join $y + c_2 = c_1$, ten reads off the flipped letters, and the six folds' sum $(10, 6)$ reads the coroot pairs two on every key; its alternant coordinates are three doubled keys at the form $[1 : 12]$ against the dot, α_1 two on the first key with its balance partner on the second and α_2 two on the second and third keys with the doubled partner on the first, each member's coordinates its fold's read, the root-fold vector twelve on the third key against the partners of four and eight on the first two, every member's dot positive. So every fixed member's positive list holds the permutation read, the reflection grading's scope: the member's Weyl list, its grading and its ρ -fold are the third read's at that list.

The defining tables. The series' defining table is the signed keys' space, each key i with its partner key $\bar{i}$, the contents $\pm e_i$ with the unit content at B_ℓ , the pairing diagonal at one with the B_ℓ null key at $[1 : 2]$: the root operators are the moved units at the form's twist, a difference move E_{ij} with $E_{j\bar{i}}$'s balance partner, a sum move $E_{i\bar{j}}$ with $E_{j\bar{i}}$'s, a C_ℓ long move the single unit $E_{i\bar{i}}$, and a B_ℓ short move $E_{i0} + 2E_{0\bar{i}}$ at the weighted transpose. The *form* b and its *dual pair* c are the twist's two data: b pairs the partner keys at one on both sides at B_ℓ and D_ℓ , at one against its balance partner at C_ℓ , and reads the B_ℓ null square at the balance partner of $[1 : 2]$, the short move's own forcing ($b(Eu, v) + b(u, Ev)$ reads the sum's unit at every key pair, and at the null key against $\bar{i}$ it collects one against the doubled null square); c , the invariant line of the doubled defining power, pairs the partner keys at one on both sides at B_ℓ and D_ℓ , at one against its balance partner at C_ℓ , and reads the B_ℓ null square at the balance partner of 2, the annihilation's own forcing (the short move's Leibniz image collects the null coefficient against the paired coefficient's double at the sum's unit); and the two data read at the product the identity, one read at the displayed entries. Each triple display is one read, $[E, F] = H$ at the coroot pair with F the pairing's transpose, the further display $[F, H] = 2F$ the grading's own read, the consecutive and last letters' checks displayed reads.

The member alternants. A dominant content β at every simple coroot pair beyond the unit (a series member's sorted lists, the entries strictly decreasing) reads its *alternant pair*, the reflection-graded fold over the Weyl list,

$$a_\beta := \left\langle \sum_{w \text{ even}} x^{w\beta} : \sum_{w \text{ odd}} x^{w\beta} \right\rangle,$$

monomials at matched-list exponents keyed by the contents, one key per content at the member's coroot solve (the displayed cofactors', Construction 3) with a coordinate display reading its exponent lists in by the displayed coroot pairs, the fixed members' keys the coroot lists outright; an

odd element exchanges the members and a reflection-fixed list reads equal members (Construction 1's reads at the member's grading); the cleared presentation multiplies by the full monomial at the top entry, every exponent natural, and the grading reads the letters, $\text{sgn } \sigma$ against the flip count's parity at B_ℓ and C_ℓ and $\text{sgn } \sigma$ alone at D_ℓ , the flip pairs even at their letter pairs. Per coordinate the *chord* t_i enters at the deck relation $x_i^2 + 1 = t_i x_i$, and the deck families read the columns (Definition 4): the plus column the plus read $x^{2b} + 1 = x^b p_b(t)$ and the word column the word read $\langle x^{2b} : 1 \rangle = \langle x^2 : 1 \rangle x^e Q_b(t)$ at the join $e + 1 = b$, the families' degree and parity reads the definition's. The cleared alternant is the signed-column minor (Definition 3), the Weyl fold regrouping per permutation at the flip folds within columns, the minor's own splitting fold, entry (i, j) the pair $\langle x_i^{b_1+b_j} : x_i^{g_j} \rangle$ at the gaps $b_j + g_j = b_1$, and the rows factor at the word columns' reads, $\langle x^{b_1+b_j} : x^{g_j} \rangle = \langle x^2 : 1 \rangle x^h Q_{b_j}(t)$ at the join $1 + h = b_1$, the coordinate's power one read per row. At ρ , the root fold's half, the cofactor at two per key, the symbol degrees are consecutive and the minor column-reduces over the chord carrier to the chord alternant, the assembly's pair product at the chords (Theorem 1): $\prod_{i < j} \langle t_i : t_j \rangle$ at C_ℓ , and $\prod_{i < j} \langle t_i^2 : t_j^2 \rangle$ at B_ℓ and D_ℓ , the columns even, the D_ℓ read on the plus-column minor with the last column's two, the signed minor's last column at equal members and the doubled alternant's two clearing by the product's injectivity. The chords clear to the coordinates at the pair identities

$$x_i x_j \langle t_i : t_j \rangle = \langle x_i : x_j \rangle \langle x_i x_j : 1 \rangle, \quad x_i^2 x_j^2 \langle t_i^2 : t_j^2 \rangle = \langle x_i^2 : x_j^2 \rangle \langle x_i^2 x_j^2 : 1 \rangle,$$

each one cross-multiplied read at the deck relations, and the power bookkeeping closes exactly, the coordinate's shift the factor count, $g + 1$ at C_ℓ , $2g + 1$ at B_ℓ and $2g$ at D_ℓ over the pair count g at $g + 1 = \ell$: the *assembly identity* is

$$a_\rho = \prod_{\alpha} \langle x^{\alpha+} : x^{\alpha-} \rangle,$$

the fold over the member's positive list at the moves' two sides, one factor per positive root, the difference family at $\langle x_i : x_j \rangle$ or its squares, the sum family at $\langle x_i x_j : 1 \rangle$ or its squares, and the per-coordinate family at $\langle x_i^2 : 1 \rangle$, every identity one minor computation per series.

Lemma 2 (Tops). *A top is a content vector whose every raising image is the sum's unit. (i) A vector off the sum's unit, of a content maximal among its invariant subspace's occupied contents, is a top. (ii) A top's content is dominant: at a simple index i , the triple $(E, F, H) := (E_i, F_i, H_i)$ reads $[E, F] = H$ and $[F, H] = 2F$ with H reading a content at its coroot pair $\langle p : q \rangle$ (Construction 3; at the matrix units the site datum at $E_{ii} = E_{i+1, i+1} + H$ over consecutive letters, the pair $\langle \mu_i : \mu_{i+1} \rangle$), and the commutator's Leibniz fold runs over the word's splittings,*

$$E F^m = F^m E + \sum_{a+b+1=m} F^a H F^b;$$

a lowering strictly descends the finite content set, so at a least $m \geq 1$ the iterate $F^m v$ is the sum's unit, and at a top v of content μ the least m reads the coroot pair's gap, $q + m = p + 1$: the pair on its upper side or equal members at every simple index, the content dominant. At a positive pairing the pair reads its side through the pairing as well, the crossed exchange's read: the lowered image's self-pairing walks across the transpose to the exchange at the top, $\langle Fv, Fv \rangle + q \langle v, v \rangle = p \langle v, v \rangle$ at the raising's unit, and at v off the sum's unit the positive pairing forces q at or below p , the consecutive letters' pair read at the matrix units. (iii) The column exhibit t_λ is a top at content λ .

Proof. (i) A raising's image is homogeneous at a content above the maximal occupied one, so its component in every occupied summand, and with them the image, is the sum's unit. (ii) Reading

the display at the top v of content μ , both outer reads are that unit and each summand reads its site's coroot pair $\langle p:q+2b \rangle$, two down per lowering at $\alpha_i(\alpha_i^\vee) = 2$, on one shared iterate $F^a F^b v$, the splitting's word one letter short of F^m and its read off the sum's unit at the least m : the collected pair's members are equal, $mp+m = mq+m^2$ at the splitting count read $\sum_{a+b+1=m}(2b+1) = m^2$, and the product's injectivity at the factor m reads the coroot pair's gap, $q+m = p+1$. The crossed exchange: $\langle Fv, Fv \rangle = \langle v, EFv \rangle$ at the transpose, and $EFv = FEv + Hv$ reads Hv at the raising's unit, the coroot pair's two members against the self-pairing. (iii) A raising E_{ij} reads each column by Leibniz, sending a v_j to v_i inside a wedge already holding v_i when $j \leq \lambda'_c$, a repeat at equal members, and reading the sum's unit outright when $j > \lambda'_c$, the wedge free of v_j (Construction 1). $\square$

Lemma 3 (The lowering span). *The invariant subspace generated by a top v is the span W of the lowering monomials read at v . Every raising and H -read maps W to itself, and every operator reading the grading with them, W graded by its words. At the matrix units the letter pairs compose from the adjacent ones: a pair at a gap beyond one is the crossed composite at an interior letter, $E_{ij} = [E_{ik}, E_{kj}]$ (Construction 2), raisings and lowerings both; so the words run at the adjacent lowerings alone, a further lowering the descent's composite, and a vector reading the sum's unit at every adjacent raising reads it at every raising: the raising kernel is the adjacent stack's. The words stabilize at the finite dimension (length $k d_f$ at the matrix units). Consequently the generated subspace has every content at or below the top's, and its top-content summand is the line $\mathbb{P}v$, a lowering word of positive length reading strictly below. A span membership is one decidable read: at a list of members independent in the pairing, the Gram's adjugate column against its determinant solves the pairing data (Definition 3), and a vector joins the list's span exactly where the residual, the determinant-scaled vector against the solved combination, reads the sum's unit componentwise: the solve reads the residual against every list member at the sum's unit, and a member vector puts the residual inside the span, where its self-pairing reads that unit at the positive pairing; and independence is the determinant's unequal-membered read at the Gram, one decidable read. The solve reads off the descent: each leading minor of the Gram, the Gram determinant of a leading part, the list's first members, sits off the sum's unit, and the adjugate against the pairing data reads off the descent bordered by the data's balance partner, one column (Definition 3). At a stated list the membership runs at its collection, one member joined per refusal along the list, the refusal the joined Gram's determinant off the sum's unit, read at one descent of the collection's Gram grown a row per join (Definition 3); at a grading of the list's members (Construction 1's contents) the collection runs one descent per grade, its Gram the grade's own, a member joining its grade at that descent's refusal, and the grades are listed at their members' first appearance along the list with each grade's members in the list's order, the collection's own arrangement, a joined member opening its grade where the grade is vacant and joining it otherwise: the collected members are independent, every listed member sits inside the collection's span, and the collection's members are listed members, so a vector joins the list's span exactly where it joins the collection's, the two memberships one read at the collection's residual, the joined-collection read; and an independent list is its own collection. An independent list's count sits at or below the coordinate count, and an independent list at the coordinate count spans every vector of the width. Two independent lists inside one span read one count, and the count comparison runs at the first list's independence alone, the adjugate column against the determinant the membership's own decidable read at an independent second list. An operator's trace read, the operator at its matrix against the space's content coordinates (the monomial coordinates at the matrix units, Construction 1), at an orthogonal spanning list of the span, the members pairwise perpendicular with every self-pairing off the sum's unit, is the fold of the diagonal operator pairings, each against the further members' self-pairings, and two orthogonal*

spanning lists read one trace at the lists' cleared products. A span member expands over the list, the projection fold

$$\left(\prod_i \langle y_i, y_i \rangle\right) v = \sum_j \langle v, y_j \rangle \left(\prod_{i \neq j} \langle y_i, y_i \rangle\right) y_j .$$

A pairwise perpendicular list of members off the sum's unit is independent, and at a pairwise perpendicular list the Gram's determinant reads the self-pairings' product. The determinant-scaled residual then reads one value entrywise with the product-scaled residual, the self-pairings' product against the vector at the projection fold's own sum withdrawn, at every vector of the list's width. The graded dimensions are then counts at the read, one member joined per refusal.

Proof. The graded closure. At a lowering word Fw' with w' shorter, and X a raising or an H -read, $XFw' = FXw' + [X, F]w'$; the first term closes by induction on the word length, and the closure field reads $[X, F]$ as a span fold at the moved content (Construction 3; a unit or a difference of diagonals at the matrix units), closing by that induction where the move is a root or the unit move and by the span where it lowers. *The adjacent stack.* At the matrix units the crossed composite reads at the table's shared-letter read with the far delta clear at the distinct ends (Construction 2), and the descent on the gap closes every pair from the adjacent ones, raisings and lowerings both; the composite's two members each read through an adjacent image, so a vector reading the sum's unit at every adjacent raising reads it at every raising. *The words.* The words stabilize, a lowering descending the dominance order strictly over the space's occupied contents, finitely many at the finite dimension. *The residual.* The determinant clears an independent list, a determinant of equal members reading its adjugate column as a dependency perpendicular to the list, the column's fold of the members the sum's unit at the pairing read, against the independence, the descent closing at the withdrawn last member's own determinant; and the determinant reads a dependency's coefficients back: a combination reading the unit family pairs against each cofactor column at its own coefficient's multiple of the determinant, the diagonal cofactor pairing the determinant's read with the off-column pairings the sum's unit, while the combination's side reads that unit through the members' own pairings, so at a determinant of unequal members every coefficient is the sum's unit by the product's injectivity. *The descent.* Each leading part of an independent list is independent, the withdrawn tail's coefficients read at the unit family one member at a time, so each leading minor of the Gram sits off the sum's unit. *The collection.* Each join's residual refuses the span collected so far, the joined list independent at a refusal by the membership's refusal above and dependent at a passing member, the determinant reading the dependency, the Gram symmetric at the dot's exchange with every leading minor off the sum's unit at the joins' refusals (Definition 3), so the collected members are independent; every listed member sits inside the collection's span, a passing member at its residual's read and a joined member as the collection's own. An independent list's leading parts are independent, and a passing residual puts the next member inside its leading part's span, the membership's combination joined with the member at one off-unit coefficient reading the unit family against the extended part's own independence, so each residual refuses and every member joins. *The count.* At an occupied first coordinate some member's first read is off the sum's unit, the pivot, and every further member joins the pivot at the scaled pair, the member at the pivot's first read against the pivot at the member's own first read's balance partner, every moved first read the sum's unit, so the moved list reads its tails at one coordinate withdrawn, independent at the pivot's clearing, and the descent closes on the count; at every first read the sum's unit the tails alone read the list. A vector off the list's span joins it at the membership's refusal, the joined list independent one past the count, against the bound. *Two lists.* A member joins the second list's span at a clearing, a scalar off the sum's unit with the member's cleared read a combination of the second list, one coefficient row per member; a combination of

the first list reads through the rows' combination at the clearings multiplied into the coefficients, the product's injectivity reading a coefficient back at its clearing, so the rows are independent at the second list's count, the width bound closes the comparison, and either count sits at or below the other at the two lists' memberships. *The trace read.* The first list's off-unit reads clear the shared factor at the product's injectivity, at the pairing's bilinearity: the projection fold's sides' gap is perpendicular to every member at the orthogonality and a span member itself, its self-pairing the members' pairings' fold at the sum's unit and the gap that unit at the positive pairing, the display's read; pairing one list's projection fold against its own further member reads the *crossed fold*, $\langle x_i, x_{i'} \rangle \prod_j \langle y_j, y_j \rangle = \sum_j \langle x_i, y_j \rangle \langle y_j, x_{i'} \rangle \prod_{k \neq j} \langle y_k, y_k \rangle$; and the second list's diagonal operator pairings expand over the first, the double fold collecting at the crossed fold with the off-diagonal terms entering at the first list's own orthogonality's unit, so the two trace folds read one value at the cleared products. *The perpendicular lists.* A combination at the unit family pairs against each member to the member's own coefficient at its self-pairing, the orthogonality clearing the rest and the positive pairing the coefficient, so the list is independent. Every assignment off the identity reads an off-diagonal pairing's factor, at equal members by the perpendicularity, so its term enters the fold at the sum's unit and the identity assignment's product is the read. Each residual is perpendicular to every list member, each pairs the two residuals' gap at the determinant against the vector's read, so the gap's self-pairing reads the sum's unit and the positive pairing reads the gap at that unit entrywise, the two residuals one value at every coordinate. $\square$

Lemma 4 (Blocks). *Write W_v for the subspace a top v generates (Lemma 3); the clauses read at the span calculus's stated data (Lemma 3). A subspace enters as a stated list of content-homogeneous members with its membership reads, each a decidable read at the list: the content projections read a general member's components as members (Construction 3; Construction 2 at the matrix units), so the homogeneous form covers every list, and the closure read is one membership read per raising or lowering and list member, the operator's image joining the list's span, an H -read scaling a homogeneous member inside it (Construction 3's grading). A map enters as its matrix family, one matrix per occupied content and per letter-pair image of one, against the space's content coordinates (the monomial coordinates at the matrix units, Construction 1), its value at a member the matrix read with the family's linearity the matrix action's own; the moved reads are the statement's data: per raising or lowering and span member, the operator's matrix at the member's value is the family's read of the operator's image. (i) W_v is irreducible. At a stated list L of members of W_v at the closure read, every W_v member joins L 's span, or every member of L is the sum's unit; the top-content group splits the two, the group's span the line $\mathbb{P}v$ (Lemma 3). (ii) Endomorphisms are scalars. At a matrix family on W_v , the top's value a W_v member at the content λ with the moved reads at the words' spanning list (Lemma 3), the values read one cleared line,*

$$\langle v, v \rangle \varphi w = \langle \varphi v, v \rangle w \quad \text{at every member } w,$$

φw the family's value at the member w . (iii) Blocks at distinct top contents are disjoint. At a matrix family from W_v into W_u , values at the members' own contents with the moved reads and some value off the sum's unit, the top contents agree. (iv) One dimension per shape. Two tops v, u at one content λ , in powers of one degree $|\lambda|$ at the matrix units. The words' pairings read the top's content alone: two words at distinct moved contents pair at equal members outright, and at lowering words m, m' at one moved content the pairing $\langle m v, m' v \rangle$ walks the reversed word across the transposes (Construction 2), the table folds each crossing into an H -read at the content with a shorter word's pairing (Construction 3), and a raising's read at the top is the sum's unit, so the pairing is the content's own coefficient against the self-pairing $\langle v, v \rangle$, one value per word pair at every top of content λ . So, content group by content group, the word images at u of W_v 's spanning

list read the group's own Gram at the coefficient scale, each group's images independent inside W_u 's group at the moved content, the groups' counts sit at or below each other's, $\dim W_v = \dim W_u$, and $d_\lambda := \dim W_{t_\lambda}$ is one count per shape (Lemma 2(iii)).

Proof. (i) A top-content member of L off the sum's unit clears v into L 's span, and every W_v member with it, a lowering word's read at v with the closure transporting the letters across L 's combinations (Lemma 3). At every top-content member of L the sum's unit, L pairs v at equal members, distinct contents pairing at equal members outright, the grading orthogonal (Construction 1; Construction 3); a word's pairing walks across the transposes, $\langle x, E_{ij} y \rangle = \langle E_{ji} x, y \rangle$ with the transposed letter keeping L 's span (Construction 2; the lowerings the raisings' transposes, Construction 3), so L pairs every W_v member at equal members; and a member's self-pairing, the fold of its own combination's pairings, reads equal members: the member is the sum's unit at the positive pairing. (ii) The top's value sits in the top-content group's span, the line $\mathbb{P}v$ (Lemma 3), the display its residual read there; and the induction along the span's words passes the display down, a further member a lowering's image of an earlier one (Lemma 3), the moved read mapping the earlier member's display across the lowering with the matrix action clearing the scalars, the further values members by the display. (iii) (ii)'s induction reads every value as its member's own word at the top's value, so the top's value is off the sum's unit. The top u sits off the sum's unit: at a unit head every word's read and every settled value is the sum's unit, against the value off it. The value list joins (i)'s data inside W_u , its closure the moved reads across W_v 's own (Lemma 3), and (i) at an off-unit value reads every W_u member into the values' span, whose occupied contents sit among the values', at or below the top's content λ (Lemma 3): u 's content sits at or below λ , λ at or below u 's at the top value's membership, and the two dominance witnesses join to the unit fold (Construction 3): the contents agree. (iv) Two words at distinct moved contents pair at equal members outright, the grading orthogonal (Construction 1). The cross-content pairings are the grading's equal-membered reads on both lists (Construction 1). The self-pairings sit off the sum's unit at the lists' independence, so each group's images are independent inside W_u 's group at the moved content, the groups' counts sit at or below each other's, two independent lists inside one span reading one count (Lemma 3), and the occupied contents' counts sum to the dimensions at the orthogonal grading. $\square$

Definition 5 (The block count). For a table space M (Construction 3; the unit spaces among them, Construction 2) and a content λ , the shapes at the matrix units, the *block count* is

$$N_\lambda(M) := \dim\{\text{content-}\lambda \text{ vectors of } M \text{ whose every raising image is the sum's unit}\},$$

a kernel dimension of the stacked raisings on the content summand (Definition 3), and the *fusion count* at contents a, b, c is $N_{ab}^c := N_c(W_a \otimes W_b)$ over blocks at those top contents, at the matrix units the shapes' blocks $W_\lambda := W_{t_\lambda}$ (Lemma 3; Lemma 2(iii)).

Lemma 5 (The block count, and the fusion data). (i) *Exhaustion.* A table space is an orthogonal sum of blocks: at occupied M take a top v at a maximal occupied content (Lemma 2(i)), split $M = W_v \perp W_v^\perp$ (Lemma 1), and descend the dimension. The complement enters as the stated members' residuals at the block's collected content groups, the joined-collection read's own residual (Lemma 3), and a joined-span member perpendicular to the block's rows sits in the complement's span alone. The descent closes at the stated members' count. (ii) *The count.* $N_\lambda(M)$ (Definition 5) adds over orthogonal sums, the kernel dimensions adding at the kernel lists (Definition 3). A block reads $N_\lambda(W_u) = \delta_{\lambda, \text{content}(u)}$ (Lemma 3, tops of a block sitting at its own top content alone by

Lemma 4(iii)'s read). So $N_\lambda(M)$ is the number of λ -blocks of every exhaustion, and

$$\dim M = \sum_{\lambda} N_\lambda(M) d_\lambda ,$$

the sum one summand per distinct top content of the exhaustion, each a shape's row list at the dominance read (Lemma 2(ii)) with d_λ its block's stated count (Lemma 4(iv)), and a shape off the tops' contents reading its count at the unit, the vacant collected group's kernel. The display refines to the contents: a content's dimension over M is the sum at $N_\lambda(M)$ against the shape's own occupancy at the content in place of d_λ , each block reading its shape's occupancy (Lemma 4(iv)'s group counts). (iii) The fusion data. $N_{ab}^c = N_c(W_a \otimes W_b)$ (Definition 5): kernel-dimension counts by construction, the stacked raisings read against the stated pair list, the two blocks' spanning lists' tensors at complementary contents (Lemma 3), contents adding over the factors with the pairs spanning the summand (Construction 1), and the pair list is independent: two pairs pair at the factors' pairings' product, a concatenated arrangement reading its split at the first factor's degree with a block's contents at one degree, and two pairs at distinct first contents pair at equal members outright, each tensor's coordinates at the sum's unit off its own contents' splits (Construction 1), so a combination reading the unit family reads its coefficients back through the two factors in turn. The kernel dimension is the display's own count, with

$$N_{ab}^c = N_{ba}^c, \quad \sum_e N_{ab}^e N_{ec}^d = \sum_f N_{bc}^f N_{af}^d, \quad N_{a1}^c = \delta_{ac}, \quad \sum_c N_{ab}^c d_c = d_a d_b,$$

the first by the flip of factors, the arrangements' exchange mapping each pair to the exchanged pair at the commuted coordinate products, an orthogonal map at the place action's transpose identity commuting with every raising (Construction 2); the second by exhausting $W_a \otimes W_b \otimes W_c$ through either pairing; the third at the unit shape's line; and the fourth by (ii) at $M = W_a \otimes W_b$. Moreover $N_{ab}^{a+b} \geq 1$: $t_a \otimes t_b$ is a top off the sum's unit at content $a + b$ with its pair's own coefficient family a kernel member off the unit family.

Proof. (i) A joined-span member perpendicular to the block's rows sits in the complement's span alone: the block's part of its combination pairs every block row at the sum's unit, its self-pairing reads that unit through the rows' pairings, and the positive pairing withdraws it (Lemma 3). Each round's top is a member of its own block, its residual reading the unit through the solve (Lemma 3), so every round withdraws at least its own pick from the stated carrier and the count bounds the rounds. (ii) The grading and the raising kernels split over an orthogonal sum, a kernel member's two parts each a kernel member of its own stated summand, the parts' raising images inside their own summands' moved spans at the closure with the two images perpendicular and their join the sum's unit, the positive pairing withdrawing both (Lemma 3); and the kernel dimensions add at the kernel lists, the two sides' lists joining to an independent spanning list of the sum's kernel, the counts the graded dimensions' read (Lemma 3). The refinement to the contents reads at the blocks pairwise perpendicular with each content summand split over their groups and the graded dimensions the counts at the read (Lemma 3). (iii) A combination reading the unit family pairs every pair at the sum's unit and its coefficients read back through the two factors in turn, per first-factor member the contraction a combination of the second factor's group pairing every group member at that unit, its self-pairing reading that unit through the pairings' fold, the positive pairing withdrawing it and the group's independence reading its coefficients at the unit family (Lemma 3), the first factor's groups then reading the coefficients themselves. At the flip of factors the exchange is orthogonal and commutes with every raising at the Leibniz fold's places exchanged (Construction 2), so a kernel

family reads across the flip at the exchanged coefficients and the two kernel dimensions read one count, each kernel list inside the other's span (Definition 3); the second by exhausting $W_a \otimes W_b \otimes W_c$ through either pairing, the composition associative at the arrangements' join (Construction 1); the parenthesized factor's exhaustion splits the triple orthogonally at its one degree, a block's contents at one degree with the exhausted carrier's members at the factors' degrees' sum, so two blocks' tensors against the further factor pair at the factors' pairings' products, the concatenated arrangements split at the one first-factor degree (Construction 1), the blocks perpendicular ((i)'s split), spanning the composite at the collected pairs (Construction 1), the counts adding over the split ((ii)); a block at a top content reads the fused count at the shape's own block: the block's members are its words' images at its top (Lemma 3), the one word family at the shape's own top spans the shape's block group by group, the images independent at the one-value word pairings with the occupancies agreeing (Lemma 4(iv)); a stacked combination of raised pairs reads the sum's unit exactly where its self-pairing does (the positive pairing), the raised tensors' pairings collecting to the factors' pairings' folds at the raisings' split over the concatenation (Construction 2), each fold the word pairings' coefficient against its top's self-pairing (Lemma 4(iv)), so the two carriers' unit reads agree at the off-unit self-pairings by the product's injectivity, the collections' refusals the pairings' own reads matched with them (Lemma 3), and the two counts are one value: either pairing's count is the display's sum, one summand per distinct top content of its exhaustion, the second pairing read across the flip; the third at the unit shape's line: the line's exhibit is the scalar one (Construction 1), a tensor against it reads the first factor's own coordinates at the vacant second arrangement, so the stated pair list is the block's own content group and the count is (ii)'s closing display, the Kronecker delta at the head's content; and the fourth by (ii) at $M = W_a \otimes W_b$, the fused carrier closed at the letters' split folds (Construction 2) with its dimension the pair list's count at every content, the two dimensions' product. At $t_a \otimes t_b$ each raising's split reads a factor's top against the other factor tensored on (Lemma 2(iii); Construction 2), so it is a top off the sum's unit at content $a + b$, and its pair's own coefficient family is a kernel member off the unit family, the kernel list's span occupied (Definition 3). $\square$

Lemma 6 (Strings, and the content symmetry). *Fix a positive root α and its triple $(E, F, H) := (E_\alpha, F_\alpha, H_\alpha)$ on a table space M (Construction 3's root triples), the matrix units' at the letter pairs. The triple's span is transpose-closed ($E^\top = F$, $H^\top = H$), so M splits orthogonally into subspaces invariant for the triple and irreducible for it (Lemma 1, descending the dimension), each generated by a triple-top v at its H -eigenvalue h , at the join $q + (h + 1) = p + 1$ over the top's coroot pair $\langle p : q \rangle$ (Lemma 2(i),(ii) at the triple), and spanned by $v, Fv, \dots, F^h v$ with $F^{h+1}v$ the sum's unit: the string's H -eigenvalues descend from the top's h by two per lowering, the depth- b eigenvalue the pair $\langle h : 2b \rangle$ at positive $b \leq h$, each once, two depths at the join $b + b' = h$ reading member swaps. At the matrix units the splitting is the residual walk at the ordered pair's height, the first letter's count against the second's: the span's contents run at the height descending, a raising moving the height key up by two, and at a content a group member off the collected members' span names a top, the membership test's residual itself (Lemma 3), per content the members closing at the exchange bound, one top per refusal, the walk's count the group's own (Lemma 3); each deeper member's self-pairing is the raise coefficients' telescoped multiple of the top's at the fold's read $E F^{b+1}v = (b+1)(h-b) F^b v$. The triple moves contents by α alone, fixing every reflection-fixed read, so the strings split each joint grading, and summing string by string,*

$$\dim M_\mu = \dim M_{s_\alpha \mu}$$

at the root reflection s_α (Construction 3; the transposition of the two letters at the matrix units): content multiplicities are invariant under every root reflection and their compositions. At the matrix

units a block's occupied contents are permutation-closed, sit at or below the top (Lemma 3), and so sit at or above its reversal $w_0\lambda$ in the dominance order, a sorted prefix sum falling as its tail rises; the reversal is occupied once, at the top's own multiplicity.

Proof. The least m at which $F^m v$ is that unit has $h+1 = m$ (Lemma 2(ii)), so the eigenvalues descend two per lowering. *The walk.* The named top is perpendicular to the collected members by the solve's identity and inside the group's span; the top's raising image sits in the content above's group span, spanned by the walk's own members there, and perpendicular to them, so the image's self-pairing reads the sum's unit at the positive pairing and the image with it. The top's gap is the pairing's read, the lowered image's self-pairing joining the second count's multiple of the top's to the first's, $\langle Fv, Fv \rangle + \nu_j \langle v, v \rangle = \nu_i \langle v, v \rangle$, the crossed exchange's read at the letter pair (Lemma 2(ii)), the positive pairing forcing the second count at or below the first, the height their gap; each deeper member's self-pairing is the telescoped multiple at the fold's read, off the sum's unit through the height and reading it at the further lowering, the coefficient's own unit there; and per content the members close at the exchange bound. $\square$

Lemma 7 (The Casimir reads the shape). $C := \sum_{p,q} E_{pq} E_{qp}$ commutes with every unit, $C E_{ab} = E_{ab} C$ (Construction 2). On a block W_v at top content λ of degree k it is therefore one scalar (Lemma 4(ii)), read at the top:

$$C|_{W_v} = q(\lambda) + \langle \lambda, 2\rho \rangle, \quad q(\lambda) := \sum_i \lambda_i^2, \quad \langle \lambda, 2\rho \rangle := \sum_{p < q} \langle \lambda_p : \lambda_q \rangle,$$

the balance-pair fold over the place pairs, a natural at the sorted rows, collected per row as the balance-pair row weights $\sum_i \lambda_i \langle r+2:2i \rangle$. The units instantiate the Casimir field (Construction 3): the diagonal read $\sum_p E_{pp}^2$ at the row squares, the pair terms $E_{pq} E_{qp} + E_{qp} E_{pq}$ at weight one, every length two, and the top read the display's. The trace recursion. On a table space's block at top content λ (Lemma 4), write $\text{mult}(\nu)$ for the content summand's dimension and, per positive root at the root coroot pair $\nu(\alpha^\vee) = \langle p : q \rangle$, $\nu^{(l)} := \nu + l\alpha$ for the moved contents: tracing the Casimir fold over the ν -summand, the summand's trace the one value at every orthogonal spanning list (Lemma 3's trace read), the block scalar against the diagonal and the root terms' string traces (Lemma 6 at the root triples), reads

$$\begin{aligned} q(\lambda+\rho) \text{mult}(\nu) + \sum_{\alpha} [\langle \alpha, \alpha \rangle : 2] 2 \sum_{l \geq 1} q \text{mult}(\nu^{(l)}) \\ = q(\nu+\rho) \text{mult}(\nu) + \sum_{\alpha} [\langle \alpha, \alpha \rangle : 2] 2 \sum_{l \geq 1} (p+2l) \text{mult}(\nu^{(l)}) . \end{aligned}$$

The telescope. Per positive root, with D_α the coroot read per monomial, $D_\alpha x^\nu = \nu(\alpha^\vee) x^\nu$, and

$$G_\alpha := \left\langle \sum_{\nu} \left(\sum_{l \geq 1} (p+2l) \text{mult}(\nu^{(l)}) \right) x^\nu : \sum_{\nu} \left(\sum_{l \geq 1} q \text{mult}(\nu^{(l)}) \right) x^\nu \right\rangle ,$$

each member finite at the occupied contents, at the move's two sides, $\alpha = \langle \alpha_+ : \alpha_- \rangle$ per coordinate,

$$x^{\alpha_+} G_\alpha = x^{\alpha_-} (G_\alpha + D_\alpha \text{ch}) \quad \text{at} \quad \text{ch} := \sum_{\nu} \text{mult}(\nu) x^\nu ,$$

and the recursion collects to

$$q(\lambda+\rho) \text{ch} = \mathcal{L} \text{ch} + \sum_{\alpha} [\langle \alpha, \alpha \rangle : 2] 2 G_\alpha \quad \text{at} \quad \mathcal{L} x^\nu = q(\nu+\rho) x^\nu .$$

The generator list reads it through the Fierz sum (Definition 7): a matrix acts as $\sum_{pq} A_{pq} E_{pq}$, so

$$\Delta = \sum_a X_a^\dagger X_a, \quad d_f \Delta + \xi \mathcal{Z}^2 = \xi d_f C, \quad \mathcal{Z} := \sum_p E_{pp},$$

positive semidefinite by its shape, its split the fold of squares, and the block read is the additive bracket $d_f \Delta|_{W_v} + \xi k^2 = \xi d_f (q(\lambda) + \langle \lambda, 2\rho \rangle)$, the cleared content read of the weight-lattice Casimir at the form's scale ξ . At the one-box shape $\lambda = (1)$ the cleared bracket reads $1 + r(r+2) = d_f(1+r)$, the one-value join $r(r+2) + 1 = d_f^2$: so $\hat{c}_2(f) = [r(r+2) : 2d_f^2]$ against the adjoint's below. At $\lambda = (2, 1^c)$ at $c+2 = d_f$ the traceless read is $2d_f$: the adjoint Casimir is $C_2(\text{adj}) = d_f$ at $\xi = [1 : 2]$, the Fierz weight of Definition 7.

Proof. The commutation. C is a diagonal unit at the grading's read, every fold term reading each content summand to itself with the unit at its scalar there (Construction 2), and an ordered letter pair at the derivation here: per factor the table's display reads the join at the factor's letters (Construction 2), $E_{qp}E_{ab} + \delta_{bq}E_{ap} = E_{ab}E_{qp} + \delta_{pa}E_{qb}$ and $E_{pq}E_{ab} + \delta_{bp}E_{aq} = E_{ab}E_{pq} + \delta_{qa}E_{pb}$, the second read inside the first's middle term chains the two; a letter off a content's support acts at the sum's unit there, the Leibniz fold over a vacant place index (Construction 2): at the unit's own letter off the source content's support the identity's two members read that unit together, and inside the fold the governing letter is the inner factor's lowered one, each member read at its own side's content, the source on the one side and the unit's image on the other, an image-side vacancy sitting at the unit's own lowered letter alone at its one source count, the term's surviving members closing at the display's own shorter instance, a factor written at a content pair off the action's own move reading the null map, the grading's read (Construction 2), and the vacant image letter's own pair closing at the diagonal reads' square, the one count twice; so the collected reads run at the letters each side supports, the unit's letter supported in the source throughout, the chained term's two move orders one content and the collected summands' contents one spelling at the round-trip and chain reads through the supported unit letter, the degenerate letters' members at the sum's unit on both sides; and the (p, q) -fold collects each side's delta families to the one join

$$C E_{ab} + \sum_p E_{pb}E_{ap} + \sum_q E_{aq}E_{qb} = E_{ab}C + \sum_q E_{aq}E_{qb} + \sum_p E_{pb}E_{ap},$$

the two folds a summand on both members fixing the value (Definition 1): $C E_{ab} = E_{ab} C$. *The top read.* At the top v , $E_{pq}E_{qp}v$ at $p < q$ is $[E_{pq}, E_{qp}]v = \langle \lambda_p : \lambda_q \rangle v$, the row pair's value, the commutator the site datum at $E_{pp} = E_{qq} + [E_{pq}, E_{qp}]$, a vacant row p reading the term and the pair's value at the sum's unit together, the row list's descent putting the later row at that unit with it (Construction 1), and at $p > q$ is the sum's unit, the raising's value at the top. *The trace recursion.* The ρ -shift is the form's bilinearity at the weight collection $\langle \nu, 2\rho \rangle = \sum_\alpha [\langle \alpha, \alpha \rangle : 2] \nu(\alpha^\vee)$, and per string one identity: an α -string through ν at depth j from its top reads the trace coefficient $(j+1)d + j(d+1)$ at the tie $d+q = p+j$, its moved reads collect at $2 \sum_{l \leq j} (p+2l) = 2j(d+1) + 2jq$, the two folds one gap identity, and a string above ν sums its moved reads to equal members, the depth pairs at the join swapping members. *The telescope.* At the monomial $x^{\nu+\alpha}$, whose coroot pair is $\langle p+2 : q \rangle$, G_α 's first member reads $\sum_{l \geq 1} (p+2+2l) \text{mult}(\nu^{(l+1)})$, the fold at l from two, and D_α ch reads $(p+2) \text{mult}(\nu^{(1)})$, the term at $l = 1$, so their join is $x^{\alpha+G_\alpha}$'s coefficient there, $\sum_{l \geq 1} (p+2l) \text{mult}(\nu^{(l)})$, one further move dropping one l ; the second members read q at every term, the coroot pair's second member fixed by the move; and the recursion's fold over ν against x^ν collects the string sums into G_α 's members, the displayed fold. *The two reads.* At $\lambda = (1)$, $q = 1$, $\langle \lambda, 2\rho \rangle = r$, $k = 1$; at $\lambda = (2, 1^c)$, $k = d_f$, $q = r+3$, $\langle \lambda, 2\rho \rangle = 2r$, and the cleared bracket reads $k^2 + 2d_f \cdot d_f = d_f(q + \langle \lambda, 2\rho \rangle)$, both sides $3d_f^2$. $\square$

Lemma 8 (Duality, and the unit read). *(i) The dual. A unit space M defines its dual action on the underlying paired space, $E_{ij}^\vee$, the balance partner of E_{ji} , the sum's every entry of equal members (Definition 1's displayed reads), a dual image the exchanged letters' image scaled by the unit's own partner and a pairing reading the partner on either side. The dual obeys the table,*

$$[E_{ab}^\vee, E_{cd}^\vee] + \delta_{da} E_{cb}^\vee = \delta_{bc} E_{ad}^\vee ,$$

the exchanged letters' read of Construction 2's, and the transpose identity, $(E_{ij}^\vee)^\top = E_{ji}^\vee$. Each content is the original's balance partner, $\mu + \mu^\vee$ of equal members coordinatewise, the dual diagonal reading a content summand at its coordinate's partner. Invariant subspaces agree, so duals of blocks are irreducible (Lemma 4(i)), and $(M^\vee)^\vee = M$. The pairing's evaluation $\text{ev} : M^\vee \otimes M \rightarrow \mathbb{P}$ is equivariant to the unit, a letter pair's Leibniz image reading $\langle E_{ij}^\vee f, x \rangle + \langle f, E_{ij} x \rangle$ at the sum's unit. Its Gram-dual coevaluation at a stated independent spanning list y is the determinant-scaled identity datum,

$$\text{coev} := \sum_{j,k} A_{jk} y_j \otimes y_k \in M^\vee \otimes M ,$$

A the Gram's adjugate against its determinant, the list against the adjugate reading the determinant's diagonal (Definition 3), the grading splitting the Gram at one block per occupied content, the cross pairings of equal members, so the adjugate collects each block's own at the further blocks' determinants; and the coevaluation is equivariant to the unit as well, the moved coevaluation pairing every member pair at the sum's unit. And $\text{inv}(M^\vee \otimes W) \cong \text{Hom}(M, W)$ at the equivariant maps: an invariant of the dual tensor is a family $\sum_i f_i \otimes w_i$ at matched contents, the dual diagonals' unit read, whose every letter pair's Leibniz image, the dual action at the first factor, pairs every member pair of the two carriers' stated lists at the sum's unit. The invariant reads as the equivariant $x \mapsto \sum_i \langle f_i, x \rangle w_i$, the fold the argument's own content group at the matched contents with every cross-content pairing the sum's unit, the map's data one matrix per occupied content with the moved reads (Lemma 4's matrix families); and back through the coevaluation, an equivariant φ reads the invariant $\sum_{j,k} A_{jk} y_j \otimes \varphi(y_k)$, the coevaluation's second slot at the map's value, the family at matched contents with each value at its member's own content. The round trips read the determinant's scale: the composite at an equivariant φ reads $\det G \cdot \varphi$ and the composite at an invariant reads its $\det G$ -multiple, the determinant off the sum's unit at the independence (Lemma 3), every read one value across the scale (Definition 1's homogeneity principle), and the two spaces' counts read one value at the correspondence's own carriers. An invariant enters at its coefficient family over the two stated lists' matched member pairs, the moved reads one stacked linear system in the coefficients, each row the moved form's fold at a letter pair and a member pair, with its count and kernel list the elimination's (Definition 3). A family whose map reads every listed member at the sum's unit is the unit family. An equivariant map enters the grid through the second list's solve, each value's cleared combination over that list's span at the graded Gram's adjugate against its determinant (Lemma 3; the coevaluation's read at the second list), the grid's datum one block per occupied content, the content group's rows at the content's own values' solves with each member's row read at the member's place in its group, and the kernel list and a stated map list read one count, two independent lists inside one span (Lemma 3). (ii) The full-column line, and the dual block. The full wedge w_{d_f} spans the block $W_{(1^{d_f})}$, the full-column line, every unit at $i \neq j$ mapping it to the sum's unit, so $d_{(1^{d_f})} = 1$; tensoring by it maps blocks to blocks, the block $W_{\lambda+(1^{d_f})}$'s span W_λ 's span at the wedge tensored on, member for member, $W_\lambda \otimes W_{(1^{d_f})} \cong W_{\lambda+(1^{d_f})}$ with $w_{d_f} \otimes t_\lambda$ its top, the exhibits' composition at the column union reading $t_{\lambda+(1^{d_f})}$ as $w_{d_f} \otimes t_\lambda$ (Construction 1), and the dimension, the occupancies at the shifted contents, and the fused counts read the mapped span's, the pair list at a shifted content the content's own list mapped with the kernel data one family at

the two pair lists (Definition 5), the count at the shifted content the content's own; and the line reads at every power, $W_{(m^{d_f})} = W_{(1^{d_f})}^{\otimes m}$ one line at every m . The dual block reads by the content symmetry: $W_\lambda^\vee$ reads its maximal occupied content, the reversal $w_0\lambda$'s balance partner, on one line, a top (Lemma 6; Lemma 2(i)), and

$$W_\lambda^\vee \otimes W_{(1^{d_f})}^{\otimes \lambda_1} \cong W_{\lambda^*},$$

λ^* the complement shape, every column of λ at a gap against the full length read to the gap's column at $\ell + \ell' = d_f$, a full column reading to the full-column line above, the complement shape's row list at the join $\lambda^* + w_0\lambda = \lambda_1(1^{d_f})$ (Lemma 4(iv)). (iii) The unit read. For shapes a, b the full-column powers' count in $W_a \otimes W_b$ is a Kronecker delta: at the tie $m d_f = |a| + |b|$ where a natural m witnesses it,

$$N_{(m^{d_f})}(W_a \otimes W_b) = \dim \text{Hom}(W_a^\vee \otimes W_{(1^{d_f})}^{\otimes m}, W_b)$$

the Kronecker delta, one exactly at b and a^* differing by full columns, the join, b 's row list joined to a 's reversal reading the full columns ((i)'s complement at the row-list join). The count is the kernel dimension of the stacked raisings at the stated pair list, the two blocks' spanning lists' tensors at complementary contents (Definition 5; Lemma 5(iii)), and the display is (i)'s adjunction at the line $W_{(1^{d_f})}^{\otimes m}$: the kernel families and the equivariant maps read one count through the value map, per argument the second slots' combination at the coefficients against the first slots' pairings, an equivariant map at the dual action at the argument against the action at the value, (i)'s count clause at the complementary grid. Off the join the kernel is the unit family; at the join the kernel is one line, occupied at the coevaluation of the dual word family at the reversal member, (ii)'s dual top, against the word family at the target's top (Lemma 4(iv)'s word families; (i)'s coevaluation at the collected families).

Proof. (i) A vector's partner is its scale by the unit's own partner (Definition 1's displayed reads), so a dual image is a scaled image, a matrix read of a scaled vector its own read's scale, two partners compose to the original, and a pairing reads the partner on either side, the coordinate fold's scale. The table: the commutator's factors compose at the double partner to $[E_{ba}, E_{dc}]$, its display at the letters (b, a, d, c) Construction 2's, and joining $\delta_{cb} E_{da}$ to both members reads each member as a unit against its own partner, the first through that display at $\delta_{ad} E_{bc}$ and the second at $\delta_{cb} E_{da}$ outright, both at the sum's unit, a summand on both members fixing the value (Definition 1); the transpose identity is the partner read across the pairing over the original's transpose, and the dual diagonal's read is the partner scalar. A dual image is the exchanged letters' image at the partner's scale, a span member's scale inside the span, and the double partner reads the original images back, so invariant subspaces agree, Lemma 4(i) reads at the agreement, and the exchange twice at the partner twice reads each letter's read the original's own. The evaluation's letter pair's Leibniz image has its first summand the partner of $\langle E_{ji}f, x \rangle$, the transpose identity reading that pairing as the second summand, and a value joined to its partner reads the sum's unit. The coevaluation's letter pair's Leibniz image paired against a member pair (y_a, y_b) collects at the adjugate identity to the determinant against one crossed pairing per summand, the partner of $\langle E_{ji}y_b, y_a \rangle$ at the first slot's read and $\langle E_{ij}y_a, y_b \rangle$ at the second's, the transpose identity with the fold's exchange (Definition 1) joining the two at partners. The invariant's map is equivariant: the invariance read at the pair of a stated argument with a further member joins two crossed folds, the first the map's value at the moved argument through the evaluation's equivariance and the second the moved value's own pairing, so the map's two moved reads agree against every member and the moved images are one value at the pairing's definiteness. An equivariant map's family is invariant: the invariance

read at the pair of a member with a further vector collects at the adjugate identity twice, the member's own column reading the determinant against the map's moved value at the member's key with the equivariance transporting the letter across the slot, and the first slots walked across the transpose identity to the members' pairings against the member's letter image, the fold at the image's pairing data the adjugate solve (Lemma 3); the image's cleared combination over the list's span reads the solve at the determinant against the combination's coefficients, the map's value additive and scale-reading across the members, so the two reads join at partners and the clearing withdraws at the product's injectivity. The round trips: the composite at an equivariant φ reads $\det G \cdot \varphi$ at the adjugate identity at the map's argument with the members' reads extending over the list's span, both composites additive and scale-reading in the argument, a span member's cleared combination reading them through the members with the clearing withdrawn at the product's injectivity (Lemma 3), and the composite at an invariant reads the identity at each f_i over the list's span. A family whose map reads every listed member at the sum's unit is the unit family: per content the value is the second list's combination at the first list's weighted pairings, the second list's independence reads those weights back (Lemma 3), and per second member the weights pair the first list's group against its every member at the sum's unit, the perpendicular combination the unit family at the group's independence (Lemma 3). At the grid the graded Gram's adjugate collects each block's own at the further blocks' determinants, the moved reads sit at the sum's unit by the equivariance above, and the map is the determinants' scale of its own by the round trip, so the kernel list and a stated map list are two independent lists inside one span.

(ii) Every unit at $i \neq j$ reads the full wedge at a repeat at each Leibniz read, the sum's unit. Two monomial tensors pair at the factors' pairings' product, a concatenated arrangement reading its split at the first factor's degree (Construction 1), a tensored member's coordinates the factor products at the split arrangements with the unit tail exactly at the factor's own, and a letter's action splits over the concatenation by the Leibniz read, the wedge factor's term the repeat's unit, a letter vacant in the further factor leaving the joint image at that term alone, the unit tail: a lowering word's image at $w_{d_f} \otimes t_\lambda$ is the wedge tensored onto its image at t_λ , and the span memberships transport at the wedge tensored on: a combination's image is the image's combination, and two images at one value read their gap's tensor at the unit tail, the tensored coordinates' split above reading the gap at the further factor against the wedge's differing members (Construction 1), so a member joins its group's span exactly where its image joins the image group's (Lemma 3's membership at its cleared combination, the two directions the solve's own reads). The fused pool against a further block pairs the mapped members at the arrangements' join (Construction 1), so the pair list at a shifted content is the content's own list mapped; a stacked raising's image at a mapped pair is the wedge tensored onto its image at the pair, the wedge factor's term the repeat's unit, and a tensored image reads the unit tail exactly at the further factor's own read, the split coordinates at the wedge's differing members, so the stacked raisings annihilate a coefficient family's combination at the mapped list exactly where they annihilate it at the list, the kernel data one family at the two pair lists (Definition 5) with the collections two lists in one span at one count (Lemma 3). The line's powers read by the descent: a power's exhibit is the wedge tensored onto the prior power's, the column union's read, a lowering's image reading the wedge factor's term at the repeat's unit and the further factor's at the prior power's own unit read, so every lowering image of every power's exhibit reads the unit tail. W_λ 's occupied contents sit between $w_0\lambda$ and λ with the reversal occupied once (Lemma 6), so the dual's maximal occupied content is the reversal's balance partner on one line, a top (Lemma 2(i)), and the dual twisted by the line and the complement shape's block are two irreducible unit spaces sharing the top content, one block (Lemma 4(iv)).

(iii) The value map is equivariant: the stacked adjacent raisings annihilate the family's combination, and the annihilation reaches every letter pair at distinct letters at the full-column content,

the adjacent lowering's at (ii)'s line with the moved content's group vacant, and each further pair's at the crossed composite on the gap, the descent at raisings and lowerings both (Lemma 3), so the moved reads close at every such pair, (i)'s moved reads at the exchanged letters. *Off the join the kernel is the unit family.* At a kernel family off the unit family some value sits off the sum's unit: per content a value is the second group's combination at the first group's weighted pairings, and a family whose every value reads the unit tail is the unit family, the two groups' independence reading the weights and the perpendicular combination back ((i)'s count clause at the complementary groups; Lemma 3). The value list at the first block's stated members is closed at the letters inside the second block's span, a letter's image of a value the value at the argument's dual move with the moved argument's combination reading through the fold, so at a value off the sum's unit every W_b member joins the value list's span (Lemma 4(i)), the head among them; the head's group in the value list holds a value off the sum's unit, a combination of unit tails reading the unit tail, and a value off that unit reads occupied slots, the pair guard's join exact: b 's row list joins an occupied content of W_a to the full columns, and the factors' exchange reads a 's row list so as well, the arrangements' exchange at the commuted coordinate products (Lemma 5(iii)). The occupied contents sit at or above the reversals (Lemma 6), and the two joins exchange the reads: at the full-column content a prefix of the complement joins the content's own prefix to the full columns' count, and at one degree a prefix of a reversal joins its list's complementary prefix to the degree, so the first factor's floor reads the occupied content at or above a 's reversal while the second factor's floor, read through the two exchanges, sits at or below it, and the two dominance witnesses join to the unit fold (Construction 3): b 's row list is the complement's read, b and a^* differing by full columns. *At the join the kernel is one line.* The first block's reversal member is occupied once (Lemma 6) and is the dual top, every dual raising's image at the unit tail ((ii)'s dual block), and its value's content, the complement of the reversal, is the join's read of the target's own top: the value at the dual top sits in the head's group, the top line (Lemma 3), a scale of the head. A kernel family off the unit family holds a value off the sum's unit at the reversal member: the head's group read above names a first-block member whose content joins b 's row list to the full columns, as the reversal does at the join, and the sum's injectivity reads the two contents one (Definition 1), the reversal member occupied once; and the value map reads its coefficient family linearly, each coefficient entering its own weight once, so two families' cleared combination reads the two values' own at every argument (Definition 1): at that scale the unit the family is the unit family, two kernel families read one line, the cleared combination at the unit scale the unit family, and the count is at most one. And the kernel is occupied: the dual walk's H -read at a content is the partner pair, one value with the complement content's own at the full column's constant cross-added (Definition 1; (i)'s dual diagonal with (ii)'s line), a crossing of the walk reads the letter pair's site datum, two letters' reads at their gap (Lemma 7's display), so the constant withdraws at the gap's cross-add; the lowering moves the letter pair down and the dual moves it up, the joined content riding each letter's round trip at the occupied read, so the two walks' contents join to the full columns wherever both images sit off the unit tail, and every crossing of the dual walk reads the complement content's own crossing, and the dual words' pairings at the reversal member fold the walk's coefficients at those reads (Lemma 4(iv)): the dual word family at the reversal member and the word family at the target's top read one Gram at the two tops' self-pairings' scales, the source and the target one block at (ii)'s dual block, the dual twisted by the line reading the complement shape's carrier, and each dual word's *partner word* is the word at the exchanged letters, a dual move at a letter pair the complement's move at the exchanged pair ((i)'s dual action). The coevaluation at the two collected word families, the Gram's adjugate against its determinant weighting each dual word against its partner word, pairs every stacked raising's image against every pair at the sum's unit: a moved pairing is the walk's read at one further crossing, and the two slots' moved reads join

at partners across the transpose identity at the exchanged letters ((i)'s coevaluation read at the two families); the coevaluation sits off the unit family at the determinant's diagonal, the collected families independent (Lemma 3): the count is one. $\square$

Construction 5 (The label set). Two shapes are one *label* when they differ by full columns, $W_{\lambda+(1^{d_f})} \cong W_\lambda \otimes W_{(1^{d_f})}$ (Lemma 8(ii)); each label reads one reduced shape, at most r rows deep. The data descend to labels: d is class-invariant (a line's tensor), and so is the fusion count at matched degrees, $N_{a+(1^{d_f}),b}^{c+(1^{d_f})} = N_{ab}^c$, the fused counts' read at the mapped span, the pair list at the shifted content the content's own mapped with the kernel data one family (Lemma 8(ii)); a full column crosses the factors at a fixed target, the display at the factors' exchange (Lemma 5(iii)) where the target holds a full column, and at a reduced target either side's count is the sum's unit, the mapped pool occupied at the last letter member for member (Lemma 8(ii)) with the target's group vacant (Lemma 5(ii)); the label count at two representatives reads at the matched-degree lift, the degree gap a full-column multiple at one class with the deficient side grown by the gap's columns, and the sum's unit at a gap off the multiples; the dual label $\bar{a}$ is the class of a^* (Lemma 8(ii)); the full columns' own class is the *unit label* $\mathbf{1}$, the unit shape's, its multiplicity in a product one exactly at the dual label, the read of Lemma 8(iii); and the Casimir read descends, the bracket of Lemma 7 at $\lambda + (1^{d_f})$ gaining $2k + d_f$ from q and returning it to the degree term. The Cartan count sharpens to an equality on shapes, $N_{ab}^{a+b} = 1$: the content- $(a+b)$ summand of $W_a \otimes W_b$ is $(W_a)_a \otimes (W_b)_b$, one line, since a content ν of W_a with $a + b - \nu$ a content of W_b sits at and below a at once (Lemma 3; the dominance order, Construction 3).

Lemma 9 (The Casimir floor). *Every nonunit label reads $\hat{c}_2(\lambda) \geq \hat{c}_2(f) = \lceil r(r+2) : 2d_f^2 \rceil$, with equality exactly at f and $\bar{f}$; so a configuration of weight-free content at most Λ_C with L occupied links reads $L \hat{c}_2(f) \leq \Lambda_C$, one pair comparison. The floor's descent is statement data: a box move joins two dominant shapes at one row pair $a < b$, the join $\lambda + e_b = \mu + e_a$; Q falls strictly along moves; every chain from a reduced shape ends at a fundamental or at the unit label, a step reaching the unit label passing θ 's class; and the endpoints read $d_f Q(\omega_j) = j g_j (r+2)$ at the complement gaps $j + g_j = d_f$. At a member's table (Construction 4) the move is one positive root at dominant data, $\mu + \alpha = \lambda$, the box move the matrix units' instance at the letter-pair move, and the fall identity is the root arithmetic*

$$C_2(\mu) + \langle \lambda + \mu + 2\rho, \alpha \rangle = C_2(\lambda) ,$$

the join's two reads collected at the form's bilinearity, its read on its upper side at dominant data: a dominant content against a positive root reads $2 \langle \lambda, \alpha \rangle = \sum_i c_i \lambda(\alpha_i^\vee) \langle \alpha_i, \alpha_i \rangle$ over the root's fold, every summand a product at the coroot reads, and $\langle 2\rho, \alpha \rangle = \sum_i c_i \langle \alpha_i, \alpha_i \rangle$ is positive. The member's end list reads its Casimirs off the tables at the end contents' cleared folds, the displays $e\mu = \sum_k b_k \alpha_k$ at a fold key e , the coroot identities $\sum_k b_k \alpha_k(\alpha_j^\vee) = e\mu(\alpha_j^\vee)$ each display's witness with the fixed members' folds the adjugate rows (Construction 4), and the Casimir the fold's form read at the halved products,

$$2e \langle \mu, \mu + 2\rho \rangle = \sum_k b_k \mu(\alpha_k^\vee) \langle \alpha_k, \alpha_k \rangle + e \sum_\alpha \sum_k c_k \mu(\alpha_k^\vee) \langle \alpha_k, \alpha_k \rangle ,$$

the second fold over the positive list at each root's fold $\alpha = \sum_k c_k \alpha_k$, read against $C_2(\theta) = 2(r+1)$, the form and residue reads (Construction 4): at B_ℓ the vector, 2 on the first key, the fold key one at $b_k = 1$, at the reads 1 and $r+1$, so $\hat{c}_2 = [\ell : r+1]$, and the spinor, 1 on every key, the fold key two at the index list $b_k = k$, at the reads $[\ell : 4]$ and $[\ell^2 : 2]$, so $[\ell(2\ell+1) : 8(r+1)]$, the list's least member the trichotomy at $2\ell+1$ against 8; at C_ℓ the first fundamental, the fold key two at twos

with one at the last key, at $[2r+1 : 4(r+1)]$, below the short dominant root's $[r : r+1]$, the root its own fold at key one; at D_ℓ the vector, the fold key two at twos with one on the last two keys, at $[r+2 : 2(r+1)]$, and the two spinors, the fold key two at the index list with the exchanged tail pairs (c, c') at the joins $2c = \ell$ and $c' + 1 = c$ at even rank and the fold key four at the doubled index list with the exchanged tails (ℓ, w) at $w + 2 = \ell$ at odd, at $[\ell(r+2) : 8(r+1)]$, the trichotomy at ℓ against 4 with equal members at $\ell = 4$; and the fixed members' evaluations, $[1 : 2]$ at G_2 's short dominant root ($C_2 = 4$ at the reads $[2 : 3]$ and $[10 : 3]$), $[2 : 3]$ at F_4 's ($C_2 = 12$ at 1 and 11), $[13 : 18]$ at E_6 's minuscule pair ($[4 : 3]$ and 16, the join $[52 : 3]$ against 24), $[19 : 24]$ at E_7 's minuscule member ($[3 : 2]$ and 27 against 36), and at E_8 the highest root's own read, $\hat{c}_2(\theta) = 1$, the level one.

Proof. Write $d_f Q(\lambda)$ for the gap read of Definition 6 at degree k , $k^2 + d_f Q(\lambda) = d_f q(\lambda) + d_f \langle \lambda, 2\rho \rangle$, the cleared traceless bracket of Lemma 7, a label read (Construction 5: a full column adds $2k + d_f$ to q and returns it to the degree term), with $d_f Q(\text{adj}) = 2d_f^2$; the claim is $d_f Q \geq r(r+2)$ on nonunit labels, one natural comparison. A *box move* joins two dominant shapes at one row pair $a < b$, the join $\lambda + e_b = \mu + e_a$, one box read at either row; it fixes the degree and lowers Q :

$$d_f Q(\lambda) = d_f Q(\mu) + d_f \langle \lambda_a + \mu_a : \lambda_b + \mu_b \rangle + 2 d_f g,$$

the pair the summed rows' at the move's places, on its upper side or of equal members at the sorted rows' orders $\lambda_b \leq \lambda_a$ and $\mu_b \leq \mu_a$, and $2g$ the places' doubled gap at $a + g = b$: the display is the join's read at the fixed degree, the square fold moved by the summed rows' pair, each place's square difference the pair member's read at the one-box gap, and the row-weight fold by the places' doubled gap, the row weights $\langle r + 2 : 2i \rangle$ at the two keys (Lemma 7's fold), the weights' gap the places' doubled own. Every reduced shape off the fundamentals and the unit moves: at some row of length at least 2, take a the last such row and b one past the last occupied row: the target μ stays sorted, $\mu_a \geq 1 \geq \mu_{a+1}$ at $\mu_a + 1 = \lambda_a$ and $\mu_b = 1 \leq \mu_{b'}$ at $b' + 1 = b$, a full column re-reducing with Q fixed; a one-column shape is the fundamental at its length, and every move off it breaks the sorting. Q falls strictly along moves, one natural measure descending, so every chain ends at a fundamental or at the unit label, and a step reaching the unit label starts at a constant row list moved by one box, dominance forcing the box from the first row to the last: θ 's class, at $d_f Q(\theta) = 2d_f^2$. The floor is the least endpoint read, the row weights summing to $j g_j$ at the complement gap $j + g_j = d_f$, the cross-added read $j(j+1) + j g_j = j(r+2)$:

$$d_f Q(\omega_j) + j^2 = d_f j + d_f j g_j, \quad d_f Q(\omega_j) = j g_j (r+2),$$

least at $j \in \{1, r\}$ with value $r(r+2)$, strictly larger between, and $2d_f^2 = r(r+2) + (r^2 + 2r + 2)$, the unit-reaching step strictly above the floor. At the series' tables the descent closes on coordinate moves, each one positive root at a stated family with the sorted order, the parity and the class kept. At B_ℓ : an even-parity content at two or more occupied keys takes the pair move, the sum root at the last two occupied keys, both reads dropping by two at the sorted tail's end; at one occupied key it takes the short root there, the chains through the vector, the short dominant root's content, to the unit; an odd-parity content off the spinor holds a read at three or beyond, the short root at the last such key moves it, the following read one at most, below the moved read; and the spinor is move-free, every root's subtraction breaking the last read or the order. At C_ℓ : two or more occupied keys take the pair move, one occupied key at two or beyond takes the long root, and the chains end at the first fundamental or reach the unit through the two dominant roots' contents. At D_ℓ : at the last key's read equal members, the pair move at the last two occupied keys at two or more of them, and the difference root at keys one and two at one occupied key reading

four or beyond; at the last key's read off equal members, the pair move of the last key against key $\ell-1$ at the last read's magnitude two or beyond, the sign at the last read's side and the last read moved toward equal members, and against the rearmost key reading three or beyond at magnitude one, the sign flipping the last read at its magnitude; the vector and the two spinors are move-free, every root's subtraction breaking the order or the last pair's dominance. Each move's fall is the identity's read on its upper side, and the Casimirs descend strictly at the member's clearing factor, the form's displayed second members clearing every value to a natural, a strictly falling natural: every chain ends on the member's end list or at the unit through a dominant root's content, so every nonunit dominant content of a series member reads $\hat{c}_2$ at or above the least of the end list's and the dominant roots' reads, the statement's trichotomies, the member's floor. At a fixed member the descent runs in the coroot coordinates: a content with a coroot read at two or beyond moves by that simple root, the off-key Cartan reads balance partners of naturals (the member's displayed data), their subtraction raising the off-key coroot reads, so the move keeps dominance, one root with the class kept and the fall the identity's, terminating at the member's clearing factor; the contents at every coroot read at most one are finitely many, and each reads its Casimir at or above every occupied fundamental's,

$$C_2(\mu) = \sum_{i,j} \mu_i \mu_j \langle \omega_i, \omega_j \rangle + \sum_i \mu_i \langle \omega_i, 2\rho \rangle ,$$

every summand at the fundamental form reads' upper sides or equal members (the adjugate rows' form reads) with $\langle \omega_i, 2\rho \rangle = 2 \sum_j \langle \omega_i, \omega_j \rangle$ positive at the root fold's all-twos coroot list (Construction 4): the member's floor is the least fundamental read, one trichotomy fold over the member's fundamental Casimir list with the displayed minima the statement's, and at E_8 the fold reads the adjoint, the least nonunit label the highest root at level one. The link count follows, each occupied link contributing at least $\hat{c}_2(f)$ to the content. $\square$

Theorem 1 (The alternant identity). *Write $\text{mult}_\lambda(\nu) := \dim(W_\lambda)_\nu$ and $\text{ch}_\lambda := \sum_\nu \text{mult}_\lambda(\nu) x^\nu$ for a block's content generating polynomial, q the standard square, and, at a list β of naturals, a_β the alternant pair, the swap-graded sum over S_{d_f} (Construction 1), a balance pair of polynomials (Definition 1):*

$$a_\beta := \left\langle \sum_{w \text{ even}} x^{w\beta} : \sum_{w \text{ odd}} x^{w\beta} \right\rangle,$$

each monomial on its permutation's side; an odd permutation of the exponent list exchanges the members and an even one fixes them (the grading additive, Construction 1), and a repeated exponent makes the members equal, the repeat's transposition mapping either side onto the other, the wedge's repeat read. Then, at the beta-sets of Construction 1, U the unit set with sorted display u and X_λ the shape's with sorted display $\lambda + u$,

$$a_u \text{ch}_\lambda = a_{\lambda+u} .$$

Proof. *The Casimir trace, string by string.* Trace the Casimir over the content summand $(W_\lambda)_\nu$, the summand's trace the one value at every orthogonal spanning list (Lemma 3's trace read): the block scalar reads $(q(\lambda) + \langle \lambda, 2\rho \rangle) \text{mult}(\nu)$ (Lemma 7), the diagonal part reads $q(\nu) \text{mult}(\nu)$, and each pair $p < q$ traces $E_{pq}E_{qp} + E_{qp}E_{pq}$ over its strings (Lemma 6): a string through ν at depth j from its top, of top eigenvalue h , at the depth's raise factor d , the pair $\langle h : j \rangle$, reads the ν -line coefficient $(j+1)d + j(d+1)$, reading Lemma 2(ii)'s fold twice, at the words F^{j+1} and F^j , at the string's tie $d + \nu_q = \nu_p + j$ the coefficient the identity

$$(j+1)d + j(d+1) + \nu_q = 2j(d+1) + \nu_p ,$$

every read the balance pairs' own. The pair's value at a content is the pair $\langle \mu_p : \mu_q \rangle$ (Definition 1); writing $\nu^{(l)}$ for the pair's l -th move of ν , the occupied content at the joins $\nu_p^{(l)} = \nu_p + l$ and $\nu_q^{(l)} + l = \nu_q$ with every other coordinate equal (Construction 2's letter-pair moves iterated), the value at $\nu^{(l)}$ is the pair $\langle \nu_p + 2l : \nu_q \rangle$; the string's values from its ν -line up to its top sum at the tie, $2 \sum_{l \leq j} (\nu_p + 2l) = 2j(d+1) + 2j\nu_q$, while a string above ν sums its values to equal members: the value at depth m is the pair $\langle h : 2m \rangle$, the top's its one-member instance, two depths at the join $m + m' = h$ reading member swaps, the summed members equal. Collecting string by string, the string coefficients' ν -terms collecting to the two side weights, ν_i weighted per place by the gaps at $1 + o = i$ and at $i + o' = d_f$, the equal-degree tie $\sum_i \lambda_i = \sum_i \nu_i$ with the shared reads $q(u)$ and twice the degree closing both sides, and the block's row read closing at $\langle \lambda, 2\rho \rangle + (d_f + 1) \sum_i \lambda_i = 2 \sum_i \lambda_i u_i$ (Lemma 7's fold), gives

$$q(\lambda + u) \text{mult}(\nu) + 2 \sum_{p < q} \sum_{l \geq 1} \nu_q \text{mult}(\nu^{(l)}) = q(\nu + u) \text{mult}(\nu) + 2 \sum_{p < q} \sum_{l \geq 1} (\nu_p + 2l) \text{mult}(\nu^{(l)}),$$

the pair sums over the places $p < q$ at their moves; the display runs at every content list of the block's width, the recursion family's keys the read's own finite carrier, and a key at the count's unit reads the display as the moved folds' one value, the above-string clause's read alone. *The generating telescope.* With $D_i := x_i \partial_i$ and the pairs' reads entering as balance pairs,

$$G_{pq} := \left\langle \sum_{\nu} \left(\sum_{l \geq 1} (\nu_p + 2l) \text{mult}(\nu^{(l)}) \right) x^{\nu} : \sum_{\nu} \left(\sum_{l \geq 1} \nu_q \text{mult}(\nu^{(l)}) \right) x^{\nu} \right\rangle,$$

the outer sum over the natural exponent lists, each member finite (an entry at ν reads occupied move targets alone, the occupied contents finitely many, the multiplicity the occupancy family's own count at the span's contents (Lemma 3's counts at the read), the recursion's keys the family joined to every pair move's sources, and a key off that join reading the count's unit on both sides): a balance pair of polynomials, and the inner sums telescope coefficientwise, one further move dropping one l , to

$$\langle x_p : x_q \rangle G_{pq} = x_q \langle D_p \text{ch} : D_q \text{ch} \rangle, \quad \text{and the recursion reads} \quad q(\lambda + u) \text{ch} = \mathcal{L} \text{ch} + 2 \sum_{p < q} G_{pq},$$

$\mathcal{L} := \sum_i (D_i + u_i)^2$ reading $q(\nu + u)$ on x^{ν} . At a key with $\nu_p = 0$ the x_p -shift sits off the natural keys and the identity's line is the string's own palindrome: ch symmetric (Lemma 6) pairs the string's targets, two shifts at the join $l + l' = \nu_q$ reading one multiplicity, so the moved weights collect to the second exponent's count, $\sum_{l \geq 1} 2l \text{mult}(\nu^{(l)}) = \nu_q \sum_{l \geq 0} \text{mult}(\nu^{(l)})$. *The assembly.* $a_u = e V$ at the full monomial $e := \prod_k x_k$ and the pair product $V := \prod_{i < j} \langle x_i : x_j \rangle$ (Definition 1): the graded sum is graded, an odd variable swap exchanging its members (Construction 1), so at $x_i = x_j$ its members are equal, the swap fixing the evaluation while exchanging the members; the division at $\langle x_i : x_j \rangle$, monic in x_i (Definition 2), leaves that evaluation as remainder, members equal, so each factor's division is exact, the factors' in turn, a product of polynomial pairs of equal members exactly at a factor with equal members (the leading reads multiply), and e 's cofactor reads monomial by monomial, every arrangement reading every variable at least once, the factors and the variables pairwise coprime (each factor monic linear in its higher variable, the variables prime); the two sides match at degree, $\sum_j u_j$ against $\binom{d_f}{2} + d_f$, and at the leading monomial x^u , on the even side at coefficient one: $a_u = e V$. So, at the cofactors Q_{ij} at $\langle x_i : x_j \rangle Q_{ij} = V$, one value each at the factor of unequal members with the index swap exchanging the members, the cofactor's coefficients are the monic letter's division descent (Definition 2), V 's at the keys moved along the pair, and $D_i e = e$

with the derivative read $D_i V = \sum_{j \neq i} x_i Q_{ij}$ closing per key: a pair's moved line through a key reads V at its two arrangements alone, the two fillings of the line's open values at the exchanged grading; the monic descent opens at the key itself, one head read per variable beyond i , the swapped cofactors moving the letter strictly; at an arrangement with exponents v the collected reads close on the value's places split,

$$v_i + \#\{j > i : v_i < v_j\} = \#\{j > i\} + \#\{j < i : v_j < v_i\},$$

v_i the count of the values below the letter's over the places' trichotomy; and at every further key the lines' reads close in pairs: a line matched at both fillings joins its two arrangements' reads at the exchanged grading within its own fold, and at a one-sided hit the pair's value in the key occupies a second place, whose line keeps the hit's upper filling with its further filling at the receding value below the letter's, one one-sided hit per partner line, an involution off its own fixed points; the two hits' reads enter at opposite members, the places' transposition exchanging the grading at partner places on one side of the letter and the swapped cofactor's memberwise exchange reading the join at places the letter splits, so the collected reads join at equal members. Per pair the operator identity $\langle x_p D_p : x_q D_q \rangle = x_q \langle D_p : D_q \rangle + \langle x_p : x_q \rangle D_p$ reads

$$Q_{pq} \langle x_p D_p \text{ ch} : x_q D_q \text{ ch} \rangle = V G_{pq} + V D_p \text{ ch}, \quad V G_{pq} = Q_{pq} x_q \langle D_p \text{ ch} : D_q \text{ ch} \rangle,$$

the second the cleared telescope at the cofactor, and summing pairs, a pair's two $D_i V$ -terms collecting to the first display's left side, $\sum_i (D_i V)(D_i \text{ ch}) = V \sum_{p < q} G_{pq} + V \sum_{p < q} D_p \text{ ch}$. With the square's Leibniz rule $D_i^2(fg) = (D_i^2 f)g + 2(D_i f)(D_i g) + f(D_i^2 g)$, $\mathcal{L}_0 := \sum_i D_i^2$ reading q at every arrangement, one value $q(u)$, on each monomial of a_u , the read $D_i a_u = a_u + e D_i V$, and the collection $\sum_{p < q} D_p + \sum_i D_i = \sum_i u_i D_i$,

$$\mathcal{L}_0(a_u \text{ ch}) = 2 a_u \sum_{p < q} G_{pq} + a_u \left(\mathcal{L}_0 + 2 \sum_i u_i D_i + q(u) \right) \text{ ch} = 2 a_u \sum_{p < q} G_{pq} + a_u \mathcal{L} \text{ ch} = q(\lambda + u) a_u \text{ ch},$$

the first equality the Leibniz square's collection at those reads, the second $\mathcal{L}$'s display, the third the telescoped recursion. *The conclusion.* $a_u \text{ ch}_\lambda$ is graded, an odd variable swap exchanging its members (ch symmetric, Lemma 6; Construction 1), with natural coefficients on each member, hence a balance-pair combination $\sum_\mu c_\mu a_{\mu+u}$ over dominant μ : a repeated exponent's read has equal members, the repeat's transposition's, each strictly decreasing exponent orbit is one graded sum, and the walk to the representative is the exchanges' own: a key reading one value at two places reads the repeat's transposition, a distinct-valued key's first place pair off the descending order is an ascent whose exchange lowers the ascending place pairs' count by exactly one, and Construction 1's descent at the reversed order closes the walk; its coefficient c_μ is the pair of the two members' reads at $x^{\mu+u}$. The support sits at $\mu \preceq \lambda$: $\nu \preceq \lambda$ (Lemma 3) and $wu \preceq u$ add, the permuted display's prefix read. A leading prefix of wu joins the assignment's to $k(d_f+1)$, entry against entry at the display $u_{w_j} + w_j = d_f + 1$; an assignment's k leading values, distinct positive naturals at most d_f , sum at or beyond the count's triangle $\sum_{i \leq k} i$, the top value withdrawn per step at the length capped by the bound; and the sorted display's prefix is the triangle's complement at $k(d_f+1)$, so it clears every permuted prefix. $c_\lambda = 1$: the top content sits on one line (Lemma 3) against $wu = u$ alone, every prefix equal forcing the assignment's prefixes to the triangle and each entry to the sorted order's own, the identity assignment. $\mathcal{L}_0$ reads each graded sum at $q(\mu+u)$, so the display forces the equality $c_\mu q(\lambda+u) = c_\mu q(\mu+u)$, and at dominant $\mu \preceq \lambda$, $\mu \neq \lambda$,

$$q(\lambda+u) = q(\mu+u) + \sum_{k < d_f} \delta_k g_k \quad \text{at} \quad \sum_{i \leq k} \mu_i + \delta_k = \sum_{i \leq k} \lambda_i, \quad \lambda_{k+1} + \mu_{k+1} + g_k = \lambda_k + \mu_k + 2,$$

the prefix gaps δ_k the dominance order's witnesses with some δ_k positive at $\mu \neq \lambda$, every g_k at least two at sorted shapes: so c_μ 's two members are equal off λ , such a summand entering every value read equally on its two sides, and $a_u \text{ch}_\lambda = a_{\lambda+u}$ is the one-value read. $\square$

Theorem 2 (The member assembly). *At a member's table with its positive list at the permutation read (Construction 4), the alternant pair at ρ assembles as the positive list's product: at the side fold h , the halved join $2h = \sum_\alpha (\alpha_+ + \alpha_-)$ over the displayed sides,*

$$x^h a_\rho = \prod_\alpha \langle x^{\alpha_+} : x^{\alpha_-} \rangle ,$$

monomials at the alternant's content keys (Construction 4), the positive-parts fold reading the shift, $\sum_\alpha \alpha_+ = \rho + h$. The series' chord-minor computations are the identity's displayed instances at their stated shifts, and the fixed members' side folds read 46, 11 and 6 on every key at E_8 , F_4 and G_2 , with E_7 's (26, 26, 26, 26, 26, 26, 17, 17) and E_6 's (16, 16, 16, 16, 16, 8, 8, 8) at the shared coordinates.

Proof. The subset fold. The product expands over the occupancy families S at the positive list, reads at most one per member: the S -term takes the second side at S 's members and the first at the rest, its side S 's count and its key h against the join's datum e_S at $e_S + \sum_{\alpha \in S} \alpha = \rho$, the shift the positive-parts fold, and the product collects factor by factor, each further factor reading every collected term twice, the term kept and the term at the key moved by the factor's root with the side exchanged, the collection a fold over the running families read per key at the two graded counts, the read the families' counts' sum at every key: the product is $x^h \Phi$ at the graded fold $\Phi := \langle \sum_{S \text{ even}} x^{e_S} : \sum_{S \text{ odd}} x^{e_S} \rangle$. *The grading.* Per simple index the pairing T_i joins α_i to the s_i -image of S at α_i off S , and reads the image of S 's further members at α_i in S : an involution moving the count by one, with $s_i e_S = e_{T_i S}$ at the letter's read $s_i \rho + \alpha_i = \rho$ and the permutation read, so Φ is graded, an odd element exchanging its members. *The support.* Every image of a key sits at ρ against a natural fold: at an element w , the join $w e_S + \sum_{\alpha \in S} w \alpha = w \rho$ splits at the images, a member at a positive image contributing it and a member at a partner image withdrawing its partner from the inversion list, one member each (the list at the images of the balance partners), so the ρ -fold collects

$$w e_S + \sum_{\beta \in N, \beta \text{ kept}} \beta + \sum_{\alpha \in S, w \alpha \text{ positive}} w \alpha = \rho$$

at N the element's inversion list, the kept members those off the withdrawals: a natural fold of positives. *The pin.* A dominant key at a simple coroot pair the unit is the letter's own fix, $s_i \mu = \mu$ at one odd letter, so a graded fold's read there is its own member swap, equal members. A dominant key at every simple pair beyond the unit is $\nu + \rho$ at dominant ν , and its support read $\nu + \rho \preceq \rho$ solves ν against the unit fold: pairing the join $\nu + \sum_i c_i \alpha_i$ at the unit fold against ρ reads $\langle \nu, \rho \rangle + \sum_i c_i \langle \alpha_i, \rho \rangle$ at equal members; $\langle \nu, \rho \rangle$ sits at its upper side or equal members, twice its read the fold of $\langle \nu, \alpha \rangle$ over the positive list at 2ρ the root fold, each member's read the halved products' fold over its own natural fold, $2 \langle \nu, \alpha \rangle = \sum_k c_k \nu(\alpha_k^\vee) \langle \alpha_k, \alpha_k \rangle$ (Lemma 9's read), every summand the length's natural multiple at a coroot pair on its upper side or equal members at dominant data, while every $\langle \alpha_i, \rho \rangle$ sits positive at its fold, so the join's fold is the unit family and ν with it: the key is ρ . *The conclusion.* Φ 's read at a key is its dominant image's up to the walk's parity, the dominance walk raising at a lower-side simple pair, the raised key's root-fold dot the key's own joined to the pair's margin at the letter's positive dot, an occupied key's dot joining its family's or its witness fold's positive dots to ρ 's own, and the raises closing at the gap's count; a key off both folds reads each graded fold at equal members, the two equal-membered reads one value. At the image ρ the read is one: $e_S = \rho$ forces $\sum_S \alpha$ the unit fold, every member's root-fold dot positive,

so S is the unit family at side even; the element at each orbit key is one and ρ 's images are one per element (the regular reads, Construction 4), so on ρ 's images Φ reads the Weyl fold's own members, and at every further dominant image the pin reads both Φ and a_ρ at equal members: $\Phi = a_\rho$, key by key. $\square$

Theorem 3 (The member character identity). *At a member's table, on a table space's block at top content λ , with ch_λ the content generating polynomial (Lemma 7's ch),*

$$a_\rho \text{ch}_\lambda = a_{\lambda+\rho} ,$$

the reads at cleared presentations with stated shifts, every display one value across the shifts (Definition 1's homogeneity principle), the A-series identity (Theorem 1) the first instantiation.

Proof. The eigen-read. Work at the coordinate dot, the form one scalar against it (Construction 4's form displays), every read one value across the rescaling: the trace recursion and telescope (Lemma 7) read $q_0(\lambda+\rho) \text{ch} = \mathcal{L}_0^\rho \text{ch} + \sum_\alpha \tilde{w}_\alpha 2G_\alpha$ at the dot weights $\tilde{w}_\alpha = [\text{dot}(\alpha, \alpha) : 2]$, $\mathcal{L}_0^\rho$ reading $q_0(\nu+\rho)$ per monomial. Three reads close the product's recursion. The *cofactors*: at the assembly identity (Theorem 2), Q_α is the cofactor at $\langle x^{\alpha+} : x^{\alpha-} \rangle$ $Q_\alpha = a_\rho$, and the telescope clears against it,

$$a_\rho G_\alpha = Q_\alpha x^{\alpha-} D_\alpha \text{ch} .$$

The *cross term*: a factor's Euler reads collect at the part-weighted folds $D_{\alpha+}$ and $D_{\alpha-}$, their gap the dot read, $D_{\alpha+} = \tilde{w}_\alpha D_\alpha + D_{\alpha-}$ per monomial, so per factor $\sum_i (D_i F_\alpha)(D_i \text{ch}) = x^{\alpha+} \tilde{w}_\alpha D_\alpha \text{ch} + F_\alpha D_{\alpha-} \text{ch}$, the first term clearing through the telescope at the cofactor, $Q_\alpha x^{\alpha+} D_\alpha \text{ch} = a_\rho G_\alpha + a_\rho D_\alpha \text{ch}$, and the fold over the factors collects to

$$\sum_i (D_i a_\rho) (D_i \text{ch}) = a_\rho \left(\sum_\alpha \tilde{w}_\alpha G_\alpha + D_P \text{ch} \right) \quad \text{at} \quad P := \sum_\alpha \alpha_+ ,$$

the positive-parts fold, $D_\alpha + D_{\alpha-} \tilde{w}_\alpha$ -collected to $D_{\alpha+}$ per factor. The *fold reads the shift*: $P = \rho + h$ at the side fold (Theorem 2), the series' displayed checks per coordinate its instances at $h = b_1 \mathbf{1}$: the positive-part counts $g_i + g + 2$ at C_ℓ , $2g_i + 2g + 2$ at B_ℓ and $2g_i + 2g$ at D_ℓ , over the joins $i + g_i = \ell$ and $g + 1 = \ell$, against ρ 's key reads $g_i + 1$, $2g_i + 1$ and $2g_i$ with the shifts $g + 1$, $2g + 1$ and $2g$. So, at the shifted Euler reads $\tilde{D}_i$, the diagonal pairs $\langle \nu_i : h_i \rangle$ per monomial at the side fold's keys, and $\tilde{\mathcal{L}} := \sum_i \tilde{D}_i^2$: the cleared a_ρ 's exponents read $w\rho + h$, the dot Weyl invariant, and the read rides the dominance walk at the graded fold: a key reading the fold off equal members raises along the walk to ρ , the walk and pin's read (Theorem 2), with each letter keeping the form (Construction 4), so q_0 reads one value at ρ and at every such key, $\tilde{\mathcal{L}} a_\rho = q_0(\rho) a_\rho$ at the graded counts key by key; and the square's Leibniz rule at the three reads collects to

$$\tilde{\mathcal{L}} (a_\rho \text{ch}) = q_0(\lambda+\rho) (a_\rho \text{ch}) .$$

The reads close at the ρ -dot's own heights: the Gram's ρ -column reads each entry's double as the length's multiple of the key's coefficient count over the positive folds (Construction 4), every entry at its upper side or equal members against the unit fold, so a dominant key at every coordinate beyond the unit sits at or beyond ρ 's own dot, the join at $\lambda + \rho$ capping its fold at the simple dots, the coefficients' fold against the simples' ρ -dots at or below $\langle \lambda, \rho \rangle$'s read, and a member on a root's line beyond a key sits within the two dots' gap at the root's positive dot. *The conclusion.* $a_\rho \text{ch}$ is reflection-graded, ch invariant (Lemma 6) and a_ρ graded, so it is a balance-pair combination $\sum_\mu c_\mu a_{\mu+\rho}$ over dominant μ at the cleared shifts: an exponent key whose dominant image holds a

unit simple pair reads equal members at that letter's own fix, the series' flips at an equal-membered coordinate and swaps at a repeated magnitude the displayed witnesses, and every further orbit reads one strictly dominant-shifted representative, the dominance walk's image at the regular reads (Construction 4). A shifted alternant is determined at four reads of a reflection-graded family: each letter joins a key to its image at the exchanged grading, every key sits at $\mu + \rho$ against a natural fold of simple roots, the read at $\mu + \rho$ is one at the even grading alone, and every key off $\mu + \rho$ at each simple coroot pair beyond the unit reads its two graded counts at one value (the regular reads, Construction 4): an occupied key raises along the walk to its dominant image, the raises closing at the fold's count and the counts mapped letter by letter, an image at a unit simple pair reads equal counts at its fixing letter, and the family reads $a_{\mu+\rho}$'s graded fold key by key. The support sits at $\mu \preceq \lambda$ ($\nu \preceq \lambda$ at the block, Lemma 3, and $w\rho \preceq \rho$ at the inversion list's fold, Construction 4, add), and $c_\lambda = 1$: a pair joining to $\lambda + \rho$ reads its two witness folds' ρ -dots joined at the sum's unit, every simple's ρ -dot positive at its fold (Theorem 2's pin), so both folds are the unit family and the pair is the two tops': the Weyl fold's one even read at ρ against the top content's one line (Lemma 3), the count one at the even grading alone. Each family is $\tilde{\mathcal{L}}$ -diagonal at $q_0(\mu+\rho)$, so the eigen-read forces $c_\mu q_0(\lambda+\rho) = c_\mu q_0(\mu+\rho)$; at dominant $\mu \preceq \lambda$ off λ the two reads' gap is the dominance fold's,

$$q_0(\mu+\rho) + \text{dot}(\lambda+\mu+2\rho, \sum_i c_i \alpha_i) = q_0(\lambda+\rho) \quad \text{at} \quad \mu + \sum_i c_i \alpha_i = \lambda ,$$

positive at dominant data, the fall identity's own read (Lemma 9): c_μ 's members are equal off λ , and the display is the one-value read. $\square$

Corollary 1 (The dimension is the Weyl product). $d_\lambda = \text{ch}_\lambda(1, \dots, 1)$, and the substitution $x_j := t^{u_j}$ at the unit set's sorted display maps the character identity's two graded sums (Theorem 1) to full monomials against pair products, the assembly's one-variable read, each graded sum reading its full monomial against its pair product, and the two full monomials' reads clear by the product's injectivity at the degree tie:

$$\text{ch}_\lambda(t^{d_f}, \dots, t) \prod_{j < k} \langle t^{u_j} : t^{u_k} \rangle = t^{|\lambda|} \prod_{j < k} \langle t^{l_j} : t^{l_k} \rangle \quad (l := \lambda + u).$$

Each factor clears by $\langle t^a : t^b \rangle = t^b \langle t : 1 \rangle^{\sum_{c+e+1=g} t^e}$ at $b < a$ and its witness gap g at $b + g = a$, the geometric word one term per splitting, the two sides with one $\langle t : 1 \rangle$ factor per pair, which drop at the shared monic factors' withdrawal, two products of one value at the division's descent (Definition 2), and the polynomial identity read at $t = 1$ gives

$$d_\lambda \prod_{j < k} g_{jk} = \prod_{j < k} g_{jk}^\lambda \quad \text{at} \quad u_k + g_{jk} = u_j, \quad l_k + g_{jk}^\lambda = l_j ,$$

the unit set's member gaps against the shape's, every right factor at least its left partner. At a member the identity specializes at the principal point, the substitution x^ν to $t^{\text{dot}(\nu, \rho)}$, and the assembly identity reads at either list (Theorem 2; Theorem 3): the member dimension is the gap product

$$d_\lambda \prod_{\alpha} g_{\alpha} = \prod_{\alpha} g_{\alpha}^\lambda \quad \text{at} \quad \text{dot}(\rho, \alpha) = g_{\alpha}, \quad \text{dot}(\lambda + \rho, \alpha) = g_{\alpha}^\lambda ,$$

the folds over the member's positive list, every right factor at or beyond its left partner.

Proof. The pair product expands over its occupancy families at the place pairs (Theorem 2's subset fold at the powers' gaps), the full monomial joining one display entry per place, so each family's

key is the fold of the display against its *taken counts*' successors, per place the count of its pairs taken there with the joined entry's own one, and its side the family's own count; at two places of one taken count the family joins its exchange at its first tied place pair, the pairs enumerated at the later place with the earlier beneath it, one read of the family kept by the exchange: each further pair reads to its partner at the places' transposition, sorted at the trichotomy, its taken place the letter swap's image with the membership flipped exactly where the sorting crosses the taken position, and the shared pair's take flipped, one key at an odd move of the side, the joined graded reads entering at equal members; a family at pairwise distinct taken counts takes every pair at one place against each further place in turn, so its counts fill the degrees below the width and the family is an assignment's, its side the inversion list's parity (Construction 1): each graded sum reads its full monomial against its pair product. At a member the substitution maps a graded fold to the exchanged list's, $\text{dot}(w\beta, \rho) = \text{dot}(\beta, w^{-1}\rho)$ at the kept form with the reversed word's element at its parity class (Construction 4), so $a_{\lambda+\rho}$'s principal read is a_ρ 's at the list $\lambda + \rho$'s own point, the series' role-symmetric signed and plus minors the displayed instances with the D_ℓ doubled alternant's two at both sides clearing by the product's injectivity, and each factor clears by the geometric word at its gap as above. $\square$

Corollary 2 (The product's block counts, by the alternant pairs). *For shapes a, b, c ,*

$$N_{ab}^c + \sum_{\sigma \text{ odd}} \text{mult}_a(\nu_\sigma) = \sum_{\sigma \text{ even}} \text{mult}_a(\nu_\sigma) \quad \text{at } \nu_\sigma + \sigma(b+u) = c+u ,$$

σ over S_{d_f} at the swap grading (Construction 1), u the unit set's sorted display, ν_σ the tie's lattice solution and the multiplicity read the contents' occupancy read, the counts descending to labels at matched degree (Construction 5). At a member the identity runs at the member alternants: at every member,

$$N_{ab}^c + \sum_{w \text{ odd}} \text{mult}_a(\nu_w) = \sum_{w \text{ even}} \text{mult}_a(\nu_w) \quad \text{at } \nu_w + w(b+\rho) = c+\rho ,$$

w over the member's Weyl list at the reflection grading (Construction 4), the ties' solutions matched lists.

Proof. $a_u \text{ch}_a \text{ch}_b = \text{ch}_a a_{b+u}$ and $a_u \text{ch}_a \text{ch}_b = \sum_{c'} N_{ab}^{c'} a_{c'+u}$ (Theorem 1 twice, the second at the product's content data, the pair list's count at every content by Lemma 5(iii), read over the exhaustion's blocks at Lemma 5(i),(ii)'s graded display), and the two reads match at the one monomial x^{c+u} : the product $\text{ch}_a a_{b+u}$ reads it at the displayed ties, one summand per permutation at its side and its solution, while a graded sum $a_{c'+u}$ reads it at $\sigma(c'+u) = c+u$ alone, the identity assignment at $c' = c$, two strictly decreasing displays at distinct entries: the coefficient pair is the display's. At a member the content generating polynomials multiply over a tensor product, the contents' sums, and the tensor exhausts into blocks at the member counts (Lemma 5(i),(ii) at table spaces), so the product $a_\rho \text{ch}_a \text{ch}_b$ reads per key twice (Theorem 3 twice): as $\text{ch}_a a_{b+\rho}$, a key's two counts the identity $a_\rho \text{ch}_b = a_{b+\rho}$'s fold over the a -family, one instance at each member's withdrawn key, and as the channels' fold $\sum_{c'} N_{ab}^{c'} a_{c'+\rho}$, each channel's identity at the key itself. At the key $c+\rho$, dominant at every simple pair beyond the unit, a further channel's fold is vacant, the orbit's one dominant image its own shifted top, and c 's channels read one at the even grading alone, the shifted key's top (the regular reads, Construction 4): the member display. $\square$

Lemma 10 (The member count and involution). *At a member's table (Construction 3; Construction 4) the fusion count and the label involution are constructions of the table's data, one each.*

(i) *The multiplicity family is the recursion's one solve. A block's family reads each occupied content at its summand's count, and the family closes key by key down the ρ -dots from the top, the recursion's one solution: the top's summand is one line with the occupied contents at or below the top (Lemma 3), every count is its dominant image's (Lemma 6), and at a dominant content below the top the trace recursion's display (Lemma 7) reads the count's multiple of the two Casimir reads' gap against the moved string folds' join. The member's fusion count is the member display's gap read at the family, the two graded sums over the member's Weyl list at the first factor's family (Corollary 2); the display folds at the family: each occupied content's moved key, the target's shifted key joined to the content's balance partner, raises along the dominance walk to its dominant image (Theorem 3's conclusion walk), one letter per raise with the letter count reading the tie element's parity, the walk's word that element's reversal at one parity class (Construction 4's grading and transport), a walk ending at the shifted second factor naming the tie's one element at the regular reads (Construction 4) with every further end off the ties, the shifted factor its orbit's one dominant image, so the count is the family's fold at the walks' parities, one walk per occupied content; and the fusion row at two labels is the dominant targets at occupied counts, the targets at or below the labels' sum (the factors' contents adding over the tensor product, Construction 3, each at or below its own top, Lemma 3).*

(ii) *The involution is the shifted walk's read. The dual action is the transposes' partner read (Lemma 8(i)'s displays at Construction 3's fields, each commutator's transpose the exchanged pair's with two partners composing to the original), so a dual block's occupied contents are the block's at their balance partners. The reversal is the element at the full inversion list, grown one letter at a member off the list (Construction 4's creation read), its image of every positive member the member's partner, the list's own read, and its image of ρ is ρ 's partner, the root fold's image at the partner list; its negated image of a dominant content is dominant, the coroot pairs read across the partner list. The dual block's top is the balance partner of the top's reversal image, maximal among the dual's occupied contents and off the sum's unit (Lemma 2(i)). So the dual label reads at the shifted walk: the dual's top joined to ρ is the dominant image of the top-with- ρ 's balance partner, one image at the regular reads (Construction 4), and the involution's construction is that walk, uniqueness by construction.*

Proof. (i) The dominance walk raises at a lower-side simple pair with each raise's ρ -dot positive (Theorem 3's conclusion walk; Theorem 2's pin), so a dominant image's ρ -dot sits at or beyond its orbit's every member's; the trace recursion's gap is positive at dominant data (Theorem 3; Lemma 9's fall identity), each moved count in the string folds' join read at its dominant image of strictly larger ρ -dot: the product's injectivity reads the count off the higher keys' (Definition 1), the descent closes on the finite window at the dominant keys' ρ -dots capped by the top's own (Theorem 3), and the family is the recursion's one solution. (ii) The reversal's image of a dominance join is the exchanged join at the partnered folds, so the image order flips and a block's occupied contents sit at or above the reversal image of the top, occupied once at the top's line (Lemma 6; Lemma 3); the dual block's top is that image's partner. $\square$

Lemma 11 (One box fuses up the shapes). $N_{\mu f}^c = 1$ at every dominant $c = \mu + e_i$, the row's members exactly those targets, f the one-box shape, its contents the e_i once each. At the complement's one-column shape f^* , the depth at the gap g at $g + 1 = d_f$ (Lemma 8(ii)'s complement at the one-box shape), the contents are the full column's withdrawn letters, one letter withdrawn per content at one count each, and $N_{\mu f^*}^c = 1$ at every dominant c at the join $c + e_i = \mu + (1^{d_f})$, the row's members exactly those targets.

Proof. Corollary 2 at $a = f$ reads the ties $\nu_\sigma + \sigma(\mu + u) = c + u$ at u the unit set's sorted display,

the one-box contents being the e_i once each, so a contribution needs $c + u = \sigma(\mu + u) + e_p$ at one place p : the display $c + u$ sorts $\mu + u$ with one value raised by one, a raise at an occupied successor reading a repeat ($\mu_h = \mu_{h+1}$ at $h + 1 = i$ puts the raised value at its neighbor's), which the strictly decreasing $c + u$ misses at every σ , and a raise at a vacant successor keeping its own place ($\sigma = \text{id}$, one count, on the even side), the target $\mu + e_i$ dominant exactly there. At f^* a count read transports across the letters' exchanges through the shared sorted representative, so each withdrawn letter's content reads the top line's count, one, and a content holding a letter twice reads its sorted representative's count, vacant at the dominance cap (Lemma 3; Lemma 6), and the tie analysis reads the complement's row: a contribution needs $c + u + e_p = \sigma(\mu + u) + (1^{d_f})$ at one place p , the tie read at the inverse permutation with the displays transported across it, a lowering of the sorted display the flipped assignment's raise (Construction 1's inverse permutation): the display $c + u$ sorting the raised display $\mu + u + (1^{d_f})$ with one value lowered by one, a lowering at a vacant gap reading a repeat ($\mu_p = \mu_{p+1}$ puts the lowered value at its follower's), which the strictly decreasing $c + u$ misses at every σ , and a lowering at an occupied gap keeping its own place ($\sigma = \text{id}$, one count, on the even side), the target dominant exactly there. $\square$

Lemma 12 (The path count's square sum). *The path count f^λ of a shape (Construction 1) is the count of its descents to the unit shape by one box removed at a time, the recursion $f^\lambda = \sum_{\mu \rightarrow \lambda} f^\mu$ over the shapes with one box removed, the unit shape's count one; at $k \leq d_f$, over the shapes of k boxes,*

$$\sum_{|\lambda|=k} (f^\lambda)^2 = k! .$$

Proof. On the free $\mathbb{P}$ -space of the shapes of at most k boxes, the unit shape among them, let U add a box over the covers and D remove one, transposes for the shape pairing, so $f^\lambda = \langle \lambda, U^k \mathbf{1} \rangle$, the recursion read at D 's transpose. At a shape of fewer than d_f rows, every shape below the degree d_f among them, $DU = UD + \text{id}$: two distinct shapes of one size share a cover exactly where they share a covered shape (their union and their intersection, one each, where they differ by one moved box), and one shape's addable corners exceed its removable ones by one (each is read off the distinct row lengths, the additions one more at the rows' end). So $D U^{j+1} \mathbf{1} = (j + 1) U^j \mathbf{1}$ at every $j < d_f$ by induction, $D \mathbf{1}$ the sum's unit and $U^j \mathbf{1}$ at the degree j , and

$$\sum_{|\lambda|=k} (f^\lambda)^2 = \langle U^k \mathbf{1}, U^k \mathbf{1} \rangle = \langle U^{k-1} \mathbf{1}, D U^k \mathbf{1} \rangle = k \langle U^{k-1} \mathbf{1}, U^{k-1} \mathbf{1} \rangle = k! .$$

$\square$

Lemma 13 (The mixed power's invariants are the permutation span). *At $k \leq d_f$ the invariants of $V^{\otimes k} \otimes (V^{\otimes k})^\vee$ are spanned by the $k!$ place permutations, whose Gram is the cycle count $d_f^{\#\text{cycles}(\sigma\tau^\top)}$, the transpose the inverse permutation's matrix (Construction 1), the pairing's index sums closing each cycle to a factor d_f .*

Proof. The block count is the path count. $M \otimes V$ exhausts as $\bigoplus_\mu N_\mu(M) (W_\mu \otimes V)$ (Lemma 5), and $W_\mu \otimes V$ by one boxes (Lemma 11), so at every power $N_\lambda(V^{\otimes k+1}) = \sum_{\mu \rightarrow \lambda} N_\mu(V^{\otimes k})$, the unit power the scalars' line, over the shapes with one box removed: $N_\lambda(V^{\otimes k}) = f^\lambda$, the path count (Lemma 12), every shape of k boxes at most $k \leq d_f$ rows deep. The invariant count. An invariant of $M \otimes M^\vee$ is the dual tensor's family at matched contents with every letter pair's Leibniz image null (Lemma 8(i)), its data one matrix per occupied content with the moved reads (Lemma 4's matrix families), and its form coordinates the matrices' entries at the contents' monomials; the families

combine at the entries, the action's row fold reading each entry once (Definition 3), so a member's value, the moved reads and the coordinates are additive and scale-reading over the members. At an exhaustion of the power's carrier (Lemma 5(i)) every top's value under the family is a top at the top's own content, a raising image of the value the moved read at the top's null image; and a family whose every top value is the sum's unit is null: Lemma 4(ii)'s cleared line reads every value at the top's value the sum's unit, the blocks span the carrier, and a null map's matrices are null at the carrier's own coordinate vectors. So the stacked top values map an independent list of invariants to an independent list inside the raising kernels' joined span, a null image's family null with the coefficients read back at the independence, and the invariants' count sits at or below the kernels' count (Lemma 3's exchange clause), N_λ tops per content (Lemma 5(ii)) at kernel dimension N_λ each (Definition 5):

$$\sum_{\lambda} N_{\lambda}^2 = \sum_{|\lambda|=k} (f^{\lambda})^2 = k!$$

(Lemma 12 at the path-count clause above). *Independence, the span, and the Gram.* A place permutation is equivariant (the units' Leibniz sum is place-symmetric, Construction 1), its family the permutation matrices per content, and at $k \leq d_f$ the $k!$ of them are independent, reading distinct monomials on $v_1 \otimes \cdots \otimes v_k$ at distinct indices; an invariant off their span would extend the independent list one past $k!$, against the count above: the permutations span. Two permutations' forms pair at the count of the monomials their two reads agree on, and the reads agree exactly at a monomial constant along each cycle of $\sigma\tau^T$, one free letter per cycle: the Gram is $d_f^{\#\text{cycles}(\sigma\tau^T)}$. $\square$

2 The form, the carrier, and the setting

Definition 6 (Labels, the Casimir, and its weight-free read). Lab is the label set of the calculus, one reduced shape per label (Construction 5) and one block at each (Lemma 4(iv)); and $C_2(\lambda) = \langle \lambda, \lambda + 2\rho \rangle$ the quadratic Casimir, the block read of Lemma 7 at the invariant form's scale (Proposition 1), with $\langle \cdot, \cdot \rangle$ the induced weight-lattice form, the reads $q(\lambda)$ and $\langle \lambda, 2\rho \rangle$ Lemma 7's display's. θ denotes the highest root, and

$$\begin{aligned} \hat{c}_2(\lambda) &:= \left[d_f Q(\lambda) : 2d_f^2 \right] \quad \text{at} \quad k^2 + d_f Q(\lambda) = d_f q(\lambda) + d_f \langle \lambda, 2\rho \rangle, \\ d_f Q(\lambda) &= \sum_{p < q} \langle \lambda_p : \lambda_q \rangle^2 + d_f \langle \lambda, 2\rho \rangle, \end{aligned}$$

the second display the join's constructed witness: the square identity $d_f q(\lambda) = k^2 + \sum_{p < q} \langle \lambda_p : \lambda_q \rangle^2$ collects the row pairs' squares against the degree (k^2 reading $q(\lambda)$ with the place pairs' cross products twice, the pairs' squares the complement), every summand a natural at the sorted rows (Lemma 7's fold), and at every nonunit label two coordinates differ (equal coordinates are full columns, the unit label's class), their pair's square positive by its shape; the dual label reads one value, $\hat{c}_2(\bar{a}) = \hat{c}_2(a)$, the complement's pair gaps the label's own at reversed places, one fold of squares, and the complement's row weights the balance partners there, the full weight list's fold of equal members (Construction 5; Lemma 8(ii)); the definition reads at the nonunit labels, the unit label's line Δ 's kernel, the kernel point's read; the clearing factor the fundamental count (Construction 1), the cleared traceless bracket at degree k against the adjoint's $Q(\theta) = 2d_f$ (Lemma 7): one quadratic form on the weight lattice read against itself, the form's components $\langle \lambda, \lambda + 2\rho \rangle$ and $\langle \theta, \theta + 2\rho \rangle$ rescaled by the one factor at every scale of the form, the pair one value across them (Definition 1; Proposition 1). The development runs on one object defined on the label index alone, $\Delta\chi_{\lambda} = C_2(\lambda)\chi_{\lambda}$.

Proposition 1 (One value on the form line). *The pairings of the θ -block against its dual are one line, the form line: θ is self-dual ($\theta^* = \theta$, Construction 5) and an equivariant map between blocks is the unit of its map space at distinct labels and a scalar at one (Lemma 4(ii),(iii)). The invariant form is a point ξ of that line, a scalar pair, and rescaling the point rescales C_2 with it while fixing every character. Hence a pair of two reads, each of degree one in the form, is one value at every point of the form line. Every energy-type read below $(E, \varepsilon_0, \text{the coupling weights, the constants } c_i)$ accordingly denotes its read against $C_2(\text{adj})$, and every surviving constant is a count.*

Proof. Rescaling the form's point rescales Δ 's block read (Lemma 7), hence $C_2(\lambda) = \langle \lambda, \lambda + 2\rho \rangle$ at every λ , and a pair of two such reads is one value at every point, its two components rescaled by the one factor. A character is a label-index generator, defined without the form (Proposition 8). Δ , content and E are each of degree one in the form and χ_θ, V, M are form-free, so each displayed pair of two such reads is one value at every form point. $\square$

Definition 7 (Generators, where a proof reaches for them). X_a are anti-Hermitian, the trace pairing reading $\text{tr}(X_a X_b) + \xi \delta_{ab}$ at the sum's unit and the Fierz read cleared at the count,

$$d_f \sum_a (X_a)_{pq} (X_a)_{rs} + \xi d_f \delta_{ps} \delta_{rq} = \xi \delta_{pq} \delta_{rs},$$

at the form's point ξ ; the Hermitian companions T_a at $X_a = iT_a$ read ξ throughout; then $\Delta = \sum_a X_a^\dagger X_a$. A generator list is displayed at entries pairs over the ground at the formal i , the remainder lists at $z^2 + 1$ (Definition 2): per place pair $p < q$ the swap combination, entry one at (p, q) with its balance partner at (q, p) , and the i -weighted sum, entry i at both places; per consecutive place pair the diagonal combination, entry i at the place's diagonal with i 's balance partner at the successor's; each anti-Hermitian by its shape, the list independent (the off-diagonal families by their disjoint place pairs and components, the diagonal telescope by its leading place), its count at the join $2\binom{d_f}{2} + r + 1 = d_f^2$. The sums run over the list against its metric dual, $\sum_a X_a(\cdot) X^a(\cdot)$, pair data at the Gram's adjugate against its determinant; the orthonormal spelling is that sum's evaluation.

Construction 6 (The fusion interface). The calculus's outputs the chain reads are one interface, its fields: the *label data*, a set with its equality read, involution and unit, each label equal to itself at that read; the *count laws*, the fusion counts at commutativity, associativity and the unit, with the unit read $N_{ab}^1 = \delta_{a\bar{b}}$ and the Cartan strictness $N_{ab}^{a+b} = 1$; the *dimension identity* $\sum_c N_{ab}^c d_c = d_a d_b$; the *weight-free Casimir*, positive at every nonunit label, with its floor at the named least labels, its dominance growth along a row, and the drift identity; the *self-dual θ* , with the character's two-sided form cap, its square's channel list at multiplicities, dimensions and Casimirs, the base c_1 , and θ 's own content list; the *class data*, the vertex law's classes over the root folds with the winding floors per class, the classes a finite group at the fold's sum with the unit class its unit; the *magnetic list L* , a list of nonunit labels, its *involution closure* $\bar{L}$ the label set of its members with their duals, the *plaquette character* $\chi_L := \sum_{\lambda \in \bar{L}} \chi_\lambda$, the closure's characters' fold, involution-fixed by the closure, and the list *faithful* where its classes' folds under the class group's sum reach every class, the adjoint's list $\{\theta\}$ faithful at the unit class group alone; the *evaluation*, the unit-coefficient read with orthonormal characters and the coefficient carrier's evaluation identity and integration by parts; and the *presentation*, the evaluation's computation at stated factor lists. A *series* enters with one further tier: the declared residue with every field a polynomial-pair read in it at stated reach clearances, the banded families the fusion-closed interiors with the θ content list among their data, and the generic lift's discipline; a fixed member enters at the member fields alone, its reads displayed finite evaluations. The label calculus above instantiates every field, each instantiating

statement the field's own derivation, and a further member's or series' instantiation is its own construction of the field data, the chain reading the fields alone: the classification's members each enter with every field a derived read at their weight tables, their positive lists, alternants, assembly, character identity, fusion rows, pairing spans and defining tables the fields' own instantiations, the count and involution fields at the member constructions (Lemma 10), the member's *stated list* a faithful magnetic list, the *A*-series' fundamental f , the *B* series' spinor, the *C* series' first fundamental, the *D* series' spinor at odd rank and its vector beside the spinor at even rank, E_6 's ω_1 , E_7 's ω_7 , and the member floor label at G_2 , F_4 and E_8 , whose class group is the unit's, each list's classes reaching every class at the member's class group, the classes the remainder families of Construction 4 (two at *B*, the spinor's odd parity generating; two at *C*, the first fundamental's odd sum; four at *D*, the spinor of order four at odd rank and the vector with the spinor the two generators of the two-by-two group at even rank; three at E_6 at ω_1 's remainder one; two at E_7 at ω_7 's occupied remainder), and every chain derivation reads the fusion data through the fields, at every member at once.

Construction 7 (The lattice interface). The carrier's region data are one interface, and every derivation reads the lattice through its fields: a set of *links* and a set of *vertices*, each link with one tail and one head, the orientation the labels' duality read; the *plaquettes*, each boundary a cyclic word of four distinct oriented links; *simplicity*, two vertices sharing at most one link; the *even-cycle read*, every cycle's length even, a two-coloring of the vertices its witness datum; the *d directions*, a partition of the links with a transverse-cut family per direction, each cut crossed by its own direction's links alone; the *boundary read*: a *cycle* at coefficients in a finite group A (the class group of Construction 6) is a family over the links with the vertex law, each vertex's outgoing coefficients' fold equal to its incoming coefficients', a *boundary fold* the fold of plaquette boundaries at coefficients, each boundary's links at the coefficient and its balance partner at the traversal signs, and the read is that every cycle whose *flux*, its fold over any transverse cut's crossing links at the crossing signs, reads A 's unit at every cut is a boundary fold, derived at a *tree*: a link subset joining every two vertices by one path, the count one below the vertices', an off-tree link's *fundamental cycle* the link closed by its tree path, a cycle's off-tree coefficients naming its fold of fundamental cycles with the gap a cycle inside the tree, the unit family at the tree's one path per vertex pair, so the read is the fundamental cycles' at the stated fluxes; the *translation action* on a torus window, permuting links, vertices and plaquettes and preserving directions, two shifts composing to the keys' one composite with the directions' composites one map at either order; and the signed coordinate permutations B_d , permuting the link and plaquette sets with orientation reversal dualizing labels. The fields read at two instantiation families. The region fields, the links, the vertices, the plaquettes, simplicity, and the even-cycle read with its two-coloring witness, are every carrier window's, the boxes of the directed family; the direction data with their transverse cuts read at both families. The translation action and the signed coordinate permutations are the torus family's, the windows of the momentum fibering: there a contractible cycle's wrap count is even, so its length reads even at the coordinate parity, while a winding cycle reads the side's parity and is odd at odd side, so a torus is a carrier of the interface's fields off the even-cycle read; the boundary read holds at the box and at the torus: at the box the tree is the *comb*, the last direction's links off the last position joined to the comb of the further directions at the first position, one direction at a time, the tree path between two vertices descending the directions' columns to the first positions in turn and ascending, the two descents' shared segment withdrawn, and every fundamental cycle bounds its *strip*, the plaquettes between the link and the tree path: at an off-tree link in direction i the strip is the fold of the plaquettes of the (i, j) -planes along the two descents' columns in the later directions j , one column per direction at the link's transverse coordinates,

joined to the direction- i segment's plaquettes at the first positions, each at its traversal sign, the fold's boundary the fundamental cycle with every interior link traversed twice at exchanged signs; so every cycle is a boundary fold, the fluxes reading the unit outright; at the torus the comb sits inside one box of the fundamental domain, a box link's fundamental cycle its strip's boundary fold at the box's read, a wrap link's fundamental cycle crossing its direction's cut at the wrap position once and every further cut at the sum's unit, so a cycle at unit fluxes reads its wrap coefficients per direction folding to the unit, its wrap part the fold of the differences of the direction's wrap cycles against one of them, each difference the boundary fold of the strip between the two wrap links, a boundary fold as well; the momentum fibering's stated family is the odd tori at side five or beyond. A *label-graph isomorphism* between two supports is a bijection of their links and vertices preserving incidence, orientation and labels. The dimension enters through the direction count alone, its reads (the neighbor count $2d$, the per-link and per-vertex plaquette counts, the per-direction momentum products, the charged direction sums) polynomial in it: at the family's boxes at two directions or beyond a link sits on at most $2(d-1)$ plaquettes, two in each plane through its direction, and a vertex on at most $2d(d-1)$, four in each plane at the vertex, and at one direction each sits on at most two, the consecutive squares sharing one link.

Definition 8 (The carrier). The sector is an indexed family with multiplicities, built from the fusion interface's fields alone (Construction 6). A *configuration* a assigns a nonunit label to each link of its *support*, finitely many links at a natural count (Construction 5); its *content* is $\sum_{\ell \in \text{supp } a} C_2(a_\ell)$, positive at every configuration; a vertex of $\text{touched}(a)$, the support's endpoints, is *occupied* where the incident labels' product, dualized on incoming links, reads an invariant count of at least one, and a is *occupied* where every touched vertex is. A *region* is a *box*, the vertices at coordinates within stated bounds in every direction with the links and plaquettes among them (Construction 7), the square window of one side its equal-sided instance, the boxes directed under inclusion, and every finite link set sits inside its *hull*, the least box holding it, its bounds the set's coordinate extremes. The window index is the occupied configurations at the region and the cutoff, and the window is the unit's line with their fibers:

$$\begin{aligned} \text{Idx}(\Lambda, C) &= \left\{ a \text{ occupied} \mid \text{supp } a \subseteq \Lambda, \sum_{\ell \in \text{supp } a} C_2(a_\ell) \leq C \right\}, \\ \text{Sect}(\Lambda, C) &= \mathbb{P} \mathbf{1} \oplus \bigoplus_{a \in \text{Idx}(\Lambda, C)} \text{Fib}(a), \quad \text{Fib}(a) = \bigotimes_{v \in \text{touched}(a)} \text{inv}(v, a), \end{aligned}$$

with $\text{inv}(v, a)$ the vertex's intertwiner space, the incident labels' product's invariants at a stated independent spanning list, incoming links dualized (Lemma 3's collected reads at Definition 5's count), and $\text{vmult}(v, a)$ its count, the vertex intertwiner multiplicity, a natural at every occupied vertex. The pairing is definitional as well: gram is the orthogonal direct sum of the unit line's pairing at one and the fibers' own pairings, per configuration the invariant lists' power pairing at the links' cleared block scale, one factor $[1 : d_{a_\ell}]$ per support link: positive definiteness is by construction, the power pairing a coordinate fold at positive scales. The windows are directed under $(\Lambda, C) \leq (\Lambda', C')$, the inclusions $\text{Sect}(\Lambda, C) \hookrightarrow \text{Sect}(\Lambda', C')$ are the summand injections mapping the unit line and each fiber to themselves, compatible under composition, and the carrier is their colimit

$$\text{Sect}_\infty = \varinjlim_{(\Lambda, C)} \text{Sect}(\Lambda, C).$$

Proposition 2 (A window is finite-dimensional, and its dimension is a count). *For a region Λ of finitely many links, the labels at content at or below C are finitely many, $\text{Idx}(\Lambda, C)$ is finite, each*

$\text{Fib}(a)$ is finite-dimensional, the dimension is one at the unit line alone, and at an occupied index

$$\dim \text{Sect}(\Lambda, C) = 1 + \sum_{a \in \text{Idx}(\Lambda, C)} \prod_{v \in \text{touched}(a)} \text{vmult}(v, a) ,$$

a count computed from the structure constants N_{ab}^c , the vertex multiplicity being $m\{a, b, c\} = N_{ab}^c$ (Proposition 8).

Proof. For dominant λ the read $\langle \lambda, 2\rho \rangle$ is a fold of natural products by its shape (Lemma 7's display), so $C_2(\lambda) \geq \langle \lambda, \lambda \rangle$ and the labels of content at most C satisfy $\langle \lambda, \lambda \rangle \leq C$: finitely many. At a reduced shape, at most r rows deep, the fold $\langle \lambda, 2\rho \rangle$ reads at or beyond the degree, the last place's pairs the rows' own reads, so $d_f Q(\lambda)$ clears $d_f k$ (Definition 6) and an enumeration at a stated cutoff on $d_f Q$ closes on the degrees whose d_f -multiple sits at or below the cutoff, at most the cutoff's quotient at the count. A configuration of $\text{Idx}(\Lambda, C)$ assigns to the links of its support, within Λ 's finitely many, labels from that finite set, so $\text{Idx}(\Lambda, C)$ is finite; and the enumeration reads the cutoff per link: a configuration's content is the fold over its links' reads, each a natural, so a partial assignment's content sits at or below every completion's and the enumeration refuses a partial beyond the cutoff with every completion refused beside it, the pruned walk and the whole product's filter one list; and the enumeration reads two cutoffs as one list: the nonunit labels at content at or below a cutoff are those at content at or below a larger cutoff filtered at the smaller, in the larger enumeration's order, so the index at the smaller cutoff is the index at the larger filtered at content at or below the smaller, the product lists' filter at the pointwise read, and Definition 8's inclusion of the window into the larger one reads at the window list's own positions. Each $\text{vmult}(v, a)$ is a natural at an occupied configuration and the product defining $\text{Fib}(a)$ has finitely many factors, so $\text{Fib}(a)$ is finite-dimensional and the displayed sum is a finite sum of finite products of structure-constant reads. Both indices are needed: the content bound alone leaves a loop free to sit at every translate, all at one content, so the region is what makes the count finite. $\square$

Proposition 3 (Fusion rows lie in content balls). *Every constituent of $a \otimes b$ has highest weight below the Cartan product $a + b$ in the dominance order, and $\hat{c}_2$ (Definition 6) is increasing there (at $\mu \preceq \lambda$ dominant, $C_2(\mu) + \sum_k \delta_k h_k = C_2(\lambda)$ at the dominance order's prefix gaps δ_k and the gaps h_k of the sorted list $\lambda + \mu + 2u$ at u the unit set's sorted display (Construction 1)), so*

$$\hat{c}_2(c) \leq \hat{c}_2(a+b) = \hat{c}_2(a) + \hat{c}_2(b) + [2d_f \langle a, b \rangle : 2d_f^2] \quad \text{for every nonunit member } c \text{ of the row,}$$

the unit member, one exactly at $b = \bar{a}$ (Proposition 8), the Casimir's kernel point, inside every content ball outright, the cross term twice the bilinear read ($C_2(a+b) = C_2(a) + C_2(b) + 2\langle a, b \rangle$). So a fusion row lies in a content ball, and content balls are finite, the labels at weight-free Casimir at or below a stated pair finitely many (Proposition 2). The attained maximum $\Phi(C, C')$ at two contents is the largest content of a product of two configurations at contents at or below C and C' : over two label lists at a shared link count at or below the two link caps' sum, the caps L at $L \hat{c}_2(f) \leq C$ and L' at $L' \hat{c}_2(f) \leq C'$ at the member's floor $\hat{c}_2(f)$ (Lemma 9), each list's labels in its content's ball with its fold at or below its content, and over every shared link's fusion target at the two labels, a label one list leaves vacant kept at the other's and a link vacant in both lists reading the unit label, the targets' contents' fold, one trichotomy fold over finitely many reads. This bound and Φ do different jobs: the bound makes the row a finite set, while Φ is the attained maximum over it, and the window index uses Φ , since a bound composed from the two contents' root sum forces a rounding whose composition depends on the order taken.

Proof. A constituent's top content is a content of the product, contents of a tensor product add over the factors, and a block's contents sit at or below its top (Lemma 3). In the displayed identity every summand is two naturals' product, so C_2 and $\hat{c}_2$ rise along the dominance order. In the display $d_f \langle a, b \rangle$ is the weight lattice's integer read, every term one form read against itself, the pair at one value across the form's scales (Definition 1; Proposition 1). The labels at Casimir at or below a stated pair are finitely many (Proposition 2); a configuration at content at or below C holds at most L links at $L \hat{c}_2(f) \leq C$, every occupied link's read at or beyond the floor (Lemma 9), so a shared link count at or below the caps' sum holds every pair of configurations with their vacant links padded, the label lists at each count are finitely many, each shared link's row finite by the ball, and the fold's values finitely many, the largest at the trichotomy fold. $\square$

Proposition 4 (Every element has a least window, read off its index support). *An $A \in \text{Sect}_\infty$ is represented by some $x \in \text{Sect}(\Lambda, C)$, a unit coordinate with its family of fiber components over the index. The index support $\text{ix}(A)$, the member configurations, is representative-free; the unit coordinate is at home in every window, multiplying as the scalar it is, every window its own. An element whose index support has members reads its window data over them:*

$$\Lambda_A = \text{the hull of } \bigcup_{a \in \text{ix}(A)} \text{supp } a, \quad C_A = \max_{a \in \text{ix}(A)} \text{content}(a)$$

are one value at every representative (Definition 8's hull), and $\text{supp}(A) = (\Lambda_A, C_A)$ is the least window whose image contains A 's components. Both are counts off the index: the links the element occupies and its members' largest content. A probe is an element of the carrier, and its window is this one.

Proof. The index support is representative-free because the inclusions are summand injections and map the unit line and each component to themselves, so two representatives read one member family and the window data, reads of that family, one value. $\square$

Definition 9 (The product and the involution). Sect_∞ is a unital $*$ -algebra. A factor of the unit coordinate alone multiplies as its scalar, every window its own (Proposition 4); at A and u whose index supports have members, at the least windows $\text{supp}(A) = (\Lambda_A, C_A)$ and $\text{supp}(u) = (\Lambda_u, C_u)$ (Proposition 4), the *target window* is $\text{Sect}_0 := \text{Sect}(\Lambda_A \vee \Lambda_u, \Phi(C_A, C_u))$ at the boxes' hull, its *window list* the unit line's vector with the fibers' stated lists, the pairing's Gram of unequal-membered determinant at the lists' independence (Definition 8; Lemma 3), and the product $A \cdot u$ is the target window's element at the reads

$$\text{gram}(A \cdot u, v) = \text{Eval}(\bar{A} \bar{u} v) \quad \text{at every window-list member } v,$$

the element the reads' solve, the adjugate column against the Gram's determinant (Definition 3), one finite ground computation per read, the right side the *unit read*: Eval reads its argument's unit coordinate in the label index (Definition 8), a finite ground computation at the product's exhaustion, the unit block's component (Lemma 5(i)), at a loop window the invariant count's own read (Proposition 8). Its matrix elements between summands, $\text{gram}(x, A \cdot y)$ for x, y window-list members, are the label-index reads of Proposition 8 on the changed edge alone, and the magnetic form is built from them. The involution is contour reversal, descending from the words' own involution, Eval one value under it.

Proposition 5 (The product reads the pairing identity, and the algebra laws are structure-constant identities). *The product of Definition 9 reads the pairing identity against every element, $\text{gram}_\infty(A \cdot$*

$u, v) = \text{Eval}(\bar{A}\bar{u}v)$ at every v , at the colimit pairing gram_∞ , two elements' read at a shared window, one value at every window; it is the target window's one element at that identity, and every window at or beyond the target window solves the identity to it; the product of two window elements sits in a window, the window bound stated on the index alone,

$$A \cdot \text{Sect}(\Lambda, C) \subseteq \text{Sect}(\Lambda_A \vee \Lambda, \Phi(C_A, C)),$$

$\Lambda_A \vee \Lambda$ the hull of the two boxes' union and Φ the attained maximum (Proposition 3); and in the colimit commutativity, associativity and the unit are the identities

$$N_{ab}^c = N_{ba}^c, \quad \sum_e N_{ab}^e N_{ec}^d = \sum_f N_{bc}^f N_{af}^d, \quad N_{a1}^c = \delta_{ac}.$$

Proof. Take A and u whose index supports have members, with $\text{Sect}_0 = \text{Sect}(\Lambda_A \vee \Lambda_u, \Phi(C_A, C_u))$ the target window of Definition 9 at the least windows (Proposition 4); a factor of the unit coordinate alone multiplies as its scalar, its identity the scalar's read. The window bound: at $u \in \text{Sect}(\Lambda, C)$ the least window sits at $\Lambda_u \subseteq \Lambda$ and $C_u \leq C$ (Proposition 4), so the target window's box sits inside $\Lambda_A \vee \Lambda$ and its content $\Phi(C_A, C_u)$ at or below $\Phi(C_A, C)$, the attained maximum nondecreasing in each argument, the larger content's lists holding the smaller's (Proposition 3); the product, the target window's element, sits in the displayed window.

The functional reads equal members off Sect_0 . Eval reads the unit coordinate, so $\text{Eval}(\bar{A}\bar{u}v)$ is of equal members unless the labels of v occur among the constituents of $\bar{A}$'s against $\bar{u}$'s, dual labels at one link set and one content each (Definition 6). Those constituents sit on the links of $\Lambda_A \vee \Lambda_u$ and, by Proposition 3, at content at most $\Phi(C_A, C_u)$: they are summands of Sect_0 . So $v \mapsto \text{Eval}(\bar{A}\bar{u}v)$ is of equal members on every summand beyond Sect_0 .

On Sect_0 the identity is the definition's solved read. Sect_0 is finite-dimensional (Proposition 2) and its window list is independent with its pairing Gram of unequal-membered determinant (Definition 8; Lemma 3), so the identity's reads against the list solve the one $x \in \text{Sect}_0$ with $\text{gram}(x, v) = \text{Eval}(\bar{A}\bar{u}v)$ at every $v \in \text{Sect}_0$, the adjugate column against the determinant (Lemma 23(ii); Definition 3); two stated lists spanning Sect_0 solve one element, each solution reading the identity against the other's members, the solutions' gap at equal members against every member with its self-pairing that unit at the positive pairing, so a presentation's wiring list reads the determination at its own Gram. For v in any larger window, $\text{gram}(x, v)$ reads only v 's Sect_0 -summands, gram being the orthogonal direct sum, and both sides are of equal members off Sect_0 as above. So the product reads the identity against every v ; a window at or beyond the target reads the identity at its own list to one further solved element, each reading the identity against every v , the two agreeing against every element, one element at the positive pairing on the larger window; and the product sits in Sect_0 and descends to the colimit.

The laws. Sect_∞ 's elements are character polynomials and Eval reads their product's unit coefficient, so the product just constructed is the character polynomials' own, commutative and associative with the unit label as its unit. Read in the label index those three statements are the displayed identities on N (Proposition 8). The pairing and the link Laplacians are compatible with the summand injections, so they descend to Sect_∞ and restrict to every window: a pairing of two colimit elements is read at any common window, one value at every window. $\square$

Lemma 14 (Matrix elements are reads at the support's reach). *Let x, y be elements of a window with joint support $S := \Lambda_x \cup \Lambda_y$. Then $\text{gram}(x, y)$ and $\langle x, Ey \rangle$ read the index data on S alone, and the magnetic row reads only the plaquettes meeting S 's vertex neighborhood: at ∂p sharing a vertex*

with S , the entry $\langle x, \chi_L(U_{\partial p}) y \rangle$ at the plaquette character (Definition 10) is the changed-edge read of Definition 9 at the labels incident to p ; at ∂p off that neighborhood it factorizes,

$$\langle x, \chi_L(U_{\partial p}) y \rangle = \text{Eval}(\bar{x}y) \cdot \text{Eval}(\chi_L), \quad \text{Eval}(\chi_L) = \sum_{\lambda \in \bar{L}} \delta_{\lambda 1} \quad (\text{Proposition 8}),$$

the fold of equal members at the list's nonunit members (Construction 6). Consequently a label-graph isomorphism of the supports' vertex neighborhoods transports every pairing, electric and magnetic read across windows: each entry is one polynomial-pair read in the residue, window-free once a window's region contains the neighborhood. The locality band is that reach, the plaquette neighborhood's: a magnetic row's entries off it are the factorized reads at the unit character, the sum's unit, so every pencil row reads within one band of its support, at every window.

Proof. gram is the orthogonal sum of the fibers' pairings at matched configurations and E scales each summand by its content, both index reads on S (Definitions 8, 10); the product $\bar{x}y$ is supported on S at bounded content (Proposition 5). For the factorization: two elements on disjoint links and vertices have Fib data at disjoint vertex sets and label data at disjoint links, so the product's index summand at (a, b) is the concatenation of the two summands, each vertex's vmult reading its own incident labels; a concatenation occupies a link exactly where a part does, so the product's unit coordinate is the unit coordinates' product, and $\text{Eval}(\chi_L)$, a fold of Kronecker deltas at two labels each at a nonunit member (Construction 6), is of equal members, the far plaquette's term with it. The near case is Definition 9's read, a fusion computation at the labels incident to p . $\square$

Construction 8 (The coefficient carrier). The carrier's vertex-free companion drops the invariance constraint: per link the *coefficient family* $\bigoplus_{\lambda} W_{\lambda} \otimes W_{\lambda}^{\vee}$, one block per label (Construction 5; the blocks and duals of Lemmas 4, 8), per window the tensor product of its links' families over the window index of Definition 8, the involution the blocks' dual exchange with the unit block fixed. Eval extends as the per-link unit coefficient: a product of coefficient elements decomposes by the fusion data, and its unit coefficient is the invariant line's read on both data, the vector datum's component at the orthogonal split (Lemma 5(i); Lemma 1) against the functional datum's value on the line, the component the projection fold's read at the line's member (Lemma 3). Two one-block elements pair at the *block orthogonality*,

$$d_{\lambda} \text{Eval}(\overline{u \otimes f} (u' \otimes f')) = \langle u, u' \rangle \langle f, f' \rangle ,$$

the product a coefficient of the dual block against the block, its unit channel one line (Proposition 8's unit read; Lemma 4(ii),(iii)) at the coevaluation's tensor, and the display the projection read at two adjugate identities: the coevaluation against a member pair reads the factors' pairing, once per datum, and its self-pairing reads the block's count against the determinants (Lemma 8(i)); two blocks' elements pair at equal members, the unit channel vacant at distinct labels (Proposition 8). The *evaluation identity* $\sum_j U_{ij} \bar{U}_{kj} = \delta_{ik}$ is the dual pairing's read: the evaluation's tensor is invariant (Lemma 8(i)'s equivariance), so its coefficient family is its unit component's constants, the Kronecker read entry by entry. gram is the orthogonal direct sum of the blocks' standard inner products at the cleared block scale $[1 : d_{\lambda}]$: positive definiteness is by construction at the positive scales, the gram is the evaluation's own pairing, $\text{gram}(x, y) = \text{Eval}(\bar{x}y)$, the block orthogonality collected over the summands, the multiplication walks across it at a coefficient element w , $\text{gram}(wx, y) = \text{gram}(x, \bar{w}y)$, the products' exchange and regrouping at Eval's one value under the involution (Definition 9; Proposition 5), and the carrier sits inside isometrically at its own gram (Definition 8): a configuration's state contracts the links' block coefficients at the vertices' invariant

lists, Eval factors over the support's links at disjoint variables (Lemma 14), and two states' pairing collects the block orthogonality link by link, the links' scales against the slot pairings' product regrouped at the vertices' lists (Definition 1), the fiber gram's own display. A generator acts within each block (a matrix coefficient of λ differentiates to one), so $\text{Eval} \circ X_a$ reads equal members at every argument, the unit block the constants' line, and the pairing integrates by parts, the two reads $\langle X_a x, y \rangle$ and $\langle x, X_a y \rangle$ balance partners (X_a a derivation at one value under the involution, $X_a \bar{x} = \overline{X_a x}$); the *parts identity* is that read at a product: at two coefficient elements F, P the evaluation of the derivation's Leibniz read at FP is the sum's unit, $\text{Eval}((X_a F) P) + \text{Eval}(F (X_a P))$ at that unit, $\text{Eval} \circ X_a$ read at the product. Consequently every finite family of coefficient vectors' Gram is positive semidefinite (Lemma 30): a coefficient vector's form at the Gram is the family's combination at that vector paired against itself, the pairing bilinear (Definition 1), a self-pairing at or beyond the sum's unit at the positive pairing, so every split of the Gram reads the upper side throughout; and a sum of squares $\sum_i \|T_i x\|^2$, each T_i a generator action or a coefficient multiplication, is a fold of squares, and where the summed operator maps the carrier to itself the read restricts to every window. A generator action joined to a coefficient multiplication at its own involution image, $T := \eta X_a + G$ at a pair η , squares to the operator read

$$T^\dagger T = \eta^2 X_a^\dagger X_a + G^2 + \langle \eta (X_a G) \rangle^\vee,$$

the last member the balance partner of the multiplication by the generator's read of the multiplier: the two cross pairings $\langle X_a x, Gy \rangle + \langle Gx, X_a y \rangle$ move the generator across the pairing at the balance partner and split the product at the derivation's Leibniz read, $X_a(Gy) = (X_a G)y + G(X_a y)$, the two multiplier terms joining to the sum's unit; so a fold of such squares over the links and the list reads the electric member at the pair's square against the multipliers' squares and the balance partner of the pair's multiple of their *divergence* $\sum_{\ell,a} X_a^\ell (G_{\ell a})$, one multiplication. The pairing's squared Cauchy–Schwarz is the two-member instance at the coordinate folds, its gap a fold of squares displayed once at the cross pairs c_{ij} , each joined to its exchange's balance partner at the memberwise swap (Definition 3),

$$\left(\sum_i u_i v_i \right)^2 + \sum_{i < j} (c_{ij} + \check{c}_{ji})^2 = \left(\sum_i u_i^2 \right) \left(\sum_i v_i^2 \right) \quad \text{at } c_{ij} := u_i v_j,$$

the cross monomials $u_i u_j v_i v_j$ entering each member once per place pair, so two vectors' squared pairing sits at or below their self-pairings' product, the comparison's witness the displayed fold; a finite family of paired slots reads it at the slots' joined coordinate list, one list. A *positive element* is a fold of products of coefficient elements' squares at coefficients of positive pairs, a stage's adjoined roots among them (Lemma 37); sums and products of positive elements are positive elements, the evaluation reads a positive element's product with a square at or beyond the sum's unit, the pairing's positivity collected over the fold, and a *comparison* $a \leq b$ of coefficient elements is the read $b = a + P$ at a positive element P , two comparisons adding and a comparison multiplying by a positive element; the evaluation's squared Cauchy–Schwarz, $\text{Eval}(fg)^2 \leq \text{Eval}(f^2) \text{Eval}(g^2)$ at involution-fixed f, g , is the pairing's own at $\text{gram}(f, g)$. The *flat read* maps every block's coefficient matrix to the identity and a product of elements to its reads' product, the tensor's identity the blocks' projectors' sum at the exhaustion (Lemma 5(i)), so it reads a character at its dimension; an element's flat read is the fold over its occupied configurations of the coefficient tensors' traces, each trace at or below the dimensions' product's root against the tensor's fold of squares, so the read's square sits at or below the square's evaluation against the *support weight*, the fold over the occupied configurations of the links' dimensions' squares, the squared Cauchy–Schwarz at the configurations' fold with the cleared block scales.

Definition 10 (The electric and magnetic operators, and the pencil). E is the electric operator, diagonal in the label index, scaling a configuration's summand by its content and reading the unit line as its kernel, so E is positive semidefinite, every content positive (Definition 8); Δ_ℓ is its per-link summand, scaling by $C_2(a_\ell)$, so $E = \sum_\ell \Delta_\ell$; M is the plaquette multiplication $\sum_p \chi_L(U_{\partial p})$ at the magnetic list's plaquette character (Construction 6), the fold over the list's involution closure $\bar{L}$ of the labels' characters, involution-fixed, at the member's stated list where a statement names one and at every magnetic list otherwise, the sum over the region's plaquettes, $\#p$ their count; χ_p abbreviates the plaquette's term $\chi_L(U_{\partial p})$. The *pencil* is the family of form pairs $(\alpha E : \beta M)$ with H their site datum at the join

$$\alpha E = \beta M + H([\alpha : \beta]), \quad \alpha, \beta \in \mathbb{N}$$

over the end-weight pairs of Proposition 6 with the two ends' one-member instances, an end's pair reading its one member alone, the contact end's datum the magnetic member's balance partner. On a window list with positive-definite Gram G (Definition 8), the symmetric matrices E , M and H are forms; their actions E , M and K are the Gram solves

$$GE = E, \quad GM = M, \quad GK = H, \quad \alpha E = \beta M + K.$$

Each column is the Gram's adjugate applied to the form's column against $\det G$, the solved witness of Definition 3. The ground level ε_0 is the least located root of the pencil (Lemma 33). Its *level-gap form* is the site datum S at $\varepsilon_0 G + S = H$, positive semidefinite; its *level-gap action* is D at the Gram solve $GD = S$. Thus $K = \varepsilon_0 I + D$. A window ground ψ reads $H\psi = \varepsilon_0 G\psi$, $K\psi = \varepsilon_0 \psi$ and $D\psi = S\psi = 0$, the last glyph the sum's unit; every read at a window ground quantifies over the grounds. On these coordinates the pairing is $\langle x, y \rangle_G := x^\top G y$. A map A from a list of Gram G to a list of Gram G' has its pairing adjoint at the solve $GA^\dagger = A^\top G'$. Its squared-action form is $A^\top G' A$. In particular,

$$\begin{aligned} D^\top G &= S, & SD &= D^\top G D, \\ \mu_1(v) &= v^\top S v = \langle v, Dv \rangle_G, & \mu_2(v) &= v^\top S D v = \langle Dv, Dv \rangle_G. \end{aligned}$$

The first identity is the transpose of $GD = S$ at S symmetric; the further identities are its products. A form's restriction to columns L is $L^\top F L$, with the pairing restricted to $L^\top G L$. At a unit Gram the form and its action read one matrix. The state reads actions, $\omega(A) = \langle \psi, A\psi \rangle_G$; an observable's form F reads $\psi^\top F \psi$. At a positive pair h and an orthogonal two-block list, $G = G_1 \oplus G_2$, write D_{ij} for the action blocks. Block multiplication of $SD = D^\top G D$ gives the whole detector's second block,

$$(SD - hS)_{22} = D_{12}^\top G_1 D_{12} + D_{22}^\top G_2 D_{22} - hG_2 D_{22}.$$

The first summand is the positive Gram of the action's image in the first block, and $S_{22} = G_2 D_{22}$ is the restricted first-moment form. At two nested cutoffs in one region, write J for the smaller list's inclusion and $J_\perp$ for its orthogonal complement, with $G = J^\top G' J$ and $G_\perp = J_\perp^\top G' J_\perp$. Let e', D', S' be the larger window's data and e, D, S the smaller's, with the gap $\delta \geq 0$ at $e' + \delta = e$ (Lemma 30). Then

$$J^\dagger D' J = D + \delta I, \quad B := J_\perp^\dagger D' J,$$

and inserting the two orthogonal projections into the square gives

$$J^\top (S' D' - h S') J = SD - hS + 2\delta S + \delta(\delta - h)G + B^\top G_\perp B.$$

The exterior image's Gram and the ground's shift enter the whole-square compression at these displayed terms. The *ground-relative level read* at an offset a is

$$\text{rev}(S : aG) = \text{rev}(H : (\varepsilon_0 + a)G).$$

The site datum R at $S = aG + R$ reads $H = (\varepsilon_0 + a)G + R$ at the ground's join, so the two counts read one matrix and one split. At positive offsets $\ell < h$, with $\ell + w = h$, the two absolute lines are $\varepsilon_0 + \ell$ and $\varepsilon_0 + h$, and their gap is w at the displayed sum's associativity and injectivity. The ground is the window's least located root, with its designation read at that window and coupling; the offsets are the fixed data in a statement over windows. The unit vector's electric form reads the sum's unit and its magnetic form reads $\text{Eval}(\chi_L)$ per plaquette, the fold of the nonunit labels' unit coefficients (Proposition 8), so the unit vector's pencil form reads the sum's unit and the least root sits at or below it (Lemma 33). The gap chain reads a coupling in its *root coordinate*: the weights $[q^2 : p^2]$ at the pair $\tau = [p : q]$, $q\tau = p$ the cross-multiplied tie, a cell's reads taken at pair τ , the pair- τ rays the coordinate's own points; toward the contact end the dual coordinate weights the couplings $[\sigma : 1]$ at pair σ . The *word sector* is $\text{Sect}_\infty \psi$, the image of multiplication by ψ (Definition 9).

Proposition 6 (The base is a closed segment). *A coupling is an end-weight pair $[\alpha : \beta]$ at $\alpha, \beta \in \mathbb{N}$, taken up to rescaling: α weights the free end's electric member E and β the contact end's magnetic member M , the coupling pencil the pair $(\alpha E : \beta M)$'s site datum at the displayed join. The base $\mathcal{C}$ is the couplings with the two ends, the one-member weightings: the free end, the electric member alone, $H = E$ there, and the contact end, the magnetic alone, H the member's balance partner. Every coupling pair is interior, both weights present by the shape of its data, and the base is the closed segment, the free and the contact end its two endpoints. A spectral quantity of the pencil homogeneous of degree one scales by one factor along a ray, so the pair of two such reads is one value on the ray, the homogeneity principle's read (Definition 1), and descends to $\mathcal{C}$.*

Proof. The pair's site datum is additive over its members and degree-one in each weight by the join's shape, so rescaling the weights by a natural u rescales the pencil by u , and scaling an operator by u scales its spectrum by u : a degree-one spectral quantity acquires that one factor, so both components of a pair of two such reads acquire it and the pair reads one value across the ray, the homogeneity principle's read, every rescaling a ground rescaling and so order-preserving by the type. The weight data up to that rescaling are the two-member rays and the two one-member classes: the segment's interior and its two ends. $\square$

3 The statement

Write r for the member's derived residue, the pairing's own fold (Construction 4; Proposition 12), the A -series' declared scalar its instance with the successor name the fundamental count $d_f = r + 1$ (Construction 1), $\mathcal{C}$ for the closed segment of Proposition 6, E_0 for the free-end level (Proposition 7), and $\mathcal{K}$ for the invariant (Definition 11).

Proposition 7 (The free end's level is the floor loop). *At the free end the floor loop, a member floor label f_0 , a nonunit label of the least weight-free Casimir (Lemma 9, f at the A -series), around one plaquette's boundary, is an eigenvector of the pencil at the least occupied content, the free-end level*

$$E_0 := 4\hat{c}_2(f_0),$$

$[2r(r+2) : d_f^2]$ at the A -series, at every member and on either carrier, positive by its shape.

Proof. At the free end the pencil is its electric member E alone (Definition 10; Proposition 6), and E is diagonal in the label index, scaling the summand at a by $\text{content}(a) = \sum_{\ell} C_2(a_{\ell})$ (Definition 10). In that index the floor loop is the configuration assigning f_0 to each link of ∂p at the traversal's orientation, occupied at every corner, one outgoing f_0 against one incoming at $\text{vmult} = N_{f_0 \bar{f}_0}^1 = 1$ (Proposition 8; Definition 8, incoming links dualized), so it is an eigenvector of E with eigenvalue its content, the fold over ∂p 's four links (Construction 7), $4 C_2(f_0)$, read against $C_2(\text{adj})$ (Proposition 1) the display. It is the least occupied content: at an occupied configuration every touched vertex reads an invariant count of at least one (Definition 8); at a vertex meeting exactly one support link an invariant line would be an equivariant line between the unit block and a nonunit one, and an equivariant line forces one label (Proposition 8; Lemma 4(iii)), against the support's nonunit datum, so the support's links meet each of their vertices at least twice; a finite graph of minimum degree two holds a cycle (a forest has a leaf), every cycle is even and at least four, the graph simple (the even-cycle and simplicity reads of Construction 7); and each occupied link reads content at or beyond the member floor (Lemma 9): every occupied content sits at or beyond four times the floor, the floor loop's own. On the coefficient carrier the floor loop is the f_0 -blocks' coefficient at the boundary's four links, each link's Δ_{ℓ} scaling its block at $C_2(f_0)$ (a matrix coefficient of λ differentiates to one, Construction 8) with every further link's summand at the sum's unit, so the eigen-read is the four-link fold there as well, one value on either carrier. $\square$

Lemma 15 (The free end reads the invariant at one). *At the free end the window cut of Definition 11 is a diagonal read, and*

$$\mathcal{K} = 1 \quad \text{at the free end,}$$

at every member, with one admitted by every finite-window cut (Definition 8; Definition 11), the level the free-end level E_0 itself (Proposition 7).

Proof. At the free end the pencil is its electric member E alone, diagonal in the label index with the contents as eigenvalues (Definition 10). Every configuration's content is positive (Definition 8), so the ground is the unit's line, ψ the algebra's unit (Definition 9) and E 's kernel point (Definition 10), the floor of equal members there and the level-gap form reading $S = E$ and its action $D = \mathbf{E}$, and the word sector is the window itself. The cut's pair $(SD : E_0 \kappa S)$ is block diagonal, the unit summand its kernel's and each configuration's block its Gram at the multiple $c \langle c : E_0 \kappa \rangle$, c the configuration's content: the cut at κ holds exactly where every occupied configuration's content is at least $E_0 \kappa$. Every occupied content sits at or beyond E_0 , the floor loop's content, and the floor loop attains it (Proposition 7), so the cut holds at every κ at or below one. In a region containing the floor plaquette, every cutoff at or beyond E_0 holds its loop and refuses a floor beyond one, the floor loop's entry $E_0 \langle E_0 : E_0 \kappa \rangle$ on its lower side there: the contents are window-free data, so the read is one diagonal computation at every window. Every window cut admits one, and a window holding the floor loop refuses every pair beyond one. Their meet therefore has edge one and admits it (Definition 11). $\square$

Definition 11 (The invariant). $\mathcal{K}$ is the pencil's level read against the free-end level E_0 of Proposition 7, given by its window cuts: at a coupling v , a window Λ and a pair $\kappa = [p : q]$,

$$\mathcal{K}_{\Lambda}(v) \geq \kappa \quad \text{is} \quad (qSD)_{\Lambda} : (E_0 pS)_{\Lambda} \text{ positive semidefinite,}$$

the pair clearing its pencil to integer coefficients, a positive rescaling fixing the count (Lemma 30), with S the level-gap form and D its action from the Gram solve (Definition 10), read on the word sector (Theorem 9), and $(\cdot)_{\Lambda}$ the compression to the window's word sector, taken of the

detector form whole: the count reads the square's compression, a compression keeping the positive-semidefinite read (Lemma 30). The display quantifies over the window grounds (Definition 10), the ground freedom's quotient read: at a ground the compression is the detector pair's pullback along word images spanning the sector, a spanning list by elimination (Definition 3), a combination reading its image vector's pair and a kernel combination reading the sum's unit, so the pulled-back pair is positive semidefinite exactly with the pair on the word sector, its entries quadratic in the ground's coordinates over the located stage (Lemma 32, the kernel's spanning list Lemma 33's), one reversal count per ground, exact by LDL^T there (Lemma 30; Corollary 4), settled by integers; a ground line reads one count, the pullback quadratic's scale the homogeneity principle's read (Definition 1; Lemma 30); a kernel of one line reads its quotient at a point, and a deeper kernel indexes the display over its ground space, the window pair's read covering every member at once; and the window pair, the detector form whole, reads every ground at once, a compression keeping the read (Lemma 30), the spectral read fixed at Theorem 7. Each window cut is available at every point of the *closed* segment: a window compression is finite-dimensional (Proposition 2), so a least eigenvalue and an exact reversal count exist at either end and at every coupling ray, an end's pencil its one member's own (Proposition 6). For a fixed region Λ , write $\mathcal{K}_{\Lambda,C}(v)$ for this finite cut at total cutoff C . The cut of $\mathcal{K}(v)$ is the meet over every region and every cutoff of Definition 8:

$$\mathcal{K}(v) \geq \kappa \quad \text{is} \quad \mathcal{K}_{\Lambda,C}(v) \geq \kappa \quad \text{at every region } \Lambda \text{ and every cutoff } C.$$

Every finite read uses that window's ground and whole detector. The *regional cut* $\mathcal{K}_{\Lambda}^{\text{reg}}(v)$ reads at every positive strict subfloor: its membership at κ requires, for every positive pair $\kappa' < \kappa$, a constructed cutoff key $C_0(\Lambda, v, \kappa')$ and the finite-window read

$$\mathcal{K}_{\Lambda,C}(v) \geq \kappa' \quad \text{at every } C \geq C_0(\Lambda, v, \kappa').$$

The key and the splits at a stated cutoff are certificate data; the assertion at every further cutoff is the theorem at that cutoff, each test the finite read above. A floor holding at every cutoff from one constructed key gives its regional membership by the downward read. Every member of $\mathcal{K}(v)$ belongs to every regional cut: the finite reads at every cutoff give every strict subfloor's reads from the cutoff key one. Each cut is downward closed, the window read weakening as its positive level falls and the regional read using the strict subfloors' keys. A member pair is the positivity read, every pair positive by its shape. A value of a cut is its located edge where one is named. Each finite-window upper cap caps the global edge, the meet's members admitted by that window. A regional upper cap is a cap at every cutoff beyond a constructed key. It caps the global edge through the finite-window cap at the key itself.

Two scalings, and which one each clause reads. Moving along a ray, $[\alpha : \beta] \mapsto [t\alpha : t\beta]$, is the freedom the pair reads through, one value on the ray (Proposition 6); rescaling the family, $E \mapsto uE$ and $M \mapsto uM$, is a second freedom, and a level of the pencil is degree one in each. $\mathcal{K}(v)$ takes E_0 at the free end against the level at v , so it is free of the second and degree one in the first: its *side* is a property of the ray, and its value at a window reads at the coupling's stated pair.

Theorem 4. *Let the member run over the classification's stated domain, each member once: the A-series at the declared residue $r \geq 1$ (Construction 1), and B_ℓ at $\ell \geq 2$, C_ℓ at $\ell \geq 3$, D_ℓ at $\ell \geq 4$ and the five fixed members at their weight tables (Construction 4), the index floors reading each member once with the low-index coincidences the classification's own identifications; let the magnetic list be the member's stated list, faithful (Construction 6); let r be the member's derived residue, $d \in \mathbb{N}$ the direction count (Construction 7), and $[\alpha : \beta]$ the coupling's end-weight pair at $\alpha, \beta \in \mathbb{N}$, interior by the shape of its data (Proposition 6). Then:*

1. a pair is a member of the cut of $\mathcal{K}([\alpha : \beta])$, positive by its shape; and
2. at every window and every coupling of the closed segment whose cut names an edge, the edge is the compressed pair's least root, a located root of the window pencil's symbol (Theorem 26(i); Lemma 20), attained at a word image with the two moments at equality and the first positive (Lemma 22(i),(ii)); and
3. every continuum datum, a finite list of supports, separations and Euclidean pairs at pair coordinates against the corner scale, reads at the chain's height: at every scale the height is located between the ray's cut member of clause (i) and its caps, every probe's two moments' read at any window and the two-member cap line at a simple ground, the spacing that height's read under the unit map, and at two directions or beyond the spacing falls under every floor beyond the floor's key (Theorem 23(iii)); and each cell's window cut's edge, where one is named, is the compressed pair's least root read at a word image at the corner scale (Theorem 23; Lemma 47; Lemma 48; Lemma 55; Lemma 56; Lemma 22(i),(ii)).

Proof. Clause (i) is Theorem 14 at the cells' closed union, the root coordinate's located segment from the free end to the validity top: the cells' reads at the root coordinate's rays, the counting sandwich of its clause (v) at every further end-weight ray of the union, the cutoffs reaching the cut within Theorem 30's priced modulus at the unit electric representative. Clause (ii) is Lemma 22: the window cuts' edges located roots of their symbols read at word images (Theorem 26). Clause (iii) is Theorem 23: the located brackets, the scale closure, and the chain's height located at every scale between the cut's member and its caps with its fall beyond the floor's key at two directions or beyond (Lemma 48; Lemma 55), the cells' edges read at word images (Lemma 47). $\square$

Clause (i) is a count of $\mathcal{K}$'s cut (Definition 11) and clause (ii) a located root at a word image; each is weight-free, one value at every form point (Proposition 1), the cut's level read against the free-end level. Clause (iii)'s located values are the family's continuum reads, the spacing among its outputs (Theorem 23); and the state's space is the word sector's own (Lemma 21). Remark 1 records the correspondence between these objects and their usual names.

Theorem 5 (The vacuum sector's cut at every magnetic list). *Let the member, the residue and the direction count run as in Theorem 4, and let the magnetic list be any involution-closed list of nonunit labels (Construction 6), the pencil at its plaquette character (Definition 10); let the vacuum sector be the unit label's fusion closure at every window (Construction 21) and the sector's cut Definition 11's read at the detector pair compressed to the sector's word images, the sector's own ground its kernel point (Lemma 19). Then at every interior end-weight ray a pair is a member of the sector's cut, positive by its shape, the cell list's reads at the sector's depth chain (Theorem 14(vi)); at a faithful list the sector's cut is the unit charge's part of Theorem 4(i), the sectors the charges (Construction 21); and at the adjoint list $\{\theta\}$ the sector is the adjoint walk (Construction 15), whose one-plaquette sector compresses to the tower at every residue and is the tower itself at the residue one (Proposition 13), its data the walk's own reads, the tower matrix (Proposition 13), the channel divisor (Theorem 13; Definition 12) and the walk residue (Theorem 28; Definition 13), the pair $(3, H_r)$ their assembly (Definition 14; Theorem 29), the scalar certificate the block certificate's instance (Lemma 57; Lemma 56), and its scheme bridge the count $2d_f$ (Theorem 27; Remark 3).*

Proof. The cell list runs on the vacuum sector's depth chain at every magnetic list: the caps and the deficit reads at the closure (Lemma 17; Lemma 18), the truncation (Theorem 30; Lemma 44), the grading and the decimation on the sector's chain (Lemma 41; Theorem 19), the dual chain's cells (Lemma 45) and the corner certificate at the list's constants (Lemma 47; Construction 19(iii));

Lemma 57; Lemma 56) read the sector's pencil alone, the faithfulness read entering the sectors' identification with the charges alone (Construction 21); so Theorem 14's cells cover the interior on the sector and its clause (vi) reads the sector's cut at every ray. The adjoint list's instance is Construction 15's chain, its one-plaquette residue-one sector the tower labels alone, every label of $SU(2)$ a Cartan power of the adjoint at even degree, so the walk's reads are the sector's own there. $\square$

Corollary 3 (The strong-coupling cut). *Let the member, the residue, the direction count and the magnetic list run as in Theorem 4. The free cell's certificate reads two located outputs per member and direction count (Lemma 64): its floor κ_1 , the certificate's located datum, the pair at $E_0\kappa_1$ within the flat window's width less the brackets' width and the band block's cap, read outright at the free end (Theorem 14(iv); Lemma 15), and its extent $\tau_1 = [n_1 : d_1]$ at that pair, each positive by its shape. Then at every ray $[1 : \tau^2]$ at $\tau \leq \tau_1$ the cut of $\mathcal{K}([1 : \tau^2])$ holds κ_1 as a member, the window cuts' meet over every region and every cutoff (Definition 11). Every such window cut holds κ_1 , and the ground multiplicity is one on the segment from the free end (Theorem 10). At every further end-weight pair $[\alpha : \beta]$ at $\beta d_1^2 \leq n_1^2 \alpha$ the cut at that representative holds $\alpha[\kappa_1 : 2]$ (Theorem 14(v)); and at every window and every such coupling the finite-window cut's edge, where one is named, is the compressed pair's least root attained at a word image with the two moments at equality and the first positive (Lemma 22(i),(ii)). The free end reads the invariant at one (Lemma 15), and the extent's positivity is the free end's interiority read (Lemma 64).*

Proof. The free-cell certificate's two counts read at its fixed offsets at every region and every cutoff (Lemma 64). Its counts are the original pencil's total counts through the pivot witnesses (Theorem 19(i)), with the separated product at its actual cutoff selection (Lemma 42(iv)). At the upper offset h , $\ell + E_0\kappa_1 \leq h$; count monotonicity gives one at $e + \ell + E_0\kappa_1$, with e that window's ground. Monotonicity and the ground below $e + E_0\kappa_1$ give the count one at that line, hence the window cut and simple ground (Lemma 20). Taking every window gives the meet's membership (Definition 11). At a further pair ray, fix one region and write P for its plaquette count, $W := P d_L$ for its magnetic form cap, and $\gamma := E_0\kappa_1 = [g : h]$. The natural

$$q := \alpha + 4P d_L(2n_1 + d_1)h + 1$$

sits beyond α and beyond $[4W(2n_1 + d_1)h : d_1g]$, the latter comparison at $d_1g \geq 1$. Let p be the least natural from one at $\beta q^2 \leq \alpha(p+1)^2$, the stepping read with witness βq^2 (Definition 1). Its predecessor refuses, with the start's strict read at $q > \alpha$, so

$$\alpha p^2 < \beta q^2 \leq \alpha(p+1)^2, \quad p d_1 < n_1 q.$$

The second comparison follows by joining the first to $\beta d_1^2 \leq n_1^2 \alpha$. Thus the pair $t := [p : q]$ sits inside the free cell. At $b := [\beta : \alpha]$ the weight gap δ at $t^2 + \delta = b$ satisfies

$$\delta \leq [2p + 1 : q^2] \leq [2n_1 + d_1 : d_1 q], \quad 4W\delta < \gamma.$$

Fix any cutoff of the stated region. The two window pencils' difference is capped two-sidedly at $\rho := W\delta$ against G (Theorem 30), so the count sandwich (Theorem 14(v)) puts every ordered root within ρ of its root at t . At a window of order at least two, its first two roots therefore read a gap at or beyond $\langle \gamma : 2\rho \rangle$, beyond $[\gamma : 2]$; the ground at t is simple by the free cell's count one. At order one the level gap is the sum's unit and the detector read holds outright. Scaling the pencil by α gives the cut member $\alpha[\kappa_1 : 2]$ at $[\alpha : \beta]$ (Lemma 20). The natural q reads the stated region's P , while this cut member depends on the member, direction count and end weights alone, so the read holds at every region and every cutoff and belongs to the window cuts' meet. The finite-window edge is Lemma 22's at the window pencil's symbol (Theorem 26(i)). $\square$

Remark 1 (Correspondence with the usual formulation). The objects above are recorded here under their standard names, so that Theorem 4’s lattice statements and the spacetime construction of Pending 2 have their conventional readings.

1. *The theory.* $H(1, b) = E + bM$ is the lattice gauge Hamiltonian, E its electric term, a sum of link Casimirs, and M its magnetic term, a sum of plaquette characters at the magnetic list (Definition 10), the Kogut–Susskind Hamiltonian of the Wilson action at a faithful character: at the A -series the list is the fundamental with its dual and the plaquette character is $\chi_f + \chi_{\bar{f}} = 2 \operatorname{Re} \operatorname{tr} U$, the standard action’s own term, and at every member the list’s classes reaching every class is the character’s faithfulness, the classes the center’s characters (Construction 6). Its space is the carrier $\operatorname{Sect}_\infty$, a colimit of windows each finite-dimensional (Definition 8, Proposition 2); its observables are the $*$ -algebra of Definition 9, whose products obey a window bound stated on the index alone (Proposition 5) and whose local algebras have $\mathfrak{A}$ as their limit (Definition 15). That bound and that limit are locality, and both are read off the index. H is a symmetric rational matrix on every window, so each level read there is linear algebra (Theorem 17). Classically the carrier’s elements are the gauge-invariant polynomial functions of the link holonomies, Eval is the Haar integral of the compact group, and the pairing is its L^2 one: the compact group, its invariant mean, and its coordinate ring are one classical reading of the fusion data, which the calculus above computes exactly. The label calculus’s classical identities are listed under their usual names: the blocks are the irreducible representations at highest weights, the orthogonal complement of Lemma 1 is what Weyl’s integration buys as complete reducibility, the scalar endomorphisms of Lemma 4 are Schur’s lemma, the alternant identity of Theorem 1 is the Weyl character formula, the alternant pair a_β reading as the determinant $\det(x_i^{\beta_j}) = \sum_w \operatorname{sgn}(w) x^{w\beta}$, the pair’s value its members’ difference and the swap grading the sign of the permutation, the full-column line of Lemma 8(ii) the determinant representation, and Corollary 1 its dimension formula, Lemma 11 the Pieri rule at one box, Lemma 12’s path count f^λ the standard tableau count at the shape (the raised glyph the classical one, the unraised f the fundamental label’s), Corollary 2 the alternating fusion count (Brauer, Klimyk), and Lemma 13 the Schur–Weyl invariant span. That reading extends to the local objects: the characters are the blocks’ traces on holonomies, the product reading $\operatorname{tr}(A \otimes B) = \operatorname{tr} A \cdot \operatorname{tr} B$ with $d_a = \chi_a(1)$; the adjoint character runs pointwise over $[-1, n^2 - 1]$ and the fundamental trace’s modulus over $[0, n]$, the tops at $U = 1$, the form caps of Lemmas 17 and 18 read pointwise; the form line of Proposition 1 is the invariant symmetric bilinear forms on $\mathfrak{su}(n)$; the two-plaquette fiber of Construction 9 is $L^2(G \times G)^G$, the functions of the two plaquette holonomies invariant under simultaneous conjugation, the theta graph’s gauge reduction; and $\mathfrak{A}$ (Definition 15) is the quasi-local C^* -algebra, the norm closure of the inductive limit of the local algebras.
2. *The vacuum.* ψ is a window ground and ε_0 its eigenvalue, so $(H - \varepsilon_0)\psi = 0$ and $H - \varepsilon_0 \succeq 0$ on $\psi^\perp$ by construction (Definition 10); on a window the ground multiplicity is a count, a divisor read (Theorem 10), and the vacuum’s correlation reads sit in named window brackets, Euclidean pairs included (Theorem 21). A reader who doubts uniqueness in the limit keeps every read: the cut is stated on counting functions, every comparison a count of the pencil itself rather than a read at a chosen state (Definition 11; Theorem 20), so Theorem 4 holds at every ground.
3. *The lattice.* The carrier is stated over the interface of Construction 7, so the derivations run on its fields alone. The reading here takes $\mathbb{Z}^d$, whose plaquettes are its unit squares, at $d \geq 2$, and the plaquette chain at $d = 1$, consecutive squares sharing one link. On either, a plaquette’s

boundary is four links, the count Proposition 7 reads, and a link sits in the plaquettes of its direction's pairs, the per-link plaquette count the gap g at $g + 2 = 2d$ here, the per-vertex count dg , a straight cell's stencil $[g^2 : 2]$, and each at most two on the chain. d counts the space directions and time is the pencil's own, so $d = 3$ is the case of $\mathbb{R}^4$.

4. *The lattice symmetries.* E and M run over all links and all plaquettes at coefficient one, so H commutes with every symmetry of $\mathbb{Z}^d$ permuting those two sets: the translations, and the hyperoctahedral group B_d of signed coordinate permutations (Theorem 22), whose permutation factor is the coordinate exchanges and whose sign flips are the reflections. B_d is what Theorem 22 is stated over, the translations entering separately at the fibering (Lemma 40). Theorem 22 reads the invariant polynomial spaces at each argument count: their degree-2 spans coincide, and the degree-4 space holds an additional invariant beyond the isotropic span. The theorem's isotropic span reads classically as the $O(d)$ -invariant ring, the first fundamental theorem for $O(d)$ the classical side's own identification. The transformation comparisons for the continuum correlations are Pending 2's data.
5. *The coupling.* The pencil's parameter is the coupling. The free end's pencil is the electric term alone and the contact end's the magnetic alone (Proposition 6), so the two ends are the strong- and weak-coupling limits; the free cell's cut at the strong end (Corollary 3) reads as a floor of the Kogut–Susskind Hamiltonian's gap at every window at magnetic weight at or below the extent's square, $\beta d_1^2 \leq n_1^2 \alpha$, the classical strong-coupling regime read at the chain's outputs.
6. *The continuum's points.* Clause (i) is stated over every interior end-weight ray, the root coordinate's rays the cells' working points (Definition 10). A reader's continuum point (a Cauchy or Dedekind real, a nested-bracket datum) reads the cut in through its own rational comparisons: the counting sandwich of Theorem 14(v) transports the rational reads to every located coupling at its stated modulus, so “at every interior coupling” holds at whatever continuum the reader brings, each point reading itself in through the rationals beside it. The chain's own points are the rational rays, and the sandwich is the whole of a further point's read.
7. *The scalar pairs.* The translation of the colon notation is a bijection and reads one way only: a scalar pair $[a : b]$ (Definition 1) is the positive rational a/b , two pairs at the cross-multiplication identity being one rational, so the stage glyph $\mathbb{P}$ reads as the positive rationals, with $\mathbb{Q}$, the field, this remark's own; under that map the displayed sum, product and order of Definition 1 are the field's own on the positive rationals, the signed readings and the zero this remark's own, a pair of equal members reading zero and an absent summand a zero term, and $H_k = \sum_{j \leq k} [1 : j]$ is the classical harmonic number $\sum_{j \leq k} 1/j$. A coupling's end-weight pair $[\alpha : \beta]$ (Proposition 6) is the point $[\alpha : -\beta]$ of the real projective line in the classical (E, M) -coefficient coordinates, the minus this remark's own; the scalar instance is the affine reading at positive second component, and the division is the remark's own reading of a clearing. A balance pair $\langle u : v \rangle$ (Definition 1) is the difference $u - v$, the subtraction this remark's own: its side is the value's sign, equal members the value zero, and a bracket of balance-pair endpoints is the signed interval its endpoints name.
8. *The names.* The document's terms of art read against their standard names, each at its defining site. The reversal count $\text{rev}(A : B)$ (Lemma 30) is the negative index of inertia of the difference $A - B$, the subtraction this remark's own, and its two-splits-one-count clause is Sylvester's law of inertia. The division descent (Theorem 16) is the Euclidean algorithm, its greatest common divisor taken monic over the pair carrier and, at the integer representatives,

primitive at the stated positive top with the clearing multipliers the remark's own rescaling; the cleared division at a stated top is the classical pseudo-division. The remainder lists at a monic S (Definition 2) are the quotient ring $K[x]/(S)$ read as data. The cofactor c at $Yc = X$ (Definition 1) is the exact quotient X/Y , computer algebra's own cofactor, and the gap g at $q + g = p$ the difference $p - q$, the remark's own operations (the classical matrix cofactors enter as the adjugate's entries, under that name). The deck relation $z^2 + 1 = wz$ (Definition 4; Theorem 12; Lemma 39) is the double cover $w = z + 1/z$, the Joukowski map, its deck involution the cover's deck transformation $z \mapsto 1/z$, and the chord $t = \zeta + \zeta^{-1} = 2 \cos k$ at the classical momentum k is the Chebyshev variable of the momentum fiber (Lemma 40). The decimation (Theorem 19) is the classical block decimation, each step a Schur complement onto the head. A located element (Lemma 32) is located in Bishop's constructive sense, the bracket its datum. The d_f -ality (Lemma 43) is the classical N -ality at $N = n$, the box count's class at full columns. The beta-set (Construction 1) is the partition's classical beta-set at the positive shift, $x_i = \lambda_i + (d_f - i) + 1$ against the classical $\lambda_i + d_f - i$. A content (Construction 1) is the classical weight of the grading, 'weight' in the chain the end and form data's word; a configuration's content (Definition 8) is its total Casimir, the electric term's total. The calculus at the symbolic residue is the unitary group $U(n)$'s, the determinant relations per-residue data, the presentation kernel's (Remark 2).

9. *The members.* The classification's members read as the simply connected compact simple groups at $r + 1 = h^\vee$, the derived residue the dual Coxeter number's read, the residue folds of Construction 4 its per-member evaluations; the title reads here. The A -series is $SU(n)$ at $n := r + 1 = d_f$, whose fundamental representation has dimension n and whose Lie rank is $n - 1$, the declared names reading the matrix size as the fundamental count d_f and the Lie rank as the residue r (Construction 1); the member weight tables read B_ℓ , C_ℓ and D_ℓ as $\text{Spin}(2\ell + 1)$, $\text{Sp}(\ell)$ and $\text{Spin}(2\ell)$, and the five fixed members the exceptional groups. Theorem 4's prefix instantiates the classical statement at every simply connected compact simple group and every $d \geq 1$, each group once: the index floors read each low-index coincidence at one member, $\text{Spin}(3)$ and $\text{Sp}(1)$ at $SU(2)$, $\text{Sp}(2)$ at $\text{Spin}(5)$, and $\text{Spin}(6)$ at $SU(4)$, with $\text{Spin}(4)$ off the simple list, the identifications the classification's own.
10. *The weak-coupling comparison, and the root coordinate.* Classically the weak-coupling theory is read against lattice Maxwell theory per Lie-algebra direction: the link field is $\log U_\ell$, the magnetic energy is the abelianized flux squared, and the comparison diagonalizes in Fourier with dispersion $\varepsilon(k)^2 = 2|b| \hat{h}(k)$ at the graph Laplacian $\hat{h}$ of $\mathbb{Z}^d$, massless at $k \rightarrow 0$. That is the classical identity of the gap chain's objects: the root coordinate τ ($b = -\tau^2$, the remark's reading of Definition 10's weights) is the harmonic frequency scale, and the soft momenta the flat step reads cell by cell (Lemma 40; Theorem 20) are the sector the massless comparison leaves ungapped: the lift above the comparison is what the counts read, the corner scale its ruler (Lemma 47).
11. *The gap.* E_0 is the free end's level, the energy of the lightest one-plaquette excitation there, the floor label's loop, an eigenvector of the pencil (Proposition 7), and $\mathcal{K}$ is the uniform lower cut for excitation gaps at every region and every cutoff, above each window's own ground, read against that level (Definition 11), one at the free end (Lemma 15). Since $E_0 > 0$, clause (i) is the lattice Hamiltonian gap: on the word sector (Definition 10) the spectrum starts *strictly* above the vacuum. The window's energy positivity, $H - \varepsilon_0 \succeq 0$, is item 2's and holds by construction; clause (i) is the strict statement above it. Clause (ii) locates the level: at every window the cut's edge is a root of the window's symbol met at a word image (Theorem 26;

Lemma 22(i),(ii)), the lightest excitation a state the local observables create from the vacuum. A reader who wants an edge as an energy restores the two scales at which the cut reads one value: it is $E_0 \kappa C_2(\text{adj})$ at the edge's pair κ , with E_0 moving with the family's scale and $C_2(\text{adj})$ with the invariant form (Proposition 1), as the level does with each, which is what makes the product an energy.

12. *The continuum reading.* The usual formulation reads the contact end's approach through the continuum construction: the spacing a sets the units, the electric weight is $g^2/2a$ and the magnetic $1/(ag^2)$ at the usual coefficients, and the construction holds the physical gap Δ fixed while $a \rightarrow 0$ along a coupling trajectory $g \rightarrow 0$, so its premise is that the gap in lattice units, $a\Delta$, falls under every floor toward the contact end, the diverging correlation length. The corner scale maps that premise onto the chain's data exactly: the two weights' product is $1/2a^2$ at every g , one value across the couplings, so the corner height of Lemma 47 reads classically as $\sqrt{2}a\Delta$ at every coupling of the approach. The premise therefore reads: the gap falls below every corner-scale floor toward the contact end, the corner cell's outputs under every floor. The chain's height at every scale is a located bracket, its floor the ray's own cut member and its caps the window's cap line at the tail read (Lemma 55) and the cutoff-tail probe caps (Lemma 22(iv)); so the premise reads on the chain at its windows and fibers as the height's fall under every floor along the scale, the tail read at two directions or beyond with one direction the floor (Lemma 55), and the formulation's spacing reads through that datum. The spacetime action on the correlation family is the construction of Pending 2.
13. *The unit map.* The classical construction chooses a trajectory $g(a)$ by hand; here the spacing is a read: at a physical gap Δ the unit is $a = h/(\sqrt{2}\Delta)$ at the chain's height h at the scale, the located bracket of clause (iii), one equation per scale, so the physical gap reads Δ at every scale of the trajectory and the trajectory's spacings are the heights' reads.
14. *The reconstruction data.* The spectral reconstruction of Theorem 31 reads the finite-window Euclidean families as folds over located energy roots at the probes' spectral weights. The state pairing is the word sector's own (Lemma 21), positive at Definition 8's pairing. The lattice correlators' brackets, clustering comparisons and Euclidean pairs are Theorem 21's displays, with $e^{-u(H-\varepsilon_0)}$ this remark's reading of the Euclidean bracket families; their moment identities are Lemma 62's, and the spacing reads the height (Theorem 23(iii)). A Wightman reconstruction requires spacetime covariance, the spectral condition, locality, the distributional growth conditions and the vacuum properties on one continuum family. Its covariance and the identification with the lattice family are Pending 2's construction.
15. *The spectral condition.* $\text{spec}(H) \subset \{\varepsilon_0\} \cup [\varepsilon_0 + E_0\kappa, \infty)$ is the window cut (Definition 11; Lemma 20), its edge met at a word image window by window with the weight at the level where the inclusion is read (Lemma 22). The lattice momentum decomposition is the fibering (Lemma 40), and the roots' spatial momentum reading is isotropic at degree two (Lemma 63). The relativistic spectral condition is support of the joint energy-momentum spectrum in the forward light cone, with the mass gap read there. It requires the spacetime translation action and its Lorentz transport on the continuum carrier, Pending 2's construction. The physical energy scale reads through the unit map at the chain's height, between the ray's cut member and its caps (Theorem 4; Lemma 48; Lemma 55; Lemma 22(iv)).
16. *Locality, and the net.* The regions' lattice algebras obey the window bounds (Proposition 5; Lemma 14; Definition 15), and equal-time products commute (Proposition 5). The propagation bracket (Theorem 21(vi)) prices the commutator beyond its stated reach at the remainder's

width; the clustering brackets are Theorem 21(iv)'s. A relativistic net requires an algebra at each spacetime region, transported by the spacetime action, with commutation at spacelike separation. The support transport is Pending 2's construction; the spacelike commutation read requires the propagation widths to close at those supports under the unit map.

17. *Invariance.* Spacetime covariance is the construction of Pending 2. Its classical reading is a strongly continuous unitary representation U of $\mathbb{R}^{d+1} \rtimes \text{SO}^+(d, 1)$ on the continuum state space, at the identities

$$U(g)\Omega = \Omega, \quad U(g)\mathfrak{A}(O)U(g)^* = \mathfrak{A}(gO)$$

for each group element g and bounded open spacetime region O . The pairing identities at the transported words give the isometries, the composition and reversed maps give the group law, and the word moduli give strong continuity. The spacetime translations' time subgroup must read the Hamiltonian evolution whose Euclidean brackets enter the construction. The lattice's signed spatial permutations and the degree-two identities are Theorem 22's finite reads.

18. *Nontriviality.* The theory is interacting at every member: the connected four-point read of the pair probe at the theta-graph pair, the probe the characters' fold at a magnetic list member with its dual and their Cartan squares, holds a positive located floor on a cell at the free end's side, the leading coefficient a displayed member read at the member's dimension, Casimir and adjoint channel (Lemma 65), while a generalized free field's four-point function is its three two-point pairings' sum, Wick's read, so its connected fold reads equal members at every argument list: the floor separates the state from every generalized free field's.
19. *The conventional target.* The Yang–Mills problem asks, for each compact simple gauge group, a nontrivial relativistic quantum field theory on $\mathbb{R}^4$ with a positive mass gap. The direction count is $d = 3$ in these coordinates. The lattice claims are Theorem 4's, with spectral brackets from Theorem 31 and spacing reads from Theorem 23. The conventional conclusion requires their identification with one continuum theory satisfying the axiomatic requirements: the spacetime action and the transported correlations of Pending 2, the joint spectral condition, spacelike locality, the vacuum and growth properties, and nontriviality of that continuum theory. The group domain of the lattice claims is the simply connected domain stated in the members item.
20. *The representation.* M multiplies by the magnetic list's plaquette character (Definition 10), a faithful representation's character at every member, the fundamental's character joined to its dual's at the A -series, the theory item's action; c_1 counts the cubic couplings of the adjoint (Lemma 24).

4 Fusion data

Proposition 8 (A fusion computation is a representation computation). *The fusion data are counts of the calculus: N_{ab}^c is the number of c -blocks in a product of blocks and d_a the block dimension (Lemma 5, on labels by Construction 5), $m\{a, b, c\}$ the invariant count of the triple product, the labels the label set's (Construction 5). Each is read in the label index: χ_a is the index's generator at a , the product of two expanding over the tensor product's blocks,*

$$\chi_a \chi_b = \sum_c N_{ab}^c \chi_c,$$

the coefficients the block counts (Lemma 5(iii), on labels by Construction 5); the dimension d_a is the block's count (Lemma 4(iv)); and the unit label $\mathbf{1}$ has $\chi_{\mathbf{1}} = 1$, the constants' line's.

Four reads follow, and they are what the label index runs on. Write $\text{Hom}(x, y)$ for the linear maps from x to y ; then $\text{Hom}(x, y) \cong x^* \otimes y$, the intertwiners are its invariants, and the unit label's multiplicity in any x is its invariant count. First,

$$N_{ab}^1 = \delta_{a\bar{b}},$$

the unit read of Construction 5. Second, the characters are orthonormal for the evaluation, the unit-label coefficient read $\text{Eval}(\chi_c) = \delta_{c1}$, the block's invariant count, so

$$\text{Eval}(\chi_a \overline{\chi_b}) = \sum_c N_{ab}^c \text{Eval}(\chi_c) = N_{ab}^1 = \delta_{ab},$$

and a product expanded in the label index therefore has unique coefficients. Third, $m\{a, b, c\} = N_{ab}^{\bar{c}}$, with the rotation $N_{ab}^{\bar{c}} = N_{bc}^{\bar{a}}$: the rotation and the factors' exchange, the transposition, generate the symmetry of m in its three arguments, and that is the symmetry of N at self-dual labels. Fourth, the involution's index read: the dual label's matrix is the transpose, $N_{ab}^c = N_{ac}^b$.

Proof. The first read: the count of the unit label in the product at a and b is the full-column-power count of Lemma 8(iii), one exactly at the dual label. The second display reads the product's display at the unit-label coefficient with the first read at the dual label. The third: $m\{a, b, c\} = \sum_d N_{ab}^d N_{dc}^1$ with the first read keeps the sum at the dual label's summand alone, $N_{ab}^{\bar{c}}$; the display reads at either pairing of the associativity at the unit target (Lemma 5(iii)), $\sum_e N_{ab}^e N_{ec}^1 = \sum_f N_{bc}^f N_{af}^1$, the first read keeping each sum at the dual label's summand alone, so $N_{ab}^{\bar{c}} = N_{bc}^{\bar{a}}$. The fourth: the triple count reads one value at every argument dualized, duals of blocks being blocks with the unit's count kept (Lemma 8(i)), and the third read's symmetry closes the exchange. $\square$

Lemma 16 (The adjoint character). *Write f for the fundamental label, the one-box shape's, whose block is V itself at the declared count d_f (Construction 1), and $\bar{f}$ for its dual, at the residue at or beyond one. Then*

$$f \otimes \bar{f} = \theta \oplus \mathbf{1}, \quad \text{hence} \quad \chi_\theta(U) + 1 = |\text{tr } U|^2,$$

and the adjoint's contents are the letter-pair moves, one place raised and one lowered, each ordered pair of distinct places once, with the unit-monomial content at multiplicity r , the product's unit-monomial count splitting as the unit line's one with the adjoint's, $r + 1 = d_f$.

Proof. $\bar{f}$ is the class of the complement f^* (Construction 5), the one-column shape at the complement gap, r rows of one box (Lemma 8(ii)), and $\bar{f} \otimes f$ exhausts by the one-box row (Lemma 11): the dominant one-box additions of (1^r) are $(2, 1^{r-1})$, the label θ (Construction 5), and the full column (1^{d_f}) , the unit label's, one block each with every further label vacant, and the factors exchange (Lemma 5(iii)): $f \otimes \bar{f} = \theta \oplus \mathbf{1}$. The product's contents are the factors' contents' sums (Construction 1): the one-box block reads each letter once and $\bar{f}$'s block each withdrawn letter once (Lemma 11's complement read), so the product reads the unit-monomial content at d_f , one per letter, and each letter-pair move once, a letter against a withdrawn one. The content refinement of Lemma 5(ii) reads the product's occupancies as the two blocks' at their counts, and the full-column line is occupied at the unit-monomial content alone (Lemma 8(ii)): θ 's occupancies are the product's with the line's withdrawn, each move once and the unit-monomial content at $d_f - 1 = r$. Hence $\chi_f \chi_{\bar{f}} = \chi_\theta + 1$ (Proposition 8), and $\chi_f(U) = \text{tr } U$ reads the display. $\square$

Lemma 17 (The fusion matrix is capped by its dimension). *On one link's character index, multiplication by χ_λ is the fusion matrix $N_{\lambda\nu}^\mu$ against the orthonormal characters (Proposition 8),*

row-finite on every window (Proposition 3), with count entries, symmetric at self-dual λ , and the dimension vector is its exact positive eigenvector:

$$\sum_{\mu} N_{\lambda\nu}^{\mu} d_{\mu} = d_{\lambda} d_{\nu},$$

dimensions multiplying over a tensor product (Lemma 5). Hence at self-dual λ the fusion form is capped two-sidedly at the dimension (Lemma 30),

$$N_{\lambda} \preceq d_{\lambda} \quad \text{with} \quad d_{\lambda} + N_{\lambda} \text{ positive semidefinite :}$$

the two squares $(d_{\nu}x_{\mu} + d_{\mu}x_{\nu})^2$ and $\langle d_{\nu}x_{\mu} : d_{\mu}x_{\nu} \rangle^2$, each positive by its shape, price each entry's read $2x_{\mu}x_{\nu}$ two-sidedly at $[d_{\nu} : d_{\mu}]x_{\mu}^2 + [d_{\mu} : d_{\nu}]x_{\nu}^2$, the comparison at its rescaling (Definition 1's homogeneity principle), the cleared weights the window's co-products, the dims' product at the pair's two factors withdrawn, every weight natural. At every label the dual pair's fold $N_{\lambda} + N_{\bar{\lambda}}$ is symmetric, the dual label's matrix the transpose (Proposition 8's involution read, $N_{\lambda\nu}^{\mu} = N_{\bar{\lambda}\mu}^{\nu}$), and capped two-sidedly at twice the dimension,

$$N_{\lambda} + N_{\bar{\lambda}} \preceq 2d_{\lambda} \quad \text{with} \quad 2d_{\lambda} + N_{\lambda} + N_{\bar{\lambda}} \text{ positive semidefinite,}$$

the self-dual display its one-member instance. Hence the magnetic list's closure fold $N_L := \sum_{\lambda \in \bar{L}} N_{\lambda}$, the multiplication by the plaquette character χ_L on the index (Construction 6), is symmetric and capped two-sidedly at the closure's dimension fold

$$d_L := \sum_{\lambda \in \bar{L}} d_{\lambda}, \quad N_L \preceq d_L \quad \text{with} \quad d_L + N_L \text{ positive semidefinite,}$$

the dual pairs' caps and the self-dual members' caps summed over the closure, a sum of two-sided caps a two-sided cap of the sum (Lemma 30), $2d_f$ at the A-series' list, the fundamental with its dual. There the fundamental's multiplication squares to the adjoint's: its pairing-adjoint is multiplication by $\chi_{\bar{f}}$, so

$$M_{\chi_f}^{\dagger} M_{\chi_f} = M_{1+\chi_{\theta}} \preceq 1 + r(r+2) = d_f^2,$$

at the display's $\lambda = \theta$, self-dual with $d_{\theta} = r(r+2)$.

Proof. The two squares price every entry and the eigen-identity sums the rows: the self-dual cap. At the dual pair's fold the form is $2 \sum_{\mu, \nu} N_{\lambda\nu}^{\mu} x_{\mu}x_{\nu}$, each entry's read priced two-sidedly by the two squares as above, and the two weighted folds read the eigen-identity at the two labels, $\sum_{\nu} N_{\lambda\nu}^{\mu} d_{\nu} = \sum_{\nu} N_{\bar{\lambda}\mu}^{\nu} d_{\nu} = d_{\lambda} d_{\mu}$ at $\bar{\lambda}$ and $\sum_{\mu} N_{\lambda\nu}^{\mu} d_{\mu} = d_{\lambda} d_{\nu}$ at λ , the dual's dimension the label's own (Lemma 8(ii)), so the priced form sits between $2d_{\lambda}$ times the coordinate fold of squares and its balance partner. $\square$

Lemma 18 (The plaquette character is capped by its dimension fold). *On every window, at every label λ , multiplication by a plaquette's character squares to the cap d_{λ}^2 ,*

$$\langle \chi_{\lambda} x, \chi_{\lambda} x \rangle \leq d_{\lambda}^2 \langle x, x \rangle$$

at every window vector x , χ_{λ} abbreviating $\chi_{\lambda}(U_{\partial p})$, and the term's form is two-sided at d_{λ} , the form and its memberwise swap each at or below $d_{\lambda} \langle x, x \rangle$. At the magnetic list's closure (Construction 6) the plaquette term $\chi_p = \sum_{\lambda \in \bar{L}} \chi_{\lambda}(U_{\partial p})$ (Definition 10) is capped at the closure's dimension fold $d_L = \sum_{\lambda \in \bar{L}} d_{\lambda}$ (Lemma 17): the term is two-sided at d_L as a form on every window, and the

multiplication's square is capped, $M_{\chi_p}^2 \preceq d_L^2$ as forms on every window, the term symmetric at the closure's involution, $2d_f$ the A -series' read at the fundamental with its dual (Lemma 16's $\chi_\theta + 1 = |\chi_f|^2$ the pair's square there). The gradient deficit. At one link and one label the block's coefficient is the matrix U_λ over the located pairs (Definition 7' remainder lists at $z^2 + 1$; Lemma 37's located pairs), the block's trace pairing $\langle A, B \rangle_\lambda$ of two block matrices the first member of the located pair $\text{tr}(A^\dagger B)$, the entries' members' dot, symmetric, positive at the entries' members' squares' fold, the pairing of Lemma 3's reads, its fold $|A|^2 := \langle A, A \rangle_\lambda$, a product's fold at or below the factors' folds' product, a sum's at or below the summand count times the summands' folds' sum (Construction 8's Cauchy-Schwarz at the entries), and a product against a block coefficient the factor's own, $|MU_\lambda^\dagger|^2 = |M|^2$ at the evaluation identity at the two invariants, $U_\lambda U_\lambda^\dagger = 1$ and $U_\lambda^\dagger U_\lambda = 1$ (Construction 8; Lemma 8(i)'s evaluation and coevaluation). The generator list acts at the block by its matrices $X_a^{(\lambda)}$, the table's read at the block (Construction 3; the derivation action, Construction 8), each anti-Hermitian at the dagger read with its trace the sum's unit, the identity perpendicular to the list, and the list is orthogonal at the block's trace pairing at one value κ_λ , the block's index, its trace the Casimir's at the block against the generator count, $\dim G \kappa_\lambda = C_2(\lambda) d_\lambda$ (Lemma 7; Definition 7), the adjoint block's matrices the brackets' own, $X_a^{(\theta)} = \text{ad } X_a$ at $[X_a, X_b] = \sum_c (\text{ad } X_a)_{cb} X_c$, κ_θ their index. The index floor: the index reads the highest root's coroot pairs over the block's contents, $\kappa_\lambda = [1 : 2] \sum_\mu \text{mult}_\lambda(\mu) \mu(\theta^\vee)^2$ at every block (Construction 3's root triples), the adjoint's at $\kappa_\theta = C_2(\theta) = 2(r + 1)$ (Proposition 12), and it sits at or beyond one at every nonunit label, so κ_Σ , the joined index $\kappa_\Sigma := \sum_{\lambda \in \bar{L}} \kappa_\lambda$, sits at or beyond one at every list. The character's gradient reads the generator's trace against the coefficient, $X_a \chi_\lambda = \text{tr}(X_a^{(\lambda)} U_\lambda)$ at the derivation action on the block's coefficients (Construction 8), its involution image the dual block's read $X_a \chi_{\bar{\lambda}}$ (the dual action, Lemma 8(i)), so at the closure the list's gradient $X_a \chi_L = \sum_{\lambda \in \bar{L}} X_a \chi_\lambda$ is involution-fixed and reads at the joined block $R := \bigoplus_{\lambda \in \bar{L}} U_\lambda$, the closure's coefficients' block diagonal, with its joined generators $X_a^R := \bigoplus_\lambda X_a^{(\lambda)}$, its trace pairing the blocks' fold, and its gap $E_R := \langle R : 1 \rangle$, entrywise the balance pair against the identity, the blocks' gaps $E_\lambda := \langle U_\lambda : 1 \rangle$ joined: $\langle X_a^R, E_R \rangle_R$ is the gradient's balance partner at the dagger read, the generators' traces the sum's unit. The joined generators are orthogonal at the joined index, and the block projectors P_λ , the identity on one block, are pairwise perpendicular at the self-pairings d_λ and perpendicular to every joined generator, so the projectors with the joined generators are one orthogonal list, and the gap's fold reads at the pairing's projection fold at it (Lemma 3) as its components' squares against the members' self-pairings joined to the spare ϱ_p , the residual's own fold, the sides' gap perpendicular to every member and a fold of squares. The projector's component reads the block's deficit $\varepsilon_\lambda := [1 : 2] |E_\lambda|^2$ at its balance partner, $\langle P_\lambda, E_R \rangle_R$; $\varepsilon_\lambda = \varepsilon_{\bar{\lambda}}$ is involution-fixed, a fold of squares, and the closure's deficits fold to the plaquette's, $\sum_{\lambda \in \bar{L}} \varepsilon_\lambda = \delta_p := \langle d_L : \chi_p \rangle$ at the closure's own involution, the joined gap's fold $|E_R|^2 = 2\delta_p$, with the complement a fold of squares as well, $|1 + R|^2 = 2(2d_L - \delta_p)$; the joined generator's component the gradient: the gradient deficit

$$\sum_a (X_a \chi_p)^2 + \kappa_\Sigma \sum_{\lambda \in \bar{L}} [1 : d_\lambda] \varepsilon_\lambda^2 + \kappa_\Sigma \varrho_p = 2\kappa_\Sigma \delta_p, \quad \sum_a (X_a \chi_p)^2 + [\kappa_\Sigma : d_L] \delta_p^2 \leq 2\kappa_\Sigma \delta_p,$$

one pair comparison at every link coefficient, the A -series' fundamental pair at $\kappa_\Sigma = 2\kappa_f$ and $d_L = 2d_f$; at a plaquette the read runs at the boundary word's coefficient, the character's gradient at a boundary link the base link's moved along the boundary arc by the arc word's adjoint coefficient at the link's traversal sign, the move identity below at the arc's coefficient (Proposition 9's joint insertion across the arc at the A -series' fundamental), either arc one value, the gradient fixed under the plaquette's own adjoint coefficient. The move identity, and the adjoint tie. The generator map

$X \mapsto X^{(\lambda)}$ is equivariant to the unit, the closure field's read at the block (Construction 3), the generators' span inside the product $W_\lambda \otimes W_\lambda^\vee$ one θ -block of the product, the generators' copy (Lemma 4(i),(iii)), and the product's coefficient at the link, the tensor $U_\lambda \otimes \bar{U}_\lambda$ reading a block matrix M to $U_\lambda M U_\lambda^\dagger$ (Lemma 8(i)), reads it at the adjoint coefficient at the generator list, the move identity

$$U_\lambda X_a^{(\lambda)} U_\lambda^\dagger = \sum_b (U_\theta)_{ba} X_b^{(\lambda)}, \quad [E_\lambda, X_a^{(\lambda)}] = \sum_b (E_\theta)_{ba} X_b^{(\lambda)} (1 + E_\lambda),$$

one datum $(U_\theta)_{ba}$ at every block, involution-fixed at the self-dual adjoint (Lemma 16) with the dagger read its transpose and the evaluation identity $U_\theta U_\theta^\top = 1$ there, $E_\theta := \langle U_\theta : 1 \rangle$ the adjoint's gap, its fold $|E_\theta|^2 = 2\delta_\theta$ at the adjoint's deficit $\delta_\theta := \langle d_\theta : \chi_\theta \rangle$, the adjoint character the coefficient's trace at the generator list. The first display's trace reads the adjoint deficit as the brackets' fold,

$$2\kappa_\lambda \delta_\theta = \sum_a |[E_\lambda, X_a^{(\lambda)}]|^2,$$

and the brackets' fold is the Casimir's form at the bracket action: a generator acts on a block matrix as its bracket, so $\sum_a |[M, X_a^{(\lambda)}]|^2$ is the form of $\sum_a X_a^\dagger X_a$ at M , the Casimir at the form's point (Definition 7); the product exhausts into blocks (Lemma 5(i)) and the fold reads the Casimirs' weighted fold of the components' squares, $\sum_a |[M, X_a^{(\lambda)}]|^2 = \sum_\mu C_2(\mu) |M_\mu|^2$ over the exhaustion's blocks at M 's components (Lemma 7; Definition 6), at or below $C_2(\lambda + \bar{\lambda}) |M|^2$ (Proposition 3), the gap the Casimir gaps' fold against the components' squares. At $M = E_\lambda$ this is the adjoint tie

$$\kappa_\lambda \delta_\theta \leq C_2(\lambda + \bar{\lambda}) \varepsilon_\lambda, \quad \delta_\theta \leq t_\lambda \varepsilon_\lambda \quad \text{at} \quad t_\lambda := [C_2(\lambda + \bar{\lambda}) : \kappa_\lambda],$$

one member pair per label, and at the closure $\delta_\theta \leq t_L \delta_p$ at t_L the least t_λ over $\bar{L}$, one trichotomy fold (Definition 1), every member's deficit within the plaquette's. The residual cap. The identity with the block's generators is an orthogonal list, and the block's gap splits at its projection fold (Lemma 3),

$$E_\lambda = \alpha_\lambda 1 + \sum_b \beta_{b\lambda} X_b^{(\lambda)} + E_\lambda^\perp \quad \text{at} \quad \alpha_\lambda d_\lambda = \langle 1, E_\lambda \rangle_\lambda, \quad \kappa_\lambda \beta_{b\lambda} = \langle X_b^{(\lambda)}, E_\lambda \rangle_\lambda,$$

the identity's coefficient the deficit's balance partner against the dimension, the gradient coordinates β_λ , and the residual $E_\lambda^\perp$ perpendicular to the list, the folds adding, $|E_\lambda|^2 = \alpha_\lambda^2 d_\lambda + \kappa_\lambda |\beta_\lambda|^2 + |E_\lambda^\perp|^2$ at $|\beta|^2 := \sum_b \beta_b^2$. The bracket against a generator keeps the list's span and its perpendicular complement, and the residual's brackets are priced at the adjoint gap against the block's Casimir,

$$\sum_a |[E_\lambda^\perp, X_a^{(\lambda)}]|^2 \leq |E_\theta|^2 C_2(\lambda) |E_\lambda|^2 = 4C_2(\lambda) \delta_\theta \varepsilon_\lambda \leq 4C_2(\lambda) t_\lambda \varepsilon_\lambda^2,$$

the last at the adjoint tie. The residual splits at the unit block's second key, the identity at the located pair's second key, $z1$ (Definition 7' remainder lists at $z^2 + 1$), perpendicular to the identity and to every generator at the first member with its bracket against every generator the sum's unit,

$$E_\lambda^\perp = z[\tau_\lambda : d_\lambda] 1 + E_\lambda^{\perp\perp}, \quad |E_\lambda^\perp|^2 = |E_\lambda^{\perp\perp}|^2 + [\tau_\lambda^2 : d_\lambda],$$

at the second member τ_λ of $\text{tr } E_\lambda$, the character's second member at the located pair, the sum's unit at a self-dual label (the character involution-fixed there) and the dual label's its balance partner; the further residual $E_\lambda^{\perp\perp}$, perpendicular to the identity at both keys and to the list, sits in the invariant

complement (Lemma 1) that exhausts into nonunit blocks, each at $C_2(\mu)$ at or beyond the member floor's $C_2(f_0)$ (Lemma 9), so

$$C_2(f_0) |E_\lambda^{\perp\perp}|^2 \leq \sum_a |[E_\lambda^\perp, X_a^{(\lambda)}]|^2.$$

The second member is second order in the gap at the block's determinant: the block's determinant is the unit block's coefficient, one (Theorem 25), the dagger is the adjugate (Definition 3) at the evaluation identity, and the adjugate's trace is the elementary fold at the key below the dimension,

$$\text{tr } U_\lambda^\dagger = \sum_{k < d_\lambda} (d_\lambda - k) e_k(E_\lambda),$$

e_k the fold of the principal k -minors (Definition 3) at $e_0 = 1$ and $e_1 = \text{tr } E_\lambda$, so at the second members $d_\lambda \tau_\lambda$ is the balance partner of the second member of $\sum_{2 \leq k < d_\lambda} (d_\lambda - k) e_k(E_\lambda)$, and the minors' folds price the second member at the deficit's square:

$$\begin{aligned} \tau_\lambda^2 &\leq c_\lambda^z \varepsilon_\lambda^2, & c_\lambda^z &:= \left[4 \sum_{j \leq d_\lambda - 2} j^2 \sum_{2 \leq k < d_\lambda} \binom{d_\lambda}{k} (4 d_\lambda)^{k-2} : d_\lambda^2 \right], \\ |E_\lambda^\perp|^2 &\leq c_\lambda^\perp \varepsilon_\lambda^2, & c_\lambda^\perp &:= [4 C_2(\lambda) C_2(\lambda + \bar{\lambda}) : \kappa_\lambda C_2(f_0)] + [c_\lambda^z : d_\lambda], \end{aligned}$$

one member pair per label, the second summand read at a label distinct from its dual and vacant at a self-dual label, the second member the sum's unit there; c_λ^z reads $[4 : 3]$ at the A -series' fundamental at the residue two. The coordinates' tie. The adjoint's gap reads the block's gradient coordinates through the brackets' matrix ad , the adjoint block's own generator read, at the generator combination $\beta_\lambda := \sum_c \beta_{c\lambda} X_c^{(\lambda)}$,

$$E_\theta = \text{ad } \beta_\lambda + \varrho^{(\lambda)} \quad \text{at} \quad \kappa_\lambda \varrho_{ba}^{(\lambda)} := \langle X_b^{(\lambda)}, [E_\lambda, X_a^{(\lambda)}] E_\lambda^\dagger \rangle_\lambda, \quad |\varrho^{(\lambda)}|^2 \leq 4 t_\lambda \varepsilon_\lambda^2,$$

one read at every label, the adjoint's own reading its split, $E_\theta = \alpha_\theta 1 + \text{ad } \beta_\theta + E_\theta^\perp$ at $X_b^{(\theta)} = \text{ad } X_b$. Two labels' coordinates then tie at the adjoint's gap, $\text{ad}(\beta_\lambda - \beta_\theta) = \alpha_\theta 1 + E_\theta^\perp - \varrho^{(\lambda)}$, the brackets' matrix rescaling the fold at the adjoint index, $|\text{ad } y|^2 = \kappa_\theta |y|^2$ at the adjoint block's orthogonality, so

$$\begin{aligned} \kappa_\theta |\beta_\lambda - \beta_\theta|^2 &\leq 3 (\delta_\theta^2 [1 : d_\theta] + c_\theta^\perp \delta_\theta^2 + 4 t_\lambda \varepsilon_\lambda^2) \leq \kappa_\theta c_{\beta, \lambda} \varepsilon_\lambda^2, \\ c_{\beta, \lambda} &:= [3 (t_\lambda^2 ([1 : d_\theta] + c_\theta^\perp) + 4 t_\lambda) : \kappa_\theta], \quad c_\theta^\perp = [4 C_2(2\theta) : C_2(f_0)], \end{aligned}$$

the adjoint's own deficit at the adjoint tie, its residual cap at $\kappa_\theta = C_2(\theta)$, the index floor's read above. The closure's second-order read. The spare is the joined gap's residual off the joined list, the blocks' residuals' folds joined to the coordinates' dispersion, the index-weighted fold of squares about the weighted mean, which collects pairwise at the index-weighted differences,

$$\varrho_p = \sum_{\lambda \in \bar{L}} |E_\lambda^\perp|^2 + [1 : \kappa_\Sigma] \sum_a \sum_{\lambda < \mu} \kappa_\lambda \kappa_\mu (\beta_{a\lambda} - \beta_{a\mu})^2;$$

a dual pair's coordinates are one value, $\beta_{\bar{\lambda}} = \beta_\lambda$, so a dual pair's term is the sum's unit, and at two labels off one dual class the difference's square folds through the coordinates' tie, so

$$\varrho_p \leq \sum_{\lambda \in \bar{L}} (c_\lambda^\perp + [2 \kappa_\lambda \kappa'_\lambda : \kappa_\Sigma] c_{\beta, \lambda}) \varepsilon_\lambda^2 \leq c_\varrho \delta_p^2 \quad \text{at} \quad \kappa'_\lambda := \sum_{\mu \in \bar{L} \text{ off } \lambda's \text{ dual class}} \kappa_\mu,$$

κ'_λ the sum's unit at a closure of one dual class, at the member pair

$$c_\varrho := \text{the largest of } c_\lambda^\perp + [2\kappa_\lambda\kappa'_\lambda : \kappa_\Sigma]c_{\beta,\lambda} \text{ over } \lambda \in \bar{L} ,$$

one trichotomy fold over the closure, the residual caps alone at every stated list off D_ℓ 's even rank. The deficits' weighted squares read at the closure's least dimension d_- , one trichotomy fold, $\sum_\lambda [1 : d_\lambda] \varepsilon_\lambda^2 \leq [1 : d_-] \sum_\lambda \varepsilon_\lambda^2 \leq [1 : d_-] \delta_p^2$. So the deficit's second-order read holds at every plaquette coefficient,

$$2\kappa_\Sigma \delta_p - \sum_a (X_a \chi_p)^2 = \kappa_\Sigma \left(\sum_{\lambda \in \bar{L}} [1 : d_\lambda] \varepsilon_\lambda^2 + \varrho_p \right) \leq \kappa_\Sigma s_L \delta_p^2 \quad \text{at } s_L := [1 : d_-] + c_\varrho ,$$

the second-order pair s_L one member pair, the gap a positive element at or below the deficit's square at the pair, the gradient square within second order of the deficit's double at the joined index.

Proof. The cap. Writing $\chi_\lambda(U_{\partial p}) = \sum_{i,j} A_{ij} (U_4)_{ji}$ over a cut of the loop in the block's coefficients, $A := U_1 U_2 U_3$ the complementary path,

$$\langle y, \chi_\lambda(U_{\partial p}) x \rangle^2 \leq \left(\sum_{i,j} \|\bar{A}_{ij} y\|^2 \right) \left(\sum_{i,j} \|(U_4)_{ji} x\|^2 \right) = d_\lambda^2 \langle x, x \rangle \langle y, y \rangle$$

by the squared Cauchy–Schwarz of the coefficient carrier's pairing (Construction 8), every square a pairing, each sum contracting by the evaluation identity at the block to the trace of the identity there, $\text{tr}(A^\dagger A) = \text{tr}(U_4^\dagger U_4) = d_\lambda$, times the pairing; at $y = \chi_\lambda(U_{\partial p}) x$ the display reads the comparison at the factor $\langle y, y \rangle$: at the factor occupied the order reads back (Definition 1), and at the factor of equal members the conclusion is the pairing's own side read, the two self-pairings at or above the sum's unit (Construction 8); and one further squared Cauchy–Schwarz at a window vector against its image, $\langle x, \chi_\lambda x \rangle^2$ at or below $\langle x, x \rangle \langle \chi_\lambda x, \chi_\lambda x \rangle$, at or below $d_\lambda^2 \langle x, x \rangle^2$, reads the term's form two-sided at d_λ at the trichotomy of squares (Definition 1). At the closure the term's form is the members' forms' fold, each two-sided at its dimension, so the term is two-sided at d_L ; and its square reads the fold at the two squares' price, $2 \langle \chi_\lambda x, \chi_\mu x \rangle$ at or below $[d_\mu : d_\lambda] \langle \chi_\lambda x, \chi_\lambda x \rangle + [d_\lambda : d_\mu] \langle \chi_\mu x, \chi_\mu x \rangle$ (the squares $\|d_\mu \chi_\lambda x + d_\lambda \chi_\mu x\|^2$ and its swap's, Lemma 17's device at the pairing), the fold over the closure's pairs collecting to $d_L \sum_\lambda [1 : d_\lambda] \langle \chi_\lambda x, \chi_\lambda x \rangle$, at or below $d_L^2 \langle x, x \rangle$ at the members' caps.

The index. Each generator matrix is a commutator of two block matrices (the triple displays, $2E_i = [H_i, E_i]$ and $H_i = [E_i, F_i]$, every raising a commutator word), so its trace reads the sum's unit at the trace's exchange; two generators' pairing at the block is an invariant form on the generator span, a point of the form line (Proposition 1), so it reads the generators' own orthogonality at one scale, its trace the Casimir's at the block against the generator count. The highest root's triple reads its coroot H_θ on a content summand at the pair $\mu(\theta^\vee)$ (Construction 3's root triples), so the block's trace pairing reads $\langle H_\theta, H_\theta \rangle_\lambda = \sum_\mu \text{mult}_\lambda(\mu) \mu(\theta^\vee)^2$ over the block's contents, and the orthogonality at the index reads it at κ_λ times H_θ 's square in the generators' form, one square at every block; at the adjoint the contents are the roots at one each with the unit content at the coordinate count (Proposition 13), a positive root reads $\theta(\theta^\vee) = 2$ at θ and one or the sum's unit off it (a positive root β at the pair two off θ reads its reflection $\langle \beta : 2\theta \rangle$'s balance partner $2\theta - \beta$ a root, θ joined to the positive fold $\theta - \beta$, against the highest root; a positive root at the pair's balance partner reads its reflection $\beta + \theta$ a root beyond θ , against the highest root), the residue fold $(2\rho)(\theta^\vee) = 2r$ (Construction 4) counts the positive roots at the pair one as $2r - 2$, and the reflection s_θ pairs each root with a root at the partner pair, so $\langle H_\theta, H_\theta \rangle_\theta = 8 + 2(2r - 2) = 4(r + 1) = 2\kappa_\theta$ at $\kappa_\theta = C_2(\theta) = 2(r + 1)$ (Proposition 12): H_θ 's square is 2, the displayed index. A θ -string at top h reads its pairs $\langle h : 2b \rangle$ at the depths b at or below h (Lemma 6), their squares' fold

$[h(h+1)(h+2):3]$, at or beyond two at h at or beyond one; and a nonunit block's top λ reads its pair at or beyond one: λ is dominant off the unit fold (Lemma 2), so some simple pair $\lambda(\alpha_i^\vee)$ sits at or beyond one, and at the highest root's displayed fold $\theta = \sum_i c_i \alpha_i$, every c_i at or beyond one at every member (Construction 4), the pair reads the halved products' fold at $\langle \theta, \theta \rangle = 2$, $\lambda(\theta^\vee) = \langle \lambda, \theta \rangle = \sum_i c_i [\langle \alpha_i, \alpha_i \rangle : 2] \lambda(\alpha_i^\vee)$, every summand a natural product and the occupied key's positive, so the top's θ -string has its height at or beyond one and $\kappa_\lambda \geq 1$.

The gradient deficit. The pairing's projection fold at the orthogonal list (Lemma 3) reads $|E_R|^2 = \sum_\lambda [1:d_\lambda] \langle P_\lambda, E_R \rangle_R^2 + [1:\kappa_\Sigma] \sum_a \langle X_a^R, E_R \rangle_R^2 + \varrho_p$ at the members' self-pairings d_λ and κ_Σ , the components the deficits' and the gradient's balance partners, and multiplied through by κ_Σ at $|E_R|^2 = 2\delta_p$ it is the identity. $|E_\lambda|^2 = \text{tr}(E_\lambda^\dagger E_\lambda)$ reads $2d_\lambda$ against $\text{tr} U_\lambda + \text{tr} U_\lambda^\dagger$ at the evaluation identity, the dual block's character the trace's dagger read, so the deficit is involution-fixed and the closure's deficits fold to δ_p , the joined fold $2\delta_p$ the blocks' folds' sum; the comparison reads the deficits' weighted squares, $d_L \sum_\lambda [1:d_\lambda] \varepsilon_\lambda^2$ at or beyond δ_p^2 (the two squares' device at the weights d_λ , Lemma 17).

The move identity, and the adjoint tie. The generator map is an invariant of $\theta^\vee \otimes W_\lambda \otimes W_\lambda^\vee$ (Lemma 8(i)), occupied at a nonunit block, whose contents sit off the unit fold with an H -read off the sum's unit (Construction 3's grading), so the generators' span inside $W_\lambda \otimes W_\lambda^\vee$ is one θ -block of the product (Lemma 4(i),(iii)); the product's coefficient at the link, the tensor $U_\lambda \otimes \bar{U}_\lambda$ reading a block matrix M to $U_\lambda M U_\lambda^\dagger$ (the dual action on the second factor, Lemma 8(i)), reads on an invariant subspace that subspace's own block's coefficient at the subspace's coordinate list (Construction 8, a product of coefficient elements decomposing by the fusion data), so at the generators' copy it is the adjoint coefficient at the generator list, the first display; the second display is the first at $U_\lambda = 1 + E_\lambda$ multiplied through by U_λ on the right at the evaluation identity, the generator's own term withdrawn on both members. The diagonal entry $(U_\theta)_{aa}$ is the pairing $\langle X_a^{(\lambda)}, U_\lambda X_a^{(\lambda)} U_\lambda^\dagger \rangle_\lambda$ against the index at the orthogonality, the moved generator's fold is the generator's own at the evaluation identity, $|U_\lambda X_a^{(\lambda)} U_\lambda^\dagger|^2 = \kappa_\lambda$, and the two-member square $|X_a^{(\lambda)} - U_\lambda X_a^{(\lambda)} U_\lambda^\dagger|^2 = 2\kappa_\lambda - 2 \langle X_a^{(\lambda)}, U_\lambda X_a^{(\lambda)} U_\lambda^\dagger \rangle_\lambda$ collects over the list, the moved difference $(X_a^{(\lambda)} U_\lambda - U_\lambda X_a^{(\lambda)}) U_\lambda^\dagger$ keeping its fold at the evaluation identity and the identity's own commutator the sum's unit, to the brackets' fold display. On $W_\lambda \otimes W_\lambda^\vee$ at the Leibniz rule with the dual action on the second factor (Lemma 8(i)) a generator acts on a block matrix as its bracket, the product exhausts into blocks (Lemma 5(i)) with the Casimir reading $C_2(\mu)$ on a block at top μ (Lemma 7; Definition 6), and every top sits at or below the Cartan product $\lambda + \bar{\lambda}$ in the dominance order with the Casimir increasing there (Proposition 3), so the fold sits at or below $C_2(\lambda + \bar{\lambda}) |M|^2$.

The residual cap. The identity's bracket is the sum's unit, a generator's bracket is the closure field's fold over the list, and at an anti-Hermitian generator the trace pairing exchanges the bracket at the balance partner, $\langle [M, X_a^{(\lambda)}], N \rangle_\lambda + \langle M, [N, X_a^{(\lambda)}] \rangle_\lambda$ the sum's unit at the trace's exchange, so a matrix perpendicular to the list has its bracket perpendicular to it. The second display's projection off the list's span then reads $[E_\lambda^\perp, X_a^{(\lambda)}]$ as the projection of $\sum_b (E_\theta)_{ba} X_b^{(\lambda)} E_\lambda$, the generators' own fold withdrawn, the projection's fold at or below the vector's own (the residual's read, Lemma 3), the squared Cauchy-Schwarz at the fold over b prices it at the adjoint gap's fold against the moved gaps' fold, and the moved gaps' fold is the Casimir's read at the block, $\sum_b |X_b^{(\lambda)} E_\lambda|^2 = \text{tr}(E_\lambda^\dagger \sum_b X_b^\dagger X_b E_\lambda) = C_2(\lambda) |E_\lambda|^2$, the Casimir the block's scalar (Lemma 7; Definition 6), the displayed price. The list's span is the unit block joined to one θ -block of the bracket action's exhaustion, the unit block once in the product (Lemma 8(iii)); the projection fold reads at the trace pairing's first member, so the unit block's second key $z1$ sits in the list's complement, and the residual splits at it as displayed; the further residual's form reads its components' folds at the blocks' Casimir scales while the bracket reads the further residual alone, the displayed floor. The

wedge of the block's whole list is a one-dimensional invariant subspace, the unit label the one label of dimension one (Theorem 25), so the block's determinant is the unit block's coefficient, one, and the adjugate identity (Definition 3) at the evaluation identity reads the dagger as the adjugate, the count $d_\lambda - k$ in the adjugate's trace the identity's places joining a k -minor of the gap to a principal minor of $1 + E_\lambda$ at the key below the dimension; the dagger's trace reads τ_λ 's balance partner at the second member, so at the second members $d_\lambda \tau_\lambda$ is the balance partner of the second member of the keys from two. A k -minor is the assignment fold, and the k -minors' fold over the $\binom{d_\lambda}{k}$ subsets holds $\binom{d_\lambda}{k} k!$ products of k entries, so its fold of squares sits at or below that count against the products' folds (Construction 8's Cauchy–Schwarz at the count), a product's fold its factors' folds' product, and the products' folds over the subsets and their assignments enter the k -th power of the entries' folds' sum $|E_\lambda|^2 = 2\varepsilon_\lambda$ once per assignment order, $k!$ times each: $e_k(E_\lambda)$'s fold sits at or below $\binom{d_\lambda}{k} (2\varepsilon_\lambda)^k$, a second member's square at or below the fold; $2\varepsilon_\lambda$ sits at or below $4d_\lambda$ as elements, the gap the fold of squares $|1 + U_\lambda|^2$, so $(2\varepsilon_\lambda)^k \leq (4d_\lambda)^{k-2} 4\varepsilon_\lambda^2$ at k at or beyond two, and the fold over the keys prices at the squared Cauchy–Schwarz at the count, the displayed c_λ^z ; the residual's cap joins the further residual's floor to the second member's price, $|E_\lambda^\perp|^2 = |E_\lambda^{\perp\perp}|^2 + [\tau_\lambda^2 : d_\lambda] \leq [1 : C_2(f_0)] 4C_2(\lambda) t_\lambda \varepsilon_\lambda^2 + [c_\lambda^z : d_\lambda] \varepsilon_\lambda^2$, the displayed $c_\lambda^\perp$.

The coordinates' tie. Pairing the second display against $X_b^{(\lambda)}$ splits the adjoint tie at the two members of $1 + E_\lambda$: the identity's member is the bracket's exchange, $\langle X_b^{(\lambda)}, [E_\lambda, X_a^{(\lambda)}] \rangle_\lambda = \langle E_\lambda, [X_a^{(\lambda)}, X_b^{(\lambda)}] \rangle_\lambda = \langle [\beta_\lambda, X_a^{(\lambda)}], X_b^{(\lambda)} \rangle_\lambda$ (the bracket inside the list's span, the identity's and the residual's pairings against it the sum's unit), the coefficient of $X_b^{(\lambda)}$ in the bracket $[\beta_\lambda, X_a^{(\lambda)}]$ against the index, $\kappa_\lambda (\text{ad } \beta_\lambda)_{ba}$; the remainder's fold reads at the projection fold over the list against the products' folds: $\sum_b \langle X_b^{(\lambda)}, M \rangle_\lambda^2 = \kappa_\lambda |\Pi M|^2 \leq \kappa_\lambda |M|^2$ at the list's projection Π (Lemma 3), a product's fold at or below the factors' folds' product (Construction 8's Cauchy–Schwarz at the entries), so $\kappa_\lambda^2 |\varrho^{(\lambda)}|^2 \leq \kappa_\lambda \sum_a |[E_\lambda, X_a^{(\lambda)}]|^2 |E_\lambda|^2 = \kappa_\lambda \cdot 2\kappa_\lambda \delta_\theta \cdot 2\varepsilon_\lambda$ at the brackets' fold identity. At the three-member fold of squares (Construction 8's Cauchy–Schwarz at the count three) the identity's summand is $\alpha_\theta^2 d_\theta = [1 : d_\theta] \delta_\theta^2$ at $\alpha_\theta d_\theta = \langle 1, E_\theta \rangle_\theta$ the deficit's balance partner, the residual's is the residual cap at $\lambda = \theta$, at $\varepsilon_\theta = \delta_\theta$ and $\kappa_\theta = C_2(\theta)$, the remainder's is the displayed $4t_\lambda \varepsilon_\lambda^2$, and $\delta_\theta^2 \leq t_\lambda^2 \varepsilon_\lambda^2$ at the adjoint tie collects the displayed $c_{\beta,\lambda}$.

The closure's second-order read. The spare's generator components are the blocks' coordinates' index-weighted folds, $\langle X_a^R, E_R \rangle_R = \sum_\lambda \kappa_\lambda \beta_{a\lambda}$, so at the projection fold's identity twice, over the joined list and over each block's own, the spare reads the blocks' residuals' folds joined to the dispersion, the mean's fold read at the squares' identity $\kappa_\Sigma \sum_\lambda \kappa_\lambda \beta_\lambda^2 - (\sum_\lambda \kappa_\lambda \beta_\lambda)^2 = \sum_{\lambda < \mu} \kappa_\lambda \kappa_\mu (\beta_\lambda - \beta_\mu)^2$, one cross product per pair. The dual block's generators and gap are the block's own at the located pairs' key swap (the dual action, Lemma 8(i); the key swap, Lemma 37), and the trace pairing's first member is kept by the swap, so $\beta_{\bar{\lambda}} = \beta_\lambda$; at two labels off one dual class the two-member fold of squares at the coordinates' tie reads $(\beta_{a\lambda} - \beta_{a\mu})^2 \leq 2(\beta_{a\lambda} - \beta_{a\theta})^2 + 2(\beta_{a\mu} - \beta_{a\theta})^2$, folding to $2c_{\beta,\lambda} \varepsilon_\lambda^2 + 2c_{\beta,\mu} \varepsilon_\mu^2$, the displayed cap; the closing comparison reads the deficits' squares' fold at or below the deficit's square, every deficit a fold of squares, the deficit's square the squares' fold joined to the cross products' doubled fold. $\square$

5 The requirement, in two moments

Theorem 6 (The divisor-separation detector). *The level- γ requirement is the probe family $\mu_2(A) \geq \gamma \mu_1(A)$ at the two moments $\mu_j(A) := \langle v_A, D^j v_A \rangle_G$ at $v_A := \Pi A \psi$, the word image projected to the window (Theorem 8's reads), and its detector pair at a spectral point λ is $(\lambda^2 : \gamma \lambda)$, the moments' spectral read: a probe's two moments read the weight fold over the located spectrum (Lemma 33), so*

the requirement's read at a probe is the fold of the detector pairs' at the probe's weights. The pair's members are equal exactly at the kernel point and at the edge $\lambda = \gamma$, the two neutral points, its site datum the factor read $\lambda \langle \lambda : \gamma \rangle$, and off them its side is λ 's against γ at positive λ : weight below γ drives failure, weight beyond passes, weight at the edge is neutral, and the vacuum direction, the kernel point, is neutral as well, the vacuum's saturation untouched.

Proof. The split's congruence diagonalizes the gram and the level gap at once at the located roots λ_j with the gram's diagonal entries g_j on their upper sides (Lemma 33), so at the probe's vector at coordinates y_j there the moments read $\mu_1(A) = \sum_j w_j \lambda_j$ and $\mu_2(A) = \sum_j w_j \lambda_j^2$ at the weights $w_j = g_j y_j^2$, and the requirement's pair $\langle \mu_2(A) : \gamma \mu_1(A) \rangle$ is the fold $\sum_j w_j \lambda_j \langle \lambda_j : \gamma \rangle$, the detector pairs' at the weights. The pair $(\lambda^2 : \gamma \lambda)$'s site datum is the product $\lambda \langle \lambda : \gamma \rangle$, the distributed read $\lambda \langle \lambda : \gamma \rangle = \langle \lambda^2 : \gamma \lambda \rangle$ (Definition 1), of equal members exactly where a factor is, λ at the kernel point or $\langle \lambda : \gamma \rangle$ at the edge, the product's injectivity at an occupied factor; off them the factor λ is positive and the side is $\langle \lambda : \gamma \rangle$'s, λ 's against γ . $\square$

Theorem 7 (The level condition over probes is the spectral condition). *On the word sector's window projection, the level- γ requirement (one condition per probe) is equivalent to the form comparison*

$$SD \succeq \gamma S \quad \text{on the word sector} \quad (\text{the kernel point with every further level at or beyond } \gamma).$$

Proof. ($\Leftarrow$) With the spectrum at the kernel point and at or beyond γ , every detector pair reads its upper side or equal members (Theorem 6), and every probe's weight fold with it; read at the pair's certificate, the positive-semidefinite split prices every sector vector's form through the congruence at its witness, the blocks' fold of squares (Lemma 30), and the site datum's decomposition reads the priced form as the two moments' comparison, the requirement's own display. ($\Rightarrow$) On a window the reversal count settles it: at $\text{rev}(SD : \gamma S) \geq 1$ a lower-side pivot sits among the elimination datum's $\text{LDL}^\top$ pivots, and Theorem 15(2) reads off the elimination a pair-entried v with $\gamma v^\top S v > v^\top S D v$, the reversal's own read. The witness is a vector of that projection, hence $\Pi A \psi$ at some word A (Definition 10), and that word's probe fails the requirement outright, the witness's two reads its two moments (Theorem 8). $\square$

Theorem 8 (The two-moment form). *At a window ground and a multiplication probe A , write $v_A = A_\Pi \psi$ for the projection of its word image to the window and J_A for the window commutator's vector (Theorem 9). The actions and pairing are the Gram solves of Definition 10. The moments and their operator read are*

$$\begin{aligned} \mu_1(A) &= \langle v_A, J_A \rangle_{G_\Pi} = \omega(A_\Pi^\dagger [K_\Pi, A_\Pi]), \\ \mu_2(A) &= \langle J_A, J_A \rangle_{G_\Pi} = \omega(Q_2(A)), \quad Q_2(A) := [K_\Pi, A_\Pi]^\dagger [K_\Pi, A_\Pi]. \end{aligned}$$

In coordinates, at $C_A := [K_\Pi, A_\Pi]$, the two forms read $\mu_1(A) = v_A^\top S_\Pi v_A$ and $\mu_2(A) = \psi^\top C_A^\top G_\Pi C_A \psi$; the action $Q_2(A)$ has form $G_\Pi Q_2(A) = C_A^\top G_\Pi C_A$.

Their entries are finite pairing reads of $J_A = \alpha \Pi[E, A] \Pi \psi + \beta \mathcal{B}_A \psi$, the boundary maps Theorem 9's. At $\alpha = 1$ and $\beta \mathcal{B}_A \psi$ the sum's unit, the moments read the electric commutator $\Pi[E, A] \Pi$ at the projected probe. On the full product carrier, at the residual r the sum's unit and $\alpha = 1$, the full product's vector is $(K - \varepsilon_0) A \psi = [E, A] \psi$ and its two pairing reads are $\omega(A^\dagger [E, A])$ and $\omega([E, A]^\dagger [E, A])$. The family $\mu_2(A) \geq \gamma \mu_1(A)$ is the positive-semidefinite read of the detector pair $(S_\Pi D_\Pi : \gamma S_\Pi)$ compressed whole to these projected word images. Both moments depend on the probe through v_A alone; the map $A \mapsto \Pi A \psi$ is linear and its image is the word sector's window projection (Lemma 19). The full word images have their pairing in their product windows (Definition 9).

The gradient read. At an involution-fixed multiplication probe f , define the coefficient element and the boundary map

$$D_f := \sum_a (X_a f)^2 = f(Ef) + \langle [1 : 2] E(f^2) \rangle^\vee,$$

$$W_f := \Pi'[E, f]\Pi = E_{\Pi'} V_f - V_f E_{\Pi}.$$

The generators are Definition 7's, and the first moment reads

$$\mu_1(f) = \alpha \omega(D_f) + \left\langle [\alpha : 2] \omega(V_f^\dagger W_f + W_f^\dagger V_f) \right\rangle^\vee$$

$$+ [\beta : 2] \omega([f_\Pi, \mathcal{B}_f]).$$

The two boundary terms read their displayed pairings, and at $V_f \psi$ and $\beta \mathcal{B}_f \psi$ the sum's unit this is $\mu_1(f) = \alpha \omega(D_f)$. At a character $\chi_c(W)$ on a loop of L distinct links the electric member reads the loop's own content, $E \chi_c(W) = L C_2(c) \chi_c(W)$, each link's Casimir at the block's coefficient (Construction 8; Proposition 7), and the square fuses over the channel list, so the gradient square is the channel fold

$$\sum_a (X_a \chi_c(W))^2 = L \sum_e N_{cc}^e \langle C_2(c) : [1 : 2] C_2(e) \rangle \chi_e(W),$$

a character polynomial of the loop's holonomy; at the adjoint character its read at the unit holonomy is the sum's unit, $\sum_e m_e d_e \hat{c}_2(e) = 2 d_\theta^2$ against $d_\theta^2 \hat{c}_2(\theta)$ (Lemma 24's second closure; Theorem 25), and at a difference of two translates of one loop at disjoint links the gradient squares add, a generator at one loop's link reading the other's character at the sum's unit.

Proof. The ground equation gives $J_A = D_\Pi v_A$ (Theorem 9), so the two moments are the displayed pairing and square. Linearity of multiplication and projection makes every combination of projected word images a projected word image; the detector's form on that combination is the moments' comparison, the two quadratic reads at that vector. For the gradient read, the derivation's Leibniz rule and $E = \sum_a X_a^\dagger X_a$ give $[f, [E, f]] = 2D_f$ and $E(f^2) = 2f(Ef) - 2D_f$ (Construction 8). The projected double commutator reads

$$[f_\Pi, \Pi[E, f]\Pi] = 2\Pi D_f \Pi - V_f^\dagger W_f - W_f^\dagger V_f,$$

by inserting $\Pi + \Pi' = 1$ in both products; $[E, f]$ has its adjoint at its balance partner, so $\Pi[E, f]\Pi' = -W_f^\dagger$. The first commutator is $[K_\Pi, f_\Pi] = \alpha \Pi[E, f]\Pi + \beta \mathcal{B}_f$ (Theorem 9). Its double commutator at the ground has expectation $2\mu_1(f)$, the two terms with K_Π at an outer slot reading $\varepsilon_0 \omega(f_\Pi^2)$ each. Combining these displays gives the stated gradient read. At $V_f \psi$ the sum's unit the paired electric boundary terms read it; at $\beta \mathcal{B}_f \psi$ the sum's unit the magnetic term reads it as well, $\mathcal{B}_f$ having its adjoint at its balance partner. $\square$

Corollary 4 (Two moments against a divisor). *Combining Theorem 8 with Theorem 7, on a window:*

$$(\forall A : \mu_2(A) \geq \gamma \mu_1(A)) \iff (SD)_\Lambda : (\gamma S)_\Lambda \text{ positive semidefinite,}$$

the compression to the window's word sector taken of the detector form whole, since $\mu_2(A) = v_A^\top S D v_A$ and $\mu_1(A) = v_A^\top S v_A$ at $v_A = \Pi A \psi$, the projected word image of Theorem 8 (Theorem 9): one reversal count on a form pair over the located stage, exact by $\text{LDL}^\top$ there (Lemma 32), whose read is the kernel point with every further level at or beyond γ .

Proof. Right to left by Theorem 6's side reads, the split's transport pricing every sector vector at the site datum's decomposition (Lemma 30); left to right by the count's own witness, which Theorem 15(2) reads off the elimination. $\square$

Lemma 19 (The sector's span read). *Multiplication by a window ground is adjoint to its involution image's: at carrier elements w, x and a window ground ψ (Definition 10; the involution Definition 9's),*

$$\langle w, x \cdot \psi \rangle = \langle w \cdot \bar{\psi}, x \rangle ,$$

and a ground's involution image is a ground. The span read at a window and a ground is the product matrix's rank at the window's dimension: the matrix $\text{Eval}(\bar{v} w \psi)$ at w over the window list and v over the product window's (Proposition 4), every entry a ground computation, the rank the elimination's read (Definition 3), one decidable read per window and ground. The span read holds at every window and every ground. The word sector's window projection fills the window, and the compression to that projection is the window pair whole. At a window list (w_i) the probes $w_i \psi$ give a spanning list of projected word images; their coefficient products and Gram solves read in their finite product windows (Definition 9; Proposition 5).

Proof. Both sides of the display are the read $\text{Eval}(\bar{w} \psi x)$, the products' one value under commutativity with the involution invariance (Definition 9; Proposition 5). The involution keeps the pairing (Eval one value under the involution, Definition 9) and the pencil, E at the dual label's Casimir (Definition 6) and M at the plaquette character, its own involution image at the closure (Construction 6; Definition 10), so it keeps the level gap's kernel: a ground's image is a ground.

Multiplication by a coefficient element off the sum's unit is injective. The computation uses the coefficient carrier's positive pairing and derivations (Construction 8), with 0 the sum's unit in the displays. First, $h^2 = 0$ forces $h = 0$: at $z = \bar{h}h$ the involution and the product give $\bar{z} = z$ and $z^2 = \bar{h}^2 h^2 = 0$, so $\langle z, z \rangle = \text{Eval}(z^2) = 0$ forces $z = 0$, and $\langle h, h \rangle = \text{Eval}(z) = 0$ forces $h = 0$. At $fg = 0$ a link generator's Leibniz identity reads

$$f(X_a g) + (X_a f)g = 0, \quad f^2(X_a g) = 0, \quad (f X_a g)^2 = 0.$$

The square read therefore gives $f(X_a g) = 0$. Iteration gives $f(X_b X_a g) = 0$ and then $f(Eg) = 0$, since the electric action is the generator pairs' fold, the pairing adjoints the generators' balance partners at the metric dual (Construction 8; Definition 7; Lemma 7). Sums and scalar products keep this identity, so $fP(E)g = 0$ at every polynomial P .

Apply this read to $\bar{g}g$, whose product with f is the sum's unit. Its occupied positive electric contents form a finite list $\mathcal{E}$, the finite product window's index reads (Proposition 5; Construction 8); every content off the unit coefficient is positive, the links' Casimirs positive at their nonunit labels (Definition 6; Construction 6). The polynomial and its scalar read are

$$P(t) := \prod_{c \in \mathcal{E}} \langle c : t \rangle, \quad d := \prod_{c \in \mathcal{E}} c, \quad P(E)(\bar{g}g) = d \text{Eval}(\bar{g}g) \mathbf{1}.$$

At a vacant list both products read one. At an occupied content the corresponding factor reads the sum's unit, and at the unit coefficient every factor reads its content, which gives the display. Thus $d \text{Eval}(\bar{g}g) f = 0$; at g off the sum's unit the scalar is positive by the pairing, and scalar product injectivity gives $f = 0$ componentwise (Definition 1). Every product and generator image in this computation has a finite coefficient window. The argument also reads over the ground's located stage: the coefficient Grams' positive splits persist there (Lemma 30; Lemma 32).

The window list's products $z_i := w_i \bar{\psi}$ are consequently independent: a combination at the sum's unit reads $(\sum_i a_i w_i) \bar{\psi} = 0$, hence $\sum_i a_i w_i = 0$ at $\bar{\psi}$ off the sum's unit, and every coefficient is the sum's unit at the window list's independence. Their Gram is positive definite, with its adjugate solve (Lemma 3; Definition 3). Given a window vector y , solve

$$\sum_j \langle z_i, z_j \rangle b_j = \langle w_i, y \rangle \quad \text{at every } i, \quad A := \sum_j b_j z_j.$$

The opening identity gives $\langle w_i, \Pi A \psi \rangle = \langle z_i, A \rangle = \langle w_i, y \rangle$, so $\Pi A \psi = y$ by the window's positive pairing. The products' independent columns give the product matrix's rank at the window dimension, the span read; the probes z_i give the displayed spanning list. $\square$

Lemma 20 (A window spectral read is the cut at that window). *At a window and a pair κ , the spectral read*

$$\text{spec}(H) \subset \{\varepsilon_0\} \cup [\varepsilon_0 + E_0 \kappa, \infty)$$

is the window cut $\mathcal{K}_\Lambda$ of Definition 11 at κ , at every window ground at once: at the read the pair $(SD : E_0 \kappa S)$ is positive semidefinite on the window, and a compression keeps that read (Lemma 30): every word-sector compression reads it, the cut's own object, the detector form compressed whole. At every ground the projection fills the window (Lemma 19), so the identification holds both ways, the window cut's edge the spectral edge there. The reading back rides the gap's own floor: the levels' positivity, S 's positive-semidefinite certificate (Definition 10), is the datum locating every root's side, the side's one source. At the ground multiplicity $g := \dim \ker S$, the equivalent count read is

$$\text{rev}(H : (\varepsilon_0 + E_0 \kappa)G) = g.$$

At every positive pair h , the joined count identity is

$$g + \text{rev}(SD : hS) = \text{rev}(H : (\varepsilon_0 + h)G).$$

Proof. The split of Lemma 33 maps the level-gap form $S(\varepsilon_0 G + S = H)$, Definition 10) to a diagonal at the kernel point's read and at reads at or beyond $E_0 \kappa$, the detector pair at each reads its upper side or equal members (Theorem 6, the two neutral points the ground's and the edge's), so the pair $(SD : E_0 \kappa S)$ is positive semidefinite on the window and every compression reads it. At every ground the projection fills the window (Lemma 19), so the compressed pair is the window pair whole. The reading back: the detector pair's side is λ 's against γ at positive λ (Theorem 6), and the certificate puts every root at the kernel point or on its upper side, so a root reading the pair's upper side sits at or beyond the edge. The count at $\varepsilon_0 + E_0 \kappa$ is the kernel dimension joined to the count of occupied gaps strictly below $E_0 \kappa$, by the split's diagonal read; thus it equals g exactly at the spectral read. The detector count reads the positive gaps strictly below h : its diagonal at a gap λ is the positive Gram weight against $\lambda \langle \lambda : h \rangle$, on its lower side exactly at $0 < \lambda < h$. The ground and the gap h read equal members. Adding the ground's g positions gives the shifted count, including its read at the gap h . At a level on its lower side the pair's read holds outright, the square sitting on its upper side from either side of its entry: the floor's certificate is the side's one source. $\square$

Theorem 9 (The window commutator and its boundary reads). *Let Π be a window's orthogonal projection and Π' its complement in a window holding the input and the product images in the displays below (Definition 8; Proposition 5). A multiplication word A reads as $A_\Pi := \Pi A \Pi$ on the input window; its full product is read in its target window. Write $K_\Pi := \Pi K \Pi$ for the compressed*

action at $\alpha E = \beta M + K$, and let ψ be a ground of K_Π at ε_0 , with D_Π its level-gap action. At its image-list columns L , the forms are $G_\Pi = L^\top GL$ and $H_\Pi = L^\top HL$. They satisfy $G_\Pi K_\Pi = H_\Pi$ and $G_\Pi D_\Pi = S_\Pi$ at $H_\Pi = \varepsilon_0 G_\Pi + S_\Pi$; all actions and pairing adjoints are the Gram solves of Definition 10. The dagger is the adjoint in the stated pairing (Construction 8). The boundary maps and their paired read are

$$U := \Pi' M \Pi, \quad V_A := \Pi' A \Pi, \quad \mathcal{B}_A := U^\dagger V_A - V_A^\dagger U.$$

Then

$$\begin{aligned} [K_\Pi, A_\Pi] &= \alpha \Pi[E, A] \Pi + \beta \mathcal{B}_A, \\ J_A &:= D_\Pi A_\Pi \psi = [K_\Pi, A_\Pi] \psi = \alpha \Pi[E, A] \Pi \psi + \beta \mathcal{B}_A \psi. \end{aligned}$$

The vector identity at its additive join is $K_\Pi A_\Pi \psi = A_\Pi K_\Pi \psi + D_\Pi A_\Pi \psi$. At $\alpha = 1$ and $\beta \mathcal{B}_A \psi$ the sum's unit, the line reads $J_A = \Pi[E, A] \Pi \psi$. Each boundary product reads one pairing, $\langle x, U^\dagger V_A y \rangle = \langle Ux, V_A y \rangle$, so its entries read on the independent collection of the boundary images at their Gram (Lemma 3). The full product's line reads the ground's residual r :

$$K\psi = \varepsilon_0 \psi + r, \quad r = (\beta U \psi)^\vee, \quad (K - \varepsilon_0)A\psi = \alpha[E, A]\psi + Ar,$$

the superscript the balance partner. At r the sum's unit and $\alpha = 1$ this reads $(K - \varepsilon_0)A\psi = [E, A]\psi$. The boundary maps' electric and product recurrences are

$$\begin{aligned} V_{[E, A]} &= E_{\Pi'} V_A - V_A E_\Pi, \quad E_{\Pi'} := \Pi' E \Pi', \quad E_\Pi := \Pi E \Pi, \\ (AB)_\Pi &= A_\Pi B_\Pi + \Pi A \Pi' V_B, \quad V_{AB} = V_A B_\Pi + \Pi' A \Pi' V_B. \end{aligned}$$

At input and output contents c, c' , the electric recurrence multiplies the boundary entry by $\langle c' : c \rangle$.

The disjoint-support cutoff read. Let A, B be multiplications on disjoint link sets R, T of one region, and let $P_{a, c, q}$ select the configurations at partial electric contents a on R , c on T and q on the further links (Definition 8). All selections and products read in a finite target holding both product orders (Proposition 5). At input and output blocks with

$$q + a + c \leq C, \quad q + a' + c' \leq C,$$

let u, v be the counts of the respective comparisons

$$q + a' + c \leq C, \quad q + a + c' \leq C,$$

each one where its comparison holds and the sum's unit otherwise. Then

$$P_{a', c', q} [B_\Pi, A_\Pi] P_{a, c, q} = \langle u : v \rangle P_{a', c', q} B A P_{a, c, q}.$$

At $a \leq a', c \leq c'$ or $a' \leq a, c' \leq c$, one has $u = v = 1$, and this holds at $a = a'$ or $c = c'$ as well. At $a < a', c' < c$ one has $v = 1$, and at $a' < a, c < c'$ one has $u = 1$. At these two move pairs, put

$$L := \max\{a + c, a' + c'\}, \quad U := \max\{a' + c, a + c'\}.$$

Then $L < U$, with gap the lesser of the two moved contents' positive gaps. The multiplier $\langle u : v \rangle$ is occupied exactly on the cutoff interval

$$q + L \leq C < q + U,$$

at the balance partner of one for $a < a', c' < c$ and at one for $a' < a, c < c'$. For $B = M_p$ at a plaquette disjoint from R , its boundary summand is

$$\mathcal{B}_A^{(p)} = \Pi B \Pi' A \Pi - \Pi A \Pi' B \Pi = [B_\Pi, A_\Pi]^\vee,$$

so its block reads the pair $\langle v : u \rangle$ at the displayed product block. These are map identities at the window's actual pairing and restrict to its invariant subspace (Construction 8).

At a total-content projection Π in one fixed region, let (A_i) be a finite family of maps raising content by positive data e_i : an input of content c has its image at content $c + e_i$. With $a_i := \Pi A_i \Pi$ and $V_i := \Pi' A_i \Pi$, their adjoint-creation commutators read

$$[a_i^\dagger, a_j] + V_i^\dagger V_j = \Pi [A_i^\dagger, A_j] \Pi.$$

The matrix of boundary blocks $(V_i^\dagger V_j)_{ij}$ is positive semidefinite in the direct-sum pairing: at vectors (x_i) in the input window its form is

$$\sum_{i,j} \langle V_i x_i, V_j x_j \rangle = \left\langle \sum_i V_i x_i, \sum_j V_j x_j \right\rangle.$$

Every image and composition reads in a finite content window of that region holding the indicated images.

The plaquette split. For a plaquette p , let P_p be the index projection onto configurations at the unit label on every link of ∂p , and let Q_p be its complementary projection. Write M_p for multiplication by χ_p , and, at the magnetic list's closure, set

$$C_{p,\lambda} := M_{\chi_\lambda(U_{\partial p})} P_p, \quad R := \sum_p Q_p M_p Q_p.$$

Then

$$M = R + \sum_p \sum_{\lambda \in \bar{L}} (C_{p,\lambda} + C_{p,\lambda}^\dagger), \quad [E, C_{p,\lambda}] = 4\hat{c}_2(\lambda) C_{p,\lambda},$$

$$C_{p,\lambda}^\dagger C_{p,\mu} = \delta_{\lambda\mu} P_p.$$

The creation maps commute: disjoint plaquette boundaries give commuting factors and projections, and intersecting boundaries give a product at the sum's unit in either order. Their total-content compressions commute as well. At the member's per-link floor $f_* := \hat{c}_2(f_0)$ (Lemma 9) and the per-link plaquette count $\nu = 2$ at $d = 1$, $\nu = 2g$ at $g + 1 = d$ for $d \geq 2$ (Construction 7), the remainder reads

$$f_* |\langle x, Rx \rangle_G| \leq \nu d_L \langle x, Ex \rangle_G.$$

This read holds at every region and every total-content cutoff, with $\Pi R \Pi$ and E_Π at the cutoff; the magnitude is Theorem 16's read. Index the plaquette creators by $i = (p, \lambda)$ and write $e_i = 4\hat{c}_2(\lambda)$, $a_i = \Pi C_i \Pi$, $V_i = \Pi' C_i \Pi$ and $R_\Pi = \Pi R \Pi$. Their excitation commutator is the additive identity

$$[K_\Pi, a_j] + \beta [R_\Pi, a_j] + \beta \sum_i \Pi [C_i^\dagger, C_j] \Pi = \alpha e_j a_j + \beta \sum_i V_i^\dagger V_j.$$

The full commutator $[C_i^\dagger, C_j]$ reads the sum's unit at disjoint plaquette boundaries and acts on their boundary union otherwise.

The summed creation commutator. Write $m := \#\bar{L}$, $A_p := \sum_{\lambda \in \bar{L}} C_{p,\lambda}$, $A := \sum_p A_p$, $J_p := A_p A_p^\dagger$, and let N be the plaquette count. Write $p \sim q$ at distinct plaquettes whose boundaries share a link, and let b_p count such q at p . The neighbor cap η is two on the square chain and $4k$ on a box, with k the positive predecessor of ν (Definition 1); then $b_p \leq \eta$. At

$$T_{pq} := [A_p^\dagger, A_q], \quad \mathbf{L} := NmI - [A^\dagger, A],$$

the action $\mathbf{L}$ is self-adjoint in the pairing and reads

$$\mathbf{L} = \sum_p (mQ_p + J_p) - \sum_{p < q, p \sim q} (T_{pq} + T_{pq}^\dagger).$$

Each of $mQ_p - J_p$ and $m(Q_p + Q_q) \pm (T_{pq} + T_{pq}^\dagger)$ is positive semidefinite in the stated pairing. The identities

$$\begin{aligned} \mathbf{L} + m \sum_p b_p Q_p &= \sum_p (mQ_p + J_p) + \sum_{p < q, p \sim q} (m(Q_p + Q_q) - T_{pq} - T_{pq}^\dagger), \\ m \sum_p (2 + b_p) Q_p - \mathbf{L} &= \sum_p (mQ_p - J_p) + \sum_{p < q, p \sim q} (m(Q_p + Q_q) + T_{pq} + T_{pq}^\dagger) \end{aligned}$$

are positive folds. At every total-content cutoff, with $a := \Pi A \Pi$, $V := \Pi' A \Pi$ and $\mathbf{L}_\Pi := \Pi \mathbf{L} \Pi$, they give

$$\begin{aligned} [a^\dagger, a] + V^\dagger V + \mathbf{L}_\Pi &= NmI_\Pi, \\ f_* \mathbf{L}_\Pi + m\nu\eta \mathbf{E}_\Pi \succeq 0, \quad m\nu(\eta + 2) \mathbf{E}_\Pi - f_* \mathbf{L}_\Pi &\succeq 0. \end{aligned}$$

The remainder at the cutoff boundary. Let $U_R := \Pi' R \Pi$, $a_* := \nu d_L$, and let Z be the electric solve on the complementary index,

$$\mathbf{E}_{\Pi'} Z = U_R.$$

Every complementary configuration has positive content, so the solve reads its column at each content c at the pair $[1 : c]$. Then

$$f_*^2 U_R^\dagger Z \preceq a_*^2 \mathbf{E}_\Pi, \quad \begin{pmatrix} a_*^2 \mathbf{E}_\Pi & f_* U_R^\dagger V \\ f_* V^\dagger U_R & V^\dagger \mathbf{E}_{\Pi'} V \end{pmatrix} \succeq 0,$$

the block read in the direct-sum pairing. Here $U = U_R + V$, the full magnetic boundary map above. With $a_e := \sum_i e_i a_i$, the summed excitation identity is

$$[K_\Pi, a] + \beta NmI_\Pi + \beta \Pi [R, A] \Pi = \alpha a_e + \beta \mathbf{L}_\Pi + \beta U^\dagger V.$$

The full commutator $[R, A]$ is the fold of $[Q_p M_p Q_p, A_q]$ over $p = q$ and $p \sim q$.

The creator's input cutoff. At an output cutoff C and a creator $i = (p, \lambda)$ with $e_i \leq C$, let Π_i^- project onto the input contents c at $c + e_i \leq C$. This is the window at the gap between e_i and C where $e_i < C$, and the unit line where $e_i = C$. Write

$$\mathcal{C}_i := C_i \Pi_i^-, \quad K_i^- := \Pi_i^- K \Pi_i^-.$$

Then $\Pi \mathcal{C}_i = C_i \Pi_i^-$, and the rectangular action reads

$$K_\Pi \mathcal{C}_i - \mathcal{C}_i K_i^- = \alpha e_i \mathcal{C}_i - \beta \Pi [M, C_i] \Pi_i^-.$$

At the two windows' ground levels $e := \varepsilon_0$ and e_i^- , let δ_i be the gap at $e + \delta_i = e_i^-$, the vacant read at equality. It sits on its upper side or at equal members, and the two level-gap actions satisfy

$$D_\Pi \mathcal{C}_i - \mathcal{C}_i D_i^- = (\alpha e_i + \delta_i) \mathcal{C}_i - \beta \Pi[\mathbf{M}, \mathcal{C}_i] \Pi_i^-.$$

The full magnetic commutator reads at the local fold

$$[\mathbf{M}, \mathcal{C}_i] = \sum_{q: \partial q \cap \partial p \neq \emptyset} [\mathbf{M}_q, \mathcal{C}_i].$$

At $C < e_i$ the output $\Pi \mathcal{C}_i$ reads the sum's unit.

Proof. The two multiplication orders agree before projection (Proposition 5). Inserting $\Pi + \Pi' = 1$ between their factors gives

$$\mathbf{M}_\Pi A_\Pi + \Pi \mathbf{M} \Pi' A \Pi = A_\Pi \mathbf{M}_\Pi + \Pi A \Pi' \mathbf{M} \Pi.$$

The two boundary terms are $U^\dagger V_A$ and $V_{A^\dagger}^\dagger U$, since $\mathbf{M}$ is self-adjoint. The electric member preserves the cutoff, so $\Pi \mathbf{E} \Pi' = \Pi' \mathbf{E} \Pi$ reads the sum's unit and $[\mathbf{E}_\Pi, A_\Pi] = \Pi[\mathbf{E}, A] \Pi$. The weighted difference of the two commutators is the first display. Applying it to the ground uses $K_\Pi \psi = \varepsilon_0 \psi$ and gives J_A . The complementary part of $K\psi$ is $(\beta U \psi)^\vee$, the electric image inside the window. The full multiplication commutator is $[K, A] = \alpha[\mathbf{E}, A]$, so applying it to ψ gives the residual display. Inserting $\Pi + \Pi'$ into AB gives both product recurrences; the electric recurrence follows from $\mathbf{E}$ preserving both projections, its action at each content the content's scalar (Definition 10). Every product is evaluated at its finite target window, with a larger window holding the required images reading its entries at their summand injections (Proposition 5).

For the disjoint-support read, the three partial-content selections commute, each a selection of the configuration index. Multiplication by A fixes the labels on T and on the further links, and multiplication by B fixes those on R and on the further links (Definition 9; Construction 8). Thus an input at (a, c, q) reaching (a', c', q) in the order BA has intermediate content (a', c, q) , and the order AB has intermediate content (a, c', q) . Inserting the total-content projection between the two factors multiplies the first block by u and the second by v . The full products agree (Proposition 5), giving the displayed balance pair. Distinct further-link contents give blocks at the sum's unit in both orders.

For two increasing moves, each intermediate total is at or below the output total. For two decreasing moves, each is at or below the input total. At one unchanged partial content the intermediate totals are the input and output totals. Both comparisons therefore hold in these cases. At $a < a', c' < c$ the second intermediate total is at or below the input total, giving $v = 1$; at $a' < a, c < c'$ the first is, giving $u = 1$. In the first case write $a + x = a'$ and $c' + y = c$ at positive gaps x, y . The larger intermediate content is $U = a' + c$. Its gaps above the input and output contents are x and y , respectively, so its gap above their larger member L is $\min\{x, y\}$. The smaller intermediate content is at or below the input, so the two cutoff counts differ exactly where the larger intermediate content is beyond the cutoff, while the input and output are admitted: $q + L \leq C < q + U$. The second case exchanges the two intermediate contents, with the two counts exchanged and the gap again the lesser moved gap. At the right endpoint both counts read one; at the left the input and output are admitted and the larger intermediate content is beyond the cutoff. Finally, inserting $\Pi + \Pi' = I$ into $BA = AB$ gives

$$[B_\Pi, A_\Pi] + \Pi B \Pi' A \Pi - \Pi A \Pi' B \Pi = 0.$$

This is the boundary summand's stated read at the plaquette, its Hamiltonian weight the factor β in $[K_\Pi, A_\Pi]$.

At the raising family, $\Pi A_i \Pi' = 0$, since $C < c$ gives $C < c + e_i$ at the cutoff C ; the adjoint read is $\Pi' A_i^\dagger \Pi = 0$. The two products therefore read

$$\Pi A_i^\dagger A_j \Pi = a_i^\dagger a_j + V_i^\dagger V_j, \quad \Pi A_j A_i^\dagger \Pi = a_j a_i^\dagger.$$

Their difference is the displayed commutator identity. The boundary form collects by the pairing's additivity to the self-pairing of the images' sum, on its upper side or at equal members; its split reads the upper side throughout (Definition 8; Lemma 30).

At P_p , multiplication by a closure character puts its nonunit label on all four boundary links. Thus $P_p M_p P_p = 0$. Inserting $P_p + Q_p = 1$ on both sides of M_p gives its three displayed terms, with $M_p P_p = \sum_\lambda C_{p,\lambda}$ and $P_p M_p = \sum_\lambda C_{p,\lambda}^\dagger$ at the closure's involution (Construction 6; Definition 10). The created boundary's content is $4\hat{c}_2(\lambda)$, joined to the input's content at its further links, so its commutator with the electric action is the displayed multiple (Definition 8; Proposition 7). The inputs at P_p are independent of the boundary links' coefficient variables, so their product pairings factor over those links; character orthogonality gives $\langle C_{p,\lambda} x, C_{p,\mu} y \rangle = \delta_{\lambda\mu} \langle P_p x, P_p y \rangle$ (Construction 8; Proposition 8). This is the adjoint product read. At disjoint boundaries the character multiplications and index projections commute. At a shared link, the first creation puts a nonunit label there, which the second creation's P_p reads at the sum's unit. The compressed product is the full product's compression, $\Pi C_i \Pi C_j \Pi = \Pi C_i C_j \Pi$, since $\Pi C_i \Pi' = 0$ at the positive content shift, so the commutation holds at the cutoff.

The plaquette form cap gives $|\langle Q_p x, M_p Q_p x \rangle_G| \leq d_L \langle Q_p x, Q_p x \rangle_G$ (Lemma 18). The projections Q_p are diagonal on the configuration index. Their sum reads at a configuration the count of plaquettes whose boundaries meet its support, at or below ν times its occupied link count: each such plaquette contains an occupied link, and each occupied link belongs to at most ν plaquettes. The content is at or beyond f_* times that link count (Lemma 9), so

$$f_* \sum_p \langle Q_p x, Q_p x \rangle_G \leq \nu \langle x, \mathbf{E} x \rangle_G.$$

The plaquette caps sum to the remainder's read by the magnitude triangle read (Theorem 16). The projections P_p, Q_p commute with the total-content projection, all three diagonal on the index, so the displayed comparisons restrict to every cutoff by compression (Lemma 30). Substitute the plaquette split into $K_\Pi = \alpha \mathbf{E}_\Pi - \beta M_\Pi$. The compressed creators commute, their electric commutator is the content shift, and their adjoint commutator is the full commutator less $V_i^\dagger V_j$, giving the excitation display. At disjoint boundaries the adjoint's multiplication and projection commute with the creator's, each reading the other boundary's labels unchanged; at intersecting boundaries both compositions act on the union's links alone.

The character orthogonality gives $A_p^\dagger A_p = m P_p$. Its image sits in Q_p , each closure label nonunit on every boundary link, so $0 \leq J_p \preceq m Q_p$: at occupied m , the map $[1 : m] J_p$ is the orthogonal projector onto that image, its square its own at $A_p P_p = A_p$ and $A_p^\dagger A_p = m P_p$ (Lemma 30). At a vacant closure the creation and remainder maps read the sum's unit, and the displayed comparisons read equal members.

Fix $p \sim q$. A shared link gives $P_p A_q = P_q A_p = A_p^\dagger P_q = A_q^\dagger P_p = 0$. Put $X := A_p^\dagger A_q$ and $Y := A_q A_p^\dagger$. Their support reads are

$$X = P_p X P_q, \quad Y = Q_p Y Q_q.$$

The input parts and output parts are orthogonal. Each map's squared action is capped at m^2 on its input part, the two factors each capped at m in squared action by the adjoint product read. Thus for $T := X - Y = T_{pq}$, $T^\dagger T \preceq m^2 I$, the two output squares adding. Also

$$TP_p = 0, \quad P_q T = 0, \quad T = Q_q T Q_p.$$

For the first read, $YP_p = 0$ by $A_p^\dagger P_p = 0$. In XP_p , a boundary link of p outside q starts at the unit label and keeps it under A_q ; multiplication in $A_p^\dagger$ puts a nonunit label there, whose P_p projection is the sum's unit. Such a link exists for distinct square plaquettes of the chain or box (Construction 7). For the second read, $P_q Y = 0$ and a boundary link of q outside p keeps its created nonunit label under $A_p^\dagger$, giving $P_q X = 0$. It follows that $m^2 Q_p - T^\dagger T \succeq 0$. At either side $s = 1$ or its balance partner, the identity

$$m(m(Q_p + Q_q) + s(T + T^\dagger)) = (mQ_q + sT)^\dagger(mQ_q + sT) + m^2 Q_p - T^\dagger T$$

is a positive fold, giving the two stated comparisons at occupied m by positive rescaling (Lemma 30).

The diagonal terms of $[A^\dagger, A]$ are $mP_p - J_p$ and the disjoint-boundary terms read the sum's unit. The intersecting pair's two terms are $T_{pq} + T_{pq}^\dagger$, so the displayed expression for L follows at $P_p + Q_p = I$. Regrouping the neighbor counts gives the two positive folds. A square's four links each meet at most $\nu - 1$ further plaquettes; on the square chain only the preceding and following squares meet it, so $b_p \leq \eta$. With $f_* \sum_p Q_p \preceq \nu E$, the earlier support count, the folds give the two electric comparisons. Compression keeps their positive reads (Lemma 30). Summing the raising-family commutator identity over both indices gives its stated instance at a , V and L_Π .

For the remainder solve, read in one finite target window holding the input and its R-image (Proposition 5). The remainder and its adjoint read the unit line at the sum's unit, since each term has Q_p at both ends. On the occupied index let E_o be the electric action and R_o the target's remainder compression. The electric action is positive definite there (Proposition 7), and $a_* E_o \pm f_* R_o \succeq 0$ by the remainder comparison. The block

$$\begin{pmatrix} a_* E_o & f_* R_o \\ f_* R_o & a_* E_o \end{pmatrix}$$

therefore reads positive at the sum and difference of its two input vectors. At the positive electric pivot its deflation, rescaled by a_* , gives

$$a_*^2 E_o - f_*^2 R_o W \succeq 0, \quad E_o W = R_o,$$

the solve of Definition 3 at each positive content. The middle electric solve preserves the cutoff and its complement. Its pairing at the remainder image splits into the two positive parts, so restricting to the complementary part gives the displayed comparison at U_R and Z , with the unit input's read the sum's unit. The block in the statement is the $E_{\Pi'}$ -Gram of the two maps $(f_* Z, V)$ joined to its upper diagonal defect $a_*^2 E_\Pi - f_*^2 U_R^\dagger Z$: the solve reads $Z^\dagger E_{\Pi'} Z = U_R^\dagger Z$ and $Z^\dagger E_{\Pi'} V = U_R^\dagger V$. Both terms are positive semidefinite (Lemma 30).

Every creation raises content, so $\Pi' A^\dagger \Pi = \Pi A \Pi' = 0$. The plaquette split gives $U = U_R + V$, and inserting $\Pi + \Pi' = I$ in the remainder's commutator gives

$$[R_\Pi, a] + U_R^\dagger V = \Pi[R, A]\Pi.$$

Substitute this identity and $\Pi[A^\dagger, A]\Pi = NmI_\Pi - L_\Pi$ in the sum of the excitation identities. The two boundary terms join to $(U_R + V)^\dagger V = U^\dagger V$, giving the stated identity. The factors of $Q_p M_p Q_p$

and A_q act on disjoint link variables where p and q have disjoint boundaries, so their commutator reads the sum's unit there.

For the input-cutoff read, a creator's image at input content c has content $c + e_i$. The output projection therefore gives $\Pi C_i = C_i \Pi_i^-$, including the unit input at $C = e_i$, with every image beyond C at $C < e_i$. The input projection is a subprojection of the output's. Inserting these identities gives

$$K_\Pi C_i - C_i K_i^- = \Pi K C_i \Pi_i^- - \Pi C_i K \Pi_i^- = \Pi [K, C_i] \Pi_i^-.$$

The electric commutator is $e_i C_i$, giving the stated identity at the pencil's weights. Compression raises the least root (Lemma 30; Lemma 33), so $e_i^- \geq e$ and its gap δ_i has the stated side. Substitute $K_\Pi = eI + D_\Pi$ and $K_i^- = e_i^- I + D_i^-$ to obtain the level-gap identity. A plaquette multiplication on links disjoint from ∂p commutes with both the character factor of C_i and its projection P_p , giving the displayed local fold. Every composition reads in its finite target window (Proposition 5). $\square$

Lemma 21 (The state space). *The word sector is the state's own space: at a window ground ψ and carrier elements A, B ,*

$$\omega(\bar{A}B) = \langle A\psi, B\psi \rangle,$$

the pairing on the word sector the state's, and the sector is $\text{Sect}_\infty \psi$, its own image (Definition 10), cyclic by construction.

Proof. Both sides are the read $\text{Eval}(\bar{A}B\bar{\psi}\psi)$, the left at the pairing identity and the right at the involution's product read, the products' one value under commutativity joining the two (Definition 9; Proposition 5). $\square$

Lemma 22 (The level is attained at a word image). *At a window, a window ground ψ (Definition 10) and the located edge γ of the level condition over the word sector (Theorem 7; the window cut's own read at $\gamma = E_0\kappa$, Definition 11): (i) the edge is the compressed pair's least root; (ii) a word image reads the two moments at equality there, the first positive; (iii) at the window's spectral read at the edge the state's weight sits at the level, a divisor point.*

(i) *The edge is the compressed pair's least root. The compression is finite-dimensional with pair entries (Proposition 2; Theorem 17), and $(S)_\Lambda$, the compression of the level gap, is positive semidefinite, a compression keeping the read (Lemma 30), with the ground inside its kernel block; off the kernel block, on the split's upper-side columns, $(S)_\Lambda$ is positive definite (Lemma 30), and the level condition's members are the levels at or below the least root of the pair $((SD)_\Lambda : \gamma(S)_\Lambda)$ there, the located edge that root, the window cut's own edge at $\gamma = E_0\kappa$.*

(ii) *Its kernel point is a word image, and the two moments read equal members there. At the edge the site datum Z at the join $\gamma(S)_\Lambda = (SD)_\Lambda + Z$ has a kernel point y off the sum's unit (Lemma 33's kernel block at a located root), a vector of the word sector's window projection and hence $\Pi A\psi$ at a word A , whose moments read that projection, and*

$$\mu_2(A) = \gamma \mu_1(A), \quad \mu_1(A) \text{ positive},$$

the moments read at coefficient one on the multiplication probes' vectors (Theorem 9; Corollary 4). The level- γ requirement therefore holds at equality at that one word, the detector pair's edge its neutral point (Theorem 6), and the floor over the probes (Theorem 8) is met at a probe of its own. The edge sits off equal members with it.

(iii) *At the window's spectral read the weight sits at the level, a divisor point. Where the window reads $\text{spec}(H) \subset \{\varepsilon_0\} \cup [\varepsilon_0 + \gamma, \infty)$, the jump comparisons' own object (Theorem 20) and the cut's identification at every ground (Lemma 20; Lemma 19), the two moments are the weight folds over*

the located spectrum (Theorem 6), the split's congruence diagonalizing the gram and the pencil at once (Lemma 33), the level gap acting at the gaps λ_j there and its square at their squares, the weights w_j the coordinates' squares against the gram's upper-side diagonal entries, and

$$\mu_1(A) = \sum_j w_j \lambda_j, \quad \mu_2(A) = \sum_j w_j \lambda_j^2, \quad \langle \mu_2(A) : \gamma \mu_1(A) \rangle = \sum_j w_j \lambda_j \langle \lambda_j : \gamma \rangle;$$

at every occupied weight λ_j is the kernel point's read or the level γ exactly, and $\mu_1(A)$ positive puts an occupied weight at γ . The level is then a located root of the window pencil, a point of its divisor at that root's multiplicity (Lemma 33; Theorem 26(i)), the word image's weight there positive, and its state is the word sector's own, the state's space (Lemma 21).

(iv) *The probe cap.* At a probe A whose two moments read the pair $(\mu_2(A) : \gamma \mu_1(A))$ on its lower side, a located root of the compressed pair sits off the kernel point below γ ; the edge, the least such root ((i)), sits at or below $\mu_2(A)$ against $\mu_1(A)$ at every probe with positive first moment, one pair comparison per probe. At its read against E_0 this caps the global edge, the meet at or below the window cut (Definition 11). At a fixed region, a pair u and a constructed cutoff key, a probe family with positive first moments and $\mu_2(A_C) \leq E_0 u \mu_1(A_C)$ at every further cutoff gives a regional upper cap u , hence a cap on the global edge (Definition 11). Each moment is read at that cutoff's actual ground.

Proof. (i) ψ is the unit's own word image and $D\psi$ the sum's unit (Definition 9; Definition 10), so the ground sits inside the kernel block. A kernel point y of $(S)_\Lambda$ reads its self-pairing at equal members, so Sy and its Gram solve Dy read the sum's unit: S is positive semidefinite (Definition 10) and its split reads the pairing as the fold over the blocks (Lemma 30), a fold of equal members putting the coordinates off the kernel block at the sum's unit, so y sits in that block's span and its image with it. Then $(SD)_\Lambda$ reads its row at y at the sum's unit as well, the pencil symmetric with $x^T SDy = \langle Dx, Dy \rangle_G$ (Theorem 17): both compressions sit on the split's own columns off the kernel block, the upper-side columns, where $(S)_\Lambda$ is positive definite. There Lemma 33 splits the pair $((SD)_\Lambda : \gamma (S)_\Lambda)$ over located roots, count(a) the roots below a : at γ at or below the least root the roots sit at or beyond the level and the pair is positive semidefinite, the congruence's diagonal $\langle x : \varepsilon_j \rangle g_j$ reading its upper side or equal members there, the least root's own read the sharpened one (Theorem 15(1)), while beyond that root the count is at least one and Theorem 15(2) reads the count's own witness off the elimination. So the level condition's members are the levels at or below the least root and the located edge is that root. (ii) The kernel point is a vector of the word sector's window projection, the image of $A \mapsto \Pi A \psi$, a linear map (Definition 10; Theorem 8); y sits on the split's upper-side columns, where the compressed level gap is positive definite, so its first moment is positive, and the kernel read gives the second at the level's multiple. At equal members the second moment reads them, its display $\langle Dy, Dy \rangle_G$ putting Dy at the sum's unit at the positive pairing (Definition 8) and the first moment at equal members with it, against its positivity: the edge sits off equal members. (iii) Every λ_j is the kernel point's read or sits at or beyond γ , so every summand of the fold reads its upper side or equal members, while the display above reads its left side at equal members: gaps add over a fold, and a natural fold reads the sum's unit exactly where every summand does (Definition 1), so at every occupied weight λ_j is the kernel point's read or the level γ exactly. (iv) The level condition fails at γ at that one probe, so the operator inequality fails on the sector (Theorem 7) and the split's congruence reads a root of the level gap's own pencil off the kernel point below the level (Lemma 33; Lemma 20). $\square$

6 The window ground

Theorem 10 (The window multiplicity is a count, and the count is a divisor read). *On a window the ground multiplicity $\dim \ker S$ is one integer, a count at a designation's data; it is the ground root's multiplicity, one integer per cell of the crossing divisor, reading 1 at the free end and over its attached cell, and moving across a located boundary only at a ground collision, the boundary's own two-sided read (Theorem 26(ii),(iii)). At a root-coordinate segment from the free end on which the window cut admits one fixed pair κ , the ground multiplicity is one throughout (Definition 11; Lemma 20). Where the window read is 1, a symmetry of the window, an orthogonal map commuting with the pencil, maps the ground line to itself at a graded read, $g\psi = \psi$ at even and $g\psi + \psi$ the sum's unit at odd, the reads of ω blind to the grading.*

Proof. The multiplicity is a count. A window compression is finite-dimensional with pair entries (Proposition 2), and $\text{count}(a) = \text{rev}(H : aG)$ counts its eigenvalues with multiplicity below a , exact over the pairs by LDL^T (Theorem 15; the multiplicity read is Lemma 33). Theorem 16 gives a pair δ below the separation of the pencil pair's site-datum determinant's distinct located roots (Lemma 33's χ), and a designation at that width (Theorem 15) reads levels $l < h$ at a gap at most δ , the pencil pair at l positive semidefinite and $\text{count}(h) \geq 1$. The bracket is narrower than the separation, so it admits the ground root alone and

$$\dim \ker S = \text{count}(h),$$

one count at the designation's data.

The count is a divisor read. The multiplicity is the ground root's order in the pencil determinant, the split's diagonal read (Lemma 33), and Theorem 26(ii),(iii) lock it: one integer per cell of the crossing divisor, read 1 from the free end's level bracket over the attached cell (every configuration's content is positive, the unit's line the ground, Lemma 15), and one two-sided bracket read at each located boundary.

The interval read. Fix a pair $t = [a : b]$ in the segment and write $\gamma := E_0\kappa = [g : h]$ and $W := \#p d_L$ for the window's magnetic form cap (Theorem 30). The natural $J := 1 + 8Wa^2h$ satisfies $8Wt^2 < J\gamma$, the comparison cleared at $b^2g \geq 1$. Subdivide from the free end to t at $t_j = [ja : Jb]$ for $1 \leq j \leq J$, with t_0 the free end. Consecutive pencils differ by a form capped two-sidedly at

$$\rho_j := W \langle t_{j+1}^2 : t_j^2 \rangle \leq [2Wt^2 : J] < [\gamma : 4] \quad (j < J).$$

Their least roots e_j, e_{j+1} satisfy $e_j \leq e_{j+1} + \rho_j$ and $e_{j+1} \leq e_j + \rho_j$, the form comparisons read at the least roots (Lemma 30; Lemma 33). The count sandwich from those comparisons is

$$\text{count}_j(x - \rho_j) \leq \text{count}_{j+1}(x) \leq \text{count}_j(x + \rho_j).$$

At $x = e_j + [\gamma : 2]$, both outer levels sit strictly between e_j and $e_j + \gamma$, while x sits strictly between e_{j+1} and $e_{j+1} + \gamma$. The cut's spectral read (Lemma 20) makes the outer counts the ground multiplicity at j and the middle count the ground multiplicity at $j + 1$, so they agree. The free-end multiplicity is one and induction over the J steps gives that value at t . The subdivision reads the window's cap; the returned count is one at every window holding the interval read.

The character. At a window read of 1, a symmetry g has $Kg = gK$ at $GK = H$ (Definition 10), so it preserves the ground eigenspace, which is the line; an orthogonal map on a line is a scalar c at $c^2 = 1$ ($g^\dagger g = 1$), so $c = 1$ or $c = -1$ reads equal members; and $\omega(g^\dagger Ag) = \langle g\psi, Ag\psi \rangle_G = c^2 \omega(A) = \omega(A)$: the state reads its ground twice, an even count, blind to the grading. $\square$

7 The integer constants

With every pair at one value across the form's scales (Proposition 1), every constant below is literally an integer or a fusion pair: 4 (links per plaquette, the module energy); $2d$ (the neighbor count, Construction 7's direction read); the winding floor $\hat{c}_2(\omega_{c_i})$ per cell and charged direction, the charged margin's growth (Lemma 43); and the cells' entries and levels, located roots and jump distances of the decimation's own divisors, outputs rather than constants. The residue enters through one pair, $\hat{c}_2(f) = [r(r+2) : 2d_f^2]$, the minimal content, its two reads the floor's product (Lemma 9) and the clearing's count (Definition 6) at the one-value identity $r(r+2) + 1 = d_f^2$, the A -series' read with the member floors the end lists' (Lemma 9); the adjoint read against it is the invariant $[2d_f^2 : r(r+2)]$, and every further residue read of the chain is a coordinate read of the cell calculus (Lemma 25(iii)). What reads $4n, 8n, 2n, n$ in the dictionary's units (Remark 1) was that integer times the Casimir.

8 The two-plaquette object

Construction 9 (Two-plaquette fiber). Fix the theta-graph of two square plaquettes sharing one link. The *invariant fiber* is the span of the permutation-presentation states in the two plaquette variables (Construction 11), with the pairing identity of Proposition 5 through the presentation Gram (§11); the two variables are the plaquette holonomies read as link matrices: the graph's evaluation is Construction 8's over its links, a holonomy is its links' product in the boundary's order, the product's coefficient the fold over the joined indices (Definition 3), so a monomial in the holonomies' coefficients and their daggered reads evaluates at one unshared link of each holonomy at that link's own contraction sum (Proposition 11), the holonomy's outer wirings the link's own and the inner wirings folding the further links' coefficients against their daggered reads to the evaluation identity's Kronecker read link by link (Construction 8), the further links' evaluation then the unit coefficient's, one: the presentation at the two holonomies is the evaluation's own read. The carrier's fibers read in the presentation at the links' variables at the member's generating table (Construction 13): a link's label enters at its fundamental fold, its block the lowering span at the fundamentals' tops' tensor in the power at the fold's degree, an incoming end at the dual action (the dual block, Lemma 8(i)), and the direct tag's kernel list, the two-end coevaluation, the link list and the boundary coevaluation below read at the table's keys and its stated matrices, the letters the instance at the matrix units, with the symbolic tag, the interpolants, the word pairs and the star the letters' own: at Proposition 11's two tags, a vertex's tag its row-end count against the fundamental count. At the letters a link's label enters at its word pair, its block inside the slot power at those degrees (Construction 12); an undaggered factor's row end and a daggered factor's column end sit at the link's tail with the further ends at its head (Definition 8), the row and column ends exchanged there; at a vertex the row ends pair against the column ends, and the vertex reads at the symbolic tag where its row-end and column-end counts match at or below the fundamental count (Lemma 13's independent span) with every incident label at a valid interpolant, and at the direct tag otherwise, an occupied vertex at unmatched counts, a full-column join, reading its kernel list there; every vertex list sits in the ends' slot power (Construction 12). The fiber's stated list (Definition 8) is the vertex lists' product. A vertex of two ends reads its block's Gram-dual coevaluation, the display $\sum_{j,k} A_{jk} y_j \otimes y_k$ at the block's stated list and its Gram's adjugate against the determinant (Lemma 8(i)), the second factor at the exchanged actions on the one coordinate family at two ends of opposite orientations, one label, and read through the identification at two ends of one orientation, the labels dual, at a member the dual pair at every slot, cW at an

outgoing second end and its solved witness $c^\top W$ at an incoming one, each at the clearing ν per slot (Construction 13), and at the letters the star at the tie columns (Construction 12), the pairing kept at the identification's clearing, the member read at its primitive representative, the coefficients' shared count withdrawn with the clearing (Theorem 16's strip), its self-pairing the block's count at the clearing, with two labels off the pair reading the vacant list; the through pairing's projection to the fused span reads one value with it (Construction 12): at two ends of opposite orientations the through pairing is the weight's cleared partner at every slot pair, the identity's tensor at the weight's clearing to the degree, and at two ends of one orientation it is the dual pair's tensor across the matched factors at the pair's clearing to the degree, the coevaluation read through the identification. A further vertex at the symbolic tag reads its *wirings*, the contractions of the row ends against the column ends, the place permutations' span (Lemma 13), each link's factors projected to the label's block at the link Casimir's interpolant through the mixed power's occupied Casimir values (Proposition 9's insertion; Construction 2' interpolant), the label once at a separated value there, the fundamentals and θ among them, and collected at the wirings' Gram, the slot power's coordinate pairing at the cycle counts (Lemma 13; Construction 12); and a further vertex at the direct tag, or at a label off the interpolant's reach, reads the kernel list of the stacked raisings on the fused span at its unit weight (Construction 12), each member read at its coordinates in the slot power and at its primitive representative there, the coordinates' shared count withdrawn with the sign the back solve's own at the member's pivot-free coordinate (Definition 5; Definition 3's back solve; Theorem 16's strip). A fiber's coordinates are the members' positions vertex by vertex, the configuration's states the vertex members' products, and the fiber's pairing is the presentation Gram of its states, the links' cleared block scales against the vertex lists' pairings' product at the slot power's coordinate pairing (Construction 8's display; Construction 12), one at a loop. A window's gram reads its values there, block by block within the locality band (Lemma 14), and a plaquette term's entry between two states of the window list reads the evaluation at the plaquette character's product, $\text{Eval}(\bar{x} \chi_L(U_{\partial p}) y)$ (Definition 9), the character the closure members' traces' fold at the boundary word's block coefficients (Construction 6), at the fundamental pair the boundary word's trace joined to its dagger's; at a member the boundary word's factor at a boundary link is the member's block's Gram-dual coevaluation at the link's two slots (Lemma 8(i)), the dual block's at a reversed traversal, its slots the exchanged signature's, the fundamental pair's read at the letters the trace joined to its dagger, and the adjoint character's, a further loop state's, the trace against its dagger less the unit, $|\text{tr } U|^2 = \chi_\theta + 1$ (Lemma 16), the unit's coevaluation the withdrawn term: the trace against its dagger holds one undaggered and one daggered factor at every boundary link, and reads at each boundary vertex the wiring pairing, in the vertex's own signature with a head's slots exchanged, the entering link's undaggered slot with the leaving link's daggered and the leaving link's undaggered with the entering link's daggered (Construction 11' cycle words); the evaluation factors over the links (Construction 8), and a link's evaluation of the three coefficient factors, the source state's at the link's slots with the dagger reads flipped, the boundary word's pair at a boundary link, and the target's, is the Gram-dual coevaluation of the link's invariant list at those combined slots (Lemma 8(i)), the list the place permutations' wirings where the undaggered and daggered counts match at or below the fundamental count (Lemma 13) and the stacked raisings' kernel list at the unit weight beyond (Definition 5; Construction 12), the coevaluation's two members read at the link's two ends, one at the tail's slots and one at the head's on the exchanged signature, one coordinate family at both, the adjugate against the determinant weighting each member pair; so the entry is one contraction: per vertex the source member at the flipped dagger reads, the boundary wiring and the target member paired against the incident links' coevaluation members at the vertex's combined slots, the coordinate pairing on the vertex's slot power, and the pairings' product folded over the links' member pairs at the adjugate weights, one

value at the members' cleared representatives, and off the plaquette's changed-edge rows the sum's unit (Definition 9; Lemma 14).

Construction 10 (The two-plaquette data). On the invariant fiber of Construction 9, the electric operator's link weights are $(3, 1, 3)$, *forced by the graph*: each plaquette has four links, one of which is shared, so three unshared each,

$$E = 3\Delta_U + \Delta_L + 3\Delta_V,$$

Δ_U, Δ_V the plaquette Laplacians, Δ_L the shared-link Casimir (Proposition 10, §10). Energies and the coupling are read against $C_2(\text{adj})$ (Proposition 1), so $\hat{c}_2(f) = [r(r+2) : 2d_f^2]$, $\hat{c}_2(\text{adj}) = 1$, while the section's displayed operators read at the trace form itself, $C_2(\text{adj}) = d_f$ there, the weight clause below the conversion. The magnetic operator is the plaquette character χ_L per plaquette (Definition 10). Module states: $\Phi_{q_1} = \chi_{\lambda_0}(U)$, $\Phi_{q_2} = \chi_{\lambda_0}(V)$ at a closure member λ_0 , on which this graph's E reads $4C_2(\lambda_0)$, the plaquette content, the floor label's the free-end level E_0 (Proposition 7), here the weights reading $3C_2(\lambda_0) + C_2(\lambda_0)$ with the Δ_V read the sum's unit (Φ_{q_1} free of V); the adjoint loop states $\chi_{\text{adj}}(U)$, $\chi_{\text{adj}}(V)$ at $4C_2(\text{adj})$ are the adjoint walk's modules (Construction 15). A Feshbach coefficient (Construction 14) on this graph reads

$$\langle \Phi_{q_2}, Mx \rangle, \quad x \text{ the solve of Construction 14 at } M \cdots \Phi_{q_1},$$

one solve per resolvent step at the pair $(E : 4C_2(\lambda_0))$'s site datum, Q the projection off the two module states (the site datum's kernel the module space, Construction 14), the inner product the pairing identity's read (Proposition 5). The magnetic multiplications read per plaquette: $\chi_L(W)$ at a state extends its factor list at the closure members' factors, the fundamental pair's at the variable's pair (Construction 11's product at the pair's presentation, Lemma 16), so a k -step image is one state per plaquette word of length k , each at its own factor list, the solves at each word's stated list and the displayed coefficient the pairing identity's fold over the words, a presentation state sitting at one factor list with the algebra's sums reading through the folds; a k -solve coefficient pairs with $C_2(\text{adj})^k$ to a weight-free read, one value at every form point (Proposition 1).

9 The permutation presentation

Construction 11 (States). Fix a factor list $F = (v_1, \dots, v_m)$, each *factor* a variable v_i with its *dagger read*, undaggered or daggered, a variable a link's matrix at the coefficient carrier (Construction 8). To a permutation $\pi \in S_m$ associate

$$\Phi_\pi = \prod_{\text{cycles } (i_1 i_2 \cdots i_k) \text{ of } \pi} \text{tr}(M_{i_1} M_{i_2} \cdots M_{i_k}),$$

M_i reading v_i at an undaggered factor and $v_i^\dagger$ at a daggered. States are the Φ_π 's combinations at polynomial-pair coefficients in the residue (Lemma 23). The states' product is the sites' concatenation: two generators multiply at the joined factor list, the second wiring's places shifted by the first list's count, the joined wiring the two blocks, and the coefficients multiply, the trace polynomials' own product; the two block orders read one word multiset, the blocks' exchange a letter-preserving relabeling of the joined list with the joined wiring composed at it, the product's identities one value at either order (Remark 2). Equivalently $\pi(i) = j$ records that the row index of factor i is contracted against the column index of factor j . The *evaluation contraction* withdraws an undaggered factor and a daggered factor of one link wired at the product, the one's row against

the other's column, splicing the wiring across the pair, the evaluation identity's read at the wiring (Construction 8), the word index's relation (Remark 2). This is the trace-polynomial reduction of the walled Brauer category: the Schur–Weyl dual presentation, in which the group enters only as the loop value d_f of the calculus.

Construction 12 (The slot power). A factor list's *slot power* is the power $V^{\otimes m}$ at one slot per factor (Construction 11), the units acting on an undaggered factor's slot by the standard action and on a daggered factor's slot by the dual action, the balance partner of the exchanged letters' (Lemma 8(i)), the two actions read on one coordinate family, the power's monomials (Construction 1). A label enters at its *word pair*: each column of the reduced shape at a length at or below its complement's enters the top word at that many undaggered factors, and a further column the bottom word at the complement length's daggered factors (Lemma 25's words; the fundamental at one undaggered factor, θ at one of each from the count three and at two undaggered at the count two, its two one-box columns at the *tie*, a column at its complement's length). A monomial's *weight* is its undaggered letter counts against its daggered, one balance pair per letter, a *graded vector* holds one weight at every occupied monomial, its data one coordinate family per content, and the *unit weight* is the full columns' class, every letter's pair at one value, the invariants' weight (Lemma 8(iii)'s join read at the slots). A label's *block* at its word pair is the lowerings' span at its top: the top-word columns' wedge exhibits on the undaggered slots (Construction 1) tensored with the bottom-word columns' *dual wedges* on the daggered slots, the wedge at the last letters of the complement's length, the lowerings the units' at both actions (Lemma 3's span read at the slot power's coordinates), the block's list the collection from the top round by round, a round the prior round's joined members' lowerings at the simple indices in order, each joined at the collection's refusal (Lemma 3) with a vacant lowering withdrawn, and an incoming end reads its label's block at the exchanged actions, the dual block on the one coordinate family. The *fused span* at a vertex is its ends' blocks' tensors at the unit weight, Lemma 5(iii)'s pair list at the presentation's slots, the list the tensors in the arrangements' order, an earlier end's member the outer index, collected at the ends' weight lists, each list's tensors at the place of its first arrangement, and the *stacked raisings* over the span the rows per simple index in order, one row per image monomial at its first occupancy over the members' images in order, each row the members' image coefficients at the monomial, so the span with its rows is stated data and the kernel list at the back solve (Definition 3) one list; its Gram splits at the ends' weight lists: a weight is one value per undaggered and daggered content pair at an end's slots, each end's undaggered slots' power and daggered slots' power graded by content on their own (Construction 1), so two tensors at distinct weight lists hold, at some end, pieces at distinct content pairs there and pair at equal members through the concatenation's split (Lemma 5(iii)), and a Gram-dual solve over the span runs one list's group at a time, the adjugate blockwise (Definition 3). A wiring's tensor reads one letter per row end, read again at the column end it pairs, one monomial per letter assignment, the place permutation's coordinate form (Lemma 13), two wirings pairing at the cycle count. The *through pairing* of two ends at dual labels pairs each column's slots with its complement column's: an undaggered slot against a daggered one as a wiring, and at a tie column undaggered against undaggered at the *star*, the monomials at complementary letter sets on the two columns at the swap grading's side, cleared at the column length's factorial (Construction 1), the full-column line's read of the dual block at the tie (Lemma 8(ii)): the star reads each arrangement of the tie column's wedge to every arrangement of the complement column's, the factorial's count, so cleared at the factorial it identifies the tie column's block with its complement's with the pairing kept, and the through pairing is the slot power's identity at the matched slots read across the pairings; a member of the block against its dual, a map of the block, pairs the identity at its trace, and the fused span's member pairing every

map at its trace is the block's own projector, the Gram-dual coevaluation (Lemma 8(i)), so the through pairing's projection to the fused span, the span's Gram-dual solve, is the block's coevaluation there, self-paired at the block's count, the projector's own trace; columns of one length match in either order, their exchange fixing the top and exchanging with the lowerings (Lemma 3).

Construction 13 (The member generating tables). A member's *generating table* is a table space (Construction 3) over a stated key list with the pairing the coordinate fold at a stated weight, the weight one value with its transpose entry by entry, one decidable read (the pairing symmetric at the exchange), and positive definite at a stated split, every block on its upper side at the orders summing to the key count (Lemma 30), one decidable read (the pairing positive, the reads of Lemma 3 at it), the weight's cleared partner at its clearing a stated datum, the two reading the clearing's diagonal at their product (Definition 3's adjudgate identity, the dual pairing's solved witness; at the identity weight the identity at one), the simple triples stated matrices over the keys at the transpose identity $E_i^T W = W F_i$ with the two triple displays, the *closure read*, every positive root's raising a commutator word of the simple raisings along the fold descent at occupied coefficients, the words stated data: per positive fold off the simple list a simple key occupied in the fold whose withdrawn fold is a member of the positive list, the fold's word the commutator of the key's raising against the withdrawn fold's word, a simple fold's word its raising, one decidable read at the stated matrices and the stated words with their keys, each word one value with its commutator and occupied, so that a vector reading the sum's unit at every simple raising reads it at every raising (Lemma 3's adjacent stack) and the stated triples generate the member's every root operator, the *dual pair*, and the *reach list*. The dual pair c is a stated matrix over the keys with its *clearing* ν , a natural, at three decidable reads: each entry sits at two keys whose coroot lists join to the unit fold, the pair at the unit content; the *invariance*, $E_i c + c E_i^T$ at the sum's unit at every simple raising, the tensor of the doubled table at the standard action twice; and the *isometry*, the pair's transpose against the weight against the pair reading the weight's cleared partner at the clearing squared,

$$W_{\det} c^T W c = \nu^2 \bar{W} \quad \text{at} \quad W \bar{W} = W_{\det} .$$

A vector at the unit content reading the sum's unit at every simple raising is a top of the unit label, its block the line (Lemma 2; Lemma 3), so the pair is invariant under every field operator, and the pair against the weight, $c W$, intertwines the standard action with the dual action (Lemma 8(i)), the transpose identity's read at the invariance, with $c^T W$ at the clearing ν^2 its solved witness by the isometry: the pair is the identification of the dual block on the coordinate family with the dual label's block, an isometry of the pairing at the clearing. The letters' table is stated without a dual pair, the letters' block against its dual's read at the star (Construction 12). The reach list names per fundamental the table itself or two earlier fundamentals: the fundamental's *top* is the one kernel member of the stacked raisings at its content among the table's keys, or in the tensor of the two fundamentals' blocks, the kernel's count one the *uniqueness read*, one decidable read per fundamental (Definition 5; Lemma 2); a label's top is the fundamentals' tops' tensor in the index order at the label's fold, the Cartan product, a top at the summed content (Lemma 5), and its block the lowering span there (Lemma 3): every count of the calculus reads at the table's powers (Definition 5; Lemma 5), every pairing at the powers is positive definite: at a positive definite weight every leading minor is of unequal members (Lemma 30's compression clause, the leading compression definite with the datum), so the weight's cleared LDL^T is diagonal at positive pivots (Jacobi's rule), and the power's pairing, the weight's tensor power at the slots (Construction 12), is congruent through the congruence's tensor power to the pivots' products' diagonal, a coordinate fold at positive scales; and the tops fix each block's copy inside the power, the *dual coherence*: the identification $c W$ slot by slot maps a label's block into its dual label's block's span with the factors

matched to the dual fundamentals, one decidable read per label, at a fundamental the uniqueness read's own (the identification maps a top of the fundamental inside the stated product to a top of the dual fundamental inside the dual product, the count one there) and at a label the fundamentals' passed to the tensor. The A -series' table is the power's letters (Construction 1), the reach the column exhibits (Lemma 2(iii)). C_ℓ 's is the defining table with its dual pair (Construction 4) at the clearing one: the class group is the unit's with the first fundamental's, the table's own block, the box count's parity at the doubled coordinates, and the reach list the first fundamental among the keys with each further fundamental in the prior's block against the first's. B_ℓ 's and D_ℓ 's are the defining table joined to the *spinor table*: the keys the occupancy families of the unprimed keys at one bit each, 2^ℓ keys, the pairing the coordinate fold at one; a vector key acts at the *wedge*, an unprimed key i raising its bit on the side of the count of occupied keys below i and reading the sum's unit at an occupied bit, at the *contraction*, a primed key $\bar{i}$ lowering the bit i on that side and reading the unit at a vacant bit, and the B_ℓ null key on the side of the bit count's parity; the simple raisings are the quadratic reads, the difference move at $(i, i+1)$ the wedge at i composed with the contraction at $i+1$, B_ℓ 's short move the wedge at ℓ composed with the null key, D_ℓ 's last move the wedge at $\ell-1$ composed with the wedge at ℓ ; the lowerings the transposes and the coroot diagonals the bit reads, every coordinate one or its balance partner at the doubled scale, the spinor's contents; the spinor table's dual pair pairs each bit family with its complement at the side of the fold $\sum_{j \in n} (\ell - j)$ over the family's occupied keys, at the clearing one, an occupied key moved down by one and the last key joined each flipping the side (the invariance's read at the quadratic reads, the wedge and the contraction reading one side at a key pair), and a table's join reads the dual pairs each scaled at the other's clearing at the clearings' product; the reach list the vector and the spinors among the keys with each further fundamental in the prior's block against the vector's. At D_ℓ the table's two parity summands are the two spinors, and a spinor-class label reaches its top at a spinor's Cartan product with a vector-class label, the last coordinate's parity at the doubled scale (Construction 4). The fixed members read one table each. E_8 reads the *adjoint table* at the displayed root list: the keys the eight simple keys with the 240 root keys, the pairing the coordinate fold at the Cartan reads on the simple keys and at one on the root keys, its split a stated split of the Cartan joined to the root keys' identity (Lemma 30's block diagonal); the simple raising at α_i reads a root key β to the key $\beta + \alpha_i$ on the *cocycle* side $\varepsilon(\alpha_i, \beta)$ where the sum is a root and to the sum's unit where it sits off the list, the key $\check{\alpha}_i$ to the simple key i on the side $\varepsilon(\alpha_i, \check{\alpha}_i)$, and a simple key j to the key α_i at the balance partner of the Cartan read $\alpha_i(\alpha_j^\vee)$; the cocycle is bimultiplicative over the root lattice at the rule $\varepsilon(\alpha_i, \alpha_j)$ the balance partner of one at $i < j$ on an edge and at $i = j$, one otherwise, so $\varepsilon(\alpha, \alpha)$ reads the side of the halved square and $\varepsilon(\alpha, \beta) \varepsilon(\beta, \alpha)$ the side of the form read, the two identities the cocycle's reads at the parities of the fold coefficients; the lowerings the transposes; the dual pair the Cartan's adjugate on the simple keys with each root key against its partner at the determinant's balance partner, the clearing the determinant (the invariance at a simple raising reads, at a simple root's partner key against a simple key, the Cartan read against the pair's entry, and at a root pair the cocycle's product $\varepsilon(\alpha_i, \beta) \varepsilon(\alpha_i, \check{\beta} - \alpha_i)$ against $\varepsilon(\alpha_i, \alpha_i)$, one side throughout); and the reach list the highest root among the keys, ω_1 and ω_7 in the adjoint's square, and $\omega_2, \omega_6, \omega_3, \omega_5, \omega_4$ in the adjoint against $\omega_1, \omega_7, \omega_2, \omega_6, \omega_5$ in turn. The triple displays, the closure read, the dual reads and the reach reads are decidable reads at the one member's stated matrices, and two cocycle rules differ by a side relabeling of the root keys, a diagonal unit congruence fixing every count (Lemma 30), the within-column order's own gauge (Construction 12). E_7 reads the E_8 table at the root keys pairing one against the highest root, the 56, closed under the first seven simples (the pairing kept by them, the highest root orthogonal to those simples), with the dual pair at the pairs of keys joining to the highest root at the cocycle's balance partner: the highest root's line is fixed by the

seven simples, the sum off the list, so the bracket's coefficient on that line is an invariant form, its tensor the form's solved witness at the cocycle's identity $\varepsilon(\beta, \theta - \beta) \varepsilon(\theta - \beta, \beta)$ the balance partner of one; the reach list ω_7 among the keys, ω_1 and ω_6 in the 56's square, and $\omega_2, \omega_5, \omega_4, \omega_3$ in the 56 against $\omega_1, \omega_6, \omega_5, \omega_2$ in turn. E_6 reads the E_8 table at the root keys at the complementary pair's dominant grade, one against the last simple root and equal members against the highest root less it, together with their partners, the 27 with its dual, closed under the first six simples (the two grades kept by them), the dual pair the adjoint table's at those keys; the reach list ω_1 and ω_6 among the keys, ω_2 in ω_1 against ω_6, ω_3 in ω_1 's square, ω_5 in ω_6 's and ω_4 in ω_2 's. A *fold* reads a table at the orbit sums of the simple raisings under a stated key permutation of the simple list, each folded triple a display at the summed coroots, the orbit's members pairwise orthogonal, the keys and the dual pair kept: F_4 reads E_6 's adjoint table, the adjoint table at E_6 's displayed root list, folded at the chain's exchange, the ends' orbit and the inner pair's orbit against the two fixed simples, the reach list ω_1 and ω_4 among the keys (the adjoint's split at the member, $78 = 52 + 26$), ω_2 in ω_1 's square and ω_3 in ω_4 's; and G_2 reads D_4 's defining table folded at the three outer simples' orbit against the center's, the reach list ω_1 among the keys and ω_2 in its square.

Remark 2 (The word index, and the presentation kernel). The states are indexed gauge-free by the multisets of oriented cyclic words in the letters, a variable with its dagger read: Φ_π reads only its cycles' words, each up to rotation, so the positions of repeated factors are a gauge mode of the $(\pi, \text{positions})$ spelling, and a state's word coefficient is the class sum of its π -coefficients, one finite projection at every residue. Identities of character polynomials close on the word index outright, coefficient comparisons per word multiset before any pairing, holding at every residue; the Gram below enters only at the state reads. The word index's one further datum is the *kernel*: the map from word monomials to functions has one, the evaluation identity $\sum_j U_{ij} \bar{U}_{kj} = \delta_{ik}$ (Construction 8) imposing polynomial-pair relations among the Φ_π within a factor list. All function-level linear algebra is therefore performed through the pairing's Gram (§11); the null space of that Gram is the presentation kernel, computed once per factor list. Determinant (ε -tensor) relations are invisible at the symbolic residue; this is precisely the statement that the calculus computes the generic theory, and it is why $r = 1$ is its own window read while $r \geq 2$ reads generically.

10 The electric operators as group-algebra insertions

Proposition 9 (Plaquette Laplacian). *The factor list's W -factors split by their dagger read (Construction 11), $W \in \{U, V\}$, and $c_f = C_2(f) = [r(r+2) : 2d_f]$ is the trace form's own read, the section's displayed operators' scale. On states of Construction 11 the insertion identity reads d_f -cleared, each sum the fold over its stated family of W -factors and the single-index sum over them all:*

$$\begin{aligned}
& d_f \Delta_W \pi + \sum_{\substack{i < j \\ \text{undaggered}}} \pi + \sum_{\substack{i < j \\ \text{daggered}}} \pi + d_f \sum_{\substack{i \text{ undaggered} \\ j \text{ daggered}}} C_{rc}(i, j) \pi \\
&= d_f c_f \sum_i \pi + \sum_{\substack{i \text{ undaggered} \\ j \text{ daggered}}} \pi + d_f \sum_{\substack{i < j \\ \text{undaggered}}} \pi \circ (i j) + d_f \sum_{\substack{i < j \\ \text{daggered}}} (i j) \circ \pi, \\
& C_{rc}(i, j) \pi = \begin{cases} d_f \pi, & \pi(i) = j, \\ \pi \circ (i k) & \text{at the witness } k \text{ at } \pi(k) = j, \text{ else,} \end{cases}
\end{aligned}$$

the witness k at $\pi(k) = j$ off the permutation's own transpose data, the inverse permutation (Construction 1). The joint insertion across a word. At two factors i and j of the list and a

word P , a factor list of links with their dagger reads, the joint insertion $\sum_a (P X_a P^\dagger)^{(i)} X_a^{(j)}$ reads the generator moved by the word's adjoint coefficient at the first factor against the generator at the second, $P X_a P^\dagger = \sum_b \text{adj}(P)_{ba} X_b$ the coefficient's read at the metric dual (Definition 7): each generator inserts at its factor's row index at an undaggered factor and at its column index with the memberwise swap at a daggered one, the derivation's own read, the word and its dagger read, the reversed list at the dagger reads flipped, about the first, and the two inserted generators contract by the Fierz display, d_f -cleared, to two wirings on the joined list $(F, P, \bar{P})$: the word member, the first factor's row against the word's row with the word's column at the second factor's row's contraction, and the second factor's row against the dagger word's row with the dagger word's column at the first factor's row's; and the identity member at the count's cofactor, the list's own wiring with the word withdrawn against its dagger read at the evaluation contraction (Construction 11); the joint insertion is the word member against the identity member's cofactor read, the pair at the display's own sides, so the unit word reads the display's transposition and identity terms at the factor pair, and a link's joint insertion against itself at the unit word, folded over its factor pairs, is Δ_W 's display. Two reads close it. A letter wired to itself withdraws: the Fierz fold at one letter's row against its own column reads the sum's unit, the transposition's loop at the count against the identity's count at the cofactor, the list traceless. The arc read: at a loop's character the joint insertion of a boundary link's factors against the base link's across the boundary arc from the base's tail to the link's tail, folded over the factor pairs, reads the base link's own Laplacian display at the evaluation contraction at the link's traversal sign, the display's balance partner at a link traversed backward, the daggered factor's insertion at the memberwise swap, the two one value at the word index's relation (Remark 2): the word runs adjacent to the loop's own factors and withdraws against them factor by factor, the moved generator meeting the loop at the base, and the mixed pairs' cut of the joined cycle sits at the link's own place against the base's, the two cycles one value at the contraction.

Derivation and verification. Δ_W , the per-variable summand of Definition 10's diagonal E , is read through the coefficient carrier's derivation action (Construction 8; Definition 7's Δ at the W -factors): each generator differentiates each W -factor once by the Leibniz rule, an undaggered factor reading its insertion at the row index and a daggered at the column index with the memberwise swap, the transpose identity's side, so Δ_W folds over the factor pairs, and each distinct pair's generator fold is the Fierz display of Definition 7, d_f -cleared. A within-factor pair reads the factor's own Casimir, the count c_f per factor; a one-sided pair reads the Fierz's transposition member, right multiplication at the undaggered pairs and left at the daggered; a mixed pair reads the memberwise-swapped fold, its transposition term joining the display's left side, the contraction C_{rc} at the wiring through π , and the closed case $\pi(i) = j$ reads the loop value d_f of the calculus (Construction 11). Every case coefficient is the Fierz display's own read; the identity instances: $\Delta_U \text{tr } U = c_f \text{tr } U$; the unit $\mathbf{1}$ is Δ_U 's kernel point on the presentation $\mathbf{1} = [1 : d_f] \text{tr}(UU^\dagger)$; $\Delta_U \chi_{\text{adj}} = d_f \chi_{\text{adj}}$; and the graded squares,

$$\begin{aligned} d_f \Delta_U s + 2s &= d_f (d_f + 1) s \quad \text{at} \quad s = (\text{tr } U)^2 + \text{tr } U^2, \\ d_f \Delta_U w + (d_f + 2) w &= d_f^2 w \quad \text{at} \quad w = \langle (\text{tr } U)^2 : \text{tr } U^2 \rangle \end{aligned}$$

all cleared polynomial identities in the residue. □

Proposition 10 (Shared-link Casimir). Δ_L is the Casimir of the simultaneous right translation, Definition 7's Δ read through the joint derivation moving both variables at once, each factor of either variable differentiated once (Construction 8). Its insertion identity is Proposition 9's at the sides exchanged, the column index for an undaggered factor and the row for a daggered, all factors

of both variables participating, and a factor pair joins the display at its dagger reads: a matched pair, two undaggered or two daggered, reads its transposition term on the display's right side, and a mixed pair on its left. The identity instances: $\Delta_L \text{tr } U = c_f \text{tr } U$, $\Delta_L \chi_{\text{adj}}(U) = d_f \chi_{\text{adj}}(U)$, and the cross-variable graded reads,

$$\begin{aligned} d_f \Delta_L s + 2s &= d_f (d_f + 1) s \quad \text{at } s = \text{tr } U \text{tr } V + \text{tr}(UV), \\ d_f \Delta_L w + (d_f + 2) w &= d_f^2 w \quad \text{at } w = \langle \text{tr } U \text{tr } V : \text{tr}(UV) \rangle \end{aligned}$$

with the module energy $E\chi_{\text{adj}} = 4d_f \chi_{\text{adj}}$ at $E := 3\Delta_U + \Delta_L + 3\Delta_V$, the plaquette Laplacians at each plaquette's three unshared links and the shared-link Casimir once, the adjoint loop's content 4 at the weight. The link translation acts through the site conjugation at g_s , $U \mapsto U g_s X g_s^\dagger$ at the site unitary g_s , its dagger read $g_s^\dagger g_s = 1$ the evaluation identity's (Construction 8); on the fiber invariant under that site conjugation the conjugation is absorbed, the read the proposition takes.

Proof. The joint derivation inserts one generator at every factor of both variables, at the right translation's slot: the column index at an undaggered factor and the row index at a daggered one, the sides of Proposition 9's single-variable insertion exchanged, and the Fierz display contracts the two inserted generators pair by pair at that display, Proposition 9's collection at the exchanged sides: two factors of one dagger read contract at the transposition wiring, on the display's right side, and two of distinct dagger reads at the identity contraction C_{rc} , on its left. The instances are the display's evaluations at the stated factor lists, one word-index computation each (Remark 2); the module energy joins the plaquette Laplacian's adjoint read d_f at each of the three unshared links, $3d_f$ (Proposition 9), to the shared link's d_f , the further plaquette's read the sum's unit at a state free of its variable. A state of the invariant fiber reads one value under every site unitary at its vertex (Definition 8's invariants at the vertices), so the translation's action on it is the read at X alone. $\square$

11 The pairing in the permutation presentation

Lemma 23 (The generic lift). *A polynomial pair in the residue is a pair of ground polynomials named as Definition 1's constructed pairs are, read at every natural argument through the members' Horner reads and across representatives by the cross-multiplied identities; every entry below is one such pair. (i) A polynomial pair whose reads at infinitely many naturals are of equal members reads equal members at every argument. (ii) A matrix of polynomial pairs whose evaluation at infinitely many residues is the Gram of an independent list has its determinant pair off equal members at each such residue and as a polynomial pair, and every unit read is solved by the adjugate column against the determinant (Definition 3's minors), the identity $\text{M adj}(\text{M}) = \det(\text{M}) 1$ the solved witness's whole verification. (iii) A polynomial pair keeps one side beyond its members' largest located root, the leading-coefficient read at Theorem 16's leading-term bound: the settled side, one decidable read per pair. (iv) A claim whose per-residue reads are the sides and equal-member reads of finitely many polynomial pairs in the residue therefore holds at every residue beyond their computed radii once its symbolic reads hold, and the finitely many residues at or below the radii are each their own window read. (v) The rank read. An exponential-polynomial pair in a natural ℓ is a finite fold $\sum_{k \leq K} p_k(\ell) 2^{k\ell}$ at polynomial pairs p_k , read at every natural, two such folds' sums and products folds again at the keys' sums; at p_K off equal members it keeps one side beyond a computed natural, the top polynomial's settled side: at B the radius of p_K and c its leading margin at the radius, the leading term's magnitude there against the further terms' magnitudes' fold (Theorem 16's leading-term bound), F the tail coefficients' magnitudes' fold over the keys below K and m the tail's largest*

degree, the computed natural is the least natural at or beyond B and $4m^2$ holding the comparison $c2^\ell > F\ell^m$, its witness the larger of $4(m+1)^2$, B and the natural at or beyond $[F : c]$, the division's quotient at its successor (Definition 1), and beyond it the tail's magnitude sits below the top's, the settled side, one decidable read per pair; and (iv) holds at exponential-polynomial pairs, the polynomial instance's derivation at the rank read, the naturals at or below the computed one each their own read.

Proof. (i) The pair's cross datum against the pair of equal members, its first member against its second, is one ground polynomial reading equal members at every stated natural, so its located roots are infinitely many where its coefficients are off the sum's unit; the roots sit inside Theorem 16's bound, finitely many, so every coefficient reads the sum's unit and the pair reads equal members at every argument. (ii) At each such residue the determinant is the Gram's, off equal members at the independence (Lemma 3), so the determinant pair's members differ at infinitely many naturals and (i) lifts the read to the pair: a pair of equal members as a polynomial reads them at every argument. (iii) The leading term dominates at every point of margin at or beyond the radius (Theorem 16), so beyond the largest located root each member keeps its leading coefficient's side and the pair its leading coefficients'. (iv) Each pair keeps its settled side beyond its radius by (iii), and a finite family's radii have a largest; the residues at or below it are finitely many, each an evaluation of the pairs. (v) At a natural ℓ at or beyond the radius B the top's magnitude sits at or beyond its leading margin: at the degree d of p_K the leading term against the further terms is ℓ^d times the margin read at the powers ℓ^{i-d} , each at or below B^{i-d} (Theorem 16's magnitude carrier), so the magnitude of $p_K(\ell)$ sits at or beyond $c[\ell : B]^d$, at or beyond c . Clearing $2^{K\ell}$ the fold reads $p_K(\ell)$ joined to the tail $\sum_{k < K} p_k(\ell) [1 : 2^{(K-k)\ell}]$, each summand's magnitude at or below its coefficients' fold at ℓ^m against 2^ℓ , so the tail's magnitude sits at or below $[F\ell^m : 2^\ell]$, below c at the comparison. The witness: 2^ℓ clears every power ℓ^m from the natural $4m^2$ on, the seed 2^{4m^2} at or beyond $(4m^2)^m$ at the exponents' comparison $4m^2 \geq 2m + 2m^2$ with $m \leq 2^m$, and the step $2^{\ell m} \geq (\ell + 1)^m$ at $\ell \geq 2m$, the binomial identity per key collecting to $(\ell + 1)^m \leq \ell^m + m(\ell + 1)^{m-1}$ (Theorem 16's binomial identity) with $(\ell + 1)$ at or beyond $2m$; at the degree $m + 1$ this reads $2^\ell \geq \ell \ell^m$ from $4(m + 1)^2$ on, and at ℓ beyond $[F : c]$ the factor ℓ clears F against c , so the comparison holds at the witness. It steps: at $\ell \geq 2m$ the step reads $2^{\ell+1} \ell^m \geq 2^\ell (\ell + 1)^m$, so $c2^{\ell+1}$ against $F(\ell + 1)^m$ holds where $c2^\ell$ against $F\ell^m$ does, the start $4m^2$ at or beyond $2m$, and the least natural is the stepping read at the witness (Definition 1). $\square$

Proposition 11 (Pairing). *For states A, B on factor lists F_A, F_B , the pairing $\langle A, B \rangle = \text{Eval}(\overline{\Phi_A} \Phi_B)$ of Proposition 5 is computed in the presentation by: (i) the involution flips daggers and transposes permutations, the transpose the inverse permutation (Construction 1); (ii) per variable, with k undaggered and k daggered factors (else the pairing reads equal members), the contraction sum*

$$\sum_{\sigma, \tau \in S_k} \text{Wg}(\sigma\tau^T; d_f) \prod_r \delta_{i_r l_{\sigma(r)}} \delta_{j_r k_{\tau(r)}}, \quad \text{Wg}: \quad Gw = e_{\text{id}}, \quad G_{\alpha\beta} = d_f^{\#\text{cycles}(\alpha\beta^T)};$$

(iii) the δ -wirings close the permutation chains into loops, each loop contributing a factor d_f . The contraction sum reads the wiring's cycle words alone: the Gram's entry at two keys, each composed between one relabeling and its inverse permutation, is the two keys' own entry, the composition's cycle count the place composition's read (Construction 1), and the identity key its own image, so the adjugate column at a composed key against the one determinant reads the key's own value, the solve read across the relabeling. Every output of the computation reads at one of two tags: symbolic, one polynomial pair in the residue at its clearance radius, the factor count, read at every residue whose

fundamental count is at or beyond the radius (Lemma 13; Lemma 23); or direct, the residue's own contraction read through its Gram with its presentation kernel (Remark 2), at each residue whose count sits below the radius. A symbolic read enters a residue's comparison at a count clearing its radius alone, so a window below a radius reads directly, the kernel's relations read where they occupy its factor lists. At a B , C or D member the presentation runs at the defining factors' pair partitions with the member Gram (Lemma 27): a daggered factor reads its slots exchanged at the form's twist, its entry the fold of the dual pair's coefficient, the factor's own entry and the form's, $\sum_{a,b} c_{xa} U_{ba} b_{by}$ at the dual pair c and the form b , the defining table's two data (Construction 4; Lemma 27), and per variable the contraction sum runs over the partition pairs,

$$\sum_{P,Q} \text{Wg}(P,Q) \prod_{(u,v) \in P} c_{x_u x_v} \prod_{(u,v) \in Q} c_{y_u y_v} \prod_j w_{y_j}, \quad \text{Wg} : \mathbf{G}^{\text{ser}} \text{Wg} = 1,$$

the row slots at the dual pair's coefficients alone and the column slots at the pairing's weights w beside them (a matrix entry is a coordinate, so the pairing's weight enters at the paired member alone), $\mathbf{G}^{\text{ser}}$ the member Gram at the loops' fold with the solve the adjugate against the determinant (Lemma 27; Lemma 23(ii)); the wirings close the chains into loops as the A -series' display's do, each loop one value: the dimension $r+3$ at B and D , the twist pair's weighted product reading the identity at the dual pair against the form, and the doubled rank $2r$ at C at the loop's parity, half the loop's edge count joined to the count of dual-pair edges read at their displayed order and of form edges read against theirs, each edge the form's multiple with the form's square reading the unit's balance partner; and an odd factor count's symbolic read equal members, the even powers the span's own carrier (Lemma 27); the two tags at the rank clearance, the factor count with the D series at its successor; a fixed member's outputs read direct outright, the finite arithmetic of its displayed member data (Construction 4).

Derivation and verification. Per variable, a product of k coefficients against k involution images is a monomial coefficient of the mixed power $V^{\otimes k} \otimes (V^{\otimes k})^\vee$, and Eval reads its unit component (Construction 8): the family splits orthogonally over the blocks (Lemma 1), so that component is the (a,b) entry of the invariant projection, a spanning invariant list's adjugate solve at its Gram, the membership residual's own read (Lemma 3). At $k \leq d_f$ the invariants are the permutation span (Lemma 13), the projection is the span's Gram projection, and its entries are the displayed contraction sum at the Gram's solved column, Wg the adjugate column against the determinant with $\mathbf{G} \text{Wg} = e_{\text{id}}$ its whole verification: the Gram's entries are residue naturals reading $\sigma\tau^\top$ alone, the transpose the inverse permutation (Construction 1), the solve Lemma 23(ii)'s at the per-residue Grams of the independent permutation list, and the δ -wirings are the permutations' monomial entries; a product at unequal daggered and undaggered counts reads equal members there, its two degrees differing by less than a full column beyond the factor count. A scheme output at a fixed factor list is one polynomial pair in the residue agreeing with the per-residue read at every $k \leq d_f$, so the identities hold as cleared polynomials in the residue (Lemma 23(i)), the generic theory of Remark 2. *The direct tag's read.* At a residue at or below the radius the read runs at the residue's own carrier. A variable's cycle word reduces at the evaluation identity (Construction 8): the daggered factors withdraw against the undaggered along the cycle, the word reads the net power, and a cycle at equal counts reads the residue's own count, a scalar factor; a state's variable datum is its *winding pair*, the net powers' occupancy family at the undaggered side against the daggered side's, the read at every wiring whose cycles each sit at one variable. The power word expands over the alternants at the residue's count: the product $p_\mu a_u$ is graded and regroupes over the strictly decreasing exponents (Theorem 1's conclusion walk at the residue's variables), the *character read*

$\chi^\lambda(\mu)$ its extraction,

$$p_\mu a_u = \sum_\lambda \chi^\lambda(\mu) a_{\lambda+u},$$

one balance pair per shape of the degree at rows within the residue, so $p_\mu = \sum_\lambda \chi^\lambda(\mu) \text{ch}_\lambda$ over those shapes, the division at a_u exact, $a_u = e V$ a product of monic factors (Theorem 1's assembly). The characters are orthonormal for the evaluation (Proposition 8), so two states pair at the residue at the *character fold*: per variable one shape read at both winding sides, $\sum_\lambda \chi^\lambda(\alpha) \chi^\lambda(\beta)$ at the matched totals, the off-diagonal shape pairs withdrawing at the orthonormality, the scalar factors' count multiplied through, and an unmatched total reading equal members. The direct member read is the at-residue Gram's own row read: at a factor list whose factors sit at one variable, a combination at coefficients occupied at the residue is the presentation kernel's member exactly where its character-fold pairing reads the sum's unit against every wiring of the list: the generators span the carrier, so rows at that unit read the self-pairing's fold at it and the pairing's definiteness closes the member, while a member's rows are the null function's own (Remark 2; Construction 8); a coefficient whose second member reaches the unit at the residue sits off the read, the occupancy its own conjunct, and a wiring at a crossing cycle reads at the factor list's own Gram, Remark 2's computation. The scheme's identity instances read at the fusion reads of Proposition 8: $\langle \text{tr } U, \text{tr } U \rangle = 1$, $\langle \mathbf{1}, \mathbf{1} \rangle = 1$, $\langle \chi_{\text{adj}}, \chi_{\text{adj}} \rangle = 1$, $\langle \mathbf{1}, \chi_{\text{adj}} \rangle$ of equal members, $\text{Eval}(|\text{tr } U|^4) = 2$, and $\langle (\text{tr } U)^2, \text{tr } U^2 \rangle$ of equal members; and the closed form

$$\text{Wg}(\text{id}; S_4) = [\langle d_f^4 + 6 : 8d_f^2 \rangle : d_f^2 g_1 g_2 g_3] \quad \text{at the gaps } k^2 + g_k = d_f^2,$$

the gaps' existence the four-factor read's own validity ($k \leq d_f$ at the independent span), is reproduced by the adjugate solve. All are cleared polynomial identities in the residue. $\square$

12 Resolvent, module deflation, and the presentation kernel

Construction 14 (The Feshbach resolvent at polynomial pairs). Given a state y on factor list F :

1. *The deflated side*: for each module presentation m in F (built as $\chi_{\text{adj}}(W)$ times paired $[1:d_f]$ -contractions of the other factors), the deflation y' at the join $\langle m, m \rangle y' + \langle m, y \rangle m = \langle m, m \rangle y$, its m -pairing reading equal members, the fold over the presentations in list order, each join at the prior deflation.
2. *The support*: the union of the π -support of y' and of the presentation kernel (Remark 2), closed under the E -insertions (Proposition 9; Proposition 10), the closure after the union (the insertions map the joined support into it).
3. *The solve*: $Sx + k = y'$ at a presentation-kernel member k , the member's read the symbolic tag's with a window at or below the clearance radius reading its own kernel's members (Proposition 11; Remark 2), and S the site datum at the module level, the section's pair $(E:4d_f)$'s, one exact adjugate solve over the polynomial pairs, null-space augmented with the member the augmentation's read (Lemma 23(ii)), then
4. *The solution's deflation*: the solution's component off the module space, the join of step 1: E is diagonal in $\hat{c}_2$, so at the generic residue every non-module gap reads unequal members as a polynomial pair and the pair $(E:4d_f)$'s site datum has the module space as its kernel. The result is the Feshbach Q -resolvent applied to y .

The solve's error is the sum's unit *as a function*: the residual w 's self-pairing reads equal members as a polynomial pair.

13 The adjoint walk

The adjoint's fusion closure of the unit label on a loop window is one chain of the calculus, the *adjoint walk*, the pencil E against $M_X := \sum_i \chi_X(U_i)$ at $\chi_X = \chi_{\text{adj}}$ on the loop labels; its data, its tower and its channel divisor are the statements of this section and the next, a chain of single-label shells at unit bonds, the scalar instance of the depth chain's certificate (Lemma 57; Lemma 56). The theory's magnetic member is the magnetic list's plaquette character (Definition 10), and the walk enters the derivations at that instance alone.

Remark 3 (The adjoint walk's reads). χ_X keeps every link's label class, the adjoint at the unit class (Construction 5), so the adjoint's fusion closure of a configuration keeps every link's class; the walk's two data are Construction 15's, its block reads and structural identities the displays of Theorem 11, Theorem 24 and Theorem 25, and the scheme bridge of Theorem 27 rescales its coupling coordinate at $2d_f$.

Lemma 24 (The adjoint square's channels). *The blocks of $\theta \otimes \theta$ are, at every residue, the labels*

$$\begin{aligned} \mathbf{1}, \quad \theta, \quad 2\theta = (4, 2^a), \\ 2\omega_1 + \omega_2^* = (3, 1^b), \quad \omega_2 + 2\omega_1^* = (3, 3, 2^b), \quad \omega_2 + \omega_2^* = (2, 2, 1^e), \end{aligned}$$

named by their weights over the fundamentals and the complements, ω_j and ω_i^ (Lemma 9; Lemma 8(ii)), at the run counts a, b, e at $a + 2 = d_f$, $b + 3 = d_f$ and $e + 4 = d_f$, at multiplicities $1, c_1, 1, 1, 1, 1$, with $c_1 = 2$ at $r \geq 2$ and 1 at $r = 1$; the outer pair are duals, the complement arithmetic of Construction 5 the manifest swap of the names at $\omega_j^* = \omega_{j'}$, every index pair joining to $r + 1$, and the other rows are self-dual. The list runs at every residue with each row entering at its count m : the closed forms read the drops themselves, the dual pair's d with the factor g at $g + 1 = r$, the residue's gap from one, and the last row's the root $r = 2$, while at $r = 1$ the row list is the alternant read's, the run $4, \dots, d_f + 1$ of y occupied exactly at $r \geq 2$. The unit channel enters at count one on the unit line, the Casimir's kernel point, and the nonunit channels close in the residue:*

label	m	d	$\hat{c}_2$
θ	c_1	$r(r + 2)$	1
2θ	1	$[(r + 1)^2 r(r + 4) : 4]$	$[2r + 4 : r + 1]$
$2\omega_1 + \omega_2^*, \omega_2 + 2\omega_1^*$	1	$[gr(r + 2)(r + 3) : 4]_{g+1=r}$	2
$\omega_2 + \omega_2^*$	1	$[(r + 1)^2(r + 2)w : 4]_{w+2=r}$	$[2r : r + 1]$

with the two closure identities

$$\sum_R m_R d_R = r^2(r + 2)^2, \quad \sum_R m_R d_R \hat{c}_2(R) = 2r^2(r + 2)^2,$$

the displayed reads of the proof, the sums over all blocks with the unit channel's count one among them and its Casimir term's members equal. The module's distances close in the residue as well: against $\hat{c}_2(\theta) = 1$, the gaps of the Casimir column read 1 at the unit channel and at the dual pair, $[g : r + 1]$ at $\omega_2 + \omega_2^$ over its gap witness $g + 1 = r$, and $[r + 3 : r + 1]$ at 2θ , at the cross-multiplied witnesses $2r = (r + 1) + g$ and $2r + 4 = (r + 1) + (r + 3)$: each gap is positive at its row's residues, one symbolic read at every one of them.*

Proof. Corollary 2 at $a = b = \theta$: a permutation contributes where the tie $\nu_\sigma + \sigma(\theta + u) = c + u$ meets a shifted content of θ , u the unit set's sorted display, (1^{d_f}) at multiplicity r or its letter-pair moves

at one each (Lemma 16), so, writing $y := \theta + u + (1^{d_f})$, of value multiset $\{2, 4, 5, \dots, d_f+1, d_f+3\}$, a label contributes only where the strictly decreasing $c + u$ rearranges y or y with one entry raised and one lowered by 1, the result's entries distinct. Equality is $c = \theta$ at matched degree, the count 2 at $2 + g = r$ against the run's adjacent value pairs, g their count: the identity's read r at the unit-monomial content on the even side against one odd transposition per adjacent value pair, the pairs sitting in the run $4, \dots, d_f+1$. The modifications with distinct entries are exactly five: $2 \rightarrow 3$ with $d_f+3 \rightarrow d_f+2$ (the unit label at matched degree); $2 \rightarrow 1$ with $d_f+3 \rightarrow d_f+4$ (2θ); $4 \rightarrow 3$ with $d_f+3 \rightarrow d_f+4$ ($2\omega_1 + \omega_2^*$); $2 \rightarrow 1$ with $d_f+1 \rightarrow d_f+2$ ($\omega_2 + 2\omega_1^*$); $4 \rightarrow 3$ with $d_f+1 \rightarrow d_f+2$ ($\omega_2 + \omega_2^*$); every other raise or lowering collides with an occupied value, the adjacent swaps returning the equality case. Each modification preserves the decreasing order, so the aligning permutation is the identity and each count is one, on the even side. At $r = 2$ the run $4, \dots, d_f+1$ is the single value 4, so $\omega_2 + \omega_2^*$'s two modified entries coincide: the row enters at $r \geq 3$; the dual pair's modifications each move one value of the run $4, \dots, d_f+1$, occupied at $r \geq 2$ alone, so the pair enters at $r \geq 2$, $\omega_2 + \omega_2^*$ at $r \geq 3$ as above, while equality reads $r = 1$.

The dimensions are Corollary 1's telescopes. At $2\theta = (4, 2^a)$ the factors from either end read one pair $[g + 2 : g]$ over the gaps $1 \leq g < r$ and telescope to $[(r + 1)r : 2]$ each, the pair $(1, d_f)$ reads $[r + 4 : r]$, and the middle pairs read 1: the cofactor at $4d_{2\theta} = (r + 1)^2 r (r + 4)$. At $2\omega_1 + \omega_{r-1} = (3, 1^b)$, at the gap g at $g + 1 = r$, the five collected reads are $[rg : 2]$, $[r + 2 : g]$, $[r + 3 : r]$, g and $[r : 2]$: the cofactor at $4d = gr(r + 2)(r + 3)$, the pair's second count equal, the dual's contents the balance partners (Lemma 8). At $\omega_2 + \omega_{r-1} = (2, 2, 1^e)$, at the gap w at $w + 2 = r$, the six collected reads are $[w + 1 : 2]$, $[(r + 1)(r + 2) : r(w + 1)]$, w , $[r(r + 1) : w(w + 1)]$, w and $[w + 1 : 2]$: the cofactor at $4d = (r + 1)^2(r + 2)w$.

The Casimirs are Lemma 7's bracket at the reduced shapes: $\hat{c}_2(2\theta) = [2r + 4 : r + 1]$, the $m = 2$ read of the tower law (Proposition 12); at $2\omega_1 + \omega_{r-1}$, $k = d_f$, $q = r + 7$, and the top row's weight $3\langle r + 2 : 2 \rangle$ with the run's collecting reads $\langle \lambda, 2\rho \rangle = \langle 4r : 2 \rangle$, so the cleared join $d_f + Q + 2 = (r + 7) + 4r$ reads $Q = 4d_f$: $\hat{c}_2 = 2$ at both members of the dual pair; at $\omega_2 + \omega_{r-1}$, $k = d_f$, $q = r + 5$, and the top rows' weights $2\langle r + 2 : 2 \rangle + 2\langle r + 2 : 4 \rangle$ with the run's summing to equal members read $\langle \lambda, 2\rho \rangle = \langle 4r : 4 \rangle$, so the cleared join $d_f + Q + 4 = (r + 5) + 4r$ reads $Q = 4r$: $\hat{c}_2(\omega_2 + \omega_{r-1}) = [2r : r + 1]$.

The closure identities. At $r \geq 2$, with $v := r(r + 2)$ and $x := d_f^2$ at the join $v + 1 = x$ (Lemma 16): the self-dual outer rows' counts Σ_o and the dual pair's Σ_d read the cross-added identities $2\Sigma_o + 3x = x^2$ and $2\Sigma_d + 5x = x^2 + 4$, so, with the unit's one and the θ row's $2v$,

$$\sum_R m_R d_R + 2x = x^2 + 1;$$

and the self-dual outer rows' weighted reads collect at their common factor to xv , the weights' tie $(r + 4) + g = 2d_f$ at the gap g at $g + 2 = r$ closing it, with the dual pair's weighted read twice its count, so

$$\sum_R m_R d_R \hat{c}_2(R) + 4x = 2x^2 + 2,$$

the displayed sums v^2 and $2v^2$, the statement's two closes. At $r = 1$ the surviving rows read $1 + 3 + 5 = 9 = r^2(r + 2)^2$ and $3 \cdot 1 + 5 \cdot 3 = 18 = 2r^2(r + 2)^2$ directly. $\square$

Lemma 25 (The window data are residue-stable on banded families). *A banded family is a pair of weight words at the fundamentals' two ends: the top word P , a finite occupancy family at $\omega_1, \omega_2, \dots$ indexed from the first fundamental, and the bottom word Q , one at the complement fundamentals $\omega_1^*, \omega_2^*, \dots$ indexed from the last (Lemma 8(ii): ω_i^* 's column at the gap ℓ at $i + \ell = d_f$), with reaches*

a and b , the words' last keys; the member at the residue r is

$$\lambda = \sum_{j \leq a} P_j \omega_j + \sum_{i \leq b} Q_i \omega_i^*,$$

one label per cleared residue, the reach clearance the residue at or beyond $a + b + 2$, at least two keys off the words between the reaches, and the member's rows are the weight family's tail sums, $\lambda_i = \sum_{j \geq i} c_j$ at the coordinates c , each row one fold over the member keys at or beyond it. The vacuum is the unit pair, a member at every residue outright, the magnetic list's members are one box on one word each (P at ω_1 the fundamental, Q at ω_1^* its dual, Construction 6), θ is one box on each word, and the families are the vacuum's fusion closure's labels: the fusion-closed interior at any cutoff and depth is one word-pair list at every cleared residue.

(i) The Casimir and the dimension are single polynomial-pair reads. The words read the rows: the height m , the bottom word's coordinate total, is the row across the run between the reaches, a top row adds its tail sum, $\lambda_i = m + p_i$ at $p_i = \sum_{j \geq i} P_j$ for $i \leq a$, and the rows below the run are the bottom word's prefix sums, $\sigma_t = \sum_{i \leq t} Q_i$ the row at the join $\ell + t = d_f$, the run count u at $a + b + u = r + 1$. $\hat{c}_2(\lambda)$ is one polynomial pair in the residue per family (Lemma 23), its second member a product of integer shifts of the residue and its powers. The dimension is one read as well, d_λ one polynomial pair in the residue per family: in Corollary 1's product the run-run factors are 1, a word-against-run factor telescopes over the position gaps g , at the joins $i + h = a$ and $i + b + g_2 = d_f$, to the binomial pair

$$\prod_{h < g \leq g_2} [p_i + g : g] = \left[\binom{p_i + g_2}{p_i} : \binom{p_i + h}{p_i} \right],$$

a factor within one word is residue-free, and a cross factor, the top row i against the bottom row σ_t , is the pair $[p_i + \tau_{t+1} + g : g]$ at the position gap's join $i + t + g = d_f$ and the bottom tail sum $\tau_t = \sum_{i \geq t} Q_i$, one factor per key pair with both members linear in the residue.

(ii) The fusion rows are word reads, one row at every residue. The list's rows are the one-box rows (Lemma 11): the fundamental's row at a word pair λ reads one at every dominant $\lambda + e_i$, a box added at a top row $i \leq a + 1$ (the top word moved, P_i raised with P_{i-1} lowered at $e_i = \omega_i - \omega_{i-1}$, the reach grown by at most one) or at a bottom row (the bottom word moved so, at the complement keys); and the dual's row reads one at every dominant $\lambda + (1^{d_f}) - e_i$, a box withdrawn so; every target a word pair at each reach grown by at most one, every count one, the diagonal the sum's unit at every family, and the row a word read, one row at every residue clearing the family's reach. The adjoint's rows run on $x := \lambda + u$, the beta-set's sorted display, on Lemma 24's alignment: the θ -contents are the letter-pair moves at multiplicity one and the unit-monomial content at r (Lemma 16), the diagonal $N_{\theta\lambda}^\lambda = \# \text{supp } P + \# \text{supp } Q$, the member-key count, residue-free, and the off-diagonal targets within one key of the words at count one each, the identity alignment's tie. A depth- D fusion closure of the vacuum therefore has as members word pairs at reaches summing to at most D , one box per step, and its whole data list, the labels, multiplicities, Casimirs, dimensions and the network weights $w_k = [m_k d_k : r^2(r+2)^2]$, is one list of polynomial-pair reads at every residue clearing the reach sum by two, each family with its reach clearance there, the recoupling entries included through Proposition 11's generic computation, one polynomial pair per entry agreeing with the per-residue read at every residue beyond the factor count.

(iii) The residue is a cell coordinate. At fixed content cutoff, region and fiber, the window pencil's entries are the reads of (i)–(ii), so they are one list of polynomial-pair data in (t, τ) and the residue at every residue clearing the reach sum by two; the cutoff comparisons $\hat{c}_2 \leq \Lambda_C$ selecting the interior are eventual-sign reads (Lemma 23(iii)); and clearing the second members is a positive

rescaling fixing every count (Lemma 30). The fiber pencils' entries then sit in $\mathbb{P}[t, \tau, r]$ and every count of the gap chain is Lemma 31's cell function with r a coordinate: a divisor's roots in r are located outright (Theorem 16, the leading-term bound the coarser certificate), and each root reads against the compared families' reach clearances: a comparison's residues are those clearing its families' reaches, so at every root below them the divisor keeps its leading side at every one of its residues and one symbolic count is every residue's at the cleared families; a family below its clearance reads its finitely many residues' data directly, the calculus's displays at the residue's label, one evaluation each; and Lemma 24's $r = 1, 2$ degenerations are each their own window read.

Proof. (i) $q(\lambda)$ splits into the top rows' squares, the run term um^2 at the run count, and the bottom rows' squares, each row a word read; $\langle \lambda, 2\rho \rangle = \sum_i \lambda_i \langle r + 2 : 2i \rangle$, the row weights balance pairs, splits into word reads and the run's arithmetic sum; and the degree is $k = \sum_j j P_j + \sum_i \ell_i Q_i$ at the complement columns ℓ_i , $i + \ell_i = d_f$, linear in the residue, so the degree's square against the count, $[k^2 : d_f]$, is one pair term: hence $\hat{c}_2(\lambda)$ is one polynomial pair in the residue per family. In the gap product the telescoped word-against-run factor is the top row's case at its tail sum p_i , with the bottom's its reflection, the run against σ_t at the row gap τ_{t+1} , the bottom tail sum at the join $\sigma_t + \tau_{t+1} = m$, a pair of two integer binomials polynomial in the residue each; a factor within one word reads its rows' and its positions' gaps at the word's own data; and the cross factors are linear pairs: d_λ is one polynomial pair in the residue per family. (ii) Interior run rows $i - 1$, i read $\lambda_{i-1} = \lambda_i = m$, so $\lambda + e_i$ reads λ_i above λ_{i-1} , off the sorted read: a box is dominant at the words' ends alone, the targets the word pairs displayed; and a box added or withdrawn moves the label, so the diagonal reads the sum's unit. (iii) The second members, integer shifts of the residue and its powers, keep one side at every residue clearing the reach sum by two, so clearing them is a positive rescaling fixing every count (Lemma 30). $\square$

Lemma 26 (The series' window data are rank-stable). *At a B_ℓ , C_ℓ or D_ℓ member (Construction 4) the magnetic list is the spinor at B_ℓ , the first fundamental at C_ℓ , and the spinor at odd rank with the vector beside it at even rank at D_ℓ (Construction 6), every member minuscule: its contents one orbit of the Weyl list at multiplicity one, the signed keys at C_ℓ and D_ℓ 's vector and the half-one lists at the spinors (the defining and spinor tables, Construction 13), every content's coroot pair at every positive root at magnitude at most one. A leading family is a finite weight word $W_1 \geq \dots \geq W_a$ at positive reads on the leading keys, its member at the rank ℓ the word's own occupancy family over those keys, and its reach clearance the rank at or beyond $a + 2$; a spinor family at B_ℓ and D_ℓ is a leading word joined to a spinor, the coordinates the word's raised by one half on the leading keys and one half on the run, the keys past the reach, the run count u at $a + u = \ell$, at D_ℓ the last key's read one half or its balance partner, the two spinors' families.*

(i) *The Casimir is a single polynomial-pair read in the rank; the dimension is one at a leading family and an exponential-polynomial pair at a spinor family. $\hat{c}_2$ is one polynomial pair per family, the spinor's $[\ell(2\ell + 1) : 8(r + 1)]$ at B_ℓ the instance (Lemma 9), and a spinor family's read sits at or beyond the spinor's, the fall identity's read (Lemma 9). The dimension is the member gap product (Corollary 1), at a leading family a product of finitely many binomial, rank-free and linear pairs in the rank, and at a spinor family the pure spinor's own dimension at the rank u times a polynomial pair,*

$$d_\lambda = 2^u R_\lambda(\ell) \quad \text{at } B_\ell, \quad d_\lambda = 2^{u-1} R_\lambda(\ell) \quad \text{at } D_\ell, \quad u + a = \ell,$$

R_λ a polynomial pair in the rank, the family's dimension an exponential-polynomial pair (Lemma 23(v)).

(ii) The rows are the minuscule rows, one row at every rank. At a list member λ_0 and a dominant λ ,

$$N_{\lambda_0\lambda}^{\lambda+\nu} = 1$$

at every content ν of λ_0 with $\lambda + \nu$ dominant, and every further target reads the sum's unit. The targets read the families: at a leading family the vector's contents $\pm e_i$ move one coordinate by one, the targets leading families at the reach grown by at most one; a spinor's contents move every coordinate by one half, the targets from a leading family spinor families at the word's keys moved by a half each and from a spinor family leading families with a run of ones, the coordinates $(W_i + [1 : 2] \pm [1 : 2])$ on the word's keys and one or the sum's unit on the run, sorted, a leading family at the reach grown by the ones' count k , its Casimir at or beyond $[k : 4]$ in the weight-free read, so at a stated content bound the targets within it sit at reaches grown by at most four times the bound. The diagonal reads the sum's unit at every family, a minuscule member without the unit content. So the fusion closure from the unit label is one family list at every rank beyond the reach clearances, its members at a stated depth the words of at most that many moves.

(iii) The rank is a cell coordinate. At fixed content cutoff, region and fiber, a series member's pencil entries are the reads of (i) and (ii) with the recoupling entries Proposition 11's at the member's tables, one list of exponential-polynomial pairs in the rank at every rank clearing the reach sums by two, the dimensions and the spinor table's loops entering at powers of two (Lemma 27; Construction 13), a leading family's data polynomial pairs outright. The head at a content bound Λ_h , the classes at every band component's content below it, holds, beyond the head rank ℓ_h , leading families of bounded reach alone: the head rank is $2\Lambda_h$, and beyond it the head's classes are leading families at the unit class, and at D_ℓ at the vector class beside it, at reaches below $a + 4\Lambda_h$, the bound's read, the head's count one value at every further rank. The head's magnetic entries are the vector's rows at C_ℓ and at D_ℓ 's even rank, polynomial pairs, and the sum's unit at the spinors' terms, whose targets sit in the tail; the decimated head gains the tail's Schur reads at the first tail shell, the spinor families joined to the head at one plaquette term, their contents polynomial pairs, their vertex counts one (the minuscule rows) and their entries exponential-polynomial pairs at (i)'s dimensions and the recouplings' Grams. The tail's dominance comparison, each tail class's content against its row's magnetic magnitude fold, runs at the scaled magnetic weight $\tilde{\beta} := \beta d_L$, the closure's dimension fold absorbed into the ray's coordinate, the map $[\alpha : \beta] \mapsto [\alpha : \tilde{\beta}]$ a bijection of the rays at every member, and the comparison at the $\alpha=1$ representative holds at every tail class from the rank $16d\tilde{\beta}$ on at the per-link plaquette count $2g$ (Construction 7), the head rank joined. So the cutoff comparisons are eventual-sign reads, the divisors' rank roots read against the clearances with the rank read (Lemma 23(v)) at the exponential-polynomial entries, one symbolic count at every cleared rank beyond the head rank and the comparison's computed natural, with the ranks at or below read directly, each a finite member (Construction 4): Lemma 25's three reads at the series' own coordinates, the A-series' the first instance, its list's data polynomial pairs outright.

Proof. (i) $q(\lambda)$ is the word's square fold, at a spinor family the raised word's joined to the run's quarters, u at $[1 : 4]$; $\langle \lambda, 2\rho \rangle = \sum_i W_i(2\rho)_i$ folds the word against the displayed root-fold key reads, each linear in the rank, at a spinor family the raised word's joined to the run's half fold $[1 : 2] \sum_{i>a} (2\rho)_i$, the arithmetic sum a quadratic pair in the rank; so $\hat{c}_2$ is one polynomial pair per family. In the member gap product at a leading family the tail factors read one, the word-against-tail factors telescope over the position gaps to binomial pairs in the rank, the word-against-word difference factors are rank-free, and the word sum, short and long factors are single linear pairs in the rank, finitely many. At a spinor family the word-against-word factors are rank-free; a word-against-run difference factor telescopes over the run to a binomial pair in the rank, and a word-against-run sum factor, $(W_i + 1 + \rho_i + \rho_j)$ against $(\rho_i + \rho_j)$ over the run keys j , telescopes

at the successor steps to a product of $W_i + 1$ linear pairs in the rank, the run's sums consecutive naturals; and the run-against-run factors read the pure spinor's own dimension at the rank u : the run's shifted reads $\rho_i + \rho_j$ and the run's raised reads at two run keys are the rank- u member's own at the keys shifted by a (the root-fold key reads $2\ell - 2i + 1$ at B_ℓ and $2\ell - 2i$ at D_ℓ , the shift withdrawing at the difference), the difference factors one and the sum factors with the short factors the spinor's gap product at the rank u , 2^u at B_u and 2^{u-1} at D_u (Construction 13's spinor table at 2^u keys, the two parity summands at D_u), the further factors' product the polynomial pair R_λ .

(ii) The alignment's member ties $\nu_w + w(\lambda + \rho) = c + \rho$ over the member's Weyl list (Corollary 2) run at the contents ν_w of λ_0 , every content at multiplicity one; at a tie the key $\lambda + \rho + \nu_w$ reads its coroot pairs at or beyond the sum's unit at every positive root, $\lambda + \rho$'s at or beyond one and the content's at magnitude at most one, and w maps it to the strictly dominant $c + \rho$, so its every coroot pair sits beyond the sum's unit (a key at a unit pair is fixed by that root's reflection, its image off the strictly dominant reads, Construction 4's regular reads), the key is strictly dominant, its image is itself at the regular reads and w is the identity: one tie per target at the even side, the count one at $c = \lambda + \nu$ dominant and the sum's unit at every further c . A leading family at the reach grown by a run of ones of count k reads its Casimir at or beyond ω_{a+k} 's (the fall identity, Lemma 9), $k(2\ell - k + 1)$ at B_ℓ against $C_2(\theta) = 2(2\ell - 1)$ and $k(2\ell - k)$ at D_ℓ against $2(2\ell - 2)$, each at or beyond $[k : 4]$ in the weight-free read at k at or below ℓ .

(iii) A spinor-class link's label reads $\hat{c}_2$ at or beyond the spinor's, $[\ell(2\ell + 1) : 8(r + 1)]$ at or beyond $[\ell : 8]$ at B_ℓ and $[\ell(2\ell - 1) : 8(r + 1)]$ at or beyond $[\ell : 8]$ at D_ℓ , a spinor-class configuration of the vacuum sector holds a cycle of such links, four or beyond, so its content clears Λ_h at ℓ beyond $2\Lambda_h$, and a leading family with a run of ones of count k reads its Casimir at or beyond $[k : 4]$ ((ii)), so at content below Λ_h the reach sits below $a + 4\Lambda_h$: the head rank is $2\Lambda_h$. At the scaled weight the stencil caps read $\tilde{\beta}$ times the term count (Lemma 18 at the closure's dimension fold), and a tail class's content per spinor-class link clears $[\ell : 8]$, so the comparison at the $\alpha=1$ representative holds at every tail class from the rank $16d\tilde{\beta}$ on at the per-link plaquette count $2g$ (Construction 7). $\square$

Lemma 27 (The member pairing span). *At a B , C or D member with its defining table (Construction 4), the invariants of the defining power $V^{\otimes 2k}$ at the rank at or beyond k , the D series at or beyond $k+1$ (the doubled-rank alternating invariant sits off the span at the tie $2\ell = 2k$), are the pair-partition span: one vector per pair partition of the $2k$ places, the partition's fold of the form's dual pairs, at the count $(2k-1)!!$.*

Proof. The invariant read. An invariant of the power is a vector at the unit content with every raising image the sum's unit: a unit-content top's block is the unit label's line, and a top at a further content sits at its own label (Lemma 2; Lemma 5), so the invariants are the unit-content summand's raising kernel, the two reads' conjunction. *The defining rows.* V 's contents are the signed keys with the unit content at B_ℓ , and its fusion rows at a block are one box up or one box down at the cleared ranks: the alignment's form read (Proposition 13's alignment form read, Corollary 2's member ties) reads the B_ℓ stay term against its correction, the unit content's one against the last letter's read at $(\lambda + \rho)(e_i^\vee) = 2\lambda_i + 2g + 1$ over the join $i + g = \ell$, one exactly at the last key off the support: equal members at every label whose support clears the last key, and the moved ties at the box targets alone; at C_ℓ and D_ℓ the signed keys sit off the root list and the unit content off the table, the rows the box moves outright. *The count.* Iterating the rows, the unit-label count of $V^{\otimes 2k}$ is the closed path fold $\langle \mathbf{1}, (U+D)^{2k} \mathbf{1} \rangle$ over the shapes at most k rows deep (the clearance; a B_ℓ stay reads at a full-depth shape, its boxes the step count's read, and the return closes beyond the walk's $2k$ steps, so the closed folds read the box moves alone), with $DU = UD + \text{id}$ (Lemma 12's displayed argument at the member's shapes) and the commutator

telescope $X^{m-1}U = U X^{m-1} + (m-1) X^{m-2}$ at $X := U+D$, so the fold obeys $m_m = (m-1) m_{m-2}$ from $m_0 = 1$: the double factorial, the matchings' count, and the unit-label count is the invariant count (Definition 5; Lemma 5 at table spaces). *Independence, and the Gram.* A pair partition's vector reads a monomial the further partitions miss at the rank clearing k , one key pair per part at distinct keys, so the $(2k-1)!!$ vectors are independent and span. The Gram is the loops' fold: two partitions' union closes into loops, each loop contracting the form's dual pairs around its cycle; at B_ℓ and D_ℓ a loop reads the dimension, $r+3$; at C_ℓ a loop reads the doubled rank at its *crossing parity*, the loop's pair count at either side with its against-order crossings along one traversal, the antisymmetric form's memberwise swap per crossing: the entry is the loops' product, $(2r)$ per loop with the parities' fold its side, one displayed traversal read per entry. $\square$

Construction 15 (The adjoint walk's data). $\chi_X = \chi_{\text{adj}}$ at $\chi_X + 1 = \chi_f \chi_{\bar{f}} = |\text{tr } U|^2$ (Lemma 16), so the adjoint walk's calculus is tensoring by the adjoint, and its data are exactly two: the fusion matrix N_{adj} and the weight-free Casimir $\hat{c}_2$. The product expands as $\chi_{\text{adj}} \chi_\lambda = \sum_\mu N_{\text{adj}, \lambda}^\mu \chi_\mu$ with unique coefficients, and the characters are orthonormal for Eval (Proposition 8), so M_X acts on the label index by N_{adj} itself, symmetric because $m\{a, b, c\} = N_{ab}^c$ is symmetric and the adjoint is self-dual, and E is diagonal there with entries $4\hat{c}_2(\lambda)$. Every anchor of the walk is a read of that pair: $c_1 = \langle \text{adj} | \chi_X | \text{adj} \rangle$ is the multiplicity of the adjoint in $\text{adj} \otimes \text{adj}$, the $m=1$ alternant read of Proposition 13, 2 at $r \geq 2$ and 1 at $r = 1$, and the member-key count of θ 's word pair, one box on each word, at the residues clearing the pair's reach (Lemma 25(ii)); $\langle q, M_X \text{ vac} \rangle$ is the multiplicity of the vacuum, 1; and the adjoint loop's content is 4 against the free-end level $E_0 = 4\hat{c}_2(f_0)$ at the floor label (Proposition 7).

The channel list is Lemma 24's, one word-pair list at every residue with each channel entering at its count (Lemma 25), the two root-drops and the $r = 1$ row list that lemma's own reads, so every quantity below is a sum over the channels at their counts.

Theorem 11 (The adjoint walk's block data). *With $q_i = \chi_X(U_i)$ at the adjoint loop's content 4: $\langle q_1, M q_1 \rangle$ has the integer coefficients $\langle q_1, M \text{ vac} \rangle = 1$ and $c_1 = \langle q_1, X q_1 \rangle$, the fusion multiplicity of Construction 15 (2 at $r \geq 2$, 1 at $r = 1$); the vacuum Q -dressing reads equal members as a polynomial pair, $M \text{ vac}$ lying in the module space Q projects off (Construction 10); the module is isolated, the channel distances the Casimir gaps of Lemma 24's list. Structural identities on the constituent data: $\sum_R m_R \dim_R = r^2(r+2)^2$ and $\sum_R m_R \dim_R \hat{c}_2(R) = 2r^2(r+2)^2$, cleared polynomial identities in the residue.*

Proof. The two coefficients are the walk's anchors, the vacuum's multiplicity one and the adjoint's own multiplicity c_1 (Construction 15). $M \text{ vac}$'s component at a plaquette is the module state itself, the adjoint character at that plaquette (Construction 10's magnetic multiplication at the vacuum), so Construction 14's first step, the deflation at the module presentation, withdraws it whole and the residual's self-pairing reads equal members as a polynomial pair. The channel distances are positive at their rows' residues (Lemma 24), the module isolated. The first identity is the block count of the square (Lemma 5(ii) over Lemma 24's list), the second the channel table's displayed second closure (Lemma 24), the uniform drift of Theorem 25 at $R = \text{adj}$, where $d_{\text{adj}} = r(r+2)$ and $\hat{c}_2(\text{adj}) = 1$ make $d_R d_{\text{adj}} (\hat{c}_2(R) + 1)$ read $2r^2(r+2)^2$. $\square$

14 The charged anchor

Proposition 12 (Charged anchor for the X -chain). *Define charge as Cartan-weight density along the highest root θ . The charged tower is the $m\theta$ -string, the Cartan powers of θ (Construction 5),*

and the residue is the pairing's own read at the highest root: the row-weight fold reads $\langle \theta, 2\rho \rangle = 2r$ against the pair-square fold at $2d_f$, the declared scalar's own value, and the tower reads

$$\hat{c}_2(m\theta) = [m(m+r) : r+1] \quad \text{at the derived residue } r,$$

an $l(l+1)$ in shape; at $r=1$ it is $[m^2+m : 2]$, the integer-spin tower's weight-free read. At a member's weight table the residue is Construction 4's one fold, $\langle \theta, 2\rho \rangle = 2r$ at the form's read $\langle \theta, \theta \rangle = 2$, and the tower reads the member's form at its two degrees, the form additive in each argument (Construction 3), $C_2(m\theta) = m^2 \langle \theta, \theta \rangle + m \langle \theta, 2\rho \rangle = 2m(m+r)$ against $C_2(\theta) = 2(r+1)$: the display at the member's derived residue. The linear coefficient r is fusion strictness: the abelian tower reads $C_2 = m^2$ and this one $m(m+r)$.

Proof. The join's constructed witness (Definition 6) reads a label at two folds, $d_f Q(\lambda) = \sum_{p<q} \langle \lambda_p : \lambda_q \rangle^2 + d_f \langle \lambda, 2\rho \rangle$, each homogeneous in the rows, degree two and degree one ($\langle \mu u : \mu v \rangle = m \langle u : v \rangle$, Definition 1's displayed reads); the tower label's rows are m times θ 's, $m\theta = (2m, m^c)$ at $c+2 = d_f$, so the tower reads off θ 's two folds,

$$d_f Q(m\theta) = m^2 \sum_{p<q} \langle \theta_p : \theta_q \rangle^2 + m d_f \langle \theta, 2\rho \rangle.$$

At the reduced shape $\theta = (2, 1^c)$ the square identity $d_f q(\theta) = k^2 + \sum_{p<q} \langle \theta_p : \theta_q \rangle^2$ (Definition 6) at $q(\theta) = r+3$ and $k = d_f$ reads the pair squares at $2d_f$; and the row weights $\langle r+2 : 2i \rangle$ sum to equal members over the run $2 \leq i \leq r$, so $\langle \theta, 2\rho \rangle = 2r$, the top row's read, the pairing's read the residue doubled. So $Q(m\theta) = 2m^2 + 2mr$, and against $Q(\theta) = 2d_f$, the display's own $m=1$ read, $\hat{c}_2(m\theta) = [m(m+r) : r+1]$; at $r=1$ the value is $[m^2+m : 2]$. $\square$

15 The X -tower matrix and the walk

Proposition 13 (The X -tower matrix). *On the orthonormal character index the multiplication by χ_θ is the symmetric matrix of fusion counts $N_{\theta\lambda}^\mu$ (Proposition 8), and its compression to the tower labels $m\theta$ is the tower matrix J , fusion counts alone. Every neighbor entry is one,*

$$N_{\theta, (m+1)\theta}^{m\theta} = m\{\theta, (m+1)\theta, m\theta\} = N_{\theta, m\theta}^{(m+1)\theta} = 1,$$

upward the Cartan equality, $\theta + m\theta = (m+1)\theta$ as reduced shapes (Construction 5), downward the m -symmetry at the self-dual tower labels (Proposition 8; $(m\theta)^* = m\theta$ by the complement arithmetic of Construction 5); and the diagonal is the tower's own fusion multiplicity $N_{\theta, m\theta}^{m\theta}$, one alternant read at every m (Corollary 2 at $a = \theta$, $b = c = m\theta$): the vector $x := m\theta + u$, at u the unit set's sorted display, has coordinates $2m + r + 1$; the middle members $m + i$ at $2 \leq i \leq r$, the run; and 1, pairwise distinct, and at $m \geq 1$

$$N_{\theta, m\theta}^{m\theta} + g = r \quad (r \geq 2, \quad g \text{ the run's unit-gap count at } g+2=r), \quad N_{\theta, m\theta}^{m\theta} = r = 1 \quad (r=1),$$

the count 2, one value at every $m \geq 1$, and at the vacuum the count $N_{\theta \mathbf{1}}^{\mathbf{1}}$ the Kronecker delta at two labels (Proposition 8): the vacuum defect. So J is the free Jacobi matrix with a single boundary defect, neighbor entries one, the diagonal c_1 at every $m \geq 1$ and the vacuum's the Kronecker delta at two labels: integer data, c_1 the one parameter, the shape's reads the fusion interface's own, the Cartan strictness and the unit read at the self-dual θ (Construction 6). At a member's table the diagonal is the alignment's form read (Corollary 2's member ties at $a = \theta$, $b = c = m\theta$), the member's θ content list the root list with its balance partners at one each and the unit content at

the coordinate count, the interface field's instantiation (Construction 6), derived on a block at top θ . The dimension is the content list's own fold, twice the positive count with the coordinate count,

$$d_\theta = [(r+2)(r+3):2] \text{ at } B_\ell \text{ and } D_\ell, \quad r(2r+1) \text{ at } C_\ell, \quad r(r+2) \text{ at the A-series,}$$

and 14, 52, 78, 133, 248 at the fixed members, the member gap product's evaluation at θ its cross-check (Corollary 1). The diagonal joins the count z of the simples at θ 's equal-membered coroot pairs to the coordinate count, $N_{\theta, m\theta}^{m\theta} + z = \ell$ at every $m \geq 1$: the value is θ 's occupied coroot support, one at every member off the A-series, the displayed tables' single keys, and the A-series' the run's read above, two at $r \geq 2$ and one at $r = 1$; the vacuum's ties read every simple at the pair one, the count the Kronecker delta at two labels, the defect's read at every member. With strictly positive off-tower weight: the X-tower leaks, and the family operator is the tower-compressed Jacobi pencil plus off-tower coupling, the dimension weights riding the off-tower channels alone.

Proof. The diagonal at the letters. A permutation contributes only where $\sigma x = x$, the adjoint's unit-monomial content, or where the tie $\nu + \sigma x = x$ holds at a root ν , the adjoint's occupied contents (Lemma 16). At $\sigma = \text{id}$ the read is r , the adjoint's multiplicity at the unit-monomial content, on the even side; a root tie forces σ to fix every coordinate off some $\{i, j\}$ (the coordinates are distinct), so $\sigma = (ij)$ at the two coordinates' gap one, an odd transposition at the root's multiplicity one. At $m \geq 1$ the unit gaps sit in the run alone, $x_{k+1} + 1 = x_k$ at $2 \leq k < r$, a run occupied at $r \geq 2$, the two boundary gaps reading $m+1 \geq 2$, the displayed count; at the vacuum the boundary gaps close to 1 and the two further odd transpositions even the two sides' counts. *The content list.* A content's dominant image is a dominant root or the unit (the descent, Lemma 9, with the content symmetry, Lemma 6), the top's line reads one (Lemma 3), the short dominant root's read is the trace recursion's own fold there (Lemma 7), one display per two-length class,

$$(2\ell-2)m = 2(\ell-1) \text{ at } B_\ell, \quad 2m = 2 \text{ at } C_\ell, \quad 6m = 6 \text{ at } F_4, \quad 4m = 4 \text{ at } G_2,$$

each right side the fold of the form reads at the contents above the short dominant root, and the unit content's read is the recursion's unit fold, $m(r+1)$ the halved lengths' fold over the root list, at the member join $\sum_\alpha \langle \alpha, \alpha \rangle = \ell(r+1)$ over the positive list, one displayed fold per member ($2\ell^2 - \ell$ at B_ℓ , $\ell^2 + \ell$ at C_ℓ , $2\ell^2 - 2\ell$ at D_ℓ , and 8, 36, 72, 126, 240 at the fixed members' length folds): the unit content at the coordinate count. *The diagonal at a member.* A tie $\nu_w + w\kappa = \kappa$ at $\kappa := m\theta + \rho$ keeps the form's square, so $2\langle \kappa, \nu_w \rangle$ reads $q_0(\nu_w)$. At ν_w the unit content the element is the identity, κ dominant at every simple pair beyond the unit with its images one per element (the regular reads, Construction 4), the read the coordinate count on the even side. At ν_w a root read, a partner's case reads the doubled pairing on its lower side against the square, so ν_w is a positive β at $\kappa(\beta^\vee) = 1$, the element the root's reflection alone, $w\kappa = s_\beta \kappa$ at the regular reads, odd at the transport with the fold descent reading β a letter's image; and $\kappa(\beta^\vee) = m\theta(\beta^\vee) + \rho(\beta^\vee)$ reads one exactly at θ 's coroot pair of equal members with β simple, the root fold's pair two at the simples and beyond two at every further positive member (the fixed members' displayed reads; the series' one fold per family), each such β one odd read at the root content's one: the diagonal joins the count z to the coordinate count. $\square$

16 The tower channel: chord map and divisor

Theorem 12 (The chord map and the channel's base). *Pass to the dual of the walk: the spectral side, in the deck coordinate z , at the walk's level the balance pair $y = \langle y_+ : y_- \rangle$, its members*

cross-added onto the two sides (Theorem 15's levels), the deck relation

$$z^2 + y_+ z + 1 = (c_1 + y_-) z,$$

one display at every representative (a summand on both members fixing the value, Definition 1), degree two in z with the deck involution ι exchanging its two roots, their product the constant coefficient one and the level read jointly by both, Definition 4's relation at the chord w joining the level's members, $w + y_+ = c_1 + y_-$. The relation's two roots are equal exactly at the crossings, ι 's fixed points, a double root squaring to the constant coefficient one at the display

$$(c_1 + y_-)^2 + y_+^2 = 2 y_+ (c_1 + y_-) + 4,$$

the double root's read cross-added onto the two sides, one display at every representative (Definition 1); the chord is a member of the carrier at the lower crossing alone, and the display holds exactly at the two crossing joins $y_+ + 2 = c_1 + y_-$ and $y_+ = c_1 + y_- + 2$; the band is the segment between the two crossing levels; and the defect's root is z_* at $c_1 z_* = 1$, where the relation reads the bound-state level at the join $c_1 y_+ + 1 = c_1 y_-$.

The chord and the defect read distinct integers. The relation's two roots are the half-line walk's neighbors, the coordination count that fixes the band's width 4, the level read jointly by the two roots. The base of the channel's value is c_1 at $c_1 z_* = 1$, the mass point's own witness. The two integers agree exactly at base two, the member-key count's two-key read (Proposition 13), and separate at base one, where the chord has two terms.

Proof. At the chord w the relation is $z^2 + 1 = w z$ (Definition 4), so its two roots are a root pair at product one, exchanged by ι , and they are equal exactly where the relation has a double root z at $2z = w$ and $z^2 = 1$, the chord's square 4: the display is $w^2 = 4$ cross-added at $w + y_+ = c_1 + y_-$, and $w^2 = 4$ holds exactly at $w = 2$, the lower crossing $y_+ + 2 = c_1 + y_-$, and at w the balance partner of 2, the upper crossing $y_+ = c_1 + y_- + 2$, the chord a carrier member at the lower alone. At z_* with $c_1 z_* = 1$ the relation reads $z_*^2 + y_+ z_* + 1 = 1 + y_- z_*$ at $c_1 z_* = 1$, the constant withdrawn at the sum's injectivity and the factor z_* at the product's, $z_* + y_+ = y_-$, and the c_1 -multiple is the bound-state join $c_1 y_+ + 1 = c_1 y_-$. The band's width is the chords' gap between the crossings, 2 against its balance partner, 4; and the two integers, the coordination count two and the base c_1 , agree exactly at $c_1 = 2$, where the mass point's chord $c_1 w = 1 + c_1^2$ reads $[5 : 2]$ and the walk's diagonal the member-key count two (Proposition 13), and separate at $c_1 = 1$, where the chord at the mass point reads $w = 2$, the relation's two terms $1 + 1$, and the mass point sits at the lower crossing. $\square$

Theorem 13 (The channel divisor). *The walk's boundary entry G_0 in the deck coordinate is the renewal witness $G_0 = z + c_1 z G_0$'s series, its divisor the witness's own at the mass point z_* at $c_1 z_* = 1$, and the channel's coordinate is the order pair, the determination: the mass point a simple pole, order 1, the ι -symmetrized weight at order 2, both orders independent of the base and summing to 3; the base c_1 is the orders' read, the single-key member's display at base one, the mass point at the crossing.*

The divisor. The walk's resolvent reads in the deck coordinate, the level the balance pair $y = \langle y_+ : y_- \rangle$ of Theorem 12. The defect-free tail's boundary entry $m^{(L)}$ at the tail's L -site cut J_L is a solved witness of Lemma 23(ii)'s shape, the boundary entry of the adjugate column against the determinant at the pair $(J_L + y_- : y_+)$'s site datum (Lemma 30; Theorem 15's levels, the datum at $J_L + y_- = y_+ + S_L$), and the boundary entry solves the pivot identity,

$$m^{(L+1)} (c_1 + y_-) = 1 + m^{(L+1)} (y_+ + m^{(L)}), \quad m^{(1)} (c_1 + y_-) = 1 + m^{(1)} y_+,$$

the one-site read its seed. The truncations generate one series m_1 , whose fixed-point identity is the deck relation in the boundary entry, $m_1^2 + y_+ m_1 + 1 = (c_1 + y_-) m_1$: its two roots are the root pair, at product one, and the member at the z -multiple series is the coordinate itself, the errors e_L at $m^{(L)} = z + e_L$ obey $e_{L+1} = z^2 e_L + z e_L e_{L+1}$, so $m^{(L)}$ agrees with z beyond z -order $2L$, and $m_1(z) = z$. The defect site then reads one further pivot, its diagonal the vacuum's (Proposition 13), and the boundary read solves it at the renewal witness

$$G_0 = z + c_1 z G_0,$$

the recursion's own series $\sum_k c_1^k z^{k+1}$ (Definition 2), coefficients naturals: its divisor is the witness's own, the least occupied degree one and the mass point z_* at $c_1 z_* = 1$, the clearing factor $\langle 1 : c_1 z \rangle$'s root. The deck involution exchanges z_* with its ι -image c_1 , the orbit's product one: z_* is the orbit's interior member, $z_*^2 \leq 1$ one pair comparison. The mass point's margin is the crossing read: the relation at the mass point reads the symbol $c_1 w = 1 + c_1^2$, its crossing pair $\langle w^2 : 4 \rangle = [\langle c_1^2 : 1 \rangle^2 : c_1^2]$, equal members exactly at $c_1 = 1$, and the margin $\langle 1 : z_*^2 \rangle$ is the mass weight's own pair, its square the crossing pair's z_*^2 -multiple (Lemma 39): $[9 : 4]$ at base two, the member-key count's two-key read (Proposition 13). Every channel datum is a read of it: the bound-state level at the join $c_1 y_+ + 1 = c_1 y_-$ (Theorem 12); the mass weight, the pair $[\langle c_1^2 : 1 \rangle : c_1^2]$ at the complement $[1 : c_1^2]$, the two summing to one; and the crossing levels at their joins (Theorem 12), with c_1 Proposition 13's. The lower crossing's level reads equal members exactly at $c_1 = 2$, the infrared contact forced by the integer counting the cubic couplings, and at $c_1 = 1$ the mass point sits at the crossing, $z_* = 1$, the divisor count unchanged there, as below.

What the channel contributes, and at which base. The divisor's data are two integer orders and one base. The mass point is a simple pole of G_0 , order 1. The weight reads the clearing factor ι -symmetrized, the ι -image's cleared read and the symmetrized pair

$$z \cdot (\langle 1 : c_1 z \rangle \circ \iota) = \langle z : c_1 \rangle, \quad [\langle 1 : c_1 z \rangle \langle z : c_1 \rangle : z],$$

whose two ι -image factors each read the base once, $\langle 1 : c_1 z \rangle$ at its leading coefficient and $\langle z : c_1 \rangle$ at its root c_1 , so the weight's read at the base is order 2. Both orders are independent of c_1 , and the base collects at the orders' sum,

$$c_1^1 \cdot c_1^2 = c_1^3.$$

So the channel's coordinate is the integer $3 = 1 + 2$ and its base is c_1 : the two orders are what the theory determines, the base what they are read at. The read is one divisor count at every position of the ι -image root, and $c_1 = 1$ places that root at the crossing, the one count throughout.

Proof. The boundary couples to the tail by the single entry one, and the unit-triangular congruence of Lemma 30 splits the pair $(J_{L+1} + y_- : y_+)$'s datum as the boundary pivot against $(J_L + y_- : y_+)$'s, the pivot the boundary diagonal $c_1 + y_-$ against y_+ joined to the coupling's square, one, times $m^{(L)}$, its unit witness the boundary entry, the seed the one-site diagonal, so the boundary entry solves the displayed pivot identity. $\square$

Definition 12 (The channel reads). At a base c_1 the channel's derived objects (Theorem 13) are ground reads (Definition 2): the boundary entry G_0 at its renewal witness $G_0 = z + c_1 z G_0$, as the witness's two coefficient lists; the mass point z_* at $c_1 z_* = 1$; the *pole order* of G_0 at z_* , the gap of the clearing factor's root multiplicity there against the series member's; the ι -symmetrized weight's factor list $(\langle 1 : c_1 z \rangle, \langle z : c_1 \rangle)$ over the second member z , each factor reading the base at degree one; and the *order count*, the pole order added to the factor list's count.

17 The cell certificate

Theorem 14 (Gap positivity on the interior, assembled). *At every interior end-weight ray of $\mathcal{C}$ the cut of $\mathcal{K}([\alpha : \beta])$ holds a member pair, positive by its shape: the interior is a union of located cells, the root coordinate's rays $q\tau = p$ read their cells directly, the counting sandwich of (v) hands every further ray its cut at the sandwich's stated width, and on each cell the cut is finitely many counts at a positive level of the cell's own.*

(i) *The count has an object. On a window the compression is finite-dimensional (Proposition 2), so “the number of eigenvalues below a line” denotes something and $\text{count}(a) = \text{rev}(H : aG)$ is an exact reversal count.*

(ii) *The interior is covered by located cells. At any interior coupling and window, Theorem 30's modulus puts the cut's verification on a finite compression at a computable cutoff, at every coupling, and the decimation presents that compression's counts per cell of the named divisor list (Theorem 19; Lemma 42; Lemma 43): the cells over the coupling, the momenta and the residue (Lemma 31; Lemma 25(iii); Lemma 26(iii)), read at each dimension through stencil caps and momentum sums polynomial in it, cover the interior, the free cell at the free end (Lemma 15; Lemma 64), the emitted cells of the decimation's divisor list between, and the contact cell at the other end (Lemma 45; the approach the corner cell's, Lemma 47), every boundary a located root.*

(iii) *Per cell the cut is a flat step, and the cutoff is priced. On each cell the flat step verifies the cut at a level κ , finitely many anchored jump comparisons, each two counts at located pairs (Theorem 20), and Theorem 30 prices the cutoff at the $\alpha=1$ representative: at an admissible cutoff's order $E_{\text{low}} < \Lambda_C$ with its witness gap ς at $E_{\text{low}} + \varsigma = \Lambda_C$,*

$$E_0 \varsigma \mathcal{K}_{\Lambda, \Lambda_C} \leq E_0 \varsigma \mathcal{K}_{\Lambda}^{\text{reg}}(v) + \beta^2 \#p^2 d_L^2,$$

at the theorem's admitted cutoffs and simple grounds: given a pair ϑ , taking $\vartheta \varsigma > \beta^2 \#p^2 d_L^2$ transports the window reads to the regional cut with a modulus rather than along a subsequence, at a rate the closure's dimension fold and the region's plaquette count fix (Lemma 17). At a certified cutoff C , the upper count at $e_C + \ell + E_0 \kappa$ equal to one and the comparison

$$\ell C > \ell W + \ell E_0 \kappa + W^2, \quad W = \beta \#p d_L,$$

give the cut at κ and ground multiplicity one at every larger cutoff, the cutoff-tail certificate with $h = E_0 \kappa$ and $\eta = \ell$ (Theorem 30). This record gives the regional membership (Definition 11).

(iv) *The level is positive per cell, and the whole read closes on finitely many cells. The cell's level is a divisor read: on a cell the original pencil's first two jumps stay separated, the ground jump at the sum's unit in the level gap (Definition 10), read 1 from the free end's side (Theorem 10; Theorem 19(i)). At every region and every cutoff the count is the original pencil's total count through its pivot witnesses, with the separated product read at the total-content selection (Lemma 42(iv)); the counting functions are flat across $(\varepsilon_0 + \ell, \varepsilon_0 + \ell + E_0 \kappa)$ at the window's ground ε_0 , the fixed pair $(\ell, \ell + E_0 \kappa)$ the cell's offsets, the two counts read through the ground-relative level identity (Definition 10), each count a cell function read once (Lemma 31; Theorem 20), and a pair κ named by the fixed offsets' positive gap is a located datum of the cell, a content cell's the $H([1 : \tau^2])$ level distance, the free and contact cells reading theirs outright (Lemma 15; Lemma 45; the contact approach at the corner scale's heights, Lemma 47) and each interior boundary one located two-sided read of the divisor, and*

$$\mathcal{K}(v) \geq \kappa$$

on the whole cell, the level at or beyond $E_0 \kappa$ above the ground (Definition 11), located point by located point, each cell read at one pair (Lemma 31). The level lines, the fixed offsets joined to each

window's designated ground, sit off the divisor at a cell's boundaries at the margin, their distance from the first two jumps there, each line's counting function one cell function on a located segment through the boundary (Lemma 31), two consecutive cells' segments overlap, and the cells' closed union is the segments' union, every root-coordinate point inside one segment. Each cell's count reads at the cell's own clearance: a cell's head is one family list at every rank clearing its reach sum by two and its head bound's rank (Lemma 25(ii),(iii); Lemma 26(ii),(iii)), its cutoff a read of the ray at the scaled magnetic weight (Theorem 30 at the closure's dimension fold; Lemma 26(iii)), so on the cell one symbolic verification is every cleared rank's, the rank divisors at their leading sides at every one of their ranks off the classifications' displayed merged roots, those ranks reading their tied pairs at the multiplicity (the classifications of Lemma 28 and Lemma 29 at the member's stated list; the settled sides, Lemma 23(iv),(v)), the ranks below the clearance read the cell directly, each a finite member (Construction 4), and a further list's cells read at the member outright, its member data finite (Construction 6); a ray's cell at a member is the member's own, its cutoff read at the ray's scaled weight, so the member's cell list is its rays' cutoffs', finite at every member with every ray read at one cell; and the channel-degeneration residues are each their own read. The quantifier over the dimension closes at the derivations: the fibering, the decimation, the charged grading and the flat step are derived at every d , with the stencil caps and the momentum sums polynomial in it, and the head's jump reads are counts at every window there (Theorem 10; Theorem 19(i)), so each cell's κ is a positive located datum at every dimension, the divisor's own read; a dimension's cell list is one instance of those derivations, as is a channel-degeneration residue's, residues one and two at their own channel data (Lemma 24).

(v) The reads close across each cell, at the sandwich's width. The cut's comparisons are counting-function reads, and counts obey a coupling sandwich: at couplings v, v' of a cell whose pencils' site datum S at $H(v) = H(v') + S$ is capped two-sidedly at ρG (Lemma 30) on the stated list, against G 's stored factorization, every count obeys

$$\text{count}_v\langle a : \rho \rangle \leq \text{count}_{v'}(a) \leq \text{count}_v(a + \rho)$$

(Lemma 30, counts monotone under $\leq$), so every jump bracket travels between the two couplings at width ρ . At every interior end-weight pair the cut of $K([\alpha : \beta])$ holds a positive member, the ray's own cell's floor: the root coordinate's rays read their cells directly at the cells' located floors, the corner rays at the decimated head's located gap at the ray (Lemma 47; Lemma 48), one positive pair per ray, and every further pair ray is a located coupling between them whose margin absorbs the sandwich's width. At the window pencil the two couplings' site datum is the magnetic weights' gap against the magnetic member, capped two-sidedly at that gap's multiple of the magnetic member's own cap W , Theorem 30's cap at the unit magnetic weight, so a further pair ray $[\alpha : \beta]$ reads its window at a root-coordinate ray $[q^2 : p^2]$ with $p^2\alpha < \beta q^2 \leq (p+1)^2\alpha$, p the least natural at the squared comparison, and q beyond α and at or beyond $4W(2\tau^* + 1)$ at a natural τ^* at or beyond the segments' union's top, the cell's levels cleared to naturals with W the cap at that clearing, the cell's level gap a natural at or beyond one there, and at a pair κ whose double sits at or below the neighbor's cell's level gap against E_0 , the cell holding the root-coordinate ray $[q^2 : p^2]$, the ray's cut holds $\alpha\kappa$ at its representative $[\alpha : \beta]$, one positive pair per ray.

(vi) The sector's read at every list. Clauses (i)–(v) read the vacuum sector's depth chain (Construction 21) at every magnetic list, the faithfulness of the member's stated list entering the charged sectors alone (Lemma 43; Theorem 20), so on the vacuum sector the cut of Definition 11 compressed to the sector holds a positive member at every interior ray at every list, and at a faithful list the sectors are the charges and the window's cut is the sectors' joined (Construction 21; Lemma 30's counts at a block diagonal).

Proof. (iv) Root collisions sit on the crossing divisor alone (Theorem 26(ii)), so on a cell the original pencil's first two jumps stay separated. At each window the ground-relative level identity (Definition 10) reads the fixed offsets at the absolute lines joined to that window's ground, their gap the offsets' own. The level lines sit off the divisor at a cell's boundaries, so each line's counting function is one cell function on a located segment through the boundary (Lemma 31). At each window the two total counts of Theorem 20 give its cut by Lemma 20. The reads at every region and every cutoff give the meet's membership (Definition 11).

(v) The cap: the decimated entries' derivative coefficient folds price the site datum entrywise over the pair distance, two points' values differing by at most the distance against the fold at the bound (Lemma 32's priced side read at Theorem 16's arithmetic), and at the head's order the scaled identity caps the priced datum two-sidedly where ρ clears the order's multiple of the entry price, each row's off-row magnitude fold within that multiple at the dominance comparison (Theorem 19(ii)). At κ below a cell's level, the level's margin names a pair distance whose sandwich width it absorbs, the pair reads hold across the cells' closed union by (ii)–(iv), the segments' union, and the sandwich transports the jump comparisons over (Theorem 20). The least natural p steps up at the square's order, its witness the weight βq^2 at α at or beyond one, a natural's square at or beyond itself (Definition 1), the gap of βq^2 against $p^2 \alpha$ at or below $(2p + 1) \alpha$ and p at or below $q \tau^*$ at the ray inside the segments' union, so the sandwich's width at the scale αq^2 sits at or below a quarter of the cell's level gap and the transported window keeps at least half of it. Fixing the region fixes the magnetic cap independently of the content cutoff. The neighbor ray's certificate reads at each cutoff, and the displayed sandwich gives that window's transported cut. Their meet gives the transported membership (Definition 11). $\square$

Lemma 28 (The rank divisors). *The rank divisors of the chain's comparisons (Lemma 25(iii)) close, per head bound, on the records of the unit label's fusion closure under the fundamental pair at the bound's depth (Construction 21; Lemma 25(ii)), the divisors read at $n := d_f$, the fundamental count (Construction 1): per band family the excess gap, the cleared read $\text{gap}_\lambda := d_f Q(\lambda)$, the pair $\hat{c}_2(\lambda)$ of Lemma 25(i)'s word data cleared at $2d_f^2$ (Definition 6) against the unit label's kernel point, one polynomial in n per family; the binding comparisons, the balance pairs $\langle \text{gap}_i : \text{gap}_j \rangle$ over the family pairs, each a polynomial in n whose equal-members reads are located roots (Theorem 16), each read at its rank; and the stencil cap $3d_L = 6n$ (Lemma 17). The depth-two instance closes on 29 records, the gaps each a product of naturals at a gap witness,*

family	gap_λ	witness	clearance
ω_1, ω_1^*	$g(n+1)$	$1+g=n$	4
$2\omega_1, 2\omega_1^*$	$2(n+2)g$	$1+g=n$	4
ω_2, ω_2^*	$2(n+1)g$	$2+g=n$	5
$\omega_1 + \omega_1^*$	$2n^2$		5

the seven families at the second fusion depth, the fundamental with its dual, their squares and their second fundamentals, and the adjoint, a dual pair reading one gap at every rank (Definition 6); the binding comparisons, twenty-one; and the cap, the clearance column the family's reach clearance read at n , the clearance residue's successor (Lemma 25(ii)). A comparison's ranks are those clearing its families' reaches, and the classification is the equal-members reads in the rank domain: the three dual pairs read one merged level at every rank, their tied pairs reading the merged level with the counts at the multiplicity (Lemma 31's boundary clause); and eight isolated reads, every one below a compared family's clearance, one cross-multiplied evaluation each: the second fundamentals' pair against the kernel point at $n = 2$ (the gap $2(n+1)g$ at the vacant witness, the second fundamental the full column there, the unit label, Construction 5) against the clearance 5, two reads; $2\omega_1$ and

$2\omega_1^*$ against the adjoint at $n = 2$ (the gaps $2n^2 + 2n - 4$ against $2n^2$) against the clearance 4, and the fundamental pair against the second fundamentals' pair at $n = 3$ (the gaps $n^2 - 1$ against $2n^2 - 2n - 4$, the difference $(n - 3)(n + 1)$) against the clearance 5, four comparisons; the further comparisons and the gaps keep one side at every rank, the differences $4n$, $2n + 4$, $n^2 + 1$ and $(n + 3)(n - 1)$ and the gaps $g(n + 1)$, $2(n + 2)g$ and $2n^2$ at n at or beyond two, with $2(n + 1)g$ beyond two. So every divisor keeps its leading side at every one of its ranks off the merged levels, every cell's count one read along the rank direction, and one symbolic verification is every rank's at the cleared families; a family below its clearance reads its finitely many ranks' data directly, the calculus's displays at the rank's label (Lemma 7; Corollary 1; Corollary 2), one evaluation each.

Proof. The gap is Definition 6's second display, $d_f Q(\lambda) = \sum_{p < q} \langle \lambda_p : \lambda_q \rangle^2 + d_f \langle \lambda, 2\rho \rangle$ at the row weights $\langle n + 1 : 2i \rangle$ (Lemma 7), two folds per family at the word data: ω_1 reads $(n - 1) + n(n - 1)$, the product $(n - 1)(n + 1)$; $2\omega_1$ reads $4(n - 1) + 2n(n - 1)$, the product $2(n + 2)(n - 1)$; ω_2 reads $2(n - 2) + 2n(n - 2)$, the product $2(n + 1)(n - 2)$; θ reads $2n + 2n(n - 1) = 2n^2$; and a dual pair reads one gap (Definition 6). The clearance is the reach clearance $a + b + 2$ at $n = r + 1$ (Lemma 25), four at the reaches $1 + 0$ and five at $2 + 0$ and at $1 + 1$. The records count $7 + 21 + 1 = 29$. The twenty-one binding comparisons read, at the dual pairs' one gap, as the three dual-pair ties and six differences, the second fundamentals' gap reading the sum's unit at $n = 2$ at its vacant witness and the gaps $g(n + 1)$, $2(n + 2)g$ and $2n^2$ positive at n at or beyond two with $2(n + 1)g$ beyond two: $2\omega_1$ against θ at $2n^2 + 2n - 4$ against $2n^2$, the difference $2n - 4$ at its root $n = 2$; ω_1 against ω_2 at $n^2 - 1$ against $2n^2 - 2n - 4$, the difference $(n - 3)(n + 1)$ at its root $n = 3$; and ω_1 against $2\omega_1$, ω_1 against θ , ω_2 against θ and $2\omega_1$ against ω_2 at the differences $(n + 3)(n - 1)$, $n^2 + 1$, $2n + 4$ and $4n$, each off equal members at n at or beyond two, so every further comparison keeps one side at every rank. $\square$

Lemma 29 (The series' rank divisors). *The rank divisors of the chain's comparisons at a B , C or D member (Lemma 26(iii)) close on the depth-two closures of the sectors' base members under the magnetic list (Construction 21; Lemma 43) at the head bound, the divisors read at the derived residue r (Construction 4). The sectors are the charges: the unit class's, based at the unit label, and one per nonunit class, based at the class's end member (Lemma 43(ii)): the spinor at B_ℓ , the first fundamental at C_ℓ , and the vector with the two spinors at D_ℓ ; the exchange of the last two simple letters is a relabeling of the D table, the last coordinate read to its balance partner, exchanging the two spinor classes, each sector's base with its closure, so the spinor pair's records are one displayed list.*

The families. A sector's families are the labels at its class within two list fusions of the base, the minuscule rows' targets (Lemma 26(ii)), each label a leading or a spinor family with its gap against the base one linear read at the derived residue, the form's bilinear read at the member's displayed data (Construction 3's additive form; Construction 4),

$$C_2(\beta) + \text{gap}_W = C_2(\beta + W) , \quad \text{gap}_W = \langle W, W \rangle + 2 \langle W, \beta \rangle + \langle W, 2\rho \rangle ,$$

β the sector's base and W the added word; every gap is displayed as a natural fold or a doubled gap at its witness, positive at every domain rank, every listed family strictly above its base; the depth-one targets at the further class sit at the moved string, the class read by the plaquette's parallel link (Lemma 43(iii)'s link-sharing entries), their records the unit tables' gaps at the offset $3C_2(\lambda_0)$, the three further links' floor, one display per member. The unit sector's members are the loops at every label, a loop's flux the unit at every cut (Lemma 43(i)), so its depth-two closure is the labels within two list fusions of the unit label whatever their class, and a charged sector's members are the string's labels at the sector's class. At C_ℓ the unit sector's depth-two closure is ω_1 , its square's two

members $2\omega_1 = \theta$ and ω_2 , and the first fundamental's sector's, based at ω_1 , the three-box labels; at B_ℓ the unit sector's depth-two closure is the spinor with the leading fundamentals ω_k of the spinor's square, k at or below the head bound's read (Lemma 26(iii)); at D_ℓ it is the spinor with the leading fundamentals ω_k , the even keys of the spinor against its dual and the odd keys of the spinor's square at odd rank and, at even rank, the vector ω_1 at depth one with the even keys of the spinor's square (Construction 13's spinor table at the wedge's parity), joined at even rank by the vector's square's member $2\omega_1$ and by the vector against the spinor, the further spinor s' and the spinor family $\omega_1 + s$ (Lemma 26(ii)), the table's ω_k row read at the listed keys; a spinor sector's, based at the spinor, the spinor families at one run of ones added, $e_1 + \cdots + e_j$ at j at or below the head bound's read, joined at even rank by the vector's square against the spinor, $2\omega_1 + s$; and the vector sector's at D_ℓ the vector against the unit sector's depth-two members at the vector's class, ω_{2j+1} and $\omega_1 + \omega_{2j}$ at j at or below the head bound's read (the vector's row at ω_{2j} , Lemma 26(ii)), joined at even rank by the vector's cube's members $3\omega_1$ and $\omega_1 + \omega_2$. The unit tables, the gaps against the unit label's kernel point at their doubled reads, the spinor rows pairs at four:

family	2 gap	witness	clearance
$C_\ell : \omega_1$	$2r + 1$		3
ω_2	$4r$		4
$2\omega_1 = \theta$	$4(r + 1)$		3
$B_\ell : \omega_k$	$2k(r + 3 - k)$		$k + 2$
the spinor	$[(r + 2)(r + 3) : 4]$		
$D_\ell : \omega_k$	$2k(r + 3 - k)$		$k + 2$
a spinor	$[(r + 2)(r + 3) : 4]$		
$2\omega_1$ (even rank)	$4(r + 3)$		3
$\omega_1 + s$ (even rank)	$[(r + 3)(r + 10) : 4]$		3

the leading fundamentals' rows one display at every k and the spinor's row read at both spinors, a dual pair at odd rank and the two self-dual spinors at even rank, one gap at every rank. The charged tables, the gaps against the base's Casimir, the spinor list one table for B_ℓ and D_ℓ with a word of odd box count at D_ℓ read at the flipped base, the class kept:

spinor word	gap	clearance	C_ℓ, ω_1	gap	wit.	cl.
$e_1 + \cdots + e_j$	$j(r + 4 - j)$	$j + 2$	$(1, 1, 1)$	$2g$	$g+1=r$	5
$2e_1$ (D_ℓ , even rank)	$2r + 8$	3	$(2, 1)$	$2r + 1$		4
			(3)	$2(r + 2)$		3

D_ℓ , vector	gap	cl.
(1^{2j+1})	$2j(r + 1 - 2j)$	$2j + 3$
$(2, 1^{2j-1})$	$2j(r + 3 - 2j) + 2$	$2j + 2$
(3) (even rank)	$2(r + 5)$	3

the spinor words' row and the vector sector's rows one display at every j .

The records. Per sector, the family gaps and the binding comparisons, the balance pairs $\langle \text{gap}_i : \text{gap}_j \rangle$ over the sector's family pairs; and per member the stencil cap $3d_L$ at the closure's dimension fold (Lemma 17), $3 \cdot 2\ell$ at C_ℓ , $3 \cdot 2^\ell$ at B_ℓ , $3 \cdot 2^\ell$ at odd-rank D_ℓ at the spinor with its dual and $3(2\ell + 2^{\ell-1})$ at even, an exponential-polynomial pair in the rank (Lemma 23(v)); the record count per head bound is the family count's read, the leading fundamentals and the spinor words at or below the bound's read.

The classification is the equal-members reads of the displayed divisors in the rank domains, the domains the residue folds' reads (Construction 4), every even residue from two at B_ℓ , every residue

from three at C_ℓ and every odd residue from five at D_ℓ , each one cross-multiplied evaluation: at C_ℓ the three unit gaps keep one side at every residue ($2r+1$ below $4r$ below $4r+4$ from $r=3$) and the three charged gaps as well ($2r-2$ below $2r+1$ below $2r+4$); at B_ℓ and D_ℓ two leading fundamentals' gaps $2k(r+3-k)$ and $2k'(r+3-k')$ at $k < k'$ differ by the doubled product $(k'-k)(r+3-k-k')$, positive at every rank clearing both families, $k+k'$ at or below $2\ell-3$ against $r+3$ at or beyond 2ℓ ; the spinor's gap against a leading fundamental's, $[(r+2)(r+3):4]$ against $2k(r+3-k)$, reads its upper side beyond the residue $8k$ (the difference $(r+3)(r+2-8k)+8k^2$) and its equality at the roots of $(r+2)(r+3)=8k(r+3-k)$, at most two per k , each a merged level read at its rank with the counts at the multiplicity (Lemma 31's boundary clause), the root at $k=1$ the residue five, D_4 's vector against its spinors; at D_ℓ 's even rank the square's gap $4(r+3)$ against ω_k 's ties exactly at $(k-2)(r+3)=k^2$, $k-2$ a divisor of four, every solve below the clearance or at an even residue, and against the spinor's at the residue fourteen, even; the two spinors read one gap at every rank, a merged level with the counts at the multiplicity, and the spinor family ω_1+s at even rank reads its gap beyond the spinors' at every rank, off the square's at the residue six, even, and against ω_k 's on its upper side beyond the residue $8k$ (the difference $(r+3)(r+10-8k)+8k^2$, rootless at $k=1$) and at the roots of $(r+3)(r+10)=8k(r+3-k)$ below it, at most two per k , each a merged level read at its rank; two spinor words' gaps $j(r+4-j)$ and $j'(r+4-j')$ at $j < j'$ differ by $(j'-j)(r+4-j-j')$, positive at every rank clearing both, $j+j'$ at or below $2\ell-3$ against $r+4$ at or beyond $2\ell+1$, and the square's word at even rank ties a spinor word exactly at $r(j-2)=j^2-4j+8$, $j-2$ a divisor of four, every solve even or below the word's clearance; and the vector sector's rows at D_ℓ keep one side at every cleared rank: (1^{2j+1}) against $(2, 1^{2j-1})$ at the gap $4j+2$, two rows of one shape at the fundamentals' read above, a cross tie (1^{2j+1}) against $(2, 1^{2j-1})$ at j off j' reading its residue at the join $r(j-j')=2j^2-j-2j'^2+3j'+1$, the gap $j-j'$ a divisor of $2j'+1$ with an odd quotient and the residue $2(j-j')-1+4j'$ joined to it, even, off the odd domain, and the cube's rows at even rank tying (1^{2j+1}) at $r(j-1)=2j^2-j+5$ and $(2, 1^{2j-1})$ at $r(j-1)=2j^2-3j+4$, $j-1$ a divisor of six or of three, every solve even or at odd rank. So every displayed divisor keeps its leading side at every further cleared rank, one symbolic read every such rank's, a comparison's ranks those clearing its families' reaches; and a family below its clearance reads its finitely many ranks' data directly, the calculus's displays at the rank's label (Lemma 7; Corollary 1; Corollary 2), one evaluation each.

Proof. The spinor pair. The exchange of the last two simple letters reads the last coordinate to its balance partner, fixing the form and the root fold (the last key's read equal members) and exchanging the two spinor classes, each sector's base with its closure, so the spinor pair's records are one displayed list. *The families.* The gap is three folds over the word's occupied keys: the square fold and the doubled base fold are rank-free, the word's own form square and its doubled dot against the base's list, and the root fold's read is the word's fold against the displayed key reads, linear in the rank. *The unit tables.* The leading fundamentals' rows read the gap $2C_2(\omega_k)$ at the displayed key reads $(k(2\ell-k+1))$ at B_ℓ and $k(2\ell-k)$ at D_ℓ , the residues $r+2=2\ell$ and $r+3=2\ell$, the spinor's the doubled display of Lemma 9 at the rank's residue, $2C_2(s)=[\ell(2\ell+1):2]$ at B_ℓ , the square's $2C_2(2\omega_1)=8\ell$ and the spinor family's $2C_2(\omega_1+s)=[\ell(2\ell+7):2]$ at D_ℓ . *The charged tables.* The rows of e_1 , e_1+e_2 , $e_1+e_2+e_3$ and $e_1+\dots+e_4$ read $r+3$, $2(r+2)$, $3(r+1)$ and $4r$, the square's word at $4+2+2(2\ell-2)$ over the three folds, and the vector sector's rows read at the form's reads $C_2(\omega_k)=k(2\ell-k)$ and $C_2(\omega_1+\omega_k)=2\ell(k+1)-k^2+1$ against $C_2(\omega_1)=2\ell-1$ at $2\ell=r+3$, the rows of $(1,1,1)$ and $(2,1)$ reading $2(r-1)$ and $2(r+2)$. $\square$

18 The count certificates

The statements below are the cell calculus's arithmetic, made finite and constructible: the counts, the located roots, and the window reads the decimation and the flat step take.

Lemma 30 (The reversal count). *Over a located stage (Lemma 32), where every entry's members' equality and side are stage reads, the reversal count of a form pair, written $\text{rev}(A:B)$ at symmetric A, B of order N , is the side count of the pair: the dimension of a maximal list on whose span B exceeds A at every combination off the sum's unit, the two lists of the proof its witnesses. The count is read off a pivot split of the pair's elimination datum S , the site structure at $A = B + S$ read entrywise (Definition 3): a unit congruence, T at its witness T' at $TT' = 1$ and $T'T = 1$, the two reads the statement's data, $T^\top ST = D$ with D block diagonal in nonsingular blocks of orders 1 and 2 and one kernel block, its columns' T -image spanning the datum's kernel; a nonsingular block reads its sides, one per unit: an order-1 block its entry's side, and a symmetric 2×2 block two side reads off its data (the determinant on its lower side reads one unit per side, the mixed block; on its upper side the identity $S_{11}S_{22} = \det + S_{12}^2$ puts the diagonal product on its upper side, both entries of unequal members at one shared side, which both units read), so a split reads a lower-side count, the order-1 blocks' sides with the 2×2 reads. Every symmetric S has a split, closed in at most N pivot steps, and every two splits read one lower-side count: $\text{rev}(A:B)$ is that shared count, the split's pivot ordering certificate data, and $\text{rev } S$ abbreviates the count at S its own elimination datum. A split enters cleared: over pair data the congruence reads at an integer T , square at the datum's order, its determinant of unequal members with the adjugate against the determinant the witness's solve (Definition 3), and the block diagonal absorbs the columns' rescaling at every side fixed, the cleared read the unit congruence's by the homogeneity principle (Definition 1), one count at either spelling. The split reads the rank with it, the kernel block's complement, one count at every split and blind to unit congruence with the reversal count: a split (T, D) of S is the split $(T, \check{D})$ of the memberwise swap $\check{S}$ (each entry's member swap) at the block sides exchanged, so the rank is the two counts' sum, $\text{rev}(S) + \text{rev}(\check{S})$. A datum is positive semidefinite where a split's nonsingular blocks read the upper side throughout, the split the statement's witness, and positive definite where those blocks' orders sum to the datum's; a form pair $(A:B)$ is positive semidefinite where its elimination datum is, the glyphs $B \preceq A$ and $A \succeq B$ denote that read, and $S \succ 0$ denotes S positive definite; a datum S is capped two-sidedly at C where $S \preceq C$ holds and the sum $C + S$ is positive semidefinite, two splits as the statement's witnesses. Consequently: $\text{rev}(T^\top ST) = \text{rev}(S)$ at T with its witness; $\text{rev}(qS) = \text{rev}(S)$ at a pair q above the sum's unit, a split (T, D) of S being the split (T, qD) of qS with every block's sides and determinant side fixed; a symmetric $\text{LDL}^\top$ reads rev as its count of lower-side pivots, and when the leading principal minors read unequal members the pivots are their successive cofactors ($m_j d_{j+1} = m_{j+1}$), so rev is the minor sequence's own orientation-reversal count, the reversals along $1, m_1, \dots, m_N$ (Jacobi's rule), the name's read; $S \preceq S'$ gives $\text{rev}(S) \geq \text{rev}(S')$; a compression's counts are at most the full form's; inertia adds at a nonsingular principal pivot,*

$$\text{rev}(S) = \text{rev}(P) + \text{rev}(S_P) \quad \text{at} \quad S = \begin{bmatrix} P & B \\ B^\top & Q \end{bmatrix}, \quad B^\top C_P + S_P = Q \quad \text{at the solve} \quad PC_P = B,$$

S_P the pivot's deflation, its witness C_P the adjugate column against the determinant, the deflation symmetric at a symmetric datum, the coupling's transpose against the witness the witness's pairing against the pivot, $B^\top C_P = C_P^\top PC_P$ at the solve's transpose, its every multiple symmetric with it, a pivot at stated places entering at the leading position by the congruence at the places' permutation matrix, its transpose the witness (Construction 1's transpose identity at the matrix carrier) with the count the congruence read's own, and the determinant splits with the inertia at the cleared scale: each

trailing row enters at the pivot's determinant against the leading rows' combination at the solve's witness column, one scale per trailing row, the minor's scale and combination reads (Definition 3) reading the scale's power at the deflation's order against the reduced list's determinant, the coupling's exchange withdrawn at the solve so the reduced list is block triangular with its trailing block the deflation at one further scale per row, and a block-triangular list's determinant is its diagonal blocks' product, every assignment crossing into the vacant slab reading an entry at the sum's unit: $\det P \cdot \det ((\det P)^2 S_P) = (\det P)^{2o} \det S$ at the deflation's order, $\det S = \det P \cdot \det S_P$ its read at the off-unit scale's injectivity; two square lists' product, its entries the first's rows against the second's columns at the coordinate fold, reads its determinant at the factors' own, one joined list's two reads: at the join of the identity's memberwise swap beside the second list over the first list beside a vacant block, the trailing slab clears against the leading rows at the first list's own coefficients, one combination per row with the further summands repeated-row reads of equal members (Definition 3), onto the vacant slab beside the two lists' product, and the cleared join's determinant is the block-triangular read at the swapped identity against the product's; the slabs' exchange, one transposition per row pair (Definition 3), reads the joined list block triangular at the factors instead, and the exchange count joins the swapped identity's diagonal to an even family (Construction 1): the signs withdraw and the product's determinant is the determinants' product; inertia adds at a block diagonal outright,

$$\text{rev}(P \oplus Q) = \text{rev}(P) + \text{rev}(Q),$$

the addition's instance at a coupling of equal members: the two splits concatenate to a split of the block diagonal, the congruences joined at that off-block with the block lists appended, the kernel columns collected to the tail and the kernel orders added; and a symmetric matrix of unequal-membered determinant has a nonsingular principal submatrix of order at most two.

Proof. Existence. At a nonsingular principal block P of order 1 or 2, the unit-triangular congruence at upper block $\check{C}_P$, the witness's memberwise swap ($PC_P = B$), maps S to $P \oplus S_P$, and the recursion descends to the deflation. With every principal minor of orders 1 and 2 of equal members, the diagonal reads equal members entrywise, each order-two minor then reads S_{ij}^2 , and S_{ij} with it, of equal members: a symmetric block either has such a pivot or is the kernel block, so the leftmost admissible pivot closes a split in at most N steps, the per-step congruences composing to T ; in particular a symmetric matrix of unequal-membered determinant has a nonsingular principal submatrix of order at most 2.

The mixed blocks' two witnesses. At $P = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ with the determinant $\langle ac:b^2 \rangle$ on its lower side, a lower-side vector is read off the entries: e_1 at a on its lower side; $(b, \check{a})$ at a on its upper side, $\check{a}$ the member swap, reading $a \langle ac:b^2 \rangle$ on the lower; and at a of equal members, where b 's members differ, the vector $(w, b+b)$ at w the balance partner of $c+1$, its read the balance partner of $(b+b)^2$, the cross terms collecting at the constant's swap, on the lower side at b 's occupied square. An upper-side vector of P is a lower-side vector of the memberwise swap $\check{P}$, the recipe read there.

The two lists of a split, and the exchange read. From a split (T, D) at lower-side count ν , the lower list takes, per lower-side unit, the T -image of one block-supported vector: each lower-side order-1 column, both columns of each 2×2 at the shared lower side, and each mixed block's lower-side vector. The list's ν members are independent, the strict read's own consequence (the forcing clause's read below), and S reads the lower side at every combination off the sum's unit: distinct blocks' pairings read equal members through D , so a combination reads a per-block sum, every term on its lower side or of equal members and the terms at coefficients of unequal members on the lower side. The complement list takes the upper-side order-1 columns, both columns of each 2×2 at the shared upper side, each mixed block's upper-side vector, and the kernel block's columns:

its member count μ at $\nu + \mu = N$, the block orders' total, independent, with every S -read of a combination on its upper side or of equal members. For two splits at counts ν_1, ν_2 , join the first's lower list to the second's complement list: the joint rank is one computation, and a rank defect exhibits a dependency whose read is one stage element with $x^\top Sx$ on its lower side and off it at once (a dependency whose lower-list part is the sum's unit sits inside the independent complement list), against a stage read's one side (Lemma 32): the joint list is independent, $\nu_1 + \mu_2 \leq N$ at $\nu_2 + \mu_2 = N$, and the exchange symmetrized reads $\nu_1 = \nu_2$. The joint-rank read at an arbitrary first list is the *forcing clause*: a list of k vectors with every S -read of a combination off the sum's unit on the lower side, joined to any split's complement list, forces $\text{rev}(S) \geq k$, the strict read forcing the list's independence outright: a combination reading the sum's unit at coefficients off the unit family reads its own lower side besides, against the trichotomy. The *dual clause* runs at the complement side with the independence its own stated clause, an at-or-above read admitting a dependency at the sum's unit: an independent list of k vectors with every S -read of a combination at its upper side or equal members, joined to any split's lower list, forces $k + \text{rev}(S) \leq N$.

The consequences. A split of $T^\top ST$ at U is a split of S at TU : the count is invariant. Splits of P and of S_P concatenate through the displayed congruence to a split of S : inertia adds, and the two forcing clauses price the identity besides: the pivot's and the deflation's lower lists, padded and lifted through the solve with every cross pairing the witness's own unit read, force one side, and the complement lists at the dual clause the other. At a coupling of equal members the lift is a padding besides, and the pricing runs without the solve: the two lower lists, each padded into its own half of the join, read the block diagonal at the two forms' sum with the cross pairings at the sum's unit, and the complement lists at the dual clause read the other side. An $\text{LDL}^\top$ is a split at order-1 blocks, the congruence by the unit-triangular witness T at $L^\top T = 1$, one back-substitution, onto the pivot diagonal, whose k -th leading minor is the product of the first k pivots. At $S \preceq S'$ the datum R at $S' = S + R$ is positive semidefinite, so $x^\top Sx + x^\top Rx = x^\top S'x$ at every x through the congruence with the R -read on its upper side or of equal members; on S' 's lower list the S -read sits below the S' -read, on the lower side with it, and the forcing clause reads $\text{rev}(S) \geq \text{rev}(S')$. For a compression $P^\top SP$, the compression's lower list maps through P to an independent list with every S -read of a combination off the sum's unit on the lower side (a combination with Pc the sum's unit at c off it reads $c^\top P^\top SPc$ on its lower side and of equal members at once, against the trichotomy), and the forcing clause closes it. $\square$

Theorem 15 (The designation, and the witness). *Let H, G have entries of a located stage (Lemma 32) with G positive definite, and let $\text{count}(a) := \text{rev}(H : aG)$; at a balance-pair level $\langle x : y \rangle$ (Definition 1) the count is*

$$\text{count}\langle x : y \rangle := \text{rev}(H + yG : xG),$$

the level's second member cross-added onto the pencil's own side, so a root on its lower side reads its bracket as one on its upper does; each count exact by symmetric $\text{LDL}^\top$ (Lemma 30).

1. (The designation.) *A designation of a jump of the counting function is the data (l, h, w) at $l + w = h$ with the two counts read at a gap, $\text{count}(l) + g = \text{count}(h)$ at $g \geq 1$, the width w at most the δ of Theorem 16: the bracket admits one distinct root alone, at multiplicity g , and the reads are its whole verification. At the least root the lower read sharpens to the pair at l positive semidefinite, one split and one count. Designations exist at every stated width, the refinement closing in the least k at $w_0 \leq 2^k \delta$ from the starting gap w_0 at $l_0 + w_0 = h_0$, each step one count. The refinement runs at any stage read whose trichotomy exchanges sides along a segment, a polynomial's side read at Theorem 16's separation among them, the equality outcome naming*

a root outright with the recentering emitting its bracket. An emitted designation enters as certificate data, its reads re-read at every verification.

2. (The witness u .) For any form pair $(A:B)$ of pair entries with $\text{rev}(A:B) \geq 1$, the symmetric $\text{LDL}^\top$ of its elimination datum has a lower-side pivot, and the corresponding column of the accumulated pair-entried transform is a vector u with $u^\top Bu > u^\top Au$, the reversal's read off the elimination. At $(H:aG)$ that is the certificate's witness whenever $\text{count}(a) \geq 1$.

Proof. The bracket is narrower than the separation, and two distinct roots inside it sit closer than δ , refused by Theorem 16: it admits one distinct root alone, and the counts' gap g is that root's multiplicity, the count identity's read (Lemma 33). The existence: the radius of Theorem 16 reads two starting levels, the lower below every root at the split and the upper beyond every root at the full count; from two levels at counts at a gap, the mean's count sits at a gap against one end's, the half at that gap the next bracket, the width halving, so the refinement closes in the least k at $w_0 \leq 2^k \delta$; and at the least-root instance a mean at the lower end's count either is positive semidefinite, moving the lower level, or is singular there, the mean naming the root and one recentering at $2w' \leq \delta$ emitting the designation. The witness: at $\text{rev}(A:B) \geq 1$ the split reads a lower-side unit (Lemma 30), u its column of the congruence, and the congruence's read $u^\top Su$ at the elimination datum S at $A = B + S$ is that unit's entry on its lower side, so $u^\top Au$ sits below $u^\top Bu$. $\square$

Lemma 31 (The count is a cell function). *Let $(H(x), G(x))$ be a symmetric pencil whose entries are polynomial in $x = (x_1, \dots, x_m)$ over $\mathbb{P}$, with $G(x) \succ 0$ on a box X , let a be a pair, and write $D(x)$ for the pair $(H(x):aG(x))$'s site-datum determinant. Then $\text{count}(x) := \text{rev}(H(x):aG(x))$ is constant on every segment between stage points of X avoiding D 's root locus (the stages of Lemma 32), the avoidance's certificate the proof's pivot cover: finitely many subintervals chained at shared endpoints, each with a designated principal minor of order at most two whose determinant keeps its side along the subinterval at the priced side read (Lemma 32) at a bound containing the subinterval, at order two its leading entry so as well where the determinant sits on its upper side, and beneath it the cleared deflation's own cover, the deflation cleared once at the designated minor's magnitude, the minor's multiple of the deflation at a pivot whose determinant keeps its upper side and its memberwise swap at a pivot keeping its lower side or at the mixed block, and at a pivot whose coupling to the trailing places reads the sum's unit the trailing block's own cover in the deflation's place, the cover at the vacant order holding outright, the determinant's root-freeness the cover's read; the locus's points are located over D 's squarefree part, the division descent's iterated divisions with the Bézout witnesses the certificate data (Theorem 16's descent at Lemma 37's squarefree read), a bracket's root count the segment count at the cleared variable (Lemma 38 at Lemma 39's clearing); hence the count is one integer on each piecewise-linearly connected cell of X off that locus; at a box of X of two variables, one bracket per variable, the cover is the box cut at stated points, one variable per cut, each point strictly inside its variable's bracket, into pieces, boxes, a cut's two boxes meeting along it, each piece with its designated minor keeping its side over the piece at the box's priced side read (Lemma 32), at order two its leading entry so as well where the determinant sits on its upper side, and beneath it the once-cleared deflation's own box cover over the piece, the deflation or its memberwise swap at the designated minor's side, so the count is one integer on the box; and at a stage point of the locus adjacent to a cell whose pair is positive semidefinite, every pencil root is at least a , the point's read entering as its own split (Theorem 15's certificate entry).*

Proof. Write $S(t)$ for the pair's site datum at $H = aG + S$ along the segment, entries polynomial in t , its determinant with every root off the segment; the claim is by induction on the order. At order

one the count is the side's, and the entry keeps its side along each piece of the cover at the priced side read (Lemma 32), the endpoint value clearing the piece's width against the derivative's coefficient fold at a bound containing the piece. At order N : inertia adds at a nonsingular principal pivot of order at most 2, and $S(t)$, its determinant of unequal members, has one at every t (Lemma 30). The principal minors of orders 1 and 2 with a coefficient of unequal members have finitely many roots; at each root point, and at the segment's ends, the unequal-membered determinant names a minor of unequal members there, whose own roots sit off that point at a distance a bracket certifies, so the segment is covered by finitely many subintervals of pair endpoints, each with one designated minor whose members differ throughout, consecutive ones overlapping at pairs. On one subinterval with pivot $P(t)$: the pivot's inertia is constant ($\det P$ keeps one side, the order-one read; at order two, the determinant on its lower side reads one unit per side, the mixed block, and on its upper side both diagonal entries keep one shared side, their roots off the subinterval, Lemma 30); the induction beneath the pivot reads two data of the object it descends to, polynomial entries at a stated degree and the count $\text{rev}(S_P)$, the deflation's entries the cofactors at $\det P$ (the solve, Definition 3), so the entries read polynomial at the determinant's multiple, and a count passes through a multiple above the sum's unit alone (Lemma 30's rescaling), so the multiple is the determinant's magnitude (Theorem 16's magnitude carrier), one object at the two sides: $\det P \cdot S_P$, the deflation cleared once at the pivot's determinant, the trailing block's multiple at the determinant less the coupling's transpose against the pivot's adjugate against the coupling, the solve $PC_P = B$ read at the adjugate against the determinant ($P \text{ adj } P = \det P \cdot 1$, Definition 3's minors), has polynomial entries, its degree within $k + 1$ times the datum's at the pivot order k (the products' tops summing, Definition 2), and is symmetric where S is at every point of the subinterval: its determinant's multiple $(\det P)^2 S_P$ is symmetric, the deflation symmetric at the solve's transpose (Lemma 30), and the factor $\det P$, off the sum's unit on the subinterval, withdraws entry by entry at the product's injectivity (Definition 1); the determinant splits at the cleared scale, $\det P \cdot \det(\det P \cdot S_P) = (\det P)^o \det S$ at o the deflation's order ($k + o = N$; Lemma 30's split, the minor's scale read at the once-cleared multiple), so at $\det P$ off the sum's unit its roots sit off the segment with $\det S$'s and the induction reads its count constant; at a pivot whose determinant keeps its upper side the positive factor fixes every count, $\text{rev}(\det P \cdot S_P) = \text{rev}(S_P)$ (Lemma 30), and at a pivot whose determinant keeps its lower side or at the mixed block the memberwise swap of $\det P \cdot S_P$ is the determinant's magnitude against S_P (Theorem 16's magnitude carrier), collected at the displayed reads as the trailing block's multiple at the determinant's swap joined to the coupling's transpose against the witness, the swap of a sum the swaps' sum and the swap of the correction's swap the correction itself (Definition 1), the positive factor fixing every count there, so $\text{rev}(S) = \text{rev}(P) + \text{rev}(S_P)$ is one integer on the subinterval; and at a pivot whose coupling to the trailing places reads the sum's unit the deflation is the trailing block at the pivot's determinant's scale, its swap at the magnitude's, inertia adding at the block diagonal (Lemma 30), so the trailing block's own cover reads beneath the pivot at the datum's own clearing. The overlaps chain the subintervals, so the count is one integer along the segment, and cells connected by such segments read one count. At a box the box's priced read keeps the designated minor's side at every point of a piece (Lemma 32), so at every point the count is the pivot's block read joined to the cleared deflation's count, the deflation's cleared evaluation at a point the evaluated blocks' own deflation with the inner evaluation reading sums to sums and products to products (Definition 2), its degree in each variable within $k + 1$ times the datum's at the pivot order k , the deflation at a designated minor keeping its upper side and its memberwise swap at one keeping its lower side or at the mixed block, and the deflation's count is one integer on the piece at its own box cover, the induction's read; a cut's two boxes read one count through a point of the cut, so the count is one integer on the box. At a boundary point x^* beside a positive-semidefinite cell, a root below a gives u with

the read $u^\top S(x^*)u$ on its lower side; along a segment into the cell, $t \mapsto u^\top S(x(t))u$ is a polynomial, on its lower side at the boundary point and so on it over an initial piece its coefficient folds name, putting a lower-side direction inside the cell: against the cell's count. $\square$

Lemma 32 (The located stage). *The arithmetic. A located root of a polynomial over a located stage ($\mathbb{P}$, or a field built here, every element with its bracket) is a root ε of its squarefree part S (the division descent, Theorem 16), with an isolating bracket of balance-pair endpoints (Definition 1) off S 's roots, narrower than half the separation of Theorem 16, whose derivation is stage arithmetic outright, division reads, coefficient folds and bracket comparisons. The stage $K(\varepsilon)$ is $K[x]$ read at ε : at $p \in K[x]$ with g the greatest common divisor of (p, S) (Theorem 16's descent), the value $p(\varepsilon)$ reads equal members exactly where g exchanges sides on the bracket; at $p(\varepsilon)$ of unequal members the root sits on the cofactor S_1 at $gS_1 = S$, coprime to p , and the Bézout read $up + vS_1 = 1$ reads at the root to the unit witness, $p(\varepsilon)u(\varepsilon) = 1$. Sums and products are remainder reads, the division at S , a value's unit is the Bézout read's witness, and a value's side at the root is a count read (Definition 1), the bracket the certificate naming which root, every entry an elimination read of the stage below, so the adjunction stays grounded and the recursion ends at the ground pairs' own comparisons. At $p(\varepsilon)$ of unequal members the unit witness floors the value: at U_1 the coefficient fold of u over the bracket (Theorem 16's magnitudes), the witness identity read cross-multiplied floors the margin of $p(\varepsilon)$ at $\text{margin} \cdot U_1 \geq 1$; and the priced side read: at bracket endpoints inside a stated bound's segment, the bound entering at a second datum beyond the sum's unit, the radius segment among them at the leading coefficient's margin, two bracket points' values of p differ by at most the bracket's width against the derivative's coefficient fold at the bound (Theorem 16's magnitudes, the binomial identity per occupied key), so an endpoint value of p clearing that price keeps its side at every bracket point, the root's among them: the cleared comparison is the side read's certificate, the endpoint value's side the comparison's own read at the width price at or beyond the sum's unit, p 's image width over the bracket priced by the fold. The read runs on a box besides: a polynomial in two variables over the stage, the inner variable's polynomials the outer variable's coefficients (Definition 2's polynomials as the entry carrier), at one bracket per variable inside its own bound, keeps its side at every point of the box where two priced reads hold, the inner variable's read above at the corner's outer point, the coefficients read there, and the outer variable's read at every inner point at once: the outer derivative's coefficient fold at the outer bound, each coefficient read at its own fold at the inner bound, against the outer width clears the corner value less the inner width against the inner derivative's fold at the two bounds, so at every inner point the outer read's endpoint value sits within the inner price of the corner value and the outer read holds there, the display above at that inner point's coefficients. The construction closes on itself: $K(\varepsilon)$ is a located stage and adjunction iterates; a finite root list enters as a tower, one extension per root, each extension's polynomial over the stage the earlier extensions built, the extension order certificate data: a stored tower names its order, and every read runs through the arithmetic above.*

Proof. g is a factor of the squarefree S , so its one possible root on the bracket is simple, and it is ε , the bracket isolating: $p(\varepsilon)$ reads equal members exactly where g exchanges sides there. At $p(\varepsilon)$ of unequal members the cofactor S_1 is coprime to p since S is squarefree, so the Bézout read exists and reads the unit witness at the root. $K(\varepsilon)$'s elements each have their bracket (the isolating data, their image widths priced above) and its division reads, coefficient folds, endpoint evaluations and count reads are the display's, so $K(\varepsilon)$ is a located stage. $\square$

Lemma 33 (The pencil's split). *For symmetric H and $G \succ 0$ of order N over a stage, with $Z(x)$*

the level pair's site datum at x $G = H + Z(x)$ and $\chi(x) := \det Z(x)$,

$$\chi(x) = \det G \prod_{j=1}^N \langle x : \varepsilon_j \rangle, \quad \text{count}(a) = \#\{j : \varepsilon_j < a\} \quad \text{at } a \text{ off the roots,}$$

the factors Definition 2's linear factors at the ε_j , located roots, and a unit congruence S_χ over the stage they generate maps the pencil to diagonal form,

$$S_\chi^\top Z(x) S_\chi = \text{diag}(\langle x : \varepsilon_j \rangle g_j), \quad g_j \text{ on their upper sides.}$$

Proof. Over the pairs the counting function climbs to N from the positive-semidefinite read between the radius reads of Theorem 16, and its jumps sit at the located roots of χ , each with its isolating bracket at the separation (Theorem 16; Lemma 32). At a located root ε , the kernel of $Z(\varepsilon)$ over $K(\varepsilon)$ has a spanning list by elimination (Lemma 32), of some dimension $m \geq 1$, G -orthogonalized within, and its G -complement W another; at z in the kernel and $w \in W$, $z^\top Z(x) w = \langle x : \varepsilon \rangle z^\top G w$, of equal members at the complement's defining read, so the congruence by the two lists is block diagonal, the kernel block $\langle x : \varepsilon \rangle G_0$ at the kernel's Gram $G_0 \succ 0$ diagonal, and

$$\chi(x) = c \langle x : \varepsilon \rangle^m \chi_W(x), \quad c \text{ on its upper side,}$$

χ_W the deflated pencil's. $\chi_W(\varepsilon)$ is of unequal members: a kernel vector of the deflated pencil at ε has its pairings against W and against the kernel both of equal members, so it is a kernel vector of the full pencil lying inside W , and the kernel meets W at the sum's unit alone. Induction on W splits χ over located roots, each deflation's root adjoined over the stage the earlier roots built (Lemma 32), and concatenates the per-step lists to S_χ , block diagonal for G and for H at once by the pairings above; the count identity is then the congruence read (Lemma 30): at a off the roots, the pair $(H : aG)$'s elimination datum is the displayed datum's balance partner, so $\text{rev}(H : aG) = \text{rev} \text{diag}(\langle \varepsilon_j : a \rangle g_j) = \#\{j : \varepsilon_j < a\}$. The completed congruence returns the factorization read: the congruence identity's determinant collects the product read at the exchanged lists (Lemma 30; Definition 3) to $\det S_\chi^2 \chi(x)$ against the diagonal blocks' product $\prod_j \langle x : \varepsilon_j \rangle g_j$; the identity's linear coefficient, $S_\chi^\top G S_\chi$ at the diagonal of the scales, collects its own determinants to $\det S_\chi^2 \det G$ against $\prod_j g_j$, and at a located root's pair data the cleared factor enters at the scale, the diagonal's entry $g_j (d_j x - n_j)$ at the root $[n_j : d_j]$ with the linear coefficients collecting to $\prod_j d_j g_j$ and the clearing $\prod_j d_j$ riding onto the factorization's first member, one value at every representative (Definition 1); and the shared factor $\det S_\chi^2$, off equal members at the unit congruence, withdraws at the products' one value key by key (Definition 2): the displayed factorization is the congruence certificate's own read. $\square$

Lemma 34 (Slab pivots: the factorization, the counts, and the Green products). *Let S be symmetric over a located stage with a slab split $1..L$, entries at the diagonal and neighboring slab pairs alone, $A_i := S[i, i]$, $B_i := S[i, i+1]$, and the two pivot recursions at solved witnesses, each solve the adjugate column against the determinant,*

$$\begin{aligned} X_L &= A_L, & X_i + B_i R_{i+1} &= A_i \quad \text{at} \quad X_{i+1} R_{i+1} = B_i^\top; \\ Y_1 &= A_1, & Y_{i+1} + B_i^\top C_i &= A_{i+1} \quad \text{at} \quad Y_i C_i = B_i, \end{aligned}$$

at pivots of unequal-membered determinant (a singular slab pivot re-orders within its slab by Lemma 30's principal pivots of order at most two, the count unchanged). Then: (i) the Schur complement of the tail $[i+1, L]$ onto $[1, i]$ maps A_i to X_i and fixes every other block: eliminating

slab L touches A_{L-1} alone, the entries at $i < L-1$ against L off the band, and downward induction extends it, so one recursion datum holds the whole tail, and every pivot is symmetric: the seed is a diagonal block of the symmetric datum, and at a symmetric X_{i+1} the witness identity transposed reads $B_i = R_{i+1}^\top X_{i+1}$, so the withdrawn term $B_i R_{i+1} = R_{i+1}^\top X_{i+1} R_{i+1}$ is symmetric with X_i , the induction downward from the seed; and the head pivots are symmetric at their own induction: the seed Y_1 is a diagonal block of the symmetric datum, and at a symmetric Y_i the witness identity transposed reads $B_i^\top = C_i^\top Y_i$, so the withdrawn term $B_i^\top C_i = C_i^\top Y_i C_i$ is symmetric with Y_{i+1} , the induction upward from its seed; (ii) $\text{rev}(S) = \sum_i \text{rev}(X_i)$ and $\det S = \prod_i \det X_i$, each elimination a unit-triangular congruence (Lemma 30); (iii) with Z_j the join's datum at $Z_j + A_j = X_j + Y_j$, read off the two recursions, the solution of $Su = v$ at v supported in slab j reads blockwise, at $i \leq j$,

$$Z_j u_j = v_j, \quad u_i = T_i T_{i+1} \cdots T_h u_j \quad \text{at } h+1 = j,$$

at the transfer factors T_k , each the memberwise swap of the witness C_k , so $Y_k T_k + B_k$ is the sum's unit: every Green block is a product of transfer factors and one corner solve; (iv) at scalar slabs the datum assembled from the slab data is square at the diagonal's count, at a bond list one shorter, and its form at a walked vector reads the display's two folds, the diagonal entries at the squares with the bonds at the doubled consecutive products.

Proof. (i) and (ii) are the displayed inductions and congruences. (iii): solve $Su = v$ at v supported in slab j . The rows above j read $X_{i+1} u_{i+1} + B_i^\top u_i$ at the sum's unit, the tail closing from L downward as in (i); the rows below read $Y_i u_i + B_i u_{i+1}$ at the sum's unit at every i below j , the induction from 1 on the Y recursion; and substituting both into row j leaves $Z_j u_j = v_j$. The display telescopes the descent, and at $L = 2$ it is the two-slab solve outright. (iv): the leading slab's row read splits the head row's fold and the block rows' folds into the slab's own entries and the tail chain's fold, the leading square with the doubled consecutive product the slab's own read, and the descent closes on the slab count, the shape read the descent's own length read. $\square$

Lemma 35 (The spectator law). *Let S, S' be two families as in Lemma 34 on one slab split, with equal data beyond slab w : $A'_i = A_i$ at $i > w$ and $B'_i = B_i$ at $i \geq w$. (i) One sandwich identity. The tail pivots are equal outright, $X'_i = X_i$ at $i > w$; the deviation sits in the boundary-direction pivots alone, and at $i \geq w$ the deviation datum D_i at $Y'_i = Y_i + D_i$ obeys the exact sandwich*

$$D_{i+1} = T_i^\top D_i T_i$$

at the two families' transfer factors (Lemma 34(iii)), telescoping to a product form: the deviation at depth i beyond the perturbation is D_{w+1} sandwiched by the transfer factors of Lemma 34(iii). Re-convergence of the recursion and the Green products' falloff are the one sandwich identity, stated twice. The identity reads at the reversed slab order at a stated diagonal tie besides: at two families with equal couplings below slab w , $B'_i = B_i$ at $i < w$, and the diagonals tied there at one datum, $A'_i = A_i + \delta 1$ at $i < w$, the tail deviation E_i at $X'_i = X_i + E_i$ obeys, at $i < w$,

$$E_i = \tilde{R}_{i+1}^\top E_{i+1} \tilde{R}'_{i+1} + \delta 1$$

at the tail witnesses' transfer factors $\tilde{R}_{i+1}$ and $\tilde{R}'_{i+1}$, the memberwise swaps of R_{i+1} and R'_{i+1} (Lemma 34(iii)); at the tie's datum of equal members the families' data are equal below the slab, the head pivots equal outright at $i < w$ and the display the sandwich $E_i = R_{i+1}^\top E_{i+1} R'_{i+1}$; the deviation at depth i below the perturbation is E_w sandwiched down to the head with the tie's datum entering once per step, and (ii)'s caps propagate along it at the tail factors' certificates. (ii) Caps propagate geometrically. The gram's slab blocks are $G_i := G[i, i]$, the cross blocks of equal members at the

index's orthogonal sum (Definition 8), each display's gram its own slab's block. At certified pairs λ_i, λ'_i with $T_i^\top G_i T_i \preceq \lambda_i^2 G_{i+1}$ and $T_i^\top G_i T'_i \preceq \lambda_i'^2 G_{i+1}$ at the two families' transfer factors, a cap D_i two-sided at $c_i G_i$ (Lemma 30) propagates to $2 D_{i+1}$ two-sided at $c_i (\lambda_i^2 + \lambda_i'^2) G_{i+1}$. Iterating, the bracket at depth i is named and geometric. At the tied families ((i)'s reversed display) each step joins the tie's datum, $D_{i+1} = T_i^\top D_i T'_i + \delta 1$, and at the datum's identity capped two-sidedly at $s G_{i+1}$ the display gains its read, $2 D_{i+1}$ two-sided at $c_i (\lambda_i^2 + \lambda_i'^2) G_{i+1} + 2 s G_{i+1}$; iterating, the bracket is the seed's geometric read joined to the tie datum's fold over the steps' certificates, one summand per step. (iii) The bulk pivot is the deck-interior solvent. At bulk data $A_i = A$, $B_i = B$ at B of unequal-membered determinant, read in a congruence mapping G to the identity (G 's factorization a stage datum, comparisons then the spectral reads of Lemma 33), the deck polynomial p , the determinant of the quadratic pair $(B^\top z^2 + B : Az)$'s site datum, obeys the deck symmetry, its coefficients equal at every index pair joining to $2m$, the display of its deck symbol (Lemma 39), and a solution pair (Y, X) of the pivot identities $YX = B$ and $Y + B^\top X = A$ has its solvent X satisfying $B^\top X^2 + B = AX$, and the pair's site datum factors exactly,

$$U(z) = U_1(z)U_2(z) \quad \text{at} \quad B^\top z^2 + B = Az + U(z), \quad zB^\top + B^\top X = A + U_1(z), \quad z = X + U_2(z),$$

so X 's polynomial is a factor of p . Off the band (Lemma 39's band read, the deck relation's crossing locus), $p = c p_{\text{int}} p_{\text{ext}}$ over a stored tower; the interior kernel $V := \ker p_{\text{int}}(C)$ at the companion action C , the shift $(x, y) \mapsto (y, y')$ at its second component's witness $B^\top y' + Bx = Ay$, has a spanning list by elimination over the tower, each companion step one solved read at $B^\top$, stacked $\begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$, and the graph read $\det V_1$ is one tower sign. Where it reads unequal members, X_* at $X_* V_1 = V_2$ satisfies the solvent identity with polynomial p_{int} and Y_* at $Y_* X_* = B$ solves the pivot identities; and every solution whose solvent polynomial is p_{int} equals it, its graph filling the one kernel V . Three counts at a located symmetric $\hat{Y}$ and pairs $\lambda < 1$, y_0 , ρ , ρ_0 , at the pivot map $Y \mapsto Y^\sharp := A + B^\top T$ over the transfer factor T , the memberwise swap of the witness at $YC = B$, so $YT + B$ is the sum's unit (Lemma 34(iii)),

$$\hat{T}^\top \hat{T} \preceq \lambda^2, \quad \hat{Y} \succeq y_0 + \rho, \quad D_\circ \text{ two-sided at } \rho_0 \quad \text{at} \quad \hat{Y}^\sharp = \hat{Y} + D_\circ,$$

with $\lambda_\circ$ at y_0 $\lambda_\circ = \lambda(y_0 + \rho)$, $\mu := \lambda_\circ^2 < 1$ and $\rho_0 + \mu\rho \leq \rho$, make the map send the ball at deviation data D two-sided at ρ , $Y = \hat{Y} + D$, to itself and contract its deviations at modulus μ , a ball member's certificate $\lambda_\circ$. A ball member prices its form at the floor, the member's form at or beyond y_0 's multiple of the gram's at every vector, so every split of a ball member reads the upper side throughout (Lemma 30). The graph read joins the validity data: where it reads unequal members, the bracket comparison D_* two-sided at ρ at $Y_* = \hat{Y} + D_*$ is one count at the tower datum, and where it holds Y_* is the ball's one solution of the pivot identities. The graph read's side and the three counts are each cell data of the family's parameters (Lemma 31; Lemma 32). The scalar instance, the walk pencil's bulk data at $B = 1$: the boundary entry solves the pivot identity $mY = 1$, the solvent, $p_{\text{int}} = \langle z : z_* \rangle$ at the interior member z_* , and (i)'s sandwich is the error law $e_{L+1} = z^2 e_L + z e_L e_{L+1}$, contraction factor z^2 , marginal exactly at the crossings, $z^2 = 1$ there.

Proof. (i) The tail pivots are equal at the recursion running on equal data from an equal seed. The sandwich is the two recursions' joins at equal A, B read against each other at the transposed witness $B_i^\top = C_i^\top Y_i$ of the symmetric head pivot (Lemma 34(i)). At the reversed order the join $X_i + B_i R_{i+1} = A_i$ read against its primed copy at the equal couplings and the tie collects the deviation to $B_i (R_{i+1} + \check{R}'_{i+1})$ beside the tie's datum, the witness identity $X_{i+1} R_{i+1} = B_i^\top$ transposed at the symmetric pivot reads $B_i = R_{i+1}^\top X_{i+1}$, and the two witness identities at the pivots read $X_{i+1} (R_{i+1} + \check{R}'_{i+1}) = E_{i+1} R'_{i+1}$, so $E_i = R_{i+1}^\top E_{i+1} R'_{i+1} + \delta 1$, the two swaps withdrawn in pairs and

the tie's datum riding each step. (ii) At $u := T_i x$ and $v := T'_i x$, the cap's two splits at $u + v$ and at $u + \check{v}$, $\check{v}$ the memberwise swap, collect to $2 u^\top D_i v$ capped two-sidedly at $c_i (u^\top G_i u + v^\top G_i v)$, and the certificates price the two squares; at the tied families the display gains the tie datum's read at the two splits' own polarization at the identity's datum. (iii) The deck symmetry holds at A symmetric, the transpose identity exchanging B with $B^\top$, and X 's polynomial is a factor of p at the adjugate identity reading Cayley–Hamilton. At the pivot map a ball member's certificate is $\lambda_\circ$, its transfer factor priced through the floor by the identity $Y D_T = D \hat{T}$ at $\hat{T} = T + D_T$, the image deviation is (i)'s sandwich capped by (ii)'s certificates, and the center's displacement tops up to ρ ; the deviation's cap withdraws against the center's floor at the gram's read, so the member's form sits at or beyond y_0 's multiple of the gram's at every vector. Two ball solutions read their deviation datum D through (i)'s sandwich, the two-sided cap at cG returning at μcG by (ii) and iterating below every pair, and a located entry below every pair is of equal members; the solvent is capped inside by the first count ($X^\top X \preceq \lambda_\circ^2$ prices every polynomial root's square below 1), so its polynomial is p_{int} and its graph fills the one kernel V . $\square$

Theorem 16 (The window is a root separation, and is explicit). *For a polynomial P over $\mathbb{P}$ (Definition 2), or over a located stage, every step below one of its operations, a division, a coefficient fold or a bracket comparison: the stage's distinct roots have a computed pair separation floor and a computed radius, by ground arithmetic alone.*

The magnitude carrier. The magnitude of a balance pair is its members' larger against smaller, the sorted pair $|\langle u : v \rangle| := \langle \max\{u, v\} : \min\{u, v\} \rangle$ at the trichotomy read, one value at every pair (a summand on both members fixing it), its value at unequal members the margin, the exceeding member's gap (Definition 1). A sum's magnitude sits at or below the magnitudes' sum, and a product's magnitude is the magnitudes' product, a memberwise swap of either datum swapping the product's members and fixing every magnitude. So a polynomial's evaluations at points of magnitude at or below a bound are capped by the coefficient fold, the coefficients' magnitudes folded against the bound's powers, one read over the occupancy family.

The squarefree part, and the radius. The division descent, iterated divisions of Definition 2, computes the greatest common divisor $c(P, P')$: the descent's last occupied remainder, monic at the rescaling by its leading coefficient (each coefficient the cofactor at the leading one), a factor of P and of P' and greatest as named. At the integer representatives the rescaling enters as data: the divisor at its primitive list with the stated positive top, the cofactors at their clearing multipliers, and the division at a stated top the cleared descent, $\ell^k P = qD + r$ at the top ℓ , one scale per eliminated key, the exactness the remainder's read; each remainder strips to its primitive list at its coefficients' one shared count, the walk's rows at their stated multipliers with the strip absorbed there, and every certificate reads at the cleared data. The squarefree part S is the cofactor at $c(P, P') S = P$, and the Bézout pair $uS + vS' = 1$ arrives with the descent at (S, S') ending at a unit: P 's roots are S 's, each simple there. Every root over the stage sits in the radius segment, between B 's balance partner and B at the pair $B := [c + \mathcal{H} : c]$, c the leading coefficient's margin (the top key's coefficient of unequal members, the degree's own read) and $\mathcal{H}$ the height, the largest coefficient magnitude, the leading term dominating at every point of margin at or beyond B .

The separation. Between two roots over the stage, at the stage gap G at $\alpha + G = \beta$, the binomial identity per occupied key at $i \geq 2$, the constant key's read equal at both points and the key-one read the gap's own join $\beta = \alpha + G$,

$$\beta^i = \alpha^i + i \alpha^h G + G^2 \sum_{a+b+2=i} (a+1) \alpha^a \beta^b \quad (h+1=i),$$

summed against the coefficients, reads $S(\beta) = S(\alpha) + G S'(\alpha) + G^2 T$ at the collected tail $T :=$

$\sum_i s_i \sum_{a+b+2=i} (a+1) \alpha^a \beta^b$; at V the coefficient fold of the Bézout witness v at the radius and the curvature fold $\Lambda := \sum_i \binom{i}{2} |s_i| B^e$ over the occupied keys at the joins $e+2=i$, $|s_i|$ the coefficient's magnitude, the stage's distinct roots satisfy

$$\Lambda V G \geq 1, \quad 2 \max\{\Lambda V, 1\} \delta = 1,$$

one pair comparison and one cofactor per polynomial (Definition 1), δ below the separation and at most $[1 : 2]$; the radius B is the leading-term bound. The object is a distance between roots of ground polynomials, and such a distance is a separation bound, which is explicit.

Proof. The magnitudes. At $\langle u : v \rangle$ and $\langle u' : v' \rangle$ with $u = v + a$ and $u' = v' + b$ the product $\langle uu' + vv' : uv' + vu' \rangle$ reads $uu' + vv' = uv' + vu' + ab$, its magnitude ab , the magnitudes' product, and at $u' + b = v'$ the product's members swap, the magnitude ab again; the sum's members differ by $a + b$ at like sides and by the gap of a and b at unlike sides, at or below $a + b$, the further cases at the members' trichotomies. *The descent.* Each remainder joins the two above it through its division, so the last is a factor of every earlier one, the two seeds among them; and a common factor g of the two seeds is a factor of every remainder down the descent, $P = SQ + R$ at the cofactors $gP_1 = P$ and $gS_1 = S$ reading $R = g(P_1 + \tilde{T})$ at $T = S_1Q$, the check the memberwise swap over the balance pairs, the last remainder among them. An irreducible factor q of S has some multiplicity $m \geq 1$ in $P = q^m R$, the remainders of q' (the derivative's degree drop) and of R (the exact multiplicity) at q occupied and their Bézout witnesses composing to an occupied remainder of $m q' R + q R'$ at the count m a natural, $q R'$'s summand the sum's unit there; at $m = 1$ the derivative is that read outright, $P' = m q' R + q R'$, and at the join $m' + 1 = m$ it is $P' = q^{m'} (m q' R + q R')$ at multiplicity exactly m' : the greatest common divisor's q -part sits one power below P 's, the unit at $m = 1$, so the cofactor S reads q at one, and at $S = qT$ the remainders of T and of $S' = q'T + qT'$ at q are occupied: P 's roots are S 's, each simple there. *The radius.* At a point of margin $y = 1 + g$ at or beyond B , with k the top key, the geometric telescope $g \sum_{i < k} y^i + 1 = y^k$ reads the tail's cap $\mathcal{H} \sum_{i < k} y^i$ below $c y^k$ at the radius' read $c g \geq \mathcal{H}$, so the leading term dominates and the evaluation reads unequal members on the leading term's own side, the tail joined at a magnitude below the term's. *The separation.* At a root α the unit witness $v(\alpha) S'(\alpha) = 1$ reads at the magnitudes: V the coefficient fold of v at the radius, so the margin of $S'(\alpha)$ cross-multiplies to margin $\cdot V \geq 1$. The two roots' equal-membered reads leave $G(S'(\alpha) + GT)$ of equal members, so at G 's unit witness $S'(\alpha)$ and GT are balance partners, and the margin of $S'(\alpha)$ sits at or below $G\Lambda$ at the curvature fold, the splitting count $\sum_{a+b+2=i} (a+1) = \binom{i}{2}$ and both roots capped by the radius: so $\Lambda V G \geq 1$. $\square$

Lemma 36 (The box read at several coordinates). *At a polynomial P over the pairs in the coordinates $y_1, \dots, y_k$ (Definition 2, one variable joined at a time) and a box, per coordinate a segment $[a_i, b_i]$ of pairs at $a_i \leq b_i$, the box read is the comparison $P(y) \geq 0$ at every point y of the box, y_i between a_i and b_i at every i . At a cut list per coordinate, finitely many pairs from a_i to b_i in order, the pieces are the products of consecutive cut segments, each with its base corner, the lower ends, and its widths w_i , the segments' gaps; at B_i the larger of the magnitudes of a_i and b_i (Theorem 16's magnitude carrier) the derivative fold M_i is the coefficient fold of the derivative $\partial_i P$ in y_i (Definition 2) over its monomials at the bounds $B_1, \dots, B_k$, one read per coordinate. Then the box read holds where every piece's base corner clears the priced widths,*

$$P(\text{base}) \geq \sum_{i \leq k} w_i M_i \quad \text{at every piece,}$$

finitely many cross-multiplied comparisons, and holds strictly above the sum's unit on the box where each comparison does. The Taylor fold of a polynomial P in one coordinate at a piece $[a, a + w]$

is $\tau_P := \sum_{j \geq 1} |P^{(j)}(a)| w^j [1 : j!]$, the magnitudes of the finite Taylor read's keys at the foot (Theorem 16's magnitude carrier), and P 's every value on the piece sits between $P(a)$ less τ_P and $P(a)$ joined to τ_P ; at a polynomial in several coordinates the fold joins coordinate by coordinate along the telescope of the proof, each step's fold at the earlier coordinates' pieces. The interval read of a formula on a piece composes its atoms' segment reads, each a segment of pairs holding the atom's every value on the piece, through the displayed operations: a sum's segment is the ends' sums, a product's the least and the largest of the four cross products, a natural power's the ends' powers at a segment on the upper side, and a quotient's the least and the largest of the four cross quotients at a divisor segment off the sum's unit on one side; the atoms are a coordinate at its own segment, a series with positive coefficients at an argument segment on the upper side between its ends' values, and a polynomial in a coordinate at its Taylor fold; the box read holds on a piece whose interval read sits on its upper side or at equal members. The halving cover of a box at a stated region of the box, a stated depth, a natural, a stated clip, a read of a piece returning a piece inside it holding the region's points of the piece, or the vacant read at a piece off the region, and a stated per-piece comparison, is the recursion: a piece reads its clip, the vacant read holding outright; a clipped piece at which the comparison holds is a leaf; a clipped piece at which the comparison refuses at an occupied depth is cut at the midpoint of the coordinate whose width against the coordinate's range in the box is largest, the earliest at a tie, into two pieces read at the depth's predecessor; and at the vacant depth the refusal is the cover's own. The cover holds where every piece is vacant or a leaf, one decidable read, and then the box read holds at every point of the region at the comparison each leaf clears, the priced read or the interval read. The leading-key read at a polynomial P in the box's coordinates joined to a further coordinate ε on the segment from the sum's unit to a pair ε_0 : at j the least key of ε occupied in P , $P = \varepsilon^j (P_j + \varepsilon R)$ at P_j the key's coefficient, a polynomial in the box's coordinates, and R the further keys' fold, and the read $P \geq 0$ holds at every point of the box and every ε at or below the least of ε_0 and $[m : F]$ where P_j less a positive pair m holds the box read (at one coordinate the segment count vacant with one endpoint value positive, Lemma 38) and F is the coefficient fold of R at the box's bounds and ε_0 (Theorem 16's magnitude carrier); and the grid read at a positive pair m : at the cut lists cutting every coordinate at pieces of widths at or below $[m : \sum_i M_i]$, the priced comparison holds at every piece whose base corner's value sits at or beyond m , so the box read holds where P sits at or beyond m at the finitely many base corners. On a range unbounded in one coordinate the read joins the box read to the leading-term bound: at P read in y_1 with the further coordinates in their segments, at a top key n whose coefficient c_n , a polynomial in the further coordinates, keeps its upper side on their box at a margin c at or beyond a stated pair, and at $\mathcal{H}$ the largest coefficient fold of the further keys over their box, P keeps its upper side at every y_1 at or beyond the radius $[c + \mathcal{H} : c]$ (Theorem 16's leading-term bound), so the read on $[a_1, \infty)$ is the box read on $[a_1, [c + \mathcal{H} : c]]$ joined to the radius's comparison.

Proof. At a point y of a piece with base corner a , the difference $P(y) - P(a)$ telescopes through the coordinates, the i -th step moving y_i alone at the earlier coordinates at y and the later at a ; each step is a one-variable difference, the binomial identity per occupied key of y_i (Theorem 16), $y_i^n - a_i^n = (y_i - a_i) \sum_{j+l+1=n} y_i^j a_i^l$, the fold's magnitude at or below $n B_i^{n-1}$ at both points of magnitude at or below B_i , so the step's magnitude sits at or below $(y_i - a_i)$ against the coefficient fold of $\partial_i P$ at the bounds, at or below $w_i M_i$ (the further coordinates' powers at their bounds), and the difference's magnitude at or below the widths' fold. So $P(y)$ sits at or beyond $P(a)$ less the fold, at or beyond the sum's unit at the piece's comparison, and strictly above it where the comparison is strict. The Taylor fold: $P(a + h) = \sum_j P^{(j)}(a) h^j [1 : j!]$ is one identity of the polynomial at every h (Definition 2's derivative, the binomial identity per key), and at h at or below w the keys from one on sit within their magnitudes' fold at w . The interval read: an atom's every value on

the piece sits in its segment read (a series with positive coefficients rises with its argument key by key; a polynomial at its Taylor fold), and the displayed operations at two segments read their members' segments, the sum at the ends' sums, the product and the quotient at their extremes over the two segments, attained at the ends since each is monotone in each member at the other fixed, the power so; so the formula's value at every point of the piece sits in its interval read. The halving cover: by induction on the depth, every point of the region inside a piece sits inside a leaf's clip, the cut's two pieces holding the piece's points between them and the clip holding the region's, so the comparison at that leaf reads the point. The leading-key read: at every point of the box and every ε at or below ε_0 the fold R 's magnitude sits at or below F , so P sits at or beyond $\varepsilon^j (m - \varepsilon F)$, at or beyond the sum's unit at $\varepsilon F \leq m$. The grid read: at every piece the widths' fold $\sum_i w_i M_i$ sits at or below $[m : \sum_i M_i] \sum_i M_i = m$, at or below the base corner's value. The unbounded range: at y_1 of magnitude at or beyond the radius the top term's magnitude $c y_1^n$ clears the further keys' fold $\mathcal{H} \sum_{j < n} y_1^j$ at the geometric telescope, $\langle y_1 : 1 \rangle \sum_{j < n} y_1^j = \langle y_1^n : 1 \rangle$, the gap of y_1 above one at or beyond $[\mathcal{H} : c]$ at the radius (Theorem 16's leading-term bound), at every point of the further coordinates' box, the coefficient bounds read at the box. $\square$

Lemma 37 (The stage splits every polynomial). *Every polynomial of degree at least 1 over a located stage splits, over a stored tower (Lemma 32), into located linear factors and definite quadratic factors at pair data (h, q) , the root relation $z^2 + q = hz$, at the definiteness gap g at $h^2 + g = 4q$, the data tower elements, multiplicities by iterated divisions.*

Proof. Factorization closes over finite cofactor lists. Over $\mathbb{P}$ a polynomial reads at its primitive integer form, the pair coefficients cleared at one shared second member with the content collected, and a factor reads primitive as well, the contents splitting off (Gauss); a primitive factor of degree d is determined by its values at $d + 1$ integers of occupied value, a point of equal-membered value reading its linear factor outright, and each value is one of the finitely many cofactor reads of the primitive form's value there at either side, a natural or its balance partner: the value lists are finite, each selection interpolates to one polynomial, and the division tests it, the remainder of equal members exactly at a factor (Kronecker). Over an extension $K(\varepsilon)$ the norm $\text{Res}_x(S_\varepsilon(x), f)$ (Definition 3's resultant, in the variable x) maps f to the stage below; at a shift $z \mapsto z + \lambda\varepsilon$ whose norm is squarefree (the failing λ are roots of one resultant, finitely many, a scalar-pair λ off them read directly), the greatest common divisor of f with each factor of the shifted norm, the division descent's end, recovers f 's factors over the extension (Trager), descending to $\mathbb{P}$. The remainder lists at an irreducible factor (Definition 2) are a discrete field, every occupied element at its Bézout unit witness, so iterating the adjunction reads a full formal root list, the *splitting data*: finite algebra over the stage, in which a root of a product is a root of one factor.

The quadratic extensions. A located positive c adjoins one extension, the remainder lists at the relation $z^2 = c$'s polynomial (Definition 2); the *adjoined root* at $z^2 = c$ is the linear monomial's class, located at the bracket of c 's sorted display against 1, the two side reads the refinement's seed (Theorem 15(1)) with the equality outcome naming the root outright, so the extension is a located stage again (Lemma 32). The *located pairs* are the remainder lists at $z^2 + 1$ (Definition 2), (a, b) written $a + ib$ at i the linear monomial's class, the *key swap* the linear key's memberwise swap, an automorphism of the carrier's reads. A located pair at b of unequal members has a square root $u + iv$ on three such extensions: w the adjoined root at $z^2 = a^2 + b^2$, then over $K(w)$ u the adjoined root at $z^2 = [1 : 2](w + a)$ and v the adjoined root at $z^2 = [1 : 2]g$ at the gap g at $a + g = w$, the order's witness (w^2 exceeds a^2 at b 's occupied square, so w exceeds a and its balance partner), v 's side one bracket read against b 's; the tower reads $u^2 = v^2 + a$ and $(2uv)^2 = b^2$ outright, so $(u + iv)^2 = a + ib$. A quadratic over located pairs at the root relation $z^2 + q = hz$

splits over located pairs, the trichotomy of its *discriminant datum* D at $D + 4q = h^2$: at D 's second component of unequal members the extensions read its square root $u + iv$, the two roots the data at $2z = h + (u+iv)$ and $2z + (u+iv) = h$; at the second component of equal members the first component D_1 's trichotomy reads: on its upper side, e the adjoined root at $z^2 = D_1$ with the roots the data at $2z = h + e$ and $2z + e = h$; at equality, the double root, the datum at $2z = h$; and on its lower side, e the adjoined root at $z^2 = m$ at D_1 's margin m , the roots the data at $2z = h + ei$ and $2z + ei = h$.

The Laplace descent. The claim: a stage polynomial p of degree $k \geq 1$ has a located root or a definite quadratic divisor, verified by the division. Induct on the degree's 2-adic valuation ν . At odd degree the leading term dominates beyond Theorem 16's radius, the side reads at the radius and at its balance partner differ, and a located root's designation follows at every width, Theorem 15(1)'s refinement at the entry's side reads. At positive ν , at a scalar pair t the resolvent

$$B_t(x) := \prod_{l < l'} \langle x : z_l + z_{l'} + t z_l z_{l'} \rangle$$

over p 's formal root list has symmetric coefficients, hence stage coefficients through the elementary reads (the lexicographic descent on symmetric monomials, one generic identity), and degree $\binom{k}{2}$ at the valuation join $\nu' + 1 = \nu$: the induction hands each t a located root or a definite divisor, either way a located pair β_t (the second component of equal members, or the adjoined root at $z^2 = [1 : 4]g$ at the definiteness gap g), a witness-verified root of B_t . Take the splitting data over the one tower with every β_t and i : there B_t is the displayed product and β_t , a root of B_t in a field, is one factor's own root: β_t meets a pair (l, l') . Over $\binom{k}{2} + 1$ distinct scalar pairs t two meet one pair, and there, the lesser parameter written t' at the gap w at $t' + w = t$,

$$wq + \beta_{t'} = \beta_t, \quad h + tq = \beta_t$$

name the located pairs q and h , each one value by the positive gap's rescaling, the displays' own joins at the meeting pair ($\beta_t = z_l + z_{l'} + t z_l z_{l'}$ at both parameters), with the relation $z^2 + q = hz$'s quadratic a factor of p ; the finitely many parameter pairs are each tested by the division, and the display guarantees a passing test. At a passing pair the discriminant trichotomy splits the quadratic, giving a root $a + ib$ of p with a, b tower elements. At b of equal members, a bracket read, a is a located root; at b of unequal members the key swap fixes p and maps the root to its swap image, the division of p at $z^2 + (a^2 + b^2) = 2az$ reads its remainder $Rz + T$ with $Ra + T$ and Rb of equal members at the two roots, so R and T read equal members at b 's unit witness: a definite divisor. Iterated divisions close the split. $\square$

Lemma 38 (The located sign is a count). *Over a located stage K , let S be squarefree of degree N with its split over a stored tower (Lemma 37). A stage value at a root is a balance pair, its side the members' order read, the upper at the first member exceeding and the lower at the second (Definition 1); the pairing below is additive over a value's site, $H_u = H_v + H_p$ at $u = v + p$, so the weight p is the site datum and the side is its read. The carrier is S 's remainder lists (Definition 2), length- N coefficient lists with the convolution's remainder as product, and M_f , the remainder multiplication at a polynomial f , is the $N \times N$ matrix whose column j is the remainder of $f x^j$. At a weight $p \in K[x]$ the Hermite pairing H_p is the symmetric form*

$$H_p[i, j] = \text{tr } M_{x^{i+j}p}, \quad \text{the keys } i, j \text{ below } N,$$

tr the matrix trace, every entry one elimination read over K (Definition 3).

(i) The trace adds over the split's blocks. The split's factors are coprime (S squarefree), their Bézout reads yield idempotent remainders, e^2 and ee' reading e and equal members, and the CRT list is a unit congruence's base change mapping every M_f to blocks, one per factor, each the factor's own remainder multiplication, and the trace adds over the blocks: a linear factor's block reads $f(\varepsilon_i)$, and a definite factor's block reads f 's remainder $\alpha + \beta x$ there at $2\alpha + \beta h$, M_1 the identity and $\text{tr } M_x = h$ at the factor's data (h, q) , its companion trace and constant (Lemma 37).

(ii) The pairing splits, and a block is one orientation read. The CRT list is a unit congruence of H_p onto the blocks' forms over the tower, the reversal count and the rank preserved (Lemma 30, the rank the kernel block's complement; a stage split is a tower split, so the tower's count is the stage's own). A linear block at ε_i is the one entry $p(\varepsilon_i)$ at a list member's square: at $p(\varepsilon_i)$ of unequal members it is one reversal exactly where the value's side is the lower, the reversal count at order one and the stage's own comparison; at p on the root it is the kernel block. A definite block, at its factor's data (h, q) and the definiteness gap g at $h^2 + g = 4q$ (the definite read of Lemma 37), is 2×2 , and it reads one unit per side, the mixed block, exactly at p off the factor, by one additive identity of the block's traces at p 's remainder $\alpha + \beta x$ there,

$$4 \text{tr}(M_{xp})^2 = 4 \text{tr}(M_p) \text{tr}(M_{x^2p}) + ((2\alpha + h\beta)^2 + g\beta^2)g,$$

and at p on the factor the block is the kernel block.

(iii) The bracket's pencil pairs the reads, and the counts close. At a located root ε with isolating bracket (l, l') of balance-pair endpoints off S 's roots, its width w at $l + w = l'$, and $p(\varepsilon)$ of unequal members, the localized read is the form pair

$$\mathbf{B} := (H_{x^2p} + ll' H_p : (l + l') H_{xp}),$$

the bracket's pencil at the three pairings, a balance-pair coefficient reading cross-added with its second member's multiple on the pencil's other side (Definition 1), its reversal count and rank its split's (Lemma 30), the pencil splitting per factor. The block at ε reads its one reversal exactly at p on its upper side at ε , at every other located root its reversal read is p 's block's own, and at a definite factor the pencil's block is (ii)'s at the remainder shifted by the bracket comparison, $x^2 + ll'$ against $(l + l')x$, the kernel block exactly with p 's. The bookkeeping closes on two natural identities:

$$p \text{ sits on its upper side at } \varepsilon \text{ exactly at } \text{rank } H_p + 2 \text{rev } \mathbf{B} = \text{rank } \mathbf{B} + 2 \text{rev } H_p + 2,$$

and on its lower exactly at $\text{rank } H_p + 2 \text{rev } \mathbf{B} + 2 = \text{rank } \mathbf{B} + 2 \text{rev } H_p$. At the unit weight and any endpoint pair off S 's roots the split reads the segment count, the count of the located roots inside the closed segment, the gap at $\text{rev } H_1 + \# = \text{rev } \mathbf{B}$, the isolating bracket the one-root instance; at a polynomial off the squarefree site the three pairings read its distinct roots, the multiplicities entering as positive weights at the roots' moment folds, so the segment count reads the distinct roots inside, the squarefree part's own count. With the entries elimination data over K and the two counts one split each, a located sign is finitely many reads of the stage below, descending to the ground pairs' own comparisons; Jacobi's minor rule is the read's nonsingular instance (Lemma 30).

Proof. (i) The trace reads one value at every spanning list ($\text{tr } AB = \text{tr } BA$), so it adds over the blocks. (ii) Cross-block entries are traces of actions by ee' , whose remainder is the sum's unit, so the CRT list is a unit congruence of H_p onto the blocks' forms over the tower. A definite block reads one unit per side exactly where its mixed entry's square exceeds its diagonal entries' product (Lemma 30's 2×2 read), and the displayed identity is one quadratic-form identity in the remainder (each side quadratic, $\text{tr } M_{x^k p}$ linear in it), settled at the three remainders 1, x and $1 + x$ by (i)'s companion rule: the excess is a square with the gap's multiple of a square, positive by its shape

at a remainder off the sum's unit, and at p on the factor every entry is the sum's unit action's trace. (iii) The three pairings admit one CRT congruence (the blocks' multiplication matrices are shared), so the pencil splits per factor. At a linear factor the datum is $p(\varepsilon_i)$ times the *bracket comparison* at ε_i , of $\varepsilon_i^2 + ll'$ against $(l + l')\varepsilon_i$; the sides' gap is one gap product at every position, $d(w + d)$ for the first side at $t = l' + d$ or $t + d = l$, and $c c'$ for the second at $t = l + c$, $c + c' = w$: above exactly off the closed bracket, below exactly inside, so the block at ε reads its one reversal exactly at p on its upper side there and every other located root's reversal read is p 's block's own. At a definite factor a common factor with the bracket's own data would read the sum $g + w^2$ at the sum's unit at the endpoints' pair $((l + l')^2$ against $4ll'$ at the width's square), or a stage root t , whose equality $4t^2 + h^2 + g = 4ht$ reads $d^2 + g$ of equal members at $2t = h + d$ or $2t + d = h$, against the summands' shapes both times; so the shift is coprime to the factor, its Bézout read mapping p 's remainder to the shifted one by a unit. In the two identities every other located root at p 's support adds one rank to each and equal reversal reads, every definite factor off p two ranks and one reversal to each, the kernel blocks off both counts, and ε adds one rank to each with the pencil's reversal at p 's upper side and H_p 's at its lower. At the unit weight every located root inside the closed segment adds one reversal to B beyond H_1 's, every root outside and every definite factor equal reversals to both, so the count is the displayed gap. $\square$

Lemma 39 (The deck symbol, and the interior factor). *A polynomial p over a located stage with the deck symmetry, its coefficients equal at every index pair joining to $2m$ and its keys beyond $2m$ at the sum's unit, and its unit-monomial coefficient of unequal members is the display of its deck symbol, a stage polynomial P of degree m in the deck's spectral variable:*

$$p = \sum_{i+j=m} P_j z^i (z^2 + 1)^j,$$

the symbol one exact back-substitution and the involution the identity on the symbol. A root pair of the deck relation $z^2 + 1 = wz$ (Definition 4) at a located w -datum is one quadratic extension (Lemma 37), the two roots at product one, the constant coefficient, and the interior member is the one at $z^2 \leq 1$, one pair comparison, strict off the crossing values, the relation's double-root values at $w^2 = 4$, the interior margin the crossing read's own, $\langle 1 : z^2 \rangle^2 = z^2 \langle w^2 : 4 \rangle$; a definite w -datum's roots enter over its pair's quadratic extensions, the interior member at the modulus comparison. The band read is P 's segment count at the crossing endpoints: the bracket pencil of Lemma 38 at those endpoints reads every root of P off the closed crossing segment, reversal-count reads throughout, with the endpoint reads P 's own, off equal members. At an integer display off the monic carrier the count runs at the cleared variable: the display scales to the monic at the top's magnitude, each coefficient against the magnitude's power at its key's gap below the top's predecessor with the members swapped at a top on its lower side, and the scaled display reads its roots at the display's own rescaled by the magnitude, the display's evaluation and the cleared one composing by formula at the homogeneity principle (Definition 1): the segment count reads at the crossing endpoints rescaled to the doubled magnitude. With the band read clear, over P 's split at the stored tower (Lemma 37),

$$p = c \, p_{\text{int}} \, p_{\text{ext}},$$

p_{int} the monic product over the located interior members, one per w -datum at its multiplicity, degree m , existence by construction, and p_{ext} the product over their ι -images, the pairing $z z' = 1$ per extension, the degrees split evenly, m each. The definite factors are the point: they enter at their pair data, and every side is Lemma 38's count.

Proof. Each deck monomial is its own coefficient join $(z^i(z^2 + 1)^j$ reading one at the top pair $i + 2j$ against i , equal ends joining to $2m$), the map from P 's coefficients to p 's top coefficients unitriangular, so the symbol is one exact back-substitution. The interior margin's identity is the relation's square at the sum $z^4 + 1 + 4z^2 = w^2z^2 + 2z^2$. On the relation each symbol monomial reads $z^i(z^2 + 1)^j = z^m w^j$, so p is z^m against the symbol's read at w ; each split factor $\langle w : w_k \rangle$ clears at z to the root pair's monic quadratic, $z \langle w : w_k \rangle$ reading $z^2 + 1$ against $w_k z$; so the pair quadratics' product against c , the symbol's leading coefficient and the display's unit-monomial read, is p : the degrees split evenly, m each. $\square$

Theorem 17 (The pencil is a symmetric pair matrix on every window). *Every level read the development takes is on a window compression, finite-dimensional with pair entries (Proposition 2): E is diagonal in the label index (Definition 10) and M is multiplication by $\sum_p \chi_L(U_{\partial p})$, involution-fixed at the closure (Construction 6; Definition 10), against the pairing identity of Proposition 5, hence symmetric, with gram positive definite by construction (Definition 8). So the pair $(\alpha E : \beta M)$'s site datum $H([\alpha : \beta])$ is symmetric there at every end-weight pair, its entries pair data at pair weights. The pencil interface. A window's pencil enters every derivation through its fields alone, one interface at every window, the label index's product their instantiation (Definition 9): the order, the window list's count (Proposition 2); the electric member, block diagonal over the index at the contents' multiples of the gram's blocks with the unit line at its head, the operator's form against the pairing (Definition 10); the gram, block diagonal over the index at the fibers' pairings with the unit line at one, positive definite by its shape (Definition 8), each block the presentation Gram of the fiber's stated list (Construction 9), the window's pencil at one positive rescaling to integer entries fixing every count (Lemma 30); and the magnetic member, the plaquette terms' sum, each term symmetric and capped two-sidedly as a form at the closure's dimension fold d_L (Lemma 18), its row at a configuration within the plaquette's changed-edge targets (Definition 9; the far reads factorized at the unit character, Lemma 14) and its entries the plaquette's multiplication read at the fibers' stated lists in the presentation (Construction 9); and the terms' transport: a label-graph isomorphism of two windows' plaquette neighborhoods (Construction 7) moves a state of the one window to a state of the other, the configuration moved along the links, the label dualized at a reversed traversal, and each vertex member relabeled along the incident ends' permutation, the slot power's place action (Construction 12; Construction 1) at the ends' slots matched column by column: at a traversal kept a column against the column of its own length, at a reversed traversal a column against the dual label's complement column, the moved slot's action the source slot's own on the one coordinate family (the dual block at the exchanged actions, Construction 12), and a tie column at a reversed traversal read at the star, the source letters against the complementary set at the swap grading's side cleared at the column length's factorial (the through pairing's star, Construction 12); at a member's table the columns are the fundamentals' factors, matched at a traversal kept by index and at a traversal reversed to the dual fundamental, and every slot of a reversed end is read at the dual pair's identification, cW from an outgoing source end and $c^T W$ from an incoming one, each at the clearing ν per slot with the pairing scaled at the clearing's square, the isometry read (Construction 13), the star the letters' instance; the relabeled member is a state of the moved window, in the moved vertex list's span at the Gram-dual solve (Lemma 3), the moved states are independent at the pairing kept and at the fiber's count, one stated list of the moved window against the fiber's own (Proposition 2), and every pairing read between two moved states is the read between the states, the gram's and each term's, so the moved window's pencil is the pencil read at the moved states, the two stated lists' pencils congruent at the solve's coefficients, and every count is one value (Lemma 30); a region's own action is the instance at every plaquette, the plaquette list permuted; at a loop window the fields read off the fusion counts, $M[x, y] = \sum_{\lambda \in \bar{L}} N_{\lambda y}^x$,*

the closure fold at the orthonormal characters (Proposition 8; Lemma 17), and in every window two loop states of one plaquette, the characters at the coevaluations' counts (Construction 9), read that plaquette's term at the fusion count and the gram at one, the characters' product read at the window's evaluation (Proposition 8; Lemma 14).

Proof. The transport. The relabeled member is a member of the moved vertex's block tensor with every raising image the sum's unit, the units acting slotwise at the matched actions with the star and the identification each intertwining the two actions at its slots (Construction 12; Construction 13; Lemma 8(ii)), so it lies in the moved vertex list's span at the Gram-dual solve, a state of the moved window; and the evaluation is one value under the relabeling at the place action's transpose identity (Construction 8; Construction 1; Definition 9; Lemma 14), so every pairing read between two moved states is the read between the states. $\square$

19 The chain's invariant form

Theorem 18 (The fiber family is a bounded chord pencil). *At the banded rows, every pencil entry within the locality band of its support (Lemma 14), and the module-pair coupling $|v|^2 = 2 + t$ at the chord t of the translation, its deck relation $T^2 + 1 = tT$ (Definition 4), the momentum-fiber pencil is a chord pencil of bounded degree per grade: the fiber enters each graded block through finitely many translation powers per direction, T_i^j at the direction's own translation, the power bounded by the grade's spatial support, each adjoint-paired read a deck-family polynomial at the direction's chord (Definition 4): the momentum variables are the chords $t_1, \dots, t_d$, one per direction, the momentum point K their list (Lemma 40). The chain object is therefore one algebraic family over the $(K; [\alpha: \beta])$ base at the chords, with integral chord data; its divisor records the band, the gap, and every "assembled" coefficient as reads at the ends of $\mathcal{C}$. Its fiber symbol is the family's characteristic Laurent polynomial $\mathcal{S}(K; k, r)$, the pair $(k: A(K))$'s site-datum determinant at the level k scalar against the gram's determinant, one value at every stated list of the fiber, of finite Newton polytope, with polynomial-pair coefficients in the residue; A is the momentum fiber of a symmetric pair pencil (Theorem 17), so a separation's block is the reversed separation's transpose, the symbol's own deck symmetry.*

Proof. A pencil entry between two states reads within the locality band of their supports (Lemma 14), so a graded block's entries at two translates read the sum's unit beyond the grade's spatial support, finitely many translation powers per direction, and on a chord block the adjoint-paired reads $T^b + (T^*)^b$ are the deck families' values at the located chord (Lemma 40), integral in the chords. Two stated lists of the fiber read congruent pencils (Theorem 17's transport), and at a unit congruence the determinant of the level pair's site datum against the gram's determinant reads one value, the congruence's determinant's square clearing against the grams' determinants at the product read (Lemma 30); the polytope is finite at the localities above, and the coefficients are polynomial pairs in the residue at the entries' reads (Lemma 25). The pencil is symmetric (Theorem 17), so the block at a separation is the transpose of the block at the reversed separation, and the deck relation exchanges T with T^* at the chord fixed (Definition 4): the symbol's deck symmetry. $\square$

Lemma 40 (The window floor fibers over momentum). *On a torus window of odd side L , the fibering's family the odd tori at $L \geq 5$ and the commuting and chord reads below at every odd side, the translations (Construction 7) commute with H (Definition 10; Theorem 17's interface); a translation T permutes the label index isometrically, an isometry of the pairing whose pairing*

adjoint is the complementary step's translation, the witness at $T^*T = 1$ at T^* the pairing adjoint, gram $T^* = T^\top$ gram at the stated list, with $T^L = 1$ along its direction, the window list moved with the index, a moved state the source's members relabeled along the incident permutation (Theorem 17's transport field) and the contents fixed at every slot. Its chord operator is $S := T + T^*$, symmetric, commuting with H and with the gram, and T reads the deck relation at it,

$$T^2 + 1 = ST$$

(Definition 4). The geometric word reads its deck symbol at the operator,

$$\sum_{a+b+1=L} T^b = T^m P_L(S) \quad \text{at } 2m+1=L, \quad P_L := 1 + \sum_{j \leq m} p_j,$$

P_L monic integer of degree m at the deck families (Definition 4; Lemma 39), and the word's read at $T^L = 1$ is

$$S P_L(S) = 2 P_L(S).$$

Divided at the monic $\langle x:2 \rangle$, P_L leaves its Horner read there, $P_L = q \langle x:2 \rangle + L$ (Definition 2), the remainder L the word's value on the unit fiber, the fixed vectors' block: membership in the kernel of the pair $(S:2)$'s site datum is the fixed read $Tv = v$. The window splits orthogonally into the chord blocks: the kernel of the pair $(S:2)$'s site datum, its members the reads $Sv = 2v$ at the join, and the kernels of the pair $(S:t_j)$'s site data at P_L 's roots t_j , the roots located and the factorization linear (Lemma 33), each root separated and bracketed by ground arithmetic (Theorem 16) and located on the stage (Lemma 32), the tower the split stores Lemma 37's; each kernel is a spanning list by elimination (Definition 3) with the idempotents the division displays' witnesses (Lemma 38's device), the splitting is orthogonal with every complement invariant (Lemma 1), and each block is invariant for H and for the gram. A fiber is a chord block. The count splits with the splitting (Lemma 30). The directions' chord operators commute pairwise and each maps every further direction's chord blocks into themselves, so the window splits orthogonally into the joint chord blocks: the momentum point $K := (t_1, \dots, t_d)$ is the chord list, one coordinate per direction, each the chord 2 or a located root of P_L along its own direction, the K -fiber the joint block at it, and the counts split with the joint splitting as with each step's (Lemma 30). At each window the decimation reads the fiber's total counts through its stated pivot witnesses, with every local entry transported across windows at the label-graph isomorphism (Theorem 17's interface; Lemma 14). This family runs at odd side $L \geq 5$, where the torus's own comparisons clear its classes off the free orbits, the configurations fixed by a translation off the identity: a fixed configuration holds a support piece, its links joined at shared vertices, that the translation fixes, a winding support, or holds a piece the translation moves beside that piece's image, two pieces at translates of each other; a winding support's content is at least $L \hat{c}_2(f)$ at the member floor $\hat{c}_2(f)$ (Lemma 9) against the plaquette level's $4 \hat{c}_2(f)$ (Proposition 7; Lemma 15), the one comparison $5 > 4$ at every member, and two pieces at translates read twice the plaquette level at the least, each piece an occupied configuration at content at or beyond the level (Proposition 7); so the fixed classes read at or beyond the level at every window of the family, and the directed windows are free of them outright, a winding support wrapping a cycle of the torus (Definition 15) and a finite support moving under every translation off the identity. ε_0 is the least first jump over the fibers, its designation certificate data (Theorem 15), the unit's line sitting in the unit fiber, and the window cut of Definition 11 holds at level κ once each fiber's counting function is flat across $(\varepsilon_0, \varepsilon_0 + E_0\kappa)$, a fiber at the first jump reading its second jump at or beyond the top and every other fiber k its first,

$$\text{rev} \left(H^{(k)} : (h + E_0\kappa) G^{(k)} \right) = 0$$

at a pair $h \geq \varepsilon_0$ (Theorem 15): together the reads are $\text{spec}(H) \subset \{\varepsilon_0\} \cup [\varepsilon_0 + E_0\kappa, \infty)$ on the window, the cut at every window ground at once (Lemma 20). On a chord block at a located root t of P_L , S reads the root itself and T the deck relation at the located chord, $T^2 + 1 = tT$; the adjoint-paired reads collect at the deck families, $T^b + (T^*)^b = p_b(t)$ on the block (Definition 4), so every such entry is a chord polynomial, its degree bounded by the grade's spatial support (Lemma 14). The block is doubled: its datum A at $T = T^* + A$ squares against the band read, $A^2 + 4 = S^2$, hence $A^2 + 4 = t^2$ on the block, and A 's adjoint is its memberwise swap, $A^* = \bar{A}$; P_L 's roots sit interior to the band, every root reading t^2 on its lower side against 4, so A^2 reads the pair $\langle t^2 : 4 \rangle$ off equal members, A 's determinant with it, and the band root s at $4s^2 + t^2 = 4$ adjoins one quadratic extension (Lemma 37), that read cleared; and the block's order is even, the doubling's read.

The deck families hold the doubling's own identity: at a free chord x ,

$$p_L(x) = \langle x : 2 \rangle P_L(x)^2 + 2,$$

one identity of the deck families, the chord free over the base and both members its polynomials (Definition 2; Definition 4); the partial sums read the word read's telescope additively (Definition 4), one join per step from the seed,

$$\langle x : 2 \rangle P_m + p_m = p_{m+1},$$

and the key list's fold collects to the symbol's own multiple,

$$\sum_{b < L} p_b = p_m P_L.$$

Per free translation orbit the doubling is witnessed and squeezed. Take a free orbit of the label index, its L translates $T^b w$ independent, and at a root t_j the deck-family column

$$u_j := \sum_{b < L} p_b(t_j) T^b w.$$

At the located root the powers wrap, $p_{L+b}(t_j) = p_b(t_j)$ and $p_{L-b}(t_j) = p_b(t_j)$ with $p_L(t_j) = 2$; $S u_j = t_j u_j$, the column occupied, and its A -image is its independent partner at the pairing reads, $\langle u_j, A u_j \rangle$ of equal members and $\langle A u_j, A u_j \rangle$ reading $\langle 4 : t_j^2 \rangle \langle u_j, u_j \rangle$ on its upper side, so $A u_j$ is occupied and perpendicular to u_j , the two independent members of the root block, the datum mapping each chord block to itself with $S A u_j = t_j A u_j$. The independence reads over the remainder lists at every side at once: the column's self-pairing reads $2L$, the image's self-pairing reads $\langle 4 : x^2 \rangle$ against it at the relation on the column, and the pair's Gram determinant reads $4L^2 \langle 4 : x^2 \rangle$ and is occupied over the remainder lists at every side, its witness against the symbol reading the constant $4L^3$ at the balance partner of the parity member, the symbol's remainder at the chord 2's partner, the product's unit at even m and its balance partner at odd; and the band symbol $\langle 4 : x^2 \rangle$'s remainder at every factor of the symbol in the stored tower is occupied. The unit fiber reads one member of the orbit, the orbit's own sum, and the orbit's count $1 + 2m = L$ squeezes the reads to equality: each root block reads exactly two members of the orbit, the pair $(u_j, A u_j)$, the block at doubled multiplicity, and per fiber the classes are the free orbits at the pair base, each member against its A -image, the row structure the gradings below read off this splitting. Over the residue the squeeze is one splitting read at stated lists: the pair base enters as the column's coefficient rows, one row per key below the residue's top against its image under the doubling datum, and the orbit sum joins them to an independent list at the orbit's own count, $1 + 2m = L$, so every orbit vector reads through the splitting at the elimination read (Definition 3), a root-datum kernel member through the pairs alone, its sum coordinate clearing at the word's value L off the unit, the division display's read.

The band congruence takes the block's list at pairs, each member against its A -image at the elimination's pair descent (Definition 3), the image columns rescaled at the band datum: a unit congruence off the crossings, mapping the block's entries to polynomials in t and s^2 , substituting at the root's identity, every count fixed (Lemma 30) and the gram's rescaled block s^2G positive definite: the fiber pencils' entries are one polynomial-pair list in (t, τ) over the residue at the root coordinate's magnetic weight τ^2 (Definition 10): each fiber floor is Lemma 31's count, a cell function of (t, τ) , constant off the root divisor of one explicit determinant of ground data and persisting at the divisor's points beside a positive-semidefinite cell. At fixed content cutoff the entries are one polynomial-pair list in (t, τ) over the residue at every L beyond the locality band (Lemma 14), the window entering through its sampled momenta and its separation range alone. The momenta are the geometric word's symbol roots, the word identity above read at the operator: the sampled momenta are the chord 2, the unit fiber, with P_L 's m located roots, one doubled block each, every root a designated bracket datum at the separation of Theorem 16.

Proof. The translations. E and M run over all links and plaquettes at coefficient one (Definition 10), E fixed at the contents and M at the terms' transport with the plaquette list permuted (Theorem 17's interface), so the translations commute with H . A fiber's coordinates are one digit per vertex of the region below the vertex's multiplicity, an untouched vertex at the product's unit (Definition 8), the moved configuration's multiplicities the source's along the induced vertex permutation (Lemma 14), and a moved state reads every pairing at one value across the move (Theorem 17's transport field), so the translation is an isometry with the complementary step its adjoint. The deck relation is the product $T^*T = 1$ collected at the chord (Definition 4); the word's symbol reads the partial sums keeping the top at the last family joined (Lemma 39), the powers matched about the middle collecting at the plus read $T^j + (T^*)^j = p_j(S)$, the root pair (T, T^*) of product one. Write $X := T^m P_L(S)$ for the word: at $T^L = 1$ the word is its own T -multiple, $TX = X$, and T^* against that read gives $T^*X = X$, so $SX = 2X$, and the orthogonal T^m clears at its witness $(T^*)^m$.

The unit fiber and the chord blocks. p_j at the chord 2 reads 2 at every key, the recursion's own read from the seeds, so P_L at 2 reads $1 + 2m = L$, the word's value on the unit fiber: at the datum D_1 of $1 + D_1 = T$ the product $T^*T = 1$ expands to $1 + D_1 + D_1^* + D_1^*D_1 = 1$ and the chord to $S = 2 + D_1 + D_1^*$, the two joining at $D_1^*D_1 + S = 2$, so at $Sv = 2v$ the read $\langle D_1v, D_1v \rangle$ is the sum's unit and D_1v with it at the positive pairing (Definition 8), the fixed read. That value is occupied at the odd side; the division display is the Bézout read, and the two factors of the split polynomial $\langle x : 2 \rangle P_L(x)$ are coprime, the display above reading it an annihilating polynomial of S . S is symmetric against the positive gram (Definition 8), so its site-datum determinant splits over located roots with a unit congruence to diagonal (Lemma 33), the located linear factorization the lemma's own display, and P_L 's roots are the occurring ones at that annihilation. A repeated root descends at the symmetry, a datum's square read against a member being the image's pairing against itself, so the distinct roots' linear factors split the window on their own, coprime at their separation, the idempotents the division displays' witnesses (Lemma 38's device, e^2 and ee' reading e and equal members). S is transpose-closed, so the splitting is orthogonal with every complement invariant (Lemma 1), and each block is invariant for H and for the gram at the commuting; the count splits with the splitting, the LDL^T of a block diagonal being its blocks' at the concatenated lists (Lemma 30). The directions' translations commute, the action's one composition read (Construction 7), and a chord operator is the join of its direction's translation with the complementary step's, the transpose's own witness, so the chord operators commute pairwise and each maps every further direction's chord blocks into themselves; the splitting iterates direction by direction within the prior steps' blocks, each step this lemma's own

at its direction's operator. A winding support occupies at least L links, one at each of its direction's transverse cuts (Construction 7's cut family), each reading content at or beyond the member floor $\hat{c}_2(f)$ (Lemma 9), so its content is at least $L \hat{c}_2(f)$. A configuration fixed by a translation off the identity reads so: its support splits into its pieces at shared vertices, the translation permutes them, and a piece the translation fixes holds a path from a vertex to the vertex's image, whose translates join to a closed walk crossing every transverse cut of a moved direction, a winding support; a piece the translation moves sits beside its image, a second piece at disjoint support, each piece an occupied configuration, the incident labels at every touched vertex its own (Definition 8), so its content sits at or beyond the least occupied content, the plaquette level (Proposition 7), and the two pieces' contents join to twice it.

The doubling. The adjoint-paired reads on a root block are the plus read's instance at the root pair (Definition 4). The datum's square is the products $TT^* = 1$ and $T^*T = 1$ collected at the band read. The adjoint of $T = T^* + A$ reads $T^* = T + A^*$, so A 's adjoint is its memberwise swap, and at a member v of the block the adjoint walk reads $\langle v, A^2v \rangle$ as the swap of $\langle Av, Av \rangle$; the block read against v is then

$$t^2 \langle v, v \rangle + \langle Av, Av \rangle = 4 \langle v, v \rangle,$$

the pair $\langle t^2 : 4 \rangle$ at its lower side or at equal members (Definition 8's positive pairing), each root block occupied at the doubling read below. At equal members $\langle Av, Av \rangle$ is the sum's unit at every member, Av with it at that pairing, so $T = T^*$ on the block and $T^2 = 1$ at the product $TT^* = 1$; $T^L = 1$ at odd L then reads $T = T(T^2)^m = T^L = 1$ at $2m + 1 = L$, the coprimality of 2 and L spelled as that word, and $S = 2$ on the whole block, so t reads 2 against $Sv = tv$ there, against t a root of P_L and the two factors coprime at the division display above (P_L at 2 reading the occupied L): every root of P_L reads t^2 on its lower side against 4, strictly interior. An antisymmetric matrix of odd order reads its determinant at its own balance partner, the assignment fold blind to the transpose with each summand's members swapped at the odd factor count (Construction 1's swap grading): the block's order is even.

The deck families' identity. At a root pair (z, z') of the relation $z^2 + 1 = xz$ at a free chord x , product one, write $W := \sum_{a+b+1=L} z^b = z^m P_L(x)$ for the word's read (Definition 4). The geometric telescope is $zW + 1 = W + z^L$, its square

$$z^2 W^2 + 2zW + 1 = W^2 + 2z^L W + z^{2L}.$$

Join $W^2 + 1$ to both members: the relation collects the left to $xzW^2 + 2zW + 2$ at $(z^2 + 1)W^2 = xzW^2$, and the plus read $z^{2L} + 1 = z^L p_L(x)$ collects the right to $2W^2 + 2z^L W + z^L p_L(x)$. The telescope reads $2W^2 + 2z^L W = 2W(W + z^L) = 2zW^2 + 2W$ and $2zW + 2 = 2(W + z^L)$, so $2W$ sits on both members and clears, leaving $xzW^2 + 2z^L = 2zW^2 + z^L p_L(x)$; at $W^2 = z^{2m} P_L(x)^2$ with $2m + 1 = L$ the read $zW^2 = z^L P_L(x)^2$ puts the occupied z^L on both members, and clearing it gives $x P_L(x)^2 + 2 = 2P_L(x)^2 + p_L(x)$, the displayed identity, every step remainder-list arithmetic at the relation (Definition 2; Definition 4). The partial sums' and the key list's displays close by one induction per side: the step joins the two new keys at the telescope, the doubling display, and the products' square $p_{m+1} p_{m+1} = p_{2m+2} + 2$, its three collectors.

The orbit. The word's read at the root pair of the located chord t_j has $W = z^m P_L(t_j)$ the sum's unit at $P_L(t_j)$'s, so the telescope reads $z^L = 1$: the powers wrap, with $p_L(t_j) = 2$ the displayed identity's read at the located root. Hence $Su_j = t_j u_j$: the coefficient of $T^c w$ in Su_j is the family join $p_{c+1}(t_j) + p_{c-1}(t_j) = t_j p_c(t_j)$ (Definition 4), the recursion closing cyclically, its two wrapped keys the reads $p_{L-1}(t_j) = p_1(t_j)$ and $p_{L-2}(t_j) = p_2(t_j)$ against the seed $p_0 = 2$. The join is the recursion's class read at the symbol: over the remainder lists at P_L (Definition 2) the column's coordinates are the families' own classes, the chord multiplies as the free chord, and the

wrapped keys read the wrap identities, $p_L = 2$ with $p_{L-b} = p_b$, one read at every side at once, a located root's reads the classes' evaluations, and a polynomial in the chord reads the column at its own multiple, the membership iterated along the Horner recursion (Definition 2). That seed against the independent translates reads u_j occupied. $\langle u_j, Au_j \rangle$ reads its own memberwise swap at $A^* = \check{A}$, hence equal members, and $\langle Au_j, Au_j \rangle$ reads $\langle 4 : t_j^2 \rangle \langle u_j, u_j \rangle$ on its upper side at the strict interiority above; the membership is the exchange's read, the datum and the chord two words in the one translation at the product $T^*T = 1$, so they exchange with every polynomial in the chord across it, the datum maps each chord block to itself, and the image row reads the moved display, $SAu_j = ASu_j = t_j Au_j$, the symbol's read at the image the member's own image. The column's self-pairing collects at the product families, $p_b^2 = p_{2b} + 2$, the doubled keys reading the key list once (the even keys outright, the odd keys at the upper wrap), and the key list's fold is the symbol's own multiple, the display above, so the self-pairing reads $2L$; the image's self-pairing reads $\langle 4 : x^2 \rangle$ against it at the relation on the column. At the band's two ends the symbol divides at the linear factors: at the chord two the remainder is L , the division display above, and at the partner chord the family reads the doubled parity member key by key, 2 at the even keys and its partner at the odd, so the remainder is the unit at even depth and its partner at odd; the two cofactors' product is the determinant's witness: the band pair is the two linear factors' product's balance partner, each end's factored read is its remainder's partner at the division display, and the three partners compose, the cleared read $4L^3$ at the balance partner of the depth's parity member, occupied at every side, so the pair is independent over the remainder lists, one read per side, the located roots' reads the evaluations; and the composed witness reads the band at every stored factor besides: an occupied-length factor of the symbol dividing the band pair $\langle 4 : x^2 \rangle$ divides the composed constant through the identity, the division's descent at the monic factor against the constant's occupied read, so the band symbol's remainder at every stored factor is occupied, the equal-members case refused at every stored root with the block pairing's lower side closing the strict interiority above. The unit fiber's fixed read $Tv = v$ forces equal coefficients along the orbit at the elimination read (Definition 3), the orbit's own sum, and the orbit's count against the blocks' pairs reads each root block at exactly two members of the orbit. Over the residue the sum pairs every row at the sum's unit, against a coefficient row at the key's coefficient of the key list's fold, the symbol's own multiple at the residue, and against an image row through the datum's memberwise-swap transpose at the sum's own image, the fixed read's unit; a coefficient row pairs an image row at two crossed folds, the cyclic shift reading either fold as the other's exchange and the wrap's reflection returning the exchange: equal members. The coefficient rows are independent at the keys below the residue's top, the families there their own below-top displays at monic tops from the seed two, a dependency reading its coefficients at the unit key by key up the tops; an image-row dependency is the doubling datum's read at a coefficient-row combination, the combination's two translation images at one value, the orthogonality doubling the iterate's fix and the member reading the translation through the wrap, $Tw = T^{2m+2}w = (T^2)^{m+1}w = w$ at $T^L = 1$ over the count, so the combination sits in the unit fiber's kernel and joins the sum's span while the sum pairs it at the sum's unit, its self-pairing reading that unit with the positive pairing withdrawing it; and a joined dependency's sum part reads its own square at the count with each block part its self-pairing against the whole, the positive pairing withdrawing both: the joined list is independent at the orbit's count. $\square$

Lemma 41 (The fiber index is its grading, and the pencil is bordered). *Per fiber on the odd tori (Lemma 40) the configuration classes read two counts off the index (Proposition 4): the band components, the components of the class's support at adjacency within the locality band, two links band-adjacent where one plaque's boundary meets both links' ends, the reach of Lemma 14, the*

components read at the adjacency's closures, from the least link off the earlier components its closure within the support, the band steps' saturation from the link, each occupied step joining a support member off the reached list, so the walk closes within the support's count of steps at the support's distinct keys, one component per closure in the links' order, each with its content, the Casimir fold at the component's links, the class's content the components' contents' sum at the closures' partition of the support; and the separation profile, the band components' relative positions up to translation, read at the band neighborhood of a configuration, a class or a component, its vertex neighborhood joined to the vertices of the plaquettes meeting it (Lemma 14's reach), two components separated beyond the band where every plaquette of the region meets at most one of their band neighborhoods, and within it at a plaquette meeting both. The grading is total, a class being its datum, and the regions are the datum's own coordinate ranges: the head, every content below a head bound Λ_h (a parameter of the presentation, as a window's cutoff is) and every separation within the band; the tower direction, a band component's content at or beyond the bound; and the slab direction, a separation beyond the locality band, the repeating shape of Lemma 34. The pencil's rows read the grading in three shapes: the diagonal, the stencil rows, and the border rows, and the tower rows repeat off the walls' collar:

$$N_{\lambda_0\lambda}^{\lambda+\nu} = \text{mult}_{\lambda_0}(\nu) \quad \text{at every closure member } \lambda_0 \text{ and every dominant } \lambda \text{ at the list's clearance,}$$

one row per member, its content list, the list's clearance the simple coroot pairs of λ at or beyond the list's coroot magnitude m_L , the largest magnitude of a closure member's content against a positive root's coroot, one member read (the string caps, Lemma 6), one at the A -series' fundamental pair and at every minuscule list, the collar the labels below that clearance.

Proof. *Totality* is the index's: a class's support decomposes into its band components with contents and relative positions, counts off the index (Proposition 4), and the regions are ranges of that datum. A band component below content C with L occupied links reads $L \hat{c}_2(f) \leq C$, one pair comparison (Lemma 9), and as many within-band gaps, so its diameter is priced by C and the head is a finite index (Proposition 2).

The rows. E is diagonal at the content sum (Definition 10). A plaquette term meeting the class's band neighborhood reads its changed edge there (Lemma 14; off the vertex neighborhood the far factorization, its loop within one band step of the component), the target's content within the fusion stencil (Proposition 3, Proposition 5) and its support within the class's support joined to the plaquette's links (a target keeps every label off the boundary, Definition 9), band components joining or parting within one locality-band step: a *stencil row*. A plaquette term off the class's band neighborhood factorizes (Lemma 14), creating its own loop class, one new band component: the loop's links are pairwise band-adjacent at the plaquette's own boundary, and a loop link band-adjacent to a support link sits at a plaquette meeting the support's vertex neighborhood, its ends in the band neighborhood against the plaquette's position off it, so the loop's closure is its own links and the further closures are the class's; at one entry per relative position beyond the locality band, a label-graph isomorphism transporting it (Theorem 17's interface): a *border row*, one repeated datum, the boundary coupling of the walk's shape (§16). Every entry is one polynomial-pair list in (t, τ) over the residue beyond the locality band and the reach bound (Lemma 40; Lemma 25(iii)), at the root coordinate's magnetic weight τ^2 .

The tower rows repeat. At the A -series' fundamental pair the rows are the one-box rows at every dominant λ (Lemma 11), $N_{f\lambda}^{\lambda+e_i} = 1$ at every dominant $\lambda + e_i$ and the complement's row at the dual, the fundamental's contents the e_i once each, so the display holds at every dominant λ there with the collar vacant; the diagonal reads the unit content's multiplicity in the member, the sum's unit at the fundamental pair and at every minuscule list.

The member rows. At a member's table the deep exclusion is the coroot pairing's: a contribution at w off the identity holds an inversion list's member γ , $w\beta = \check{\gamma}$ at a positive β , and the tie reads $w(\lambda + \rho) + \nu_w = \mu + \rho$ at the target μ and a content ν_w of the member λ_0 (Corollary 2's member ties); pairing the tie at γ 's coroot moves the image across the kept form, $w^{-1}\gamma = \check{\beta}$ at the transport, and collects

$$(\lambda + \rho)(\beta^\vee) + (\mu + \rho)(\gamma^\vee) = \nu_w(\gamma^\vee) .$$

The right side sits at or below the list's coroot magnitude m_L , the string's own cap (Lemma 6); the left summands read the simple pairs at the roots' natural coroot folds, so at the list's clearance the first sits beyond m_L and the second beyond the sum's unit, the target's pair the base's joined to one content move: the join clears the cap, the tie refused off the identity. The identity's term reads the display at every content ν whose target $\lambda + \nu$ is dominant, the deep pairs absorbing each move at the clearance: the member rows at simple pairs at or beyond it are the member's content list's own, the fundamental pair's the one-box rows and a minuscule member's its contents at one each (Lemma 26(ii)). $\square$

Lemma 42 (The separated sector fibers over relative momentum). *Per fiber on the odd tori (Lemma 40), at every $d \geq 1$:*

(i) *Beyond the locality band the components factorize. At a class of $\varrho \geq 2$ band components with every separation beyond the locality band, the pencil's rows are per component, every entry the moved component's own read at one relative position (Lemma 41). The ϱ -sector's pencil therefore agrees, beyond the collar (the classes with some separation within the locality band, a finite border block of the head, Proposition 2), with the components' tensor-sum pencil extended over every relative position,*

$$H^{(\varrho)} = \sum_{a+b+1=\varrho} G^{\otimes a} \otimes H^{(1)} \otimes G^{\otimes b} \quad \text{against} \quad G^{\otimes \varrho},$$

the fold over the factor splittings at the unit power the scalars' line, $H^{(1)}$ the single-component sector's intensive pencil at Lemma 41's index, the sector's own pencil the join of the vacuum diagonal with it, and the two pencils' site datum supported on the collar.

(ii) *The relative fibering. The extended pencil's entries read the relative positions through translation differences alone, so the relative translations act on it and it block-diagonalizes over their chord operators by Lemma 40's chord splitting, run here at the relative translations: their blocks read at the chord coordinate (Lemma 40), and the relative momenta, the sampled chord values, are the chord 2 with the geometric word's symbol roots, each root a designated bracket datum at the separation of Theorem 16.*

(iii) *The separated bottom adds, and the sector clears the bracket. A single-component head pair positive semidefinite at the level ℓ is the form floor $H^{(1)} \succeq \ell G$ (Lemma 30), with its site datum S_1 at $H^{(1)} = \ell G + S_1$, and the tensor sum reads it ϱ times:*

$$H^{(\varrho)} = \varrho \ell G^{\otimes \varrho} + \sum_{a+b+1=\varrho} G^{\otimes a} \otimes S_1 \otimes G^{\otimes b},$$

each summand positive semidefinite against the positive-definite tensor Gram, and the multiset symmetry and the window compress it, a compression keeping the split (Lemma 30). So on a bracket with top at or below $\varrho \ell$ the extended sector's pair is positive semidefinite at every relative fiber, the collar is a head border, and the border rows between component counts are the walk-shaped border chain: the separated sector contributes its collar to the head and its jumps sit beyond the bracket, the comparison's validity divisor the single-component level's own, located per cell (Lemma 31). At $d \geq 2$ the clearance is the slab direction's step, the relative coordinates fibered and

the comparison Lemma 31's cell read, per relative fiber the single-component level's own, so the direction count enters through the fiber list alone (Lemma 40).

(iv) The far product at the total cutoff. On occupied separated head and far lists whose pairing and pencil entries factor as in (i), write their forms as (H_A, G_A) and (H_F, G_F) , with electric contents a_i, c_j on their configuration blocks. The full product has forms

$$G_{AF} = G_A \otimes G_F, \quad H_{AF} = H_A \otimes G_F + G_A \otimes H_F.$$

Its roots are the factors' roots' sums, and its count at a level is the root pairs' count below it. At total cutoff C , let J_C be the inclusion of the product keys at $a_i + c_j \leq C$. The admitted product's forms are

$$G_C = J_C^\top G_{AF} J_C, \quad H_C = J_C^\top H_{AF} J_C.$$

At first-factor content a_i , the far selection is $c_j + a_i \leq C$. Let e_A, f_C be the factors' least roots, with the far factor taken at cutoff C , and put $S_A = H_A - e_A G_A$, $S_F = H_F - f_C G_F$. Then

$$H_C - (e_A + f_C)G_C = J_C^\top (S_A \otimes G_F + G_A \otimes S_F) J_C \succeq 0.$$

At an occupied admitted list its least root is $e_A + f_C + \Delta_C$ at a stage gap $\Delta_C \geq 0$. This gap reads the sum's unit exactly where the image of J_C intersects $\ker S_A \otimes \ker S_F$ at an occupied vector. Its level-gap form is

$$H_C - (e_A + f_C + \Delta_C)G_C = J_C^\top (S_A \otimes G_F + G_A \otimes S_F) J_C - \Delta_C G_C.$$

At a head floor $H_A = \ell G_A + T_A$ with $T_A \succeq 0$, and a full-window ground $e \leq f_C$, the read at a positive offset $h \leq \ell$ is the positive fold

$$H_C - (e + h)G_C = J_C^\top (T_A \otimes G_F + G_A \otimes S_F) J_C + (\ell - h + f_C - e)G_C \succeq 0.$$

Its count is the sum's unit; at $\ell - h + f_C - e > 0$ it is positive definite. The comparison $e \leq f_C$ holds where extension by unit labels identifies the far list's pencil with the full window's restriction to that image. In particular, this is the read on the invariant carrier when the far region is the links outside the head neighborhood and its plaquettes are the full region's plaquettes contained in those links.

At a positive-definite admitted block P and coupling B to the further places, the whole count reads

$$\text{rev} \begin{pmatrix} Q & B \\ B^\top & P \end{pmatrix} = \text{rev}(Q - BX), \quad PX = B^\top$$

(Lemma 30). Here B is the block of the actual cutoff pencil, with every admitted column. The correction is $BX = X^\top PX$, at the block's Gram solve.

Proof. (i) E adds the contents, a plaquette term meets at most one component's neighborhood (Lemma 14), and a stencil move moves one component within its own locality band with the others spectators, so the rows are per component. (iii) A product of two tensors reads factorwise, the product index's fold collecting to the factors' folds' product at the ground identities (Definition 1), so the gram's pivot congruence at its positive pivots, the split at order-one blocks a positive-definite datum reads off its leading minors of unequal members (Lemma 30's minor sequence), and the site datum's split compose to the summand's own congruence, its image the split's blocks at the pivots' positive scales, and the image's form at every vector is the fold of the blocks' reads at the scales, on its upper side or of equal members (Lemma 30). (iv) The factors' congruences' tensor maps the full

tensor sum's site datum at a level to the diagonal at the summed roots against the scales' products ((iii)'s tensor congruence; Lemma 33), and the count at a level is the root pairs' count below it (Lemma 30's congruence read at the diagonal). The contents add at a product configuration, so the cutoff's index is precisely the displayed selection (Definition 8). The two factor level-gap forms are positive semidefinite. Their tensor folds and their restriction through J_C have that read by (iii) and Lemma 30. In the factors' split coordinates, the folded form's kernel consists of the root pairs at both least roots: every other pair has a positive gap at a positive Gram weight. Thus its restriction has an occupied kernel exactly at the stated intersection. The least-root read gives $\Delta_C \geq 0$, with equality exactly there (Lemma 33). Substitution of the head floor and the two ordered gaps gives the displayed fold at $e + h$.

Extension by unit labels preserves the pairing and the electric form. A plaquette contained in the far link set has its far entry. Every further plaquette has a link outside that set, whose two state labels are unit and whose character factor is nonunit; the link's unit evaluation reads the sum's unit (Construction 8; Proposition 8). The magnetic restriction is therefore the far pencil's, and compression puts the full least root at or below f_C (Lemma 30). The last count and its correction are the positive principal pivot's deflation, with $BX = X^\top PX$ at the solved identity's transpose. $\square$

Theorem 19 (The decimation: the counts live on the decimated head). *Fix a fiber, its grading (Lemma 41), a head bound Λ_h , and a pair bracket $[\ell_1, \ell_2]$ of spectral levels. The pencil decimates onto its head in three located steps, each step a Schur complement (Lemma 34; Lemma 30's deflation), and the decimation is the count's own presentation.*

(i) *The depth direction is the walk's shape. The fiber's sector is its fusion closure from the vacuum class, the depth chain (Construction 21), and the pencil blocked at the shells is block tridiagonal over the depth with the vacuum class the boundary shell, the pencil at the depth order its own count at the places' congruence (Lemma 30), and the count at every level splits over the chain's pivot recursion, $\text{rev} = \sum_\kappa \text{rev}(X_\kappa)$ (Lemma 34(i),(ii)): the walk's shape at matrix blocks (§16), the boundary pivot the vacuum's dressing. The split is order-free, each slab ordering of the elimination reading the one count, and an ordering is certificate data, its validity divisors entering the emitted list as any boundary's do. The ground jump is a divisor point of the decimated head: the level is the distance of the first two jumps of one counting function, the walk's own read, its boundary defect the vacuum's diagonal, the plaquette character's evaluation at the sum's unit (Lemma 14; Proposition 8).*

The counted pivot. At a level datum Z , a principal pivot P of unequal-membered determinant has the read

$$Z = \begin{pmatrix} P & B \\ B^\top & Q \end{pmatrix}, \quad PC = B, \quad Q = B^\top C + Z_P, \quad \text{rev } Z = \text{rev } P + \text{rev } Z_P$$

(Lemma 30). *The head count equals the whole count at the pivot's count the sum's unit. At two stated lines z_-, z_+ and a prescribed count g , the certificate reads the two total counts,*

$$\text{rev } P_- + \text{rev } Z_{P_-} = g = \text{rev } P_+ + \text{rev } Z_{P_+}.$$

The pivot's split and its solved witness are the data at each line. For the detector datum $SD - hS$ at a positive pair h , the prescribed count is the sum's unit (Lemma 20), its blocks the whole-square restrictions of Definition 10.

(ii) *The tower tails decimate on their dominance cells. A class's diagonal is intensive, the far plaquettes reading equal members on it (Lemma 14): the content sum against the local magnetic*

stencil, the stencil rows one capped list (Lemma 41; Lemmas 17, 18). A tail block's positive-pivot datum is a split with every block on its upper side and its kernel block vacant (Lemma 30). A dominance certificate reads the following comparisons on that block: per row, the gap $d_i := a_{ii} + \check{f}_i$, the diagonal's entry joined to the balance partner of the off-row entries' magnitude fold f_i (Theorem 16's magnitude carrier), at the comparison $f_i \leq a_{ii}$, the gap on its upper side or at equal members. The form sits at or beyond the gaps' fold of squares $\sum_i d_i x_i^2$ at every vector, at or beyond the sum's unit where the comparison holds; every split of the block then reads the upper side throughout, at every gap beyond the sum's unit the kernel block is vacant, so the determinant sits off equal members with located brackets and the adjugate its solve (Definition 3); the block is a nonsingular principal pivot of the datum at its designated places, inertia adds at it with its count the sum's unit, and the datum's count is its deflation's, the decimated head's (Lemma 30), at a chain's tail the slab recursion's own with the count telescoping over the head slabs' pivots (Lemma 34(ii)). The fold f_i runs over every off-diagonal column of the stated block. A creation at a disjoint plaquelette contributes where its output is a member of that block, at the coefficient read of Theorem 9. The support comparison is the remainder's form read,

$$f_* |\langle x, R_\Pi x \rangle_G| \leq \nu d_L \langle x, E_\Pi x \rangle_G, \quad M_\Pi = R_\Pi + \sum_i (a_i + a_i^\dagger)$$

(Theorem 9). The full block's dominance comparison reads its full fold f_i . At either representative, on a cell of the entries' side reads, its validity is the side read of that comparison at the stated matrix entries, with its own located divisor (Lemma 31). The constant-coefficient instance is the walk's boundary recursion, the deck quadratic's interior root (Theorem 13).

(iii) The slabs at the deck pivot. Beyond the locality band the slab direction's bulk datum repeats (Lemma 41); its pivot is the deck-interior solvent of the pivot identities (Lemma 35(iii); Lemma 39), the deck orbit's interior member at $m = 1$, Theorem 13's read, valid exactly off the bulk block's own band, the crossing locus of its deck relation, and at a graph read of unequal members (Lemma 35(iii)): the decimation's validity divisors there are the tail's band and the graph read's side, located data of the bulk block alone. At $d \geq 2$ the slab step is Lemma 42's clearance, the relative coordinates fibered and the deck pivot the $d = 1$ sharpening. At two slab families with equal entries on the stated range, the pivot differences obey the sandwich identity and its geometric brackets at their transfer certificates (Lemma 35(i),(ii)). A total-content cutoff reads the product selection of Lemma 42(iv). Its positive fold gives the separated block's lower read at the actual ground-translated line. Every coupling to the further places enters the deflation at its actual matrix and solved witness, and (i)'s total count is the whole count.

The decimated symbol. On the bracket, within the validity cells of (ii) and (iii), the count of the full fiber pencil is the decimated head's, the dropped places the tower block's, every class at a component's content at or beyond the head bound, joined to the far block's, the classes at a separation beyond the band off the tower block, a class in both ranges the tower block's, its row (ii)'s comparison, where (iii)'s ball reads the slab direction's bulk at contents below the bound alone; the decimated head a finite symmetric pencil over the located stage (Lemma 32), entries pair data of the deck coordinates and the located pivots, cell functions over (t, τ, r) (Lemma 31; Lemma 25(iii)). The spectrum on the bracket is where the decimated determinant reads equal members, Theorem 26 per cell concrete; and the cells the validity divisors cover are the decimation's own output, every boundary a located root.

The divisor list, named. The decimation's validity divisors are the stated certificates' comparison fields: the pivot divisors, the pivot and congruence determinants and the side comparisons of the nonsingular split blocks, with the dominance divisors at a dominance certificate's full row folds ((ii)); the deck divisors, the bulk blocks' crossing loci with the graph read's side, Lemma 35(iii)'s counts

((iii)); and the jump divisors, the decimated determinant's roots at the bracket's levels (Lemma 31). The list is finite per stage, its cells partition the base, counts constant per cell and persisting at a boundary beside a positive-semidefinite cell (Lemma 31); the emitted record, the located root list with its per-cell counts and the binding read of which comparison binds at each boundary, is the decimation's displayed object, every entry one stage read of the crossing divisor (Theorem 26).

Proof. (i) A pencil row joins two classes at a plaquette term's entry off the sum's unit, a border row or a stencil row of Lemma 41, and a joined class sits at most one depth beyond the class it joins, so the pencil blocked at the shells is block tridiagonal over the depth. Inertia adds at every nonsingular principal pivot (Lemma 30), so each slab ordering of the elimination reads the one count. At a counted principal pivot, Lemma 30's unit-triangular congruence reads $P \oplus Z_P$. Its two counts add, giving the displayed total at each stated line. The positive-pivot instance has its first count at the sum's unit. The detector's prescribed count is the spectral read of Lemma 20; its compression includes the action's image in both blocks at $SD = D^TGD$. (ii) The peel telescopes at the symmetric datum (Theorem 17): a vector's form peels at the leading row, the leading square with the doubled cross fold against the trailing block's form, and each cross term joins its magnitude's squares at one square's multiple,

$$2axy + |a|(x^2 + y^2) = |a|(x + y)^2 \text{ at } a \text{ its own magnitude, and } |a|\langle x:y \rangle^2 \text{ at the swap,}$$

at or beyond the sum's unit by its shape, so the leading row's magnitudes withdraw from the trailing rows' folds into their gaps and the descent closes on the order: the form sits at or beyond the gaps' fold of squares. A split's lower list at an occupied count would name a vector pricing the form strictly below the sum's unit against the fold (Lemma 30), so every split reads the upper side; at every gap beyond the sum's unit a kernel column's image, off the unit family at the congruence witness's clearing, would read its form at equal members against the strict floor (Lemma 30), so the kernel block is vacant. *The decimated symbol.* The tower block and the far separated block are nonsingular principal pivots of the datum at their designated places, every split of each reading the upper side with the kernel block vacant ((ii); at (iii) the tail block's determinant is its recursion pivots' product, each pivot inside the ball at the deck pivot's floor with its every split on the upper side, and the extended sector's pair positive semidefinite below the bracket's top (Lemma 34(ii); Lemma 35(iii); Lemma 42(iii))), so inertia adds at them at counts the sum's unit and the datum's count is its deflation at the dropped places (Lemma 30). $\square$

Lemma 43 (The charged cells). *Fix a fiber on the odd tori (Lemma 40). The d_f -ality of a label is its chain endpoint, the fundamental or the unit label Lemma 9's descent ends at, one value at every chain: a chain to the unit label reads the label's degree at full columns outright (Construction 5), $|\lambda| = m d_f$, and one to ω_j at the two-sided join $|\lambda| + m' d_f = j + m d_f$ at naturals m, m' , the fundamentals' degrees their indices. Two labels are one class where their degrees join to full columns, $|\lambda| + m d_f = |\mu| + m' d_f$ at naturals m, m' , the endpoint the class's one fundamental or the unit; the dual's endpoint is the complement at the join $a + a' = d_f$ at nonunit endpoints and the unit at the unit's, the dual's degree joining the label's to full columns, $|\lambda^*| + |\lambda| = d_f \lambda_1$.*

(i) *Charge is a cut-free class, and the pencil grades by it. At an occupied configuration every touched vertex reads an invariant count of at least one at the incident product, incoming links dualized (Definition 8), and reads the vertex law, the outgoing against the incoming degree folds at the unit class,*

$$\sum_{\text{out}} |a_\ell| + m d_f = \sum_{\text{in}} |a_\ell| + m' d_f \quad \text{at naturals } m, m'.$$

The flux through a cut is the class of its crossing links' degree fold, one class per direction, its endpoint the class's read, and two parallel transverse cuts' fluxes are one class; a cut transverse to direction i is crossed by direction- i links alone (Construction 7's direction reads), so the directions' classes are independent data, and the charge c is the occupancy family over the directions at the flux endpoints, a direction occupied at a nonunit class by its fundamental's index c_i , the charged directions its members; one direction at $d = 1$. At a member's table the vertex law reads at the member's class group, the classes over the root folds, finite data per member (Construction 4): the d_f -ality at the matrix units, the degree at full columns; the series' classes the contents' parity data against the root folds' spans (every even list at B_ℓ , the short family's doubled keys generating; the even-sum lists at C_ℓ , the letter-pair moves with the long family's; the even lists at key-four sums at D_ℓ , the pair moves'), two classes at B_ℓ , the spinor's odd parity against the unit's, two at C_ℓ , the first fundamental's odd sum against the unit's, and four at D_ℓ , the parity with the sum's key-four remainder, the vector's and the two spinors' with the unit's; and the fixed members' classes the adjugate rows' remainder families at the fold key, the unit's alone at G_2 , F_4 and E_8 at key one, three at E_6 , the key-three remainders pairing ω_1 with ω_5 and ω_3 with ω_6 , and two at E_7 , the key-two remainders occupied at ω_2 , ω_5 and ω_7 . At a member whose class group is its unit alone every configuration reads the unit class, the charged sectors are the one sector, the winding clauses read at the unit class's one instance, and the cell list's tension family drops, its root list the unit class's, every other read fixed: the one-sector read is the class fold's own derivation. The fiber pencil (Definition 10) is block diagonal over the charges and its counts split (Lemma 30), M of the unit charge class and the vacuum line sitting there.

(ii) The winding floor, attained. Every label reads $\hat{c}_2$ at or above its endpoint's, Q falling along the chain (Lemma 9), the endpoint read $\hat{c}_2(\omega_j) = [j g_j (r+2) : 2d_f^2]$ at the complement gap $j + g_j = d_f$ (Lemma 9), its first component the $j g_j$ count at the $(r+2)$ weight, written $w(j) := j g_j$ below. The floor merges over a cut: at a family of nonunit endpoints whose degree fold sits at a nonunit class c , two members merge at their sum's trichotomy against d_f : below, to the fundamental at the sum, at the cost identity $w(a_1) + w(a_2) = w(a_1 + a_2) + 2 a_1 a_2$; at d_f , the pair drops outright at $w(a_1) + w(a_2) = 2 a_1 a_2$, the rest still at c ; and beyond, to the fundamental at the gap g at $d_f + g = a_1 + a_2$, at $w(a_1) + w(a_2) = w(g) + 2 g_1 g_2$ over the complement gaps $a_i + g_i = d_f$, the cost at or beyond $2 g_2$ at $a_1 \leq r$: every family descends to the one member c , so its content is at least $\hat{c}_2(\omega_c)$, one value with its complement's at the dual pair $c + c' = d_f$, at or above $\hat{c}_2(f)$. Two distinct endpoints' floors tie exactly at the dual pair, the sum's trichotomy against d_f their read at $j' + a = j$: below, $w(j') + a b = w(j)$ at the join $j + j' + b = d_f$; at d_f , the dual pair's one value; and beyond, $w(j) + a b' = w(j')$ at $d_f + b' = j + j'$, so off the dual pairs the floors' trichotomy is one symbolic read at every residue. At a charged direction i every direction- i cut is crossed at a nonunit net class, so the charged diagonal sits at or above the summed floor

$$L \sum_{i \text{ charged}} \hat{c}_2(\omega_{c_i}) \geq L \hat{c}_2(f) .$$

At a member's table the winding floor per class is the class's least read: a series member's is the descent's, a nonunit-class chain ending on its own class's end members (Lemma 9), the spinor's read at B_ℓ , the first fundamental's at C_ℓ , and the vector's or a spinor's at D_ℓ 's three nonunit classes; and a fixed member's is the fundamental fold's class-respecting read, at E_7 a nonunit-class end content occupying one of ω_2 , ω_5 , ω_7 with its Casimir clearing the least of their displayed list, the minuscule member's join $[57 : 2]$, and at E_6 occupying a fundamental of its own class or two of the further nonunit class, either read clearing the pair's shared least $[52 : 3]$ (Lemma 9). The class floors add over a cut's crossing family, the floor list subadditive at the class sums, a pair at the unit class outright, E_6 's nonunit pair at the shared least against its double, and D_ℓ 's spinor

pair at the vector's class, twice $[\ell(r+2) : 8(r+1)]$ against $[r+2 : 2(r+1)]$, the cross-multiplied read ℓ against 2, a spinor's with the vector's clearing the further spinor's at the spinors' one value: a family netting class c reads content at or above the class- c floor, the merge identities above the matrix units' instance. The floor is attained by the winding strings: one straight ω_{c_i} -cycle per charged direction, at charge 1 the fundamental f along its cycle, each within-string vertex reading $N_{\omega\bar{\omega}}^1 = 1$ (Proposition 8). Two strings of distinct directions share at most one vertex, meeting exactly where their fixed transverse coordinates agree, and the union is occupied at the summed floor exactly. At the free end the charged ground is the strings', and the margin above the vacuum grows by $\sum_i \hat{c}_2(\omega_{c_i})$ per cell of length.

(iii) At bounded excess the charged head is the dressed string, and its pencil is one window-free head at a length shift. Write the excess of a charge- c class for the witness gap of (ii)'s comparison, the summed winding floor with the excess joining to the content, the grading the charged decimation reads. The exclusions are extensive, read per cut, each cut's price direction-free, the direction count entering at the cut families' list alone, and every price is a read of the member's class floor table, (ii)'s floors with their subadditivity folds; the member price ϖ_0 is the least of the two cut price reads below, one trichotomy fold over the member's displayed class data, the d_f -ality the class read at the matrix units, each extra crossing's floor clearing the merge read, the unit-class read at $2d_f^2$ against $2(r+2)$ over the square's growth $d_f^2 = (r+1)^2$ and the netted pair's at $2r(r+2)$ against $2(r+2)$ at the occupied factor; the off-cut read below prices the deviation links at their own count, the dressing bound's read at the excess floors rather than the cut count's. A transverse cut with two or more nonunit classes among its crossing multiset's members prices an excess at or above the merge gap, (ii)'s subadditivity fold at the crossing pair against the net class's floor, positive at every nonunit pair of the member's class group: twice a class floor at a unit-class join, D_ℓ 's spinor pair at the cross-multiplied ℓ against 2 with the vector's own read at a mixed pair, and E_6 's shared least against its double ((ii)'s folds), and at the A-series the merge gaps at or above $[r+2 : d_f^2]$, the instance. A cut crossed beyond its net class prices the excess at its extra crossings' own floors: an extra crossing at the unit class and a nonunit label reads $\hat{c}_2$ at or above the unit class's least nonunit read, θ 's read one at the A-series, at D_ℓ , and at E_6 and E_7 , the vector's $[r+2 : 2(r+1)]$ at B_ℓ and the short dominant root's $[r : r+1]$ at C_ℓ ; and a crossing family netting to the unit class holds two members at their classes' floors, at or above twice the least class floor, the A-series' fundamental pair $2\hat{c}_2(f)$ the instance. And an occupied link off every charged direction's cuts reads the member's floor outright (Lemma 9), the A-series' $\hat{c}_2(f)$.

The head classification. At excess below Λ_h the cuts crossed beyond their one net crossing number at most L_0 , the least natural at $\varpi_0 L_0 \geq \Lambda_h$, one cross-multiplied read at the member price with its witness the cross-multiplied comparison's quotient at its successor (Definition 1), the A-series' at $(r+2)L_0 \geq \Lambda_h d_f^2$, and at $L > L_0$ the head's classes are the dressed string: the winding strings of (ii), one straight ω_{c_i} -cycle per charged direction sharing at most one vertex per pair, deviating and dressing within the bound the excess floors above set, each crossing one more localized dressing, the far local components tensor factors beyond the collar (Lemma 42). The rows read at the dressing's neighborhood (Lemma 14), the string's far links at one repeated per-link datum, and within the fiber the classes are the dressings' translation orbits at the pair base, one pair per orbit (Lemma 40), every entry a finite stencil sum at translation powers: beyond L_0 and the locality band (Lemma 14) the off-diagonal entries and the excess reads are one polynomial-pair list in (t, τ) over the residue, the label-graph isomorphism of dressed neighborhoods transporting them along the string and across windows (Theorem 17's interface). The diagonal splits off the one extensive read, uniformly over the head: at a far cell the magnetic read is off-diagonal in the label index, the far terms factorizing at $\text{Eval}(\chi_L)$, the fold of Kronecker deltas at nonunit members (Lemma 14; Proposition 8), so the

strings' per-cell diagonal at the $\alpha=1$ representative is the electric read alone,

$$\varpi := \sum_{i \text{ charged}} \hat{c}_2(\omega_{c_i}) ,$$

window-free and coupling-free, the charged head pencil is

$$H_c = L \varpi G + H_{\text{int}},$$

H_{int} the window-free intensive head at the stencil rows, and a charged count at a level λ is H_{int} 's count at the shift cross-added onto the level (Theorem 15's balance-pair levels), the two pencils one matrix at matched levels: the length enters the charged head through the shift, the sampled momenta and the separation range alone. The magnetic datum at a far cell is its dressing fold, the tower decimation's per-cell Schur read at the cell's stencil and the bracket's level, one adjugate solve per pivot (Theorem 19(ii)), each atom read at the entry's square against its gap, $[e^2 : g]$, the pivot's cofactor at the coupling's two reads, entering the dressed diagonal at the squared magnetic weight, $[p^4 : q^2]$ on the ray at $[q^2 : p^2]$ and τ^4 at the $\alpha=1$ representative: its atoms are the stencil's changed-edge entries (Proposition 8) at their content gaps, a link-sharing plaquette at the row targets μ of $\omega_{c_i} \otimes \lambda_0$ over the closure's members λ_0 , one entry per target (the rows at the list, Lemma 25(ii); Lemma 26(ii) at the further series), at the content pair $3 \hat{c}_2(\lambda_0) + \langle \hat{c}_2(\mu) : \hat{c}_2(\omega_{c_i}) \rangle$, and a vertex-sharing plaquette at its own loop, the plaquette content $4 \hat{c}_2(\lambda_0)$ (Proposition 7), its entry the vertex's own word read. A counting function is monotone in its level, so the flat step's charged comparison rides the shift against the unit fiber's anchor through the tension, the balance pair of the string's dressed per-cell floor against the unit class's,

$$\left\langle \varpi + \sum_{i \text{ charged}} D_i^{\text{vac}} : \sum_{i \text{ charged}} D_i^{\text{str}} \right\rangle,$$

D_i^{str} the direction- i cell's weighted dressing fold and D_i^{vac} the unit class's at the matched slab, its entries one at the plaquette contents $4 \hat{c}_2(\lambda_0)$ over the closure's members (Proposition 7; Lemma 14), the pair read at the stencil neighborhood alone, a located datum within Lemma 35(ii)'s brackets: at tension off its lower side on the cell the least-length read extends to every larger length, and the tension's root is one more located divisor of the list, the free end reading the electric member alone, both folds at the end's one-member read, the value $\sum_i \hat{c}_2(\omega_{c_i})$ among the integer constants. The folds' atoms are residue reads, the entries word reads and the gaps the distance classifications (Lemma 25(ii); Lemma 24; Lemma 9; the member gaps the words' bilinear reads, Construction 3's additive form), so the tension's divisor takes the residue as a coordinate with its roots against the compared families' clearances (Lemma 25(iii); Lemma 26(iii)). At the end weights the ray's factors read the pair whole, q^2 at the electric member and $[p^4 : q^2]$ at the two folds, at $q\tau = p$ (Lemma 44's seam identity): the tension's side is a ray datum. The folds' position lists read the cell's stencil, the link-sharing and vertex-sharing plaquettes at the interface's plaquette counts (Construction 7), both members linear in the term count, so the tension's divisor takes it as one further coordinate, a linear read with its root located per charge vector and count.

(iv) The charged decimation, and the cell. The grading and the decimation run per charge: every pencil row preserves the charge ((i)), so each diagonal, stencil and border row of Lemma 41 is a row of one charge sector, read there; beyond L_0 the charged index is the dressed string of (iii), one dressed ω_{c_i} -string per charged direction, its entries the window-free list there, with the far local components reading by the vacuum sector's own rows (Lemma 42); the tower and slab decimations are Theorem 19(ii),(iii), with (ii)'s dominance comparison run against the charged diagonal's floor

(ii), the head bound a parameter of the presentation. The charged counting function on a bracket is therefore the charged decimated head's, a cell function of (t, τ, r) with located boundaries (Lemma 31; Lemma 25(iii); Lemma 26(iii)), the comparison's validity divisor its own, and the charged cell's entry that divisor's first positive root, an output (Lemma 33). The head's level list, its multiplicities and its block structure are the reads of that head pencil, pinned per cell.

Proof. The endpoint. A chain's moves fix the reduced degree and each re-reduction withdraws one full column (Construction 5), so a chain to the unit label reads the label's degree at full columns and one to ω_j the two-sided join; two endpoints' joins against one $|\lambda|$ force one index by the sum's injectivity, a positive gap of full columns reaching d_f against the indices at most r .

(i) At a touched vertex the unit label is a constituent, a constituent's reduced degree joins the factors' degree fold to full columns (contents of a tensor product add over the factors and a block's contents sum to its degree, Construction 1; Construction 5), and the incoming duals read across at the join $|\lambda^*| + |\lambda| = d_f \lambda_1$: the vertex law. Summing the vertex laws over the region between two parallel transverse cuts, an interior link's degree enters both folds once and the sum's injectivity reads it off: the two cuts' pairs are one class. At a member's table the law is class additivity: a block's occupied contents are one class (Construction 3), classes add over tensor products, the contents' sums, and a dual's class joins its label's to the unit class, so the vertex law reads at the class group. E is diagonal; a plaquette term of M fuses the boundary's four links by one closure member (Definition 10), and a transverse cut meeting a plaquette's boundary meets it at two links of the cut's direction traversed oppositely (Construction 7), the two links' classes moved by the member's class and its dual, the cut's flux fixed at the class group's sum; a translation permutes the index at one charge, the relabeled configuration's flux through a cut its own through the cut it came from, the cut family's grades read along the translation, one term per link along the permutation with the directions kept (Construction 7), the conservation reading the base cut's class back: the pencil is block diagonal over the charges.

(ii) At the complement gap $j + g_j = d_f$ the endpoint count reads $w(j) + j^2 = j d_f$, and the merge identities are that read at the sum's trichotomy: below d_f , $w(a_1) + w(a_2) + a_1^2 + a_2^2 = (a_1 + a_2) d_f = w(a_1 + a_2) + (a_1 + a_2)^2$, the cross products the cost; at d_f the gaps exchange, $g_1 = a_2$, so $w(a_1) = w(a_2) = a_1 a_2$; and beyond, at $g + g_1 + g_2 = d_f$, $w(a_i) + g_i^2 = d_f g_i$ and $d_f (g_1 + g_2) = w(g) + (g_1 + g_2)^2$, the cross products the cost again, each identity read off by the sum's injectivity. The ties read at $j' + a = j$: below, at $g_j = j' + b$ and $g_{j'} = j' + a + b$, $(j' + a)(j' + b) = j'(j' + a + b) + a b$; at d_f the two reads are the one product $j j'$; and beyond, at $j' = g_j + b'$ and $g_{j'} = g_j + a$, $(g_j + b')(g_j + a) = g_j (g_j + a + b') + a b'$, the gap product the order's witness in each case. The charged diagonal's floor sums over the cuts: the direction- i cuts cross disjoint link sets, and the directions' link sets are disjoint in turn. At a series member a move is one positive root, the class kept, so a nonunit-class chain ends on its own class's end members; at a fixed member a chain keeps its class and ends at a content at every coroot read at most one, whose occupied keys' classes add to its own (the remainder families of Construction 4), the E_7 read the least of the three fundamentals' displayed list and the E_6 read the pair's shared least or, at two fundamentals of the further class, two summands of the displayed fold (Lemma 9). Over a cut's crossing family each nonunit-class member reads at or above its class's floor and a unit-class member's content is positive, so the family's content sits at or above the subadditive floor list's read at the net class. At a shared vertex of two winding strings $\text{vmult} \geq 1$, the incident product with each string's pair, whose unit constituents pair (Proposition 8; Definition 8), so the union is occupied at the summed floor exactly.

(iii) At the A -series the merge gaps are the complement-gap reads, (ii)'s below and beyond costs $2a_1 a_2$ and $2g_1 g_2$ at the complement gaps $a_i + g_i = d_f$, each at least 2, the parts nonunit,

the drop case at the dropped pair's own floors, $2w(a)$ at or beyond $2r$, and the unit conversion at $\hat{c}_2(\omega_j) = [j g_j (r + 2) : 2d_f^2]$ at $j + g_j = d_f$ (Lemma 9), each at or above $[r + 2 : d_f^2]$. An extra crossing at the unit class and a nonunit label keeps the class along its chain and ends at the unit through a dominant root's content or on a unit-class end content (Lemma 9; (ii)), the unit-class fundamentals' reads at E_6 and E_7 at or beyond θ 's at the Casimir lists' trichotomies, and a two-key end content at or beyond the nonunit pair's doubled least, the display's cross reads positive. At excess below Λ_h a two-string class with every cut crossed beyond its net crossing number exits the head at $L > L_0$ outright, so the head's classes beyond the threshold are the dressed strings. Off the dressings' neighborhoods a head class's cells are label-graph isomorphic to the straight strings' own, so the content and the magnetic stencil reads repeat cell by cell (Lemma 14) and each dressing's deviation is finitely many cells of the intensive head; at a far cell a straight cell meets a plaquette at one link at most, consecutive links collinear, the term writes a closure member on the free links, three at a link-sharing plaquette and four at a vertex-sharing one, and Eval reads the unit coordinate, so every index component of the term's product sits off the class's own configuration, the magnetic read off-diagonal. At the tension the far plaquettes' folds enter both members equally, a summand on both members fixing the value (Definition 1), so the pair reads at the stencil neighborhood alone. $\square$

Theorem 20 (The cut is a flat step of the decimated counting function). *On a coupling cell, within the decimation's validity cells (Theorem 19), the window cut of Definition 11 at level κ is reached by located jump comparisons on the decimated symbols: per fiber and per charge (the d_f -ality class of Lemma 43(i), M of the unit charge class), the counting function's first jump sits at or above $\varepsilon_0 + E_0\kappa$, and on the vacuum's own sector its second jump does, the first the window's ε_0 , the boundary pivot's own root (Theorem 19(i)). Each comparison is two counts at located pairs: the jumps are count data on the decimated head, cell functions of (t, τ, r) with located boundaries (Lemma 31; Theorem 19), the momenta read cell by cell at one bracket pair each (Lemma 40), and the charged sectors' counts the reads of Lemma 43.*

The comparisons reach the cut. They read $\text{spec}(H) \subset \{\varepsilon_0\} \cup [\varepsilon_0 + E_0\kappa, \infty)$ on the window, the cut at that window at every ground (Lemma 20). The offsets are fixed per cell, with each absolute line read at the window's own ground. The decimation reads the actual total counts, and a separated product its total-content compression (Theorem 19; Lemma 42(iv)). Per cell the verification is a finite list of anchored jump brackets, one pair of counts each: the cells' reads are the gap chain's data, each level naming its representative and transported across the seam at the factors of Lemma 44's identity.

Proof. A count at a level is the roots below it (Lemma 33), the window's roots the fibers' and the charges' joined (Lemma 40; Lemma 43(i); Lemma 30's counts at a block diagonal): a sector whose first jump sits at or above $\varepsilon_0 + E_0\kappa$ has every root there, and the vacuum's sector with its first jump at ε_0 and its second at or above $\varepsilon_0 + E_0\kappa$ has its roots at the kernel point and at or beyond the edge, so the window's spectrum is the displayed read, the cut at every ground (Lemma 20). The counts are the decimated head's on the bracket (Theorem 19), so each jump comparison is two counts at located pairs, and the head's reads move across the seam at Lemma 44's factors with their levels. $\square$

Theorem 21 (The ground reads are bracketed). *Fix a window at a coupling with the ground bracket of Theorem 10 and the slab split of Lemma 34 along the decimation's direction (Theorem 19), within the decimation's validity cells and at the spectator certificates λ_i (Lemma 35(ii)).*

(i) A ground read is a stage read. The ground ψ has a spanning list by elimination over the ground root's stage (Lemma 33), and at an observable action A the read enters as the pair of

the two pairings, $\psi^\top G A \psi$ against $\psi^\top G \psi$, its value the cofactor (Definition 1), the two members Definition 10's $\omega(A)$ and $\omega(I)$; each pairing is one stage read, its entries Lemma 14's at the supports.

(ii) The tail rides the transfer factors. At the ground root the kernel rows of the slab split close from the far end, $X_{i+1} \psi_{i+1} + B_i^\top \psi_i$ the sum's unit per row (Lemma 34(iii)'s row read), so a depth- j component is a j -fold transfer image and its weight is capped geometrically at the gram's slab blocks (Lemma 35(ii)),

$$\langle \psi_{i+1}, G_{i+1} \psi_{i+1} \rangle \leq \lambda_i^2 \langle \psi_i, G_i \psi_i \rangle,$$

the certificates' product pricing the tail's whole weight beyond any depth.

(iii) The ground transport, and the window sandwich. The transport display: at two pencils on one head at the unit gram, their site datum S at $\hat{P} = P + S$, the later ground $\tilde{\psi}$ at a root drift δ and the datum capped two-sidedly off the drift, S' at $w G$ (Lemma 30) at $S = \delta G + S'$, with the earlier ground simple at the flat window's clearance γ' , the cell's second-jump separation (Theorem 16; Theorem 26(ii)), the later ground's part φ off the earlier line obeys

$$\gamma'^2 \langle \varphi, G \varphi \rangle \leq w^2 \langle \tilde{\psi}, G \tilde{\psi} \rangle, \quad \gamma' w_\omega \leq 4 c_A w,$$

w_ω the width of the one bracket holding the two grounds' reads of an observable at the head cap c_A , the top root of the observable's form compressed to the head against the gram (Lemma 33) or any pair at or beyond it. Two windows enter at their own roots' site data at one shared clearing: below depth b the couplings are equal and the diagonals tied at the roots' drift, each window's own data beyond (Lemma 35(i)'s tie), and a ground read of a supported observable at depth j is window-free within the spectator brackets, the drift's jump bracket the decimation transports across windows (Theorem 19(iii)) and the depth- j pivot's tied sandwich capped by (ii)'s certificates, geometric in the gap g at $j + g = b$, the brackets' data the spectator certificates, cell data of the family's parameters (Lemma 35(iii)).

(iv) The separated reads cluster. At observables A, B of supports separated by j slabs, at the unit gram, the connected read is one pairing of two off-line vectors: with $\hat{B}\psi$ the connected datum at the join $B\psi = \omega(B)\psi + \hat{B}\psi$, off the ground line by its construction,

$$\omega(AB) \langle \psi, G \psi \rangle = \omega(A) \omega(B) \langle \psi, G \psi \rangle + \langle \hat{A}\psi, G \hat{B}\psi \rangle,$$

one rearrangement at the gram-symmetric multiplications; at the dual representative, the pair $(\sigma E : M)$'s site datum $P(\sigma)$ (Lemma 44), the off-line vector solves the sourced pencil,

$$D \hat{B}\psi = \sigma [E, B] \psi \quad \text{at the level gap } \varepsilon_0 G + S = P(\sigma), \quad GD = S,$$

the magnetic part commuting (the algebra commutative, Proposition 5) and the source supported at B 's reach (Lemma 14); and the two reads $\omega(AB)$ and $\omega(A)\omega(B)$ read one value within the bracket the crossing certificates, the j separating slabs' certificates (Lemma 35(ii)), and the clearance γ' price, geometric in j .

(v) The Euclidean brackets. At a pair u , the level gap S with its action D (Definition 10) and its located gaps γ_j at $\varepsilon_0 + \gamma_j = \varepsilon_j$ (Lemma 33), $\varkappa$ the largest gap, and a natural N beyond $u \varkappa$, the lower and upper members are the N -fold reads of two solved witnesses,

$$L_N := \ell^N \quad \text{at } N G \ell + u S = N G, \quad U_N := v^N \quad \text{at } (N G + u S) v = N G,$$

each solve one Green solve at a located pair (Lemma 34(iii)), the upper's datum $N G + u S$ positive definite ($G \succ 0$ with S 's split, Lemma 30); the split maps both to diagonal reads at the gaps

(Lemma 33), the lower's $[g_\gamma : N]$ at the witnessed gap $u\gamma + g_\gamma = N$ and the upper's $[N : N + u\gamma]$, and the members' gap is at most $[u^2\mathfrak{K}^2 : N]$ at every spectral point, each member's read between the two at every larger N , and a read $\omega(A \cdot B)$ at the pair of families is priced by the weighted polarization at the two probes' gram reads (Theorem 30's display), the spectral width the factor: the Euclidean read at u is the datum the nested brackets locate, one located object per pair u , the reader's continuum point reading itself in through the pair brackets beside it. The m -point reads iterate the displays, one bracket per separation and one family per pair, every entry a stage read by (i); and the iteration is capped, the growth cap: an m -point bracketed read's magnitude (Theorem 16's carrier) sits at or below the probes' head caps' fold against the separations' certificate factors, one factor per separation, every constant a located pair, the head caps the compressions' top roots (Lemma 33) and the factors the certificates' own.

(v') The decomposition display. At a convex splitting of the state into translation-invariant states, $\omega = \sum_i w_i \omega_i$ at pair weights folding to one, each component's reads agree with the state's within the connected reads' widths: at the translate fold S_n , the n -translate average of a probe A , the weighted square fold $\sum_i w_i \langle \omega_i(A) : \omega(A) \rangle^2$ sits below the widths' read at every n , one located bracket per probe.

(vi) The propagation bracket. The action D at $GD = S$ is banded at the locality band, L_N is a polynomial in the banded D , and its reach is N bands, while U_N sits within (v)'s width of it and a Green solve's beyond-band blocks are the transfer products at the certificates (Lemma 34(iii); Lemma 35(ii)). An operator reading a region's index data alone commutes with one reading a disjoint region's, the fibers' factors disjoint (Lemma 14; Lemma 42(i)), so at an involution-fixed observable A reading the leading slabs alone and its evolved probe $U_N A U_N$, the probe reads within the reach at one bracket, the beyond-reach remainder's form reads capped by the width and the crossing certificates' geometric read, and every ordering difference against an involution-fixed observable beyond the reach is the remainder's own read: the commutator reads at separation beyond N bands are priced by the bracket alone, the certificates cell data of the family's parameters (Lemma 35(iii)).

(vii) The scale sandwich. On the corner cell, at scales $\eta' < \eta$ (two equal scales reading one pencil outright) the pencils' site datum is gE at the squares' gap g , the order's witness at $\eta'^2 + g = \eta^2$, one positive-semidefinite form, the reads at the unit gram, and the transport prices at the grounds' own electric weights rather than at a head cap. The drift is a Rayleigh read at the earlier ground,

$$\varepsilon_0(\eta') \leq \varepsilon_0(\eta) \leq \varepsilon_0(\eta') + g e', \quad e' \langle \psi', G \psi' \rangle = \langle \psi', E \psi' \rangle,$$

and the off-line weight reads at the later ground's electric square weight, the electric image's self-pairing at the diagonal E (Definition 10),

$$\gamma'^2 \langle \varphi, G \varphi \rangle \leq 2g^2 \langle \tilde{\psi}, E^2 \tilde{\psi} \rangle + 2\delta^2 \langle \tilde{\psi}, G \tilde{\psi} \rangle,$$

δ the drift; and the electric square weight closes on one finite sum over the head's shells (Construction 21) at (ii)'s caps,

$$\langle \tilde{\psi}, E^2 \tilde{\psi} \rangle \leq \left(\sum_i (4C_i)^2 \prod_{j < i} \lambda_j^2 \right) \langle \tilde{\psi}_0, G \tilde{\psi}_0 \rangle,$$

C_i the i -th shell's Casimir bound (the electric read at most $4C_i$ there), $\tilde{\psi}_0$ the head-shell component, the shells finitely many at the stated cutoff: two scales' reads of one observable sit in one bracket at the priced width, (iii)'s read display at the priced cap, the weights the grounds' own reads. A support's site count at a scale is the scale count against the chain's height at the scale, the cut's edge at the corner scale (Definition 11), the count of the height's multiples at or below the support's

separation, two comparisons its read, and at every tail scale the corner floor holds, the ray's own cut member (Definition 11): the clearance is located there, the site counts stabilize with the height's brackets, and a datum's scale reads close within the stated widths, their scale sum one cofactor read at the upper family's rate.

Proof. (ii) The tail pivots are of unequal-membered determinant on the validity cells, the ground's root the decimated head's own (Theorem 19(i)), so each kernel row solves its deeper component as the transfer image of the shallower (Lemma 34(iii)), and the certificates price each step (Lemma 35(ii)).

(iii) The first display is the pencil identity read at φ with the cap's polarization (Theorem 30's display), and the second collects the reads' pairing members, three off-line terms each polarized at the cofactor pair $[w : \gamma']$, a read the cofactor at its gram's weight ((i); Definition 1), and at w beyond γ' the head cap alone reads the width at twice itself, below the display's right side. The display instantiates at the head decimated onto the depths at or below an observable's depth j : the two heads differ at the drift along the leading diagonal and at the depth- j pivot, the pivot's deviation the tied sandwich of Lemma 35(i) from depth b down to j , capped by (ii)'s certificates with the tie's datum entering once per step over the gap g at $j + g = b$, with the grounds' weights beyond the head priced by this theorem's (ii), and the drift sits within the jump bracket the decimation transports across windows (Theorem 19(iii)).

(iv) $\hat{B}\psi$'s weight at A 's side rides the Green products through the j separating slabs at the certificates (Lemma 34(iii); Lemma 35(ii)) against the clearance γ' : A 's support is the split's leading slabs at their joined width, the weight's head part the cross-cleared sum over them priced by the certificates' fold at the head's side (this theorem's (ii) at the leading slabs), and the ride anchored at the source slab's own block weight with the crossing certificates pricing the j separating steps; and the connected pairing is that weight at A 's head cap, the cap's polarization closing it (Theorem 30's display).

(v) At the two diagonal reads the cleared product identity and the cleared Bernoulli display close the bracket,

$$g_\gamma (N + u\gamma) + u^2\gamma^2 = N^2, \quad g^N + N a b^{N'} \geq b^N \geq g^N + N a g^{N'} \quad (a + g = b, \quad N' + 1 = N),$$

so the members' gap is at most $[u^2\gamma^2 : N]$ at every spectral point, each member's read between the two at every larger N (the refinement comparisons, the Bernoulli display again). The growth cap: each pair step is the weighted polarization at the probe's head cap against the prior fold's gram weight (Theorem 30's display) and each separation rides the Green products at the certificates (Lemma 34(iii); Lemma 35(ii)).

(v') Each component reads its own mean square below the fold's, $\omega_i(\bar{S}_n S_n)$ at or beyond $\omega_i(\bar{A})\omega_i(A)$ by the invariance with the pairing's positivity, while the state's read collects the pair reads, $\omega(\bar{S}_n S_n)$ within the clustering brackets of $\omega(\bar{A})\omega(A)$ at the mean's own weight and the off-diagonal widths' geometric fold (the display of (iv)), so the weighted square fold sits below the widths' read.

(vi) The gram is the fibers' block diagonal (Definition 8) and S is banded at the locality band (Lemma 14; Lemma 41's rows), so the action D is banded, and ℓ reads the join $N\ell + uD = N$ at the unit's N -fold on the right, so L_N is a polynomial in D at the reach of N bands; and at the disjoint regions' operators commuting the evolved probe reads within the reach at one bracket and every ordering difference beyond the reach is the remainder's own read.

(vii) The drift display reads the lower side at the counts monotone (Lemma 30) and the upper at the earlier ground as trial pair; the off-line weight's display is the transport display's pairing step at the two-term square (the pencil identity read at φ , (iii)); and each summand of the shell

sum is one comparison at the certificates. A scale count is one integer across a scale bracket whose two ends read it, the count's ties, the multiples at the separation, its located change points, so the site counts stabilize with the height's brackets. $\square$

Theorem 22 (Euclidean restoration is a count). *Take the carrier at the lattice interface's fields and a region mapped to itself by the hyperoctahedral group B_d of signed coordinate permutations (Construction 7).*

The symmetry passes to the fiber symbol. $E = \sum_{\ell} \Delta_{\ell}$ and $M = \sum_p \chi_L(U_{\partial p})$ run over all links and all plaquettes at coefficient one (Definition 10), B_d permutes those two sets, and $H([\alpha : \beta])$ commutes with the action. A lattice symmetry g permutes the label index isometrically and maps each direction's translation to the permuted direction's own, so it maps chord blocks onto chord blocks (Lemma 40), the direction permutation permuting the block index $(t_1, \dots, t_d)$. So the K -fiber's pencil and the gK -fiber's are one matrix at the moved states' list (Theorem 17's transport field, the moved members at the dual labels where the orientation reverses), and the symbol is one value at every stated list of the fiber (Theorem 18): their characteristic polynomials agree, coefficient by coefficient,

$$\mathcal{S}(gK; k, r) = \mathcal{S}(K; k, r)$$

with $\mathcal{S}$ the fiber symbol of Theorem 18. Every coefficient of $\mathcal{S}$ in k is therefore one value at every relabeling of the chord list, and the coefficients are the permutation-invariant chord polynomials, one coefficient identity per congruence; their momentum reading is even in each coordinate, the deck relation's two roots exchanging at a flip with the chord fixed (Definition 4). The derivation runs unchanged at every number of arguments. A lattice symmetry permutes the label index and maps the fibers isometrically, an orthogonal map commuting with H , so at a window whose multiplicity reads 1 (Theorem 10) it reads the ground line by a graded read the reads of ω are blind to: ω is B_d -invariant, and hence every read $\omega(A_1 \cdots A_m)$ is B_d -invariant, jointly in the positions and the orientations B_d permutes. The momentum transform of the read family is its translate family read at monomials: per argument the translate key list $a^{(r)}$, one key per direction below the side, the read $\omega(T^{a^{(1)}} A_1 \cdots T^{a^{(m)}} A_m)$ at the composite shifts, and the transform the occupancy family of those reads at the per-argument exponent lists, the translate keys' own matched lists (Definition 1), the momenta $K^{(1)}, \dots, K^{(m)}$ the exponent coordinates, one list per argument; a lattice symmetry moves a translate read to the g -moved keys' own, the direction permutation permuting each argument's keys and a flip reading a key to its balance partner at the wrap (Lemma 40's $T^L = 1$), so the transform is B_d -invariant with the reads, the moves the coefficient identities.

The invariant spaces, degree by degree. The momenta enter as formal coordinates at the transform's own carrier, one matched list per argument, and two moves settle every degree: a reflection, one coordinate K_i read to its balance partner across all m momenta at once, sends every monomial at an odd count of that coordinate to its balance partner, the coefficient's members then equal, and a transposition of coordinates equalizes what survives.

1. Degree 2: the rings coincide, at every m and every d . The degree-2 B_d -invariants are spanned by the pairwise inner products $K^{(r)} \cdot K^{(s)}$, each of which is $O(d)$ -invariant already, so the degree-two component is isotropic.
2. Degree 4: breaking enters. The isotropic span at a degree is the pairwise inner products' polynomial span, item 1's ring at its products, and the breaking multiplicity is the independent invariants' count beyond it, the classical $O(d)$ -invariant ring the isotropic span's reading in the dictionary alone: the derivations read the two spans and their counts. The count reads one diagonal quartic, $p_4 = \sum_i (K_i^{(1)})^4$, every coordinate total four. At $m = 1$ every invariant is a cleared combination of p_2^2 and p_4 over the balance pairs (Definition 1), and the two are

independent: multiplicity 1 at the join $1 + 1 = 2$, at every $d \geq 2$, with p_4 off the isotropic span at the paired monomial $K_i^2 K_j^2$'s coefficient. At every $m \geq 1$ the quartic p_4 is an invariant off the isotropic span at $d \geq 2$, so the multiplicity is positive there.

3. *The degree gap is two, at every $m \geq 1$. An odd degree's every invariant reads equal members at every monomial, so the first permitted positive degree is 2 and the first breaking invariant sits at degree 4, the gap at $2 + 2 = 4$, at every argument count and at every $d \geq 2$. The multiplicity at degree 4 depends on m .*

At every member and argument count the degree-2 invariant space is the inner products' span.

Proof. The symmetry. A signed permutation reverses link orientations, which dualizes labels, so E is fixed because $C_2(\bar{\lambda}) = C_2(\lambda)$ (Definition 6) and M is fixed because the closure holds every member's dual, a reversed boundary reading the dual label's character and the plaquette character its own image under the involution (Construction 6); hence $H([\alpha : \beta])$ commutes with the action. An orientation flip reads T at T^* , the chord operator $S = T + T^*$ fixed with it, so a lattice symmetry maps chord blocks onto chord blocks at the direction permutation alone; a flip fixes each chord at the operator's own read, so B_d reads the chord list through its direction permutations alone, and the coefficients are the permutation-invariant chord polynomials. *The degrees.* The reflection leaves only $\sum_i K_i^{(r)} K_i^{(s)}$, and the transposition fixes its coefficient, so the degree-2 B_d -invariants are spanned by the pairwise inner products; the degree-two component is isotropic because the invariant space at that degree is the inner products' span, the inclusion $B_d\text{-inv} \subseteq O(d)\text{-inv}$ at that degree the whole of what the coincidence asks, and the two moves above exhibit it. At degree four the reflection leaves the coordinate indices paired, so the space includes diagonal quartics alongside the inner products' products, and the count reads p_4 ; at $m = 1$ the cleared combination's clearing datum is two at the paired monomial's splitting count, $K_i^2 K_j^2$ read at its two factor orders, and the two are independent, a combination at the sum's unit reading each coefficient at the quartic and the paired monomial in turn; at every m the quartic's two reads sit at the first argument's own monomials. At an odd degree the coordinate totals join to the degree with one member odd, so every invariant reads equal members at every monomial. $\square$

Theorem 23 (The continuum reads). *Fix the member and the direction count. A continuum datum is a finite list of supports, separations and Euclidean pairs stated against the chain's height at pair coordinates (Lemma 47; Theorem 21(v)), its observables the algebra's at those supports (Definition 9).*

(i) *The data are bracketed, and the spectral heights read two-sided on the certificate's cell. At every scale of the corner cell the datum's read sits in the stated brackets (Theorem 21(i)–(vi)), the support counts read as the scale counts against the chain's height at the scale (Theorem 21(vii)), and two scales' reads sit in the scale sandwich's priced brackets (Theorem 21(vii)). At every tail scale the corner floor, the decimated head's located gap at the scale (Lemma 48), locates the clearance, the ground-level sandwich sums to one cofactor read at the upper family's rate (Lemma 46), the heights read two-sided at the floor's and the caps' rates at the scale, the site counts stabilizing with the floor, and a datum's scale reads close on the moment folds' clearance cell (Lemma 62).*

(ii) *The bracketed reads cohere. Every finite Gram of bracketed reads is positive semidefinite within its brackets; reads at disjoint supports read one value at either order, and an ordering difference beyond the reach is the remainder's own read (Theorem 21(vi)); separated reads pair to their product within the crossing certificates' brackets (Theorem 21(iv)); the certificate's head's heights are two-sided located data at its positive located floor at the floor's rate at the scale, and the spectral data sit at the ground in the corner scale there (Lemma 48; Lemma 20); and the momentum*

reads are isotropic at degree two with the direction data two degrees down, the multiplicities scale-free (Lemma 63; Theorem 22).

(iii) The spacing is an output. The chain's height at a scale is located between the ray's floor and its caps, the decimated head's two-sided datum at the floor's and the caps' rates at the scale, and the cutoff-tail probe caps (Lemma 48; Lemma 22(iv)), one output bracket per scale, and at two directions or beyond the height falls under every floor pair beyond the floor's key (Lemma 55).

Proof. (i) A support's read at a scale is a stage read within the transport bracket (Theorem 21(i),(iii)), a separation's within the crossing certificates' (Theorem 21(iv)) and a Euclidean pair's within the pair bracket (Theorem 21(v)); the site count is the scale count against the height at the scale, and two scales' reads sit in the sandwich's priced brackets (Theorem 21(vii)). At a tail scale the corner floor is positive (Lemma 48), so the flat window between the ground and the floor locates the clearance there; the drift's rate sums the halving scales' ground-level widths to one cofactor read (Lemma 46); the heights read between the floor and the caps at their rates (Lemma 48; Lemma 55), the site counts one value across a scale bracket whose ends read one count (Theorem 21(vii)), and the moment folds' solve closes the datum's scale reads on the clearance cell (Lemma 62(iv),(v)). (ii) A Gram of bracketed reads is the pairing's Gram at the reads' vectors, every split on its upper side at the pairing definitional at every scale (Definition 8; Lemma 30); operators at disjoint supports commute and the beyond-reach difference is the propagation bracket's remainder (Theorem 21(vi)); separated reads pair to their product within the crossing certificates' brackets, the connected datum solving the sourced pencil with its off-line part capped over the separating slabs (Theorem 21(iv)); the head at the ray is the decimation's own, its count the chain's (Theorem 19(ii)), the floor's clearance the spectral read at the ground (Lemma 20); and the multiplicities are scale-free at Theorem 22's count, a read of which invariant forms exist at each degree rather than of their size, the degree-two term isotropic with the direction data at degree four (Lemma 63). (iii) The height is the cut's edge at the corner scale (Lemma 47); every pair at $E_0\kappa$ below the floor is a member of the ray's cut (Lemma 48), the meet at or below each regional cut (Definition 11). The simple-ground key is the key of Lemma 48's certificate and finite witness. Joining it to the trial members' contents makes the cap line valid on that region's cutoff tail (Lemma 62(vi); Lemma 55). That cap and the cutoff-tail probe caps (Lemma 22(iv)) sit at or beyond the edge; at two directions or beyond the scale key puts the window's height below the floor pair at every further scale, the cap line against the floor line at a window of the directed family, the lines' gap falling with the window's mode read $e_W = [20 : L^2 + 2L + 2]$ at the side (Lemma 55), the chain's height under that regional cap. $\square$

20 The network constituents

Theorem 24 (Network reduction and the two-vertex coupling law). (i) A three-link path labeled R is one edge labeled R (the convolution identity), so the theta graph reduces to the three-edge Θ -graph and the state space is $\Theta(R_1, k, R_2; \alpha, \beta)$ -networks with E diagonal, eigenvalue $3C_2(R_1) + C_2(k) + 3C_2(R_2)$ at the graph's link weights (Construction 10). (ii) Couplings between sectors differing on the shared link need the label-index reads of Proposition 8 on the shared edge alone: the module-pair weights are

$$w_k = [m_k d_k : r^2(r+2)^2], \quad \sum_k m_k d_k = r^2(r+2)^2$$

(the block count over Lemma 24's list, Lemma 5(ii)),

with d_k and m_k the dimension and the multiplicity of Proposition 8 at the channel k , so the sum is the completeness of $\text{adj} \otimes \text{adj}$, whose dimension is $r^2(r+2)^2$; the weights are the channel table's reads (Lemma 24),

$$w = \left([1 : r^2(r+2)^2], [c_1 : r(r+2)], [(r+1)^2 r(r+4) : 4r^2(r+2)^2], \right. \\ \left. [g(r+3) : 4r(r+2)]_{g+1=r}, [g(r+3) : 4r(r+2)]_{g+1=r}, [(r+1)^2(r+2)w' : 4r^2(r+2)^2]_{w'+2=r} \right)$$

over the channel list, the leading-term reads in the residue: the first two weights' second members exceed in degree, and the four channel weights read $[1 : 4]$ at the top (Theorem 16's leading terms). (iii) The weights are the data. The slot-two self-energy assembled from these weights and the network energies is $\sum_k w_k [q_k : 2q_k + p_k]$ at $\hat{c}_2(k) = [p_k : q_k]$, the factor's gap the module pair's network eigenvalue against the module count, $3 + \hat{c}_2(k) + 3$ against the adjoint loop's content 4 at $\hat{c}_2(\text{adj}) = 1$ ((i)'s eigenvalue), the resolvent factor a constructed pair. The recoupling constituent's content is the finite tetrahedral list: the within-pair middle matrix $\langle (\text{adj}, k', \text{adj}) | \chi_L(U_1) | (\text{adj}, k, \text{adj}) \rangle$ at the plaquette character (Definition 10), the adjoint walk's instance at $\chi_{\text{adj}}(U_1)$ (Construction 15), the middle index the channel table's occupied Casimir values, the dual pair's shared value one joint block at the summed weight, computable per entry by the E_{link} interpolant, the projection onto one Casimir value a polynomial in E_{link} (Construction 2's Lagrange read) entered cleared at the node differences' product, one value across the clearing scale (Definition 1's homogeneity principle), at the channel distances' occupied gaps (Lemma 24).

Proof. (i) A path's two interior vertices are two-valent, each reading the unit label's count at the incident label against its dual, $N_{RR}^1 = 1$ (Proposition 8), so the path's three labels are one label and its fiber the edge's; the electric member reads each link's Casimir at the link weights $(3, 1, 3)$ (Construction 10), the fold $3C_2(R_1) + C_2(k) + 3C_2(R_2)$. (ii) The shared link's label runs over the channels of $\text{adj} \otimes \text{adj}$ at the multiplicities m_k and dimensions d_k (Lemma 24), the weight the channel's share of the dimension identity $\sum_k m_k d_k = d_{\text{adj}}^2$ (Lemma 5(ii)) at $d_{\text{adj}} = r(r+2)$; the six weights are the table's m_R against d_R cleared at $r^2(r+2)^2$, the leading terms read at the degrees, the first two weights' first members constants against second members of degrees four and two, and the four channel weights' members at one degree each with the top coefficients one against four. (iii) The module pair's network eigenvalue against the module count is $6 + \hat{c}_2(k)$ against 4, the gap $2 + \hat{c}_2(k)$ at the join $4 + (2 + \hat{c}_2(k)) = 6 + \hat{c}_2(k)$, the pair $[2q_k + p_k : q_k]$ at $\hat{c}_2(k) = [p_k : q_k]$, and the resolvent factor is its cofactor, $[q_k : 2q_k + p_k]$. The projection onto one Casimir value of E_{link} is the Lagrange interpolant through the occupied values (Construction 2), cleared at the node differences' product to a polynomial with pair coefficients (Definition 2), one value across the clearing scale, the values' gaps the channel distances (Lemma 24). $\square$

21 The drift and the divisor

Theorem 25 (The uniform drift). *At every label R the adjoint row reads $\sum_k m_k d_k \hat{c}_2(k) = d_R d_{\text{adj}}(\hat{c}_2(R) + 1)$, the fold over the row's channels at the multiplicities against the dimensions: the dimension-weighted fusion walk's Casimir drift is exactly $+1$, uniformly in R ($\sum_k w_k \hat{c}_2(k) = 2$, the channel table's second closure identity, Lemma 24). With strictness (the unit label is the one label of dimension one: a nonunit reduced shape reads an occupied head row at its positive degree against the withdrawn full column's vacant last row, the rows weakly descending (Construction 1), so a consecutive row gap $h \geq 1$ sits at some $\lambda_{i+1} + h = \lambda_i$, its gap factor $h + 1$ against the unit display's one with every further factor at or beyond its left partner (Corollary 1's display), so $d_\lambda \geq 2$),*

the closure datum is a constant rather than a decaying one. And the display holds at every second factor: at labels R and λ the row over the channels of $R \otimes \lambda$ reads

$$\sum_k m_k d_k \hat{c}_2(k) = d_R d_\lambda (\hat{c}_2(R) + \hat{c}_2(\lambda)) ,$$

the derivation below at λ 's block, so at the magnetic list the closure's rows fold to

$$\sum_{\lambda_0 \in \bar{L}} \sum_k m_k d_k \hat{c}_2(k) = d_R \left(d_L \hat{c}_2(R) + \sum_{\lambda_0 \in \bar{L}} d_{\lambda_0} \hat{c}_2(\lambda_0) \right)$$

(Construction 6).

Proof. The member tier. Trace the Casimir over $W_R \otimes W_\theta$ at the trace read (Lemma 3), one value at every orthogonal spanning list. At an exhaustion the blocks' lists join to one orthogonal spanning list, the blocks pairwise perpendicular (Lemma 5(i),(ii)), and the fold splits per block at the cleared products, each block's diagonal pairings the scalar's multiples of the self-pairings (Lemma 4(ii); Lemma 7): the trace reads $\sum_k m_k d_k C_2(k)$, one summand per channel at its count. The tensor's operators read by the Leibniz rule and the Casimir's fold collects at the factors (Construction 3),

$$C_\otimes = C_R \otimes 1 + 1 \otimes C_\theta + 2 \sum_\alpha [\langle \alpha, \alpha \rangle : 2] (E_\alpha \otimes F_\alpha + F_\alpha \otimes E_\alpha) + Z ,$$

each root term's two orders splitting at the four arrangements with the one-sided pairs collecting to the factors' own folds, and Z the diagonal fold's cross read, the form's additivity at the joint contents, $q(\mu + \nu) = q(\mu) + q(\nu) + 2 \langle \mu, \nu \rangle$ per content pair. The trace factorizes over the tensor list, two tensors pairing at the factors' pairings' product (Lemma 5(iii)), so each term's trace is the factor traces' product: the two one-sided terms read $C_2(R) d_R d_\theta$ and $d_R C_2(\theta) d_\theta$; a root term's factor trace is a fold of cross-content pairings, each of equal members at the content sum's orthogonality (Construction 3), so the root terms' traces read equal members; and Z 's trace is the double fold $2 \sum_{\mu, \nu} \text{mult}_R(\mu) \text{mult}_\theta(\nu) \langle \mu, \nu \rangle$, the form's bilinearity collecting it to $2 \langle F_R, F_\theta \rangle$ at the content folds $F_X := \sum_\mu \text{mult}_X(\mu) \mu$. The θ -block's contents are simple folds, the root list's members with the unit content (Proposition 13's content list) at each root's fold over the simple list (Construction 3), so the pairing collects at the halved products to F_R 's coroot pairs, $2 \langle F_R, \alpha_i \rangle = F_R(\alpha_i^\vee) \langle \alpha_i, \alpha_i \rangle$, and each reads equal members: the multiplicities are one value at every reflection (Lemma 6), a content joined to its s_i -image reads the coroot pair with its balance partner, and a fixed content's pair is its own partner. So

$$\sum_k m_k d_k C_2(k) = d_R d_\theta (C_2(R) + C_2(\theta)) ,$$

and the display clears at $C_2(\theta) = 2(r+1)$ (Proposition 12), one value across the clearing (Definition 1's homogeneity principle): $\sum_k m_k d_k \hat{c}_2(k) = d_R d_\theta (\hat{c}_2(R) + 1)$. At a second factor λ the trace runs over $W_R \otimes W_\lambda$: the one-sided terms read $C_2(R) d_R d_\lambda$ and $d_R C_2(\lambda) d_\lambda$, a root term's factor trace is a fold of cross-content pairings at equal members at the content sum's orthogonality, and Z 's trace is $2 \langle F_R, F_\lambda \rangle$ at the content folds: F_λ is fixed by every reflection (the content symmetry, Lemma 6, the fold's members permuted at one multiplicity each), so at every simple index $F_\lambda(\alpha_i^\vee) \alpha_i$ reads the sum's unit at the reflection's join and F_λ 's simple coroot pairs read equal members, the content itself the unit fold at the coroot reads' injectivity (Construction 3), and the cross fold reads the sum's unit: the display at λ .

The label calculus at the residue. At the matrix units the constant family holds every consecutive coroot pair at equal members, so the content folds read the constant family and the derivation runs at the alternant display's recursion (Theorem 1), the display at every content list of the block's width with a key at the count's unit reading the moved folds' one value. Sum the display over the recursion family's keys: the moved folds reindex at the move's own round trip, per place pair $a < b$ and per occupied target κ the sources at the shifts $1 \leq l \leq \kappa_a$, one source per shift at the reverse move, the second letter's weight reading the target's own $\kappa_b + l$ and the shifted first letter's $\kappa_a + l$; the two reindexed folds share the shift fold $\sum_{l \leq \kappa_a} l$, a summand on both members fixing the value (Definition 1), and the summed display reads

$$q(\lambda + u) d_\lambda + 2 \sum_{\kappa} \text{mult}(\kappa) \sum_{a < b} \kappa_a \kappa_b = \sum_{\kappa} \text{mult}(\kappa) q(\kappa + u) + 2 \sum_{\kappa} \text{mult}(\kappa) \sum_{a < b} \kappa_a^2 .$$

The swap symmetry (Lemma 6's read at the matrix units) reads each place fold place-free, the first moment M_1 , the square fold M_2 and the product fold M_{11} one value per place and place pair with $d_f M_1 = k_\lambda d_\lambda$ at the contents' degree read, so the summed display collects to $q(\lambda + u) d_\lambda + 2 p M_{11} = d_f M_2 + 2 U_1 M_1 + q(u) d_\lambda + 2 p M_2$ at p the place pairs' count and U_1 the unit display's total. The fused carrier's family splits twice, over the channels at the exhaustion's content refinement (Lemma 5(ii)) and over the factors at the contents' addition (Construction 1), so each moment fold reads both ways, the channels' counted fold against the factors' split at the further factor's count,

$$M_1(F) = d_\theta M_1(R) + d_R M_1(\theta) , \quad M_2(F) = d_\theta M_2(R) + 2 M_1(R) M_1(\theta) + d_R M_2(\theta) ,$$

with $M_{11}(F)$ at M_2 's shape, the cross read $2 M_1(R) M_1(\theta)$ shared: the collected display at the fused carrier, read against the two factors' own collected displays at the further factors' counts, withdraws the quadratic folds (a summand on both members fixing the value) and leaves

$$\sum_c N^c q(c + u) d_c + q(u) d_R d_\theta = (q(R + u) + q(\theta + u)) d_R d_\theta + 2 d_f M_1(R) M_1(\theta) ,$$

whose d_f -multiple reads the moments at the degree folds, $d_f M_1(R) = k_R d_R$ and $d_f M_1(\theta) = d_f d_\theta$ at θ 's degree d_f . The square identity (Definition 6) with the row read $\langle \lambda, 2\rho \rangle + (d_f + 1) k_\lambda = 2 \sum_i \lambda_i u_i$ (Theorem 1's collection at Lemma 7's fold) reads $d_f q(\mu + u) = d_f Q(\mu) + k_\mu^2 + d_f(d_f + 1) k_\mu + d_f q(u)$ at every shape; every channel shares the degree $k_c = k_R + d_f$, the contents adding over the factors at θ 's degree, so the dimension identity $\sum_c N^c d_c = d_R d_\theta$ (Lemma 5(iii)) withdraws the shared reads and the display closes at $Q(\theta) = 2 d_f$ (Lemma 7):

$$\sum_c N^c d_c d_f Q(c) = d_R d_\theta (d_f Q(R) + 2 d_f^2) ,$$

the display's $\hat{c}_2$ -cleared read at $[d_f Q : 2 d_f^2]$ (Definition 6); at a second factor λ the collected display runs at λ 's block, every channel at the degree $k_c = k_R + k_\lambda$, the moments' read $2 d_f^2 M_1(R) M_1(\lambda) = 2 k_R k_\lambda d_R d_\lambda$ withdrawing the degree squares' cross term at the dimension identity, and the display closes at the second factor's own cleared bracket $d_f Q(\lambda)$: $\sum_c N^c d_c d_f Q(c) = d_R d_\lambda (d_f Q(R) + d_f Q(\lambda))$. $\square$

Theorem 26 (The divisor identification, and the crossing divisor). *(i) The identification. The spectrum is the root locus of the exact fiber symbol (Theorem 18): on a window a level is a pencil root exactly where the determinant reads equal members, the factorization read of Lemma 33 at the gram's determinant off the sum's unit, and the Feshbach determinant reads equal members with it,*

one Schur factor of the whole at a complement of unequal-membered determinant (Lemma 30), at every coupling outright, each root a located divisor point.

(ii) *The crossing divisor.* For a symmetric stage pencil of the chain with $G \succ 0$, a window compression or a fiber pencil, the entries polynomial in the base coordinates over the stage's field (Lemma 40's band congruence), write $\chi(x)$ for the level pair's site-datum determinant (Lemma 33's χ) and take the squarefree split $\chi = c \prod_j S_j^j$ over the base's polynomial pairs, the division descent's iterated greatest common divisors, the equality test cross-multiplied through the iterated division, each S_j taken primitive over the base's polynomial ring with the contents collected into c , itself a base polynomial (Gauss): the split is an identity of base polynomials and specializes at every stage point. The crossing read is

$$D := \prod_j \text{Res}_x(S_j, S'_j) \cdot \prod_{i < j} \text{Res}_x(S_i, S_j) ,$$

the resultants Definition 3's at the variable x , a base element off the sum's unit. At a base point off D 's root locus the specialized split keeps the generic structure, every root of χ at that point a root of exactly one S_j , simple in it, of multiplicity j in χ , and distinct roots stay distinct: root collisions sit on D 's located root locus alone, the jump divisors of the decimation's list; a decimated head reads its counts as the fiber pencil's on its bracket, so its jumps inherit the confinement.

(iii) *The ground's read.* On a cell with pairs $l < h$, the pencil pair at l positive semidefinite and the counting function flat at h across the cell, the ground multiplicity is $\text{count}(h)$, both reads cell functions read once (Lemma 31); by (ii) it moves across a located boundary only where a ground collision sits on the boundary, the boundary's own two-sided bracket counts. On the vacuum sector's pencil at the free end's side it reads 1 over the cell attached to the free end, the least root in the free end's level bracket (Lemma 15) at multiplicity 1 there.

Proof. (ii) A factor of D of equal members reads an occupied cross pair on a squarefree or coprime pair, against its Bézout read $up + vq = 1$ (Definition 3), so D sits off the sum's unit. At a base point off D 's root locus the leading x -coefficients multiply to $\det G$ there, on its upper side by the split, so every one reads unequal members and the degrees hold; each S_j stays squarefree and the pairs stay coprime, the resultants' own reads there; so every root of χ at that point is a root of exactly one S_j , simple in it, of multiplicity j in χ , and distinct roots stay distinct. (iii) A level bracket of pairs (l, h) isolating the free ground, the unit's line against content $4\hat{c}_2(f)$ (Lemma 15), reads the pair at l positive semidefinite and $\text{count}(h) = 1$ at the free end, both reads constant over the attached cell (Lemma 31), so the least root sits in the bracket at multiplicity 1 there. $\square$

Theorem 27 (The scheme bridge is a count).

$$[d_{\text{adj}} C_2(\text{adj}) : d_f C_2(f)] = 2 d_f$$

exactly, the pair of two weight-1 objects, one value at every form point (Proposition 1), and the read is a count: the f -flux bridge's component at the contact end is $b_f^{\text{equiv}} = 2d_f b_X$, one rescaling of the coupling coordinate.

Proof. $d_{\text{adj}} = r(r+2)$ with $\hat{c}_2(\text{adj}) = 1$ (Lemma 16; Definition 6), and the fundamental count d_f with $\hat{c}_2(f) = [r(r+2) : 2d_f^2]$ (Lemma 7), so the pair reads the cross-multiplied natural identity $r(r+2) \cdot 2d_f^2 = 2d_f \cdot d_f \cdot r(r+2)$; the $r = 2$ evaluation is 6. $\square$

Definition 13 (The ruler excess). At the residue r the *ruler excess* is the polynomial

$$A(u) := \sum_{j=1}^r [1 : j] u^j$$

(Definition 2), the one polynomial supported from key one with $A' + u^r = 1 + u A'$, r the pairing's residue.

Theorem 28 (The walk residue). *The tower pairing is forced. At the residue r : for the principal adjoint tower (the $k\theta$ -string, θ the highest root, $\hat{c}_2(k) = [k(k+r) : r+1]$), the Casimir $C_2(m\theta) = m(m+r)$ of Proposition 12 splits into a square part and a row-weight part, and the pairing is the second over the whole:*

$$\varphi_m = [m r : m(m+r)] = [r : m+r], \quad \varphi_m + \psi_m = 1 \quad \text{at} \quad \psi_m = [m : m+r],$$

two weight-1 objects read as one pair, one value at every form point (Proposition 1), every datum forced, ψ_m the pairing's complement, the one-value read. Its generating object is

$$\Psi(u) := \sum_{m \geq 1} [1 : m] \psi_m u^m = \sum_{m \geq 1} [1 : m+r] u^m,$$

and the shifted series $F(u) := u^r \Psi(u) = \sum_{j \geq r+1} [1 : j] u^j$ satisfies the recursion $F' = u^r + u F'$, coefficientwise one read per degree: its divisor pair reads the least occupied degree r and the pole order 1 at $u=1$, both naturals. Against the harmonic ruler $R(u) := \sum_{m \geq 1} [1 : m] u^m$, the ruler at the recursion $R' = 1 + u R'$, its pole order 1 as well, the pole excess is one polynomial, coefficientwise exact and finitely supported:

$$R(u) = F(u) + A(u), \quad A(u) = \sum_{j=1}^r [1 : j] u^j \quad (\text{Definition 13}),$$

$$A' + u^r = 1 + u A', \quad \text{the support from key one},$$

the one polynomial with the geometric derivative, closing $F' + A' = R'$ term by term, a divisor-characterized object. The value's second coordinate is its read at $u = 1$, the polynomial evaluation

$$A(1) = H_r = \sum_{j=1}^r [1 : j],$$

one pair per residue, the ruler's excess over the walk; and the coefficientwise record at the Ψ -level is the finite telescope, exact at every M ,

$$\sum_{m=1}^M [r : m(m+r)] + \sum_{j=M+1}^{M+r} [1 : j] = H_r, \quad [1 : m] = [1 : m+r] + [r : m(m+r)],$$

the tail itself an r -term expression. One object, one pole, one read: φ fixes Ψ , Ψ 's divisor fixes F' , and the ruler's excess is the polynomial A , the construction settling the coordinate outright. The exponent is an integer at $r = 1$ alone; the pairing's own pole residue r agrees with it there and separates from it at $r = 2$ on, so the residue-one instance leaves the two together, and the construction rather than the instance is what selects.

Proof. The pairing: $C_2(m\theta) = m^2 + mr$, the square part m^2 and the row-weight part mr , and the second over the whole is $[mr : m(m+r)] = [r : m+r]$ at the shared factor m (Definition 1's one value at $ad = cb$); the complement joins at $[r : m+r] + [m : m+r] = [m+r : m+r] = 1$. Ψ 's coefficient is $[1 : m][m : m+r] = [1 : m+r]$; the shift by u^r reads F 's coefficient at the key $j = m+r$ as $[1 : j]$ from the key $r+1$ on, so F' has the coefficient one at every key at or beyond r , the recursion $F' = u^r + u F'$ key by key, the least occupied degree of F the key $r+1$, its derivative's r , and the pole order one at $u = 1$ the geometric series' own. R' has the coefficient one at every key, $R' = 1 + u R'$, and R against F reads the keys one to r alone, $R = F + A$ at the displayed A , whose derivative has the coefficient one at the keys below r , $A' + u^r = 1 + u A'$, and $F' + A' = R'$ key by key. $A(1)$ is the fold of the coefficients, H_r . The summand identity reads $(m+r) = m+r$ at the clearing $m(m+r)$, the three terms' first members there; summing it over m from one to M reads $H_M = \sum_{m \leq M} [1 : m+r] + \sum_{m \leq M} [r : m(m+r)]$, and the first sum is H_{M+r} less H_r , the shifted keys, so the telescope collects at H_r with the tail $\sum_{j=M+1}^{M+r} [1 : j]$ the shifted sum's excess over H_M . The exponent: $H_1 = 1$; at $r \geq 2$ let 2^k be the largest power of two at or below r ($k \geq 1$) and P the product of the odd naturals at or below r ; every $j \leq r$ off 2^k is $2^a b$ at an odd b and $a < k$ (at $a \geq k$ the natural j is 2^k itself or sits at or beyond 2^{k+1} , past r), so $2^k P [1 : j]$ is the even natural $2 \cdot 2^{k-1-a} P_b$ at the cofactor P_b at $b P_b = P$, while $2^k P [1 : 2^k] = P$ is odd; the fold $2^k P H_r$ is an even natural joined to an odd one, odd, while every natural's $2^k P$ -multiple is even, so H_r is off the naturals at every $r \geq 2$. The pole residue r reads one at $r = 1$ with H_1 ; at $r \geq 2$ the two separate, H_r below r (the terms beyond the first at or below $[1 : 2]$, $H_r \leq [r+1 : 2] < r$). $\square$

Definition 14 (The contact pair). At the contact end the walk is the head's compression to the tower labels, the tower matrix (Proposition 13; Lemma 45), and two divisors read it: the channel's, the boundary resolvent G_0 with its ι -symmetrized weight (Theorem 13), and the tower's, the generating object Ψ against the harmonic ruler (Theorem 28). The *contact pair* is the two divisors' reads: the first coordinate is the channel's order count at the base c_1 (Definition 12); the second is the ruler's excess over the walk read at one, $A(1)$ (Definition 13).

Theorem 29 (The contact pair's assembly). *The contact pair of Definition 14 reads factor by factor, and there are two: the walk exponent H_r on the ruler's side, the polynomial excess $A(1)$ (Theorem 28) and coupling-free; and the channel's bit count, read off the divisor of G_0 (Theorem 13). At the contact end the walk is the head's compression to the tower labels (Proposition 13; Construction 15), and its two divisors are: the vacuum's boundary resolvent G_0 with its ι -symmetrized weight (Theorem 13) and the tower's generating object Ψ against the harmonic ruler (Theorem 28); the mass weight is the pair $[\langle c_1^2 : 1 \rangle : c_1^2]$ against its complement $[1 : c_1^2]$, the two summing to one, its side the upper at base two and equal members at base one, the mass point at the crossing there, the divisor count one value throughout (Theorem 13). At every window and coupling of the closed segment, each window cut's edge, where one is named, is the compressed pair's least root, a located root of the window pencil's symbol (Theorem 26(i); Lemma 33; Lemma 20), read at a word image with the two moments at equality and the first positive, the state the word sector's own (Lemma 22(i),(ii); Lemma 21). So the pair's two coordinates read:*

$$(3, H_r),$$

and both are divisor reads.

The pair $(3, H_r)$ is what the theory determines: two orders, both naturals, summing to 3, and an exact pair, the first coordinate read at the base c_1 , a fusion multiplicity moving with the residue (Construction 15). The two reads are of two divisors, the channel's (G_0 , Theorem 13) and the tower's (Ψ , Theorem 28), and the coordinates read them alone.

Proof. The second coordinate is the walk's pole excess against the harmonic ruler, the polynomial read $A(1) = H_r$ on the ruler's side, off F' 's divisor pair, the least occupied degree r and the pole order 1 at $u=1$ (Theorem 28). The first is the channel's order count: the boundary entry at its renewal witness $G_0 = z + c_1 z G_0$ reads its divisor as the witness's own, the mass point at $c_1 z_* = 1$ a simple pole, order 1, and the ι -symmetrized weight reading the base at order 2 (Theorem 13); the two orders against the base give $c_1^1 \cdot c_1^2 = c_1^3$, the order count $1 + 2 = 3$ (Definition 12). The mass weight's side is the pair $\langle c_1^2 : 1 \rangle$'s, the upper at $c_1 = 2$ and equal members at $c_1 = 1$, where $z_* = 1$ sits at the crossing (Theorem 13); and the edge clause is Lemma 22(i),(ii) at the window's compressed pair with the state Lemma 21's. $\square$

Theorem 30 (Truncation control). *Fix a region Λ . At the coupling $[\alpha : \beta]$ the truncation at Casimir cutoff Λ_C is a projection Π , the statement homogeneous on the ray, and the off-block coupling is the magnetic form's cap. M is multiplication by $\sum_p \chi_L(U_{\partial p})$ (Definition 10), each plaquette term capped as a form, two-sided at the closure's dimension fold (Lemma 18), so the sum is capped two-sidedly and with it the pencil's magnetic part (Lemma 30),*

$$\beta M \preceq W := \beta \# p d_L \quad \text{with} \quad W + \beta M \text{ positive semidefinite,}$$

the closure's dimension fold times the region's plaquette count at the magnetic weight, $d_L = 2d_f$ the A-series read at the fundamental with its dual (Lemma 17), independent of the cutoff; E being diagonal reads its off-block part at the sum's unit ($\Pi E \Pi'$ at the complement projection Π' at $\Pi + \Pi' = 1$), so the off-block coupling of H is the magnetic one alone, under the one form cap W . The cap's compression (Lemma 30) bounds the magnetic part inside the removed block, whose electric diagonal sits at or above $\alpha \Lambda_C$, so that block's floor is $f := \langle \alpha \Lambda_C : W \rangle$, at the join $W + f = \alpha \Lambda_C$; writing E_{low} for the higher of the two window levels the window cut of Definition 11 reads plus the cap W , an admissible cutoff reads $E_{\text{low}} < \alpha \Lambda_C$, and at its witness gap ς ,

$$E_{\text{low}} + \varsigma = \alpha \Lambda_C,$$

the removed block's floor sits at least ς above each level. A compression lowers each count below a line (Lemma 30), raising each low level, and the cap's two splits collect to the polarization $2\langle x, \beta M y \rangle + c_W \langle x, x \rangle + \varsigma \langle y, y \rangle$ on its upper side or of equal members at x in the head and y in the removed block, at the pair weights W , ς and the shift cap c_W , the cofactor at $\varsigma c_W = W^2$ (Definition 1, ς the admissible cutoffs' own gap), so the shifted split reads $H + c_W \oplus \varsigma \succeq H_{11} \oplus H_{22}$ with the removed block's floor above every level, counts are monotone under $\succeq$ (Lemma 30), and each level's downward shift is at most c_W . At a cutoff holding two roots, with the ground simple there and at every larger cutoff, the gap of the first two roots changes by at most c_W : each root falls by a datum between the sum's unit and c_W . Thus the finite cut and the regional cut's located edge, where named, bracket each other at the priced width,

$$E_0 \varsigma \mathcal{K}_\Lambda^{\text{reg}} \leq E_0 \varsigma \mathcal{K}_{\Lambda, \Lambda_C} + \beta^2 \# p^2 d_L^2, \quad E_0 \varsigma \mathcal{K}_{\Lambda, \Lambda_C} \leq E_0 \varsigma \mathcal{K}_\Lambda^{\text{reg}} + \beta^2 \# p^2 d_L^2,$$

degree matched on the ray as the level is, the two reads' bracket priced by one pair comparison each way at every α , the closure's dimension fold ($2d_f$ at the A-series) and the region's plaquette count its data: the truncation is certifiable at every window size with exact atoms, and the window cuts approach the regional cut with a modulus rather than along a subsequence. The gap chain reads the modulus at the $\alpha=1$ representative, and the display is homogeneous on the ray (Proposition 6), every representative's read one rescaling of it.

The cutoff-tail certificate. At $\alpha = 1$ and positive pairs h, η , let C be a stated cutoff with

$$\eta C > \eta W + \eta h + W^2, \quad \text{rev}(H_C : (e_C + h + \eta)G_C) = 1,$$

where e_C is the cutoff pencil's designated least root. The cutoff comparison includes its positive gap and the count includes its pivot split. Then at every $C' \geq C$, with $e_{C'}$ that cutoff's least root,

$$\text{rev}(H_{C'} : (e_{C'} + h)G_{C'}) = 1, \quad \mathcal{K}_{\Lambda, C'}([1 : \beta]) \geq [h : E_0].$$

The ground is simple on this cutoff tail, and $[h : E_0]$ belongs to the regional cut (Definition 11). The cutoff is finite certificate data at the two displayed reads. At $\eta = h$ these are $hC > hW + h^2 + W^2$ and the count one at $e_C + 2h$.

Proof. The cap's two splits are read at the vector $Wx + \varsigma y$ and at its partner $Wx + \varsigma \tilde{y}$, $\tilde{y}$ the memberwise swap of y : the diagonal blocks' reads withdraw between the two reads, the coupling's pairings double, and the cofactor at $\varsigma c_W = W^2$ clears the doubled weight, so the two collect to the polarization. The positive removed block at a level z has its Schur correction between the sum's unit and $c_W G_{11}$, the polarization's positive split at that block's pivot (Lemma 30). Thus the larger count at z lies between the head counts at z and $z + c_W$. The ordered-root reads of Lemma 33 give the stated fall of each root. At simple grounds their gap changes by at most c_W , the two falls' difference. The lower bracket gives a floor on the whole cutoff tail where positive, and the upper bracket caps that tail, giving the regional read of Definition 11.

For the cutoff-tail certificate put $s = [W^2 : \eta]$ at $W > 0$. Fix $C' \geq C$ and set $z = e_{C'} + h$. Compression and the unit vector's pencil read give $e_{C'} \leq e_C \leq 0$ (Definition 10; Lemma 30), hence $z \leq h$. The tail at the split between C and C' has form

$$P_z \succeq (C - W - h)G_{\text{tail}} \succ sG_{\text{tail}}.$$

Its coupling B is the actual magnetic off-block. The two-sided magnetic cap gives its squared-action cap W^2 : the magnetic split puts every root between $-W$ and W , its square at or below W^2 , and the cross image is a projection of that action's image (Lemma 33; Lemma 30). At the solved witness $P_z X = B^\top$ the polarization with $\eta s = W^2$ therefore gives

$$0 \preceq BX = X^\top P_z X \preceq \eta G_C.$$

The count at z is the deflated count and satisfies

$$\begin{aligned} \text{rev}(H_{C'} : zG_{C'}) &= \text{rev}(H_C - zG_C - BX) \\ &\leq \text{rev}(H_C : (e_{C'} + h + \eta)G_C) \\ &\leq \text{rev}(H_C : (e_C + h + \eta)G_C) = 1. \end{aligned}$$

The actual ground is strictly below $e_{C'} + h$, giving the reverse comparison. The count and ground multiplicity are one, so Lemma 20 gives the cut. A vacant removed list reads the head count directly. At $W = 0$ the magnetic part reads the sum's unit, $B = 0$, and the removed electric contents exceed $C > h$; the count comparison holds at the diagonal tail outright. $\square$

Lemma 44 (The dual truncation). *At the coupling $[\sigma : 1]$ at a pair σ , the cap, the floor and the compression of Theorem 30 read with one substitution, the two terms' roles exchanged in the bound's inputs: on a fixed region, the magnetic cap at weight one is $W := \#p d_L$, the plaquette caps' fold (Lemma 18; Theorem 30). The off-block coupling and the removed block's magnetic part read that cap by compression and polarization. At an index compression with a verified two-sided cap, the argument reads that cap at the block's Gram (Lemma 30). The removed block's electric diagonal sits at or above $\sigma \Lambda_C$, and each level shifts downward under the compression by at most the shift cap c_W , the cofactor at $\varsigma c_W = W^2$ (Theorem 30), at the order $\lambda + W < \sigma \Lambda_C$'s witness*

gap ς , $\lambda + W + \varsigma = \sigma \Lambda_C$. A count at level λ is therefore pinned on the finite head whenever the bracket $[\lambda, \lambda + c_W]$ is clear of the head's divisor (Lemma 31, the level as the parameter): every count of the dual pencil, the pair $(\sigma E : M)$'s site datum $H([\sigma : 1])$, lives on a finite head pencil, at every cutoff with

$$\sigma \Lambda_C > \lambda + W + c_\vartheta \quad \text{at} \quad \vartheta c_\vartheta = W^2,$$

one pair comparison at a clearance pair ϑ , the admissible set an output whose floor rises toward the contact end. The transfer is one-sided: the compression lowers counts and raises levels, each level's downward shift capped at c_W (Theorem 30; Lemma 30), so between two head jumps the lower comparison reads at its equality, the head ground at or above the full one, and the clearance rides the upper jump alone: the admissible levels are those beyond the ground bracket's top whose sum with c_W sits below the second jump's bottom, the measured gap spent on one side.

The count beyond every cutoff. The display defines the dual pencil's count at λ over the unbounded content range: at a level whose bracket $[\lambda, \lambda + c_W]$ is clear of the head's divisor, it is the head count at any cutoff beyond the display's bound, and two admissible cutoffs read one integer.

The seam identity. The ray at end weights $[q^2 : p^2]$ names one integer pencil pair, $(q^2 E : p^2 M)$, with its site datum S at the join $q^2 E = p^2 M + S$, and the two coordinates' representatives are the datum's positive rescalings,

$$q^2 H([1 : \tau^2]) = S = p^2 H([\sigma : 1]) \quad \text{at} \quad q\tau = p, \quad p^2\sigma = q^2,$$

so every subspace read agrees at rescaled levels: a rescaling-fixed direct summand reads at rescaled levels, the fibers among them (Lemma 40), and on each,

$$\text{count}_{[1:\tau^2]}(\lambda) = \text{count}_S(q^2\lambda) = \text{count}_{[\sigma:1]}(\sigma\lambda)$$

at every level, the reversal count blind to the positive factors (Lemma 30), brackets, jumps and extents rescaled by the one factor with their levels. A stored level therefore names its representative, and a seam comparison is two reads of the one integer pencil at matched levels, the factors the display's.

Proof. The cap at the dual representative is the fold of the region's plaquette caps at magnetic weight one. Compression keeps each two-sided comparison, and Theorem 30's polarization gives the off-block read at that cap. Substituting $\alpha = \sigma$ and $\beta = 1$ gives the displayed cutoff and shift comparisons at the electric floor $\sigma \Lambda_C$. The count beyond every cutoff. The smaller head is the larger head's own truncation (the projections nest), the display runs on the larger head at the one cap W (a compression of a cap is a cap, Lemma 30) with the removed block's floor at least ς , and the clear bracket pins the count both times, so two admissible cutoffs read one integer. $\square$

Lemma 45 (The near-contact cell). *The dual pencil, the pair $(\sigma E : M)$'s site datum $P(\sigma)$, is affine in σ , and on a fiber head (Lemma 44) its entries are one polynomial-pair list in (t, σ) over the residue beyond the locality band (Lemma 40; Lemma 14), the window entering through its sampled momenta and its separation range alone, the residue a further coordinate beyond the reach bound (Lemma 25(iii); Lemma 26(iii) at the further series). Every count of the dual chain is therefore Lemma 31's cell function over (t, σ, r) with located divisors: the contact cell is the contact end's, its σ -extent σ_1 the divisor's first positive root, an output of Lemma 25(iii)'s reads, the residue direction the classifications' (Lemma 28; Lemma 29 at the further series), the divisors' roots against the families' clearances; anchors at pair σ are window reads; and $r = 1$ enters as a residue read like any other, $c_1 = 1$ flowing through the channel data (Lemma 24).*

The truncation pair. On the vacuum sector's one-plaquette chain (the unit label's fusion closure under the magnetic list, Construction 21, one label per index entry at the electric read $4\hat{c}_2$) the chain's count at a level is bracketed by two finite divisors. The first is the plain truncation, a compression, its counts at or below the chain's (Lemma 30). The second is the cutoff-bordered pencil: the head with one auxiliary row per cutoff label (a head label whose fusion by a closure member leaves the cutoff, Proposition 3), coupling one and auxiliary diagonal the pair $[\langle 4\sigma\Lambda:d_L + \lambda \rangle : \varkappa_B]$ at the cutoff Λ on the labels' Casimir, its first member the seed floor $F := \langle 4\sigma\Lambda:d_L \rangle$ at the level on the second member, the floor of the whole excluded block, every excluded label at Casimir at or beyond Λ by the cutoff's own read and its loop's content at four times it (Proposition 7); d_L is the magnetic form cap, kept by every principal block (Lemma 17); and the border cap $\varkappa_B$ is the product of the coupling's largest row and column sums, $BB^\top \preceq \varkappa_B$ on the cutoff labels' support, the two-sided Schur read. At every level below F the chain's count sits between the pair's two counts at every cutoff, and a level whose two counts agree is a located count of the chain itself, the count beyond every cutoff (Lemma 44). At $r = 2$ the label lattice is two-row, a reduced shape at most r rows deep (Construction 5). At every member the depth chain's head shells (Construction 21) are the chain's leading block, the chain's counts at or beyond the head's at every level (Lemma 30's compression read). A shell ordering of the elimination is certificate data (Theorem 19(i)), its validity divisors (the seed's dominance root and the envelope's own positivity ceiling) entering the emitted list as any boundary's do; the deep tower rows repeat off the collar (Lemma 41), the decimation's constant-stencil direction.

The contact end is interior to its cell. At the contact end the head pencil is the magnetic member's balance partner (Proposition 6), its level pair's datum the balance partner of $M + \lambda G$, and the counts are the closure fold's reads, N_L against the level at the orthonormal characters (Lemma 17; Theorem 17): the contact end sits inside the contact cell and its counts are the cell's, direct reads; at a level on the closure fold's divisor, Lemma 31's boundary clause extends the positive-semidefinite cells into the closed endpoint.

Proof. The truncation pair. Between any two cutoffs the excluded block sits at or above F (σ times its Casimir floor against the magnetic cap), so at every level below F the chain dominates the shifted split, the head less the border cap's multiple of the cutoff labels' support projector at the gap's cofactor beside the excluded block less the gap (Theorem 30's shifted split at the border cap's Schur read, $BB^\top \preceq \varkappa_B$ on the support), the excluded member positive semidefinite there, and the chain's count sits at or below the shifted head's (Lemma 30, counts monotone); the bordered pencil's auxiliary pivots are positive below F and inertia adds at them, its deflation the shifted head (Lemma 30): the chain's count sits between the pair's two counts. *The contact end.* At a level off the closure fold's divisor the cell's determinant reads unequal members at the contact end, so the contact end sits inside the contact cell. $\square$

Lemma 46 (The contact drift). *At the dual representative, the pair $(\sigma E:M)$'s site datum on a fiber head (Lemma 44; Lemma 45), the ground's electric weight rides the contact drift: the ground identity reads*

$$\begin{aligned} \sigma \langle \psi, E \psi \rangle &= \varepsilon_0(\sigma) \langle \psi, G \psi \rangle + \langle \psi, M \psi \rangle, \\ \sigma \langle \psi, E \psi \rangle &\leq g \langle \psi, G \psi \rangle \quad \text{at the drift } g \text{ at } \varepsilon_0(\sigma) + m_+ = g, \end{aligned}$$

the magnetic read at or below the top root m_+ of $(M:G)$ on the head, the contact end's ground level the top root's balance partner (Lemma 33; the contact end's reads Lemma 45's), the order witnessed by the counts monotone in the scale (Lemma 30): the electric weight is priced by two located roots, the drift its own cap, and the scale sandwich's electric weights read it at the grounds' own data (Theorem 21(vii)). The drift falls at a certified positive rate, the upper family at the contact end.

The grade key is the highest root's coroot pair, $\mathcal{G}(\lambda) := \lambda(\theta^\vee)$, the interface's root coroot pair at θ (Construction 3), one natural per label, and three member reads price the chain's labels at the key: the key moves by at most two under fusion by a closure member; $\hat{c}_2(\lambda)$ sits at or below $c_r(\mathcal{G}^2 + 1)$ at the member pair $c_r := [F_2 + F_1 : C_2(\theta)]$, the fundamental form's total $F_2 := \sum_{i,j} \langle \omega_i, \omega_j \rangle$ joined to the fundamentals' weight total $F_1 := \sum_i \langle \omega_i, 2\rho \rangle$; and the labels at one key value with their dimensions count at or below $A(\mathcal{G} + 1)^\wp$ at $A := \ell_s r^{N_+}$ and $\wp := N_+ + \ell$, N_+ the positive list's count, ℓ the coordinate count and $\ell_s := \prod_\alpha [2 : \langle \alpha, \alpha \rangle]$ the positive list's length fold (Construction 4). At the rate's data $q := 2\wp + 3$, $C_E := 4c_r A^2 (2\wp + 2)!$ and $C_g := 2^{q+1} C_E + 2d_L + 1$, at a scale σ and a pair ξ at the gap w at $\xi + w = 1$ inside the bracket $w^{q+1} \leq \sigma \leq (2w)^{q+1}$,

$$g(\sigma)^{q+1} \leq C_g^{q+1} \sigma,$$

one cross-multiplied comparison per scale, a located C_g per member and direction count, at the certificate's data: a natural N and the vacuum sector's head holding every label at key at or below $2N$ with its fusion targets at the closure, the head's cutoff clearing the grade reach $c_r(4N^2 + 1)$, and the weighted power $2d_L A^2 (2N+1)^{2\wp} \xi^{4N}$ at or below w at the count reads $A, \wp$. The halving scales' ground-level widths of the scale sandwich are then geometric over the tail at the contact-drift display's weights, their sum one cofactor read (Theorem 21(vii)).

Proof. At the fundamental fold $\lambda = \sum_i m_i \omega_i$ the key is the dual fold's read, $\mathcal{G} = \sum_i m_i c_i^\vee$ at the halved summands $2c_i^\vee = c_i \langle \alpha_i, \alpha_i \rangle$ of the residue fold, naturals at one or beyond joining to $\sum_i c_i^\vee = r$ (Construction 4), so the coefficient sum sits at or below $\mathcal{G}$, the vacuum reads the key at the sum's unit, and ρ reads it at r , the simple coroot reads one each. Three reads price the family at the key. *The move cap:* a closure member's content ν sits at or below the member's top λ_0 (Lemma 3), the join $\nu + \sum_i d_i \alpha_i = \lambda_0$ at natural d_i reading $\nu(\theta^\vee) \leq \lambda_0(\theta^\vee)$ at the highest root's coroot dominant, every simple root's pair there at or beyond the sum's unit, and every content's dominant image so as well (the content symmetry, Lemma 6), so $\nu(\theta^\vee)$ sits at magnitude at most the member's grade key $\mathcal{G}(\lambda_0)$, the closure's largest at most two: one at a minuscule member, one at the floor labels of G_2 and F_4 (the short dominant roots' coroot pairs at θ), and two at E_8 's highest root; the key moves by at most two under fusion by a closure member. *The Casimir cap:* the fundamental display (Lemma 9) reads

$$C_2(\lambda) = \sum_{i,j} m_i m_j \langle \omega_i, \omega_j \rangle + \sum_i m_i \langle \omega_i, 2\rho \rangle,$$

every summand a natural product at the fundamental form reads (the adjugate rows' displayed folds at the fixed members, the coordinate reads at the series, Construction 4), every coefficient at or below the key at the dual fold's read $\sum_i m_i c_i^\vee = \mathcal{G}$ with $c_i^\vee$ at or beyond one, so $C_2(\lambda) \leq \mathcal{G}^2 F_2 + \mathcal{G} F_1 \leq (\mathcal{G}^2 + 1)(F_2 + F_1)$ at $\mathcal{G} \leq \mathcal{G}^2$ and the totals $F_2 := \sum_{i,j} \langle \omega_i, \omega_j \rangle$ and $F_1 := \sum_i \langle \omega_i, 2\rho \rangle$: $\hat{c}_2(\lambda)$ sits at or below $c_r(\mathcal{G}^2 + 1)$ at $c_r := [F_2 + F_1 : C_2(\theta)]$. *The counts:* the labels at one key value are the dual fold's solutions, each coefficient at or below $\mathcal{G}$, at most $(\mathcal{G} + 1)^\ell$ at ℓ the coordinate count; and the member gap product (Corollary 1) reads a label's dimension as the fold of $\text{dot}(\lambda + \rho, \alpha)$ against $\text{dot}(\rho, \alpha)$ over the positive list, each divisor $\rho(\alpha^\vee) [\langle \alpha, \alpha \rangle : 2]$ at or beyond $[\langle \alpha, \alpha \rangle : 2]$ (ρ 's coroot pairs at or beyond one, Construction 4) and each $\text{dot}(\lambda + \rho, \alpha)$ at or below the θ factor, $\text{dot}(\lambda + \rho, \alpha) \leq \text{dot}(\lambda + \rho, \theta) = \mathcal{G} + r$ at the join's read against $\lambda + \rho$'s coroot pairs, so $d_\lambda \leq \ell_s (\mathcal{G} + r)^{N_+} \leq \ell_s r^{N_+} (\mathcal{G} + 1)^{N_+}$ at $\mathcal{G} + r \leq r(\mathcal{G} + 1)$: the per-key label and dimension counts sit at or below $A(\mathcal{G} + 1)^\wp$ at $A := \ell_s r^{N_+}$ and $\wp := N_+ + \ell$, the positive list's count and the coordinate count joined. The *dimension-weighted family* is

$$\phi_\xi := \sum_{\mathcal{G}(\lambda) \leq 2N} d_\lambda \xi^{\mathcal{G}(\lambda)} \chi_\lambda,$$

the sum over the chain's labels at the grade key, the characters orthonormal (Proposition 8) and the vacuum's term reading one; at the A -series the key reads the reduced shape's top row, the pair $\langle \lambda_1 : \lambda_{d_f} \rangle$ at the reduced shape's one-member site with the full column's read of equal members (Construction 5), one value per label. The move cap prices the magnetic read termwise at the dimension eigen-identity $\sum_{\lambda_0 \in \bar{L}} \sum_{\mu} N_{\lambda_0 \nu}^{\mu} d_{\mu} = d_L d_{\nu}$ (Lemma 17), every summand a natural at pair weights with the key's weight falling as the key rises:

$$\langle \phi_{\xi}, M \phi_{\xi} \rangle + B \geq \xi^2 d_L \langle \phi_{\xi}, \phi_{\xi} \rangle,$$

B the top two key values' boundary term at the cap d_L , the two keys' part of the pairing priced at the counts' squares, below w times the vacuum's term at the weighted power comparison. The electric read prices at the key's caps, $\hat{c}_2(\lambda)$ at or below $c_r (\mathcal{G}^2 + 1)$ with the per-key label and dimension counts at or below $A (\mathcal{G} + 1)^{\wp}$, the pairing's per-key weight the fold of the dimensions' squares at or below the count's own square $A^2 (\mathcal{G} + 1)^{2\wp}$, and the weighted geometric sums close on two displayed reads. The power cap is one rising-product read,

$$(L + 1)^j \leq (L + 1)(L + 2) \cdots (L + j),$$

every factor $L + i$ at or beyond $L + 1$; and the graded sums at the rising-product weights, $S_j(M) := \sum_{L \leq M} (L + 1) \cdots (L + j) x^L$ at $x + y = 1$, descend by Pascal's identity at the rising products, $(L + 1) \cdots (L + j + 1) = L \cdots (L + j) + (j + 1)(L + 1) \cdots (L + j)$, the exact displays

$$y S_{j+1}(M) + (M + 1) \cdots (M + j + 1) x^{M+1} = (j + 1) S_j(M), \quad y S_0(M) + x^{M+1} = 1,$$

so $y^{j+1} S_j(M) \leq j!$ at every M . At $x = \xi^2$ the gap reads $y = w(1 + \xi)$ at or beyond w , the vacuum's unit term floors the pairing, and

$$w^{2\wp+3} \langle \phi_{\xi}, E \phi_{\xi} \rangle \leq 4 c_r A^2 (2\wp + 2)! \langle \phi_{\xi}, \phi_{\xi} \rangle :$$

the rate's data are $q = 2\wp + 3$ and $C_E = 4 c_r A^2 (2\wp + 2)!$, member reads. The Rayleigh read at ϕ_{ξ} then closes the drift's rate against the head's top root m_+ at its cap d_L (Lemma 17): at the scale's w , the bracket $w^{q+1} \leq \sigma \leq (2w)^{q+1}$ the certificate's own read, one power comparison,

$$g(\sigma)^{q+1} \leq C_g^{q+1} \sigma,$$

one cross-multiplied comparison per scale at the collected read $C_g = 2^{q+1} C_E + 2 d_L + 1$, the join read at the count carrier: the Rayleigh read puts the count occupied at the assembled margin's unit shift, the scale's electric power at the bracket's upper side with the boundary read over the cap's weight; the head's first occupied level $\varepsilon_0(\sigma)$ reads the count vacant at its own level and at every level below; the monotone count read (Lemma 30) holds the vacant level strictly below the occupied one, the two levels' difference site certified positive semidefinite at the unit gram's scalar copy with the equal-level reading refused at the vacant certificate, so $\varepsilon_0(\sigma)$ sits at or below the assembled margin; the top root's cap prices m_+ at d_L (Lemma 17); and the drift $\varepsilon_0(\sigma) + m_+ = g$ obeys the margin comparison, the power comparison's own drift datum, a located C_g per member and direction count: the contact drift falls at a certified positive rate, and the halving scales' ground-level widths of the scale sandwich are geometric over the tail at the contact-drift display's weights, their sum one cofactor read (Theorem 21(vii)). $\square$

Lemma 47 (The corner pencil). *Toward the contact end the chain reads in the corner coordinate: the ray $[\eta^2 : 1]$ at the scale pair η , its levels at the corner scale, a level the product $\eta \eta'$ at a height*

η' , itself a balance pair (Definition 1). The corner data are the dual pencil's at the substitution $\sigma = \eta^2$, $\lambda = \eta\eta'$ (Lemma 44), and the corner count is the reversal count of the form pair

$$\text{rev}(\eta^2 E + \eta\eta'_- G : M + \eta\eta'_+ G) \quad \text{at } \eta' = \langle \eta'_+ : \eta'_- \rangle,$$

the height's members cross-added onto the pair's two sides (Lemma 30; Theorem 15), every entry's weight a natural product of the scale's members and the height's. The corner's shifted scaling: at the magnetic form cap d_L (Lemma 17) the shifted pencil $Q(\sigma) := \sigma E + S_\theta$, at S_θ the pair $(d_L G : M)$'s site datum, is a sum of two positive-semidefinite forms (the electric diagonal's; the cap's, Lemma 17), and at scales $\sigma' < \sigma$ (two equal scales reading one pencil outright) with the cofactor ρ at $\rho\sigma = \sigma'$,

$$\rho Q(\sigma) \preceq Q(\sigma') \preceq Q(\sigma), \quad \rho \mu_j(\sigma) \leq \mu_j(\sigma') \leq \mu_j(\sigma),$$

μ_j the shifted roots; the comparison is cutoff-free and window-free, a form comparison compressing. The ground's electric weight rides the contact drift at a certified positive rate (Lemma 46), and the tail's count read is the sector's tail certificate's (Construction 19), the block chain's at bounded shells (Lemma 57) and the band-one walk's at the single-plaquette window (Lemma 56). On a fiber head the entries are Lemma 45's list at the substitution, one polynomial-pair list in (t, η, η') over the residue beyond the locality band, so every corner count is Lemma 31's cell function over the (t, η, η') base with located boundaries, and the corner divisor is Theorem 26's crossing read at that base. The corner cell is the contact end's, and its data are one located output per member and direction count (Construction 7's direction reads), the extent η_1 , at a chain of bounded shells the corner divisor's first positive root in the scale over the cell, at its leading-term radius in the residue (Theorem 16; Lemma 25(iii)), and at a chain whose shells grow with the depth the tail certificate's ceiling (Construction 19(iii)), and one located datum per ray of the cell, the floor κ at the ray, the least of the certificate's three positive lower margins for the regional records' located gaps above their grounds at the ray: the unit fiber's second root's, the further fibers' least first root's over the band's chords, t^2 below 4 (Lemma 40), and the charged sectors' first roots' (Lemma 31 at the chord; Lemma 43(iii),(iv)), one trichotomy fold read ray by ray, its residue and rank direction the cell coordinate's read (Lemma 25(iii); Lemma 26(iii)). The ray coordinate reads the outputs back at squares: the ray $[a : b]$ sits in the extent exactly at $a d_1^2 \leq n_1^2 b$ at $\eta_1 = [n_1 : d_1]$, and a level reads through its height, the cofactor at $\eta\eta' = \lambda$ (Definition 1), the height against the floor one pair comparison: the level's square against the weight product, the two end weights' own scale.

The boundary families. The corner cell's boundaries are the located roots of four families over the (t, η, η') base, each family one of the chain's comparisons read at the substitution and entering the list as any boundary does (Theorem 19, the emitted record), the corner divisor's own root list in the scale beside them, the extent its first positive member: the jump family, the pencil determinant's roots at the bracket's heights, the decimation's own jump divisors at the substitution (Theorem 19); the cutoff family, the tail's dominance comparison at the line (Theorem 19(ii)): at a line the labels whose diagonal clears their off-row magnitude fold are dominant at every cutoff, the chain's count at the line sits at or below the floored head's, the head less the tail coupling's solve at the gaps (Construction 19(i)), one finite pencil per scale and line, and the dominance root in the scaled square keys the head, the plateau read $[2 : 3](s - 2\eta)$ against the line's rate at the residue two (Construction 18), so the cutoffs enter through the floored head alone, cutoff-free at every scale of the tail at once, d_L the member's dimension fold with the A-series' $2(r + 1)$ its read at the fundamental pair (Lemma 17); the deck family, the bulk blocks' crossing loci with the graph read's side and the spectator caps' three counts, each a cell datum of the corner base (Lemma 35(iii); Lemma 39), the $r = 1$ reads among them, and the contraction cap read on the cell at the boundary root's isolation

datum: the cell is a component of the divisor list's complement, the located root's bracket its margin, and the cap's margin squares to the crossing read, $\langle 1:z^2 \rangle^2 = z^2 \langle w^2:4 \rangle$ (Lemma 39); and the tension family, the charged sectors' tension, the winding shift's dressed floor against the unit class's dressing folds, and the separated sectors' clearance at the height floor (Lemma 43(iii),(iv); Lemma 42(iii)), one root list per charge vector. Every window read sits in the list's cells: the momenta are sampled in t (Lemma 40), the slabs read within the spectator brackets (Theorem 19(iii)), the cutoffs within the floored head, and the sectors at their families; so the extent and the floor are outputs of the one list, and the cell's t -section at the outputs is the whole range exactly where the list's t -roots sit off it, itself a located read. The rank direction inherits: the substitution fixes every entry's residue dependence, so the families' divisors are the chain's own read at corner base points, the deck base the member-key count, one value at every cleared residue (Proposition 13; Lemma 25(ii)).

The chain's height at a scale is its cut's edge at the corner scale (Definition 11; Lemma 22(i)), the ray's floor a member pair of its cut, the cut's meet at or below each regional cut (Definition 11), and each cutoff-tail probe cap a cap on it (Lemma 22(iv), the first moment at the gradient and boundary reads of Theorem 8). Each cell reads its edges at word images: the flat window's upper end is the cell's second root, a point of its own divisor (Lemma 33; Theorem 26(i)), and the window cut's edge there, where one is named, is the compressed pair's least root, read at a word image with the two moments at equality and the first positive, the state the word sector's own (Lemma 22(i),(ii)), one word image per window read on the cell at the corner scale. The extent's positivity is the contact end's interiority read, the σ_1 brackets Lemma 45's own outputs.

Proof. The corner count is the seam pencil's own: the display is cleared by the scale's square at the seam identity's integer pencil (Lemma 44), the reversal count blind to the positive factor (Lemma 30). At the shifted scaling $Q(\sigma')$ is the sum $\rho Q(\sigma) + w S_\theta$ at the gap w , the scales' order's witness at $\rho + w = 1$, read against its second summand positive semidefinite, and the root reads the comparison's counts (Lemma 30). The extent's bracket at bounded shells is the corner divisor's first positive root's, the divisor the crossing read over the cell's base (Theorem 26), and the contact end sits inside its cell (Lemma 45), so the extent is positive; at growing shells the extent is the certificate's ceiling, a halving of one, positive by its shape. $\square$

Lemma 48 (The cell floor). *At a corner ray (Lemma 47) the corner floor is positive at every tail scale, the certificate's common lower margin at the ray. The depth chain's decimated head at the dual representative at the ray's scale (Construction 21; Theorem 19), finite at the ray at Lemma 44's cutoff and Theorem 19(ii)'s head bound, reads its ground ε_0 and its three located gaps above it (Lemma 33), each at or beyond the corresponding certificate margin: the unit fiber's second root's, the least first root's over the further fibers' band, the chords at t^2 below 4 (Lemma 40; Lemma 31 at the chord), and the charged sectors' first roots' (Lemma 43(iii),(iv)). Per fiber the vacuum sector's depth chain reads the count at the certificate's line at most one at every tail scale, the sector's tail certificate at the fiber's shells (Construction 19; at a chain of bounded shells the block chain's certificate, Lemma 57, the band-one walk at the single-plaquette window its scalar instance, Lemma 56(v),(vi); the count a cell function of the chord, Lemma 31), the line above the unit fiber's lower line by the certificate's lines' gap at the scale: at the unit fiber the crossing is the ground's, the lifted witness reading the ground at or below the lower line (Lemma 56(v)), so the ground is one level at its multiplicity one (Theorem 10) and the unit fiber's second root sits at or beyond the line; at a further fiber the crossing is vacant, the count at the line the sum's unit, so the fiber's first root sits at or beyond the line; and the charged sectors' first roots clear the ground at the winding floor and the tension's side (Lemma 43(ii),(iii)): the floor, the least of the three positive margins, is positive, and every pair κ at $E_0\kappa$ below the floor is a member of*

the ray's cut (Lemma 20). The certificate's lines' gap is the common lower read at the ray, the individual roots read at their windows. The total counts are the decimation's (Theorem 19(i)), the charged lower read the tension's (Lemma 43(iii)). In each region the finite lower witness's content and the stated head's contents give a cutoff key from which the upper count is one and the ground simple, with every positive gap at or beyond the certificate's lines' gap. The deeper sectors' couplings enter their actual deflations (Theorem 19; Lemma 42(iv)). The heights read two-sided at the ray: the floor's clearance below, and above the window's cap line at the tail read (Lemma 55) and the cutoff-tail probe caps (Lemma 22(iv)), the chain's height at the ray between them, one bracket per ray. At the adjoint list's one-plaquette sector at the residue one the head is the band-one walk (Proposition 13), the single-plaquette window at the unit fiber alone, and the gap is the band-one certificate's (Lemma 56(v),(vi)), the lines' gap on η at every tail scale; at every further residue the band-one walk is that sector's compression to the tower labels, its lower line and its cap line lifting to the sector (Lemma 30), and the sector's count at the line is the tail certificate's at the sector's shells, growing with the depth (Construction 19(iii)), the residue two at Construction 18's profile at the certificate's data and their reads (Construction 20; Lemma 61) and every further member and residue at its own (Pending 1).

Proof. The head at the ray is a finite symmetric pencil over the located stage (Lemma 44; Theorem 19(ii); Lemma 32), so its roots are located with the split's congruence (Lemma 33) and each gap is one located comparison; the further fibers' least first root over the chord range is the divisor's read along the chord, the count constant off its located roots (Lemma 31), and the charged sectors' first roots are the dressed strings' head reads at the ray (Lemma 43(iv)). The certificate's two lines are corner levels at the vacuum sector's depth chain, the head's shells its blocks (Construction 19; Theorem 19(i)), and the count splits over the fibers, the chain block diagonal at the chord splitting with the counts adding (Lemma 40; Lemma 30), each fiber's count at the line the certificate's own read there, a cell function of the chord (Lemma 31): at the unit fiber the ground sits at or below the lower line at the lifted witness and the count reads at most one at the line, so the counting function is flat between them, the ground the one root below the line at its multiplicity one (Theorem 10; Theorem 26(iii)) and the unit fiber's second root at or beyond the line; at a further fiber the certificate's crossing is vacant, the count at the line the sum's unit, so the fiber's first root sits at or beyond the line; the lines' gap is cleared by the first two gaps at every tail scale. Fix one region and scale. Write l, u for the certificate's lower and upper lines, with $l + h = u$ at its positive gap. Its verified total count at u is at most one on its stated cutoff tail, and its lower witness x has finite content. Let C_s be the largest of one, the stated head's contents and the witness's occupied contents, a finite trichotomy fold (Definition 1; Proposition 4). At every $C \geq C_s$ the witness's form is its original read, giving

$$e_C \leq l < u, \quad 1 \leq \text{rev}(H_C : uG_C) \leq 1.$$

Thus the ground is simple on this cutoff tail. Since $e_C + h \leq u$ and $e_C < e_C + h$, count monotonicity gives

$$\text{rev}(H_C : (e_C + h)G_C) = 1.$$

This is the regional floor $[h : E_0]$ (Lemma 20; Definition 11). The two-component and deeper sectors' admitted product forms read the positive fold of Lemma 42(iii),(iv) at the stated floors. Their couplings enter the certificate's deflations, whose pivot counts join to its total count (Theorem 19(i)); every comparison of regional records reads Construction 19(ii)'s cap and margin counts at those actual data. The caps are Lemma 55's cutoff-tail lines and Lemma 22(iv)'s cutoff-tail probe caps, each at or beyond the chain's height (Definition 11, the cut's meet at or below each regional

cut). Its simple-ground key is C_s above; joining that key to the two trial members' contents gives the key of Lemma 55's cap. At the adjoint list's one-plaquette sector at the residue one the head is the band-one walk's, its count at the lines the certificate's ((v) the ground below the lower line, (vi) the count at most one at the line), the gap the lines' own; at a further residue the compression's lower line and cap line lift (Lemma 30), the count at the line the tail certificate's at the growing shells (Construction 19(iii)). $\square$

Construction 16 (The corner window's data). At two directions or beyond the *square window* of side L has the vertices at the coordinates from one to $L + 1$ in every direction with the links and plaquettes among them, one window of the directed family (Definition 8), $\#p$ its plaquettes' count, $\#\ell$ its links' count, and g the direction gap at $g + 1 = d$; every plaquette's boundary word runs from its corner along the lower direction's link forward, its *base link*, then along the higher direction's link at the corner moved one step in the lower direction forward, and returns along the lower direction's link at the corner moved one step in the higher direction and along the higher direction's link at the corner, both backward (Construction 7). The window's data are the reads below, each a read of the window's finite lists.

The counts. A plaquette's boundary holds four links; a link sits on at most $2g$ plaquettes, two in each plane through its direction, the plane's plaquette at the link's tail, holding the link forward as its base link or as its corner's higher link, and the plaquette at the tail moved back one step in the plane's other direction, holding it backward, the second occupied exactly where the tail's coordinate in that direction sits beyond one; a *cube* is three directions in order with a corner whose three coordinates sit below the side, its six *faces* the plaquettes at the corner in each two of its directions and at the corner stepped along the third, and a plaquette is a face of at most $2(g - 1)$ cubes, two in each direction off its plane; and the lists' counts are box counts, the sites whose coordinates at k stated directions sit below the side and are free at the further directions numbering $L^k(L + 1)^{g+1-k}$, one digit per direction: the links' count $\#\ell = (g + 1)L(L + 1)^g$, the plaquettes' count $\#p = [g(g + 1) : 2]L^2(L + 1)^{g-1}$, and, at three directions or beyond at the gap h at $h + 2 = g$, the cubes' count $[g(g + 1)(g - 1) : 6]L^3(L + 1)^h$. The window's *tree* is the comb rooted at the *root*, the vertex at coordinate one in every direction (Construction 7's boundary read): a vertex's *root path* ascends the directions in their order, at most L links per direction and at most $(g + 1)L$ links in all; the *path word* $R_{v \leftarrow w}$ between two vertices is the tree path's adjoint coefficient, the ordered product of the links' adjoint coefficients at the traversals' dagger reads (Lemma 18's move identity), orthogonal at the evaluation identity, two paths composing at their join, $R_{u \leftarrow v} R_{v \leftarrow w} = R_{u \leftarrow w}$, a link traversed both ways withdrawing at the evaluation identity (Construction 8), and the transposed word the reversed path's, $R_{v \leftarrow w}^\top = R_{w \leftarrow v}$; a plaquette's *root path* is its base link's tail's, and a link sits on the root paths of at most $\#p$ plaquettes, the count's own read. At a plaquette p and a boundary link ℓ the *loop region* $R_{\ell p}$ is the region between the boundary arc off ℓ and the tree path from the base link's tail to ℓ 's tail, the plaquettes of the columns descended in the later directions at the plaquette's corners, one column of width one per descent at a height at most the side, so at most gL plaquettes, the *region's count* $n_{\ell p}$; the region is occupied at the two links off the corner alone, the tree path from the corner to the corner moved by one step in a direction a of the plane descending the directions beyond a to the first positions and ascending; and a plaquette sits in at most $[2 : (g + 1)L]\#p$ regions: a plaquette of the plane (a, k) at $a < k$, at the corner y , sits in the region of (p, ℓ) exactly where p 's plane holds the direction a , ℓ 's tail is p 's corner moved by one step in that direction, y sits at the first position in every direction beyond k (a corner off the first position there sits off every region), and p 's corner agrees with y at every direction below k , sits beyond y_k at k and is free beyond k , the g planes through a , at most L positions at k and $L + 1$ at each later direction, at the plaquette count

$\#p = [g(g+1) : 2] L^2(L+1)^{g-1}$; so the regions' counts fold to at most $[2g : g+1] \#p$ over the pairs whose region holds one plaquette, at or below $2 \#p$.

The incidence. The *incidence* K is the plaquette-link list at the traversals' signs, one at a boundary link traversed forward, its balance partner backward, and the sum's unit off the boundary (Construction 7); the plaquettes' *Gram* $\Delta := KK^T$ reads 4 on the diagonal, the boundary's link count, at two plaquettes sharing a link the two traversals' signs' product, the balance partner of one at two plaquettes of one plane (the shared link traversed at exchanged directions), and the sum's unit at two plaquettes off a shared link, symmetric at the product's exchange and positive semidefinite as a Gram (Lemma 30), its form the links' folds' squares' fold,

$$W^T \Delta W = \sum_{\ell} \left(\sum_{p \ni \ell} K_{p\ell} W_p \right)^2.$$

The kernel and the range. At two directions the kernel is the unit family: a plaquette combination pairing every link at the sum's unit reads that unit at every coefficient, a plaquette at the window's boundary holding a link of one plaquette, its coefficient that link's read, the withdrawn window a window and the descent closing at the count. At three directions or beyond the *cubes' boundaries* N are the plaquette-cube list at the faces' signs, per cube one sign at each of its six faces, the signs those at which the six boundary words fold to the sum's unit at every link, each link of a cube on two of its faces and the two faces' signs at the link's two traversals joining to that unit, one sign family per cube at the sign one at the face at the cube's corner in the cube's two lower directions, so $K^T N$ reads the sum's unit entrywise; and a plaquette combination pairing every link at the sum's unit is a combination of the cubes' boundaries by the descent in the last direction: its read at a top face, a plaquette at the last position whose cube is one, withdraws at that cube's boundary's multiple, and a top link, a link at the last position in an earlier direction, then sits on one occupied plaquette, the face below it in the link's plane with the last direction, whose coefficient the link's pairing reads at the sum's unit, so the combination sits on the window one position shorter, the descent closing at the plane window at the two directions' read. The *kernel list* N_0 is the cubes' boundaries' collection (Lemma 3, one member joined per refusal along the cubes in the corners' order), vacant at two directions, spanning the kernel with its Gram $N_0^T N_0$ positive definite at the independence, its count the kernel dimension, the descent's kernel list and the collection two independent lists inside one span (Definition 3; Lemma 3), and the *range projector* is

$$\Pi := 1 - N_0 (N_0^T N_0)^{-1} N_0^T,$$

the solve the adjugate against the determinant (Definition 3), one value at every spanning list of the kernel, the projector onto the range at the kernel's own span, the identity at two directions, symmetric at the solve's symmetry and idempotent at the solve's identity, with $\Delta \Pi = \Pi \Delta = \Delta$ at ΔN_0 the sum's unit, Π 's diagonal a self-pairing at or below one, and a vector's *range part* Πu against its *kernel part*; the *shifted Gram* $\Delta + N_0 N_0^T$ is positive definite, its form the Gram's joined to the kernel list's pairings' squares, the sum's unit at a vector in the kernel perpendicular to the kernel list, the unit vector alone, and the *range solve* at a range vector u is $\Delta^{-1} u := (\Delta + N_0 N_0^T)^{-1} u$, the adjugate against the determinant: the solve's kernel part reads the sum's unit (the kernel list's Gram against it at that unit, the Gram definite), so $\Delta \Delta^{-1} u = u$ and $\Pi \Delta^{-1} u = \Delta^{-1} u$.

The roots' bracket. Every root of the Gram sits at or below $8g$, $\Delta \preceq 8g$: each link's fold's square sits at or below its plaquette count $2g$ against the plaquettes' squares (Construction 8's Cauchy–Schwarz), and the fold collects each plaquette's square at its four links. And every root off the sum's unit sits at or beyond $[2 : L^2]$, the *Poincaré read* $[2 : L^2] \Pi \preceq \Delta$, at the *Hodge fold*: the Gram joined to the cubes' boundaries' Gram, $\Delta + N N^T$, reads plane type by plane type, on the

plaquettes of one plane at their corners' coordinates, the two in-plane *Dirichlet folds* joined to the *Neumann folds* in the further directions, a Dirichlet fold the consecutive differences' squares' fold along the plane direction at the sum's unit beyond both ends, the list 2 on the diagonal with the balance partner of one at consecutive keys, and a Neumann fold the consecutive differences' squares' fold along a further direction, 2 on the interior diagonal, one at the two end keys and the balance partner of one at consecutive keys; the identity reads entry by entry: at one plaquette the diagonal 4 joined to the count of cubes holding it, two per further direction less one at an end position; at two plaquettes of one plane at consecutive positions in a plane direction the shared link's exchanged traversals, the balance partner of one, with the cubes' Gram at the sum's unit there, the two off every common cube, a cube's two faces of one plane type sitting opposite; at two plaquettes of one plane type at consecutive positions in a further direction the incidence at the sum's unit, the two holding distinct links, and the cube between them at its two opposite faces' signs, the balance partner of one at the faces' words running the one way; and at two plaquettes of distinct plane types sharing a link, the pair in one cube alone, the cube at the shared link's corner in the two planes, the incidence's read $K_{pl}K_{ql}$ against the cube's two faces' signs' product, the two joining to the sum's unit at the faces' signs' read $N_{pc}K_{pl} + N_{qc}K_{ql}$ at that unit; and the sum's unit at every further pair, a pair off a shared link and off a common cube, the Gram and the cubes' Gram both reading the unit there: two plaquettes sharing a link are of one plane type at consecutive positions in a plane direction or of distinct plane types in the cube at the shared link's corner, and a cube's two faces of one plane type sit opposite at consecutive positions in the further direction while its faces of distinct plane types share a link, so every pair sits in one class above. So at a range member W , the cubes' boundaries pairing it at the sum's unit, the Gram's form is the Hodge fold's, the Neumann folds' read at or beyond the sum's unit by their shape, and the Dirichlet folds telescope: along one plane direction a plaquette's read is the differences' fold from the end, its square at or below the side against the differences' squares (Construction 8's Cauchy–Schwarz at the count), so the self-pairing sits at or below the side's square against either direction's differences' fold and at or below half the side's square against the two folds' join, $W^T W \leq [L^2 : 2] W^T \Delta W$. The identity is one mechanism at every cell degree: the k -cells' Gram joined to the $(k+1)$ -cells' boundaries' Gram reads, cell type by cell type, k Dirichlet folds in the cell's directions joined to Neumann folds in the further directions, the four entry classes above at a cell in place of a plaquette and its faces in place of its links; at $k = 2$ it is the display above, and at $k = 3$ the cubes' Gram $N^T N$ joined to the four-cells' boundaries' Gram $N_4 N_4^T$, a *four-cell* four directions in order with a corner whose four coordinates sit below the side, its eight cubes the cubes at the corner in each three of its directions and at the corner stepped along the fourth, and N_4 the cube-cell list at the signs at which each four-cell's eight cubes' boundaries fold to the sum's unit, one sign family per four-cell at the sign one at the cube at the four-cell's corner in the four-cell's three lower directions, each plaquette of a four-cell a face of two of its eight cubes with the two cubes' signs at the plaquette's two face signs joining to that unit, so $N N_4$ reads the sum's unit entrywise, and at three directions or beyond, at the *cubes' range projector* Π_3 , the projector off the four-cells' boundaries' collection and the identity at three directions, a cube chain in the cubes' range has its self-pairing at or below a third of the side's square against its Gram read, $[3 : L^2] \Pi_3 \preceq N^T N$.

The tower and the modes' kernel. The Gram splits over its located roots e_K with a unit congruence to diagonal (Lemma 33; the roots located and separated at Theorem 16), the roots off the sum's unit inside the bracket above, and the *blocks* Π_K , one per distinct occupied root, are the interpolant polynomials in Δ through the distinct roots at the unit at e_K and the sum's unit at every further root and at the kernel (Construction 2' Lagrange read), symmetric, idempotent and pairwise at the sum's unit at the congruence's diagonal read, with $\Delta \Pi_K = e_K \Pi_K$ and $\sum_K \Pi_K = \Pi$.

The *window's tower* adjoins, per distinct occupied root, the root w_K at

$$w_K^2 2\kappa_\Sigma e_K = 1$$

(Lemma 37's quadratic extension at the located positive $[1 : 2\kappa_\Sigma e_K]$), κ_Σ the closure's joined index (Lemma 18), at or beyond one, one extension per root, a located stage (Lemma 32); every tower element enters with its bracket, and its comparisons against the pairs are the stage's side reads. The *modes' kernel* is

$$\mathbf{S} := \sum_K w_K \Pi_K ,$$

a matrix over the tower, symmetric, $\mathbf{S} \Pi = \Pi \mathbf{S} = \mathbf{S}$, with the *kernel identity*

$$\mathbf{S} \Delta \mathbf{S} = [1 : 2\kappa_\Sigma] \Pi$$

at the blocks' reads and the adjoined relation, its form at or beyond the sum's unit and at or below $[L : 2]$ times the coordinate fold of squares (w_K^2 at or below $[L^2 : 4]$ at the Poincaré read and $\kappa_\Sigma \geq 1$, the trichotomy of squares), the *roots' floor* $w_K \leq [L : 2]$; its *diagonal*, the *kernel's diagonal* per plaquette,

$$D_p := (\Delta \mathbf{S})_{pp} = \sum_K [1 : 2\kappa_\Sigma w_K] (\Pi_K)_{pp} ,$$

a tower element between the sum's unit and $D_+ := 2g$, each block's diagonal a self-pairing, the blocks' diagonals folding to Π 's at or below one, and $[1 : 2\kappa_\Sigma w_K]^2 = [e_K : 2\kappa_\Sigma] \leq [4g : \kappa_\Sigma] \leq D_+^2$ at every root at or below $8g$; its *corner fold* $\sum_{\ell \in \partial p} (K^\top \mathbf{S})_{\ell p}^2 \leq \sum_{\ell} (K^\top \mathbf{S})_{\ell p}^2 = (\mathbf{S} \Delta \mathbf{S})_{pp} \leq [1 : 2\kappa_\Sigma] \leq 1$, the boundary's fold a part of the links'; and at a pair ε at or below one the *shifted kernel's solve* on the range is

$$(\mathbf{S} + \varepsilon)^{-1} := \sum_K [1 : w_K + \varepsilon] \Pi_K , \quad (\mathbf{S} + \varepsilon) (\mathbf{S} + \varepsilon)^{-1} = \Pi ,$$

its diagonal at or below $W_0 := 1 + 8\kappa_\Sigma g$ (every root at or below $8g$ reads $[1 : w_K]^2 = 2\kappa_\Sigma e_K \leq 16\kappa_\Sigma g$, and a pair sits at or below the successor of its square's half), and at a range vector u the *dual weights* over the links

$$c^{(u)} := K^\top \Delta^{-1} (\mathbf{S} + \varepsilon)^{-1} u ,$$

at the two identities $K c^{(u)} = (\mathbf{S} + \varepsilon)^{-1} u$ (the Gram against the range solve the range vector itself) and $\sum_{\ell} c_{\ell}^{(u)} (K^\top u')_{\ell} = u'^\top (\mathbf{S} + \varepsilon)^{-1} u$ at a further range vector u' , and the fold $|c^{(u)}|_2^2 = u^\top (\mathbf{S} + \varepsilon)^{-1} \Delta^{-1} (\mathbf{S} + \varepsilon)^{-1} u \leq 2\kappa_\Sigma |u|^2$ ($\Delta^{-1} = 2\kappa_\Sigma \mathbf{S}^2$ on the range at the adjoined relation, and $\mathbf{S}$ against $\mathbf{S} + \varepsilon$ at or below one block by block), so $|c^{(u)}|_1^2 \leq \#\ell 2\kappa_\Sigma |u|^2$ and the fold $|c^{(u)}|_1$ sits at or below $C_1 := 1 + 2\kappa_\Sigma \#\ell$ at a unit range vector and at or below $C_1 |u|^2$ at $|u|^2$ at or beyond one (a value at or below the successor of its square, halved).

The profile. The *profile* W sits on the plaquettes of the first two directions' plane at every position of the further directions, $W_p := w_x w_y$ at the plane coordinates (x, y) of the plaquette's corner, $1 \leq x, y \leq L$, and $w_b := b(L + 1 + b^\vee)$ at b 's balance partner, the parabola at the sum's unit at $b = 0$ and $b = L + 1$; the transposed incidence against the profile reads, at a link of the first direction at the vertex (x, y) , $1 \leq x \leq L$ and $1 \leq y \leq L + 1$, the value $w_x \langle w_y : w_{y-1} \rangle$, the plaquette at the corner (x, y) forward at its base link and the plaquette at the corner $(x, y - 1)$ backward at its lower direction's link at the corner moved one step in the higher direction, each occupied exactly where its corner's coordinates sit at or below L and the parabola at the sum's unit at the ends reading the further ones; at a link of the second direction at (x, y) , $1 \leq y \leq L$ and

$1 \leq x \leq L+1$, the value $w_y \langle w_{x-1} : w_x \rangle$, the plaquette at the corner $(x-1, y)$ forward at its higher direction's link at the corner moved one step in the lower direction and the plaquette at the corner (x, y) backward at its higher direction's link at the corner; and the sum's unit at a link of a further direction; so at the $(L+1)^{g-1}$ positions of the further directions the Gram read and the self-pairing are $W^\top \Delta W = 2(L+1)^{g-1} \sum_{b=1}^L w_b^2 \sum_{b=0}^L \langle w_{b+1} : w_b \rangle^2$ and $W^\top W = (L+1)^{g-1} \left(\sum_{b=1}^L w_b^2 \right)^2$; it sits in the range at three directions or beyond, a cube's two faces in the plane reading one value at the exchanged signs and its further faces off the support, so the cubes' boundaries pair it at the sum's unit. Its Gram read against its self-pairing is the two directions' consecutive differences' folds at two closed sums,

$$30 \sum_{b=1}^L w_b^2 = L(L+1)(L+2)(L^2+2L+2), \quad 3 \sum_{b=0}^L \langle w_{b+1} : w_b \rangle^2 = L(L+1)(L+2),$$

so $W^\top \Delta W = e_W W^\top W$ at the *mode read* $e_W := [20 : L^2 + 2L + 2]$, falling with the side; $|W|^2$ denotes its self-pairing and $|W|_\infty$ its largest entry, and the *profile's solve* is $\sigma_W := W^\top (S + \varepsilon)^{-1} W$, between $\sigma_- := |W|^2 [2 : L+2]$ and $\sigma_+ := W_0 |W|^2$ at the roots' floor and the diagonal's cap, with the *profile's weight vector* $\omega := (S + \varepsilon)^{-1} W$ at $|\omega|^2 \leq W_0^2 |W|^2$.

The fluxes. The *moved flux* of a plaquette is its base link's gradient moved to the root along the root path,

$$f_p := R_{0 \leftarrow p} \nabla \chi_p,$$

one coefficient element per generator index (the character's gradient, Lemma 18); the *dumbbell* $D_{pq} := \sum_a (f_p)_a (f_q)_a$ is two moved fluxes' dot, a carrier element, the root's site action (Proposition 10) moving both lists by one adjoint coefficient and the dot its own read across it; two fluxes' pairing at a vertex reads the tree path from the one base to the other, the transports' orthogonality, $\langle R_{v \leftarrow p} \nabla \chi_p, R_{v \leftarrow q} \nabla \chi_q \rangle = D_{pq}$ at every vertex v ; the *mode coordinate* at a plaquette vector u is $y_u := \sum_p u_p f_p$, one coefficient element per generator index at the root; the *certificate's fluxes* are the kernel's reads of the moved fluxes at each link's tail,

$$G_{\ell a} := \sum_p (K^\top S)_{\ell p} (R_{\ell \leftarrow p} \nabla \chi_p)_a = (R_{\ell \leftarrow 0} y_{u_\ell})_a \quad \text{at} \quad u_\ell := (K^\top S)_\ell,$$

the mode coordinate at the kernel's row moved from the root to ℓ 's tail at the paths' composition; and at a scale η the *certificate's squares* are the generator actions joined to the balance partners of the certificate's fluxes,

$$T_{\ell a} := \eta X_a^\ell + \langle G_{\ell a} \rangle^\vee$$

over the links and the generator list, a fold of squares restricting to every fiber (Construction 8).

The loops. At a plaquette p and a boundary link ℓ the *loop* $\text{Hol}_{\ell p}$ runs from ℓ 's tail back along the boundary arc off ℓ to the base link's tail and forward along the tree path to ℓ 's tail, at the two links at the corner the arc vacant and the loop the unit word, its adjoint coefficient the ordered product of the loop region's plaquettes' adjoint coefficients each moved along the region's column and back (the interior links traversed both ways withdrawing at the evaluation identity), and its *adjoint deficit* is $\delta_{\text{Hol}} := \langle d_\theta : \chi_\theta(\text{Hol}_{\ell p}) \rangle$ at the adjoint dimension $d_\theta = \dim G$, the orthogonal adjoint coefficient's trace, with $\sum_{a,b} \langle \text{Hol} : 1 \rangle_{ab}^2 = 2 \delta_{\text{Hol}}$ at the entrywise balance pair against the identity (Lemma 18's adjoint gap) and the cap

$$\delta_{\text{Hol}} \leq t_L n_{\ell p} \sum_{q \in R_{\ell p}} \delta_q \leq t_L g L \sum_{q \in R_{\ell p}} \delta_q$$

at the deficits δ_q and the adjoint tie's pair t_L (Lemma 18), each deficit at or below $2d_L$: $\langle \text{Hol} : 1 \rangle$ telescopes over the factors, the i -th summand the product of the earlier factors against the i -th factor's own $\langle A_i : 1 \rangle$, the fold of squares reads the count against the summands' folds (Construction 8's Cauchy–Schwarz at the count), and each summand's fold is the moved coefficient's, the plaquette's own adjoint gap fold $2\delta_\theta(q)$, the trace one value at the moved list (Lemma 3's trace read), at or below $2t_L\delta_q$ at the adjoint tie (Lemma 18). The *deficits' fold* is $D := \sum_p \delta_p$. The *Casimir-weighted character* $\chi'_p := \sum_{\lambda \in \bar{L}} C_2(\lambda) \chi_\lambda(U_{\partial p})$ has the flat read the *Casimir fold* $K_L := \sum_{\lambda \in \bar{L}} C_2(\lambda) d_\lambda = \dim G \kappa_\Sigma$ (Lemma 18) and the *weighted deficit* $\delta'_p := \langle K_L : \chi'_p \rangle = \sum_\lambda C_2(\lambda) \varepsilon_\lambda$, the blocks' deficits at their Casimirs, between $C_2^-\delta_p$ and $C_2^+\delta_p$ at the closure's least and largest Casimirs C_2^-, C_2^+ ; and the *second gradient* of a plaquette at a boundary link is the generator pairs' reads $(\nabla_{\ell p}^2)_{ab} := X_a^\ell X_b^\ell \chi_p$ at the link's tail, the moved generator's index first.

The off-range fold. The moved fluxes' *off-range fold* $|(1 - \Pi)f|^2 := \sum_p |y_{(1-\Pi)e_p}|^2$ sits at or below $c_{\text{off}} D^2$ at the *cubes' boundaries' fold*

$$c_{\text{off}} := \kappa_\Sigma (g - 1) L^2 \left(21888 (1 + d_L)^8 + 72 (1 + 2 d_L) (g^2 L \# p + 2) \right),$$

the sum's unit at $d = 2$ by its shape: the off-range part of a plaquette vector is its projection to the cubes' boundaries' span, the incidence's kernel at three directions or beyond, the cube chain's boundary Nz at the chain $z := \Pi_3 \hat{z}$ in the cubes' range, $\hat{z}$ the kernel list's solve $(N_0^\top N_0)^{-1} N_0^\top f$ at the sum's unit off the list's cubes, the four-cells' boundaries' part withdrawing at $N N_4$ the sum's unit; the cubes' Poincaré read puts the chain's self-pairing at or below a third of the side's square against its Gram read, $[3 : L^2] |z|^2 \leq |Nz|^2$, a positive-semidefinite form's read at a list of elements a fold of squares at its split's positive pivots (Lemma 30), and at $|Nz|^2 = \langle z, N^\top f \rangle (N^\top \Pi \text{ the sum's unit})$ the two-member square at the pair $[L^2 : 3], 2 \langle z, N^\top f \rangle \leq [3 : L^2] |z|^2 + [L^2 : 3] |N^\top f|^2$, reads $|(1 - \Pi)f|^2 \leq [L^2 : 3] \sum_c |\langle \partial c, f \rangle|^2 \leq [L^2 : 2] \sum_c |\langle \partial c, f \rangle|^2$ over the cubes. A face's flux reads at the joined block (Lemma 18): the character's gradient is the joined generators' trace against the joined block's gap, and the root path's adjoint coefficient moves the generator index as the joined block's path coefficient moves the gap (the move identity), so the cube's signed flux fold is $\langle \partial c, f \rangle_a = \text{tr} (X_a^R M_c)$ at $M_c := \sum_{q \in \partial c} s_q E_q^0$, E_q^0 the face's gap moved to the root, a face traversed backward entering at its transposed gap, whose trace against an antisymmetric generator is the gap's own trace's balance partner, so $s_q E_q^0$ denotes the gap at a forward traversal and its transpose at a backward one, and the generators read a matrix's fold at the index, $\sum_a \text{tr} (X_a^R M)^2 \leq \kappa_\Sigma |M|^2$. Per cube the boundary's six coefficients, each moved to the cube's corner along the cube's own links, multiply to the identity, so at the gaps E'_q of the corner-moved coefficients from the identity, each at the fold $|E'_q|^2 = 2\delta_q$ (Lemma 18), at or below $4d_L$ and at or below $(1 + d_L)^2$ (the trichotomy of squares), the signed fold $\sum_q s_q E'_q$ reads the elementary products' fold from the second order on, at most 57 products, and its fold of squares sits at or below 57 times the products' folds' sum (Construction 8's Cauchy–Schwarz at the count), a product's fold at or below its factors' folds' product, a product's two least factors read at the deficits and the further factors at the cap $(1 + d_L)^2$, each pair the least pair of at most 16 products, and the pairs' products' fold at or below three times the folds' squares' sum (the six-member fold of squares):

$$\left| \sum_{q \in \partial c} s_q E'_q \right|^2 \leq 57 \cdot 16 \cdot 4 \cdot 3 (1 + d_L)^8 \sum_{q \in \partial c} \delta_q^2 = 10944 (1 + d_L)^8 \sum_{q \in \partial c} \delta_q^2.$$

The root-moved gap differs from the corner-moved one at the *cube loop* L_q , the two paths from the root to the face's corner, the tree's own and the corner's root path joined to the cube's link, whose region is the columns descended in the later directions from the cube's corner at the link's

direction, at most gL plaquettes as a loop region's, with the loop's gap G_q at the joined block, $E_q^0 = E'_q + G_q E'_q + E'_q G_q^\top + G_q E'_q G_q^\top$ read in the corner's own coordinates, its *loop deficit* $\delta_{L_q} := \langle d_L : \chi_L(\text{Hol}_{L_q}) \rangle$ at $|G_q|^2 = 2\delta_{L_q}$, at or below $4d_L$, and at the telescope over the region's factors $\delta_{L_q} \leq gL \sum_{q' \in R(L_q)} \delta_{q'}$ (the joined block's gaps at $2\delta_{q'}$ each, Cauchy-Schwarz at the count); the difference's fold of squares sits at or below $3|E'_q|^2 |G_q|^2 (2 + |G_q|^2)$ (the three-member fold of squares and a product's fold at its factors' folds' product), at or below $24(1 + 2d_L) \delta_q \delta_{L_q}$. The two-member fold of squares doubles, $|M_c|^2 \leq 2|\sum_q s_q E'_q|^2 + 2 \cdot 6 \sum_q |E_q^0 - E'_q|^2$, and over the cubes each plaquette is a face of at most $2(g-1)$ cubes, the products $\delta_q \delta_{L_q}$ at or below the halved squares' sum, and the cube loops' deficits' squares fold at the regions' count: a plaquette sits in the region of at most $[g-1 : (g+1)L] \#p$ cube loops (the $[g(g-1) : 2]$ cube types through the loop's direction at the corner's positions, as the loop regions' count), so $\sum_{(c,q)} \delta_{L_q}^2 \leq (g-1)g^2 L \#p D^2$ at each deficit at or below $gL D$; and the fold over the cubes at the count $[L^2 : 2] \kappa_\Sigma$ collects to the displayed c_{off} .

The member reads. The *adjoint index* $\kappa_\theta = C_2(\theta)$ (Lemma 18), the brackets' matrices' fold $\sum_a \text{ad} X_a (\text{ad} X_a)^\top = \kappa_\theta$ at the adjoint block's orthogonality (Lemma 18's coordinates' tie), the closure field's read $(X_a X_b \chi_p) + (X_b X_a \chi_p)^\vee = \sum_c f_{ab}^c X_c \chi_p$ at $f_{ab}^c = (\text{ad} X_a)_{cb}$ (Construction 3); the *second-gradient cap* h_L , the least natural whose square sits at or beyond d_L times the generators' second-order trace fold $\sum_{a,b} \text{tr}((X_a^R X_b^R)^\dagger X_a^R X_b^R)$ at the joined block's trace pairing (Lemma 18's joined generators), the stepping read by halving from one at the witness $d_L t$ at the fold's pair $[t : t']$ (Definition 1), so that the second gradient's fold $\sum_{a,b} (X_a X_b \chi_L)^2$ over the joined block sits at or below h_L^2 (Construction 8's Cauchy-Schwarz at the block coefficient's fold d_L); the *deviation pair* $m_L := \kappa_\theta \kappa_\Sigma + h_L + 1$; and the *covering count*

$$\Omega := m_L (4 + 2t_L \#p) ,$$

a pair, the plaquette count against the deviation pair: a fold over the plaquettes and their boundary links of $m_L (\delta_p + \delta_{\text{Hol}})$ sits at or below $\Omega \sum_q \delta_q$, the deficits' part at the four links and the loop deficits' part at the cap $t_L n_{\ell p} \sum_{R_{\ell p}} \delta_q$ folded at the regions' counts, at most $[2g : g+1] \#p \leq 2 \#p$ over the pairs whose region holds a plaquette.

Lemma 49 (The squares' certificate). *At two directions or beyond, at the square window's data (Construction 16) and the scale η (Lemma 47), the squares' certificate reads the window's pencil at the dual representative as the certificate's squares' fold joined to a multiplication ζ (Construction 8), at three reads. (i) The local square. The certificate's fluxes' fold collects at the kernel identity to the plaquettes' gradient squares against the off-range residual,*

$$2\kappa_\Sigma \sum_{\ell,a} G_{\ell a}^2 + |v + (\Pi v)^\vee|^2 = \sum_p \sum_a (X_a \chi_p)^2 \quad \text{at } v := (R_{0 \leftarrow p} \nabla \chi_p)_p ,$$

the second summand the self-pairing of the fluxes moved to one vertex off the incidence's range, at or beyond the sum's unit at the positive pairing. (ii) The cross read. At a plaquette p and a link ℓ the generators' trace of the moved gradient reads the Casimir-weighted character's balance partner joined to the deviation at a boundary link and the sum's unit at every further link,

$$\sum_a X_a^\ell (R_{\ell \leftarrow p} \nabla \chi_p)_a = K_{p\ell} \langle \text{dev}_{\ell p} : \chi_p' \rangle \quad \text{at } \ell \in \partial p ,$$

$$\text{dev}_{\ell p} := \sum_{a,b} \langle \text{Hol}_{\ell p} : 1 \rangle_{ab} (\nabla_{\ell p}^2)_{ab} ,$$

each deviation and its balance partner at or below $m_L (\delta_p + \delta_{\text{Hol}})$ at the loop's adjoint deficit; folded at the certificate's fluxes, the divergence (Construction 8) reads the kernel's diagonal against the

Casimir-weighted characters joined to the deviations,

$$\sum_{\ell,a} X_a^\ell(G_{\ell a}) = \left\langle \sum_p \sum_{\ell \in \partial p} K_{p\ell} (K^\top S)_{\ell p} \text{dev}_{\ell p} : \sum_p D_p \chi'_p \right\rangle ,$$

the deviations' weighted fold over the plaquettes and their boundary links at or below $\Omega \sum_q \delta_q$ at the covering count. (iii) The floor line. ζ sits at or beyond the floor line: at the deficits δ_p at the closure's dimension fold (Lemma 18), the kernel's diagonal D_p per plaquette, a tower element between the sum's unit and $D_+ = 2g$, the covering count Ω , a pair, the closure's largest Casimir C_2^+ and its dimension-weighted mean c_L at $c_L d_L = K_L$, the Casimir fold $\sum_{\lambda \in \bar{L}} C_2(\lambda) d_\lambda$,

$$\begin{aligned} \zeta &\geq \sum_p [1 : 2d_L] \langle \delta_p : \eta d_L (C_2^+ D_p + \Omega) \rangle^2 + \ell_0 , \\ \ell_0 &:= \sum_p d_L \left\langle \eta c_L D_p : 1 + [1 : 2] \eta^2 (C_2^+ D_p + \Omega)^2 \right\rangle , \end{aligned}$$

the floor line, a tower element, the window's pencil at or beyond it over the window's tower (Lemma 30 at the located stage): the ground sits at or beyond ℓ_0 , the kernel's read $\eta K_L \sum_p D_p$ against the flat read $d_L \# p$, the loss the scale's square against the count at the covering count's square.

Proof. The certificate's data are the window's (Construction 16). *The local square.* Two moved fluxes' pairing at one link reads the tree path from the one base to the other, the link's own path backtracking in the tree, so the squares' fold over the links collects at the kernel's identity, $\sum_\ell (K^\top S)_{\ell p} (K^\top S)_{\ell p'} = (S \Delta S)_{pp'}$, to the display, the range's projector symmetric and idempotent, the residual the sum's unit at one and two directions, the range the whole, and at three directions or beyond the cubes' boundaries' fold, each cube's six moved fluxes at the traversals' signs joined. *The cross read.* The divergence $\sum_{\ell,a} X_a^\ell(G_{\ell a})$ (Construction 8) reads at each plaquette and each boundary link the generator against the moved gradient, the transport's own derivative the sum's unit (a link sits on its own tree path at most as the last link into its tail, where the generator differentiates the path word at its endpoint, the letter wired to itself withdrawing, Proposition 9), and at a boundary link's factor the read is the joint insertion across the loop $\text{Hol}_{\ell p}$ (Construction 16) at the link's traversal sign $K_{p\ell}$: the arc's part is the base link's own Laplacian, the joint insertion across the arc at the evaluation contraction (Proposition 9's arc read), the generator moved through the loop's own coefficient and back at the trace read's one value at the moved list (Lemma 3), the block read of the class function (Lemma 7), $\sum_a (X_a X_a \chi_p)$ the balance partner of the Casimir-weighted character χ'_p at the flat read K_L and the weighted deficit δ'_p (Construction 16), and the tree path's part enters through the loop's adjoint coefficient against the second gradient $\nabla_{\ell p}^2$: the cross read is the first display and its fold at the certificate's fluxes the second, the deviation the pairing of $\langle \text{Hol} : 1 \rangle$ against $\nabla_{\ell p}^2$, and the main read the kernel's diagonal D_p at the incidence's Gram, the signed boundary fold $\sum_{\ell \in \partial p} K_{p\ell} (K^\top S)_{\ell p} = (\Delta S)_{pp}$. *The assembly.* The fold of squares joined to the electric member's balance partner reads the pencil at the dual representative as the squares' fold joined to ζ , the balance partner of the multiplication $\sum_{\ell,a} G_{\ell a}^2 + \sum_p \chi_p + \eta \sum_{\ell,a} X_a^\ell(G_{\ell a})$ (Construction 8's joined square at the pair η), and ζ reads at or beyond a floor by three comparisons at the deficits. The local square against the gradient deficit (Lemma 18) puts $\sum G^2$ at or below $\sum_p \langle \delta_p : [1 : 2d_L] \delta_p^2 \rangle$. The deviation prices at the member reads, the adjoint index κ_θ , the second-gradient cap h_L and the deviation pair m_L , with the corner fold putting each weight's magnitude at or below one (Construction 16). The pairing of $\langle \text{Hol} : 1 \rangle$ against $\nabla_{\ell p}^2$ splits at the two matrices' symmetric and antisymmetric parts, the cross pairings the sum's unit. The antisymmetric part

of $\nabla^2 \chi_p$ is the structure constants at the gradient, half the closure field's read, its fold of squares the adjoint read $[1 : 4] \kappa_\theta \sum_c (X_c \chi_p)^2$ at the brackets' matrices' fold (Construction 16's member reads), at or below $[1 : 2] \kappa_\theta \kappa_\Sigma \delta_p$ at the gradient deficit; the antisymmetric part of $\langle \text{Hol} : 1 \rangle$ has its fold of squares at or below the whole's, $\sum_{a,b} \langle \text{Hol} : 1 \rangle_{ab}^2 = 2 \delta_{\text{Hol}}$ at the loop's adjoint deficit (Construction 16); so the antisymmetric pairing and its balance partner each sit at or below $[1 : 2] (\kappa_\theta \kappa_\Sigma \delta_p + \delta_{\text{Hol}})$, the two-member square at the pair $[1 : 2]$, twice the pairing at or below half the one fold joined to twice the other. The symmetric part of $E := \langle \text{Hol} : 1 \rangle$ is the balance partner of half the square $E^\top E$ at the unitary read, so the symmetric pairing is the balance partner of half the fold over E 's rows E_k of $E_k^\top S E_k$ at S the symmetric part of $\nabla_{\ell p}^2$, whose fold sits at or below h_L^2 ; a symmetric matrix at a natural h with h^2 at or beyond its fold reads $x^\top S x$ and its balance partner at or below $h |x|^2$ at every list x (the two-member square $2h x^\top S x \leq h^2 |x|^2 + |Sx|^2$ at $|Sx|^2 \leq h^2 |x|^2$, Construction 8's Cauchy-Schwarz, the factor $2h$ withdrawn at the positive pair $[1 : 2h]$), so at h_L the symmetric pairing and its balance partner each sit at or below $[1 : 2] h_L |E|^2 = h_L \delta_{\text{Hol}}$: the deviation at (p, ℓ) and its balance partner each sit at or below $m_L (\delta_p + \delta_{\text{Hol}})$ at the deviation pair m_L . Folded over the plaquettes and their boundary links the deviation sits within $\Omega \sum_q \delta_q$ at the loop deficits' cap and the covering count (Construction 16). And the square completes at the scale, the weighted deficit read at the largest Casimir, $\delta'_p \leq C_2^+ \delta_p$,

$$\zeta \geq \sum_p [1 : 2d_L] \langle \delta_p : \eta d_L (C_2^+ D_p + \Omega) \rangle^2 + \ell_0 ,$$

$$\ell_0 := \sum_p d_L \left\langle \eta c_L D_p : 1 + [1 : 2] \eta^2 (C_2^+ D_p + \Omega)^2 \right\rangle \quad \text{at} \quad c_L d_L = K_L ,$$

the *floor line*, the window's pencil at or beyond it: the ground sits at or beyond ℓ_0 , the kernel's read $\eta K_L \sum_p D_p$ against the flat read $d_L \# p$, the loss the scale's square against the count at the covering count's square. $\square$

Construction 17 (The corner trial). At the square window's data (Construction 16), the scale $\eta = [n : s]$ at or below one and a stated pair ε at or below one, the *mass*, the trial's data are the reads below, D the deficits' fold (Construction 16). *The exponent* is the balance partner of the kernel's form at the moved fluxes joined to the mass's multiple of the deficits' fold,

$$\Phi := \left\langle [1 : 2\kappa_\Sigma] \sum_{p,q} S_{pq} D_{pq} + \varepsilon D \right\rangle^\vee , \quad \Phi_+ := (L + 2\varepsilon) d_L \# p ,$$

at or below the sum's unit and at or beyond the balance partner of Φ_+ : the form is the fold over the generator index of the kernel's form at the lists $(f_p)_a$, a fold of squares at the blocks' coefficients w_K , each at or beyond the sum's unit and at or below the roots' floor $[L : 2]$, so the form sits between the sum's unit and $[L : 2]$ times the lists' self-pairing, at or below $[L : 2] \sum_p 2\kappa_\Sigma \delta_p$ at the gradient deficit (Lemma 18), and the deficits' fold at or below $2d_L \# p$. The two summands read the sum's unit at distinct configurations: the kernel's form reads it exactly where every moved flux does, and a plaquette's flux reads it at every coefficient whose adjoint block is symmetric at the orthogonal read, the gradient's trace pairing an antisymmetric generator against a symmetric matrix, while the deficit there reads half the fold of squares of $1 - R$ at the joined coefficient (Lemma 18), occupied off the identity; the deficits' fold reads the sum's unit at the flat configurations alone, so the mass fixes the trial's weight at the flat configurations at every scale below the pair's own, its mismatch against the kernel's form at the modes priced at the cap. *The near data*. At the count reads A , $\wp$ of Lemma 46, the *support exponent* $e := (3\wp + 1) \# \ell$ and the *support coefficient* $c_W := A^{3\# \ell} (9g)^e$, the natural $j := e + 1$, the pair $\tau := [n : 4d_L \# p (L + 2\varepsilon) s]$ and the *concentration key* $k := k(\tau)$, the

least natural at $k\tau^j \geq c_W j^j$, its witness the division's quotient of $c_W j^j b^j$ at a^j at its successor at $\tau = [a : b]$ (Definition 1), the *near mass* is the pair $\vartheta := [1 : 2c_W k^e]$, one read of the scale, the side and the mass. The *mode square* is $Q := |y_W|^2$, the mode coordinate's self-pairing at the profile (Construction 16), a positive element, with its *pairs*

$$Q_+ := 4\kappa_\Sigma d_L \#p|W|^2, \quad Q_- := \eta\kappa_\Sigma \dim G|W|^2 \vartheta [1 : 4(L+2)] ;$$

the *stated weight pairs* are the unit's $(1, \vartheta)$, the mode square's (Q_+, Q_-) and its square's (Q_+^2, Q_-^2) ; the *bit count* j' is the largest, over the three pairs (P_+, P_-) and the powers k at or below four, of the least natural at or beyond one at $2^{j'} \geq P_+ [1 : \varepsilon^k P_-]$, one trichotomy fold, each least natural the stepping read by halving from one at the witness the compared pair's first member (Definition 1); and the *near scale* is $K := [48(4+j') : \varepsilon]$. *The exponent count.* The natural m is the larger of the division's quotient of $2(L+2)d_L \#pss$ at nn at its successor and $4+j'$, one trichotomy fold (Definition 1), so m sits at or beyond $2\Phi_+ ss$ against nn at the mass at or below one and at or beyond $4+j'$. *The trial.* The *trial's factor* and the *trial* are

$$B := 1 + \Phi[s : nm], \quad \Psi := B^m,$$

B a multiplication between $[1 : 2]$ and one, the exponent's read against the scale's multiple of m at or below one half $(\Phi_+[s : nm] \leq [n : 2s] \leq [1 : 2])$ at the scale at or below one), so Ψ sits between $[1 : 2]^m$ and one, positive by its shape; the generators read it at the powers' Leibniz read, $X_a^\ell \Psi = m B^{m-1} X_a^\ell B$, so

$$B X_a^\ell \Psi = [s : n] (X_a^\ell \Phi) \Psi,$$

one exact identity per link and generator. *The moment reads.* At a carrier element F the *moment reads* are $\mu[F] := \text{Eval}(F\Psi^2)$ and $\nu[F] := \text{Eval}(F B^{2m-1})$, the trial's square with one factor B withdrawn, tower elements at the tower's own evaluation, the evaluation additive and scale-reading in the coefficients; a *weight* is a positive element P at or below P_+ , a comparison of elements, with its moment $\mu[P]$ at or beyond P_- at one of the stated weight pairs, and a weight's square is a weight at the squared pair, P^2 at or below P_+^2 at the two gaps' product and $\mu[P^2] \mu[1] \geq \mu[P]^2$ at $\mu[1] \leq 1$ (the evaluation's squared Cauchy-Schwarz; $\Psi^2 \leq \Psi \leq 1$ at the factor between $[1 : 2]$ and one); the parts identity at the product $F\Psi^2$ (Construction 8) with the identity above reads

$$\mu[X_a^\ell F] + 2[s : n] \nu[F X_a^\ell \Phi] = 0, \quad \mu[F] = \nu[F] + [s : nm] \nu[F \Phi],$$

the *parts identity at the trial* and the *factor's display*, the second the square's split at one factor; at the exponent count the exponent's read against the scale's multiple of m sits at or below half the scale, $\Phi_+[s : nm] \leq [n : 2s]$, so at a *cap* of a member F , a positive element F_+ with $F_+ + F$ and $F_+ + F^\vee$ positive elements, each of $\mu[F] - \nu[F]$ and its balance partner sits at or below $[n : 2s] \nu[F_+]$ (the read $\nu[F \Phi]$ at Φ the balance partner of the positive $|\Phi|$ at or below Φ_+), and at a positive element F the two reads are ordered, $\langle 1 : [n : 2s] \rangle \nu[F] \leq \mu[F] \leq \nu[F]$, with $\nu[F] \leq 2\mu[F]$ at the scale at or below one. *The deviation.* The exponent's gradient reads the certificate's fluxes, the *mass gradient* and a deviation quadratic in the fluxes, $X_a^\ell \Phi = G_{\ell a} + \varepsilon F_{\ell a} + R_{\ell a}$ at $F_{\ell a} := \sum_{p \ni \ell} K_{p\ell} (R_{\ell \leftarrow 0} f_p)_a$, the link's plaquettes' moved fluxes at the traversal signs (Lemma 18's gradient move), its square fold at or below the link's plaquette count $2g$ against the fluxes' squares (Construction 8's Cauchy-Schwarz at the count): a generator at a link reads a moved flux at three sites, the boundary read $X_a^\ell \nabla \chi_p = K_{p\ell} \nabla_{\ell p}^2$ at $\ell \in \partial p$, the second gradient at the arc (Proposition 9's joint insertion), whose read is the count's balance partner at the arc's coefficient joined to a remainder, $\nabla_{\ell p}^2 = \langle E_{\ell p} : \kappa_\Sigma R_{\text{arc}} \rangle$ at the remainder's fold $\sum_{a,b} (E_{\ell p})_{ab}^2 \leq 2h_L^2 \delta_p$ (the second gradient at the loop's coefficient, Lemma 7's block read at the trace read's one value at

the moved list (Lemma 3), the second-gradient cap against $\langle \text{Hol}:1 \rangle$'s fold $2\delta_p$ by Construction 8's Cauchy–Schwarz); the transport read at a link of the root path, the generator inserted at the link's tail in the path word, $X_a^\ell R_{0 \leftarrow p} = \text{ad}(R_{0 \leftarrow \ell} X_a) R_{0 \leftarrow p}$, its fold over the generator index the adjoint Casimir's read κ_θ against the flux's self-pairing (the moved list $R_{0 \leftarrow \ell} X_a$ orthonormal at the adjoint coefficient's evaluation identity, the Casimir one operator at every orthonormal list, Definition 7); and the arc's coefficient against the tree path's, the loop's own, $R_{0 \leftarrow p} R_{\text{arc}} = R_{0 \leftarrow \ell} \text{Hol}_{\ell p}$ at the loop (Construction 16). Folded at the kernel the generator's read of the exponent collects the certificate's flux, the loop's coefficient against the identity, the remainder and the transport,

$$\begin{aligned} R_{\ell a} = & \sum_{p \ni \ell} K_{p\ell} \sum_q S_{pq} \langle R_{0 \leftarrow \ell} \langle \text{Hol}_{\ell p}:1 \rangle e_a, f_q \rangle + [1 : \kappa_\Sigma] \sum_{p \ni \ell} K_{p\ell} \sum_q S_{pq} \langle R_{0 \leftarrow p} E_{\ell p} e_a, f_q \rangle^\vee \\ & + [1 : \kappa_\Sigma] \sum_{p: \ell \in \text{path}(p)} \sum_q S_{pq} \langle \text{ad}(R_{0 \leftarrow \ell} X_a) f_p, f_q \rangle^\vee, \end{aligned}$$

the last fold over the plaquettes whose root path holds ℓ , each summand a product of two flux-order reads, the first at the loop's deficit, the second at the remainder's fold and the third at the adjoint index's, at $(Sf)_p := \sum_q S_{pq} f_q$: the deviation's square fold over the links and the generator index reads the three members' squares at three folds (Construction 8's Cauchy–Schwarz at the counts, at most $2g$ boundary plaquettes per link and at most $\#p$ plaquettes whose root path holds the link), the first at the loop's Frobenius read $\sum_{a,b} \langle \text{Hol}:1 \rangle_{ab}^2 = 2\delta_{\text{Hol}}$ (the unitary read), the second at the remainder's fold $2h_L^2 \delta_p$ and the third at the adjoint index κ_θ , the transport read's fold, each against $|(Sf)_p|^2$. The kernel's square reads at the blocks, $\sum_p |(Sf)_p|^2 = \sum_a \sum_K w_K^2 |\Pi_K f_a|^2$ at f_a the list $(f_p)_a$ over the plaquettes and the blocks' orthogonality, each block's coefficient at or below the roots' floor's square $[L^2 : 4]$ and the blocks' folds' sum $|\Pi f_a|^2$ at or below $|f_a|^2$, the range and kernel parts' squares, so $\sum_p |(Sf)_p|^2 \leq [L^2 : 4] \sum_p |f_p|^2 \leq [L^2 : 4] 2\kappa_\Sigma D$; the boundary pairs fold at four links per plaquette with the loop deficits at $\delta_{\text{Hol}} \leq t_L g L D$ (the loop's cap, Construction 16), and the path pairs fold at the root paths' lengths, at most $(g+1)L$ links each, so the third fold's count is the length against the plaquette count:

$$\sum_{\ell,a} R_{\ell a}^2 \leq \Theta_R D^2, \quad \Theta_R := 24 t_L \kappa_\Sigma g^2 L^3 + 24 g h_L^2 L^2 + 3(g+1) \kappa_\theta \#p L^3,$$

the *deviation's pair*, in the side and the plaquette count, with $T_R := 1 + \Theta_R$ its successor at $\Theta_R \leq T_R^2$; at a positive pair t and a list x over the generator index the pairing against a link's deviation and its balance partner each sit at or below the two-member square's read,

$$2 \langle x, R_\ell \rangle \leq [1 : t] |x|^2 + t \sum_a R_{\ell a}^2,$$

the gap the fold of squares $[1 : t] \sum_a \langle x_a : t R_{\ell a} \rangle^2$ and at the balance partner the fold at the swapped members.

Lemma 50 (The near reads). *At the corner trial (Construction 17) three reads hold at every scale at or below one. (i) The far and near reads. At $x := |\Phi| [s : nm]$ and a natural j at or below m , the comparisons of positive elements*

$$C(2m, j) x^j \Psi^2 \leq 1, \quad \Psi^2 (\varepsilon D)^j \leq j! \eta^j, \quad \Psi^2 \geq 1 - 2mx \quad \text{at} \quad 2mx = 2|\Phi| [s : n].$$

(ii) *The near mass. At the near mass ϑ (Construction 17),*

$$\mu[1] \geq \vartheta.$$

(iii) *The near scale.* At a weight P at one of the stated weight pairs, a power k at or below four and the near scale K (Construction 17),

$$\mu[P D^k] \leq 2(K\eta)^k \mu[P] .$$

Proof. The positive reads. The deficit and its complement are folds of squares, $2\delta_p$ and $2(2d_L - \delta_p)$ the self-pairings of $1 - R$ and $1 + R$ at the plaquette's joined coefficient (Lemma 18; Construction 8's evaluation identity), so the exponent's balance partner $|\Phi|$, the kernel's form joined to the mass's fold, is a positive element with Φ_+ at or beyond it at three positive gaps, the roots' floor at the kernel's form ($[L : 2]$ against each w_K at the blocks' squares), the gradient deficit's spare, the projection residual's fold of squares (Lemma 18), and the deficits' complements. So at $x := |\Phi| [s : nm]$ the trial reads $\Psi^2 = (1 - x)^{2m}$ at the positive elements x and $1 - x$, the latter the pair $1 - \Phi_+[s : nm]$, at or beyond $[1 : 2]$, joined to $(\Phi_+ - |\Phi|)[s : nm]$. Two reads at the binomial partition $1 = \sum_i C(2m, i) x^i (1 - x)^{2m-i}$, every summand positive. The *far read*: at a natural j at or below m ,

$$C(2m, j) x^j \Psi^2 \leq 1 , \quad \Psi^2 (\varepsilon D)^j \leq j! \eta^j ,$$

the first the partition's further summands joined to the j -th summand's read against $(1 - x)^j$, the gap $x \sum_{i < j} (1 - x)^i$ at the telescope $(1 - x)^i = 1 - x \sum_{i' < i} (1 - x)^{i'}$, and the second at $\varepsilon D \leq |\Phi|$ with $j! C(2m, j) \geq m^j$ at $j \leq m$: the evaluation of a positive element against $\Psi^2 (\varepsilon D)^j$ sits at or below $j! \eta^j$ against its own evaluation. The *near read*: $\Psi^2 \geq 1 - 2mx$, the gap $x^2 \sum_{i < 2m} \sum_{i' < i} (1 - x)^{i'}$ at the telescope, and $2mx = 2|\Phi|[s : n]$ at $|\Phi| \leq ([L : 2] + \varepsilon) D$. The *near mass*. At the *complement read* $u_p := (2d_L - \delta_p)[1 : 2d_L]$, a positive element with $1 - u_p$ positive and flat read one, and a natural k , the *return read* $Q := \prod_p u_p^{2k}$ is a positive element below one, the gap $\sum_p (1 - u_p^{2k}) \prod_{q < p} u_q^{2k}$ at the product's telescope, with flat read one and support weight W_k at or below $c_W k^e$ at the support exponent e and the support coefficient c_W (Construction 17): a plaquette character's power reads labels at fusion depth at most its degree on the plaquette's links, the link's plaquettes' powers fuse at the link to depth at most $4gk$, the grade key moves by at most two per fusion (the move cap, Lemma 46), and per key the labels and dimensions sit at or below $A(\mathcal{G} + 1)^\wp$ (the counts, Lemma 46), so a link's occupied dimensions' squares fold to at most $A^3(8gk + 1)^{3\wp+1}$ and the weight to at most that power's $\#\ell$ -th, at $8gk + 1 \leq 9gk$. So $\text{Eval}(Q)$ sits at or beyond $[1 : W_k]$, the flat read's square against the support weight (Construction 8). Per plaquette the reads $b_i := \text{Eval}(u_p^{2i} \prod_{q \neq p} u_q^{2k})$ fall at a slowing rate: $b_0 \leq 1$, $b_k = \text{Eval}(Q)$, $b_i^2 \leq b_{i-1} b_{i+1}$ (the evaluation's squared Cauchy-Schwarz at the positive element $\prod_{q \neq p} u_q^{2k}$), and $b_{i+1} \leq b_i$ at $(1 - u_p^2) u_p^{2i}$ positive, so the *step quotients* r_i at $r_i b_i = b_{i+1}$ ascend below one and b_k sits at or below r_{k-1}^k : at a step quotient's gap y at $r_{k-1} + y = 1$ beyond a pair τ the far read's Bernstein bound at the natural $j := e + 1$, $C(k, j) y^j r_{k-1}^k \leq 1$, reads $C(k, j) \tau^j < W_k$ at $\tau^j < y^j$, against $C(k, j) j^j \geq k^j$ at $k \geq j$ and $W_k \leq c_W k^e$, so $k \tau^j < c_W j^j$ at $j = e + 1$, against the concentration key's comparison $k \tau^j \geq c_W j^j$ at $k(\tau)$ (Construction 17) and at every k beyond it, so the gap sits at or below τ , and the deficit's read at the return read, $\delta_p Q = 2d_L (1 - u_p) u_p^{2k} \prod_{q \neq p} u_q^{2k}$ at $(1 - u_p) u_p^{2k} \leq (1 - u_p^2) u_p^{2k}$, sits at or below $2d_L (b_k - b_{k+1}) \leq 2d_L \tau \text{Eval}(Q)$ at $k + 1$ at or beyond the key, the raised member $u_p^{2k+2} \prod_{q \neq p} u_q^{2k}$ fusing at p 's links to depth at most $4gk + 2$, its key at most $8gk + 5$, at or below $9gk$ from k at or beyond five, the concentration key at or beyond $e + 1$ and e at or beyond four at the window's four links or more, so its evaluation sits at or beyond $[1 : c_W k^e]$ as $\text{Eval}(Q)$ does. The trial's mass then reads at the near read against the return read, $\Psi^2 \geq \Psi^2 Q \geq (1 - 2mx) Q$ at $1 - Q$ positive, the exponent's read against the deficits' fold and the

fold's read against the return read:

$$\begin{aligned} \mu[1] &\geq \text{Eval}(Q) \left(1 - 2[s : n]([L : 2] + \varepsilon) \#p 2d_L \tau \right) \geq [1 : 2] \text{Eval}(Q) \geq [1 : 2 W_k] \\ &\geq \vartheta \end{aligned}$$

at the stated pair τ and the concentration key $k = k(\tau)$, the near mass ϑ (Construction 17). *The near scale.* At a weight P at the pairs (P_+, P_-) and a natural k the deficits' fold reads its k -th power against a weight at the *near scale* K : at $y := D[1 : K\eta]$ and a natural j' ,

$$y^k \leq 1 + y^{k+j'}, \quad \mu[P D^k] \leq (K\eta)^k \mu[P] + \mu[P D^{k+j'}] [1 : (K\eta)^{j'}] \leq 2 (K\eta)^k \mu[P],$$

the first at the gap $(1-y)^2(\sum_{i < k} y^i)(\sum_{i < j'} y^i) + y^{j'}$, the second its moment against $P (K\eta)^k$, and the third at the far read, $\mu[P D^{k+j'}] \leq (k+j')! \eta^{k+j'} [1 : \varepsilon^{k+j'}] P_+$, against $(K\eta)^k P_-$ at $m \geq k+j'$, the exponent count's read: at the bit count j' (Construction 17) and a near scale at or beyond $[2k!(k+j') : \varepsilon]$ the comparison $K^{j'} \varepsilon^{k+j'} P_- \geq (k+j')! P_+$ holds at $(k+j')! \leq (k+j')^{j'} k!$; the stated weight pairs are the unit's, the mode square's and its square's (a weight's square a weight at the squared pair, Construction 17), the powers at or below four, and the bit count is the largest over them (the lower powers at $\varepsilon \leq 1$), so the stated near scale $K = [48(4+j') : \varepsilon]$ reads every weight at every power at or below four ($2k!(k+j')$ at or below $48(4+j')$ at every power k at or below four, the comparison monotone in the near scale). $\square$

Lemma 51 (The moment identities). *At the corner trial (Construction 17) and the window's data (Construction 16): (i) the cross read at a moved coordinate, at a range vector u' and a link ℓ ,*

$$\begin{aligned} \sum_a X_a^\ell (R_{\ell \leftarrow 0} y_{u'})_a &= \langle r_\ell : \kappa_\Sigma \dim G(K^\top u')_\ell \rangle, \\ |r_\ell| &\leq \varrho_1 |u'|_\infty D, \quad \varrho_1 := C_2^+ + m_L(1 + 2t_L g^2 L); \end{aligned}$$

(ii) the coordinate identity, at range vectors u, u' and an involution-fixed positive element Z ,

$$\begin{aligned} 2[s : n] \nu[Z \langle y_{u'}, y_u \rangle] &+ \sum_\ell c_\ell^{(u)} \mu[Z r_\ell] \\ &+ \sum_\ell c_\ell^{(u)} \mu[\langle R_{\ell \leftarrow 0} y_{u'}, X^\ell Z \rangle] + 2[s : n] \sum_\ell c_\ell^{(u)} \nu[Z \langle R_{\ell \leftarrow 0} y_{u'}, R_\ell \rangle] \\ &= \kappa_\Sigma \dim G u'^\top (S + \varepsilon)^{-1} u \mu[Z]; \end{aligned}$$

and (iii) the deficit cell: at the near scale K (Lemma 50) and the two comparisons at the scale

$$2 \#p \eta C_1 \varrho_1 \leq \kappa_\Sigma, \quad \eta K^2 (8[1 : \kappa_\Sigma] C_1^2 T_R^2 + 4[1 : \kappa_\Sigma] c_{\text{off}} + 4c'_q) \leq 1$$

at $c'_q := [s_L : 2]$ (Lemma 18's second-order pair) and c_{off} the cubes' boundaries' fold (Construction 16), the deficits' fold reads

$$\mu[D] \leq C_D \#p \eta \mu[1], \quad C_D := 2 \dim G W_0 + 1,$$

C_D a member pair in the direction count, free of the side, the mass and the near scale.

Proof. The moment identities. At a range vector u the dual weights read the exponent's gradient back to the coordinate, $\sum_\ell c_\ell^{(u)} R_{0 \leftarrow \ell}(G_\ell + \varepsilon F_\ell) = y_u$ at $(S + \varepsilon) K K^\top \Delta^{-1} = (S + \varepsilon) \Pi$, and

read a further vector's incidence fold to the shifted kernel's pairing, $\sum_{\ell} c_{\ell}^{(u)} (K^{\top} u')_{\ell} = u'^{\top} (S + \varepsilon)^{-1} u$. A generator at a link reads the moved coordinate's trace at the cross read (Lemma 49(ii)), $\sum_a X_a^{\ell} (R_{\ell \leftarrow 0} f_p)_a = K_{p\ell} \langle \text{dev}_{\ell p} : \chi_p' \rangle$ at $\ell \in \partial p$ and the sum's unit at every further link, the deviation within $m_L(\delta_p + \delta_{\text{Hol}})$; at $K_L = \dim G \kappa_{\Sigma}$ (Construction 16) and χ_p' the weighted deficit's partner at K_L the fold is

$$\sum_a X_a^{\ell} (R_{\ell \leftarrow 0} y_{u'})_a = \langle r_{\ell} : \kappa_{\Sigma} \dim G (K^{\top} u')_{\ell} \rangle ,$$

$$|r_{\ell}| \leq \varrho_1 |u'|_{\infty} D$$

at the displayed ϱ_1 , the boundary plaquettes' weighted deficits folding to $C_2^+ D$ and their loop deficits to $2g \cdot g_L D$ at the deviation's cap. The dual weights' fold sits at or below C_1 at a unit range vector and at or below $C_1 |u|^2$ at $|u|^2$ at or beyond one (Construction 16). The parts identity at the element $F_{\ell a} := c_{\ell}^{(u)} (R_{\ell \leftarrow 0} y_{u'})_a Z$, Z a further involution-fixed positive element, folded over the links and the generator index, reads the *coordinate identity*

$$\begin{aligned} & 2[s : n] \nu[Z \langle y_{u'}, y_u \rangle] + \sum_{\ell} c_{\ell}^{(u)} \mu[Z r_{\ell}] \\ & + \sum_{\ell} c_{\ell}^{(u)} \mu[\langle R_{\ell \leftarrow 0} y_{u'}, X^{\ell} Z \rangle] + 2[s : n] \sum_{\ell} c_{\ell}^{(u)} \nu[Z \langle R_{\ell \leftarrow 0} y_{u'}, R_{\ell} \rangle] \\ & = \kappa_{\Sigma} \dim G u'^{\top} (S + \varepsilon)^{-1} u \mu[Z] , \end{aligned}$$

the left member's first summand the exponent's gradient read back to y_u through the weights, its second and fourth the derivative's remainder and the deviation, and its third the generator's read of Z . *The second moments.* At $Z = 1$ the third summand is the sum's unit and the identity at $u = u' = \Pi e_p$, a unit range vector, reads the coordinates' pairings at the shifted kernel's solve, every remainder second order: the derivative's remainder sits within $C_1 \varrho_1 \mu[D]$, and the deviation's, at the two-member square at $t := C_1$ (Construction 17) with each weight's magnitude at or below the weights' fold C_1 , within $[s : n] \nu[|y_{\Pi e_p}|^2]$, absorbed by the leading read, joined to $[s : n] C_1^2 \nu[\sum_{\ell, a} R_{\ell a}^2]$, at or below $[s : n] C_1^2 T_R^2 \nu[D^2]$ at the deviation's pair; the diagonal $(S + \varepsilon)_{pp}^{-1}$ sits at or below W_0 (Construction 16); so

$$\nu[|y_{\Pi e_p}|^2] \leq \eta \kappa_{\Sigma} \dim G W_0 \mu[1] + \eta C_1 \varrho_1 \mu[D] + C_1^2 T_R^2 \nu[D^2] .$$

The fluxes' range parts fold to the deficits, $|f_p|^2 \leq 2|y_{\Pi e_p}|^2 + 2|y_{(1-\Pi)e_p}|^2$ with the off-range fold $\sum_p |y_{(1-\Pi)e_p}|^2 = |(1-\Pi)f|^2 \leq c_{\text{off}} D^2$ (the off-range fold, Construction 16), the factor's display reads $\nu[F] \leq 2\mu[F]$ on positive elements at $\eta \leq 1$, and the near scale reads $\mu[D^2] \leq 2(K\eta)^2 \mu[1]$ at the unit weight; at the deficit's second-order read (Lemma 18) the deficits' fold then reads

$$\begin{aligned} \mu[D] & \leq [1 : 2\kappa_{\Sigma}] \sum_p \mu[|f_p|^2] + [s_L : 2] \mu[D^2] \\ & \leq \#p \eta \dim G W_0 \mu[1] + [1 : \kappa_{\Sigma}] \#p \eta C_1 \varrho_1 \mu[D] \\ & + (4[1 : \kappa_{\Sigma}] C_1^2 T_R^2 + 2[1 : \kappa_{\Sigma}] c_{\text{off}} + 2c'_q) \#p K^2 \eta^2 \mu[1] \end{aligned}$$

at c'_q , the second moments' display doubled at the range parts and folded over the plaquettes, $\nu[D^2]$ read at twice $\mu[D^2]$ and every second-order member at the near scale's read, the off-range and self-read members joined at the count's multiple, the count at or beyond one; and on the *deficit cell*, the two comparisons at the scale

$$2\#p \eta C_1 \varrho_1 \leq \kappa_{\Sigma} , \quad \eta K^2 (8[1 : \kappa_{\Sigma}] C_1^2 T_R^2 + 4[1 : \kappa_{\Sigma}] c_{\text{off}} + 4c'_q) \leq 1 ,$$

the self-read withdraws at the factor two, the second-order members read at or below half the count's multiple of the scale's multiple of $\mu[1]$, and the display follows, the mass and the near scale entering the read through the deficit cell alone. $\square$

Lemma 52 (The excess). *At the corner trial (Construction 17) and the squares' certificate (Lemma 49): (i) The squares' cap. The certificate's squares at the trial fold within*

$$\sum_{\ell,a} (T_{\ell a} \Psi)^2 \leq t(D) \Psi^2, \quad t(D) := 12 (16 g \kappa_\Sigma \varepsilon^2 D + \Theta_R D^2 + [n^2 : 4s^2] D),$$

a comparison of positive elements. (ii) The slack. The window's pencil at the dual representative less the floor line's multiple of the gram is the certificate's squares' fold joined to the slack q , a multiplication, the assembly's read (Lemma 49), and

$$q \leq c_q D^2 + \eta \Omega D + W_q \eta^2, \\ c_q := [s_L : 2] + c_{\text{off}} [1 : 2\kappa_\Sigma], \quad W_q := [1 : 2] d_L \# p (C_2^+ D_+ + \Omega)^2,$$

s_L the closure's second-order pair (Lemma 18) and c_{off} the cubes' boundaries' fold (Construction 16). (iii) The excess. The excess r_u is the trial's read against the floor line, $r_u \mu[1]$ the pencil's form at Ψ less the floor line's multiple of the gram; on the deficit cell (Lemma 51) at the near reads (Lemma 50),

$$r_u \leq 192 g \kappa_\Sigma C_D \# p \varepsilon^2 \eta + P_u \eta^2 \quad \text{at} \quad P_u := 24 \Theta_R K^2 + 2 c_q K^2 + 3 C_D \# p + \Omega C_D \# p + W_q.$$

Proof. The excess. The trial's read against the floor line is the certificate's squares at the trial joined to the slack's moment, $\sum_{\ell,a} \|T_{\ell a} \Psi\|^2 + \mu[q]$, and both read at the moments. The squares: $T_{\ell a} \Psi$ reads $(\varepsilon F_{\ell a} + R_{\ell a} + G_{\ell a} \Phi^\vee[s : nm]) \Psi [1 : B]$ at the exact identity, so the fold sits at or below $12 (\varepsilon^2 8g \sum_p \mu[|f_p|^2] + \Theta_R \mu[D^2] + [n^2 : 4s^2] \mu[D])$ at $B \geq [1 : 2]$, the three-member fold of squares, the mass gradient's cap, the deviation's pair and $\Phi_+[s : nm] \leq [n : 2s]$ with $2\kappa_\Sigma \sum G^2 \leq \sum_p |f_p|^2$ at the local square. The slack: q reads the balance pair $\langle \sum_p \delta_p + [1 : 2] \eta^2 d_L \sum_p (C_2^+ D_p + \Omega)^2 : \sum G^2 + \eta \sum_p D_p \delta'_p + \eta \text{dev} \rangle$ at the assembly's reads, the deficits' fold less the squares' fold the spares' fold joined to the off-range residual at the local square, each spare at or below $[s_L : 2] \delta_p^2$ (Lemma 18's second-order read) and the residual at or below $c_{\text{off}} D^2$, the *off-range fold* (Construction 16); so $q \leq ([s_L : 2] + c_{\text{off}} [1 : 2\kappa_\Sigma]) D^2 + \eta \Omega D + [1 : 2] \eta^2 d_L \# p (C_2^+ D_+ + \Omega)^2$ at the deviation's cap and the kernel's positive diagonal. At the second moments, the gradient deficit ($|f_p|^2 \leq 2\kappa_\Sigma \delta_p$ at every plaquette, Lemma 18) and the near-scale reads the excess is the display, the squares' cap's three members at the second moments, $96 g \varepsilon^2 \cdot 2\kappa_\Sigma \mu[D]$ the leading read, $12 \Theta_R \mu[D^2]$ and $12 [n^2 : 4s^2] \mu[D]$ at the scale's cube, and the slack's three, each at the near scale's second power $\mu[D^2] \leq 2 (K\eta)^2 \mu[1]$ (Lemma 50) and the deficits' fold $\mu[D] \leq C_D \# p \eta \mu[1]$, the scale's higher powers withdrawn at the scale at or below one. $\square$

Lemma 53 (The mode member). *At the mode square Q (Construction 17) the mode mean $\bar{q}$ is the tower element at $\bar{q} \mu[1] = \mu[Q]$, the stage's cofactor at $\mu[1]$ occupied at the near mass (Lemma 50), and the mode member $v := \langle Q : \bar{q} \rangle \Psi$, perpendicular to the trial in the gram by its construction. The remainder cell is the four comparisons at the scale*

$$16 W_0^4 C_W^2 T_R^2 K^2 \eta (\kappa_\Sigma \dim G \sigma_+ + 2a_1 K \eta) + 64 W_0^8 C_W^4 T_R^4 K^4 \eta^2 \leq [1 : 1024] \kappa_\Sigma^2 \sigma_-^2, \\ 2a_1 K \eta \leq [1 : 32] \kappa_\Sigma \sigma_- , \quad a_2 \leq [1 : 16] \kappa_\Sigma \sigma_- , \\ 16 W_0^4 C_W^2 T_R^2 K^2 \eta ([17 : 16] \kappa_\Sigma (\dim G + 2) \sigma_+ + a_2) + 64 W_0^8 C_W^4 T_R^4 K^4 \eta^2 \leq [1 : 256] \kappa_\Sigma^2 \sigma_-^2$$

at $C_W := C_1 |W|^2$, $a_1 := W_0^2 C_W |W|_\infty \varrho_1$, $a_2 := 2 a_1 K \eta + 32 K \eta W_0^4 C_W^2 \varrho_2 |W|_\infty^2 [1 : \kappa_\Sigma \sigma_-]$ and $\varrho_2 := 12 g h_L^2 + 24 t_L \kappa_\Sigma^2 g^3 L + 6 \kappa_\Sigma \#p \kappa_\theta$. On the remainder cell at $\eta (\dim G + 2) \leq 1$, at the near reads (Lemma 50): (i) the mode identities, the coordinate identity (Lemma 51) at $Z = 1$ and at $Z = Q$ at $u = u' = W$ and the profile's solve σ_W ,

$$2[s : n] \nu[Q] = \kappa_\Sigma \dim G \sigma_W \mu[1] + r_1, \quad 2[s : n] \nu[Q^2] = \kappa_\Sigma (\dim G + 2) \sigma_W \mu[Q] + r_2,$$

at the remainders' caps $|r_1| \leq [1 : 16] \kappa_\Sigma \sigma_W \mu[1]$ and $|r_2| \leq [3 : 16] \kappa_\Sigma \sigma_W \mu[Q]$, and at $u = \omega$, $u' = W$ at the weight vector's self-pairing $|\omega|^2$ (Construction 16),

$$\begin{aligned} 2[s : n] \nu[\langle y_W, y_\omega \rangle] &= \kappa_\Sigma \dim G |\omega|^2 \mu[1] + r'_1, \\ 2[s : n] \nu[Q \langle y_W, y_\omega \rangle] &= \kappa_\Sigma (\dim G + 2) |\omega|^2 \mu[Q] + r'_2, \end{aligned}$$

at $|r'_1| \leq [1 : 16] \kappa_\Sigma \sigma_- \mu[1]$ and $|r'_2| \leq [3 : 16] \kappa_\Sigma \sigma_- \mu[Q]$; (ii) the mode square is a weight at its pairs (Q_+, Q_-) ; (iii) the variance floor

$$\langle v, v \rangle \geq [5 : 8] \eta \kappa_\Sigma \sigma_W \mu[Q];$$

and (iv) the mode mean's read $\mu[Q]^2 \leq \dim G \mu[1] \langle v, v \rangle$.

Proof. The mode member. The mode square is a positive element at or below Q_+ , $Q \leq |W|^2 \sum_p |f_p|^2$ (Construction 8's Cauchy–Schwarz at the count) at the gradient deficit $|f_p|^2 \leq 2 \kappa_\Sigma \delta_p$ and the deficit's complement, $\delta_p \leq 2 d_L$ (Lemma 18), comparisons of elements, the lower weight read at the first identity below. The coordinate identity (Lemma 51) at $Z = 1$ and at $Z = Q$, both at $u = u' = W$ and the profile's solve σ_W (Construction 16), reads

$$2[s : n] \nu[Q] = \kappa_\Sigma \dim G \sigma_W \mu[1] + r_1, \quad 2[s : n] \nu[Q^2] = \kappa_\Sigma (\dim G + 2) \sigma_W \mu[Q] + r_2,$$

the second's third summand the generator's read of the mode square, $\langle R_{\ell \leftarrow 0} y_W, X^\ell Q \rangle = 2 \langle (R_{\ell \leftarrow 0} y_W)^\top \epsilon_\ell (R_{\ell \leftarrow 0} y_W) : \kappa_\Sigma (K^\top W)_\ell Q \rangle$ at $X_a^\ell (R_{\ell \leftarrow 0} y_W)_b$ the pair $\langle \epsilon_{\ell, ba} : \kappa_\Sigma (K^\top W)_\ell \delta_{ab} \rangle$, the boundary read's leading member at the joint insertion and its gap ϵ_ℓ first order in the fluxes, $|\epsilon_\ell|^2 \leq \varrho_2 |W|_\infty^2 D$ (the three-member fold of squares at the counts $2g$ and $\#p$, the boundary plaquettes and the plaquettes whose root path holds the link: the second gradient's remainder $2h_L^2 \delta_p$, the loop's gap $\kappa_\Sigma^2 2\delta_{\text{Hol}}$ at $\delta_{\text{Hol}} \leq t_L g L D$, and the transport's read at the root paths' links, $\kappa_\theta |f_p|^2$ at the gradient deficit), the leading member collecting to $2 \kappa_\Sigma \sigma_W \mu[Q]$ through the weights; the remainders r_1 and r_2 are second-order reads priced at the near scale. The dual weights at W fold at or below C_W and at the profile's weight vector ω at or below $W_0^2 C_W$ (Construction 16), and the profile's solve sits between σ_- and σ_+ (Construction 16). At u' among W and ω the derivative's remainder sits within $a_1 \mu[D]$, and the deviation's, $X := 2[s : n] |\sum_\ell c_\ell \nu[\langle R y_W, R_\ell \rangle]|$, reads its square at the evaluation's squared Cauchy–Schwarz twice, each moment's square at or below $\nu[Q] \nu[\sum_a R_{\ell a}^2]$ at the generator index's joined list and the weighted fold's square at or below the weights' fold against the weighted squares' fold (Construction 8), the weights' fold at or below $W_0^2 C_W$ and the deviation's fold at or below $T_R^2 D^2$ (Construction 17), so $X^2 \leq 4[s : n]^2 W_0^4 C_W^2 T_R^2 \nu[Q] \nu[D^2]$: at the near scale's reads at the unit weight (Lemma 50), $\mu[D] \leq 2K\eta \mu[1]$ and $\nu[D^2] \leq 4K^2 \eta^2 \mu[1]$, at $M := 8 W_0^4 C_W^2 T_R^2 K^2 \eta \mu[1]$ and the first identity's own read of $\nu[Q]$, $X^2 \leq M (\kappa_\Sigma \dim G \sigma_+ + 2a_1 K \eta) \mu[1] + M X$, the two-member fold of squares reads $X^2 \leq 2M (\kappa_\Sigma \dim G \sigma_+ + 2a_1 K \eta) \mu[1] + M^2$; the second identity's remainder reads its three members at the mode square's moments, $\mu[QD] \leq 2K\eta \mu[Q]$ at the near scale, the generator's gap read at the two-member square at $t := \kappa_\Sigma \sigma_- [1 : 16 W_0^2 C_W]$, the pairing $2 \langle x, \epsilon_\ell x \rangle$ at $x := R_{\ell \leftarrow 0} y_W$ and its balance partner each at or below $t |x|^2 + [1 : t] |\epsilon_\ell x|^2$ with $|\epsilon_\ell x|^2 \leq |\epsilon_\ell|^2 Q$ (Construction 8's

Cauchy–Schwarz), and its deviation’s square at or below $4[s : n]^2 W_0^4 C_W^2 T_R^2 \nu[Q^2] \nu[QD^2]$ at the squared Cauchy–Schwarz at the weight Q . So on the remainder cell the remainders read $|r_1| \leq [1 : 16] \kappa_\Sigma \sigma_W \mu[1]$ and $|r_2| \leq [3 : 16] \kappa_\Sigma \sigma_W \mu[Q]$, and so do the identities’ remainders r'_1, r'_2 at $u = \omega, u' = W$, the leading coefficient the weight vector’s self-pairing $|\omega|^2$ at σ_- in the caps: the deviation’s square sits at or below $[1 : 1024] \kappa_\Sigma^2 \sigma_-^2 \mu[1]^2$ at the first comparison, the deviation at $[1 : 32] \kappa_\Sigma \sigma_- \mu[1]$ at the trichotomy of squares, and the derivative’s remainder at $[1 : 32] \kappa_\Sigma \sigma_- \mu[1]$ at the second comparison through $\mu[D] \leq 2K\eta\mu[1]$, the two joining to $|r_1|$ ’s read; at the mode square’s moments the derivative’s remainder with the gap’s second member sit within $a_2 \mu[Q]$, at or below $[1 : 16] \kappa_\Sigma \sigma_- \mu[Q]$ at the third comparison, the gap’s first member at $[1 : 16] \kappa_\Sigma \sigma_- \mu[Q]$, and the deviation’s square, $\nu[Q^2]$ read through the second identity at those two members, sits at or below $[1 : 256] \kappa_\Sigma^2 \sigma_-^2 \mu[Q]^2$ at the fourth comparison, the deviation at $[1 : 16] \kappa_\Sigma \sigma_- \mu[Q]$, the three joining to $|r_2|$ ’s read, and σ_- at or below σ_W . The member’s self-pairing is the mode’s variance, $\mu[Q^2]$ less $\bar{q} \mu[Q]$, and at $\langle 1 : [n : 2s] \rangle \nu[F] \leq \mu[F] \leq \nu[F]$ on positive elements (the factor’s display) the two identities read it from below,

$$\langle v, v \rangle + [\eta : 2] (\bar{q} |r_1| + |r_2|) \geq \eta \kappa_\Sigma \sigma_W \mu[Q] \langle 1 : [1 : 4] \eta (\dim G + 2) \rangle ,$$

the chi-square read at $\dim G$ degrees ($\mu[Q^2]$ at or beyond $\langle 1 : [n : 2s] \rangle \nu[Q^2]$ and the mean’s square $\bar{q} \mu[Q]$ at or below $\bar{q} \nu[Q]$); at $\eta (\dim G + 2) \leq 1$ and the remainder cell the variance sits at or beyond five eighths of its leading member, $\langle v, v \rangle \geq [5 : 8] \eta \kappa_\Sigma \sigma_W \mu[Q]$, so the mode square is a weight at Q_- ($\mu[Q] \geq [1 : 2] \nu[Q] \geq [\eta : 4] (\kappa_\Sigma \dim G \sigma_W \mu[1] - |r_1|)$ at $\mu[1] \geq \vartheta$, Lemma 50), and the *mode mean’s read* holds, $\mu[Q]^2 \leq \dim G \mu[1] \langle v, v \rangle$ ($\mu[Q] \leq \nu[Q] \leq [17\eta : 32] \kappa_\Sigma \dim G \sigma_W \mu[1]$ against the variance’s floor). $\square$

Lemma 54 (The mode read). *The mode read r_v is the mode member’s read against the floor line, $r_v \langle v, v \rangle$ the pencil’s form at v less the floor line’s multiple of the gram (Lemma 49; Lemma 53). On the deficit cell (Lemma 51) and the remainder cell at $\eta (\dim G + 2) \leq 1$ (Lemma 53), at the near reads (Lemma 50) and the trial’s factor display (Construction 17),*

$$\begin{aligned} r_v &\leq 16 \kappa_\Sigma (L + 2) e_W \eta + 384 g \kappa_\Sigma C_Z \varepsilon^2 \eta + P_v \eta^2 , \\ C_Z &:= 2 \dim G W_0 \#p + 8 W_0^2 (L + 2) , \\ P_v &:= 32 K \varrho_2 \# \ell (L + 2) + (384 g \kappa_\Sigma + 6 + \Omega) P' + (6 + \Omega) C_Z \\ &\quad + (96 \Theta_R + 4 c_q) (1 + 2 \dim G) K^2 + W_q , \\ P' &:= [32 : \kappa_\Sigma] \#p C_1^2 T_R^2 K^2 (1 + 2 \dim G) + [26 : \kappa_\Sigma] \#p C_1^2 \varrho_2 (L + 2) K \\ &\quad + 26 W_0^2 (L + 2) K + 4 (1 + 2 \dim G) \\ &\quad + (8 c_{\text{off}} [1 : \kappa_\Sigma] + 8 c'_q) K^2 (1 + 2 \dim G) , \end{aligned}$$

C_Z in the side and the direction count, P' and P_v in the side, the direction count and the near scale, all free of the mass, the leading read the mode’s own frequency read at the profile’s Gram against its self-pairing.

Proof. The member’s form against the floor line reads the squares and the slack at v , $T(\langle Q : \bar{q} \rangle \Psi) = \eta(XQ) \Psi + \langle Q : \bar{q} \rangle T\Psi$, so

$$\begin{aligned} \langle v, (H - \ell_0) v \rangle &\leq 2\eta^2 \mu[|XQ|^2] + 2\mu[\langle Q : \bar{q} \rangle^2 t(D)] + \mu[\langle Q : \bar{q} \rangle^2 q] , \\ \mu[|XQ|^2] &\leq 8 \kappa_\Sigma^2 e_W |W|^2 \mu[Q] + 8 \varrho_2 \# \ell |W|_\infty^2 \mu[QD] , \end{aligned}$$

$t(D)$ the squares’ cap and q the slack (Lemma 52), $|XQ|^2$ the fold of $2 \langle X^\ell(R_{\ell \leftarrow 0} y_W), R_{\ell \leftarrow 0} y_W \rangle$ ’s squares at the leading member $W^\top \Delta W = e_W |W|^2$, the mode read at the closed sums, and the

mode square's deficit moment $\mu[QD] \leq 2K\eta\mu[Q]$ at the near scale (Lemma 50). *The mode weight's moments.* At the *mode weight* $Z := \langle Q : \bar{q} \rangle^2$, an involution-fixed positive element at the moment $\mu[Z] = \langle v, v \rangle$, the coordinate identity at $u = u' = \Pi e_p$ reads the fluxes' range part against the weight. The mode weight's square moment reads at the near scale (Lemma 50) on the weights Q^2 and the unit, the square a weight at the squared pair (Construction 17), at the mode mean's read, $Z \leq 2Q^2 + 2\bar{q}^2$,

$$\mu[ZD^2] \leq 4(K\eta)^2(1 + 2\dim G) \langle v, v \rangle.$$

The identity's derivative's remainder sits within $C_1\varrho_1\mu[ZD]$; its deviation splits at the two-member square at $t := 2C_1$ (Construction 17) into a quarter of the leading read and $2[s : n]C_1^2\nu[Z\sum_{\ell,a}R_{\ell a}^2]$, at or below $16C_1^2T_R^2K^2(1 + 2\dim G)\eta\langle v, v \rangle$ at the deviation's pair, $\nu[F] \leq 2\mu[F]$ on positive elements and the mode weight's square moment; and its third summand is the generator's read of the mode square at the weight's root, $2\langle Q : \bar{q} \rangle X^\ell Q$, whose gap ϵ_ℓ splits at the two-member square at $t := 4C_1\eta$ into a further quarter of the leading read and $[13 : \kappa_\Sigma]C_1^2\varrho_2(L + 2)K\eta\langle v, v \rangle$ (the mode square's near-scale moment against the variance's floor, at $|W|_\infty$ at or below $|W|$), while its leading member folds over the plaquettes to $4\kappa_\Sigma\mu[(Q - \bar{q})\langle y_\omega, y_W \rangle]$ at $y_\omega = \sum_p \omega_p f_p$, read at the two identities at $u = \omega$, $u' = W$: $\nu[Q\langle y_W, y_\omega \rangle]$ at $[\eta : 2]\kappa_\Sigma(\dim G + 2)|\omega|^2\mu[Q]$ and $\nu[\langle y_W, y_\omega \rangle]$ at $[\eta : 2]\kappa_\Sigma\dim G|\omega|^2\mu[1]$ at $|\omega|^2 \leq W_0^2|W|^2$ (the mode identities at $u = \omega$, Lemma 53), their remainders at the remainder cell, and the factor's display at a cap (Construction 17), each of $\mu[F] - \nu[F]$ and its balance partner at or below $[\eta : 2]\nu[F_+]$ at the cap $F_+ := \kappa_\Sigma W_0^2|W|^2D + [1 : 2]Q$ of $F = \langle y_\omega, y_W \rangle$ and the cap QF_+ of $F = Q\langle y_\omega, y_W \rangle$, the two-member square $2\langle y_\omega, y_W \rangle$ and its balance partner each at or below $|y_\omega|^2 + Q$ at $|y_\omega|^2 \leq |\omega|^2\sum_p|f_p|^2$ (Construction 8's Cauchy-Schwarz) and the gradient deficit, so the fold sits within $4\eta\kappa_\Sigma^2|\omega|^2\mu[Q]$ joined to the remainders' reads, at or below $4\kappa_\Sigma W_0^2(L + 2)\langle v, v \rangle$ (the leading read's $[16 : 5]$ with the remainders' scale-free part at $[4 : 5]\kappa_\Sigma\langle v, v \rangle$ inside it) joined to $(13\kappa_\Sigma W_0^2(L + 2)K + 2\kappa_\Sigma(1 + 2\dim G))\eta\langle v, v \rangle$ at the variance's floor and the mode mean's read. Each plaquette's identity then reads its leading summand at the half, the two quarters withdrawn, and the fold over the plaquettes at the count, the deficit's second-order read at the fluxes' range parts against the off-range fold, the self-read withdrawing at the factor two at the deficit cell's first comparison, reads the deficits' fold at the mode weight, $\mu[ZD] \leq (C_Z\eta + P'\eta^2)\langle v, v \rangle$ at the displayed C_Z and P' . The member's form then reads at the moments: the squares' cap at the mode weight against the deficit's two moments, the slack at the mode weight against them, the mode read at the closed sums against the variance read on the cell (Lemma 53) (the variance at or beyond half its leading member, the five eighths' read at the remainder cell and $\eta(\dim G + 2) \leq 1$), and at $\sigma_W \geq \sigma_- = |W|^2[2 : L + 2]$ and κ_Σ at or beyond one the mode read is the display, the leading read the mode's own frequency read at the profile's Gram against its self-pairing, $[20 : L^2 + 2L + 2]$ at the side $(2\eta^2 \cdot 8\kappa_\Sigma^2 e_W |W|^2 \mu[Q])$ at the variance's half floor), the mass read the squares' cap's mass gradient at the mode weight's deficit moment, and the scale read the further caps, the mode square's deficit moment $\mu[QD] \leq 2K\eta\mu[Q]$ at the near scale among them $(2\eta^2 \cdot 8\varrho_2\#\ell|W|_\infty^2 \cdot 2K\eta\mu[Q])$ at the half floor and the solve's floor, the index withdrawn at or beyond one), every cap displayed. $\square$

Lemma 55 (The tail read). *At two directions or beyond, on a regional cutoff tail with simple grounds, the caps fall under every floor along the scale, read at the square region (Construction 16; Lemma 49), the cut's meet at or below the regional cut (Definition 11). Each admitted window's height reads at its two lines (Lemma 62(vi)): the floor at the squares' certificate's floor line (Lemma 49) and the cap at two stated members, the trial and its mode member, the lines' gap against the scale the height's cap, the gap falling with the window's mode read e_W at the side*

(Construction 16). At a floor pair $\varphi = [f_n : f_d]$ three located naturals, the side key L_0 , the mass ε 's halving index and the scale key η_0 's, are each the least member of its family at a comparison stepping along the family, the stepping read at its displayed witness (Definition 1), one halving read per floor pair; at every scale under η_0 the height sits below φ at every cutoff holding the two members and beyond the simple-ground key. This caps the regional height and the chain's height (Lemma 47; Definition 11). At one direction the lines' gap keeps the floor $4\kappa_\Sigma$ at every side.

Proof. The window's data are Construction 16's, the trial's Construction 17's at the scale and the mass, the trial at the exponent count m , and the reads are the five lemmas' (Lemmas 50, 51, 52, 53, 54). *The cap.* The two members are finite coefficient elements in the square region. Their least contents (Proposition 4) joined to the simple-ground key give one cutoff from which both members are admitted and the ground is simple. The floor line is valid at every cutoff in that region (Lemma 49), so Lemma 62(vi) reads the cap on the whole cutoff tail. At every line beyond $\ell_0 + r_u + r_v$ the window's count reads at or beyond two below it: the pencil at the floor line is positive semidefinite on the window (Lemma 49), the trial and the mode member are perpendicular in the gram by the member's construction (Lemma 53), and their diagonal reads against the members' gram are the excess r_u and the mode read r_v (Lemma 62(vi)); on the deficit cell and the remainder cell at $\eta(\dim G + 2) \leq 1$ the reads' sum sits at or below $\eta(16\kappa_\Sigma(L+2)e_W + g\kappa_\Sigma(192C_D\#p + 384C_Z)\varepsilon^2 + (P_u + P_v)\eta)$, the excess and the mode read joined (Lemma 52; Lemma 54). *The key.* At a floor pair $\varphi = [f_n : f_d]$ the *side key* L_0 is the least side at or beyond one at which the mode read sits under a quarter of the floor, $64\kappa_\Sigma(L+2)e_W < \varphi$, the cross-multiplied comparison $1280\kappa_\Sigma(L+2)f_d < f_n(L^2 + 2L + 2)$ at e_W 's pair, stepping up the sides (a hold at L reads its successor's, the successor's members moved by $f_n(2L+3)$ against $1280\kappa_\Sigma f_d$, on the upper side at the hold since $1280\kappa_\Sigma f_d(L+2) < f_n(L^2 + 2L + 2) \leq f_n(2L+3)(L+2)$), its witness the side $1280\kappa_\Sigma f_d$ (there and at every larger side $L(L+2)$ sits at or beyond $1280\kappa_\Sigma f_d(L+2)$, under $L^2 + 2L + 2$); the *mass* ε the first member of the halving family $[1 : 2^i]$, the least halving index, at which the two mass reads sit under a further quarter, $4g\kappa_\Sigma(192C_D\#p + 384C_Z)\varepsilon^2 < \varphi$ at the key's side, its witness the exponent $4g\kappa_\Sigma(192C_D\#p + 384C_Z)f_d$ (the power 4^i beyond i at every natural); and the *scale key* η_0 the first member of the halving family $[1 : 2^i]$, the least halving index, at which the *key's list* holds and the near scale clears $6b_K$, the growth pair below. The key's list is the cell's comparisons at the key's side and mass, the scale's square read through the scale at or below one, nine reads $\eta P \leq \gamma$ at the pairs

$$\begin{aligned}
P^{(1)} &:= 2\#pC_1\varrho_1, \quad \gamma^{(1)} := \kappa_\Sigma; & P^{(7)} &:= \dim G + 2, \quad \gamma^{(7)} := 1; \\
P^{(9)} &:= d_1, \quad \gamma^{(9)} := n_1; & P^{(4)} &:= 2a_1K, \quad \gamma^{(4)} := [1 : 32]\kappa_\Sigma\sigma_-; \\
P^{(2)} &:= K^2(8[1 : \kappa_\Sigma]C_1^2T_R^2 + 4[1 : \kappa_\Sigma]c_{\text{off}} + 4c'_q), \\
\gamma^{(2)} &:= 1; & P^{(8)} &:= 4(P_u + P_v), \quad \gamma^{(8)} := \varphi; \\
P^{(3)} &:= 16W_0^4C_W^2T_R^2K^2(\kappa_\Sigma\dim G\sigma_+ + 2a_1K) \\
&\quad + 64W_0^8C_W^4T_R^4K^4, \quad \gamma^{(3)} := [1 : 1024]\kappa_\Sigma^2\sigma_-^2; \\
P^{(5)} &:= 2a_1K + 32KW_0^4C_W^2\varrho_2|W|_\infty^2[1 : \kappa_\Sigma\sigma_-], \quad \gamma^{(5)} := [1 : 16]\kappa_\Sigma\sigma_-; \\
P^{(6)} &:= 16W_0^4C_W^2T_R^2K^2([17 : 16]\kappa_\Sigma(\dim G + 2)\sigma_+ \\
&\quad + P^{(5)}) + 64W_0^8C_W^4T_R^4K^4, \quad \gamma^{(6)} := [1 : 256]\kappa_\Sigma^2\sigma_-^2,
\end{aligned}$$

the deficit cell's two (Lemma 51), the remainder cell's four at $a_2 = \eta P^{(5)}$ (Lemma 53), $\eta(\dim G + 2) \leq 1$, the scale comparison $4(P_u + P_v)\eta \leq \varphi$ and the corner cell's extent (Lemma 47), the scale's ray $[\eta^2 : 1]$ in the extent at the squares' read, $d_1^2 \leq n_1^2 4^i$ at $\eta = [1 : 2^i]$, the read $d_1 \leq n_1 2^i$

at the trichotomy of squares, each one polynomial read of the scale and the near scale at natural coefficients cleared at the pairs' second members, of degree at most four in the near scale, the further data fixed by the key's side and mass, and the list holding puts the cell's comparisons in force. The near scale moves down the family at two reads. It grows by at most $b_K := 48b[1 : \varepsilon]$ per halving at the bit count's growth natural $b := 2 + 2(e+1)e$: the halved scale multiplies the cleared comparison's first member by 2^{e+1} at $j = e+1$, so the concentration key by at most 2^{e+1} (the least natural at a comparison's first member multiplied by a power of two sits at or below that power's multiple of the least natural at the comparison), the near mass falls by at most $2^{(e+1)e}$, and the bit count's compared pair grows by at most $2^{2+2(e+1)e}$, four at the scale's square, so the bit count by at most b . And the compared pair at least quadruples per halving, the halved scale halving the lower weight at the least, the concentration key monotone in the scale and the near mass with it; its first member sits at or beyond one, and its second member $(\kappa_\Sigma \dim G |W|^2)^2$ sits at or below 2^{d_W} at its *binary length* d_W , the least natural at that comparison, its witness the compared square's first member (the power 2^i beyond i at every natural); so at the i -th halving the pair's first member sits at or beyond 4^i , the bit count joined to d_W at or beyond $2i$, and the near scale, monotone down the family, at or beyond $48(4+2i)[1 : \varepsilon]$ less $48d_W[1 : \varepsilon]$: it clears $6b_K$ from the $(3b+d_W)$ -th halving on. Past the clearing a monomial's growth per halving sits under two, $(K+b_K)^m \leq 2K^m$ at $m \leq 4$ ($(1+[1:6])^4$ under two), so a comparison $\eta P(K) \leq \gamma$ holding at a scale holds at its half, and the key's list with the clearing holding at a scale holds at every scale under it; and at the i -th halving past the $(3b+d_W)$ -th, from the near scale K_* and the scale η_* there, the near scale sits at or below $K_*+b_K i$ and the read $\eta_*[1:2^i]P(K_*+b_K)i^4$ caps $\eta P(K)$ ($(K_*+b_K i)^m \leq (K_*+b_K)^m i^4$ at $m \leq 4$ and i at or beyond one), while 2^i clears i^5 from $i = 32$ ($2^{32} \geq 32^5$ and $(1+[1:i])^5 \leq 2$ from $i = 7$), so the comparison holds at every i at or beyond 32 and $\eta_* P(K_*+b_K)[1:\gamma]$, the witness: the scale key is the first member of the halving family at the list with the clearing, the least halving index (Definition 1), one halving read per floor pair at the witness's halving count, the $(3b+d_W)$ -th halving's index joined to 32 and to the sum over the list of the least natural at or beyond each read's pair $\eta_* P(K_*+b_K)$ against γ , the division's quotient's successor (Definition 1). At every scale under η_0 the key's list holds, so the reads' sum sits under three quarters of the floor's multiple of the scale, $4(r_u+r_v) < 3\eta\varphi$, the line $\ell_0 + [1:2](r_u+r_v+\eta\varphi)$ beyond $\ell_0+r_u+r_v$ with the window's count at or beyond two below it, so the window's height sits at or below $[7:8]\varphi$, below φ (Lemma 62(vi)), at every cutoff beyond the constructed key. The regional upper cap puts the chain's height below that pair as well (Definition 11). At one direction the plaquettes' Gram reads 2 joined to the path's differences' read at the ends' sum's unit, the two unshared links' count joined to the shared links' difference read, at or beyond 2 at every side, so every mode's frequency sits at or beyond the adjoined root at $4\kappa_\Sigma$ and the lines' gap keeps that floor at every side. $\square$

Lemma 56 (The corner disconjugacy certificate). *On a depth chain's decimated head at the dual representative (Construction 21; Theorem 19(i)) whose shells are single labels at unit bonds, the band-one walk: the adjoint list's one-plaquette sector at the residue one and that sector's compression to the tower labels at every further residue (Construction 15; Proposition 13), at a scale pair η and a probe level cross-added in the diagonal (Theorem 15's spelling), the count below the probe is the pivot sequence's lower-side count (Lemma 30), and the certificate settles the read at most one at every tail scale, its displayed comparisons at the corner scale on the depth axis: the entries are polynomials in the scale and the product $m\eta$ (the corner scale's multiples, a pair read each), and every comparison below is a two-variable polynomial read of those data.*

(i) *The pivot comparison. At a positive list v on a segment with the termwise read*

$$v_m + v_{m+2} \leq a_{m+1} v_{m+1}$$

and the entry read $p_{m_0} v_{m_0} \geq v_{m_0+1}$, every later pivot of the segment clears its entry read, $p_m v_m \geq v_{m+1}$.

(ii) The segment supersolutions are the well's own series. The well's data enter at the square scale: the probe line at a pair rate q is the level λ at the join $\lambda + c_1 + 2 = q\eta$, the walk's band top $c_1 + 2$ cross-added (Theorem 12's crossing joins), the line at the rate q , the level $\langle q\eta : c_1 + 2 \rangle$ (Theorem 15's balance-pair levels) written $2 + q\eta$ at the diagonal's spelling below, so the head's diagonal at the dual pencil, σE_m against the walk's diagonal c_1 and the level at unit bonds (Proposition 13; Lemma 44), reads $a_m = 2 + \eta W$ at the well read, the balance pair

$$W = \langle [4 : r + 1] (s + r u) : q \rangle \quad \text{at } u = m\eta, \quad s = m^2\eta,$$

its first member the tower's own electric read, $[4 : r + 1] (s + r u) = 4\eta \hat{c}_2(m\theta)$ at the substitution (Proposition 12), the two pair reads joined at the relation $u^2 = \eta s$, and the boundary's diagonal a_0 at the join $a_0 + q\eta = c_1 + 2$, the vacuum's diagonal the Kronecker delta at two labels (Proposition 13), the band read 2 the constant. A segment's list enters as a polynomial V in the scale and the depth's two pair reads, $v_m = V(\eta, u, s)$ read per depth, at u -degree at most one, a product's u -square re-entering the scale-free slab at the relation; the neighbor fold has one scale factor, $v_m + v_{m+2} = 2v_{m+1} + \eta D[V]$ at the displayed action

$$m^j + (m+2)^j = 2(m+1)^j + 2 \sum_{i \geq 1, 2i \leq j} \binom{j}{2i} (m+1)^{j-2i}$$

read per monomial of the pair coordinates, so (i)'s comparison is the polynomial read $D[V] \leq W V$, and a profile is a supersolution on a segment exactly where the remainder $W V - D[V]$ and the profile read positive there, two polynomial reads in the depth and the scale, each a sign read on the segment's region at the graded device (the certificate's data below). A profile is produced data, certificate data entered with the reads deciding it (Theorem 15's emitted designations, every produced datum below so), a polynomial in the scale and the two pair reads at u -degree at most one at a produced degree; the well's own series produce one at every rate: at a pair rate q' the scaled recurrence is the pair of coefficient lists

$$\begin{aligned} f_0 &= 1, & f_1 &= \langle 1 : 1 \rangle [q' : 2], & 2k(2k-1)f_k + q'f_{k-1} &= [4 : r + 1]f_{k-2}, \\ g_0 &= 1, & g_1 &= \langle 1 : 1 \rangle [q' : 6], & 2k(2k+1)g_k + q'g_{k-1} &= [4 : r + 1]g_{k-2}, \end{aligned}$$

balance pairs over the scalar pairs, each key the join's cofactor read at the equal-membered class as the sum's unit (Definition 1); the mixing pair is

$$\mu := [2(c_1 + 1) : 4 - c_1], \quad \nu := \langle c_1 : 2 \rangle [1 : 4 - c_1],$$

$(3, \langle 1 : 1 \rangle)$ at the base two and $([4 : 3], \langle 1 : 2 \rangle [1 : 3])$ at the base one; and at the degree N the series' profile at the rate is the series at the scale's clearing,

$$V_{q'}(\eta, u, s) := \eta \sum_{k \leq N} f_k s^k + (\mu u + \nu \eta) \sum_{k \leq N} g_k s^k, \quad \eta v_m = V_{q'}(\eta, m\eta, m^2\eta),$$

the second series at the depth's multiple through u against the scale joined to its own read, the list's reads at the scale's clearing (the termwise comparison and the entry read one value at the common factor); the square scale's action reads each series' key to the key below at the rate's well read, $4sF'' + 2F'$ against $[4 : r + 1]sF$ less $q'F$ at the first list's recursion and $4sG'' + 6G'$ against $[4 : r + 1]sG$ less $q'G$ at the second's, term by term, and the mixing pair is the boundary row's

read at its two leading keys, the profile extended to the vacuum depth at $v_0 = 1 + \nu$ pairing the boundary's diagonal to the depth one, $a_0 v_0 = v_1$, at the scale-free key and at the first key at the line's own rate. The coefficients' magnitudes fold under C^k against the factorial at the natural C the division's quotient of the rate's first datum at twice its second at its successor, so at a degree at or beyond $18C s_{q'}$ at the rate's square scale $s_{q'} := [q'(r+1) : 4]$ (a natural at the witness the division's quotient of $18C q'_n(r+1)$ at $4q'_d$ at its successor at $q' = [q'_n : q'_d]$, Definition 1) the series' tails beyond the degree sit under $2[1 : 2]^N$ at every square scale at or below $3s_{q'}$, the produced degree at or beyond it where the series produce the profile. The two profiles are produced data at the pre rate $q_a := [101 : 100]q$ and the post rate $q_b := [103 : 100]q$ at the line's rate q , the series' own where the series produce them, the pre profile V_{q_a} and the post profile the balance partner of V_{q_b} , each at its own degree; the comparison's remainder at a profile reads the rate's gap against the line's at its leading key, $WV - D[V]$ the profile's multiple of $\langle q' : q \rangle$ joined to the electric read's depth term, to the mixing key's own term $4\nu G'$, to the degree's top keys and to the action's further keys, the depth's multiple of the second series above the mixing key's term and the top keys at the tail's fold on the box, so a profile above the line's rate is a supersolution where it reads positive, the read the two polynomial reads above. The pre root s_a and the post root s_b are the first located roots in the square scale of the two profiles' u -slabs' scale-free parts, the second lists $\sum_{k \leq N} g_k s^k$ at the pre rate and at the post rate at the series' profiles, each with its isolating bracket (Theorem 16), the post root's bracket below the pre root's, one located comparison; the seam scale s_1 is the halved sum of the post root's bracket top and the pre root's bracket bottom; and the seam M_2 is the least natural whose square scale sits at or beyond twice the post rate's square scale, $M_2^2 \eta \geq 2s_{q_b}$, read by halving at the witness the successor of the division's quotient of $2ad$ at nb at $\eta = [n : d]$ and $s_{q_b} = [a : b]$, a natural's square at or beyond itself (Definition 1). The segments are a seam-chained list on each side of the crossing, each segment a profile on a box of square scales and consecutive segments joined at a seam: an entry read crosses a seam at the crossed products' read, the outgoing profile's shift against the incoming profile joined at the swap to the incoming's shift against the outgoing, positive on the seam's box at the crossed products' scale strip; the certificate's instance is two segments, the pre segment, the pre profile at the depths from the depth one whose square scale sits at or below the seam scale, and the post segment, the post profile at the depths whose square scale sits at or beyond the seam scale, up to the seam's successor, the two boxes meeting at the seam scale and the post box's top the successor's square scale, at or below $3s_{q_b}$ at every scale under the ceiling. The pre profile's entry read at the depth one, $p_1 v_1 \geq v_2$ at the boundary pivot p_1 at the join $p_1 + [1 : a_0] = a_1$, reads its first key in the scale as $[17 : 4]\eta \langle q_a : q \rangle$ at the base two and $[1 : 9]\eta \langle 22q_a : 20q \rangle$ at the base one (the series' profiles' seeds at the first key, at the base two $v_1 + q_a \eta = 4$, $v_2 + 6q_a \eta = 7$ and $p_1 + [17 : 16]q\eta = [7 : 4]$, and at the base one $v_1 + [2 : 3]q_a \eta = 2$, $v_2 + [32 : 9]q_a \eta = [10 : 3]$ and $p_1 + [10 : 9]q\eta = [5 : 3]$), positive at the pre rate beyond the line's, one polynomial read at the scale; the post profile's growth read at the seam, $v_{M_2+1} \geq v_{M_2}$, reads the depth's difference of the profile, $v_{m+1} - v_m$ at $m = M_2$, one polynomial read at the scale, its leading key the second series' growth past the post rate's square scale, the balance partner of $\mu(G + 2sG')$ at the square scale $2s_{q_b}$; and the post segment's step cap, $v_{m+1} \leq a_m v_m$ at every depth of the segment, one polynomial read on the segment. The constant-floor instance is the walk's own list, the recurrence $U_j + U_{j+2} = g_i U_{j+1}$ at the boundary's unit seed $U_1 = 1$ read from the boundary (the truncation data of Theorem 13's recursion, integer-polynomial in the floor).

(iii) The crossing, and the rebound. The first depth whose pivot reads on the lower side, where one sits, has the following pivot beyond its diagonal, the identity at a lower-side predecessor read on the exceeding side (the rebound read), and the exceeding pivot seeds the entry read of the segment it sits in at the diagonal itself, $p_{c+1} v_{c+1} \geq a_{c+1} v_{c+1} \geq v_{c+2}$ at the step cap; the pre segment's pivots read positive at (i) from the entry read at the depth one, so the first lower-side pivot sits

past the pre segment, its square scale beyond the seam scale, and past it the post segment's pivots read positive at (i) from the rebound's entry read, the seam's pivot at or beyond the growth read, $p_{M_2} \geq 1$, where the crossing sits inside the post segment, and the rebound's own read beyond its diagonal where it sits past the seam. A depth whose minor reads equal members re-orders within its slab at Lemma 30's principal pivots of order at most two (Lemma 34), the count read unchanged across the splice: the deep end takes the order-one kernel block, and an interior depth its order-two block with the following minor read through the spliced pair.

(iv) The dominance tail. At depths whose diagonal clears 2 the constant list is the supersolution, $v_m + v_{m+2} = 2v_{m+1} \leq a_{m+1}v_{m+1}$, and the pivots stay clear to the head's end from an entry read $p_m \geq 1$; at depths whose diagonal clears 2 by a stated margin and a stated allowance, a pair, the geometric list is a supersolution as well, each member the prior member's multiple at one pair of the scale's data, the pair's gap the allowance's read of the depth step at the seam depth, the constant list the vacant allowance's instance, and the entry read is the final seam's slope read, the profile's shift joined to the allowance's multiple of the step read of the profile clearing the profile on the seam's box, the growth read its read at the vacant allowance. The certificate's tail is the constant list: the seam's square scale sits beyond the line's own, $2s_{q_b}$ against $s_q = [q(r+1) : 4]$, so every depth past the seam reads its well read on the upper side, the diagonal beyond 2, and the entry read at the seam is (iii)'s, the growth read where the crossing sits before the seam and the rebound's diagonal where it sits past it; a lower-side pivot past the seam rebounds into the constant list outright. So the count reads at most one at the line: at most one lower-side pivot before the seam by (iii), and one at most in all by the tail.

(v) The ground witness, and the lower line as the family's own read. The witness family is $x_m = s \langle s_2 : s \rangle$ at the square-scale pair $s_2 = N^2\eta$, the endpoint depth N the least natural beyond one at $40N^4\eta^2 \geq 363(r+1)$, the stepping read by halving at the witness the larger of two and the successor of the division's quotient of $363(r+1)d^2$ at $40n^2$ at the scale $\eta = [n : d]$, one trichotomy fold (Definition 1), the comparison the rate's two top terms' equality below; the family's read at the endpoint is equal members and beyond it the sum's unit. The form read telescopes at unit bonds, the boundary bond's square x_1^2 the displayed first tail term,

$$\sum_{m \geq 1} a_m x_m^2 = 2 \sum_{m \geq 1} x_m x_{m+1} + x_1^2 + \sum_{m \geq 1} \langle x_{m+1} : x_m \rangle^2 + \eta \sum_{m \geq 1} W_- x_m^2 ,$$

W_- the lower line's well read, the bond pair per depth the displayed square $(2u+\eta)^2 \langle s_2 : 2s+2u+\eta \rangle^2$, and each sum is one closed read of the depth's powers at the endpoint (the power sums' binomial identities), the closed reads' top coefficients at the endpoint's powers $[8 : 315]$ at the family's square, degree nine, $[44 : 105]$ at the difference sum, degree seven, and $[4 : r+1][8 : 693]$ at the well sum's square part, degree eleven. The lower rate is the family's own cleared read,

$$q_- \eta \sum_{m \geq 1} x_m^2 = [4 : r+1] \eta^2 \sum_{m \geq 1} (m^2 + r m) x_m^2 + \sum_{m \geq 1} \langle x_{m+1} : x_m \rangle^2 + x_1^2 ,$$

one pair per scale and residue, the cofactor at the closed reads (Definition 1), its two top terms $[33 : 2]$ at the cofactor against ηN^2 and $[20 : 11(r+1)]$ at the factor ηN^2 , the tops' cofactor reads, whose equality is the key's comparison, $40N^4\eta^2 = 363(r+1)$: at the line $2 + q_- \eta$ the form read's two sides read equal members at the family, a positive-definite read excluded, so the least root sits at or below the line (Lemma 30; Lemma 33), the family its own witness. The lower rate's square at the residue's successor is capped at 121: the family's sums are η^4 times the depth's power sums at the endpoint, $x_m = \eta^2 m^2 (N^2 - m^2)$, so $q_- = [4 : r+1] \eta A + [B : \eta]$ at the polynomial pairs $A := (\Sigma_2 + r \Sigma_1) [1 : \Sigma_0]$ and $B := (\Sigma_d + (N^2 - 1)^2) [1 : \Sigma_0]$ in the endpoint and the residue over the

scale-free power sums $\Sigma_0 = \sum_m m^4(N^2 - m^2)^2$, $\Sigma_1 = \sum_m m^5(N^2 - m^2)^2$, $\Sigma_2 = \sum_m m^6(N^2 - m^2)^2$, the depths from one to the endpoint, and $\Sigma_d = \sum_m ((m+1)^2(N^2 - (m+1)^2) - m^2(N^2 - m^2))^2$ over the depths from one to the endpoint's predecessor, and the scale's square is priced by the key's bracket $40(N-1)^4\eta^2 < 363(r+1) \leq 40N^4\eta^2$, so

$$q_-^2(r+1) \leq [16 \cdot 363 A^2 : 40(N-1)^4] + 8AB + [40 : 363]N^4B^2 \leq 121$$

at every endpoint at or beyond 1257 and every residue at $1257(r+1) \leq N$, the cleared comparison one polynomial in the endpoint at the residue read at its cap N against 1257 (A increasing in the residue), whose display at $N = 1257 + t$ reads a natural coefficient at every key of t . The second root prices from above by the pair read at the witness's span: the second family is the square scale's multiple of the witness, $x'_m := s x_m$, the span of the two the witness's shape at one further degree, and the span's compressed pencil at a line $2 + q\eta$ is one 2×2 read whose entries are power-sum reads of the two lists at the endpoint, closed reads as (v) 's own, its site datum's determinant a quadratic in the rate with two located roots (Lemma 33 at the compressed gram, positive definite at the two lists' independence); the cap rate q_+ is the larger root, one located pair per scale and residue, and a compression's counts sit at or below the full form's (Lemma 30), so the compressed count of two at every line beyond the cap places two chain roots below it: the second root at or below the cap line $2 + q_+\eta$, the heights two-sided on the cell, the compressed count one value at every spanning list of the span (Lemma 30's congruence).

(vi) The probe lines, and the floor. The lower rate is the witness family's own read $((v))$, and the line rate at a residue is the produced pair at the residue's scale key s_r , the least natural at $s_r^2 \geq 100(r+1)$, the stepping read by halving from one at the witness $10(r+1)$ (Definition 1), so $10\sqrt{r+1} \leq s_r < 10\sqrt{r+1} + 1$ in the dictionary's reading, the squares' reads $100(r+1) \leq s_r^2$ and $(s_r - 1)^2 < 100(r+1)$ here: the line rate is $q_L := [125 : s_r]$ at the residues one and two, $[25 : 3]$ and $[125 : 18]$, and $q_L := [4325 : 36s_r]$, the transported rate, at every residue at or beyond three. The count reads at most one at the line $2 + q_L\eta$ at every tail scale, the certificate of (ii)–(iv) at the residues one and two, each at its own base, and at a residue r at or beyond three the residue-two certificate transported: at the second scale $\eta_2 := [173 : 10s_r]\eta$ the tie $(r+1)\eta_2^2 \leq 3\eta^2$ holds at the squares' read, the two electric members compare depth by depth,

$$[4 : r+1]\eta^2(m^2 + rm) \geq [4 : 3]\eta_2^2(m^2 + 2m),$$

the tie at the squares against $rm \geq 2m$, the rates' tie $q_L\eta = [125 : 18]\eta_2$ is the transported rate's own display, and the two boundaries' diagonals read one value at the base two, so the residue- r diagonal $2 + \eta W$ sits at or beyond the residue-two diagonal at the scale η_2 and the residue-two line at every depth, the two bonds one value, and the residue- r count at its line sits at or below the residue-two count at its line and the second scale, at most one, the reversal count monotone at a diagonal datum on its upper side (Lemma 30); the second scale sits at or below the scale, $173 \leq 10s_r$ at every residue at or beyond two, so it sits under the residue-two ceiling wherever the scale does, the residue- r ceiling's first clause below. The floor is the line's gap against the lower rate, o at $q_- + o = q_L$, positive at every residue and tail scale by the squares' read at the residue's successor: $q_L^2(r+1)$ reads $[1250 : 9]$ at the residue one, $[15625 : 108]$ at the residue two, and at or beyond three $[4325^2(r+1) : 36^2s_r^2]$, each above 121 at the scale key's squares' read $(s_r - 1)^2 < 100(r+1)$, against the lower rate's square at or below 121 $((v))$, and two pairs at ordered squares are ordered (the trichotomy of squares, Definition 1); the ground sits at or below the lower line by (v) and the count reads at most one at the line, so the counting function is flat from the lower line to the line, the gap clears $o\eta$ at every tail scale and the height's square clears o^2 , one cross-multiplication per scale, the floor per member and direction count. The residue direction reads at the two certificates,

the residue- r read the transported comparison at its line. The member's tail scale is the ceiling, the least halving of one under which the certificate's reads hold, the scale sits at or below $[s_{q_b} : 128]$ at the post rate's square scale of the residue's own certificate, $[2575 : 600]$ at the residue one and $[38625 : 7200]$ at the residue two and at every residue at or beyond three, and the endpoint key clears $1257(r+1)$, the certificate's reads at the residues one and two their own and at a residue at or beyond three the residue-two reads at the second scale, the second scale at or below the scale $(173 \leq 10s_r)$ so that the residue-two scale clause reads at the scale itself: the reads are polynomial comparisons in the scale on the segments' boxes, each read at its device at a cleared ceiling, a produced pair (c_0, d_0) with the device's reads deciding the comparison at every scale at or below $[c_0 : d_0]$ (the devices below), holding from the halving index d_0 (the power 2^i beyond i at every natural), the scale's bound holds from the halving index $7+b$ at $s_{q_b} = [a : b]$, and the key clears $1257(r+1)$ at every scale at $40 \cdot 1257^4(r+1)^4\eta^2 \leq 363(r+1)$, holding from the halving index $40 \cdot 1257^4(r+1)^3$ (the power 4^i beyond i), each a comparison stepping down the halvings, so the ceiling is the stepping read by halving at the witness the three indices displayed joined (Definition 1).

The certificate's data are the line rate at the residue's scale key with every profile comparison read at it and the scale's ceiling with every tail scale read under it, each datum produced at its residue ((vi)), the count's line read at every rate at or below the line's, the lower line among them, the transfer the diagonal's monotonicity in the rate, the rate entering the well read on its pair's second member, one cross-multiplied comparison per depth, the segment list with each boundary the cross-multiplied pair comparison of $m^2\eta$ against the end's pair and with consecutive boxes overlapping at the two further depths the termwise comparison reads, a depth inside one box's top stepping to a square scale inside the next box's, at the instance the two segments at their produced ends, the depth one, the seam scale and the seam at the doubled square scale ((ii)), the profiles at their produced degrees, the two roots' brackets at their order, the boundary seed's data, the three clearings of the boundary value, of the boundary comparison's object, the shifted diagonal's multiple of the profile against the rate's multiple of the shifted profile, and of the termwise comparison's scale strip, each floored past its origin by the ceiling's priced fold, the deep coefficients' fold against the ceiling's second member's powers at the produced degree, the floor at one power beyond the fold's and the squared floors' square at one beyond twice it, the origins vacant at the graded device's scale-free vacancy (the depth read's origin the profile's own, the comparison's origins the band clearing's multiples of vacant depth-read origins, the termwise comparison's origin vacant at every profile and rate, the two origin slabs' withdrawal, and the strip's origin the affine depth reads' second difference, the depth read's scale coefficient at a vacant slab the first inner's own joined to the depth's multiple of the u -slab's origin), the slope allowance with the dominance margin and the tail's three scale-free comparisons, the margin located at the final device's floor, one cross-multiplied read, the margin against the allowance's squared read at the geometric tail's handover cap, and the gap's half cap at the ceiling, the allowance vacant at the instance with the growth read the seam's slope read, and per segment and seam the devices' finitely many polynomial comparisons in the scale and the scaled depth on stated boxes, the profile's positivity, the termwise comparison, the crossed products at a seam, and on the post segment the step cap with the seam's growth read ((ii)), each one polynomial in the depth and the scale read at the graded device below, and the lower rate, the cap rate and the witness pair, (v)'s derived reads at the endpoint key (Theorem 16; Lemma 31). The devices. A comparison on a segment is one polynomial P in the scale and the two pair reads at u -degree at most one, read at every natural depth of the segment at every scale under the ceiling, its positivity a read on the segment's box of square scales at the scale's cleared ceiling, two naturals c_0, d_0 with the scale's bound $\eta d_0 \leq c_0$ read cross-multiplied. A box device prices a comparison occupied at the scale-free order from below on its box at the cleared ceiling, its data the composite endpoints with the cut lists, the priced side reads' bound, the box top's cleared read, the floors and the shape

clearing; and the graded device prices a comparison vacant there at the scale order, its data the composite endpoints with the one cut list, the pieces' stated caps, the tail's handover key and the shape clearing: each device reads the comparison strictly above the sum's unit at every depth whose square scale sits inside the box, the derivation below. The well's oscillation reads at the corner scale, one read at every scale, so one segment list runs the whole tail, and the one certificate settles the count at most one at every depth and every tail scale at once: the entry reads chain across the segments to the crossing seam, the crossing seam the two chains' break at the count comparison alone with the rebound seeding the later chain's entry reads, the first lower-side pivot past it with the later boxes' reads running from below it, and the count read closing beyond it at (iii)'s rebound seed and (iv)'s dominance tail at the final seam's own read, the slope read at the instance's vacant allowance the post profile's growth read at the seam, the flat window's lower end the ground bracket's own (Theorem 10).

Proof. (i) The pivot identity $p_{m+1} p_m + 1 = a_{m+1} p_m$ with the two reads $a_{m+1} v_{m+1} \geq v_m + v_{m+2}$ and $p_m v_m \geq v_{m+1}$ drives the induction, one cross-multiplied step per depth: $p_{m+1} v_{m+1}$ joined to $v_{m+1} [1 : p_m]$ reads $a_{m+1} v_{m+1}$, at or beyond $v_m + v_{m+2}$, and $v_{m+1} [1 : p_m]$ sits at or below v_m at the entry read, the positive pivot withdrawing at the product's injectivity.

(ii) *The diagonal and the boundary.* The head's diagonal at the dual pencil is σE_m against the walk's diagonal c_1 and the level at unit bonds (Proposition 13; Lemma 44), $\sigma E_m + c_1 + 2$ against $c_1 + q\eta$ at the line's join, the band top $c_1 + 2$ withdrawing to the band read 2 joined to ηW at $\sigma = \eta^2$ and $E_m = 4\hat{c}_2(m\theta) = [4 : r + 1] (m^2 + r m)$ (Proposition 12), and at the vacuum depth the walk's diagonal is the Kronecker delta at two labels, the sum's unit, so the boundary's diagonal is $c_1 + 2$ against $q\eta$. *The action.* The neighbor fold of a monomial m^j is the displayed binomial identity, the two neighbors' powers collected at $(m+1)^j$ with the odd keys withdrawing in pairs, so the fold of V reads $2v_{m+1}$ joined to the scale's multiple of the collected further keys, one scale factor at every summand at the pair reads $u = m\eta$ and $s = m^2\eta$ (u moved by η and s by $(2m+1)\eta$, the pair reads' moves each one scale factor). *The recurrence.* At the leading key in the scale the action reads a series $F(s)$ at $4sF'' + 2F'$, the two neighbors' squares $(m \pm 1)^2\eta$ moving s by $\pm 2u + \eta$ with the second-order term at $(2u)^2 = 4\eta s$, a series at the depth's multiple $mG(s)$ at $m(4sG'' + 6G')$, the depth factor's own move joining $4uG'$ to the neighbors' fold, and a series $G(s)$ at $4sG'' + 2G'$; the well read at the rate q' reads W against the profile as $([4 : r + 1] s - q') V$ joined to the electric read's depth term $[4 : r + 1] r u V$ and the line's gap $\langle q' : q \rangle V$, so the two series at the displayed recurrences read the action to the well read key by key: the coefficient of s^{k-1} in $4sF'' + 2F' + q'F$ is $2k(2k-1)f_k + q'f_{k-1}$ against $[4 : r + 1] f_{k-2}$, and in $4sG'' + 6G' + q'G$ it is $2k(2k+1)g_k + q'g_{k-1}$ against $[4 : r + 1] g_{k-2}$, the seeds at $k = 1$. *The mixing pair.* The profile extended to the vacuum depth reads $v_0 = 1 + \nu$, and the boundary row pairs it to the depth one, $a_0 v_0 = v_1$, at $v_1 = 1 + \mu + \nu$ joined to $(f_1 + (\mu + \nu)g_1)\eta$ at the first key: the scale-free key reads $(c_1 + 2)(1 + \nu) = 1 + \mu + \nu$, so $\mu = (c_1 + 1)(1 + \nu)$, and the first key at the line's own rate reads $q(1 + \nu)$ against $q[3 + \mu + \nu : 6]$, the join $6(1 + \nu) = 3 + \mu + \nu$; the two joins solve at the sum's injectivity to $(4 - c_1)\nu = \langle c_1 : 2 \rangle$ and $(4 - c_1)\mu = 2(c_1 + 1)$, the displayed pair. *The magnitude fold.* At $|f_k| \leq C^k [1 : k!]$ and $|f_{k-1}|, |f_{k-2}|$ so, the recursion reads $2k(2k-1)|f_k|$ at or below $q' C^{k-1} [1 : (k-1)!] + [4 : r + 1] C^{k-2} [1 : (k-2)!]$, against $2k(2k-1) C^k [1 : k!]$ the two summands read $q' k$ against C and $[4 : r + 1] k(k-1)$ against C^2 , each at or below $k(2k-1)$, one half of the target, at $q' \leq 2C$ (the quotient's successor) and $[4 : r + 1](k-1) \leq (2k-1)$ at $[4 : r + 1] \leq 2$; the seeds read 1 and $q'[1 : 2] \leq C$, and the second list's recursion folds so at the larger divisors. *The tail.* At a square scale $s \leq 3s_{q'}$ a key beyond the degree reads $|f_k| s^k \leq (3C s_{q'})^k [1 : k!]$, and $k! \geq (k[1 : 3])^k$ at the product's fold against the cube's puts it at or below $(9C s_{q'} [1 : k])^k \leq [1 : 2]^k$ at $k \geq 18C s_{q'}$, the tail the geometric fold under $2[1 : 2]^N$. *The remainder's leading key.* The action at the profile reads the well read at

the rate q' joined to the depth-free series' second key, $D[V]$ against $([4 : r + 1] s - q') V$ less $4 \nu G'$ at the leading key, the ν -term's action $4s G'' + 2G'$ reading the second list's recursion less $4G'$, so the remainder $WV - D[V]$ reads $\langle q' : q \rangle V$ joined to $[4 : r + 1] r u V$, to $4 \nu G'$, to the degree's top keys (the recurrence identities read at every key below the degree, the two top keys reading $q' f_N s^N$ joined to $[4 : r + 1] (f_{N-1} s^N + f_N s^{N+1})$ at the first list and so at the second, each at or below the tail's fold on the box) and to the action's further keys, the further keys at the scale and beyond against the depth's powers, at or below the scale's half power against the leading key on the segment's region; the leading key $\langle q' : q \rangle V$ is the depth's multiple of the second series at the rates' gap, at or beyond the depth's multiple of the second series' read at the box's ends, while the mixing key's term $4 \nu G'$ is depth-free and the top keys sit under $2[1 : 2]^N$, so the depth's multiple clears both under the ceiling: a profile positive on its box at a rate beyond the line's is a supersolution there under the ceiling, the read the cell read. *The seeds.* At the base two, $v_1 = 4 - q' \eta$ and $v_2 = 7 - 6 q' \eta$ at the first key ($f_1 + 3g_1$ reading q' 's balance partner and $4f_1 + 24g_1$ the partner of $6 q'$), $a_0 = 4 - q \eta$, $a_1 = 2 - q \eta$ at the first key, and $p_1 = a_1 - [1 : a_0]$ reads $[7 : 4]$ less $[17 : 16] q \eta$ ($[1 : 4 - q \eta]$ at $[1 : 4] + [q \eta : 16]$ at the first key), so $p_1 v_1 - v_2$ reads $[17 : 4] \eta \langle q_a : q \rangle$ at the pre rate; at the base one, $v_0 = [2 : 3]$, $v_1 = 2 - [2 : 3] q' \eta$ and $v_2 = [10 : 3] - [32 : 9] q' \eta$ ($f_1 + g_1$ and $4f_1 + [28 : 3] g_1$ at the mixing pair ($[4 : 3], \langle 1 : 2 \rangle [1 : 3]$)), $a_0 = 3 - q \eta$, $p_1 = [5 : 3] - [10 : 9] q \eta$, so $p_1 v_1 - v_2$ reads $[1 : 9] \eta \langle 22 q_a : 20 q \rangle$, positive at every pre rate at or beyond the line's. *The roots and the seam.* The second list at a higher rate reads its first root below the lower rate's, the located comparison of the two brackets, one read; the seam scale sits between the two brackets, so the pre profile's second series reads positive on the pre box and the post profile's on the post box, and each profile's positivity on its box is the depth's multiple of that read joined to the first series' read, positive at every depth under the ceiling, the graded device's read; and the post box's top sits at or below $3 s_{q_b}$: the seam's predecessor's square scale sits below $2 s_{q_b}$, the successor's is the predecessor's joined to $4(M_2 - 1) \eta + 4 \eta$, and at the scale under $[s_{q_b} : 128]$ the squares' read $(M_2 - 1)^2 \eta^2 < 2 s_{q_b} \eta \leq [s_{q_b}^2 : 64]$ puts $4(M_2 - 1) \eta$ under $[s_{q_b} : 2]$. *The growth read.* The depth's difference of the post profile at the seam is the action's first key, the balance partner of $\mu(G + 2 s G')$ at $s = 2 s_{q_b}$ joined to the scale's half power's multiples, the second series past the post rate's square scale reading its growth on the upper side there, the read the graded device's.

The box device. The depth's read splits as the scale-free part, the scale-free slab's origin coefficients at the square scale, joined to the scaled remainder, and the remainder's cap follows the profile's grading one order down: its origin coefficients, the scale-free slab's scale coefficients with the u -slab's origins, read on the box at the chained priced side read, the scale coefficients above their floor's balance partner and the u -slab's origins at their squared price with the square scale the box's own coordinate, each comparison at its own cut list and its floor at or beyond the sum's unit, while its deeper coefficients price at the ceiling's fold through the box top's cleared read into the floors, the fold's power the produced degree's successor at the depth shift's second clearing read once more at the scale's bound, the floor at one power beyond the fold's and the squared floors' square at one beyond twice it, the squared fold's floor at or beyond the sum's unit, so the read clears the sum's unit wherever the scale-free part clears the floors' join at the chained priced side read with the remainder inside it.

The graded device. The split follows the profile's grading: a profile occupied at the scale-free order prices there, and a profile vacant there prices at the scale order, the depth's leading part the depth multiple of the u -slab's origin reads joined to the scale-free slab's scale-key reads, one read per depth and linear in it. The depth family collects at one chain: every depth's square clears its square scale at the ceiling's cross-multiplied bound, and at origin reads above the sum's unit on the box a deeper multiple only raises the read, so the family's clearing of the deeper coefficients' caps reduces to one line, the scale-key read's above the caps' join; piece by piece along the cut list

the piece keeps the line's upper side with the origin reads' floor at or beyond the sum's unit, or the squared comparison's, the origin reads' floor beyond the sum's unit and the square scale's multiple of the floor's square at the ceiling's cross-multiplied read above the square of the scale-key read's cap joined to the deeper coefficients' caps, whose read returns the depth multiple past the caps at the trichotomy of squares, every value priced across the piece at the piece's own endpoint: the binomial identity per occupied key (Theorem 16) rewrites each slab's scale polynomial at the endpoint, the rewrite cleared at the endpoint's second datum's powers at the top key's count, one value at every representative (Definition 1's homogeneity principle), the rewritten coefficients exact cleared reads of the produced profile, a floor the endpoint's read less the deeper rewritten coefficients' magnitude fold at the piece's width and a cap their join, each at the stated clearing, the deeper coefficients capped at the ceiling's powers and joined on the line's side, below the tail's stated handover key each column's cap the piece rewrite's own and at or beyond it the column's magnitude fold against the box top's powers, the u -slab's deeper caps at the piece's stated cap's multiple, the stated cap a natural whose square scale sits at or beyond the piece's top, the depth's u -read at or below the cap's ceiling multiple at the relation $u^2 = \eta s$, the square scale inside the piece with the scale under the ceiling, and the comparison one cross-multiplied read at the priced values with the square scale read at the piece's floor, the piece's every point reading the comparison at the ordered squares.

The seams. The crossed products are vacant at the scale-free order and at the scale order's leading part, the shift acting at the leading order as the products' own exchange, so the strip's withdrawal is the crossed read's whole object, and the products' injectivity reads the crossed entry back at positive profile values.

(iii) The rebound is the pivot identity at a lower-side predecessor, read on the exceeding side: $p_{c+1} p_c + 1 = a_{c+1} p_c$ at p_c on the lower side reads p_{c+1} beyond a_{c+1} , the product's sign at the unit joined.

(iv) The constant list's neighbor sum is 2 against the diagonal at or beyond 2, the termwise read; at the entry read $p_m \geq 1$ the induction of (i) runs at the constant list, so every later pivot sits at or beyond one. At the geometric list the neighbor sum joins the arriving member's double at the gap's square against the withdrawn member, so the margin's clearing of the squared gap's doubled price, the gap at most the pair's half, reads the termwise comparison at every deeper depth, the dominance comparison's own growth, and the slope read seeds (i) at the seam's own profile value.

(v) The form read telescopes at unit bonds, the boundary bond's square x_1^2 the displayed first tail term: $\sum_m a_m x_m^2$ against $2 \sum_m x_m x_{m+1}$ reads $\sum_m (2 + \eta W_-) x_m^2$ against $2 \sum_m x_m x_{m+1}$, the squares' fold $\sum_m \langle x_{m+1} : x_m \rangle^2$ collecting $2 \sum_m x_m^2$ against $2 \sum_m x_m x_{m+1}$ with the boundary's square at the vacant depth. Each sum is one closed read of the depth's powers at the endpoint, the power sums' binomial identities, the top coefficients the integrals' reads $\int_0^1 t^4 (1 - t^2)^2$ at $[8 : 315]$, $\int_0^1 (2t - 4t^3)^2$ at $[44 : 105]$ and $\int_0^1 t^6 (1 - t^2)^2$ at $[8 : 693]$, the polynomial reads of the sums at their top keys; the lower rate's two top terms are $[33 : 2]$ at the cofactor against ηN^2 , the difference sum's top against the family's square's, and $[20 : 11(r + 1)]$ at the factor ηN^2 , the well sum's top against the family's square's, whose equality is the key's comparison, $40 N^4 \eta^2 = 363(r + 1)$: at the line $2 + q_- \eta$ the form read's two sides read equal members at the family, a positive-definite read excluded, so the least root sits at or below the line (Lemma 30; Lemma 33). The cap 121: the three summands of the displayed comparison are each increasing in A and in B , A increasing in the residue at its coefficient $\Sigma_1 [1 : \Sigma_0]$, so the residue reads at its cap N against 1257 in A , and the cleared comparison is one polynomial in the endpoint whose display at $N = 1257 + t$ reads a natural coefficient at every key of t , the comparison at every endpoint at or beyond 1257. The cap rate: a compression's counts sit at or below the full form's (Lemma 30), so the compressed count of two at the cap line places two chain roots below it.

(vi) The counting function is monotone in its level (Lemma 30), and the ground sits at or below the lower line by (v), so the counting function is flat from the lower line to the line at the count at most one there. The two electric members compare depth by depth at the tie: $[4 : r + 1] \eta^2 (m^2 + r m)$ sits at or beyond $[4 : r + 1] \eta^2 (m^2 + 2 m)$ at $r m \geq 2 m$, and at or beyond $[4 : 3] \eta_2^2 (m^2 + 2 m)$ at the tie $(r + 1) \eta_2^2 \leq 3 \eta^2$; the tie's squares' read is $173^2 (r + 1) = 29929 (r + 1)$ against $300 s_r^2 \geq 30000 (r + 1)$; the rates' tie is the display $4325 \cdot 180 = 125 \cdot 173 \cdot 36$, both 778500; and the boundaries read 4 against $q_L \eta$ and against $[125 : 18] \eta_2$, one value at the rates' tie. The floor's positivity at the residue three or beyond: the scale key's squares' read $(s_r - 1)^2 < 100 (r + 1)$ puts s_r^2 below $100 (r + 1) + 2 s_r$, and $s_r < 6 (r + 1)$ at $(6 (r + 1) - 1)^2 \geq 100 (r + 1)$ from the residue three, so $121 \cdot 36^2 s_r^2$ sits below $156816 (112 (r + 1) + 1)$, below $18705625 (r + 1) = 4325^2 (r + 1)$ at every residue, the cross-multiplied read of $[4325^2 (r + 1) : 36^2 s_r^2]$ above 121. The ceiling: the certificate's reads are polynomial comparisons in the scale on the segments' boxes, each holding at every scale at or below its device's cleared ceiling, a produced pair, and the scale's bound and the key's comparison hold at every scale under their displayed halvings, so the least halving under the three is the stepping read at the witness the three indices joined. $\square$

Lemma 57 (The block chain's certificate). *On a decimated head's block chain (Theorem 19(i); Lemma 34) whose shells beyond the collar are blocks of one stated order at every depth, the depth chain's shells at a member and list whose fusion closure of the base holds one shell shape beyond the collar, the band-one walk's single labels the order one, the corner disconjugacy certificate runs at block data, the band-one clauses (Lemma 56) its scalar instance: the count splits over the depth pivots (Lemma 34(ii)), the recursion's seed the boundary block with every later pivot one depth deeper, the band-one walk's own march; a segment's pivots stay positive at the ball certificate of Lemma 35(iii) read at per-depth stated centers, one matrix polynomial in the scale and the depth's pair reads, the three counts cleared by the centers' determinants to polynomial comparisons, each count's site datum at the dominance comparison (Theorem 19(ii)) with the two-sided count's at its two arms, one box read per row at the diagonal's entry less the off-row magnitude fold, the magnitudes the coefficients' own (Theorem 16), the comparisons read as cell functions over the corner base (Lemma 31), the center defect the drift's slot, per bond the defect the shallower center's transfer image at the deeper slab against the deeper slab's own center; the crossing, where the chain holds one, is one block's split at one lower-side unit with its witness list certificate data (Lemma 30), the split entering with its lower unit at an order-one block, a mixed block's two units parted at the congruence of its two side witnesses (Lemma 30), and the rebound is the split's rank-one read: the positive part is the split's block diagonal at the lower unit's entry withdrawn to the vacant read, the transfer is that datum's form at the recursion witness's image read in the split's own coordinates at the congruence witness's solve, and the following pivot sits at or beyond its diagonal less the transfer; the dominance tail reads at block diagonal dominance: past the crossing a stated positive floor list prices every deeper pivot's form at or beyond its floor pair's multiple of the coordinates' fold of squares, the seed floor at the crossing's following depth read off the rebound at one cross-multiplied comparison, and at a chain without the crossing at the walked segment's last depth read off its ball, the center's floor pair less the deviation's cap, the diagonal's floor at or beyond the seed floor joined to the transfer's cap, the diagonal's floor a stated pair pricing the depth's diagonal from below at that fold and the transfer's cap a stated pair pricing the transfer's form at it, and each deeper depth one cross-multiplied comparison, the arriving floor's multiple of the diagonal's floor at or beyond its multiple of the depth's floor joined to the bond's square cap, a stated pair pricing the bond's square's form at the fold, every deeper pivot clearing its floor; the certificate's two lines are the lower line at the lifted ground witness's rate and the line at the count's read, their gap the block chain's own produced pair, the band-one clauses' o its instance (Lemma 56(vi)); and*

the ground witness and the cap pair read lift through the compression unchanged (Lemma 56(v); Lemma 30). The collar's finitely many blocks read exactly, the bonds enter through their squares alone, and the certificate's data gain the center family with its pairs, the crossing split, and the dominance tail's pairs: per depth the center with its floor pair, per bond the contraction, the defect and the modulus pairs, the one pair ρ the segment's steps share, the self-map's own read, the seed's ball at the boundary block, the block's deviation from the boundary depth's center two-sided at ρ 's multiple of the gram, its two splits the statement's witnesses (Lemma 30), and past the crossing per depth the diagonal's floor pair with the depth's floor pair, per bond the square's cap pair, and the transfer's one cap pair. The chain's count closes at the certificate: the count splits over the depth pivots (Lemma 34(ii)), the walked segment's stated splits read vacant at the balls, the crossing's split reads its one lower unit where the chain holds the crossing, the dominance tail's floors read every deeper split vacant, and every split of the assembled datum reads the count at the crossings' count, one at a chain with the crossing and the sum's unit at a chain without, the vacuum sector's chain at the unit fiber with the crossing and at a further fiber without.

Proof. The handover. The defect is Lemma 35(iii)'s display at the depth's stated data, so each step hands the ball from the shallower center to the deeper outright and the centers' consecutive drift enters the defect as its own summand.

The rebound. The following pivot's read is the withdrawn term's display (Lemma 34(i)) with the congruence's form transport the one congruence identity and the lower entry's read at or below the vacant read's at every coordinate square the one comparison.

The dominance tail. The withdrawn term's form at the recursion witness's image (Lemma 34(i)) is the image's dot against the bond's transpose image, its square at or below the two images' folds of squares' product (Construction 8's displayed comparison), the arriving pivot's floor reads the image's fold into the term with the bond's square cap pricing the transpose image's, so the arriving floor's multiple of the term sits at or below the bond's square cap's read, the term's occupied factor withdrawing at the trichotomy with the vacant term's read outright, and the depth's pivot clears its floor, Lemma 56(i)'s comparison at the block reads. $\square$

Lemma 58 (The fold at a positive list). *Let Z be a symmetric matrix of pair data over a finite index whose off-diagonal entries sit on their lower side or read equal members, u a positive list over the index, and, at every list x, y the list at the cofactors $u_i y_i = x_i$ (Definition 1). Then*

$$x^\top Z x = \sum_i [(Zu)_i : u_i] x_i^2 + \sum_{i < j} \check{Z}_{ij} u_i u_j (y_i - y_j)^2 ,$$

$\check{Z}_{ij}$ the balance partner of the entry, the fold identity, one collected display at every x . At a family $\bar{w}$ of pairs over the index with $(Zu)_i + \bar{w}_i u_i$ on its upper side or of equal members at every i , the defect cap,

$$x^\top Z x \geq \sum_{i < j} \check{Z}_{ij} u_i u_j (y_i - y_j)^2 - \sum_i \bar{w}_i u_i^2 y_i^2 ,$$

and on the hyperplane $\sum_i \bar{w}_i u_i^2 y_i = 0$ the last fold is the variance $\sum_i \bar{w}_i u_i^2 (y_i - \bar{y})^2$ at the mean $\bar{y}$, the cofactor at $\bar{y} \sum_i \bar{w}_i u_i^2 = \sum_i \bar{w}_i u_i^2 y_i$; so where the Poincaré read

$$\sum_i \bar{w}_i u_i^2 (y_i - \bar{y})^2 \leq \sum_{i < j} \check{Z}_{ij} u_i u_j (y_i - y_j)^2$$

holds at every list y , Z is positive semidefinite on the hyperplane and its reversal count is at most one (Lemma 30), and at a cap whose members all read the sum's unit Z is positive semidefinite

outright. At Z the site datum of a pencil at a line over a chain's index at a cutoff (Lemma 44; Theorem 17), its off-diagonal entries at the chain's moves alone, and u positive at every label of the chain, a label's read $(Zu)_i$ at a cutoff sits at or beyond its read at the chain's whole move list, so a defect cap read at the whole list holds at every cutoff holding the label.

Proof. The second fold's summand collects to $\check{Z}_{ij}(u_j x_i^2[1 : u_i] + u_i x_j^2[1 : u_j]) + 2Z_{ij}x_i x_j$ at the cofactors, so over the pairs the second fold reads $\sum_i [1 : u_i] x_i^2 \sum_{j \neq i} \check{Z}_{ij} u_j + \sum_{i \neq j} Z_{ij} x_i x_j$, and the first fold reads $\sum_i [1 : u_i] x_i^2 (Z_{ii} u_i + \sum_{j \neq i} Z_{ij} u_j)$; the two off-diagonal folds at each i are balance partners, a summand on both members fixing the value (Definition 1), and the join is $\sum_i Z_{ii} x_i^2 + \sum_{i \neq j} Z_{ij} x_i x_j$, the form. The cap prices the first fold from below, $(Zu)_i [1 : u_i] x_i^2$ at or beyond the balance partner of $\bar{w}_i u_i^2 y_i^2$ at $x_i = u_i y_i$, and on the hyperplane the mean's fold reads the sum's unit, so $\sum_i \bar{w}_i u_i^2 y_i^2$ and the variance are one value there, the mean's square's multiple withdrawn. The count: a list of two members on whose span the form reads its lower side at every combination off the unit meets the hyperplane at a combination off the unit, one linear read on two coordinates, and the read at that member sits on its upper side or at equal members, against the span's read, so the maximal list of Lemma 30's side count holds at most one member. The cutoff: a label's read at a cutoff is its diagonal's product with u_i joined to the off-diagonal entries' products with the moved labels' members inside the cutoff, each product on its lower side or of equal members, so the join over the whole move list sits at or below the join over the cutoff's part. $\square$

Lemma 59 (The rows' split read). *Let a finite index be partitioned into rows $R_0, \dots, R_M$, each row a path, its labels in a stated order with the consecutive pairs its row moves, each at a positive row conductance; at each $b < M$ let a stated set of cross moves (i, j) at $i \in R_b, j \in R_{b+1}$ enter, each at a positive cross conductance c_{ij} ; and at positive masses m_i over the index write $\mu_b := \sum_{i \in R_b} m_i$ for the row mass, $p_b(i)$ for the row profile at $\mu_b p_b(i) = m_i$, $\bar{y}_b := \sum_{i \in R_b} p_b(i) y_i$ for the row mean of a list y over the index, and $\bar{y}$ for its mean at the masses. The row read B_b at a base $p \in R_b$ is the larger of the two folds*

$$\sum_{i \text{ after } p} m_i \sum_k [1 : c_k] \quad \text{and} \quad \sum_{i \text{ before } p} m_i \sum_k [1 : c_k] ,$$

the inner fold over the row moves k between p and i at the row conductances c_k , one trichotomy read. A coupling π_b at $b < M$ is a family of balance pairs over the cross moves at b , each on its upper side or at equal members, transporting the two rows' profiles onto each other, the two marginal reads

$$\sum_i \pi_b(i, j) = p_{b+1}(j) \quad \text{at every } j \in R_{b+1} , \quad \sum_j \pi_b(i, j) = p_b(i) \quad \text{at every } i \in R_b ,$$

the folds over the cross moves at the label, and its step is

$$\bar{\alpha}_b := \sum_{(i,j)} \pi_b(i, j)^2 [1 : c_{ij}]$$

over the cross moves, at or below $\sum_{j \in R_{b+1}} p_{b+1}(j)^2 [1 : c_j]$ at c_j the least cross conductance at j and at or below the display at the lower row's profile and least conductances, each pair's square at or below its multiple of the marginal. The row chain is the symmetric list over the rows at the diagonal $[1 : \bar{\alpha}_{b-1}] + [1 : \bar{\alpha}_b]$ joined to the balance partner of μ_b , a vacant step's cofactor the sum's

unit at the two ends, and the bonds the balance partners of the steps' cofactors at consecutive rows, its form at a list w over the rows

$$\sum_{b < M} [1 : \bar{\alpha}_b] (w_{b+1} - w_b)^2 + \left\langle \sum_b \mu_b w_b^2 \right\rangle^\vee ,$$

its reversal count (Lemma 30) at or beyond one, the form at the unit list the masses' fold's balance partner. Then at every list y over the index: (i) $\sum_i m_i (y_i - \bar{y})^2 = \sum_b \sum_{i \in R_b} m_i (y_i - \bar{y}_b)^2 + \sum_b \mu_b (\bar{y}_b - \bar{y})^2$; (ii) $\sum_{i \in R_b} m_i (y_i - \bar{y}_b)^2 \leq \sum_{i \in R_b} m_i (y_i - y_p)^2 \leq B_b \sum_k c_k (y_{k+} - y_{k-})^2$ at the row's base p , the last fold over the row moves k at their two ends; (iii) at the row chain's count one, $\sum_b \mu_b (\bar{y}_b - \bar{y})^2 \leq \sum_{b < M} [1 : \bar{\alpha}_b] (\bar{y}_{b+1} - \bar{y}_b)^2$; (iv) $(\bar{y}_{b+1} - \bar{y}_b)^2 \leq \bar{\alpha}_b \sum_{(i,j)} c_{ij} (y_j - y_i)^2$, the fold over the cross moves at b ; and (v) at the two reads

$$B_b \leq 1 \quad \text{at every } b \leq M, \quad \text{the row chain's count one ,}$$

the Poincaré read holds, $\sum_i m_i (y_i - \bar{y})^2 \leq \sum_k c_k (y_{k+} - y_{k-})^2 + \sum_{(i,j)} c_{ij} (y_j - y_i)^2$, the folds over every row move and every cross move. The plan on two paths. At two rows whose cross moves join each label i of the lower row to the two labels i' , $(i+1)'$ of the upper row at the label's place and the next, the first label of the upper row joined from the lower's first alone and its last from the lower's last alone, the cross moves form one path alternating the two rows, and a coupling on them is one family: at the partial folds $F_b(z) := \sum_{i \leq z} p_b(i)$ and $F_{b+1}(z) := \sum_{j \leq z} p_{b+1}(j)$, the lag $D(z) := \langle F_b(z) : F_{b+1}(z) \rangle$, the plan reads

$$\pi_b(i, (i+1)') = D(i) , \quad \pi_b(i, i') = \langle p_b(i) : D(i) \rangle = \langle F_{b+1}(i) : F_b(i-1) \rangle ,$$

the marginal reads at every label, and it is a coupling exactly where the lag sits on its upper side or at equal members at every z and at or below $p_b(z)$; at a lower row of n labels and an upper row of $n+1$ whose profiles are fixed by the reversal of the paths, $p_b(i) = p_b(n+1-i)$ and $p_{b+1}(j) = p_{b+1}(n+2-j)$, the two reads are one: $D(z)$ on its upper side or at equal members at every z from one to n . And at a single crossing, the places of the lower row at which the place quotient $p_{b+1}(z) [1 : p_b(z)]$ sits at or beyond one a final segment of the row, the lag sits on its upper side or at equal members at every z from one to n , so at the reversal-fixed profiles the plan is a coupling.

Proof. The device is the *weighted square*: at positive r_k and lists x_k, z_k ,

$$\left(\sum_k x_k z_k \right)^2 + \sum_{k < l} \langle r_l x_k z_l : r_k x_l z_k \rangle^2 [1 : r_k r_l] = \left(\sum_k x_k^2 [1 : r_k] \right) \left(\sum_k r_k z_k^2 \right) ,$$

one collected display (the cross monomials $x_k z_k x_l z_l$ entering each member once per pair, Construction 8's squared Cauchy-Schwarz at the weights), so a sum's square sits at or below the two folds' product. The step's cap: at a label j the squares' fold $\sum_i \pi_b(i, j)^2$ sits at or below the fold's square $p_{b+1}(j)^2$ at every pair at or beyond the sum's unit, and each cofactor $[1 : c_{ij}]$ at or below $[1 : c_j]$; the lower row's display so at the second marginal. (i) at a row, $\sum_{i \in R_b} m_i (y_i - \bar{y})^2$ collects to $\sum_{i \in R_b} m_i (y_i - \bar{y}_b)^2 + \mu_b (\bar{y}_b - \bar{y})^2$, the cross fold $\sum_{i \in R_b} m_i (y_i - \bar{y}_b)$ reading the sum's unit at the row mean, and the rows add. (ii) the first comparison is (i)'s display at one row against the point y_p in place of the mean, the added member $\mu_b (\bar{y}_b - y_p)^2$ at or beyond the unit; at a label i after p the difference $y_i - y_p$ is the fold of the row moves' differences from p to i , and the weighted square at $r_k = [1 : c_k]$, $x_k = 1$, z_k the differences reads its square at or below $\sum_k [1 : c_k]$ over those moves against the moves' fold $\sum_k c_k z_k^2$, at or below the fold over the row's moves after p ; the masses' fold over the labels after p reads the first sum of B_b against that fold, the labels before p its second

against the moves before p , and the larger of the two against the two folds' join. (iii) At T the row chain, m the masses' fold and μ the masses' list over the rows, T against the unit list reads μ 's balance partner row by row, so the *joined chain* $\tilde{T} := T + [1 : m] \mu \mu^\top$ against the unit list reads the sum's unit; a list w splits as t times the unit list joined to a list h on the hyperplane $\sum_b \mu_b h_b = 0$ at t the mean of w at the masses, and the joined chain's form at w collects to T 's form at h , the cross terms $2t h^\top T 1$ and $2t [1 : m] (\mu \cdot h) (\mu \cdot 1)$ the balance partners of one value and the unit list's two reads $t^2 1^\top T 1$ and $t^2 [1 : m] m^2$ balance partners as well. So T reads its upper side or equal members on the hyperplane exactly where the joined chain does at every list. At T 's count one the joined chain reads so: a list y_1 at which the joined chain reads its lower side puts, with the unit list, a list of two members on whose span T reads its lower side at every combination off the unit, the form at a times the unit list joined to b times y_1 reading b^2 times the joined chain's read at y_1 less $[1 : m] (a m + b \mu \cdot y_1)^2$, the first on its lower side at b off the unit and the second at equal members exactly at $a m + b \mu \cdot y_1$ the sum's unit, where b off the unit keeps the first and b at the unit puts a off it with the read $a^2 m$'s balance partner; the side count is the dimension of a maximal such list (Lemma 30), two against one. The joined chain's read at $\bar{y}$ collects to the display, the masses' fold about the mean $\sum_b \mu_b (\bar{y}_b - \bar{y})^2$ reading $\sum_b \mu_b \bar{y}_b^2$ less $[1 : m] (\sum_b \mu_b \bar{y}_b)^2$. (iv) the row means' difference collects at the coupling, $\bar{y}_{b+1} - \bar{y}_b = \sum_{(i,j)} \pi_b(i,j) (y_j - y_i)$, the upper row's mean the coupling's fold at the first marginal and the lower row's at the second, and the weighted square at $r_{(i,j)} = c_{ij}$, $x_{(i,j)} = \pi_b(i,j)$, $z_{(i,j)} = y_j - y_i$ reads the square at or below the step's multiple of the cross moves' fold. (v) (i) at (ii) and (iii), (iii) at (iv) with the steps' cofactors clearing the steps, collects to $\sum_b B_b$ times the row's moves' fold joined to the cross moves' fold, each row's coefficient at or below one at the row reads. *The plan.* The cross moves at the label i' of the upper row are $(i-1, i')$ and (i, i') at $1 < i \leq n$, so the first marginal at i' reads $\pi_b(i-1, i') + \pi_b(i, i') = p_{b+1}(i')$ and the second at i reads $\pi_b(i, i') + \pi_b(i, (i+1)') = p_b(i)$, the first label's and the last's at one summand each: the two reads descend the path from the first cross move, $\pi_b(1, 1') = p_{b+1}(1')$, each read naming the next pair as the cofactor at the sum, one value by the sum's injectivity, and the named pairs collect to the displayed lag reads, $\pi_b(i, (i+1)') = F_b(i) - F_{b+1}(i)$ and $\pi_b(i, i') = F_{b+1}(i) - F_b(i-1)$ at the folds' telescope; the last label's read, $\pi_b(n, (n+1)') = p_{b+1}((n+1)'),$ is the telescope's closure at the two profiles folding to one. So the family is a coupling exactly where every displayed pair sits at or beyond the sum's unit, the lag's two reads. At the reversal-fixed profiles the folds read $F_b(z) + F_b(n-z) = 1$ and $F_{b+1}(z) + F_{b+1}(n+1-z) = 1$, so $\langle F_{b+1}(i) : F_b(i-1) \rangle = D(n+1-i)$ at $F_{b+1}(i) = 1 - F_{b+1}(n+1-i)$ and $F_b(i-1) = 1 - F_b(n+1-i)$: the second read at i is the lag at $n+1-i$, one family. The single crossing: at the places at a quotient at or beyond one a final segment, $p_b(z) - p_{b+1}(z)$ sits on its upper side or at equal members up to a stated label and on its lower side beyond, the lag's consecutive differences, so the lag rises to the label and falls beyond it, its least value at an end, the vacant folds' unit at $z = 0$ and $p_{b+1}((n+1)')$ at $z = n$, each on its upper side or at equal members. $\square$

Construction 18 (The residue-two chain). At the one-plaquette window, the residue two and the list $f + \bar{f}$, the vacuum sector's depth chain (Construction 21) has the labels $(a, b) = a\omega_1 + b\omega_2$ at every natural pair, the depth $a + b$, and the six *moves* at count one each, the fundamental's row $(a+1, b)$, $(a-1, b+1)$, $(a, b-1)$ and its dual's, the reverses (Lemma 11; Lemma 25(ii)), a move off the chamber reading the sum's unit. At $x := \lambda + \rho$, the coordinates $x_1 = a+1$, $x_2 = b+1$, the form reads $|x|^2 = [2 : 3] (x_1^2 + x_1 x_2 + x_2^2)$, the Casimir $C_2 = |x|^2 - 2$ against $C_2(\theta) = 6$ (Proposition 12; Construction 4), the dimension is the *harmonic cubic* $d(x) = [1 : 2] x_1 x_2 (x_1 + x_2)$ (Corollary 1), and the dimension fold is $d_L = 6$ (Lemma 17). At the scale η the *scaled square* of a label is $s := \eta |x|^2$, and the dual pencil's electric read is $\sigma 4c_2 = [2 : 3] \eta (s - 2\eta)$ at $\sigma = \eta^2$ (Definition 10; Definition 6), the gram one and the magnetic member the fusion fold at the moves (Theorem 17);

at a line $\ell = \langle q\eta : d_L \rangle$ at a pair rate q the site datum Z at $\sigma E = M + \ell G + Z$ has the diagonal $[2 : 3]\eta(s - 2\eta) + 6 - q\eta$ and the balance partner of one at every move. *The hexagon folds.* At the six moves w the dimension's folds against the moved square's powers are the identities

$$\begin{aligned} \sum_w d(x+w) &= 6d(x), & \sum_w d(x+w) (|x+w|^2 - |x|^2) &= 16d(x), \\ \sum_w d(x+w) (|x+w|^2 - |x|^2)^2 &= (8|x|^2 + [80 : 3])d(x), \\ \sum_w d(x+w) (|x+w|^2 - |x|^2)^3 &= (40|x|^2 + [304 : 9])d(x), \end{aligned}$$

each one polynomial identity in the coordinates at the displayed operations, the moved square's difference a balance pair; write $S_k(x)$ for the k -th fold. *The pencil at a dimension-weighted list.* At a list $u_i := d(x_i)\Phi(s_i)$ with Φ a polynomial in the scaled square over a located stage (Lemma 32), the site datum reads

$$(Zu)_i = ([2 : 3]\eta(s_i - 2\eta) - q\eta)u_i - \sum_{k \geq 1} [\eta^k : k!] S_k(x_i) \Phi^{(k)}(s_i),$$

the sum to the polynomial's degree, $\Phi^{(k)}$ the k -th derivative (Definition 2): a moved label's scaled square is $s_i + \eta(|x_i + w|^2 - |x_i|^2)$, the polynomial's value there the derivatives' fold against the difference's powers, one identity per key, and the first fold's $6d(x_i)\Phi(s_i)$ withdraws against the diagonal's 6; so at a constant Φ the read is $([2 : 3]\eta(s_i - 2\eta) - q\eta)u_i$, on its upper side exactly at $[2 : 3](s_i - 2\eta) \geq q$, the *plateau read*. *The shell coordinates.* On the shell of depth n write $L := n + 2$ for the coordinates' sum $x_1 + x_2$ and $P := x_1x_2$ for their product, the shell's labels at $x_1 = 1, \dots, L - 1$: then $|x|^2 = [2 : 3](L^2 - P)$, $d = [1 : 2]LP$ and $s = [2 : 3]\eta(L^2 - P)$, so along a shell the list data are reads of P alone, P largest at the shell's middle and falling to $L - 1$ at the walls, and a label's two lower neighbors $(x_1 - 1, x_2)$ and $(x_1, x_2 - 1)$ on the shell below sit at the scaled squares $s - [2 : 3]\eta(L + x_1 - 1)$ and $s - [2 : 3]\eta(L + x_2 - 1)$, both below s , the swap $x_1 \leftrightarrow x_2$ exchanging the two. *The square coordinates* are the shell's *square* $y := \eta L^2$, the shell's *step* $\varepsilon := [1 : L]$ and the product's *share* $\varpi := P[1 : L^2]$, between $\varepsilon(1 - \varepsilon)$ at the walls and $[1 : 4]$ at an even shell's middle: $\eta = y\varepsilon^2$, $s = [2 : 3]y(1 - \varpi)$, $d = [1 : 2]\varpi[1 : \varepsilon^3]$, the shell above at the square $y(1 + \varepsilon)^2$ and the shell below at $y(1 - \varepsilon)^2$, and a read fixed by the swap, a polynomial in L , P and the scale, is one polynomial in the three coordinates at a stated power of the step cleared. *The exponential's series, and the sector profile.* $T_K(t) := \sum_{k \leq K} [1 : k!] t^k$ at a stated even degree K , the *degree*, is the exponential's series at the degree, its coefficients positive, $T_K \geq 1 + t$ at $t \geq 0$, its derivative the series one degree down, $T'_K = T_{K-1}$, and $T_K(t) \leq 1 + 2t$ at $t \leq 1$ (the coefficient fold $\sum_{k \geq 1} [1 : k!]$ at or below two at $k! \geq 2^{k-1}$, the geometric fold); its *product read* $T_K(a)T_K(b) \geq T_K(a+b)$ at $a, b \geq 0$, the product holding every monomial of $T_K(a+b)$ at the binomial's count with the further monomials beyond the degree besides; its *binomial read* $T_K(a+b) = \sum_{m \leq K} [b^m : m!] T_{K-m}(a)$ at every key; and its *concavity read*

$$T_k(t)^2 = T_{k-1}(t)T_{k+1}(t) + [t^k : k!]\left(1 + \sum_{j=1}^k [(k+1-j)t^j : (k+1)j!]\right),$$

one collected display, so $T_k^2 \geq T_{k-1}T_{k+1}$ at $t \geq 0$ and the pairs $[T_k : T_{k+1}]$ rise with k . The *series at the balance partner* $E_K(t) := T_K(\check{t})$ reads against the series at the *excess identity*

$$E_K(t)T_K(t) = 1 + r_K(t), \quad r_K(t) := \sum_{\substack{K < m \leq 2K \\ m \text{ even}}} 2 \binom{m-1}{K} [t^m : m!],$$

the product even in t , the key m 's fold $\sum_j (-1)^j \binom{m}{j}$ over the keys j at or below K with $m-j$ at or below K reading the sum's unit at m from one to K (the partial fold $\sum_{j \leq n} (-1)^j \binom{m}{j} = (-1)^n \binom{m-1}{n}$ at $n = m$) and, at m beyond K , the partial folds at $n = K$ and $n = m - K - 1$ joining to $\binom{m-1}{K} ((-1)^K + (-1)^{m-K})$ at $\binom{m-1}{m-K-1} = \binom{m-1}{K}$, twice the binomial at even m and the sum's unit at odd m at the even degree; so E_K is positive at every $t \geq 0$, $E_K = [1 + r_K : T_K]$ with r_K rising in t , its derivative $E'_K = \tilde{E}_{K-1}$, the partner of the series one degree down; at the odd degree the product reads $E_{K-1}(t) T_{K-1}(t) = 1 - \sum 2 \binom{m-1}{K-1} [t^m : m!]$ over the even keys m from K to $2K-2$, so E_{K-1} is positive on a segment from the unit exactly where that fold, rising in t , sits below one at the segment's top, one located comparison. The *sector profile* at a stated pair κ at or below one, the *plateau weight*, and at the *seam squares* $y_1 < y_2$, stated pairs, is

$$\begin{aligned} u_i &:= d(x_i) \Phi(x_i) , & \Phi(x) &:= E_K(cs) \chi(y) + \kappa , \\ \chi(y) &:= 1 \text{ at } y \leq y_1, & \chi(y) &:= 1 - 3t^2 + 2t^3 \text{ at } t = [y - y_1 : y_2 - y_1] \text{ between,} \\ & & \chi(y) &:= \text{the sum's unit at } y \geq y_2 , \end{aligned}$$

at the shell's square $y = \eta L^2$, the *seam shells* those at squares between the seam squares, positive at every label, over the tower at c , one polynomial in the square coordinates on each of the three square ranges, and on the head with its upper neighbors, where χ reads one, a polynomial in the scaled square alone, $\Phi(s) = E_K(cs) + \kappa$, falling in s where $E_{K-1}(cs)$ is positive; write $\theta := [\kappa : \Phi]$, rising in s there and below one, at $\Phi(1 - \theta) = E_K(cs) \chi$, and $\rho_k(s) := [E_{K-k}(cs) : E_K(cs)]$ at $k \leq K$, the *truncation reads*. The *profile's read*. At a label whose moves stay where χ reads one, the head and its upper neighbors among them, the moved label's value is the series' binomial read at the moved square, $E_K(c(s + \eta \Delta_w)) = \sum_{k \leq K} [(c\eta \Delta_w)^k : k!] \tilde{E}_{K-k}(cs)^{(k)}$, the k -th derivative the partner at odd k , so the site datum reads exactly, at the hexagon folds,

$$(Zu)_i = \eta u_i \left[\theta ([2 : 3] (s - 2\eta) - q) + (1 - \theta) (Q_K(x, \eta) - q) \right] ,$$

the *plateau read* on the first member and the *Gaussian read* on the second, at

$$\begin{aligned} Q_K(x, \eta) &:= [2 : 3] (s - 2\eta) + 16c\rho_1 - \rho_2([2 : 3]s + [20 : 9]\eta) \\ &+ \rho_3([10 : 9]c\eta s + [76 : 81]c\eta^2) - \rho_4([1 : 54]\eta s^2 + [19 : 162]\eta^2 s + [31 : 729]\eta^3) \\ &+ \rho_5([19 : 810]c\eta^2 s^2 + [31 : 729]c\eta^3 s + [92 : 10935]c\eta^4) \\ &- \sum_{k=6}^K \rho_k [(\check{c})^k \eta^{k-1} S_k(x) : k! d(x)] , \end{aligned}$$

the folds' reads at $c^2 = [1 : 6]$ and $\eta |x|^2 = s$, $S_4 = (16|x|^4 + [304 : 3]|x|^2 + [992 : 27])d$ and $S_5 = ([304 : 3]|x|^4 + [4960 : 27]|x|^2 + [2944 : 81])d$ the fourth and the fifth folds, each one polynomial identity in the coordinates as the first three, and S_k at $k \geq 6$ the k -th hexagon fold, one polynomial identity per key at the six moves; the read is the plateau read's and the Gaussian read's convex combination at the weights $\theta, 1 - \theta$, the Gaussian read at $16c - q$ joined to the scale's multiples, the rate c the one value at which the second fold's read $4c^2\rho_2 s$ withdraws the electric read $[2 : 3]s$ at ρ_2 one, the display $6c^2 = 1$, and the plateau read at or beyond the sum's unit exactly at $[2 : 3](s - 2\eta) \geq q$; the whole read, cleared at $\eta d(x)$, is one polynomial in the coordinates and the scale, its side a polynomial comparison; at the square coordinates it is one polynomial in the square, the share and the step (every explicit scale $\eta = y\varepsilon^2$, and the k -th fold's term at $\eta^{k-1} S_k$ against d , a polynomial of degree k in the place, at the step's power $k - 2$ or beyond), its value at the vacant step the *scale-free read*

$$A(s) := (E_K(cs) + \kappa) ([2 : 3]s - q) + 16c E_{K-1}(cs) - [2 : 3]s E_{K-2}(cs) ,$$

one polynomial in the scaled square, and the read joined to the cap $\Omega(1 - \theta)\Phi = \Omega E_K(cs)$ so at $A(s) + \Omega E_K(cs)$. At a label whose moves reach the seam, the moved labels read χ at their own squares, the read the display's at χ 's value at the label's square joined to the moved labels' folds at the shifts $\chi(y(1 \pm \varepsilon)^2) - \chi(y)$, each fold at the two squares on one range of χ one polynomial in the square coordinates; and beyond the seam the profile is the plateau κd , the read $\kappa \eta d([2 : 3](s - 2\eta) - q)$ outright.

Lemma 60 (The rows' data at the sector profile). *At the residue-two chain's sector profile (Construction 18) on a head H of whole shells at or below the head depth n_H , the rows the shells, the row moves the diagonal moves and the cross moves the lower-neighbor moves at the conductances $u_i u_j$, the masses $m_i = \bar{w}_i u_i^2$ at the cap family $\bar{w}_i = \Omega \eta(1 - \theta(s_i))$, and the coupling the plan on two paths (Lemma 59), the shell L 's labels at the places $x_1 = 1, \dots, L - 1$ and the shell $L + 1$'s at $1, \dots, L$, a label's two upper neighbors at its own place, the second coordinate raised, and at the next place, the first raised, the shells' profiles fixed by the swap $x_1 \leftrightarrow x_2$ (the mass a read of L and P), write $L = b + 2$ for the shell of the row R_b , $M := n_H$ for the top row, $U_b := \sum_{i \in R_b} u_i^2$ for the row's square fold, $S_H := [2 : 3]\eta(L_H^2 - L_H + 1)$ at $L_H := n_H + 2$ for the head's top scaled square, the top shell's wall read, the largest scaled square on the head, and per shell the middle square $s_{\text{mid}} := [2 : 3]\eta(L^2 - p(L - p))$ at p the division's quotient of L at two and the wall square $s_{\text{wall}} := [2 : 3]\eta(L^2 - L + 1)$, the least and the largest scaled squares on the shell. The read of Lemma 59 holds at every base, so the certificate's row read is the least of the reads at the row's labels, each label the base of its read, one trichotomy fold. The reads below hold at every scale at which $E_{K-1}(cs)$ is positive on the head, the odd degree's excess fold below one at cS_H (Construction 18), and the head's successor shell sits at a square at or below the first seam square, $\eta(L_H + 1)^2 \leq y_1$, so that χ reads one at every head label and its upper neighbors, in the naturals L , the scale, and the profile's values at the shells' two squares. (i) The row read.*

$$B_b \leq \Omega \eta(L^2 - 1)[1 : 8] .$$

(ii) *The quotient read. The plan on the two shells' paths (Lemma 59) is a coupling at the quotient read: the place quotient $m(z, L+1)[1 : m(z, L)]$ at the masses rises along the shell, at every place z from one to $L - 2$,*

$$\psi(s(z, L)) \psi(s(z+1, L+1)) \geq (1 - \mu^2)^2 \psi(s(z, L+1)) \psi(s(z+1, L)) \quad \text{at } \mu := [1 : L - z] ,$$

$\psi := (1 - \theta)\Phi^2 = E_K(cs)(E_K(cs) + \kappa)$, the two cross products of the four masses at their common factor $[\Omega \eta : 4]^2 L^2 (L+1)^2 z^2 (z+1)^2$ and at $(L - z)^4$ withdrawn, one polynomial comparison in the square, the place's share $\zeta := z[1 : L]$ and μ at the step $\varepsilon = \mu(1 - \zeta)$, the moved scaled squares $s(z+1, L+1) = s + [2 : 3]y\varepsilon(1 + \zeta + \varepsilon)$, $s(z, L+1) = s + [2 : 3]y\varepsilon(2 - \zeta + \varepsilon)$ and $s(z+1, L) = s + [2 : 3]y\varepsilon(2\zeta - 1 + \varepsilon)$ at $s = s(z, L)$; its value at the vacant key of μ is the sum's unit and its second key's coefficient is $2\psi(s)^2$ joined to $(1 - \zeta)^2$ times $\langle [4 : 9]y^2(2 - \zeta)(2\zeta - 1)(\psi\psi_{ss} - \psi_s^2) : [2 : 3]y\psi\psi_s \rangle$ at the derivatives in the scaled square. (iii) *The step's cap. At the shell's cap $\hat{c}_2 := 3$ and*

$$\begin{aligned} \hat{c}_L &:= [L + 1 : L](1 + (L - 1)e_f(L)) , \\ e_f(L) &:= [L(L - 1)(L - 2)[1 : 6] + 2L(L - 2) + 2L^2 H_{L-2}[1 : L - 1] + 2L^2[1 : L - 2] \\ &\quad : (L + 1)((L + 1)^4 - 1)[1 : 30]] \end{aligned}$$

at $L \geq 3$, and $\omega_L := 1 + \kappa T_K(c s_{\text{wall}}(L))$, at or beyond $[1 : 1 - \theta]$ at every label of the shell,

$$\bar{\alpha}_b \leq \Omega \eta \hat{c}_L \omega_{L+1}[1 : \mu_{b+1}] .$$

(iv) *The row mass. The row mass is the shell's fold of one polynomial in P ,*

$$\mu_b = \Omega \eta \sum_{x=1}^{L-1} [1 : 4] L^2 P^2 E_K(cs) (E_K(cs) + \kappa) \quad \text{at } s = [2 : 3] \eta (L^2 - P), \quad P = x(L - x),$$

one polynomial in the shell and the scale, the power sums' closed reads at every key of P , and the consecutive read $g_b := \mu_{b+1} [1 : \mu_b]$ is the two polynomials' pair, one exact read per row. The fold is the Euler–Maclaurin read: at the shell's square $y := \eta L^2$, at $\varepsilon := [1 : L]$ and at the summand at the place's share, $f(\xi) := (\xi(1 - \xi))^2 \psi([2 : 3] y (1 - \xi + \xi^2))$, a polynomial in ξ of degree $4K + 4$ with $f_n(y)$ its ξ^n coefficient, one polynomial in y per key,

$$\sum_{x=1}^{L-1} f(x [1 : L]) = L I(y) - 2 \sum_{m=1}^{2K+2} [B_{2m} : 2m] L^{1-2m} f_{2m-1}(y), \quad I(y) := \sum_n f_n(y) [1 : n + 1],$$

at the Bernoulli numbers, $B_0 = 1$ with $\sum_{i \leq m} \binom{m+1}{i} B_i$ the sum's unit at every $m \geq 1$; so $\mu_b = [\Omega y : 4\varepsilon^5] \hat{\mu}(y, \varepsilon)$ at

$$\begin{aligned} \hat{\mu}(y, \varepsilon) &:= I(y) - 2 \sum_{m=1}^{2K+2} [B_{2m} : 2m] \varepsilon^{2m} f_{2m-1}(y), \\ g_b &= (1 + \varepsilon)^7 \hat{\mu}(y(1 + \varepsilon)^2, \varepsilon [1 : 1 + \varepsilon]) [1 : \hat{\mu}(y, \varepsilon)], \end{aligned}$$

one polynomial in the shell's square and $[1 : L]$ and one quotient of two. (v) The row chain's certificate. The capped chain is the row chain at the caps' cofactors $\beta_b := [\mu_{b+1} : \Omega \eta \hat{c}'_L \omega_{L+1}]$ in place of the steps' cofactors, at the polynomial cap $\hat{c}'_L$, the shell's cap of (iii) with the harmonic number H_{L-2} read at its cap $L - 2$ (each of its terms at or below one), at or beyond $\hat{c}_L$ and a quotient of two polynomials in $[1 : L]$ at a divisor positive at every shell from three; the capped chain's form sits at or below the row chain's at every list ((iii), the bonds' cofactors at or below the steps'), so its count at or beyond the row chain's (Lemma 30). The certificate's crossing shell L_c is the least natural at $\rho_H L_c \geq L_H$ and its band top L_+ the least natural at $L_+ \geq \rho_+ L_c$, at the stated pairs ρ_H and ρ_+ , the head factor and the band factor, each beyond one, so that L_c sits between $L_H [1 : \rho_H]$ and its successor and L_+ between $\rho_+ L_c$ and its successor; the pre profile is the parabola $v^-(L) := L_c^2 - L^2$ on the shells from 2 to L_c , positive below L_c and the sum's unit there, its differences $\Delta_L^- := v^-(L) - v^-(L + 1) = 2L + 1$; the post profile is concave across the band and linear beyond,

$$\begin{aligned} v^+(L) &:= (L - L_c) - (L - L_c)^2 [1 : L_c] \quad \text{at } L_c \leq L \leq L_+, \\ v^+(L) &:= v^+(L_+) + \sigma (L - L_+) \quad \text{beyond, at } \sigma := 1 - (2(L_+ - L_c) + 1) [1 : L_c], \end{aligned}$$

the post slope σ positive, the sum's unit at L_c and positive beyond, its differences $\Delta_L^+ := v^+(L + 1) - v^+(L)$, $1 - (2(L - L_c) + 1) [1 : L_c]$ across the band and σ beyond; at the four reads

$$\begin{aligned} g_0 \Delta_2^- &\geq \Omega \eta \hat{c}'_2 \omega_3 v^-(2), \\ g_b \Delta_L^- [1 : \hat{c}'_L \omega_{L+1}] &\geq \Delta_{L-1}^- [1 : \hat{c}'_{L-1} \omega_L] + \Omega \eta v^-(L) \quad \text{at every shell } 3 \leq L \leq L_c - 1, \\ \Delta_{L-1}^+ [1 : \hat{c}'_{L-1} \omega_L] &\geq g_b \Delta_L^+ [1 : \hat{c}'_L \omega_{L+1}] + \Omega \eta v^+(L) \quad \text{at every shell } L_c + 1 \leq L \leq L_H - 1, \\ \Delta_{L_H-1}^+ [1 : \hat{c}'_{L_H-1} \omega_{L_H}] &\geq \Omega \eta v^+(L_H), \end{aligned}$$

the boundary, pre, post and top reads, the capped chain's count is one, and the row chain's with it.

Proof. Two reads run through the proof. *The rearrangement read:* at positive weights w over a finite index and two families f, h ordered oppositely,

$$\left(\sum_i w_i f_i\right)\left(\sum_i w_i h_i\right) = \left(\sum_i w_i\right)\left(\sum_i w_i f_i h_i\right) + \sum_{i < k} w_i w_k \langle f_i : f_k \rangle \langle h_k : h_i \rangle,$$

one collected display, every summand of the last fold on its upper side, so the mean of f at the weights wh sits at or below its mean at w ; along a shell the profile's factors Φ^2 and $\psi := (1-\theta)\Phi^2 = E_K(cs)(E_K(cs) + \kappa)$ rise in P , s falling in P and both falling in s on the head (E_{K-1} positive there), so a mean at the weights $u^2 = d^2\Phi^2$ of a family falling in P sits at or below its mean at P^2 , and that mean is a closed read at the shell's sum $\sum_x P^2 = L(L^4 - 1)[1 : 30]$ over $x = 1, \dots, L-1$ at $P = x(L-x)$, the power sums' closed reads of (iv). (i) Along the shell $P = x_1(L-x_1)$ rises to the middle and falls after it, s falls to the middle and rises after, $d = [1 : 2]LP$ rises to the middle, and Φ falls in s , so $u = d\Phi(s)$ rises to the middle label p , at $L = 2p + \varrho$ with the remainder ϱ below two, and falls after it (Construction 18's shell coordinates), and $\bar{w}_i \leq \Omega\eta$ at θ below one. At the base p and a label x_1 after it, every row move between them has both ends at u at or beyond $u(x_1)$, so its conductance's cofactor $[1 : u(k)u(k+1)]$ sits at or below $[1 : u(x_1)^2]$ and the inner fold over the $x_1 - p$ moves at or below $(x_1 - p)[1 : u(x_1)^2]$; the mass against it reads $m(x_1)(x_1 - p)[1 : u(x_1)^2] = \bar{w}(x_1)(x_1 - p) \leq \Omega\eta(x_1 - p)$, and the fold over the $L - p - 1$ labels after the base sits at or below $\Omega\eta(L - p)(L - p - 1)[1 : 2]$; the fold over the $p - 1$ labels before the base reads so at p in place of $L - p$, u rising up to the base, at or below it at $p \leq L - p$; and $2(L - p) = L + \varrho \leq L + 1$ with $2(L - p - 1) \leq L - 1$. The least over the row's labels sits at or below the read at p . (ii) A label at the place x_1 of the shell L has its two upper neighbors at the places x_1 and $x_1 + 1$ of the shell $L + 1$, the first upper label joined from the first lower alone and the last from the last (a move off the chamber the sum's unit), and the masses are reads of L and P , fixed by the swap, so Lemma 59's plan on two paths runs at $n = L - 1$ at the reversal-fixed profiles, its coupling read the single crossing: the profiles' place quotient is the mass quotient against the consecutive read, $p_{b+1}(z)[1 : p_b(z)] = m(z, L+1)[1 : m(z, L)][1 : g_b]$, so at a mass quotient rising along the shell the places at a profile quotient at or beyond one are a final segment. The masses read $m(x, L) = [\Omega\eta : 4]L^2x^2(L-x)^2\psi(s(x, L))$ at $d = [1 : 2]Lx(L-x)$, so the quotient's rise from z to $z+1$, $m(z+1, L+1)m(z, L) \geq m(z, L+1)m(z+1, L)$, reads at the common factor withdrawn $(L-z)^4\psi(s(z, L))\psi(s(z+1, L+1)) \geq ((L-z)^2 - 1)^2\psi(s(z, L+1))\psi(s(z+1, L))$, the display at $\mu(L-z) = 1$; the moved scaled squares are the shell coordinates' reads, $s(x, L) = [2 : 3]\eta(L^2 - x(L-x))$ at $\eta L = y\varepsilon$ and $\eta z = y\varepsilon\zeta$; and the second key: the shifts a_1, a_2, a_3 of the three moved squares read $a_1 - a_2 - a_3 = [2 : 3]\eta$'s balance partner and $a_1^2 - a_2^2 - a_3^2$ the partner of $[8 : 9]y^2\varepsilon^2(2 - \zeta)(1 - 2\zeta)$ with $a_2a_3 = [4 : 9]y^2\varepsilon^2(2 - \zeta)(2\zeta - 1)$ at the first keys, so the products' difference at the Taylor read to the second key is ε^2 times the displayed bracket at $\varepsilon = \mu(1 - \zeta)$, and $1 - (1 - \mu^2)^2$ reads $\mu^2(2 - \mu^2)$. (iii) The cross moves at b are the pairs (i, j) at j on the shell $L + 1$ and i a lower neighbor of j , and Lemma 59's cap reads $\bar{\alpha}_b \leq \sum_j p_{b+1}(j)^2[1 : c_j]$ at c_j the least of the conductances $u_i u_j$ at j . At a cross move the upper label sits at the larger scaled square, $s_j = s_i + [2 : 3]\eta(L + x_k + 1)$ at the raised coordinate x_k , and Φ falls in s , so $u_i \geq [d_i : d_j]u_j$ at $d_j = [L + 1 : L][y_k : y_k - 1]d_i$ at the lowered coordinate $y_k = x_k + 1$ of j , and $[1 : u_i u_j] \leq [d_j : d_i][1 : u_j^2]$; the larger over the two lower neighbors sits at the smaller coordinate $y_{\min}$ of j , so $[1 : c_j] \leq c'_j[1 : u_j^2]$ at $c'_j := [L + 1 : L](1 + [1 : y_{\min} - 1])$ at an interior j , at or below $[L + 1 : L](1 + (L - 1)[1 : P_j - L])$ at $[1 : y_{\min} - 1] \leq [1 : y_1 - 1] + [1 : y_2 - 1] = (L - 1)[1 : P_j - L]$ over $(y_1 - 1)(y_2 - 1) = P_j - L$ and at or below $2[L + 1 : L]$ at $y_{\min} \geq 2$, and $c'_j := [L + 1 : L][L : L - 1]$ at a wall j , its one lower neighbor at the coordinate L lowered, at or below $2[L + 1 : L]$. With $p_{b+1}(j)^2[1 : u_j^2] = (\Omega\eta)^2(1 - \theta(s_j))^2 u_j^2[1 : \mu_{b+1}^2]$ and $(1 - \theta)^2$ at or below one the step reads $\bar{\alpha}_b \leq (\Omega\eta)^2[1 : \mu_{b+1}^2]\sum_j u_j^2 c'_j$; at the family f over the shell $L + 1$, $(L - 1)[1 : P - L]$ at an interior

label and $(L-1)[1:L-2]$ at the two walls (at or beyond the walls' own $[1:L-1]$ and at the interior's read at the walls' neighbors, $P-L=L-2$ there, so f falls in P over the whole shell), $c'_j \leq [L+1:L](1+f_j)$, and the rearrangement read at $P^2[1:P-L] = (P-L)+2L+L^2[1:P-L]$ folds $\sum_j u_j^2 c'_j \leq \hat{c}_L U_{b+1}$ at $L \geq 3$, the shell's cap the interior's three folds and the two walls at $P=L$ against the shell's squares' sum, and at $L=2$ the shell 3's two labels are walls both at $c'=3$; and on the shell $L+1$ every label reads $[1:1-\theta] = 1 + \kappa[1:E_K(cs)] \leq 1 + \kappa T_K(cs) \leq \omega_{L+1}$ at $s \leq s_{\text{wall}}(L+1)$, so $U_{b+1} = \sum_j [1:1-\theta(s_j)](1-\theta(s_j))u_j^2 \leq \omega_{L+1}\mu_{b+1}[1:\Omega\eta]$ and $\bar{\alpha}_b \leq \Omega\eta\hat{c}_L\omega_{L+1}[1:\mu_{b+1}]$. (iv) The mass at a label is $\Omega\eta(1-\theta)u^2 = \Omega\eta d^2\psi$ at $d=[1:2]LP$ and $\psi = E_K(cs)(E_K(cs) + \kappa)$, $E_K(cs)$ a polynomial in s of degree K and s a polynomial in P of degree one, so the summand is a polynomial in P of degree $2K+2$, and the fold over the shell of $P^k = (x(L-x))^k$ is the binomial's fold of the power sums $\sum_{x<L} x^j$, each one polynomial in L , the *power sums' closed reads*: the telescope of $(x+1)^{j+1} - x^{j+1} = \sum_{i \leq j} \binom{j+1}{i} x^i$ over $x < L$ reads $L^{j+1} - 1$ as $(j+1)\sum_{x<L} x^j$ joined to the lower keys' folds at the binomials, so the key j 's fold is the cofactor at $j+1$ of that read less the lower keys', the descent from the key one; the consecutive read is the cofactor of two occupied polynomials. The Euler–Maclaurin read is the telescope's solve at the Bernoulli numbers: the closed read is $\sum_{x<L} x^j = [1:j+1]\sum_{i \leq j} \binom{j+1}{i} B_i L^{j+1-i}$, one polynomial identity in L per key, the telescope's read at the recursion (the fold $\sum_{i \leq j} \binom{j+1}{i}$ against the closed reads at the keys below j collecting, per power of L , to the recursion's sum's unit, the top power alone surviving); at the summand's coefficients, $\sum_{x<L} f(x[1:L]) = \sum_n f_n L^{-n} \sum_{x<L} x^n$, the key i of the Bernoulli read collects $B_i L^{1-i} \sum_n f_n \binom{n+1}{i} [1:n+1] = B_i L^{1-i} [1:i] \sum_{n \geq i} f_n \binom{n}{i-1}$, the fold the gap of the $(i-1)$ -th derivative's read at $\xi=1$ against its read at $\xi=0$ over $(i-1)!$, and f fixed by the swap $\xi \mapsto 1-\xi$ reads an odd derivative at $\xi=1$ as its balance partner at $\xi=0$, the read $f^{(i-1)}(0) = (i-1)!f_{i-1}$ at the key's coefficient, the key one's read $[f(0):2]$ less $[f(1):2]$ the sum's unit at the double roots and the odd keys beyond one at the sum's unit (B_i there); the powers of L read $L^{1-i} = L\varepsilon^i$, and μ_b collects at $L^2 P^2 = L^6(\xi(1-\xi))^2$ against $[1:4]$, the seven powers of L the cofactor $[1:\varepsilon^7]$ against $\eta = y\varepsilon^2$; the consecutive read's shell $L+1$ sits at the square $y(1+\varepsilon)^2$ and at $[1:L+1] = \varepsilon[1:1+\varepsilon]$, the two prefactors' quotient $(1+\varepsilon)^7$. (v) The capped chain's form at a list w is the row chain's less the bonds' differences' fold $\sum_b ([1:\bar{\alpha}_b] - \beta_b)(w_{b+1} - w_b)^2$, each coefficient on its upper side or of equal members at (iii), so the counts are ordered (Lemma 30). At the bonds β_b and the diagonal $a_b := \beta_{b-1} + \beta_b$ less μ_b , the vacant bonds the sum's unit at the two ends, the pivots run the identity $p_{b+1}p_b + \beta_b^2 = a_{b+1}p_b$ from $p_0 = a_0$ (Lemma 30's LDL^T at the tridiagonal list), and at a positive list v with the termwise read $a_{b+1}v_{b+1} \geq \beta_b v_b + \beta_{b+1}v_{b+2}$ and the entry read $p_b v_b \geq \beta_b v_{b+1}$ the next pivot clears its entry read, $p_{b+1}v_{b+1} \geq \beta_{b+1}v_{b+2}$: $p_{b+1}v_{b+1}$ joined to $\beta_b^2 v_{b+1}[1:p_b]$ reads $a_{b+1}v_{b+1}$, at or beyond $\beta_b v_b + \beta_{b+1}v_{b+2}$, and $\beta_b^2 v_{b+1}[1:p_b]$ sits at or below $\beta_b v_b$ at the entry read, the positive pivot withdrawing at the product's injectivity (Lemma 56(i) at the bonds). The reads divide by the row mass: at a decreasing profile the termwise read at the row of the shell L collects to $\beta_b \Delta_L^- \geq \beta_{b-1} \Delta_{L-1}^- + \mu_b v^-(L)$, and against μ_b at $\beta_b[1:\mu_b] = g_b[1:\Omega\eta\hat{c}'_L\omega_{L+1}]$ and $\beta_{b-1}[1:\mu_b] = [1:\Omega\eta\hat{c}'_{L-1}\omega_L]$ it is the pre read; the entry read at the boundary row, $p_0 v^-(2) \geq \beta_0 v^-(3)$ at $p_0 = \beta_0$ less μ_0 , reads $\beta_0 \Delta_2^- \geq \mu_0 v^-(2)$, the boundary read at g_0 ; so every pivot at a shell from 2 to $L_c - 1$ sits on its upper side or at equal members, the pre profile positive there and the sum's unit at L_c . A pivot at equal members at a shell below L_c sits at the shell $L_c - 1$ alone, the entry read $p_b v_b \geq \beta_b v_{b+1}$ at a positive profile value beyond it refusing equal members; there the descent takes the two rows $L_c - 1, L_c$ in one block of order two at the determinant β_b^2 's balance partner, the mixed block reading one lower unit (Lemma 30), and the deflation onto the row $L_c + 1$ withdraws the block's solve at the sum's unit, the block's adjugate reading the sum's unit at its corner, so the pivot at $L_c + 1$ is the diagonal a_{L_c+1} , the rebound's read below. The first pivot on its lower side, where one sits, is at a shell c at

or beyond L_c , and the following pivot sits at or beyond its diagonal, the identity at a lower-side predecessor read on the exceeding side (Lemma 56(iii)); at an increasing profile the termwise read at the row of the shell L collects to $\beta_{b-1} \Delta_{L-1}^+ \geq \beta_b \Delta_L^+ + \mu_b v^+(L)$, against μ_b the post read and at the top row the top read (the right bond vacant), so the rebound seeds the entry read at the shell $c+1$, $p_{c+1} v^+(c+1) \geq a_{c+1} v^+(c+1) \geq \beta_c v^+(c) + \beta_{c+1} v^+(c+2) \geq \beta_{c+1} v^+(c+2)$ at $v^+(c)$ at or beyond the sum's unit, and the induction runs to the top row. So the pivots' lower-side count is at most one, the capped chain's reversal count (Lemma 30), and at or beyond one at the unit list's read (Lemma 59): one. The profiles' differences: $v^-(L) - v^-(L+1) = 2L+1$, and $v^+(L+1) - v^+(L) = 1 - (2(L-L_c)+1)[1:L_c]$ across the band by the square's difference, at or beyond σ there, and σ beyond, the post profile positive beyond L_c at σ positive. $\square$

Construction 19 (The sector's tail certificate). At a member, a direction count and a window of the directed family (Definition 8), the vacuum sector's *tail certificate* on a scale segment is the data reading, at every scale of the segment and every fiber (Lemma 40), the sector's decimated head's count at a stated line one at the unit fiber and the sum's unit at a further fiber (Theorem 19; Lemma 30), the line above the ground by a stated positive pair at the scale, the *lines' gap*, so that the corner floor at every ray of the segment sits at or beyond the pair; its reads are five. (i) *The scale's read*. At one scale the count at the line is one finite read: the labels whose diagonal at the line clears their off-row magnitude fold are the *dominant tail* (Theorem 19(ii)), the tail's form at or beyond its *gaps'* fold of squares at every cutoff, the gap the diagonal less the fold, so the chain's count at the line at every cutoff sits at or below the count of the *floored head*, the head's pencil less the tail coupling's solve at the gaps' diagonal, every gap beyond the sum's unit (the dominance read's kernel block vacant there, Theorem 19(ii)), $Z_H - B \text{diag}(g)^{-1} B^\top$ at the coupling B and the gaps g : the tail's form joined to the coupling's cross read sits at or beyond the balance partner of the solve's form at every tail vector, the completed square at the gaps, so the full pencil's form at or beyond the floored head's at every vector and the counts monotone (Lemma 30); the floored head is a finite symmetric pencil at the scale and the line, one split, cutoff-free. (ii) *The comparison read*. At two regional records, let $Z, \tilde{Z}$ be the reduced site data at their stated lines, on a common list of positive Gram G through the records' specified congruences. A transport at width ρ has the two-sided cap and the two margin counts

$$Z - \rho G \preceq \tilde{Z} \preceq Z + \rho G, \quad \text{rev}(Z + \rho G) = q = \text{rev}(Z : \rho G).$$

Inertia gives $q \leq \text{rev } \tilde{Z} \leq q$, so the reduced counts agree (Lemma 30). Each original count joins its record's dropped-pivot counts to this read (Theorem 19(i)). A spectator comparison includes the slab families' equal-entry or diagonal-tie identities and their transfer certificates (Lemma 35(i),(ii)); its width enters the displayed margin counts at the actual site data. A *reach* includes these reads at every comparison beyond its stated bound. (iii) *The tail's read*. The count at the line is a split's lower-side count (Lemma 30), and the certificate over the segment is the *split family*: at every scale of the segment the splits at the line of the actual regional records' floored heads ((i)), a reduction through finitely many window shapes including (ii)'s reads at every transport. The heads' entries are one polynomial-pair list in the chord and the scale (Lemma 45; Lemma 44), the family closing in finitely many displayed comparisons over the segment, each a polynomial comparison in the scale and the head's data on a stated box with its validity divisors located (Lemma 31; Theorem 16). At a chain whose shells beyond the collar are blocks of one stated order the family is the block chain's certificate (Lemma 57), the band-one walk at the single-plaquette window its scalar instance (Lemma 56). At a chain whose shells grow with the depth the certificate's data are a *profile* u , positive at every label, a *head* H , a finite label set at the scale, a *cap family* $\bar{w}$ of positive pairs over H , a partition of H into *rows*, each a path along the chain's moves, with a *coupling* per consecutive

pair along the cross moves (Lemma 59), the *line rate* q_L and the *lower rate* q_- at $q_- + o = q_L$, a *witness* x supported on a stated finite set, and the *ceiling*, the segment's top; its reads at every scale under the ceiling are: (a) $(Zu)_i$ at the line on its upper side or of equal members at every label off H ; (b) $(Zu)_i + \bar{w}_i u_i$ so on H ; (c) the Poincaré read of the masses $\bar{w}_i u_i^2$ on H against the conductances $u_i u_j$ at the rows' moves and the cross moves, Lemma 59(v)'s two reads at the rows' data; (d) the witness's form at the lower line on its lower side or of equal members; and (e) o positive. By (a), (b) and the fold at the list (Lemma 58) the pencil at the line reads its upper side on the hyperplane $\sum_{i \in H} \bar{w}_i u_i^2 y_i = 0$ at every cutoff, the rows' moves and the cross moves among the chain's moves so that the Poincaré read's right member sits at or below the fold's own, so the count at the line is at most one; by (d) the ground sits at or below the lower line; and the lines' gap is $o\eta$. At the residue-two chain the profile is Construction 18's at the shell coordinates, the head the shells at squares at or below the *head square* y_H , a stated pair, whole, the *head depth* n_H the largest natural whose shell's square sits at or below y_H , the rows the shells of the head, the row moves the shell's diagonal moves and the cross moves the two lower-neighbor moves, every move of the chain one of the two; the coupling the plan on the two shells' paths (Lemma 59), the shells' profiles fixed by the swap $x_1 \leftrightarrow x_2$, at the quotient read (Lemma 59; Lemma 60(ii)); the cap family $\bar{w}_i := \Omega \eta (1 - \theta(s_i))$ at a stated pair Ω ; the reads (a) and (b) the profile's read at Construction 18's display, (a) the plateau read's clearance of the Gaussian read's defect at every label off the head, the seam's and the plateau's reads among them, and (b) the defect's cap on it, each one polynomial comparison in the coordinates and the scale on a stated region, exact at the polynomial profile, read at the shell L , the product P and the scale through the box read (Lemma 36) at the scale's pair reads ηL and ηL^2 and the product's share $P[1 : L^2]$, the region's unbounded shells at the leading-term bound in the scaled square, and the reads of Lemma 60 so at the shell and the scale; the seam shells at the *seam read*: at a seam shell the site datum is the display at $\chi(y)$ joined to the *shift fold*, the moved labels' values at the shifts $\chi(y(1 \pm \varepsilon)^2) - \chi(y)$, the upper moves at the one shift and the lower at the other with the diagonal moves at the sum's unit, one polynomial in the coordinates and the scale per label; the seam squares sit at y_1 beyond y_H and y_2 at or beyond $3q + 4$, so that the plateau read clears at every label at a square at or beyond y_2 at every scale at or below one, a label's scaled square at or beyond $[1 : 2] \eta L^2$ ($L^2 - P$ at or beyond $[3 : 4] L^2$) and $[2 : 3] ([1 : 2] y - 2\eta) - q$ on its upper side or at equal members at $y \geq 3q + 4\eta$; the head square and the seam squares are entered with read (a) at every label off the head at a square below $y_2 [1 : (1 - \varepsilon)^2]$ deciding them, a label at or beyond that square with its lower neighbors beyond the seam at the plateau read outright, every label off the head then at read (a); the read (c) at the rows' data over the shells, the row read, the quotient read, the step's cap, the row mass and the row chain's certificate at Lemma 60; and the witness the dimension against a polynomial in the scaled square and the shell, $x_i = d(x_i) E(s_i) \eta (L_x^2 - L_i^2)$ on the shells at or below an *endpoint shell* L_x and the sum's unit beyond, E the exponential's series at the balance partner at a produced degree, the cutoff's consecutive differences first order in the scale at every move so that the moves' fold sits at the scale's order with the profile's read; its form at the lower line reads at the site datum's diagonal and the moves within the box, and the *lower rate* is $q_- := q_L - o$ at the stated gap o , positive, read (e) its shape: the witness's *cleared read*

$$\hat{q} \sum_i x_i^2 = \sum_i [2 : 3] (s_i - 2\eta) x_i^2 + [1 : \eta] \left(6 \sum_i x_i^2 - 2 \sum_{(i,j)} x_i x_j \right),$$

the folds over the box's labels and the moves between them, each a Faulhaber read at the shell coordinates, one pair per scale, is the rate at which the form reads equal members at the witness, the form falling in the rate, so read (d) holds exactly at $\hat{q} \leq q_-$; the degree and the endpoint are produced data, entered with read (d) deciding them (Lemma 56(v)'s family at the chain's shells).

Construction 20 (The residue-two certificate's data). At the one-plaquette window, the residue two and the list $f + \bar{f}$ (Construction 18), the sector's tail certificate at the sector profile (Construction 19(iii)) enters at the data: the rate c at $6c^2 = 1$, the degree $K := 50$, the plateau weight $\kappa := [1 : 400]$, the line rate $q_L := [657 : 100]$, the cap $\Omega := [143 : 1000]$, the witness at the degree 120 and the endpoint shell L_x the least shell at $\eta L_x^2 \geq 100$, and the six data below, each the least member of a stated finite family at its read, in the order of their reads (Construction 19; Lemma 61):

1. the head square y_H , the least natural at which the scale-free read $A(s)$ (Construction 18) sits on its upper side at every s at or beyond $[1 : 2] y_H$, the segment count at Theorem 16's radius, with $A(s) + \Omega E_K(cs)$ on its upper side at $s \leq [2 : 3] y_H$, $E_{K-1}(cs)$ positive there, $\Omega y_H \leq 8$, and the quotient read's second key on its upper side on its box (Lemma 61(iii));
2. the seam squares, $y_2 := 2 y_1$ at y_1 the least natural from $y_H + 1$ to $8 y_H$ at which the seam box's and the two straddle boxes' reduced reads at the vacant step sit on their upper side on their boxes (Lemma 61(ii)), one trichotomy fold over the naturals between, $2 y_1$ at or beyond $3 q_L + 4$ at y_1 beyond $[3 : 2] q_L + 2$;
3. the lines' gap o , the largest of the halvings of one from the first to the twentieth at which read (d)'s form at the vacant step, its value at $y_x = 100$ and ε_x the sum's unit, sits on its lower side (Lemma 61(iv)), the read held at a halving and so at every further one, the form falling in the rate;
4. the head factor ρ_H and the band factor ρ_+ , each the least member of the family $1 + j [1 : 16]$ at j from one to sixteen at which the row chain's pre, post and top reads at their vacant steps hold on their boxes, the boxes' step at or below $[1 : 4]$ (Lemma 61(iii)), with the post slope positive, the band factor's fold at the head factor's member;
5. the split shell L_s , the least natural at or beyond four and at or beyond every $[F : m]$ over the boxes of Lemma 61(ii), (iii) and (iv), m per box the largest of the halvings of one from the first to the fortieth at which the box's leading key's coefficient less m holds the grid read of Lemma 36, held at a halving and so at every further one (the grid finer and the comparison weaker), and F the further keys' fold at the bounds, with the seam's width beyond $4 y_2 [1 : L_s]$ and y_H below $y_1 [1 : (1 + [1 : L_s])^2]$;
6. the ceiling, the least halving of one under which every one-variable read of the shells below L_s and of the endpoint shells below L_s holds at its located roots, the box reads at those shells' own steps holding on their boxes, and the head's successor shell sits at a square at or below y_1 (Lemma 60), the comparison $\eta (L_H + 1)^2 \leq y_1$ holding at every scale at which L_H sits at or beyond $2 y_H [1 : y_1 - y_H]$.

Its reads at a scale are finitely many polynomial comparisons in the coordinates and the scale, each exact at the polynomial profile and at the row mass's closed read (Lemma 60(iv)): (a) at every label off the head at a square below $y_2 [1 : (1 - \varepsilon)^2]$, the plateau read exact beyond; (b) at every label of the head at the stated cap; (c) Lemma 59(v)'s two reads at Lemma 60's reads, the row read at or below one at every row, the quotient read at every place of every shell, the step's cap at every row, the odd degree's excess fold below one at $c S_H$, the post slope positive and the row chain's four reads at the crossing shell; (d) the witness's form at the lower line, the line at the lower rate $q_L - o$, on its lower side or at equal members; and (e) the lines' gap o positive.

Lemma 61 (The residue-two reads at the square coordinates). *At the residue-two chain's tail certificate at the sector profile (Construction 19(iii); Construction 18), its data the profile's, the*

head square y_H , the seam squares $y_1 < y_2$, the line rate q_L , the lines' gap o , the cap Ω and the head and band factors ρ_H, ρ_+ (Lemma 60(v)), and at every scale at or below one, the reads (a) to (e) hold where the reads below hold, each one polynomial comparison in stated coordinates on a stated box (the box read, Lemma 36), joined to finitely many one-variable comparisons in the scale: at a stated natural L_s at or beyond four, the split shell, with the seam's width $y_2 - y_1$ beyond $4y_2[1 : L_s]$ and the head square y_H below $y_1[1 : (1 + [1 : L_s])^2]$, the reads at the shells at or beyond L_s are the box reads of (ii) and (iii), read (d) at endpoint shells at or beyond L_s is the box read of (iv), and every read at a shell below L_s , and read (d) at an endpoint shell below L_s , is the box read at the shell's own step, the scale and the roundings its coordinates, joined to one polynomial comparison in the scale on each of the finitely many segments of the scale at which a produced natural sits below L_s , each read at the polynomial's located roots (Theorem 16); the certificate's ceiling is the least halving of one under which every one-variable comparison holds.

(i) The read at a label. At a label the site datum's read at the line reads, at the seam values χ_-, χ_0, χ_+ , the profile's χ at the squares $y(1 - \varepsilon)^2, y$ and $y(1 + \varepsilon)^2$,

$$(Zu)_i = \eta d([2 : 3](s - 2\eta) - q)(E_K(cs)\chi_0 + \kappa) + 6dE_K(cs)\chi_0 - \chi_0 D_0 - \chi_+ D_+ - \chi_- D_- ,$$

D_0, D_+ and D_- the diagonal, upper and lower folds, the folds $\sum d(x + w)E_K(cs_w)$ over the two diagonal moves, the two upper moves and the two lower moves at the moved scaled squares $s_w = s + \eta\Delta_w$, each fixed by the swap, the upper and the lower folds positive off the shell two, so the read falls in χ_+ and in χ_- ; the read joined to the cap, read (b)'s object, is the display joined to $\Omega\eta dE_K(cs)\chi_0$. The cleared read is the read against ηd ; at the seam values one it is the profile's read (Construction 18), one polynomial in the square coordinates. At a label whose seam values differ the cleared read is the profile's read at χ_0 less the shift fold against ηd , the upper and the lower folds' summands at the shifts $\chi_{\pm} - \chi_0$, and an upper or lower move's dimension factor $d(x + w)$ against d splits as its shell part $1 \pm \varepsilon$ joined to its place part $\pm(1 \pm \varepsilon)[1 : x_k]$ at the moved coordinate x_k ; at χ falling in the square every place part's summand sits on its lower side or at equal members, so the cleared read sits at or beyond the reduced read, the cleared read at the shell parts alone, its multiple at the square one polynomial in the square coordinates.

(ii) Reads (a) and (b) at the shells at or beyond the split shell. At $\varepsilon \in [0, [1 : L_s]]$ and $\varpi \in [0, [1 : 4]]$ throughout, read (b) at every label of the head holds where the cleared read joined to $\Omega E_K(cs)$ holds at the seam values one on the box $y \in [0, y_H]$; and read (a) at every label off the head holds where four box reads of the reduced read hold, each at stated seam values, every label sitting in a box reading its own χ_0 and reading χ_+ and χ_- at or beyond the label's own, so that the box's read at the label sits at or below the label's own read: at $y \in [y_H, y_1]$ the seam values one throughout; at $y = y_1 + \varepsilon\sigma(y_2 - y_1)$ on $\sigma \in [0, 4y_1 : y_2 - y_1]$ the seam values one, $\chi(y)$ and $\chi(y(1 + \varepsilon)^2)$ at the cubic's formula; at $y \in [y_1, y_2]$ the three seam values at the cubic's formula; and at $y = y_2 + \varepsilon\sigma(y_2 - y_1)$ on $\sigma \in [0, 4y_2 : y_2 - y_1]$ the seam values $\chi(y(1 - \varepsilon)^2)$ at the cubic's formula, the sum's unit and the sum's unit; a label at or beyond y_2 with its lower neighbors at or beyond it reads the plateau outright (Construction 19). The value at the vacant step of the head's read is $A(s) + \Omega E_K(cs)$, of the first box's $A(s)$ (Construction 18), and of the seam box's reduced read, at the primes in the square,

$$\begin{aligned} \kappa([2 : 3]s - q) + \chi(E_K(cs)([2 : 3]s - q) + 16cE_{K-1}(cs) - [2 : 3]sE_{K-2}(cs)) \\ - 8y\chi''E_K(cs) - 12\chi'E_K(cs) + 8cy\chi'E_{K-1}(cs) . \end{aligned}$$

(iii) The rows' reads. The row read holds at $\Omega y_H \leq 8$; the reads of Lemma 60 run at every scale at which $E_{K-1}(cs)$ is positive at $s \leq [2 : 3]y_H$, one comparison in the scaled square; the quotient

read at the shells at or beyond the split shell holds where its display (Lemma 60(ii)), one polynomial in the square, the place's share ζ and μ , holds on the box $y \in [0, y_H]$, $\zeta \in [0, 1]$, $\mu \in [0, [1 : 2]]$, every place of every shell at a point of the box at $\varepsilon = \mu(1 - \zeta)$; and the row chain's pre, post and top reads hold where the displays of Lemma 60(v), at the polynomial cap, hold on the boxes of the factor coordinates, the crossing shell, the band top and the head's top shell read against the label's shell at the roundings θ_H , θ_+ in $[0, 1]$ and θ_c in $[0, \rho_H]$: $L_c = \ell L$, $L_H = (\rho_H \ell - \theta_c \varepsilon) L$, $L_+ = \ell_+ L$ at $\ell_+ = \rho_+ \ell + \theta_+ \varepsilon$ and $y(\rho_H \ell - \theta_c \varepsilon + \theta_H \varepsilon)^2 = y_H$, the crossing's square $y_c = \eta L_c^2 = y \ell^2$; the pre read on the box $y \in [0, y_H [1 : \rho_H^2] (1 + 2 [1 : L_s])^2]$, $\varepsilon \in [0, [1 : L_s]]$, $\delta \in [-1, \rho_H]$ at $y_c(\rho_H - \delta \varepsilon)^2 = y_H$, one polynomial in y , ε and δ at $\Omega \eta v^-(L) = \Omega(y_c - y)$ and $\Delta_L^- = 2L + 1$; the post read on the box $\varepsilon \in [0, [1 : L_s]]$, the roundings, and $\lambda \in [0, 1]$ at $\ell = (1 + (1 + \theta_c) \varepsilon) [1 : \rho_H] + \lambda(1 - \varepsilon - (1 + (1 + \theta_c) \varepsilon) [1 : \rho_H])$, the profile at $v^+(L) = L(1 - \ell)(2\ell - 1) [1 : \ell]$ across the band, $\Delta_L^+ = (3\ell - 2 - \varepsilon) [1 : \ell]$ and $\Delta_{L-1}^+ = (3\ell - 2 + \varepsilon) [1 : \ell]$ there, and beyond the band top at $v^+(L_+) = L(\ell_+ - \ell)(2\ell - \ell_+) [1 : \ell]$, $\sigma = (3\ell - 2\ell_+ - \varepsilon) [1 : \ell]$ and $\sigma(L - L_+) = \sigma L(1 - \ell_+)$, the two boxes cut at ℓ_+ against $1 + \varepsilon$, the beyond box reading Δ_{L-1}^+ at σ , at or below the band's own value at the band top's shell; and the top read on the box of the step and the roundings at $\ell = [1 : \rho_H - \theta_c \varepsilon]$ and $y(1 + \theta_H \varepsilon)^2 = y_H$, in the band or beyond it at the cut of ℓ_+ against $1 + \varepsilon$; the boundary read sits at the shell two, below the split shell.

(iv) Read (d). At an endpoint shell at or beyond the split shell the witness's form at the lower line, the folds over the shells at or below L_x and their places at the power sums' closed reads, is one polynomial in the endpoint's square $y_x := \eta L_x^2$ and step $\varepsilon_x := [1 : L_x]$ at a stated power of the step cleared, and read (d) holds where it sits on its lower side or at equal members on the box $y_x = 100(1 + \varepsilon_x \tau)$, $\tau \in [0, 6]$, $\varepsilon_x \in [0, [1 : L_s]]$.

Proof. (i) The site datum at the line is the diagonal $[2 : 3] \eta(s - 2\eta) + 6 - q\eta$ at the label against the balance partner of one at every move (Construction 18), so $(Zu)_i$ is the diagonal's multiple of $u_i = d(E_K(cs)\chi_0 + \kappa)$ less the six moved values $d(x + w)(E_K(cs_w)\chi_w + \kappa)$, the diagonal moves at the seam value χ_0 , the upper at χ_+ and the lower at χ_- ; the six κ -terms collect to $6\kappa d$ at the first hexagon fold, withdrawing against the diagonal's $6\kappa d$, and the display follows. An upper or lower fold's summands are positive at $d(x + w)$ positive off the chamber's walls and E_K positive (the excess identity); the two moves of a fold are exchanged by the swap, so the fold is fixed by it and a polynomial in L , P and the scale, one polynomial in the square coordinates at a stated power of the step cleared (Construction 18). The shift fold: the moved values at $\chi_{\pm}$ read χ_0 joined to $\chi_{\pm} - \chi_0$, the first part collecting with the diagonal moves to the profile's read at χ_0 ; an upper move's dimension factor reads $d(x + w) = d(1 + [1 : x_k])(1 + \varepsilon)$ at its raised coordinate and a lower move's $d(1 - [1 : x_k])(1 - \varepsilon)$ at its lowered coordinate, the place part's summand $\pm(1 \pm \varepsilon) [1 : x_k] d E_K(cs_w)(\chi_{\pm} - \chi_0)$ at $\chi_+ - \chi_0$ on its lower side or at equal members and $\chi_- - \chi_0$ on its upper side or at equal members at χ falling, each summand on its lower side or at equal members, and the shift fold enters the read on the lower side. (ii) The read falls in χ_+ and in χ_- at the folds' sign and every box reads χ_0 at the label's own value, so a box whose $\chi_{\pm}$ sit at or beyond the label's own reads at or below the label's read, and the reduced read at or below the cleared read. A label off the head at $y \leq y_1$ sits in the first box at every seam value one, χ_+ at or beyond the cubic's value; a label at $y \geq y_1$ whose lower neighbors sit below y_1 sits in the second box, $\sigma = (y - y_1) [1 : \varepsilon(y_2 - y_1)]$ below $y_1(2 - \varepsilon) [1 : (1 - \varepsilon)^2(y_2 - y_1)]$ at $y(1 - \varepsilon)^2 < y_1$, at or below $4y_1 [1 : y_2 - y_1]$ at $\varepsilon \leq [1 : 4]$, with $\chi_- = 1$ its own and χ_0, χ_+ the cubic's values, the upper square within the seam at the width's bound; a label at $y \leq y_2$ whose lower neighbors sit at or beyond y_1 sits in the third box, its lower and its own seam values the cubic's own and its upper value the cubic's or, beyond y_2 , the cubic's value $3t^2 + 2t^3$ at the excess t beyond one, on its upper side, at or beyond the sum's unit; and a label at $y \geq y_2$ whose lower neighbors sit below y_2 sits in the fourth box, σ so

at y_2 , its lower seam value the cubic's own (the lower square at or beyond y_1 at the width's bound) and its own and upper values the sum's unit. The head's labels sit in the head box with every seam value one at $y(1+\varepsilon)^2 \leq y_1$. The value at the vacant step: at the seam values one the display collects to the profile's read (Construction 18), the folds $\sum_w d(x+w)E_K(cs_w) = \sum_k [\eta^k : k!] S_k E_K^{(k)}(cs)$ at the hexagon folds, its value at the vacant step $A(s)$ and, with the cap, $A(s) + \Omega E_K(cs)$. At the seam box the reduced read's shift fold has the shifts $\chi_{\pm} - \chi_0 = \pm 2y\varepsilon\chi' + \varepsilon^2(y\chi' + 2y^2\chi'')$ to the second key of the step (the cubic at $y(1 \pm \varepsilon)^2 - y = \pm 2y\varepsilon + y\varepsilon^2$), the shell parts $1 \pm \varepsilon$, and $E_K(cs_w) = E_K(cs)(1 \mp \varepsilon[2 : 3]cy(1 + \xi_w)\rho_1)$ at $\Delta_w = \pm[2 : 3]L(1 + \xi_w)$ to the first key at the moved coordinate's share ξ_w , and the fold collects at the second key, the first keys withdrawing between the upper and the lower moves, to $\varepsilon^2 d E_K(cs)(8y^2\chi'' + 12y\chi' - 8cy^2\chi'\rho_1)$ at $(1 + \xi) + (2 - \xi) = 3$ over the two upper moves; cleared at $\eta d = y\varepsilon^2 d$ the reduced read's value at the vacant step is the stated one. (iii) The row read's cap $\Omega\eta(L^2 - 1)[1 : 8]$ sits below $\Omega y_H[1 : 8]$ on the head; S_H sits at or below $[2 : 3]y_H$ at $L_H^2 - L_H + 1 \leq L_H^2$; the quotient read is Lemma 59's coupling read at the quotient's rise (Lemma 60(ii)), a place z of a shell L at the box's point $\zeta = z[1 : L]$, $\mu = [1 : L - z]$ at or below $[1 : 2]$ at $z \leq L - 2$; the factor coordinates: the crossing shell sits between $L_H[1 : \rho_H]$ and its successor, so $\rho_H L_c - L_H = \theta_c$ at a rounding θ_c in $[0, \rho_H]$, the band top at $L_+ - \rho_+ L_c = \theta_+$ in $[0, 1]$, and the head's top shell's square ηL_H^2 sits at or below y_H below $\eta(L_H + 1)^2$, so $y_H = \eta(L_H + \theta_H)^2$ at a rounding θ_H , the three displays at $L = [1 : \varepsilon]$; the pre read's shells sit at or below $L_c - 1$, at $y \leq y_c$, and $y_c = y\ell^2$ reads $y_H \ell^2[1 : (\rho_H \ell - (\theta_c - \theta_H)\varepsilon)^2] = y_H[1 : (\rho_H - \delta\varepsilon)^2]$ at $\delta = (\theta_c - \theta_H)[1 : \ell]$, between the partner of one and ρ_H at $\ell \geq 1$; the post read's shells run from $L_c + 1$, at $\ell \leq 1 - \varepsilon$, to $L_H - 1$, at $\rho_H \ell - \theta_c \varepsilon \geq 1 + \varepsilon$, the segment's parameter λ ; the profiles' displays are Lemma 60(v)'s at $L - L_c = L(1 - \ell)$ and $L_+ - L_c = L(\ell_+ - \ell)$, the band's difference at the band top's shell $\sigma + 2[1 : L_c]$ at or beyond σ ; the polynomial cap reads in ε at H_{L-2} read at $L - 2$; and the top read sits at $L = L_H$, $\ell = L_c[1 : L_H]$ and $y = \eta L_H^2$. The boundary read sits at the shell two, $\varepsilon = [1 : 2]$. (iv) The witness is a polynomial in the place, the shell, the scale and L_x at every label at or below L_x and the sum's unit beyond, so the form is a polynomial in the scale and L_x at the power sums' closed reads, and at $y_x = \eta L_x^2$ the endpoint's own read puts y_x at or beyond 100 below $100[1 : (1 - \varepsilon_x)^2]$, the excess τ at $y_x = 100(1 + \varepsilon_x\tau)$ below $(2 - \varepsilon_x)[1 : (1 - \varepsilon_x)^2]$, at or below six at $\varepsilon_x \leq [1 : 2]$. The shells below the split shell: at each such shell and each of its labels the read is the box read's polynomial at the shell's own step, the square $y = \eta L^2$ one coordinate of the scale and the produced naturals entering through the roundings at or beyond the split shell, and at the finitely many scale segments on which a produced natural sits below the split shell, one segment per value of the naturals, the read is a polynomial in the scale alone. $\square$

Pending 1 (The sector certificate at the further members and residues). At every member of the classification and every residue beyond the residue-two instance (Construction 20; Lemma 61), the sector's tail certificate at the sector's shells (Construction 19(iii)) holds at a profile of the member's chain with its produced data and its reads (a) to (e); and the corner floor at the member reads the certificate's lines' gap at every tail scale.

Construction 21 (The depth chain of a window). On a window (Definition 8) the *fusion graph* joins two members of the window index, each with its fiber, where a plaquette's term reads a state of the one against a state of the other off the sum's unit (Theorem 17's magnetic member; Definition 9's changed-edge read), a symmetric relation at the term's symmetry, the unit's line one member, the graph over the region's whole index, every content at once; a *sector* is a base's fusion closure, the members the walk from the base reaches, one component of the graph, the unit's line's closure the *vacuum sector*; a member's *depth* is its distance from the base along the

graph, the least count of joined steps, one natural per member, a member joined to one at depth d sitting at depth at most $d + 1$; and the *shells* are the members at one depth, finitely many: a member at depth d has its support within d plaquettes' links of the base's and its contents within the attained maximum Φ iterated d times from the base's against the closure's four-link contents (Proposition 3's Φ ; Proposition 5), one bound per depth (Proposition 2), the occupied depths the counts from the base's up, a member first reached at a count joined to a member first reached at the count before. *The sectors are the charges.* Two occupied configurations of one *charge*, the fluxes of their class families through the transverse cuts one class per direction (Construction 7's boundary read at the vertex law of an occupied vertex, the incident labels' classes folding to the unit), sit in one sector: the two class families differ by a cycle at unit fluxes, a boundary fold of plaquettes at class coefficients (Construction 7's boundary read), and the walk from the one reaches the other at those plaquettes: per plaquette the coefficient's class is a fold of the list's classes (Construction 6), read as a product of the plaquette's closure characters, and per link of the target the target's label sits in the product of the source's label with the plaquette characters' labels around the link, every label at some depth of the closure's fusion from the unit (the *reach read* below), the further factors joined in dual pairs at the unit channel (Proposition 8); and one plaquette term joins a configuration to every admissible target: the term's read at a source state composes the state's vertex members at the boundary's four vertices with the loop's coevaluation, the loop's coefficient contracted at each vertex (Definition 9's changed-edge read), and its component at a target whose boundary labels are constituents of the source's labels against the member is the coevaluation's channel split at every boundary link (Lemma 8(i); the pair networks' split, Theorem 24(ii)), each channel at the weight $[m_b d_b : d_a d_\lambda]$, a summand of the dimension identity (Lemma 5(ii)), positive, so the component is occupied at every admissible target; the target's labels are reached along the associativity's occupied summands, each link's chain of channels at positive counts (Lemma 5(iii)), one plaquette term per step, so a walk joins the two. *The reach read:* every label sits in a product of the closure's members. Every label is a Cartan product of fundamentals at its fundamental fold, one at every step (Construction 6's Cartan strictness), and every fundamental sits in a product of the closure's members at the member's reach list (Construction 13) seeded at the list: at the A -series the fundamentals are the column exhibits, ω_k in the fundamental's k -th power (Lemma 2(iii)); at C_ℓ the reach list runs from the first fundamental, the list's own; at B_ℓ the vector sits in the spinor's square and at D_ℓ in the spinor's square at odd rank, the spinor's class of order four reading the vector's at its double, the minuscule rows (Lemma 26(ii)), and is a list member at even rank, the reach list then running from the vector; at E_6 , E_7 and E_8 the reach list's seeds are the list's members; at F_4 the reach list's seed ω_1 sits in the list member's square, one stated count of the member's table (the uniqueness read, Construction 13), and at G_2 the reach list runs from the list. So at a faithful list the sectors are one per charge, the vacuum sector the unit charge's; at a further list the vacuum sector is the unit label's closure, one component among the charge's. *The depth chain* is the sector's pencil at the shells, the window list's positions at the shells' members: the gram is the fibers' orthogonal sum and the electric member its form at the contents (Definition 8; Definition 10; Theorem 17), so their blocks are the shells' own diagonals, and the magnetic member's terms join members at depths within one of each other, the joined step's move read at the term's symmetry from both members, so the pencil is block tridiagonal over the depth, the shells its slabs (Lemma 34), its bonds the terms' reads between consecutive shells and its diagonal blocks the terms' reads within one shell joined to the shell's contents against the gram. The one-plaquette window's vacuum sector reads its shells at the labels of one fusion depth from the unit label under the closure, the word pairs at reaches summing to the depth at the A -series (Lemma 25(ii)), the pencil there the fusion closure fold N_L against the contents (Lemma 17; Theorem 17); the theta graph's fiber (Construction 9) reads its shells at the label triples of its two plaquettes' unshared links

and the shared link (Definition 8's vertex counts at the two three-valent vertices), a plaquette's term fusing a closure member into one plaquette's labels with the shared link's at the vertices' recoupling (Theorem 24).

Lemma 62 (The moment folds). *On a depth chain's head at the dual representative (Construction 21), at the ground's block recurrence $B_m^\top \psi_m + B_{m+1} \psi_{m+2} = A_{m+1} \psi_{m+1}$ (the eigen-identity's row read at the shells' blocks, the diagonal block at the ground root cross-added, Lemma 34(iii)'s row read), its scalar spelling $b_m \psi_m + b_{m+1} \psi_{m+2} = A_{m+1} \psi_{m+1}$ at single-label shells, the ground enters every datum read through two moment streams,*

$$\rho[P] := \sum_m \psi_m^2 P(m) , \quad c[P] := \sum_m \psi_m \psi_{m+1} P(m) ,$$

P over the polynomial weights, and the streams solve the recurrence's own folds, exact at every scale:

(i) *The fold identities. The recurrence folded against the middle member's $\psi_{m+1} P(m+1)$ reads*

$$c[bP] + c[bP^+] = \rho[AP] \quad (P^+(m) := P(m+1)) ,$$

and folded against $\psi_{m+2} P(m+1)$ a second identity of the recurrence at the shifted weights, its second-bond term reduced by one further fold of the recurrence: every correlator beyond the two streams re-enters them, so the family closes on ρ and c alone, one identity per weight, each a display of the recurrence at a polynomial weight.

(ii) *The graded system, and its clearance. At the corner the weights read in the pair coordinates ($u = m\eta$, $s = m^2\eta$): the bond reads the border rows' one repeated datum (Lemma 41), the folds' bond weights collecting at that datum's factor, and the diagonal A_m reads σE_m at weight-degree two, so each fold relates the streams' degree- k reads to degree- $(k+2)$ at one scale factor, the second fold's reduction withdrawing its top weight to the reads at or below the weight's degree's successor with the boundary square ψ_0^2 , the down-shift's read at the vacant head key. The identities at weights through a stated degree, with the base reads, the unit read $\rho[1]$ at the gram, the depth read $\rho[m]$ and the bond read $c[1]$, each a single-region read of clause (v), and the boundary square, are one square linear system for the truncated moment vector, the two streams' reads through the stated degree with the boundary square, over the located stage, its entries polynomial in the scale and the rates at the eigenvalue's bracket (the certificate's two lines, Lemma 56(vi)), the far edge's reads, the weights' reads at the head's last key against its square, $(n-1)^t \psi_{n-1}^2$ at the head's depth count n , entering the rows at clause (iii)'s priced tail weight: the solve is the elimination's (Definition 3), and the system's determinant clearance is one located divisor read over the corner base (Lemma 31), the validity cell the certificate's architecture.*

(iii) *The truncation cap. Beyond the stated degree the streams' tails read in the dominance region, where the ground's components ride the transfer factors (Theorem 21(ii)): the beyond-degree weight is priced by the certificates' geometric caps against the transfer factors' cap, one comparison at the stated degree, the degree held at its statement, every compared datum a gram weight or a weighted square at or beyond the sum's unit, the pairings' own side (Theorem 21(ii)).*

(iv) *The scale comparison. Two scales' systems share their shape, the coefficients differing by displayed polynomial reads of the scale and the base reads within the scale sandwich's priced widths (Theorem 21(vii)), so on the clearance cell the two moment vectors differ by the elimination's cofactor reads at those differences, one width per moment, and the widths take the scale factor: the streams' reads close across the scales on the clearance cell.*

(v) *The datum assembly. A single-region datum read is a stream read at banded weights, the band its support's reach (Lemma 14): a width-one datum reads the square stream at its diagonal weights*

with the doubled bond stream at its bond weights (Lemma 34(iv)), and a wider band's read re-enters the two streams by (i)'s folds; so its scale closure is (iv)'s on the clearance cell; separated and Euclidean data assemble from single-region reads at the crossing certificates (Theorem 21(iv)–(vi)), the certificates cell data (Lemma 31), so the assembled reads close where the parts do. The block chains read the streams at block weights through the folds of their own recurrence, Lemma 34(iii)'s row read.

(vi) *The height reads at two lines.* On a window with simple ground the height reads at two lines, one count per line: at a floor line under which the window's pencil is positive semidefinite the ground sits at or beyond the line, and at a cap line with the window's count at or beyond two below it the window's second root sits at or below the line (Lemma 33's count read), so the height sits at or below the lines' gap against the scale; the floor is a positive-semidefinite certificate of the window's pencil above the line, a fold of squares at generator actions joined to coefficient multiplications (Construction 8), and the cap the two-by-two block read at two stated members perpendicular in the gram (Lemma 30's block reads), the window's count at or beyond two at every line beyond the floor line by the members' diagonal reads' sum against the gram, the two-by-two compression's count two at or below it; the certificate's data and the members' reads are polynomial pairs in the scale and the side, their comparisons' validity divisors located over the scale and the side (Lemma 31). At a fixed region, suppose the floor line holds on every cutoff from a stated key, and the two members have finite content supports in that region. From the larger of that key and their two contents onward, their forms and Gram are their original reads. At simple grounds on this tail the lines' gap therefore caps the regional height, and hence the global height (Definition 11).

Proof. (i) The recurrence at the middle member $m + 1$ multiplied by $\psi_{m+1}P(m+1)$ and summed over m reads $c[bP^+]$ at the first bond, $c[bP]$ at the second at the index shifted, and $\rho[AP]$ on the right, the display; multiplied by $\psi_{m+2}P(m+1)$ and summed it reads the first bond at the second-bond correlator $\sum_m b_m \psi_m \psi_{m+2} P(m+1)$, the second at ρ at the shifted bond weight and the right side at $c[AP]$, and the row read once more, $b_{m+1} \psi_m \psi_{m+2} + b_m \psi_m^2 = A_{m+1} \psi_m \psi_{m+1}$, returns the second-bond correlator to the two streams. A commutator's ground read is of equal members outright, a commutator of two symmetric operators' pairing at a ground reading equal members, the folds the streams' two reads. (vi) Write $g_x, g_y > 0$ for the two perpendicular members' Gram entries and a, d, c for their two diagonal entries and cross entry in the form above the floor line. That form's positive split gives $c^2 \leq ad$ (Construction 8's squared Cauchy–Schwarz). At $\lambda > [a : g_x] + [d : g_y]$, both entries $a - \lambda g_x, d - \lambda g_y$ sit on their lower sides and

$$(\lambda g_x - a)(\lambda g_y - d) - c^2 = \lambda(g_x g_y \lambda - g_y a - g_x d) + ad - c^2 > 0.$$

Thus the compression has count two, and the full count is at least two (Lemma 30). With the ground simple and at or beyond the floor line, the gap is at most λ . The two members' least contents give their inclusion at every larger cutoff (Proposition 4), with their pairings and pencil forms fixed at that inclusion (Lemma 14). The floor read holds on the stated tail, so the cap does as well; the regional upper-cap read of Definition 11 gives the last clause. $\square$

Lemma 63 (The cone read). *The jump family's momentum structure at the corner is isotropic at degree two: on the decimation's validity cells the decimated determinant's root locus is the fiber symbol's own, one Schur factor at a complement of unequal-membered determinant (Theorem 26(i)); the fiber symbol's coefficients are permutation-invariant polynomials of the chord list at every coupling of the approach (Theorem 22, at Lemma 47's base), their momentum reading even in each coordinate at the deck relation's exchanged roots (Definition 4; Theorem 22); the degree-two invariants at that evenness are the standard square's span (Theorem 22, degree 2 at the rings'*

coincidence); and the direction data enter at degree four, the breaking multiplicities positive there. So the jump family's loci at the corner read the isotropic square at degree two with one located coefficient per root: the symbol's momentum reading is $S_0(k)$ at the unit monomial with $S_2(k)$ against the isotropic square Q at degree two, the odd degrees at the sum's unit and the direction data at degree four; at a located root ε of S_0 , simple at Theorem 16's separation, the locus at the moved level $\varepsilon + aQ$ reads the symbol's degree-two term as $S'_0(\varepsilon)a + S_2(\varepsilon)$ against Q , and the coefficient is $\bar{a}$ over the clearing c at the derivative's unit witness $uS'_0 + vS = c$ cleared at the extension's monic S (Lemma 32), $\bar{a}$ the balance partner of S_2u reduced at S , one stage element per root; and the direction reads two momentum degrees down.

Proof. At a level off the designated pivot's root the cleared deflation's determinant is the pencil's own at one Schur factor (Theorem 26(i); Lemma 30's determinant split), so the decimated determinant's root locus on the validity cells is the symbol's. The symbol's coefficients are one value at every relabeling of the chord list and even in each coordinate (Theorem 22), so its momentum reading holds the isotropic square alone at degree two, the sum's unit at the odd degrees and the direction data from degree four (Theorem 22's degree reads). At the moved level $\varepsilon + aQ$ the symbol reads $S_0(\varepsilon + aQ) + S_2(\varepsilon + aQ)Q$ through degree two, and the binomial identity per occupied key (Theorem 16) reads $S_0(\varepsilon + aQ)$ as $S_0(\varepsilon) + S'_0(\varepsilon)aQ$ with Q 's square two degrees beyond, $S_0(\varepsilon)$ of equal members at the root, so the degree-two term is $S'_0(\varepsilon)a + S_2(\varepsilon)$ against Q , the locus its equal members. At the root simple at the separation the unit witness $uS'_0 + vS = c$ reads $u(\varepsilon)S'_0(\varepsilon) = c$, $S(\varepsilon)$ of equal members (Lemma 32), so at $\bar{a}$ the balance partner of S_2u reduced at S the read $\bar{a}S'_0(\varepsilon) + cS_2(\varepsilon)$ is $S_2(\varepsilon)u(\varepsilon)S'_0(\varepsilon)$ against itself, equal members: the coefficient is $\bar{a}$ over c . $\square$

Lemma 64 (The free cell). *The pencil at the $\alpha=1$ representative, the pair $(E:\tau^2M)$'s site datum $H([1:\tau^2])$, is affine in τ^2 , and on a fiber head (Theorem 30) its entries are one polynomial-pair list in (t,τ) over the residue beyond the locality band (Lemma 40; Lemma 14), the window entering through its sampled momenta and its separation range alone, the residue a further coordinate beyond the reach bound (Lemma 25(iii); Lemma 26(iii) at the further series): every count near the free end is Lemma 31's cell function over (t,τ,r) with located divisors, and the free cell is the free end's, its τ -extent the divisor's first positive root, an output, the contact cell's construction at this end (Lemma 45). At each window and coupling the ground is the least located root ε_0 of its pencil (Definition 10; Lemma 33), and the certificate's fixed data are positive offsets $\ell < h$, common to every region and cutoff on the stated root-coordinate segment. Its two count reads are*

$$\text{rev}(H : (\varepsilon_0 + \ell)G) = \text{rev}(H : (\varepsilon_0 + h)G) = 1 .$$

The level gap reads these counts at ℓ and h , and the absolute lines read the ground at that window and coupling. The polynomial displays include a ground coordinate e : the site data at $H = (e + \ell)G + S_\ell$ and $H = (e + h)G + S_h$ are polynomial in e and the base coordinates, and their counts enter at $e = \varepsilon_0$, the least root's designation. The cell's boundary families are the decimation's divisor list at the $\alpha=1$ representative (Theorem 19): the jump family, the pencil determinant's roots at the two ground-translated lines, the validity families of the stated pivot certificates, with the dominance family at a dominance certificate, the deck family with the spectator caps' three counts each a cell datum (Lemma 35(iii)), and the tension family of the charged sectors (Lemma 43(iii),(iv)); the two total counts give ground multiplicity one, with the ground below the lower line and every further level at or beyond the upper line. The free cell's certificate is the flat window at the fixed pair of offsets, the ground-relative level read's pair, read outright at the free end (Lemma 15) with the counts at the two lines cell functions (Lemma 31). At every region, every cutoff and every root-coordinate pair of the cell, its record reads these two counts at that cutoff's ground designation. The floor

satisfies $\ell + E_0\kappa_1 \leq h$. At the specialization $\gamma = E_0\kappa_1$, $\ell = \gamma$, $h = 2\gamma$, a cutoff C with the positive comparison

$$\gamma C > \gamma W + \gamma^2 + W^2, \quad W = \tau^2 \# p d_L,$$

and the upper count give the cutoff-tail certificate of Theorem 30; the lower count is its head's count at $e_C + \gamma$, one by monotonicity and the ground below that line. The floor and extent are located outputs per member and direction count (Construction 7's direction reads): the extent τ_1 , the first positive member in the root coordinate of the families' root list, the members' root lists joined, over the band, the chords at t^2 below 4 (Lemma 40), at its leading-term radius in the residue (Theorem 16; Lemma 25(iii); Lemma 26(iii)), the cutoff and its ground designation among each window record's data, with its sampled momenta and the spectator brackets read at that record, the deck family's cell data (Theorem 19(iii); Lemma 35(iii)). A finite chained cover reads the two total counts at every point of its stated segment: each piece has its own pivot witnesses, and at a shared endpoint the two pieces read the original pencil's counts (Theorem 19; Lemma 31). The validity comparisons are the fields of those witnesses on their pieces.

The scalar dominance instance. At positive data Λ_h, d_L, p_v, f_* , a row-cap record at matrices E, M and stated tower places reads

$$\Lambda_h \leq c_i, \quad f_* m_i \leq 2d_L p_v c_i, \quad c_i := E_{ii}, \quad m_i := \sum_{j \neq i} |M_{ij}|$$

at every stated place, the full row fold at the matrix's column list. Here p_v is the per-vertex plaquette count (Construction 7). At a fixed upper level $u < \Lambda_h$, its two comparison polynomials are

$$d_0(\tau) := f_* \langle \Lambda_h : u \rangle - 2d_L p_v \Lambda_h \tau^2, \quad d_1(\tau) := f_* - 2d_L p_v \tau^2.$$

At a diagonal electric matrix, the unit Gram, a level $z \leq u$ and the displayed cap read, the row gap has the comparison

$$f_*(c_i - z - \tau^2 M_{ii} - \tau^2 m_i) \geq d_1(\tau)(c_i - \Lambda_h) + d_0(\tau) - \tau^2 f_* M_{ii}.$$

Where the right member is positive at every stated row, it gives a positive dominance pivot at those places. At $M_{ii} \leq 0$ this holds at $d_0(\tau) > 0$ and $d_1(\tau) \geq 0$. At the ground-translated line, $u = h$ gives $e + h \leq u$ at $e \leq 0$ (Definition 10). Each displayed polynomial has one positive root, its constant and quadratic coefficient at opposite sides with positive margins. This instance's root list is occupied, and its first positive member is a located read of that record. A certificate's count covers verify its actual pencil and Gram on the stated range. The cell's t -section at the extent is the whole band exactly where the list's t -roots sit off it, itself a located read.

The free end is interior to its cell. At the free end the pencil is its electric member E , diagonal at the free reads: the decimated symbol's jumps are the occupied contents and the level is the member floor label's loop read, one diagonal computation at every window (Lemma 15); the free end sits inside its cell with its counts the cell's, direct reads; at a level on the content list, Lemma 31's boundary clause extends the positive-semidefinite cells into the closed endpoint. The free-end reading of Definition 11 is those reads.

The first creation threshold. Let e_L be the least of $4\hat{c}_2(\lambda)$ over $\lambda \in \bar{L}$, the trichotomy fold over the magnetic closure (Definition 1; Construction 6). At every member and direction count, every region and every total-content cutoff $C < E_0 + e_L$, the window cut admits every positive pair κ at the read

$$\nu d_L \tau^2 + f_* \kappa \leq f_*, \quad f_* := \hat{c}_2(f_0), \quad E_0 = 4f_*.$$

The threshold reads $2E_0$ where the magnetic closure holds a floor label. Here ν is the per-link plaquette count of Theorem 9. At the window's ground ε_0 the two counts are

$$\text{rev}(H : E_0 \kappa G) = \text{rev}(H : (\varepsilon_0 + E_0 \kappa)G) = 1.$$

At the residue one and the member's stated magnetic list $\{f\}$, the window cut also holds throughout $C < 3E_0$, at every region and direction count, at

$$3\nu d_L \tau^2 + 2f_* \kappa \leq 2f_*.$$

The corresponding ground-translated count is one.

At residue one, list $\{f\}$ and cutoff $C = E_0 =: c$, the whole window at a region of N plaquettes has forms

$$G = I, \quad H = \begin{pmatrix} 0 & -\tau^2 \mathbf{1}^\top \\ -\tau^2 \mathbf{1} & cI_N \end{pmatrix}.$$

For $N \geq 1$ and $\tau > 0$, its least root is $-\rho_N$ at the positive located datum

$$\rho_N(c + \rho_N) = N\tau^4.$$

The further roots are c at multiplicity $N - 1$ and $c + \rho_N$ once. Thus the cut admits one at every root-coordinate pair. At two directions the display holds also on the odd tori of Lemma 40: the unit joint chord fiber has roots $-\rho_N, c + \rho_N$, and every further occupied fiber reads c at its order. The ground-relative distances are $c + 2\rho_N$ and $c + \rho_N$, respectively.

The creator range and its complementary count. At every member and magnetic closure, fix a region and total-content cutoff C . Let $\mathcal{R}_C$ be the span of the images of the admitted maps $\mathcal{C}_i$ of Theorem 9, each at its input selection $c + e_i \leq C$. Its orthogonal complement is

$$\mathcal{N}_C := \bigcap_i \ker \mathcal{C}_i^\dagger, \quad \mathcal{N}_C^+ := \mathcal{N}_C \cap \mathbf{1}^\perp.$$

The unit line sits in $\mathcal{N}_C$, and the window splits orthogonally as

$$\mathcal{W}_C \perp \mathcal{N}_C^+, \quad \mathcal{W}_C := \mathbb{P}\mathbf{1} \perp \mathcal{R}_C.$$

The list of $\mathcal{W}_C$ starts at the unit, followed by the creator images' collection, run per electric content. The list of $\mathcal{N}_C^+$ is the collection of the occupied window list's residuals off those images, per content (Lemma 3); the Grams are the lists' actual pairings. Let W, Z be the two lists' columns in the output window list. Their forms are the restrictions of Definition 10, in particular

$$G_N = Z^\top GZ, \quad E_N = Z^\top EZ, \quad R_N = Z^\top G R_\Pi Z, \quad H_N = E_N - \tau^2 R_N.$$

At a located line z , a pair ξ and a positive datum ω of z 's stage at

$$\nu d_L \tau^2 + \xi = f_*, \quad f_* z + \omega = \xi E_0,$$

its site datum $S_N(z) := H_N - zG_N$ has the positive fold

$$f_* S_N(z) = \tau^2 (\nu d_L E_N - f_* R_N) + \xi (E_N - E_0 G_N) + \omega G_N \succ 0.$$

At $B := W^\top HZ$, the form at the combined list is

$$(W \ Z)^\top (H - zG) (W \ Z) = \begin{pmatrix} H_W - zG_W & B \\ B^\top & S_N(z) \end{pmatrix}.$$

The count is

$$\text{rev}(H : zG) = \text{rev}(H_W - zG_W - BX), \quad S_N(z)X = B^\top.$$

The unit row of B reads the sum's unit, so the deflation fixes the unit row of $H_W - zG_W$. At a vacant $\mathcal{N}_C^+$ the count reads on $\mathcal{W}_C$ directly.

At an occupied magnetic closure and a pair $h < E_0$, this complementary pivot has one common validity divisor,

$$d_h(\tau) := f_*\langle E_0 : h \rangle - \nu d_L E_0 \tau^2.$$

It has exactly one positive located root τ_N ,

$$\nu d_L E_0 \tau_N^2 = f_*\langle E_0 : h \rangle.$$

At every root-coordinate pair $\tau < \tau_N$, every region and every cutoff, the complementary count holds at $z = e + h$, where e is that window's ground. The divisor and its root depend on the member, direction count, closure dimension fold d_L and offset.

The translated line on its upper side. Let e^+ be the largest $4\hat{c}_2(\lambda)$ over the occupied magnetic closure, the trichotomy fold over that list. At $\tau < \tau_N$ and

$$d_h(\tau)^2 \geq 4f_*\nu E_0 h(h + e^+),$$

every window whose ground satisfies $e + h \geq 0$ reads

$$\text{rev}(H : (e + h)G) = 1.$$

This holds at every total-content cutoff and every region, the comparison's data the member, direction count, closure folds d_L, e^+ , offset and root-coordinate pair.

The ground-side root. At the complementary data, write $\delta := d_h(\tau)$, $c_\rho := 4f_*\nu E_0$, and let a_ρ, b_ρ be the positive gaps at

$$a_\rho + f_*^2 = c_\rho, \quad b_\rho + 2f_*\delta = c_\rho e^+.$$

At the positive quadratic root q_ρ and its read

$$q_\rho^2 = b_\rho^2 + 4a_\rho\delta^2, \quad \rho_* := [2\delta^2 : b_\rho + q_\rho],$$

one has $a_\rho\rho_*^2 + b_\rho\rho_* = \delta^2$. At every region and cutoff whose ground reads $e^\vee \leq \rho_*$, the count is

$$\text{rev}(H : (e + h)G) = 1.$$

At $\delta^2 \geq c_\rho h(h + e^+)$ the root satisfies $\rho_* > h$, so this count includes the lines $e + h < 0$ at $h < e^\vee \leq \rho_*$. Under this comparison, at $e^\vee > \rho_*$, the unit pivot at the line $e + h$ exceeds the positive stage gap $\langle \rho_* : h \rangle$.

The comparisons' common segment. At $c_\rho := 4f_*\nu E_0$, define the positive data

$$a_0 + f_*^2 = c_\rho, \quad b_0 := 2f_*^2 E_0 + c_\rho e^+, \quad k_0 := f_*^2 E_0^2, \quad h_0 := [k_0 : 2(b_0 + a_0 E_0)].$$

Then $0 < h_0 < [E_0 : 4]$. Let q_0 be the positive root at $q_0^2 = c_\rho h_0(h_0 + e^+)$, let w_0 be the positive gap at

$$q_0 + w_0 = f_*\langle E_0 : h_0 \rangle,$$

and let T_0 be the positive root at

$$\nu d_L E_0 T_0^2 = w_0.$$

At every root-coordinate pair $\tau \leq T_0$, the complementary comparison at $h = h_0$ reads

$$d_{h_0}(\tau) \geq q_0 > 0, \quad d_{h_0}(\tau)^2 \geq c_\rho h_0(h_0 + e^+).$$

The datum h_0 and these located roots depend on the member, direction count and magnetic closure alone. The complementary count and the translated-line counts above consequently hold at these data on their stated cutoff and ground ranges.

The count on the input lists. Let $\mathcal{C}$ be the joined column list of the maps $\mathcal{C}_i$, with domain the direct sum of their input windows, each in its window list at Gram G_i^- , and let $\mathcal{J}$ be the joined list of the maps $\Pi[\mathbf{M}, \mathcal{C}_i]\Pi_i^-$. At the data of the complementary count, put

$$\Gamma := \mathcal{C}^\top G \mathcal{C}, \quad \Theta := \mathcal{C}^\top G \mathcal{J}, \quad L_N := Z^\top G \mathcal{J}, \quad S_N(z)Y_N = L_N,$$

and let $\mathcal{T}_z$ be the direct sum of the input actions $K_i^- + (e_i - z)I$. Let $\mathbf{r}$ be the column with one at each input window's unit key and the sum's unit at every further key. Then

$$\text{rev}(H : zG) = \text{rev} \begin{pmatrix} -z & -\tau^2 \mathbf{r}^\top \\ -\tau^2 \mathbf{r} & \Gamma \mathcal{T}_z - \tau^2 \Theta - \tau^4 L_N^\top Y_N \end{pmatrix}.$$

The range block is symmetric at the identity

$$\Gamma \mathcal{T}_z - \tau^2 \Theta = \mathcal{C}^\top (H - zG) \mathcal{C} = \mathcal{T}_z^\top \Gamma - \tau^2 \Theta^\top.$$

The input directions in $\ker \mathcal{C}$ give a kernel block of the displayed count. At $z = e + \gamma$, with e the output ground and $0 < \gamma < E_0$, each input action has the positive-definite read

$$K_i^- + (e_i - z)I = (K_i^- - eI) + (e_i - \gamma)I \succ 0$$

in its input pairing.

The Gram blocks read at two smaller input selections. Let Π_{ij}^- select the contents at $c + e_i + e_j \leq C$, and put $\mathcal{C}_{j|i} := \mathcal{C}_j \Pi_{ij}^-$ with output in Π_i^- , and $\mathcal{C}_{i|j} := \mathcal{C}_i \Pi_{ij}^-$ with output in Π_j^- . On their input and output lists,

$$\Gamma_{ij} = G_i^- (\mathcal{C}_{j|i} \mathcal{C}_{i|j}^\dagger + \Pi_i^- [C_i^\dagger, \mathcal{C}_j] \Pi_j^-).$$

At $C < e_i + e_j$ the first term reads the sum's unit. At disjoint plaquette boundaries the commutator term reads the sum's unit (Theorem 9).

The parity elimination at residue one. At the magnetic list $\{f\}$, the configuration index of every region and total-content cutoff has two grades, the unit in the even grade and each plaquette multiplication exchanging the grades. On their lists the forms read

$$G = G_+ \oplus G_-, \quad H = \begin{pmatrix} E_+ & -\tau^2 B \\ -\tau^2 B^\top & E_- \end{pmatrix}.$$

At every located line $z < E_0$, the odd block $D_- := E_- - zG_-$ is positive definite and the count is

$$\text{rev}(H : zG) = \text{rev}(E_+ - zG_+ - \tau^4 BX), \quad D_- X = B^\top.$$

The solve is blockwise on odd configurations: at content c_a , its block is $(c_a - z)G_a$, the positive scalar gap against the configuration Gram's adjugate solve (Definition 3). Write $D_-(z) := E_- - zG_-$ and X_z for that solve, and put $F_z := E_+ - zG_+ - \tau^4 BX_z$. At the ground $e := \varepsilon_0$ and a positive pair γ with $e + \gamma < E_0$, the two-line read is

$$\begin{aligned} \text{rev}(H : (e + \gamma)G) &= \text{rev}(F_e : \gamma W_\gamma), \\ W_\gamma &:= G_+ + \tau^4 B Y_\gamma \succ 0, \quad D_-(e + \gamma)Y_\gamma = G_- X_e. \end{aligned}$$

The form F_e is positive semidefinite, and its kernel maps bijectively to the ground kernel by $v \mapsto (v, \tau^2 X_e v)$.

Proof. At a fixed window and a level off the content list the cell's determinant reads unequal members at the free end, the diagonal's entries off the level, so the free end sits off the divisor and inside its cell, its counts the cell's. The counts at the certificate's two ground-translated lines are read at the ground's designation in the polynomial displays with coordinate e (Lemma 31). At the free end the ground reads the sum's unit and the absolute lines read the offsets (Definition 10), where the pencil is its diagonal electric member (Lemma 15) and the ground alone sits below either line, so both counts read one on the cell, the ground's multiplicity one over the cell attached to the free end (Theorem 10; Theorem 26(iii)), and every further level sits beyond the upper line.

For the scalar dominance record, subtracting the displayed lower member from the row gap's f_* -multiple gives the positive fold

$$f_*(u - z) + \tau^2(2d_L p_v c_i - f_* m_i) \geq 0.$$

The diagonal magnetic term enters both compared members at $\tau^2 f_* M_{ii}$'s balance partner. A positive right member makes the actual row gap positive; the dominance read of Theorem 19(ii) then gives the positive pivot. The principal block's off-row fold is a subfold of m_i , so its row gap sits at or beyond the full row's. At $M_{ii} \leq 0$ the diagonal term is on its upper side or at equal members, and $c_i - \Lambda_h \geq 0$ gives the stated instance. The two polynomials have positive constant margins at $u < \Lambda_h$ and positive quadratic coefficients on the second member; their positive roots are the quadratic extensions' positive members (Lemma 37). For a finite chained count cover, a point of the segment belongs to a piece, whose split reads the original pencil's total count by Theorem 19's counted-pivot identity. At a shared endpoint either piece reads that count. This gives the count on the covered union (Lemma 31).

For the cutoff read, project off the unit line to the occupied index, with projection Π_+ . Every occupied input has content at least E_0 (Proposition 7), and each creation $C_{p,\lambda}$ of Theorem 9 raises it by $4\hat{c}_2(\lambda) \geq e_L$. Thus at $C < E_0 + e_L$ its occupied-to-occupied compression is the sum's unit, and its adjoint's is as well. The occupied compression of K_Π is therefore that of $\mathbf{E} - \tau^2 \mathbf{R}$. The displayed comparison gives a positive gap a at $\nu d_L \tau^2 + a = f_*$, with $a \geq f_* \kappa$. On the occupied subspace the electric-relative read of Theorem 9 yields

$$f_* \langle x, K_\Pi x \rangle_G \geq a \langle x, \mathbf{E} x \rangle_G \geq a E_0 \langle x, x \rangle_G \geq f_* E_0 \kappa \langle x, x \rangle_G.$$

The pair at $E_0 \kappa$ is consequently positive semidefinite on a subspace of codimension one, so its reversal count is at most one (Lemma 30's dual clause). The unit vector's pencil form is the sum's unit (Definition 10), and its form after withdrawing $E_0 \kappa G$ is strictly on its lower side, so the count is at least one. It equals one. The least root satisfies $\varepsilon_0 \leq 0$, and the line $\varepsilon_0 + E_0 \kappa$ sits at or below $E_0 \kappa$ and strictly above the ground. Its count is at most one by monotonicity and at least the ground multiplicity by the split (Lemma 33), so both that count and the ground multiplicity are one. The spectral count then gives the window cut (Lemma 20). The creation threshold is attained on two plaquettes with disjoint boundaries: the f_0 -loop on the first has content E_0 and sits in P_p for the second; creation there at a closure member of loop content e_L is an isometry on that input (Theorem 9), with occupied output at content $E_0 + e_L$. A box containing the two plaquettes admits this read, and the three-square chain contains such a pair at $d = 1$.

The two-creation range at residue one. Write $c := E_0 = [3 : 2]$, $f_* = [3 : 8]$, and $b := \tau^2$. The magnetic closure is $\{f\}$, with $d_L = 2$ and one creator $C_{p,f}$ per plaquette. The next occupied content after c is at least $c' = [9 : 4] = [3 : 2]c$. An occupied support has every vertex of degree at least two (Proposition 7). Five links would give at most five vertices; at five vertices every degree would be two, an odd cycle, and at at most four vertices a simple bipartite graph has at most four links. Thus a support off the four-link cycles has at least six links, at content at least $6f_* = c'$.

A four-link cycle has one label around it, the two-valent unit read (Proposition 8); at residue one the label of row length n reads $[n(n+2) : 8]$ per link (Definition 6). Beyond $n = 1$ this is at least one, giving content at least four on the cycle. The six-link boundary of two adjacent plaquettes at the fundamental label attains c' . Four-link cycles in a box are its plaquettes: their four unit coordinate steps use two directions, each once forward and once backward; the square chain has its own four-link cycles at its plaquettes.

For $c \leq C < 3c$, let N be the plaquette count. The unit, the N fundamental loops, and the admitted products of two loops at disjoint boundaries are orthonormal, the pair norm the creation isometry's read (Theorem 9). Two distinct pair products have distinct supports: a plaquette in one pair off the other pair meets each of the other's plaquettes in at most one boundary link, so at least two of its four links lie off that pair. Orthogonality follows from the configuration Gram (Definition 8). At a positive scalar u , the action

$$K_u := uE_\Pi - b \sum_i (a_i + a_i^\dagger)$$

on these groups has matrix

$$\begin{pmatrix} 0 & -b\mathbf{1}^\top & 0 \\ -b\mathbf{1} & ucI & -bB \\ 0 & -bB^\top & 2ucI \end{pmatrix},$$

where a column of B is $e_i + e_j$ for an admitted pair. Let Q be the Gram fold $\sum (e_i + e_j)(e_i + e_j)^\top$ over the excluded unordered pairs, every pair excluded at $C < 2c$. Counting all unordered pairs gives

$$BB^\top + Q + 2I = NI + \mathbf{1}\mathbf{1}^\top, \quad Q \succeq 0.$$

The six-link sources. At residue one the occupied contents below $2c$ are c and c' alone. The four-link and five-link cases are the reads above. Seven links below $2c = 3$ would all have label f , since one label at or beyond θ raises the content to at least $[26 : 8] > 3$. The f -link subgraph has even degree at every vertex, the class read at residue one (Construction 6; Lemma 43(i)), and therefore splits into even cycles in the bipartite box; its link count is even. For six links, at most one label could be θ , leaving five f -links, which the even count excludes; a label beyond θ already exceeds the content bound. Thus all six labels are f , their even subgraph one six-cycle, since two cycles have at least eight links. Eight links have content at least $2c$.

For $C < 3c$, the action K_u consequently has the preceding block on the unit, single loops and pair products and a block from six-cycle states of content c' to their created plaquette products of content $c' + c$. Their map is $B_6\chi_s := \sum_p a_p\chi_s$, the six-cycle s at its character. Source states have identity Gram; the product images read in their own Gram. At $C < c' + c$ the map reads the sum's unit. At $C \geq c' + c$, its diagonal square is

$$(B_6^\dagger B_6)_{ss} = N - m_s,$$

where m_s counts plaquettes whose boundaries meet the six-cycle's links: the other creators are isometries on χ_s and their images have distinct supports (Theorem 9).

Two product images can share a support at three decompositions of four paths, of lengths 1, 3, 3, 3. To read this list, a six-cycle and its disjoint-boundary plaquette share at most two vertices: three square vertices would require two paths of length at least three between adjacent square vertices, the square edges excluded, and a path of at least two between diagonal vertices, exceeding six links. At two diagonal vertices the square's two length-two paths and a length-two part of the six-cycle would give three common neighbors of two grid vertices; distinct grid vertices have at

most two, their coordinate differences giving two orders of two unit steps at most. At two adjacent vertices the square gives paths of lengths one and three, and the six-cycle gives two paths of length three. These four paths have exactly three decompositions into a square and a six-cycle. At one shared vertex or at disjoint vertices the decomposition is unique. A six-cycle has at most one link joining opposite vertices: two such links give a pair of vertices with three common neighbors. Such a link is its chord, the length-one path.

On the four paths let U_0, U_1, U_2, U_3 be their coefficient products from one common end to the other. The three states are $\chi_f(U_0 U_i^\dagger) \chi_f(U_j U_k^\dagger)$, with $\{i, j, k\} = \{1, 2, 3\}$. Their diagonal Gram reads one. Each off-diagonal read is $[1 : 4]$: contracting U_0 between two traces gives $[1 : 2]$ times the joined trace, contracting the next path gives another $[1 : 2]$, and the final character pair reads one. These are the coefficient block orthogonality and character unit reads (Construction 8; Proposition 8); the three-link paths read their one-edge coefficients by convolution (Theorem 24(i)). The self-dual character fixes the trace under path reversal.

If s has a chord, its two length-three paths close at that chord to two plaquettes. Let t_s count the further plaquettes at the chord. Each gives two off-diagonal entries $[1 : 4]$ in the row of $B_6^\dagger B_6$, the other two decompositions. Each resulting six-cycle determines the added path and its plaquette, so these entries have distinct keys. At a six-cycle without a chord, $t_s = 0$ and the row is diagonal. The count t_s injects into the m_s list: translate each further chord plaquette along the detour of one of the first two plaquettes. Its chord moves to the parallel outer edge of that plaquette, a link of s . The translated plaquette is in the box; a different plane has its detour coordinate constant at either of the two admitted positions, and the opposite plaquette in the detour's plane translates to the first plaquette itself. Translation is injective, so $t_s \leq m_s$. On the square chain the chord belongs to the first two plaquettes alone, and $t_s = 0$.

It follows that

$$B_6^\dagger B_6 \preceq NI.$$

Below the target content the map is the sum's unit. At $C \geq c' + c$ the comparison's form is the fold of differences' squares at $[1 : 4]$ over the paired row keys, joined to the diagonal squares at $m_s - [t_s : 2] \geq 0$, the row's off-diagonal fold $[t_s : 2]$. This is the diagonal dominance split (Theorem 19(ii)).

The scalar u is the cofactor at

$$f_* u + \nu d_L b = f_*.$$

At $3\nu d_L b + 2f_* \kappa \leq 2f_*$ it reads $[1 : 3] < u \leq 1$. With $\gamma := c\kappa$, the parameter reads are

$$2\gamma \leq (3u - 1)c, \quad \gamma \leq uc.$$

The second follows from $u \leq 1$. The remainder comparison of Theorem 9 gives $K_\Pi \succeq K_u$ in their pairing, directly at the electric form.

The actual pencil's compression to the unit and single loops has magnetic entries one at the vacuum couplings and the sum's unit between single loops. An entry of the latter kind is the evaluation of three plaquette characters. The plaquettes occurring an odd number of times form a list of one or three members. At three members a boundary has at least two links outside the other two boundaries, since distinct plaquettes share at most one link. Thus some link has odd fundamental class, and its unit coefficient is the sum's unit (Construction 8; Construction 6). The entry at one member has that class on its boundary as well. For $N \geq 1$ and $b > 0$, the compression to the unit and the sum of the single loops therefore has least root $-a$, where $a(a + c) = Nb^2$ and $a > 0$ (Lemma 33). The actual ground satisfies $\varepsilon_0 \leq -a$ by compression (Lemma 30). Put

$$\rho := \varepsilon_0^\vee \geq a, \quad z := \varepsilon_0 + \gamma = \gamma - \rho, \quad c_u := uc, \quad \sigma := c_u - z, \quad d_u := c_u + \sigma.$$

Both σ and d_u are positive. The two reads

$$\begin{aligned}\rho(c + \rho) &\geq Nb^2, \\ \sigma d_u &= \rho(c + \rho) + \rho((3u - 1)c - 2\gamma) + (uc - \gamma)(2uc - \gamma)\end{aligned}$$

give $\sigma d_u \geq Nb^2$. At the line z , the pair block of K_u is $d_u I$. Its deflation onto the unit and single loops is

$$\begin{pmatrix} -z & -b\mathbf{1}^\top \\ -b\mathbf{1} & v_0 I + hQ - h\mathbf{1}\mathbf{1}^\top \end{pmatrix}, \quad h := [b^2 : d_u], \quad v_0 := [\delta + 2b^2 : d_u] > 0,$$

where $Nb^2 + \delta = \sigma d_u$. On the hyperplane $x_0 + [b : c_u] \sum_i x_i = 0$, its form is

$$v_0 \sum_i x_i^2 + hx^\top Qx + [b^2 \sigma(2c_u + \sigma) : c_u^2 d_u] \left(\sum_i x_i \right)^2,$$

a positive fold, with the pair-block directions joined through the deflation's congruence (Lemma 30).

For the six-cycle block put $v_1 := uc' - \gamma > 0$ and $v_2 := u(c' + c) - \gamma > 0$. Its target at z has scalar $v_2 + \rho$ in its pairing, and its deflated source reads at or beyond

$$(v_1 + \rho - [Nb^2 : v_2 + \rho])I \succ 0,$$

since $B_6^\dagger B_6 \preceq NI$ and

$$(v_1 + \rho)(v_2 + \rho) = \rho(c + \rho) + \rho((4u - 1)c - 2\gamma) + v_1 v_2 > Nb^2,$$

at $2c' + c = 4c$ and $(4u - 1)c - 2\gamma \geq uc > 0$. The orthogonal complement to these blocks, collected per electric content, reads K_u at uE_Π alone: a creation would reach $3c$ or beyond, and an adjoint image below $2c$ would be a floor loop or a six-cycle, already included among the sources. The complement's pairings against their created images are the sum's unit. Its contents exceed c , so their u -multiples exceed $z \leq \gamma \leq uc$.

The positive subspace of the whole lower pencil has codimension one, giving

$$\text{rev}(H : (\varepsilon_0 + \gamma)G) \leq \text{rev}(GK_u : zG) \leq 1.$$

The first comparison is $K_\Pi \succeq K_u$, read as forms (Lemma 30); the actual ground contributes at least one below $\varepsilon_0 + \gamma$. Thus the count and the ground multiplicity are one, and the cut follows (Lemma 20). At $N = 0$ the region is acyclic; at $C < c$ its window is the unit line; and at $b = 0$ the free diagonal read holds (Lemma 15). These give the further cases of the stated range.

At $C = c$ the support classification above leaves the unit and the fundamental plaquette loops alone. Their Gram is the identity, their electric entries $0, c, \dots, c$, and their magnetic entries are the unit couplings at one, with the loop-to-loop entries at the sum's unit by the three-character read above. Let u be the loops' sum. The coefficient lists whose sum is the sum's unit form a subspace of dimension $N - 1$, perpendicular to the unit and u , at pencil cI . On $(\mathbf{1}, u)$ the forms are

$$\begin{pmatrix} 0 & -\tau^2 N \\ -\tau^2 N & cN \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & N \end{pmatrix}.$$

Adjoin the positive root q at $q^2 = c^2 + 4N\tau^4$ (Lemma 37). It exceeds c ; the gap at $c + 2\rho_N = q$ gives the displayed product identity. The two roots of this pair are $-\rho_N, c + \rho_N$, by its determinant and Lemma 33. Its least root is strictly below the further roots, and every positive level gap is at

or beyond c , giving the cut at one (Lemma 20). The cases $N = 0$ and $\tau = 0$ are the unit line and the free-end diagonal read.

On an odd torus of side at least five, a cycle of at most four links has its coordinate steps within one unwrapped box, since an occupied winding uses at least five links. The four-link support classification and the magnetic entry computation therefore hold there as well. At two directions translations permute the plaquettes transitively, so the fixed loop list is the line at u . Every further joint chord fiber lies in its perpendicular complement and reads c . The unit line with u gives the displayed two roots (Lemma 40).

The creator range's complement. A vector pairs every creator image at the sum's unit exactly where every creator adjoint reads it there, by the pairing adjoint's identity (Definition 10). Thus the joint kernel is the range's orthogonal complement. The creators raise electric content, so their images pair the unit line at the sum's unit. Their range is preserved by the electric action: the image $\mathcal{C}_i x$ has its electric image $\mathcal{C}_i(Ex) + e_i \mathcal{C}_i x$. The orthogonal complement is preserved as well, the electric action self-adjoint. Collection per content therefore gives the displayed orthogonal decomposition, with positive Grams and full span (Lemma 3; Definition 3).

On the joint kernel, every full $C_i^\dagger$ reads the sum's unit: its image from the output window lies in its stated input window (Theorem 9). The creation pairings between two joint-kernel vectors read that unit at the adjoint identity, and the annihilation pairings do so outright. The plaquette split consequently gives $H_N = E_N - \tau^2 R_N$. The remainder comparison gives $\nu d_L E_N - f_* R_N \succeq 0$, and the electric floor gives $E_N - E_0 G_N \succeq 0$ on the part perpendicular to the unit (Theorem 9; Proposition 7). Substituting the two witnessed joins into the site datum gives the displayed positive fold; its last term is positive definite at $\omega > 0$ and the positive Gram. The block is therefore a positive principal pivot, with its solve at the adjugate against its determinant, and inertia gives the stated count (Definition 3; Lemma 30).

The unit's Hamiltonian image is $K_\Pi \mathbf{1} = -\tau^2 \sum_i \mathcal{C}_i \mathbf{1}$, the electric and remainder reads at the sum's unit and the annihilation read there as well. It belongs to $\mathcal{R}_C$, so it pairs $\mathcal{N}_C^+$ at the sum's unit. This is the unit row of B , and the unit row of BX reads that unit. The vacant complementary list contributes its vacant count, with the full window then spanned by $\mathcal{W}_C$.

For the ground-translated line, at $d_h(\tau) > 0$ its positive margin gives the pair $\xi := [f_* h + d_h(\tau) : E_0]$. The divisor identity gives $\xi + \nu d_L \tau^2 = f_*$. The ground satisfies $e \leq 0$ (Definition 10), so the second witness is

$$\omega = d_h(\tau) + f_* e^\vee > 0, \quad f_*(e + h) + \omega = \xi E_0.$$

These are the two witnesses of the complementary count, at every window. The divisor's constant is positive at $h < E_0$, and its quadratic term is the balance partner of a positive datum at the occupied closure. Its positive root is the quadratic extension's positive member (Lemma 37); the trichotomy of squares puts the divisor on its upper side exactly below that root among positive root coordinates.

The translated line on its upper side. Put $b := \tau^2$ and $\rho := e^\vee$. The active closure members are those at $e_i \leq C$, their count m_C , and the admitted creators' count is $s := Nm_C$ at the region's plaquette count N . If s is occupied, let $u := \sum_i \mathcal{C}_i \mathbf{1}$ over those creators. Its loop summands are orthonormal: at one plaquette this is character orthogonality, and at different plaquettes an unshared link reads a nonunit coefficient against the unit (Construction 8; Proposition 8). Thus

$$\langle u, u \rangle = s, \quad \langle u, Eu \rangle \leq s e^+, \quad \langle \mathbf{1}, K_\Pi u \rangle = -bs.$$

The magnetic read $\langle u, Mu \rangle$ sits on its upper side or at equal members. A term at three distinct plaquettes has an unshared link: each square shares at most one link with each other square, so at least two of its links occur there alone. At two distinct plaquettes one has such a link as well.

Those evaluations read the sum's unit. The terms on one plaquette are unit coefficients of products of characters, natural folds of fusion counts (Proposition 8). Consequently $\langle u, K_{\Pi} u \rangle \leq se^+$.

The ground form $H - eG$ is positive semidefinite. Its compression to $(\mathbf{1}, u)$ has diagonal entries ρ and $\langle u, K_{\Pi} u \rangle + \rho s$, and off-diagonal entry $-bs$. Its determinant read therefore gives

$$b^2 s^2 \leq \rho(\langle u, K_{\Pi} u \rangle + \rho s) \leq \rho s(e^+ + \rho).$$

With $z := e + h \geq 0$, one has $0 \leq \rho \leq h$, so

$$sb^2 \leq \rho(\rho + e^+) \leq h(h + e^+).$$

At a vacant s this comparison holds outright.

For a vector x perpendicular to the unit, put $n := \langle x, x \rangle$, $t := \langle x, \mathbf{E}x \rangle$ and $v := \langle x, \mathbf{A}x \rangle$, where $\mathbf{A}$ is the admitted creators' sum for this read. At one plaquette the projections $C_i C_i^\dagger$ over the active closure are pairwise orthogonal, by $C_i^\dagger C_j = \delta_{ij} P_p$ there, and their sum sits below Q_p . The earlier support count therefore gives

$$\sum_i \langle C_i^\dagger x, C_i^\dagger x \rangle \leq [\nu : f_*] t, \quad \sum_i \langle P_p x, P_p x \rangle \leq sn.$$

Here p is the plaquette of i in the second fold. The adjoint image sits in P_p , so v is the pairing of the two lists $(C_i^\dagger x)_i$ and $(P_p x)_i$. Their squared pairing read gives

$$f_* v^2 \leq s \nu t n$$

(Construction 8's squared Cauchy–Schwarz). Every inactive creator's form on the window reads the sum's unit, since it raises past the cutoff. The magnetic split and remainder comparison then give

$$f_* \langle x, (K_{\Pi} - zI)x \rangle \geq \xi t - f_* z n - 2f_* b |v|.$$

The electric floor is $t \geq E_0 n$. Write $d_z := \xi E_0 - f_* z$, so $d_z \geq d_h(\tau) > 0$ at $0 \leq z \leq h$. The identity

$$E_0(\xi t - f_* z n)^2 - d_z^2 t n = (t - E_0 n)(\xi^2 E_0 t - f_*^2 z^2 n)$$

has its right member on its upper side or at equal members: $t \geq E_0 n$ and $\xi E_0 > f_* z \geq 0$. The stated parameter comparison and the two pairing reads give

$$E_0(\xi t - f_* z n)^2 \geq d_h(\tau)^2 t n \geq 4f_* \nu E_0 s b^2 t n \geq 4f_*^2 E_0 b^2 v^2.$$

At an occupied x , the quantity $\xi t - f_* z n$ is positive at $d_z n > 0$. The trichotomy of squares therefore gives $\xi t - f_* z n \geq 2f_* b |v|$, so the form at z is positive semidefinite on the unit's perpendicular complement. Its count is at most one (Lemma 30). Since $z = e + h > e$, the actual ground contributes at least one (Lemma 33). The count is one.

The ground-side root. The identities $E_0 = 4f_*$, $\nu \geq 2$, $e^+ \geq E_0$ and $0 < \delta < f_* E_0$ give the two positive gaps a_ρ, b_ρ . The quadratic extension's positive member q_ρ exceeds b_ρ at the trichotomy of squares (Lemma 37). Substitution of ρ_* at q_ρ 's relation gives $a_\rho \rho_*^2 + b_\rho \rho_* = \delta^2$. The polynomial

$$P(\rho) := (\delta + f_* \rho)^2 - c_\rho \rho(\rho + e^+) = \delta^2 - b_\rho \rho - a_\rho \rho^2$$

therefore reads

$$P(\rho) = \langle \rho_* : \rho \rangle (a_\rho(\rho_* + \rho) + b_\rho).$$

At $\rho \geq 0$ its second factor is positive, so its side is the first factor's, with equality at the root.

Put $u := [\xi E_0 : f_*]$ and $T := h + u$. The join $\xi E_0 = \delta + f_* h$ gives $h < u$ and $\delta + f_* T = 2\xi E_0$. Also

$$c_\rho T(T + e^+) > c_\rho u^2 = 4\xi^2 E_0^2 [\nu E_0 : f_*] > 4\xi^2 E_0^2.$$

Thus $P(T) < 0$ and $\rho_* < T$. At $0 \leq \rho \leq \rho_*$ and $z = h - \rho$ one consequently has $f_*^2 z^2 < \xi^2 E_0^2$ and $d_z = \xi E_0 - f_* z = \delta + f_* \rho > 0$. For every x perpendicular to the unit, the determinant and squared-pairing reads above give

$$sb^2 \leq \rho(\rho + e^+), \quad f_* v^2 \leq s \nu t n,$$

at $\rho = e^\vee$, with $t \geq E_0 n$. The displayed square identity has its right member on its upper side or at equal members at $f_*^2 z^2 < \xi^2 E_0^2$. Since $P(\rho) \geq 0$, it gives

$$E_0(\xi t - f_* z n)^2 \geq d_z^2 t n \geq c_\rho \rho(\rho + e^+) t n \geq 4f_*^2 E_0 b^2 v^2.$$

At an occupied x the quantity before squaring is $\xi(t - E_0 n) + d_z n > 0$. The trichotomy of squares gives $\xi t - f_* z n \geq 2f_* b|v|$. The magnetic split's comparison therefore puts the form at z on its upper side or at equal members on the unit's perpendicular complement. Its count is at most one, and the ground below $e + h$ gives at least one (Lemma 30; Lemma 33). At $\delta^2 \geq c_\rho h(h + e^+)$ the read

$$P(h) = (\delta + f_* h)^2 - c_\rho h(h + e^+) > \delta^2 - c_\rho h(h + e^+) \geq 0$$

gives $\rho_* > h$. For $\rho > \rho_*$ the unit pivot $-z = \rho - h$ exceeds $\rho_* - h$ at the sum's injectivity (Definition 1).

For the common segment, $E_0 = 4f_*$ and $\nu \geq 2$ give $c_\rho = 16\nu f_*^2 > f_*^2$, so a_0 is a positive gap. The displayed b_0 exceeds $2f_*^2 E_0$, giving $0 < h_0 < [E_0 : 4]$ at the cross-multiplied read. The polynomial identity at the offset is

$$f_*^2 \langle E_0 : h \rangle^2 - c_\rho h(h + e^+) = k_0 - b_0 h - a_0 h^2.$$

At $h = h_0 \leq E_0$, its second and third terms satisfy

$$b_0 h_0 + a_0 h_0^2 \leq (b_0 + a_0 E_0) h_0 = [k_0 : 2].$$

Thus the first display exceeds the sum's unit at h_0 . The positive quadratic root q_0 is strictly below $f_* \langle E_0 : h_0 \rangle$ by the trichotomy of squares, giving $w_0 > 0$. The quadratic extension at $[w_0 : \nu d_L E_0]$ gives $T_0 > 0$ (Lemma 37). For $\tau \leq T_0$, the divisor reads

$$d_{h_0}(\tau) = q_0 + \nu d_L E_0 \langle T_0^2 : \tau^2 \rangle,$$

whose second term is on its upper side or at equal members. This gives both comparisons, with the earlier count displays read at h_0 and τ .

The count on the input lists. Join the rectangular identities of Theorem 9, at $\alpha = 1$ and $\beta = \tau^2$, over the input lists. They give

$$(K_\Pi - zI)\mathcal{C} = \mathcal{C}\mathcal{T}_z - \tau^2 \mathcal{J}.$$

Multiplication by $\mathcal{C}^\top G$ gives the stated range form; its transpose gives the further identity at the symmetry of $H - zG$. Every input window is a compression of the output window, so $K_i^- - eI$ has a positive-semidefinite read in its pairing (Lemma 30). Each $e_i \geq E_0$, the member floor's comparison, so $e_i - \gamma$ is positive at $\gamma < E_0$ and the stated input action is positive definite.

The map $(s, x) \mapsto s\mathbf{1} + \mathcal{C}x$ has image $\mathcal{W}_C$. Its pullback reads the deflated form's count: a negative subspace in the domain maps injectively to its image, since a kernel vector reads the form at the sum's unit, and every negative subspace in $\mathcal{W}_C$ lifts at a right solve, the image collection's Gram-dual solve (Lemma 3; Lemma 30). Thus the two maximal negative counts agree. The kernel of this map is the input kernel of $\mathcal{C}$, the unit perpendicular to its image, so it contributes a kernel block.

The unit row of the complementary coupling is at the sum's unit. Its range pairing at the joined creator list is $-\tau^2 \mathbf{r}^\top$: at every admitted creator,

$$\mathcal{C}_i^\dagger \sum_j \mathcal{C}_j \mathbf{1} = \mathbf{1}$$

on its input window. At its own plaquette the character orthogonality reads the matching label once; at a different plaquette an unshared boundary link puts the adjoint image at the sum's unit (Theorem 9; Construction 8). Pairing the input unit against its window list gives the stated column. Pair the joined rectangular identity against Z . Since $Z^\top G \mathcal{C} = 0$, it reads

$$Z^\top H \mathcal{C} = -\tau^2 L_N.$$

The complementary deflation's correction is therefore $\tau^4 L_N^\top Y_N$, symmetric and positive semidefinite at $L_N^\top Y_N = Y_N^\top S_N(z) Y_N$. This gives the displayed pullback matrix.

For the Gram recursion, the adjoint and creator images fit in their stated windows, giving

$$G_i^- \mathcal{C}_i^\dagger \mathcal{C}_j = \Gamma_{ij}, \quad \mathcal{C}_i^\dagger \mathcal{C}_j = \Pi_i^- \mathcal{C}_i^\dagger \mathcal{C}_j \Pi_j^-.$$

Split the last product as $\mathcal{C}_j \mathcal{C}_i^\dagger + [\mathcal{C}_i^\dagger, \mathcal{C}_j]$. The adjoint $\mathcal{C}_i^\dagger$ sends a content c in Π_j^- to a content c' at $c' + e_i = c$, so $c' + e_i + e_j \leq C$. Its image is therefore in Π_{ij}^- , and the following $\mathcal{C}_j$ sends it into Π_i^- . The product is precisely $\mathcal{C}_{j|i} \mathcal{C}_{i|j}^\dagger$. At $C < e_i + e_j$ that intermediate list is vacant. The local commutator's disjoint-boundary read gives the last assertion.

The parity elimination. At $d \geq 2$, let $x_j(v)$ be the coordinate count of a vertex v from the box's lower face in direction j . For a link in direction i from v , let η_ℓ be the parity of $\sum_{j < i} x_j(v)$. Reversing the link keeps that parity. On a plaquette in directions $i < j$, the two direction- i parities agree and the two direction- j parities differ by one, so the boundary fold reads odd. On the square chain, assign odd parity to each square's least boundary link belonging to that square alone, and even parity to the further links. Its boundaries each read one odd link. Both assignments are finite index reads (Construction 7; Definition 1).

At a configuration a , define its grade as the parity of $\sum_\ell \eta_\ell n(a_\ell)$, where n is the label's row length at residue one, and the unit label reads the vacant count. The vertex and fusion class laws are the parity of this row length (Construction 6). Fusion by f at a plaquette changes every boundary link's class by one and fixes the further link classes, so the two grades of every occupied magnetic entry differ by the odd boundary fold. Its blocks within either grade are the sum's unit. The electric and Gram forms preserve the grades, being diagonal on configurations, and the unit has even grade (Definition 8; Definition 10). The total-content projection preserves this split at every cutoff.

The odd configurations are occupied, each with content at least E_0 (Proposition 7). Thus for $z < E_0$ each block $(c_a - z)G_a$ is positive definite. Each block's solve is the Gram's adjugate against its determinant and the positive scalar gap. The direct sum gives the displayed X . The odd-block deflation is $E_+ - zG_+ - \tau^4 BX$, symmetric at $BX = X^\top D_- X$. Inertia adds at this positive pivot with its reversal count the sum's unit (Lemma 30), giving the stated count. At an occupied

plaquette count, $C \geq E_0$ and $\tau > 0$, the actual-ground compression above gives $\varepsilon_0 < 0$, so every line $\varepsilon_0 + E_0\kappa$ at $\kappa \leq 1$ lies below E_0 and has this read. The unit-only and free-end windows have their direct diagonal reads.

For the two-line read, subtract the two solved identities $D_-(e)X_e = B^\top$ and $D_-(e + \gamma)X_{e+\gamma} = B^\top$. Since $D_-(e) = D_-(e + \gamma) + \gamma G_-$, they give

$$X_{e+\gamma} = X_e + \gamma Y_\gamma, \quad F_e = F_{e+\gamma} + \gamma W_\gamma.$$

On the odd configuration a , the form BY_γ has the contribution

$$[1 : (c_a - e)(c_a - e - \gamma)] B_a G_a^{-1} B_a^\top,$$

where B_a is the corresponding column block and G_a^{-1} denotes its adjugate solve. Both scalar gaps are positive. Each contribution is positive semidefinite, the Gram-dual square at a positive scalar, so their fold joined to G_+ is positive definite (Lemma 30). The parity count at $e + \gamma$ and the displayed difference give the two-line identity. At e , the full level-gap form is positive semidefinite (Definition 10), and the odd block is positive definite; its deflation F_e has the positive read by inertia. Its lower kernel row forces the odd coordinate to be $\tau^2 X_e v$, and its upper row is $F_e v = 0$, giving the stated kernel correspondence. $\square$

Lemma 65 (The four-point floor). *Fix two plaquettes sharing one link, the theta graph (Construction 9), a closure member λ_0 of the magnetic list (Construction 6, the member's stated list's first member), the pair probe $\varphi := \sum_{c \in P} \chi_c$ over the label set $P := \{\lambda_0, \bar{\lambda}_0, 2\lambda_0, 2\bar{\lambda}_0\}$, the Cartan squares at the Cartan strictness (Construction 6), involution-fixed at the duals and free of the unit label, $q_i := \varphi(U_{\partial p_i})$ the pair's modules, and the ray $[1 : \tau^2]$ at a window ground ψ beyond the graph's locality band (Lemma 14). Write $m_a := \text{Eval}(\varphi^a)$ for the probe's free moments, $m_0 = 1$, m_1 the sum's unit at the nonunit members, $m_2 = \#P$ at the orthonormal characters (Proposition 8), and $T_a(k)$ for the k -count of the a -th power, the coefficient of χ_k in φ^a , a natural fold of fusion counts, $T_0(k) = \delta_{k1}$, $T_1(k)$ one at $k \in P$, and $T_2(\lambda_0) \geq 2$ (Proposition 8). The connected four-point read at the pair, each field twice, is the moments' partition fold at $\mu_{ab} := \omega(q_1^a q_2^b)$, one term per partition of the four fields at the blocks' moment product, the weight $j!$ at the join $j + 1 = b$ over the block count b , the one-block term at weight one, and the side the count's parity,*

$$\varkappa_4 := \langle \mu_{22} + 2\mu_{10}^2 \mu_{02} + 2\mu_{01}^2 \mu_{20} + 8\mu_{10} \mu_{01} \mu_{11} : 2\mu_{10} \mu_{12} + 2\mu_{01} \mu_{21} + \mu_{20} \mu_{02} + 2\mu_{11}^2 + 6\mu_{10}^2 \mu_{01}^2 \rangle.$$

At every member the read holds a positive located floor on a cell at the free end's side. Write $a_\lambda := [1 : 4\hat{c}_2(\lambda)]$ for a closure member's loop weight, the plaquette content's cofactor (Proposition 7); at two closure members λ, μ the pair networks $F_k^{\lambda\mu}$ are the channel split of the adjacent product $\chi_\lambda(U_{\partial p_1}) \chi_\mu(U_{\partial p_2}) = \sum_k F_k^{\lambda\mu}$ over the shared link's channels $k \in \lambda \otimes \bar{\mu}$, the electric reads $e_k^{\lambda\mu} := 3\hat{c}_2(\lambda) + 3\hat{c}_2(\mu) + \hat{c}_2(k)$ and the weights $w_k^{\lambda\mu} := \langle F_k^{\lambda\mu}, F_k^{\lambda\mu} \rangle = [m_k d_k : d_\lambda d_\mu]$ (Theorem 24(i),(ii) at the pair's labels; Lemma 5(ii)), the weights folding to one and their electric fold to the distant read $e_\infty^{\lambda\mu} := 4\hat{c}_2(\lambda) + 4\hat{c}_2(\mu)$, the two closure identities $\sum_k w_k^{\lambda\mu} = 1$ and $\sum_k w_k^{\lambda\mu} \hat{c}_2(k) = \hat{c}_2(\lambda) + \hat{c}_2(\mu)$ (the dimension identity, Lemma 5(ii); the uniform drift at the second factor, Theorem 25); and the pair excess

$$J_{\lambda\mu} := \sum_k w_k^{\lambda\mu} \langle [1 : e_k^{\lambda\mu}] : [1 : e_\infty^{\lambda\mu}] \rangle, \quad e_\infty^{\lambda\mu} J_{\lambda\mu} = \sum_{k < l} w_k^{\lambda\mu} w_l^{\lambda\mu} \langle e_k^{\lambda\mu} : e_l^{\lambda\mu} \rangle^2 [1 : e_k^{\lambda\mu} e_l^{\lambda\mu}],$$

every summand a square at pair weights, positive by its shape at distinct channel reads. Then the read sits within the priced tail's bracket of the jet's value

$$\hat{\varkappa}_4 = 2\tau^4 \sum_{\lambda, \mu \in \bar{L}} (a_\lambda + a_\mu) T_2(\lambda) T_2(\mu) J_{\lambda\mu},$$

whose side is its upper at every member, the margin at or beyond the diagonal pair's read at λ_0 ,

$$\hat{\varkappa}_4 \geq \tau^4 \left[d_\theta : 12 d_{\lambda_0}^4 \hat{c}_2(\lambda_0)^3 (6 \hat{c}_2(\lambda_0) + 1) \right] ,$$

the diagonal pair's excess read at its unit channel against its adjoint channel, the adjoint occupied once at least in $\lambda_0 \otimes \bar{\lambda}_0$ (the generators' copy, Lemma 18's move identity).

The magnetic jet. The ground's jet at the free end is the displayed second-order solve (Construction 14's shape): $\hat{\psi} := 1 + \tau^2 \psi_1 + \tau^4 \psi_2$ at $\psi_1 = \sum_p \sum_{\lambda \in \bar{L}} a_\lambda \chi_\lambda(U_{\partial p})$, each closure member's loop an eigenvector at its plaquette content $4\hat{c}_2(\lambda)$ (Proposition 7; Lemma 18's plaquette character the members' fold), and ψ_2 the solve at $\mathbf{M}\psi_1$'s components off the unit line: a single-plaquette channel $\chi_k(p)$ at the content $4\hat{c}_2(k)$, k a nonunit member of $\lambda \otimes \mu$ at closure members, an adjacent pair's product $\chi_\lambda(U_{\partial p})\chi_\mu(U_{\partial p'})$ at its channel split $\sum_k F_k^{\lambda\mu}$ over the shared link's channels at the contents $e_k^{\lambda\mu}$, entering $\mathbf{M}\psi_1$ at the coefficient $a_\lambda + a_\mu$ (the one plaquette's term against the other's loop, either way), and a distant product at $e_\infty^{\lambda\mu}$, each component's solve the content's cofactor; the unit coordinate of $\mathbf{M}\psi_1$ reads $\#p \sum_\lambda a_\lambda$, each plaquette's term against its own loops at the orthonormal characters.

The moments' displayed orders. On the theta graph $\text{Eval}(f(U)g(V))$ is the two unit coefficients' product at every pair of loop polynomials f, g (Theorem 24(i); Proposition 11), the graph's free moments are the per-plaquette reads $m_a m_b$, and the jet moments read the pair's insertions alone, the far diagonal reads entering the pairing and the gram equally (Lemma 14; Definition 1): the jet moments read

$$\begin{aligned} \hat{\mu}_{ab} &:= m_a m_b + 2\tau^2 \sum_{\lambda \in \bar{L}} a_\lambda (T_a(\lambda) m_b + m_a T_b(\lambda)) + \tau^4 D_{ab} , \\ D_{ab} &= D_{ab}^{\text{sq}} + D_{ab}^{\text{ch}} + 2 \sum_{\lambda, \mu \in \bar{L}} (a_\lambda + a_\mu) T_a(\lambda) T_b(\mu) \sum_k w_k^{\lambda\mu} [1 : e_k^{\lambda\mu}] , \end{aligned}$$

D_{ab}^{sq} the pair insertions' square reads, the fold of ψ_1 's square against the observable less the gram's own read of that square against $m_a m_b$, each summand a product of a U -read and a V -read, D_{ab}^{ch} the single-plaquette channels' solves, products so as well, and the third fold the pair networks', at $\langle q_1^a q_2^b, F_k^{\lambda\mu} \rangle = T_a(\lambda) T_b(\mu) w_k^{\lambda\mu}$.

The collection. The moments' product parts read a product state: a moment family $u_a v_b$ at two reads reads the partition fold at equal members outright, one polynomial identity in the moments; the jet moments are that family joined to the pair networks' fold at the distant read withdrawn, the pair networks' product part $\sum_k w_k^{\lambda\mu} [1 : e_\infty^{\lambda\mu}]$, the distant product's own solve, absorbed into the product family, so

$$\hat{\mu}_{ab} = u_a(\tau) v_b(\tau) + 2\tau^4 \sum_{\lambda, \mu} (a_\lambda + a_\mu) T_a(\lambda) T_b(\mu) J_{\lambda\mu} + \tau^6(\dots) ,$$

and the partition fold at the jet moments reads its first variation at the product family, $\varkappa_4(\hat{\mu}(\tau)) = \hat{\varkappa}_4 + \tau^6 R(\tau)$, R the fold's keys three to eight at the displayed orders, the collection its keys through two. The adjoint walk's instance (Construction 15), the probe χ_θ at the adjoint's own loops, reads $\hat{\varkappa}_4 = \tau^4 c_1^2 \langle S : [1 : 8] \rangle$ at $S := \sum_k w_k [1 : 6 + \hat{c}_2(k)]$ over Lemma 24' list, $T_2(\theta) = c_1$ and $2 \cdot 2a_\theta = 1$.

The tail, and the cell. The jet's identity is exact at the displayed join,

$$\mathbf{E} \hat{\psi} + \tau^4 \hat{\varepsilon}' \mathbf{1} + \tau^6 \mathbf{M} \psi_2 = \tau^2 \mathbf{M} \hat{\psi} \quad \text{at} \quad \hat{\varepsilon}' := \#p \sum_{\lambda \in \bar{L}} a_\lambda ,$$

so the jet solves the pencil to the τ^6 read, the pencil's form H at the ray and its Gram G reading

$$H\hat{\psi} = \hat{\varepsilon}G\hat{\psi} + G\rho, \quad \rho := \tau^6(\hat{\varepsilon}'\psi_1 + (\mathbf{M}\check{\psi}_2)) + \tau^8\hat{\varepsilon}'\psi_2,$$

the jet level $\hat{\varepsilon}$ the balance partner of $\tau^4\hat{\varepsilon}'$ and ρ the residual, the check the memberwise swap. The window ground ψ at its level ε_0 , the level-gap action D (Definition 10) and the clearance γ' , every further root at or beyond it on the free cell (Lemma 15; Lemma 64), prices the jet's part φ' off the ground's line (Theorem 21(iii)),

$$(\gamma' + \langle \varepsilon_0 : \hat{\varepsilon} \rangle) \langle \varphi', \varphi' \rangle \leq \langle \varphi', \rho \rangle,$$

and the drift $\langle \varepsilon_0 : \hat{\varepsilon} \rangle$ is floored at the magnetic member's cap $W := \#p d_L$ (Lemma 18 per term at the closure's dimension fold; Theorem 30's cap at the ray's magnetic weight), so

$$\langle \gamma' : \tau^2 W \rangle^2 \langle \varphi', \varphi' \rangle \leq \langle \rho, \rho \rangle \quad \text{at } \tau^2 W < \gamma'.$$

The residual's weight is the caps' fold: at the first-order weight $\langle \psi_1, \psi_1 \rangle = p_1 := \#p \sum_{\lambda \in \bar{L}} a_\lambda^2$, the free end's level $\ell := E_0 = 4\hat{c}_2(f_0)$ (Lemma 15) and the solve's cap, the cofactor Z at $\ell^2 Z = W^2 p_1$ with $\langle \psi_2, \psi_2 \rangle \leq Z$,

$$\langle \rho, \rho \rangle \leq \tau^{12} Q(\tau), \quad Q(\tau) := 4\hat{\varepsilon}'^2 p_1 + 4W^2 Z + 2\tau^4 \hat{\varepsilon}'^2 Z.$$

The moments' brackets read at two states, ω the ground's and the jet state $\hat{\omega} := \langle \hat{\psi}, \cdot \hat{\psi} \rangle$ against $\langle \hat{\psi}, \hat{\psi} \rangle$, at the observable $A = q_1^a q_2^b$'s cap $c_A := d_\varphi^{a+b}$ at the probe's dimension fold $d_\varphi := \sum_{c \in P} d_c$, the loop cap iterated over the word, each character's form two-sided at its dimension and its square at the dimension's square (Lemma 18), both states' reads at or below the cap, each moment's bracket the transport display's width read at the jet as the later ground (Theorem 21(iii)): at the pair w at $w \langle \gamma' : \tau^2 W \rangle = \gamma' \tau^6 K(\tau)$, $K(\tau) := K_0 + \tau^2 K_1$ over the root caps $K_0^2 \geq 4\hat{\varepsilon}'^2 p_1 + 4W^2 Z$ and $K_1^2 \geq 2\hat{\varepsilon}'^2 Z$ (the pair $[x+1 : 2]$ clears every x 's square at $(x+1)^2 = 4x + \langle x : 1 \rangle^2$), the width reads

$$\gamma' |\omega(A) - \hat{\omega}(A)| \leq 4c_A w.$$

The jet state's moment is the pairing at the full jet, P_A against G at $P_A := \sum_{i,j} \tau^{2(i+j)} \langle \psi_i, A\psi_j \rangle$ over $\psi_0 := \mathbf{1}$ and $G := P_1$, degree eight, while the displayed $\hat{\mu}_{ab}$ reads P_A against G through τ^4 ; the datum N_A at $P_A = \hat{\mu}_{ab} G + N_A$ sits at the keys three to six, $|\hat{\omega}(A) - \hat{\mu}_{ab}(\tau)| \leq |N_A(\tau)|$ at $G \geq 1$, and at $\hat{\mu}_{ab}$'s three displayed coefficients $\hat{\mu}_0, \hat{\mu}_1, \hat{\mu}_2$ the magnitude fold reads

$$|N_A(\tau)| \leq \tau^6 T_A(\tau), \quad T_A := n_3 + \tau^2 n_4 + \tau^4 n_5 + \tau^6 n_6$$

at the key caps $n_3 := (c_A + |\hat{\mu}_0|)(p_1 + Z) + |\hat{\mu}_1| p_1$, $n_4 := (c_A + |\hat{\mu}_0|)Z + |\hat{\mu}_1|(p_1 + Z) + |\hat{\mu}_2| p_1$, $n_5 := |\hat{\mu}_1|Z + |\hat{\mu}_2|(p_1 + Z)$ and $n_6 := |\hat{\mu}_2|Z$. The displayed moments sit at or below the caps below their own comparisons' first located roots: per index pair the magnitude fold $|\hat{\mu}_0| + \tau^2 |\hat{\mu}_1| + \tau^4 |\hat{\mu}_2|$ reads against c_A , the comparison holding at the free end and its first located root joining the cell's list (Theorem 16). The partition fold's bracket telescopes factor by factor at the moments' brackets and the caps, the fold's factor count at the block-count weights 75, so

$$|\varkappa_4(\omega) - \varkappa_4(\hat{\mu}(\tau))| \leq 75 d_\varphi^4 \left([4w : \gamma'] + \tau^6 \sum_{(a,b)} T_{q_1^a q_2^b} \right) =: \tau^6 B(\tau),$$

and with the fold at the displayed orders, $\varkappa_4(\hat{\mu}(\tau)) = \hat{\varkappa}_4 + \tau^6 R(\tau)$ at the collection, the read sits within the tail's bracket of the jet's value,

$$|\varkappa_4 - \hat{\varkappa}_4| \leq \tau^6 \text{tail}(\tau), \quad \text{tail} := B + |R|,$$

$|R|$ the magnitude fold of R 's coefficients, and two windows' reads sit within the spectator brackets (Theorem 21(iii)), the anchor window the graph's band neighborhood. The cell is the τ -segment below the cell list's least located root, the displayed comparisons' roots joined to the comparison's (Theorem 16), the comparison the leading read against twice the tail's fold at the drift's clearing,

$$\langle \gamma' : \tau^2 W \rangle \hat{\kappa}_4 [1 : \tau^4] \quad \text{against} \quad 2\tau^2 \left(75 d_\varphi^4 (4K(\tau) + \langle \gamma' : \tau^2 W \rangle \sum T) + \langle \gamma' : \tau^2 W \rangle |R|(\tau) \right),$$

one polynomial in τ , its first root below γ' against W and the cell inside the drift's clearance outright: on it κ_4 reads its upper side at the margin $[1 : 2] \hat{\kappa}_4$ or beyond, at every window beyond the band, the floor's data located member reads.

Proof. The counts and the excess. $T_2(\lambda_0) \geq 2$: the ordered pairs $(2\lambda_0, \bar{\lambda}_0)$ and $(\bar{\lambda}_0, 2\lambda_0)$ each read $N_{2\lambda_0 \bar{\lambda}_0}^{\lambda_0} = N_{\lambda_0 \lambda_0}^{2\lambda_0} = 1$, the triple count's symmetry with the involution's read (Proposition 8) at the Cartan strictness. The pair excess's identity is the product of the electric fold against the weighted cofactor fold, collected per channel pair at the square identity $e_k^2 + e_l^2 = \langle e_k : e_l \rangle^2 + 2e_k e_l$ against the weights' squared fold. The jet's value reads its upper side at every member, every summand at or beyond the sum's unit, the loop weights and the counts naturals and each pair excess a fold of squares; and the diagonal pair's excess reads its unit channel against its adjoint channel, $k = 1$ at $e_1 = 6\hat{c}_2(\lambda_0)$ and $k = \theta$ at $e_\theta = 6\hat{c}_2(\lambda_0) + 1$, at the weights $w_1 = [1 : d_{\lambda_0}^2]$ and $w_\theta \geq [d_\theta : d_{\lambda_0}^2]$, so $e_\infty J_{\lambda_0 \lambda_0} \geq [d_\theta : d_{\lambda_0}^4 e_1 e_\theta]$ at $e_\infty = 8\hat{c}_2(\lambda_0)$, and the diagonal summand's coefficient reads $2 \cdot 2a_{\lambda_0} \cdot T_2(\lambda_0)^2 \geq [4 : \hat{c}_2(\lambda_0)]$, the displayed margin.

The displayed orders. On the theta graph the evaluation splits at the shared link and the outer word: the shared read pairs the two loops' channels by the evaluation identity and the outer word reads the unit label (Theorem 24(i); Proposition 11), the two unit coefficients' product. A far insertion factorizes at its own links (Lemma 14): a plaquette off the pair holds a link off the union, the union's four-cycles the two boundaries, so the jet moments read the pair's insertions alone, the far diagonal reads entering the pairing and the gram equally, a summand on both members fixing the value (Definition 1). A product state's pairing against the pair networks reads at the labeled free links alone, the gram's configuration split (Definition 8), the displayed $\langle q_1^a q_2^b, F_k^{\lambda\mu} \rangle$.

The collection. At a product state every partition's term reads at the fold of the U -reads' and the V -reads' own partitions entering the two sides equally (a two-variable cumulant at independent reads), the partition fold at equal members outright. The fold is a polynomial in the moments, the product family reads it at equal members, and the deviation enters at the fold's partial reads at the product family at the leading order, the partial at μ_{22} one and every further partial the sum's unit or paired against a vacant deviation (μ_{11} 's, μ_{12} 's and μ_{21} 's partials at the product family's first moments u_1, v_1 , each at the order τ^2 , their products against the deviation at the order τ^6 inside R ; μ_{20} 's and μ_{02} 's against the deviation at $T_0(\mu)$, the sum's unit at a nonunit μ), the displayed first variation.

The tail. $E\psi_1 = M1$ outright (the loops' fold, its unit coordinate the sum's unit) with $E\psi_2$ against $M\psi_1$ the join at $\hat{\varepsilon}'1$, the unit coordinate of $M\psi_1$, the displayed identity; at 1 read as ψ against $\tau^2\psi_1 + \tau^4\psi_2$ the site datum's display follows. The clearance prices the jet's part by the transport display's pairing step (Theorem 21(iii)): $\gamma'\langle\varphi', \varphi'\rangle \leq \langle\varphi', D\varphi'\rangle = \langle\varphi', D\hat{\psi}\rangle$, the ground in the gap's kernel, and $D\hat{\psi}$ reads $\langle\hat{\varepsilon} : \varepsilon_0\rangle \hat{\psi} + \rho$ at the identity, the displayed comparison. The ground's level reads $\langle\psi, E\psi\rangle$ against $\tau^2\langle\psi, M\psi\rangle$ over $\langle\psi, \psi\rangle$, the electric read on its upper side or equal members and the magnetic read capped, so ε_0 sits at or beyond the balance partner of τ^2W while $\hat{\varepsilon}$ sits at or below the sum's unit, the drift's floor, and the pairing's squared Cauchy–Schwarz (Construction 8) reads the displayed comparison at $\tau^2W < \gamma'$. The magnetic member's square reads $\langle Mx, Mx \rangle_G \leq \#p \sum_p \langle \chi_p x, \chi_p x \rangle \leq W^2 \langle x, x \rangle$, the count's fold of squares, $\#p \sum_p a_p^2 =$

$(\sum_p a_p)^2 + \sum_{p < q} \langle a_p : a_q \rangle^2$ (Definition 1), at the per-term square cap (Lemma 18); the first-order weight is p_1 , the loops orthonormal at distinct plaquettes and labels; and the solve's weight is floored at the free end's level: every component of ψ_2 sits at content at or beyond ℓ (every nonunit label at or beyond the member floor, Lemma 9), so $\ell \langle \psi_2, \psi_2 \rangle \leq \langle \psi_2, \mathbf{E}\psi_2 \rangle = \langle \psi_2, \mathbf{M}\psi_1 \rangle$ (ψ_2 off the unit line) and the squared Cauchy–Schwarz reads $\ell^2 \langle \psi_2, \psi_2 \rangle \leq W^2 p_1$, the cap Z . The squares' fold twice, $(a+b)^2 \leq 2a^2 + 2b^2$, then reads the displayed $Q(\tau)$. Each moment's bracket is the transport display's width read at the jet as the later ground, one polarization each (Theorem 21(iii)): the display's first read holds, $\gamma'^2 \langle \varphi', \varphi' \rangle \leq w^2 \langle \hat{\psi}, \hat{\psi} \rangle$ at $\langle \hat{\psi}, \hat{\psi} \rangle \geq 1$ (ψ_1 and ψ_2 off the unit line), the displayed width. The datum N_A 's keys are capped at the weights: P_A 's key three, $2\langle \psi_1, A\psi_2 \rangle$, sits at or below $c_A(p_1 + Z)$ in magnitude (the polarization at weight one) and its key four, $\langle \psi_2, A\psi_2 \rangle$, at or below $c_A Z$; G 's key two is p_1 , its key three, $2\langle \psi_1, \psi_2 \rangle$, sits at or below $p_1 + Z$ and its key four at or below Z , the displayed key caps. At the free end $|\hat{\mu}_0| = m_a m_b$, one at $(0,0)$ and below the cap at an occupied power (m_1 the sum's unit, $m_2 = \#P$ at or below four against d_φ^2 at or beyond 25, the member at dimension two or beyond (Theorem 25's strictness) and its square at three or beyond, the square's pair at an occupied simple key at or beyond two so its string there holding three contents (Lemma 6), and m_a at or below the flat read d_φ^a with the unit coefficient below it at a nonunit member, Construction 8), so the comparison holds there. The telescope: at a term's moment product, one factor moved from ω 's read to $\hat{\mu}$'s at a time, $|\prod_i \mu_i - \prod_i \hat{\mu}_i| \leq \sum_i \delta_i \prod_{j \neq i} c_j$ at the moments' brackets δ_i and the caps, the ω factors' the states' reads and the $\hat{\mu}$ factors' the displayed comparisons' below the cell's list; every term has field count four, so a factor's move at (a_i, b_i) costs $\delta_i d_\varphi^{4-a_i-b_i}$ at $\delta_i \leq [4c_i w : \gamma'] + \tau^6 T_i$, and the fold's factor count at the block-count weights is 75, one per moment factor of every term (the one-block term one, the seven two-block terms two each, the six three-block terms three each at weight two, the four-block term four at weight six), the displayed $B(\tau)$ at $d_\varphi^{4-a-b} T \leq d_\varphi^4 T$. The cell's comparison is positive at the free end (the side, the jet's value's coefficient at or beyond the displayed member pair) and on its lower side at $\tau^2 W = \gamma'$ (the left member of equal members against an occupied right), so its first root sits below γ' against W . $\square$

22 The reconstruction

Theorem 31 (The spectral bracket reconstruction). *The Euclidean bracket families have the following spectral reads over the located stage. The data are the bracketed ground reads (Theorem 21), the restoration counts with the cone read (Theorem 22; Lemma 63), and the pairing's definitional positivity (Definition 8), per member and direction count at every window of the corner cell.*

(i) The spectral data are located divisor data. Per fiber the energy roots with their weights are the split's own (Lemma 33), the Euclidean families read as the weight folds over them (Theorem 21(v)), and on the clearance cell the moment folds' solve returns the truncated moment vector from the streams' reads (Lemma 62), the located divisor read its clearance: the divisor data read the Euclidean families as their weight folds, one solved witness per family, the lower's and the upper's (Theorem 21(v)).

(ii) The root loci through degree two. Per probe pair the family is the weight fold over the located roots at the fiber's chord, at the probe's spectral weights from (i). The roots' momentum reading is the cone's: at every located root the locus at the moved level reads the isotropic square at degree two with the root's located coefficient, the odd degrees at the sum's unit and the direction data at degree four (Lemma 63). The families are window-free within the spectator brackets (Theorem 21(iii)), the growth caps their m -point folds' reads (Theorem 21(v)).

(iii) The state reads close the families. The pairing's positivity is the families' own Gram read

at every finite probe list, every split of the Gram on the upper side throughout (Lemma 30); the window's energy gap is its cut (Lemma 20), the word sector is the state's space, cyclic (Lemma 21), and every read holds at every window ground at once: each output is a located bracket read of the inputs, one solve and one count per datum.

Proof. (i) The split's congruence diagonalizes the gram and the level gap at the located roots (Lemma 33), a root's weight the coordinate's square against the gram's diagonal entry, and the Euclidean members are the two solved witnesses' N -fold reads, diagonal at the gaps, the lower's $[g_\gamma : N]$ and the upper's $[N : N + u\gamma]$ per root (Theorem 21(v)), so a family is the weight fold of those reads over the roots; the moment folds' square system solves at the elimination's adjudicate against its determinant on the clearance cell (Lemma 62(ii)), the determinant's clearance the located divisor read (Lemma 31). (ii) Per probe pair (i)'s fold runs at the fiber's chord, and each root's locus at the moved level reads $S'_0(\varepsilon)a + S_2(\varepsilon)$ against the isotropic square with the direction data two degrees down (Lemma 63), the root's degree-two identity. The families are window-free because two windows' grounds differ by the transport display's off-line part, capped at the drift over the tied depths (Theorem 21(iii)), and the m -point folds' magnitudes fold the head caps against the separations' certificate factors (Theorem 21(v)). (iii) A family's Gram at a probe list, its reads at the probe pairs, prices its form at coefficient data as the family's read at the probes' combination, the pairing bilinear (Definition 1), and that read is (i)'s weight fold at the combination's coordinate squares, every weight the gram's positive scale against the root's clearing and every entry the upper's or the lower's read on its upper side at the certificate's arms (Theorem 21(v)), so every split of the Gram reads the upper side throughout (Lemma 30); the window's energy gap is its cut because a window's spectral read is its cut at every ground (Lemma 20), and the word sector is the state's space at $\omega(\bar{A}B) = \langle A\psi, B\psi \rangle$ (Lemma 21). $\square$

Pending 2 (Spacetime covariance of the continuum correlations). At each member and magnetic list of Theorem 4, each direction count $d \geq 2$, and the corner trajectory at its unit map (Lemma 47; Remark 1), construct the following data over the ground pairs (Definition 1).

The spacetime words and their identification. Ordered words of the probes with their spacetime supports, their product, involution and unit, with v_P the vector at the word P . Each component of a pairing is a compatible family of brackets at every requested positive pair width, the endpoints ground pairs or elements of a finite located stage (Lemma 32; Lemma 37), the two components the located-pair reading of Theorem 21(v). The pairing is additive in each slot, a scalar in the first slot reading at its involution image and in the second at itself. Each finite combination's self-pairing has its first component on its upper side or at equal members, and its second at equal members; the pairings agree under list inclusions, and the unit word's self-pairing reads one. These identities and orders have componentwise bracket-comparison witnesses at every positive pair width. The words include the time translates of each probe. Their time evolution has the positive-time Euclidean reads of Theorem 21(v), at the spectral weights of Theorem 31(i). The identification with the lattice probes includes each support's site data and each probe's clearing under the unit map, and a bracket at every requested positive pair width, with constructed scale keys, region keys at each admitted scale, and cutoff keys at each such region. The actual lattice correlation reads lie in that bracket at every admitted scale, region and cutoff beyond their respective keys (Theorem 23's data).

The transport. A spacetime point is a matched list of $d + 1$ balance pairs over the scalar pairs. An affine map is the displayed read

$$x' = [1 : q](Ax + b), \quad A^T J A = q^2 J, \quad \det A = q^{d+1}, \quad A_{00} \geq q,$$

at a natural q , a square list A and a matched list b over the balance pairs of counts; J is diagonal, one at the time key and its balance partner at each spatial key. The map transports the words with their supports and probe indices. The transports read identity, composition, product and involution at the words' located reads. The map at

$$q' = q^{d+1}, \quad A' = q \operatorname{adj}(A), \quad b' = (\operatorname{adj}(A)b)^\vee$$

reads the two composites at the identity, the superscript the componentwise balance partner, and the two transports compose to the identity as well.

The pairing comparisons. At every finite word list, every displayed affine map and every positive pair ϵ , construct scale keys, region keys at each admitted scale, cutoff keys at each such region, and bracket-refinement keys, with their comparison witnesses. At every admitted datum beyond its keys, the two orientations of each component of the transported-pairing difference sit at or below ϵ , with the identification's widths included. Writing P, Q for words and gP, gQ for their transports, the pairing's component comparisons at every positive pair width read

$$\langle v_{gP}, v_{gQ} \rangle = \langle v_P, v_Q \rangle,$$

with the unit word fixed. These are the full correlation reads at the actual probes' weights.

The continuity moduli. At every displayed affine map, every word and positive pair ϵ , construct a positive pair width about that map in the coefficient reads $[1 : q]A$ and $[1 : q]b$. The bracket controls the self-pairing of the difference of the transported vectors at or below ϵ^2 . Every affine datum at located coordinates obeying the displayed identities is read by a bracket family of displayed maps at every positive pair width. The word reads extend along those bracket families, with the composition, pairing and support identities throughout. The spacetime translations' time reads agree with the words' Hamiltonian time evolution. The interpretation on the completed state space is the invariance item of Remark 1.

23 The quasi-local algebra

Definition 15 (The quasi-local algebra). $\mathfrak{A}$ is the carrier of the infinite lattice with its filtration by least windows: every element has one (Proposition 4), so $\mathfrak{A}$ is the union of the regions' local algebras. A winding operator exists only on the finite torus: its index support wraps a cycle of the torus, while every configuration of the infinite lattice's index has finite support (Definition 8), so winding operators lie outside every local algebra and outside $\mathfrak{A}$ with them.